HL7 FHIR R4 — Chapter 4: Terminology Module — part 6 of 6 (sections 4.4.2.112 – 4.12.11)

Source: HL7 FHIR R4 (v4.0.1) specification, https://hl7.org/fhir/R4/ — text-only rendition for AI ingestion; page order follows toc.html.

4.4.2.112 HL7 v3 Code System RouteOfAdministration

URL: https://hl7.org/fhir/R4/v3/RouteOfAdministration/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-RouteOfAdministration) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-RouteOfAdministration
Version: | 2018-08-12
Name: | v3.RouteOfAdministration
Title: | v3 Code System RouteOfAdministration
Definition: | The path the administered medication takes to get into the body or into contact with the body.
OID: | 2.16.840.1.113883.5.112 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: Immunization Route Codes (The value set to instantiate this attribute should be drawn from a terminologically robust code system that consists of or contains concepts to support describing the administrative routes used during vaccination. This value set is provided as a suggestive example.)

• ValueSet: Enteral Route Codes (EnteralRouteOfAdministration: Codes specifying the route of administration of enteral formula. This value set is composed of HL7 V3 codes and is provided as a suggestive example.)

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• ValueSet: v3 Code System RouteOfAdministration (The path the administered medication takes to get into the body or into contact with the body.)

• ValueSet: Immunization Route Codes (The value set to instantiate this attribute should be drawn from a terminologically robust code system that consists of or contains concepts to support describing the administrative routes used during vaccination. This value set is provided as a suggestive example.)

4.4.2.112.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_RouteByMethod) Abstract | | Route of substance administration classified by administration method.
2 | SOAK | Immersion (soak) | Immersion (soak)
2 | SHAMPOO | Shampoo | Shampoo
2 | TRNSLING | Translingual | Translingual
2 | PO | Swallow, oral | Swallow, oral
2 | GARGLE | Gargle | Gargle
2 | SUCK | Suck, oromucosal | Suck, oromucosal
2 | (_Chew) Abstract | | Chew
3 | CHEW | Chew, oral | Chew, oral
2 | (_Diffusion) Abstract | | Diffusion
3 | EXTCORPDIF | Diffusion, extracorporeal | Diffusion, extracorporeal
3 | HEMODIFF | Diffusion, hemodialysis | Diffusion, hemodialysis
3 | TRNSDERMD | Diffusion, transdermal | Diffusion, transdermal
2 | (_Dissolve) Abstract | | Dissolve
3 | DISSOLVE | Dissolve, oral | Dissolve, oral
3 | SL | Dissolve, sublingual | Dissolve, sublingual
2 | (_Douche) Abstract | | Douche
3 | DOUCHE | Douche, vaginal | Douche, vaginal
2 | (_ElectroOsmosisRoute) Abstract | | Electro-osmosis
3 | ELECTOSMOS | Electro-osmosis | Electro-osmosis
2 | (_Enema) Abstract | | Enema
3 | ENEMA | Enema, rectal | Enema, rectal
3 | RETENEMA | Enema, rectal retention | Enema, rectal retention
2 | (_Flush) Abstract | | Flush
3 | IVFLUSH | Flush, intravenous catheter | Flush, intravenous catheter
2 | (_Implantation) Abstract | | Implantation
3 | IDIMPLNT | Implantation, intradermal | Implantation, intradermal
3 | IVITIMPLNT | Implantation, intravitreal | Implantation, intravitreal

3 | SQIMPLNT | Implantation, subcutaneous | Implantation, subcutaneous
2 | (_Infusion) Abstract | | Infusion
3 | EPI | Infusion, epidural | Infusion, epidural
3 | IA | Infusion, intraarterial catheter | Infusion, intraarterial catheter
3 | IC | Infusion, intracardiac | Infusion, intracardiac
3 | ICOR | Infusion, intracoronary | Infusion, intracoronary
3 | IOSSC | Infusion, intraosseous, continuous | Infusion, intraosseous, continuous
3 | IT | Infusion, intrathecal | Infusion, intrathecal
3 | IV | Infusion, intravenous | Infusion, intravenous
4 | IVC | Infusion, intravenous catheter | Infusion, intravenous catheter
4 | IVCC | Infusion, intravenous catheter, continuous | Infusion, intravenous catheter, continuous
4 | IVCI | Infusion, intravenous catheter, intermittent | Infusion, intravenous catheter, intermittent
4 | PCA | Infusion, intravenous catheter, pca pump | Infusion, intravenous catheter, pca pump
3 | IVASCINFUS | Infusion, intravascular | Infusion, intravascular
3 | SQINFUS | Infusion, subcutaneous | Infusion, subcutaneous
2 | (_Inhalation) Abstract | | Inhalation
3 | IPINHL | Inhalation, respiratory | Inhalation, oral
4 | ORIFINHL | Inhalation, oral intermittent flow | Inhalation, oral intermittent flow
4 | REBREATH | Inhalation, oral rebreather mask | Inhalation, oral rebreather mask
3 | IPPB | Inhalation, intermittent positive pressure breathing (ippb) | Inhalation, intermittent positive pressure breathing (ippb)
3 | NASINHL | Inhalation, nasal | Inhalation, nasal
4 | NASINHLC | Inhalation, nasal cannula | Inhalation, nasal, prongs
3 | NEB | Inhalation, nebulization | Inhalation, nebulization
4 | NASNEB | Inhalation, nebulization, nasal | Inhalation, nebulization, nasal
4 | ORNEB | Inhalation, nebulization, oral | Inhalation, nebulization, oral
3 | TRACH | Inhalation, tracheostomy | Inhalation, tracheostomy
3 | VENT | Inhalation, ventilator | Inhalation, ventilator
3 | VENTMASK | Inhalation, ventimask | Inhalation, ventimask
2 | (_Injection) Abstract | | Injection
3 | AMNINJ | Injection, amniotic fluid | Injection, amniotic fluid
3 | BILINJ | Injection, biliary tract | Injection, biliary tract
3 | CHOLINJ | Injection, for cholangiography | Injection, for cholangiography
3 | CERVINJ | Injection, cervical | Injection, cervical
3 | EPIDURINJ | Injection, epidural | Injection, epidural
3 | EPIINJ | Injection, epidural, push | Injection, epidural, push
3 | EPINJSP | Injection, epidural, slow push | Injection, epidural, slow push
3 | EXTRAMNINJ | Injection, extra-amniotic | Injection, extra-amniotic
3 | EXTCORPINJ | Injection, extracorporeal | Injection, extracorporeal
3 | GBINJ | Injection, gastric button | Injection, gastric button
3 | GINGINJ | Injection, gingival | Injection, gingival
3 | BLADINJ | Injection, urinary bladder | Injection, urinary bladder
3 | ENDOSININJ | Injection, endosinusial | Injection, endosinusial
3 | HEMOPORT | Injection, hemodialysis port | Injection, hemodialysis port
3 | IABDINJ | Injection, intra-abdominal | Injection, intra-abdominal
3 | IAINJ | Injection, intraarterial | Injection, intraarterial
4 | IAINJP | Injection, intraarterial, push | Injection, intraarterial, push
4 | IAINJSP | Injection, intraarterial, slow push | Injection, intraarterial, slow push
3 | IARTINJ | Injection, intraarticular | Injection, intraarticular
3 | IBURSINJ | Injection, intrabursal | Injection, intrabursal
3 | ICARDINJ | Injection, intracardiac | Injection, intracardiac
4 | ICARDINJRP | Injection, intracardiac, rapid push | Injection, intracardiac, rapid push
4 | ICARDINJSP | Injection, intracardiac, slow push | Injection, intracardiac, slow push
4 | ICARINJP | Injection, intracardiac, push | Injection, intracardiac, push
3 | ICARTINJ | Injection, intracartilaginous | Injection, intracartilaginous
3 | ICAUDINJ | Injection, intracaudal | Injection, intracaudal
3 | ICAVINJ | Injection, intracavernous | Injection, intracavernous
3 | ICAVITINJ | Injection, intracavitary | Injection, intracavitary
3 | ICEREBINJ | Injection, intracerebral | Injection, intracerebral
3 | ICISTERNINJ | Injection, intracisternal | Injection, intracisternal
3 | ICORONINJ | Injection, intracoronary | Injection, intracoronary
4 | ICORONINJP | Injection, intracoronary, push | Injection, intracoronary, push
3 | ICORPCAVINJ | Injection, intracorpus cavernosum | Injection, intracorpus cavernosum
3 | IDINJ | Injection, intradermal | Injection, intradermal
3 | IDISCINJ | Injection, intradiscal | Injection, intradiscal
3 | IDUCTINJ | Injection, intraductal | Injection, intraductal
3 | IDURINJ | Injection, intradural | Injection, intradural
3 | IEPIDINJ | Injection, intraepidermal | Injection, intraepidermal
3 | IEPITHINJ | Injection, intraepithelial | Injection, intraepithelial
3 | ILESINJ | Injection, intralesional | Injection, intralesional
3 | ILUMINJ | Injection, intraluminal | Injection, intraluminal
3 | ILYMPJINJ | Injection, intralymphatic | Injection, intralymphatic
3 | IM | Injection, intramuscular | Injection, intramuscular

4 | IMD | Injection, intramuscular, deep | Injection, intramuscular, deep
4 | IMZ | Injection, intramuscular, z track | Injection, intramuscular, z track
3 | IMEDULINJ | Injection, intramedullary | Injection, intramedullary
3 | INTERMENINJ | Injection, interameningeal | Injection, interameningeal
3 | INTERSTITINJ | Injection, interstitial | Injection, interstitial
3 | IOINJ | Injection, intraocular | Injection, intraocular
3 | IOSSINJ | Injection, intraosseous | Injection, intraosseous
3 | IOVARINJ | Injection, intraovarian | Injection, intraovarian
3 | IPCARDINJ | Injection, intrapericardial | Injection, intrapericardial
3 | IPERINJ | Injection, intraperitoneal | Injection, intraperitoneal
3 | IPINJ | Injection, intrapulmonary | Injection, intrapulmonary
3 | IPLRINJ | Injection, intrapleural | Injection, intrapleural
3 | IPROSTINJ | Injection, intraprostatic | Injection, intraprostatic
3 | IPUMPINJ | Injection, insulin pump | Injection, insulin pump
3 | ISINJ | Injection, intraspinal | Injection, intraspinal
3 | ISTERINJ | Injection, intrasternal | Injection, intrasternal
3 | ISYNINJ | Injection, intrasynovial | Injection, intrasynovial
3 | ITENDINJ | Injection, intratendinous | Injection, intratendinous
3 | ITESTINJ | Injection, intratesticular | Injection, intratesticular
3 | ITHORINJ | Injection, intrathoracic | Injection, intrathoracic
3 | ITINJ | Injection, intrathecal | Injection, intrathecal
3 | ITUBINJ | Injection, intratubular | Injection, intratubular
3 | ITUMINJ | Injection, intratumor | Injection, intratumor
3 | ITYMPINJ | Injection, intratympanic | Injection, intratympanic
3 | IUINJ | Injection, intrauterine | Injection, intracervical (uterus)
3 | IUINJC | Injection, intracervical (uterus) | Injection, intracervical (uterus)
3 | IURETINJ | Injection, intraureteral, retrograde | Injection, intraureteral, retrograde
3 | IVASCINJ | Injection, intravascular | Injection, intravascular
3 | IVENTINJ | Injection, intraventricular (heart) | Injection, intraventricular (heart)
3 | IVESINJ | Injection, intravesicle | Injection, intravesicle
3 | IVINJ | Injection, intravenous | Injection, intravenous
4 | IVINJBOL | Injection, intravenous, bolus | Injection, intravenous, bolus
4 | IVPUSH | Injection, intravenous, push | Injection, intravenous, push
4 | IVRPUSH | Injection, intravenous, rapid push | Injection, intravenous, rapid push
4 | IVSPUSH | Injection, intravenous, slow push | Injection, intravenous, slow push
3 | IVITINJ | Injection, intravitreal | Injection, intravitreal
3 | PAINJ | Injection, periarticular | Injection, periarticular
3 | PARENTINJ | Injection, parenteral | Injection, parenteral
3 | PDONTINJ | Injection, periodontal | Injection, periodontal
3 | PDPINJ | Injection, peritoneal dialysis port | Injection, peritoneal dialysis port
3 | PDURINJ | Injection, peridural | Injection, peridural
3 | PNINJ | Injection, perineural | Injection, perineural
3 | PNSINJ | Injection, paranasal sinuses | Injection, paranasal sinuses
3 | RBINJ | Injection, retrobulbar | Injection, retrobulbar
3 | SCINJ | Injection, subconjunctival | Injection, subconjunctival
3 | SLESINJ | Injection, sublesional | Injection, sublesional
3 | SOFTISINJ | Injection, soft tissue | Injection, soft tissue
3 | SQ | Injection, subcutaneous | Injection, subcutaneous
3 | SUBARACHINJ | Injection, subarachnoid | Injection, subarachnoid
3 | SUBMUCINJ | Injection, submucosal | Injection, submucosal
3 | TRPLACINJ | Injection, transplacental | Injection, transplacental
3 | TRTRACHINJ | Injection, transtracheal | Injection, transtracheal
3 | URETHINJ | Injection, urethral | Injection, urethral
3 | URETINJ | Injection, ureteral | Injection, ureteral
2 | (_Insertion) Abstract | | Insertion
3 | CERVINS | Insertion, cervical (uterine) | Insertion, cervical (uterine)
3 | IOSURGINS | Insertion, intraocular, surgical | Insertion, intraocular, surgical
3 | IU | Insertion, intrauterine | Insertion, intrauterine
3 | LPINS | Insertion, lacrimal puncta | Insertion, lacrimal puncta
3 | PR | Insertion, rectal | Insertion, rectal
3 | SQSURGINS | Insertion, subcutaneous, surgical | Insertion, subcutaneous, surgical
3 | URETHINS | Insertion, urethral | Insertion, urethral
3 | VAGINSI | Insertion, vaginal | Insertion, vaginal
2 | (_Instillation) Abstract | | Instillation
3 | CECINSTL | Instillation, cecostomy | Instillation, cecostomy
3 | EFT | Instillation, enteral feeding tube | Instillation, enteral feeding tube
3 | ENTINSTL | Instillation, enteral | Instillation, enteral
3 | GT | Instillation, gastrostomy tube | Instillation, gastrostomy tube
3 | NGT | Instillation, nasogastric tube | Instillation, nasogastric tube
3 | OGT | Instillation, orogastric tube | Instillation, orogastric tube
3 | BLADINSTL | Instillation, urinary catheter | Instillation, urinary catheter

3 | CAPDINSTL | Instillation, continuous ambulatory peritoneal dialysis port | Instillation, continuous ambulatory peritoneal dialysis port
3 | CTINSTL | Instillation, chest tube | Instillation, chest tube
3 | ETINSTL | Instillation, endotracheal tube | Instillation, endotracheal tube
3 | GJT | Instillation, gastro-jejunostomy tube | Instillation, gastro-jejunostomy tube
3 | IBRONCHINSTIL | Instillation, intrabronchial | Instillation, intrabronchial
3 | IDUODINSTIL | Instillation, intraduodenal | Instillation, intraduodenal
3 | IESOPHINSTIL | Instillation, intraesophageal | Instillation, intraesophageal
3 | IGASTINSTIL | Instillation, intragastric | Instillation, intragastric
3 | IILEALINJ | Instillation, intraileal | Instillation, intraileal
3 | IOINSTL | Instillation, intraocular | Instillation, intraocular
3 | ISININSTIL | Instillation, intrasinal | Instillation, intrasinal
3 | ITRACHINSTIL | Instillation, intratracheal | Instillation, intratracheal
3 | IUINSTL | Instillation, intrauterine | Instillation, intrauterine
3 | JJTINSTL | Instillation, jejunostomy tube | Instillation, jejunostomy tube
3 | LARYNGINSTIL | Instillation, laryngeal | Instillation, laryngeal
3 | NASALINSTIL | Instillation, nasal | Instillation, nasal
3 | NASOGASINSTIL | Instillation, nasogastric | Instillation, nasogastric
3 | NTT | Instillation, nasotracheal tube | Instillation, nasotracheal tube
3 | OJJ | Instillation, orojejunum tube | Instillation, orojejunum tube
3 | OT | Instillation, otic | Instillation, otic
3 | PDPINSTL | Instillation, peritoneal dialysis port | Instillation, peritoneal dialysis port
3 | PNSINSTL | Instillation, paranasal sinuses | Instillation, paranasal sinuses
3 | RECINSTL | Instillation, rectal | Instillation, rectal
4 | RECTINSTL | Instillation, rectal tube | Instillation, rectal tube
3 | SININSTIL | Instillation, sinus, unspecified | Instillation, sinus, unspecified
3 | SOFTISINSTIL | Instillation, soft tissue | Instillation, soft tissue
3 | TRACHINSTL | Instillation, tracheostomy | Instillation, tracheostomy
3 | TRTYMPINSTIL | Instillation, transtympanic | Instillation, transtympanic
3 | URETHINSTL | instillation, urethral | Instillation, urethral
2 | (_IontophoresisRoute) Abstract | | Iontophoresis
3 | IONTO | Topical application, iontophoresis | Topical application, iontophoresis
2 | (_Irrigation) Abstract | | Irrigation
3 | GUIRR | Irrigation, genitourinary | Irrigation, genitourinary
3 | IGASTIRR | Irrigation, intragastric | Irrigation, intragastric
3 | ILESIRR | Irrigation, intralesional | Irrigation, intralesional
3 | IOIRR | Irrigation, intraocular | Irrigation, intraocular
3 | BLADIRR | Irrigation, urinary bladder | Irrigation, urinary bladder
4 | BLADIRRC | Irrigation, urinary bladder, continuous | Irrigation, urinary bladder, continuous
4 | BLADIRRT | Irrigation, urinary bladder, tidal | Irrigation, urinary bladder, tidal
3 | RECIRR | Irrigation, rectal | Irrigation, rectal
2 | (_LavageRoute) Abstract | | Lavage
3 | IGASTLAV | Lavage, intragastric | Lavage, intragastric
2 | (_MucosalAbsorptionRoute) Abstract | | Mucosal absorption
3 | IDOUDMAB | Mucosal absorption, intraduodenal | Mucosal absorption, intraduodenal
3 | ITRACHMAB | Mucosal absorption, intratracheal | Mucosal absorption, intratracheal
3 | SMUCMAB | Mucosal absorption, submucosal | Mucosal absorption, submucosal
2 | (_Nebulization) Abstract | | Nebulization
3 | ETNEB | Nebulization, endotracheal tube | Nebulization, endotracheal tube
2 | (_Rinse) Abstract | | Rinse
3 | DENRINSE | Rinse, dental | Rinse, dental
3 | ORRINSE | Rinse, oral | Rinse, oral
2 | (_SuppositoryRoute) Abstract | | Suppository
3 | URETHSUP | Suppository, urethral | Suppository, urethral
2 | (_Swish) Abstract | | Swish
3 | SWISHSPIT | Swish and spit out, oromucosal | Swish and spit out, oromucosal
3 | SWISHSWAL | Swish and swallow, oromucosal | Swish and swallow, oromucosal
2 | (_TopicalAbsorptionRoute) Abstract | | Topical absorption
3 | TTYMPTABSORP | Topical absorption, transtympanic | Topical absorption, transtympanic
2 | (_TopicalApplication) Abstract | | Topical application
3 | DRESS | Topical application, soaked dressing | Topical application, soaked dressing
3 | IONTO
3 | SWAB | Topical application, swab | Topical application, swab
3 | TOPICAL | Topical | Topical
3 | BUC | Topical application, buccal | Topical application, buccal
3 | CERV | Topical application, cervical | Topical application, cervical
3 | DEN | Topical application, dental | Topical application, dental
3 | GIN | Topical application, gingival | Topical application, gingival
3 | HAIR | Topical application, hair | Topical application, hair
3 | ICORNTA | Topical application, intracorneal | Topical application, intracorneal
3 | ICORONTA | Topical application, intracoronal (dental) | Topical application, intracoronal (dental)
3 | IESOPHTA | Topical application, intraesophageal | Topical application, intraesophageal

3 | IILEALTA | Topical application, intraileal | Topical application, intraileal
3 | ILTOP | Topical application, intralesional | Topical application, intralesional
3 | ILUMTA | Topical application, intraluminal | Topical application, intraluminal
3 | IOTOP | Topical application, intraocular | Topical application, intraocular
3 | LARYNGTA | Topical application, laryngeal | Topical application, laryngeal
3 | MUC | Topical application, mucous membrane | Topical application, mucous membrane
3 | NAIL | Topical application, nail | Topical application, nail
3 | NASAL | Topical application, nasal | Topical application, nasal
3 | OPTHALTA | Topical application, ophthalmic | Topical application, ophthalmic
3 | ORALTA | Topical application, oral | Topical application, oral
3 | ORMUC | Topical application, oromucosal | Topical application, oromucosal
3 | OROPHARTA | Topical application, oropharyngeal | Topical application, oropharyngeal
3 | PERIANAL | Topical application, perianal | Topical application, perianal
3 | PERINEAL | Topical application, perineal | Topical application, perineal
3 | PDONTTA | Topical application, periodontal | Topical application, periodontal
3 | RECTAL | Topical application, rectal | Topical application, rectal
3 | SCALP | Topical application, scalp | Topical application, scalp
3 | OCDRESTA | Occlusive dressing technique | Occlusive dressing technique
3 | SKIN | Topical application, skin | Topical application, skin
3 | SUBCONJTA | Subconjunctival | Subconjunctival
3 | TMUCTA | Topical application, transmucosal | Topical application, transmucosal
3 | VAGINS | Topical application, vaginal | Insertion, vaginal
2 | INSUF | Insufflation | Insufflation
2 | TRNSDERM | Transdermal | Transdermal
3 | TRNSDERMD
1 | (_RouteBySite) Abstract | | Route of substance administration classified by site.
2 | TRNSDERM
2 | (_AmnioticFluidSacRoute) Abstract | | Amniotic fluid sac
3 | AMNINJ
2 | (_BiliaryRoute) Abstract | | Biliary tract
3 | BILINJ
3 | CHOLINJ
2 | (_BodySurfaceRoute) Abstract | | Body surface
3 | DRESS
3 | ELECTOSMOS
3 | IONTO
3 | SOAK
3 | SWAB
3 | TOPICAL
2 | (_BuccalMucosaRoute) Abstract | | Buccal mucosa
3 | BUC
2 | (_CecostomyRoute) Abstract | | Cecostomy
3 | CECINSTL
2 | (_CervicalRoute) Abstract | | Cervix of the uterus
3 | CERV
3 | CERVINJ
3 | CERVINS
3 | DEN
3 | DENRINSE
2 | (_EndocervicalRoute) Abstract | | Endocervical
3 | AMNINJ
3 | BILINJ
2 | (_EnteralRoute) Abstract | | Enteral
3 | EFT
3 | ENTINSTL
2 | (_EpiduralRoute) Abstract | | Epidural
3 | EPI
3 | EPIDURINJ
3 | EPIINJ
3 | EPINJSP
2 | (_ExtraAmnioticRoute) Abstract | | Extra-amniotic
3 | EXTRAMNINJ
2 | (_ExtracorporealCirculationRoute) Abstract | | Extracorporeal circulation
3 | EXTCORPDIF
3 | EXTCORPINJ
2 | (_GastricRoute) Abstract | | Gastric
3 | GBINJ
3 | GT
3 | NGT
3 | OGT
2 | (_GenitourinaryRoute) Abstract | | Genitourinary
3 | GUIRR

2 | (_GingivalRoute) Abstract | | Gingival
3 | GIN
3 | GINGINJ
2 | (_HairRoute) Abstract | | Hair
3 | HAIR
3 | SHAMPOO
2 | (_InterameningealRoute) Abstract | | Interameningeal
3 | INTERMENINJ
2 | (_InterstitialRoute) Abstract | | Interstitial
3 | INTERSTITINJ
2 | (_IntraabdominalRoute) Abstract | | Intra-abdominal
3 | IABDINJ
2 | (_IntraarterialRoute) Abstract | | Intra-arterial
3 | IA
3 | IAINJ
2 | (_IntraarticularRoute) Abstract | | Intraarticular
3 | IARTINJ
2 | (_IntrabronchialRoute) Abstract | | Intrabronchial
3 | IBRONCHINSTIL
2 | (_IntrabursalRoute) Abstract | | Intrabursal
3 | IBURSINJ
2 | (_IntracardiacRoute) Abstract | | Intracardiac
3 | IC
3 | ICARDINJ
2 | (_IntracartilaginousRoute) Abstract | | Intracartilaginous
3 | ICARTINJ
2 | (_IntracaudalRoute) Abstract | | Intracaudal
3 | ICAUDINJ
2 | (_IntracavernosalRoute) Abstract | | Intracavernosal
3 | ICAVINJ
2 | (_IntracavitaryRoute) Abstract | | Intracavitary
3 | ICAVITINJ
2 | (_IntracerebralRoute) Abstract | | Intracerebral
3 | ICEREBINJ
2 | (_IntracervicalRoute) Abstract | | Intracervical
3 | IUINJC
2 | (_IntracisternalRoute) Abstract | | Intracisternal
3 | ICISTERNINJ
2 | (_IntracornealRoute) Abstract | | Intracorneal
3 | ICORNTA
2 | (_IntracoronalRoute) Abstract | | Intracoronal (dental)
3 | ICORONTA
2 | (_IntracoronaryRoute) Abstract | | Intracoronary
3 | ICOR
3 | ICORONINJ
2 | (_IntracorpusCavernosumRoute) Abstract | | Intracorpus cavernosum
3 | ICORPCAVINJ
2 | (_IntradermalRoute) Abstract | | Intradermal
3 | IDIMPLNT
3 | IDINJ
2 | (_IntradiscalRoute) Abstract | | Intradiscal
3 | IDISCINJ
2 | (_IntraductalRoute) Abstract | | Intraductal
3 | IDUCTINJ
2 | (_IntraduodenalRoute) Abstract | | Intraduodenal
3 | IDUODINSTIL
3 | IDOUDMAB
2 | (_IntraduralRoute) Abstract | | Intradural
3 | IDURINJ
2 | (_IntraepidermalRoute) Abstract | | Intraepidermal
3 | IEPIDINJ
2 | (_IntraepithelialRoute) Abstract | | Intraepithelial
3 | IEPITHINJ
2 | (_IntraesophagealRoute) Abstract | | Intraesophageal
3 | IESOPHINSTIL
3 | IESOPHTA
2 | (_IntragastricRoute) Abstract | | Intragastric
3 | IGASTINSTIL
3 | IGASTIRR
3 | IGASTLAV
2 | (_IntrailealRoute) Abstract | | Intraileal
3 | IILEALINJ

3 | IILEALTA
2 | (_IntralesionalRoute) Abstract | | Intralesional
3 | ILESINJ
3 | ILESIRR
3 | ILTOP
2 | (_IntraluminalRoute) Abstract | | Intraluminal
3 | ILUMINJ
3 | ILUMTA
2 | (_IntralymphaticRoute) Abstract | | Intralymphatic
3 | ILYMPJINJ
2 | (_IntramedullaryRoute) Abstract | | Intramedullary
3 | IMEDULINJ
2 | (_IntramuscularRoute) Abstract | | Intramuscular
3 | IM
2 | (_IntraocularRoute) Abstract | | Intraocular
3 | IOINJ
3 | IOSURGINS
3 | IOINSTL
3 | IOIRR
3 | IOTOP
2 | (_IntraosseousRoute) Abstract | | Intraosseous
3 | IOSSC
3 | IOSSINJ
2 | (_IntraovarianRoute) Abstract | | Intraovarian
3 | IOVARINJ
2 | (_IntrapericardialRoute) Abstract | | Intrapericardial
3 | IPCARDINJ
2 | (_IntraperitonealRoute) Abstract | | Intraperitoneal
3 | IPERINJ
3 | PDPINJ
3 | CAPDINSTL
3 | PDPINSTL
2 | (_IntrapleuralRoute) Abstract | | Intrapleural
3 | IPLRINJ
3 | CTINSTL
2 | (_IntraprostaticRoute) Abstract | | Intraprostatic
3 | IPROSTINJ
2 | (_IntrapulmonaryRoute) Abstract | | Intrapulmonary
3 | EXTCORPINJ
3 | IPINJ
2 | (_IntrasinalRoute) Abstract | | Intrasinal
3 | ISININSTIL
2 | (_IntraspinalRoute) Abstract | | Intraspinal
3 | ISINJ
2 | (_IntrasternalRoute) Abstract | | Intrasternal
3 | ISTERINJ
2 | (_IntrasynovialRoute) Abstract | | Intrasynovial
3 | ISYNINJ
2 | (_IntratendinousRoute) Abstract | | Intratendinous
3 | ITENDINJ
2 | (_IntratesticularRoute) Abstract | | Intratesticular
3 | ITESTINJ
2 | (_IntrathecalRoute) Abstract | | Intrathecal
3 | IT
3 | ITINJ
2 | (_IntrathoracicRoute) Abstract | | Intrathoracic
3 | ITHORINJ
2 | (_IntratrachealRoute) Abstract | | Intratracheal
3 | ITRACHINSTIL
3 | ITRACHMAB
2 | (_IntratubularRoute) Abstract | | Intratubular
3 | ITUBINJ
2 | (_IntratumorRoute) Abstract | | Intratumor
3 | ITUMINJ
2 | (_IntratympanicRoute) Abstract | | Intratympanic
3 | ITYMPINJ
2 | (_IntrauterineRoute) Abstract | | Intrauterine
3 | IUINJ
3 | IURETINJ
3 | IU
3 | IUINSTL
2 | (_IntravascularRoute) Abstract | | Intravascular

3 | HEMODIFF
3 | IVASCINFUS
3 | HEMOPORT
3 | IVASCINJ
2 | (_IntravenousRoute) Abstract | | Intravenous
3 | IVFLUSH
3 | IVINJ
2 | (_IntraventricularRoute) Abstract | | Intraventricular
3 | IVENTINJ
2 | (_IntravesicleRoute) Abstract | | Intravesicle
3 | IVESINJ
2 | (_IntravitrealRoute) Abstract | | Intravitreal
3 | IVITINJ
2 | (_JejunumRoute) Abstract | | Jejunum
3 | GJT
3 | JJTINSTL
3 | OJJ
2 | (_LacrimalPunctaRoute) Abstract | | Lacrimal puncta
3 | LPINS
2 | (_LaryngealRoute) Abstract | | Laryngeal
3 | LARYNGINSTIL
3 | LARYNGTA
2 | (_LingualRoute) Abstract | | Lingual
3 | TRNSLING
2 | (_MucousMembraneRoute) Abstract | | Mucous membrane
3 | MUC
2 | (_NailRoute) Abstract | | Nail
3 | NAIL
2 | (_NasalRoute) Abstract | | Nasal
3 | NASINHL
3 | NASALINSTIL
3 | NASOGASINSTIL
3 | NASAL
2 | (_OphthalmicRoute) Abstract | | Ophthalmic
3 | OPTHALTA
2 | (_OralRoute) Abstract | | Oral
3 | CHEW
3 | DISSOLVE
3 | IPINHL
3 | ORALTA
3 | ORRINSE
3 | PO
2 | (_OromucosalRoute) Abstract | | Oromucosal
3 | GARGLE
3 | ORMUC
3 | SUCK
3 | SWISHSPIT
3 | SWISHSWAL
2 | (_OropharyngealRoute) Abstract | | Oropharyngeal
3 | OROPHARTA
2 | (_OticRoute) Abstract | | Otic
3 | OT
2 | (_ParanasalSinusesRoute) Abstract | | Paranasal sinuses
3 | PNSINJ
3 | PNSINSTL
2 | (_ParenteralRoute) Abstract | | Parenteral
3 | PARENTINJ
2 | (_PerianalRoute) Abstract | | Perianal
3 | PERIANAL
2 | (_PeriarticularRoute) Abstract | | Periarticular
3 | PAINJ
2 | (_PeriduralRoute) Abstract | | Peridural
3 | PDURINJ
2 | (_PerinealRoute) Abstract | | Perineal
3 | PERINEAL
2 | (_PerineuralRoute) Abstract | | Perineural
3 | PNINJ
2 | (_PeriodontalRoute) Abstract | | Periodontal
3 | PDONTINJ
3 | PDONTTA
2 | (_PulmonaryRoute) Abstract | | Pulmonary
3 | IPPB

3 | VENT
3 | VENTMASK
3 | ETINSTL
3 | NTT
3 | ETNEB
2 | (_RectalRoute) Abstract | | Rectal
3 | ENEMA
3 | RETENEMA
3 | PR
3 | RECINSTL
3 | RECIRR
3 | RECTAL
2 | (_RespiratoryTractRoute) Abstract | | Respiratory tract
3 | IPINHL
2 | (_RetrobulbarRoute) Abstract | | Retrobulbar
3 | RBINJ
2 | (_ScalpRoute) Abstract | | Scalp
3 | SCALP
2 | (_SinusUnspecifiedRoute) Abstract | | Sinus, unspecified
3 | ENDOSININJ
3 | SININSTIL
2 | (_SkinRoute) Abstract | | Skin
3 | OCDRESTA
3 | SKIN
2 | (_SoftTissueRoute) Abstract | | Soft tissue
3 | SOFTISINJ
3 | SOFTISINSTIL
2 | (_SubarachnoidRoute) Abstract | | Subarachnoid
3 | SUBARACHINJ
2 | (_SubconjunctivalRoute) Abstract | | Subconjunctival
3 | SCINJ
3 | SUBCONJTA
2 | (_SubcutaneousRoute) Abstract | | Subcutaneous
3 | SQIMPLNT
3 | SQINFUS
3 | IPUMPINJ
3 | SQ
3 | SQSURGINS
2 | (_SublesionalRoute) Abstract | | Sublesional
3 | SLESINJ
2 | (_SublingualRoute) Abstract | | Sublingual
3 | SL
2 | (_SubmucosalRoute) Abstract | | Submucosal
3 | SUBMUCINJ
3 | SMUCMAB
2 | (_TracheostomyRoute) Abstract | | Tracheostomy
3 | TRACH
3 | TRACHINSTL
2 | (_TransmucosalRoute) Abstract | | Transmucosal
3 | TMUCTA
2 | (_TransplacentalRoute) Abstract | | Transplacental
3 | TRPLACINJ
2 | (_TranstrachealRoute) Abstract | | Transtracheal
3 | TRTRACHINJ
2 | (_TranstympanicRoute) Abstract | | Transtympanic
3 | TRTYMPINSTIL
3 | TTYMPTABSORP
2 | (_UreteralRoute) Abstract | | Ureteral
3 | URETINJ
2 | (_UrethralRoute) Abstract | | Urethral
3 | URETHINJ
3 | URETHINS
3 | URETHINSTL
3 | URETHSUP
2 | (_UrinaryBladderRoute) Abstract | | Urinary bladder
3 | BLADINJ
3 | BLADIRR
2 | (_UrinaryTractRoute) Abstract | | Urinary tract
3 | BLADINSTL
2 | (_VaginalRoute) Abstract | | Vaginal
3 | DOUCHE
3 | VAGINSI

3 | VAGINS
2 | (_VitreousHumourRoute) Abstract | | Vitreous humour
3 | IVITIMPLNT

4.4.2.113 HL7 v3 Code System Sequencing

URL: https://hl7.org/fhir/R4/v3/Sequencing/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-Sequencing) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-Sequencing
Version: | 2018-08-12
Name: | v3.Sequencing
Title: | v3 Code System Sequencing
Definition: | Specifies sequence of sort order.
OID: | 2.16.840.1.113883.5.113 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System Sequencing (Specifies sequence of sort order.)

4.4.2.113.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | A | Ascending | Ascending sequence order.
1 | D | Descending | Descending sequence order.
1 | N | None | No enforced sequence order.

4.4.2.129 HL7 v3 Code System SpecimenType

URL: https://hl7.org/fhir/R4/v3/SpecimenType/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-SpecimenType) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-SpecimenType
Version: | 2018-08-12
Name: | v3.SpecimenType
Title: | v3 Code System SpecimenType
Definition: | **** MISSING DEFINITIONS ****
OID: | 2.16.840.1.113883.5.129 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System SpecimenType (**** MISSING DEFINITIONS ****)

4.4.2.129.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_SpecimenEntityType) Abstract
2 | ABS | Abcess
2 | AMN | Amniotic fluid
2 | ASP | Aspirate
2 | BBL | Blood bag
2 | BDY | Whole body
2 | BIFL | Bile fluid
2 | BLD | Whole blood
2 | BLDA | Blood arterial
2 | BLDC | Blood capillary
2 | BLDCO | Blood - cord
2 | BLDV | Blood venous
2 | BON | Bone
2 | BPH | Basophils
2 | BPU | Blood product unit
2 | BRN | Burn
2 | BRO | Bronchial
2 | BRTH | Exhaled gas (=breath)
2 | CALC | Calculus (=Stone)
2 | CDM | Cardiac muscle

2 | CNJT | Conjunctiva
2 | CNL | Cannula
2 | COL | Colostrum
2 | CRN | Cornea
2 | CSF | Cerebral spinal fluid
2 | CTP | Catheter tip
2 | CUR | Curettage
2 | CVM | Cervical mucus
2 | CVX | Cervix
2 | CYST | Cyst
2 | DIAF | Dialysis fluid
2 | DOSE | Dose med or substance
2 | DRN | Drain
2 | DUFL | Duodenal fluid
2 | EAR | Ear
2 | EARW | Ear wax (cerumen)
2 | ELT | Electrode
2 | ENDC | Endocardium
2 | ENDM | Endometrium
2 | EOS | Eosinophils
2 | EYE | Eye
2 | FIB | Fibroblasts
2 | FIST | Fistula
2 | FLT | Filter
2 | FLU | Body fluid, unsp
2 | FOOD | Food sample
2 | GAS | Gas
2 | GAST | Gastric fluid/contents
2 | GEN | Genital
2 | GENC | Genital cervix
2 | GENF | Genital fluid
2 | GENL | Genital lochia
2 | GENV | Genital vaginal
2 | HAR | Hair
2 | IHG | Inhaled Gas
2 | ISLT | Isolate
2 | IT | Intubation tube
2 | LAM | Lamella
2 | LIQ | Liquid NOS
2 | LN | Line
2 | LNA | Line arterial
2 | LNV | Line venous
2 | LYM | Lymphocytes
2 | MAC | Macrophages
2 | MAR | Marrow (bone)
2 | MBLD | Menstrual blood
2 | MEC | Meconium
2 | MILK | Breast milk
2 | MLK | Milk
2 | NAIL | Nail
2 | NOS | Nose (nasal passage)
2 | PAFL | Pancreatic fluid
2 | PAT | Patient
2 | PLAS | Plasma
2 | PLB | Plasma bag
2 | PLC | Placenta
2 | PLR | Pleural fluid (thoracentesis fld)
2 | PMN | Polymorphonuclear neutrophils
2 | PPP | Platelet poor plasma
2 | PRP | Platelet rich plasma
2 | PRT | Peritoneal fluid /ascites
2 | PUS | Pus
2 | RBC | Erythrocytes
2 | SAL | Saliva
2 | SER | Serum
2 | SKM | Skeletal muscle
2 | SKN | Skin
2 | SMN | Seminal fluid
2 | SMPLS | Seminal plasma
2 | SNV | Synovial fluid (Joint fluid)
2 | SPRM | Spermatozoa
2 | SPT | Sputum

2 | SPTC | Sputum - coughed
2 | SPTT | Sputum - tracheal aspirate
2 | STL | Stool = Fecal
2 | SWT | Sweat
2 | TEAR | Tears
2 | THRB | Thrombocyte (platelet)
2 | THRT | Throat
2 | TISG | Tissue gall bladder
2 | TISPL | Tissue placenta
2 | TISS | Tissue, unspecified
2 | TISU | Tissue ulcer
2 | TLGI | Tissue large intestine
2 | TLNG | Tissue lung
2 | TSMI | Tissue small intestine Tissue ulcer
2 | TUB | Tube, unspecified
2 | ULC | Ulcer
2 | UMB | Umbilical blood
2 | UMED | Unknown medicine
2 | UR | Urine
2 | URC | Urine clean catch
2 | URNS | Urine sediment
2 | URT | Urine catheter
2 | URTH | Urethra
2 | USUB | Unknown substance
2 | VOM | Vomitus
2 | WAT | Water
2 | WBC | Leukocytes
2 | WICK | Wick
2 | WND | Wound
2 | WNDA | Wound abscess
2 | WNDD | Wound drainage
2 | WNDE | Wound exudate

4.4.2.131 HL7 v3 Code System TableCellHorizontalAlign

URL: https://hl7.org/fhir/R4/v3/TableCellHorizontalAlign/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-TableCellHorizontalAlign) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-TableCellHorizontalAlign
Version: | 2018-08-12
Name: | v3.TableCellHorizontalAlign
Title: | v3 Code System TableCellHorizontalAlign
Definition: | These values are defined within the XHTML 4.0 Table Model
OID: | 2.16.840.1.113883.5.131 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System TableCellHorizontalAlign (These values are defined within the XHTML 4.0 Table Model)

4.4.2.131.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | center | center
1 | char | char
1 | justify | justify
1 | left | left
1 | right | right

4.4.2.132 HL7 v3 Code System TableCellScope

URL: https://hl7.org/fhir/R4/v3/TableCellScope/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-TableCellScope) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-TableCellScope

Version: | 2018-08-12
Name: | v3.TableCellScope
Title: | v3 Code System TableCellScope
Definition: | These values are defined within the XHTML 4.0 Table Model
OID: | 2.16.840.1.113883.5.132 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System TableCellScope (These values are defined within the XHTML 4.0 Table Model)

4.4.2.132.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | col | col
1 | colgroup | colgroup
1 | row | row
1 | rowgroup | rowgroup

4.4.2.133 HL7 v3 Code System TableCellVerticalAlign

URL: https://hl7.org/fhir/R4/v3/TableCellVerticalAlign/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-TableCellVerticalAlign) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-TableCellVerticalAlign
Version: | 2018-08-12
Name: | v3.TableCellVerticalAlign
Title: | v3 Code System TableCellVerticalAlign
Definition: | These values are defined within the XHTML 4.0 Table Model
OID: | 2.16.840.1.113883.5.133 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System TableCellVerticalAlign (These values are defined within the XHTML 4.0 Table Model)

4.4.2.133.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | baseline | baseline
1 | bottom | bottom
1 | middle | middle
1 | top | top

4.4.2.134 HL7 v3 Code System TableFrame

URL: https://hl7.org/fhir/R4/v3/TableFrame/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-TableFrame) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-TableFrame
Version: | 2018-08-12
Name: | v3.TableFrame
Title: | v3 Code System TableFrame
Definition: | These values are defined within the XHTML 4.0 Table Model
OID: | 2.16.840.1.113883.5.134 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System TableFrame (These values are defined within the XHTML 4.0 Table Model)

4.4.2.134.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | above | above
1 | below | below

1 | border | border
1 | box | box
1 | hsides | hsides
1 | lhs | lhs
1 | rhs | rhs
1 | void | void
1 | vsides | vsides

4.4.2.136 HL7 v3 Code System TableRules

URL: https://hl7.org/fhir/R4/v3/TableRules/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-TableRules) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-TableRules
Version: | 2018-08-12
Name: | v3.TableRules
Title: | v3 Code System TableRules
Definition: | These values are defined within the XHTML 4.0 Table Model
OID: | 2.16.840.1.113883.5.136 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System TableRules (These values are defined within the XHTML 4.0 Table Model)

4.4.2.136.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | all | all
1 | cols | cols
1 | groups | groups
1 | none | none
1 | rows | rows

4.4.2.137 HL7 v3 Code System TargetAwareness

URL: https://hl7.org/fhir/R4/v3/TargetAwareness/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-TargetAwareness) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-TargetAwareness
Version: | 2018-08-12
Name: | v3.TargetAwareness
Title: | v3 Code System TargetAwareness
Definition: | **** MISSING DEFINITIONS ****
OID: | 2.16.840.1.113883.5.137 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System TargetAwareness (**** MISSING DEFINITIONS ****)

4.4.2.137.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | D | denying | Target person has been informed about the issue but currently denies it.
1 | F | full awareness | Target person is fully aware of the issue.
1 | I | incapable | Target person is not capable of comprehending the issue.
1 | M | marginal | Target person is marginally aware of the issue.
1 | P | partial | Target person is partially aware of the issue.
1 | U | uninformed | Target person has not yet been informed of the issue.

4.4.2.139 HL7 v3 Code System TimingEvent

URL: https://hl7.org/fhir/R4/v3/TimingEvent/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-TimingEvent) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-TimingEvent
Version: | 2018-08-12
Name: | v3.TimingEvent
Title: | v3 Code System TimingEvent
Definition: | **** MISSING DEFINITIONS ****
OID: | 2.16.840.1.113883.5.139 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System TimingEvent (**** MISSING DEFINITIONS ****)

• ValueSet: EventTiming (Real world event relating to the schedule.)

• ValueSet: EventTiming (Real world event relating to the schedule.)

4.4.2.139.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | AC | AC | before meal (from lat. ante cibus)
1 | ACD | ACD | before lunch (from lat. ante cibus diurnus)
1 | ACM | ACM | before breakfast (from lat. ante cibus matutinus)
1 | ACV | ACV | before dinner (from lat. ante cibus vespertinus)
1 | C | C | Description: meal (from lat. ante cibus)
2 | CD | CD | Description: lunch (from lat. cibus diurnus)
2 | CM | CM | Description: breakfast (from lat. cibus matutinus)
2 | CV | CV | Description: dinner (from lat. cibus vespertinus)
1 | HS | HS | Description: Prior to beginning a regular period of extended sleep (this would exclude naps). Note that this might occur at different times of day depending on a person's regular sleep schedule.
1 | IC | IC | between meals (from lat. inter cibus)
1 | ICD | ICD | between lunch and dinner
1 | ICM | ICM | between breakfast and lunch
1 | ICV | ICV | between dinner and the hour of sleep
1 | PC | PC | after meal (from lat. post cibus)
1 | PCD | PCD | after lunch (from lat. post cibus diurnus)
1 | PCM | PCM | after breakfast (from lat. post cibus matutinus)
1 | PCV | PCV | after dinner (from lat. post cibus vespertinus)
1 | WAKE | WAKE | Description: Upon waking up from a regular period of sleep, in order to start regular activities (this would exclude waking up from a nap or temporarily waking up during a period of sleep) Usage Notes: e.g. Take pulse rate on waking in management of thyrotoxicosis. Take BP on waking in management of hypertension Take basal body temperature on waking in establishing date of ovulation

4.4.2.140 HL7 v3 Code System TribalEntityUS

URL: https://hl7.org/fhir/R4/v3/TribalEntityUS/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-TribalEntityUS
Version: | 2018-08-12
Name: | v3.TribalEntityUS
Title: | v3 Code System TribalEntityUS
Definition: | INDIAN ENTITIES RECOGNIZED AND ELIGIBLE TO RECEIVE SERVICES FROM THE UNITED STATES BUREAU OF INDIAN AFFAIRS
OID: | 2.16.840.1.113883.5.140 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System TribalEntityUS (INDIAN ENTITIES RECOGNIZED AND ELIGIBLE TO RECEIVE SERVICES FROM THE UNITED STATES BUREAU OF INDIAN AFFAIRS)

4.4.2.140.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_NativeEntityAlaska) Abstract | | NATIVE ENTITIES WITHIN THE STATE OF ALASKA RECOGNIZED AND ELIGIBLE TO RECEIVE SERVICES FROM THE UNITED STATES BUREAU OF INDIAN AFFAIRS
2 | 338 | Village of Afognak
2 | 339 | Agdaagux Tribe of King Cove
2 | 340 | Native Village of Akhiok
2 | 341 | Akiachak Native Community
2 | 342 | Akiak Native Community
2 | 343 | Native Village of Akutan

2 | 344 | Village of Alakanuk
2 | 345 | Alatna Village
2 | 346 | Native Village of Aleknagik
2 | 347 | Algaaciq Native Village (St. Mary's)
2 | 348 | Allakaket Village
2 | 349 | Native Village of Ambler
2 | 350 | Village of Anaktuvuk Pass
2 | 351 | Yupiit of Andreafski
2 | 352 | Angoon Community Association
2 | 353 | Village of Aniak
2 | 354 | Anvik Village
2 | 355 | Arctic Village (See Native Village of Venetie Trib
2 | 356 | Asa carsarmiut Tribe (formerly Native Village of M
2 | 357 | Native Village of Atka
2 | 358 | Village of Atmautluak
2 | 359 | Atqasuk Village (Atkasook)
2 | 360 | Native Village of Barrow Inupiat Traditional Gover
2 | 361 | Beaver Village
2 | 362 | Native Village of Belkofski
2 | 363 | Village of Bill Moore's Slough
2 | 364 | Birch Creek Tribe
2 | 365 | Native Village of Brevig Mission
2 | 366 | Native Village of Buckland
2 | 367 | Native Village of Cantwell
2 | 368 | Native Village of Chanega (aka Chenega)
2 | 369 | Chalkyitsik Village
2 | 370 | Village of Chefornak
2 | 371 | Chevak Native Village
2 | 372 | Chickaloon Native Village
2 | 373 | Native Village of Chignik
2 | 374 | Native Village of Chignik Lagoon
2 | 375 | Chignik Lake Village
2 | 376 | Chilkat Indian Village (Klukwan)
2 | 377 | Chilkoot Indian Association (Haines)
2 | 378 | Chinik Eskimo Community (Golovin)
2 | 379 | Native Village of Chistochina
2 | 380 | Native Village of Chitina
2 | 381 | Native Village of Chuathbaluk (Russian Mission, Ku
2 | 382 | Chuloonawick Native Village
2 | 383 | Circle Native Community
2 | 384 | Village of Clark's Point
2 | 385 | Native Village of Council
2 | 386 | Craig Community Association
2 | 387 | Village of Crooked Creek
2 | 388 | Curyung Tribal Council (formerly Native Village of
2 | 389 | Native Village of Deering
2 | 390 | Native Village of Diomede (aka Inalik)
2 | 391 | Village of Dot Lake
2 | 392 | Douglas Indian Association
2 | 393 | Native Village of Eagle
2 | 394 | Native Village of Eek
2 | 395 | Egegik Village
2 | 396 | Eklutna Native Village
2 | 397 | Native Village of Ekuk
2 | 398 | Ekwok Village
2 | 399 | Native Village of Elim
2 | 400 | Emmonak Village
2 | 401 | Evansville Village (aka Bettles Field)
2 | 402 | Native Village of Eyak (Cordova)
2 | 403 | Native Village of False Pass
2 | 404 | Native Village of Fort Yukon
2 | 405 | Native Village of Gakona
2 | 406 | Galena Village (aka Louden Village)
2 | 407 | Native Village of Gambell
2 | 408 | Native Village of Georgetown
2 | 409 | Native Village of Goodnews Bay
2 | 410 | Organized Village of Grayling (aka Holikachuk)
2 | 411 | Gulkana Village
2 | 412 | Native Village of Hamilton
2 | 413 | Healy Lake Village
2 | 414 | Holy Cross Village
2 | 415 | Hoonah Indian Association

2 | 416 | Native Village of Hooper Bay
2 | 417 | Hughes Village
2 | 418 | Huslia Village
2 | 419 | Hydaburg Cooperative Association
2 | 420 | Igiugig Village
2 | 421 | Village of Iliamna
2 | 422 | Inupiat Community of the Arctic Slope
2 | 423 | Iqurmuit Traditional Council (formerly Native Vill
2 | 424 | Ivanoff Bay Village
2 | 425 | Kaguyak Village
2 | 426 | Organized Village of Kake
2 | 427 | Kaktovik Village (aka Barter Island)
2 | 428 | Village of Kalskag
2 | 429 | Village of Kaltag
2 | 430 | Native Village of Kanatak
2 | 431 | Native Village of Karluk
2 | 432 | Organized Village of Kasaan
2 | 433 | Native Village of Kasigluk
2 | 434 | Kenaitze Indian Tribe
2 | 435 | Ketchikan Indian Corporation
2 | 436 | Native Village of Kiana
2 | 437 | King Island Native Community
2 | 438 | King Salmon Tribe
2 | 439 | Native Village of Kipnuk
2 | 440 | Native Village of Kivalina
2 | 441 | Klawock Cooperative Association
2 | 442 | Native Village of Kluti Kaah (aka Copper Center)
2 | 443 | Knik Tribe
2 | 444 | Native Village of Kobuk
2 | 445 | Kokhanok Village
2 | 446 | Native Village of Kongiganak
2 | 447 | Village of Kotlik
2 | 448 | Native Village of Kotzebue
2 | 449 | Native Village of Koyuk
2 | 450 | Koyukuk Native Village
2 | 451 | Organized Village of Kwethluk
2 | 452 | Native Village of Kwigillingok
2 | 453 | Native Village of Kwinhagak (aka Quinhagak)
2 | 454 | Native Village of Larsen Bay
2 | 455 | Levelock Village
2 | 456 | Lesnoi Village (aka Woody Island)
2 | 457 | Lime Village
2 | 458 | Village of Lower Kalskag
2 | 459 | Manley Hot Springs Village
2 | 460 | Manokotak Village
2 | 461 | Native Village of Marshall (aka Fortuna Ledge)
2 | 462 | Native Village of Mary's Igloo
2 | 463 | McGrath Native Village
2 | 464 | Native Village of Mekoryuk
2 | 465 | Mentasta Traditional Council
2 | 466 | Metlakatla Indian Community, Annette Island Reserv
2 | 467 | Native Village of Minto
2 | 468 | Naknek Native Village
2 | 469 | Native Village of Nanwalek (aka English Bay)
2 | 470 | Native Village of Napaimute
2 | 471 | Native Village of Napakiak
2 | 472 | Native Village of Napaskiak
2 | 473 | Native Village of Nelson Lagoon
2 | 474 | Nenana Native Association
2 | 475 | New Koliganek Village Council (formerly Koliganek
2 | 476 | New Stuyahok Village
2 | 477 | Newhalen Village
2 | 478 | Newtok Village
2 | 479 | Native Village of Nightmute
2 | 480 | Nikolai Village
2 | 481 | Native Village of Nikolski
2 | 482 | Ninilchik Village
2 | 483 | Native Village of Noatak
2 | 484 | Nome Eskimo Community
2 | 485 | Nondalton Village
2 | 486 | Noorvik Native Community
2 | 487 | Northway Village

2 | 488 | Native Village of Nuiqsut (aka Nooiksut)
2 | 489 | Nulato Village
2 | 490 | Nunakauyarmiut Tribe (formerly Native Village of T
2 | 491 | Native Village of Nunapitchuk
2 | 492 | Village of Ohogamiut
2 | 493 | Village of Old Harbor
2 | 494 | Orutsararmuit Native Village (aka Bethel)
2 | 495 | Oscarville Traditional Village
2 | 496 | Native Village of Ouzinkie
2 | 497 | Native Village of Paimiut
2 | 498 | Pauloff Harbor Village
2 | 499 | Pedro Bay Village
2 | 500 | Native Village of Perryville
2 | 501 | Petersburg Indian Association
2 | 502 | Native Village of Pilot Point
2 | 503 | Pilot Station Traditional Village
2 | 504 | Native Village of Pitka's Point
2 | 505 | Platinum Traditional Village
2 | 506 | Native Village of Point Hope
2 | 507 | Native Village of Point Lay
2 | 508 | Native Village of Port Graham
2 | 509 | Native Village of Port Heiden
2 | 510 | Native Village of Port Lions
2 | 511 | Portage Creek Village (aka Ohgsenakale)
2 | 512 | Pribilof Islands Aleut Communities of St. Paul & S
2 | 513 | Qagan Tayagungin Tribe of Sand Point Village
2 | 514 | Qawalangin Tribe of Unalaska
2 | 515 | Rampart Village
2 | 516 | Village of Red Devil
2 | 517 | Native Village of Ruby
2 | 518 | Saint George Island(See Pribilof Islands Aleut Com
2 | 519 | Native Village of Saint Michael
2 | 520 | Saint Paul Island (See Pribilof Islands Aleut Comm
2 | 521 | Village of Salamatoff
2 | 522 | Native Village of Savoonga
2 | 523 | Organized Village of Saxman
2 | 524 | Native Village of Scammon Bay
2 | 525 | Native Village of Selawik
2 | 526 | Seldovia Village Tribe
2 | 527 | Shageluk Native Village
2 | 528 | Native Village of Shaktoolik
2 | 529 | Native Village of Sheldon's Point
2 | 530 | Native Village of Shishmaref
2 | 531 | Shoonaq Tribe of Kodiak
2 | 532 | Native Village of Shungnak
2 | 533 | Sitka Tribe of Alaska
2 | 534 | Skagway Village
2 | 535 | Village of Sleetmute
2 | 536 | Village of Solomon
2 | 537 | South Naknek Village
2 | 538 | Stebbins Community Association
2 | 539 | Native Village of Stevens
2 | 540 | Village of Stony River
2 | 541 | Takotna Village
2 | 542 | Native Village of Tanacross
2 | 543 | Native Village of Tanana
2 | 544 | Native Village of Tatitlek
2 | 545 | Native Village of Tazlina
2 | 546 | Telida Village
2 | 547 | Native Village of Teller
2 | 548 | Native Village of Tetlin
2 | 549 | Central Council of the Tlingit and Haida Indian Tb
2 | 550 | Traditional Village of Togiak
2 | 551 | Tuluksak Native Community
2 | 552 | Native Village of Tuntutuliak
2 | 553 | Native Village of Tununak
2 | 554 | Twin Hills Village
2 | 555 | Native Village of Tyonek
2 | 556 | Ugashik Village
2 | 557 | Umkumiute Native Village
2 | 558 | Native Village of Unalakleet
2 | 559 | Native Village of Unga

2 | 560 | Village of Venetie (See Native Village of Venetie
2 | 561 | Native Village of Venetie Tribal Government (Arcti
2 | 562 | Village of Wainwright
2 | 563 | Native Village of Wales
2 | 564 | Native Village of White Mountain
2 | 565 | Wrangell Cooperative Association
2 | 566 | Yakutat Tlingit Tribe
1 | (_NativeEntityContiguous) Abstract | | NATIVE ENTITIES WITHIN THE CONTIGUOUS 48 STATES
2 | 1 | Absentee-Shawnee Tribe of Indians of Oklahoma
2 | 10 | Assiniboine and Sioux Tribes of the Fort Peck Indi
2 | 100 | Havasupai Tribe of the Havasupai Reservation, Ariz
2 | 101 | Ho-Chunk Nation of Wisconsin (formerly known as th
2 | 102 | Hoh Indian Tribe of the Hoh Indian Reservation, Wa
2 | 103 | Hoopa Valley Tribe, California
2 | 104 | Hopi Tribe of Arizona
2 | 105 | Hopland Band of Pomo Indians of the Hopland Ranche
2 | 106 | Houlton Band of Maliseet Indians of Maine
2 | 107 | Hualapai Indian Tribe of the Hualapai Indian Reser
2 | 108 | Huron Potawatomi, Inc., Michigan
2 | 109 | Inaja Band of Diegueno Mission Indians of the Inaj
2 | 11 | Augustine Band of Cahuilla Mission Indians of the
2 | 110 | Ione Band of Miwok Indians of California
2 | 111 | Iowa Tribe of Kansas and Nebraska
2 | 112 | Iowa Tribe of Oklahoma
2 | 113 | Jackson Rancheria of Me-Wuk Indians of California
2 | 114 | Jamestown S'Klallam Tribe of Washington
2 | 115 | Jamul Indian Village of California
2 | 116 | Jena Band of Choctaw Indians, Louisiana
2 | 117 | Jicarilla Apache Tribe of the Jicarilla Apache Ind
2 | 118 | Kaibab Band of Paiute Indians of the Kaibab Indian
2 | 119 | Kalispel Indian Community of the Kalispel Reservat
2 | 12 | Bad River Band of the Lake Superior Tribe of Chipp
2 | 120 | Karuk Tribe of California
2 | 121 | Kashia Band of Pomo Indians of the Stewarts Point
2 | 122 | Kaw Nation, Oklahoma
2 | 123 | Keweenaw Bay Indian Community of L'Anse and Ontona
2 | 124 | Kialegee Tribal Town, Oklahoma
2 | 125 | Kickapoo Tribe of Indians of the Kickapoo Reservat
2 | 126 | Kickapoo Tribe of Oklahoma
2 | 127 | Kickapoo Traditional Tribe of Texas
2 | 128 | Kiowa Indian Tribe of Oklahoma
2 | 129 | Klamath Indian Tribe of Oregon
2 | 13 | Bay Mills Indian Community of the Sault Ste. Marie
2 | 130 | Kootenai Tribe of Idaho
2 | 131 | La Jolla Band of Luiseno Mission Indians of the La
2 | 132 | La Posta Band of Diegueno Mission Indians of the L
2 | 133 | Lac Courte Oreilles Band of Lake Superior Chippewa
2 | 134 | Lac du Flambeau Band of Lake Superior Chippewa Ind
2 | 135 | Lac Vieux Desert Band of Lake Superior Chippewa In
2 | 136 | Las Vegas Tribe of Paiute Indians of the Las Vegas
2 | 137 | Little River Band of Ottawa Indians of Michigan
2 | 138 | Little Traverse Bay Bands of Odawa Indians of Mich
2 | 139 | Lower Lake Rancheria, California
2 | 14 | Bear River Band of the Rohnerville Rancheria, Cali
2 | 140 | Los Coyotes Band of Cahuilla Mission Indians of th
2 | 141 | Lovelock Paiute Tribe of the Lovelock Indian Colon
2 | 142 | Lower Brule Sioux Tribe of the Lower Brule Reserva
2 | 143 | Lower Elwha Tribal Community of the Lower Elwha Re
2 | 144 | Lower Sioux Indian Community of Minnesota Mdewakan
2 | 145 | Lummi Tribe of the Lummi Reservation, Washington
2 | 146 | Lytton Rancheria of California
2 | 147 | Makah Indian Tribe of the Makah Indian Reservation
2 | 148 | Manchester Band of Pomo Indians of the Manchester-
2 | 149 | Manzanita Band of Diegueno Mission Indians of the
2 | 15 | Berry Creek Rancheria of Maidu Indians of Californ
2 | 150 | Mashantucket Pequot Tribe of Connecticut
2 | 151 | Match-e-be-nash-she-wish Band of Pottawatomi India
2 | 152 | Mechoopda Indian Tribe of Chico Rancheria, Califor
2 | 153 | Menominee Indian Tribe of Wisconsin
2 | 154 | Mesa Grande Band of Diegueno Mission Indians of th
2 | 155 | Mescalero Apache Tribe of the Mescalero Reservatio
2 | 156 | Miami Tribe of Oklahoma

2 | 157 | Miccosukee Tribe of Indians of Florida
2 | 158 | Middletown Rancheria of Pomo Indians of California
2 | 159 | Minnesota Chippewa Tribe, Minnesota (Six component
2 | 16 | Big Lagoon Rancheria, California
2 | 160 | Bois Forte Band (Nett Lake); Fond du Lac Band; Gra
2 | 161 | Mississippi Band of Choctaw Indians, Mississippi
2 | 162 | Moapa Band of Paiute Indians of the Moapa River In
2 | 163 | Modoc Tribe of Oklahoma
2 | 164 | Mohegan Indian Tribe of Connecticut
2 | 165 | Mooretown Rancheria of Maidu Indians of California
2 | 166 | Morongo Band of Cahuilla Mission Indians of the Mo
2 | 167 | Muckleshoot Indian Tribe of the Muckleshoot Reserv
2 | 168 | Muscogee (Creek) Nation, Oklahoma
2 | 169 | Narragansett Indian Tribe of Rhode Island
2 | 17 | Big Pine Band of Owens Valley Paiute Shoshone Indi
2 | 170 | Navajo Nation, Arizona, New Mexico & Utah
2 | 171 | Nez Perce Tribe of Idaho
2 | 172 | Nisqually Indian Tribe of the Nisqually Reservatio
2 | 173 | Nooksack Indian Tribe of Washington
2 | 174 | Northern Cheyenne Tribe of the Northern Cheyenne I
2 | 175 | Northfork Rancheria of Mono Indians of California
2 | 176 | Northwestern Band of Shoshoni Nation of Utah (Wash
2 | 177 | Oglala Sioux Tribe of the Pine Ridge Reservation,
2 | 178 | Omaha Tribe of Nebraska
2 | 179 | Oneida Nation of New York
2 | 18 | Big Sandy Rancheria of Mono Indians of California
2 | 180 | Oneida Tribe of Wisconsin
2 | 181 | Onondaga Nation of New York
2 | 182 | Osage Tribe, Oklahoma
2 | 183 | Ottawa Tribe of Oklahoma
2 | 184 | Otoe-Missouria Tribe of Indians, Oklahoma
2 | 185 | Paiute Indian Tribe of Utah
2 | 186 | Paiute-Shoshone Indians of the Bishop Community of
2 | 187 | Paiute-Shoshone Tribe of the Fallon Reservation an
2 | 188 | Paiute-Shoshone Indians of the Lone Pine Community
2 | 189 | Pala Band of Luiseno Mission Indians of the Pala R
2 | 19 | Big Valley Band of Pomo Indians of the Big Valley
2 | 190 | Pascua Yaqui Tribe of Arizona
2 | 191 | Paskenta Band of Nomlaki Indians of California
2 | 192 | Passamaquoddy Tribe of Maine
2 | 193 | Pauma Band of Luiseno Mission Indians of the Pauma
2 | 194 | Pawnee Nation of Oklahoma
2 | 195 | Pechanga Band of Luiseno Mission Indians of the Pe
2 | 196 | Penobscot Tribe of Maine
2 | 197 | Peoria Tribe of Indians of Oklahoma
2 | 198 | Picayune Rancheria of Chukchansi Indians of Califo
2 | 199 | Pinoleville Rancheria of Pomo Indians of Californi
2 | 2 | Agua Caliente Band of Cahuilla Indians of the Agua
2 | 20 | Blackfeet Tribe of the Blackfeet Indian Reservatio
2 | 200 | Pit River Tribe, California (includes Big Bend, Lo
2 | 201 | Poarch Band of Creek Indians of Alabama
2 | 202 | Pokagon Band of Potawatomi Indians of Michigan
2 | 203 | Ponca Tribe of Indians of Oklahoma
2 | 204 | Ponca Tribe of Nebraska
2 | 205 | Port Gamble Indian Community of the Port Gamble Re
2 | 206 | Potter Valley Rancheria of Pomo Indians of Califor
2 | 207 | Prairie Band of Potawatomi Indians, Kansas
2 | 208 | Prairie Island Indian Community of Minnesota Mdewa
2 | 209 | Pueblo of Acoma, New Mexico
2 | 21 | Blue Lake Rancheria, California
2 | 210 | Pueblo of Cochiti, New Mexico
2 | 211 | Pueblo of Jemez, New Mexico
2 | 212 | Pueblo of Isleta, New Mexico
2 | 213 | Pueblo of Laguna, New Mexico
2 | 214 | Pueblo of Nambe, New Mexico
2 | 215 | Pueblo of Picuris, New Mexico
2 | 216 | Pueblo of Pojoaque, New Mexico
2 | 217 | Pueblo of San Felipe, New Mexico
2 | 218 | Pueblo of San Juan, New Mexico
2 | 219 | Pueblo of San Ildefonso, New Mexico
2 | 22 | Bridgeport Paiute Indian Colony of California
2 | 220 | Pueblo of Sandia, New Mexico

2 | 221 | Pueblo of Santa Ana, New Mexico
2 | 222 | Pueblo of Santa Clara, New Mexico
2 | 223 | Pueblo of Santo Domingo, New Mexico
2 | 224 | Pueblo of Taos, New Mexico
2 | 225 | Pueblo of Tesuque, New Mexico
2 | 226 | Pueblo of Zia, New Mexico
2 | 227 | Puyallup Tribe of the Puyallup Reservation, Washin
2 | 228 | Pyramid Lake Paiute Tribe of the Pyramid Lake Rese
2 | 229 | Quapaw Tribe of Indians, Oklahoma
2 | 23 | Buena Vista Rancheria of Me-Wuk Indians of Califor
2 | 230 | Quartz Valley Indian Community of the Quartz Valle
2 | 231 | Quechan Tribe of the Fort Yuma Indian Reservation,
2 | 232 | Quileute Tribe of the Quileute Reservation, Washin
2 | 233 | Quinault Tribe of the Quinault Reservation, Washin
2 | 234 | Ramona Band or Village of Cahuilla Mission Indians
2 | 235 | Red Cliff Band of Lake Superior Chippewa Indians o
2 | 236 | Red Lake Band of Chippewa Indians of the Red Lake
2 | 237 | Redding Rancheria, California
2 | 238 | Redwood Valley Rancheria of Pomo Indians of Califo
2 | 239 | Reno-Sparks Indian Colony, Nevada
2 | 24 | Burns Paiute Tribe of the Burns Paiute Indian Colo
2 | 240 | Resighini Rancheria, California (formerly known as
2 | 241 | Rincon Band of Luiseno Mission Indians of the Rinc
2 | 242 | Robinson Rancheria of Pomo Indians of California
2 | 243 | Rosebud Sioux Tribe of the Rosebud Indian Reservat
2 | 244 | Round Valley Indian Tribes of the Round Valley Res
2 | 245 | Rumsey Indian Rancheria of Wintun Indians of Calif
2 | 246 | Sac and Fox Tribe of the Mississippi in Iowa
2 | 247 | Sac and Fox Nation of Missouri in Kansas and Nebra
2 | 248 | Sac and Fox Nation, Oklahoma
2 | 249 | Saginaw Chippewa Indian Tribe of Michigan, Isabell
2 | 25 | Cabazon Band of Cahuilla Mission Indians of the Ca
2 | 250 | Salt River Pima-Maricopa Indian Community of the S
2 | 251 | Samish Indian Tribe, Washington
2 | 252 | San Carlos Apache Tribe of the San Carlos Reservat
2 | 253 | San Juan Southern Paiute Tribe of Arizona
2 | 254 | San Manual Band of Serrano Mission Indians of the
2 | 255 | San Pasqual Band of Diegueno Mission Indians of Ca
2 | 256 | Santa Rosa Indian Community of the Santa Rosa Ranc
2 | 257 | Santa Rosa Band of Cahuilla Mission Indians of the
2 | 258 | Santa Ynez Band of Chumash Mission Indians of the
2 | 259 | Santa Ysabel Band of Diegueno Mission Indians of t
2 | 26 | Cachil DeHe Band of Wintun Indians of the Colusa I
2 | 260 | Santee Sioux Tribe of the Santee Reservation of Ne
2 | 261 | Sauk-Suiattle Indian Tribe of Washington
2 | 262 | Sault Ste. Marie Tribe of Chippewa Indians of Mich
2 | 263 | Scotts Valley Band of Pomo Indians of California
2 | 264 | Seminole Nation of Oklahoma
2 | 265 | Seminole Tribe of Florida, Dania, Big Cypress, Bri
2 | 266 | Seneca Nation of New York
2 | 267 | Seneca-Cayuga Tribe of Oklahoma
2 | 268 | Shakopee Mdewakanton Sioux Community of Minnesota
2 | 269 | Shawnee Tribe, Oklahoma
2 | 27 | Caddo Indian Tribe of Oklahoma
2 | 270 | Sherwood Valley Rancheria of Pomo Indians of Calif
2 | 271 | Shingle Springs Band of Miwok Indians, Shingle Spr
2 | 272 | Shoalwater Bay Tribe of the Shoalwater Bay Indian
2 | 273 | Shoshone Tribe of the Wind River Reservation, Wyom
2 | 274 | Shoshone-Bannock Tribes of the Fort Hall Reservati
2 | 275 | Shoshone-Paiute Tribes of the Duck Valley Reservat
2 | 276 | Sisseton-Wahpeton Sioux Tribe of the Lake Traverse
2 | 277 | Skokomish Indian Tribe of the Skokomish Reservatio
2 | 278 | Skull Valley Band of Goshute Indians of Utah
2 | 279 | Smith River Rancheria, California
2 | 28 | Cahuilla Band of Mission Indians of the Cahuilla R
2 | 280 | Snoqualmie Tribe, Washington
2 | 281 | Soboba Band of Luiseno Indians, California (former
2 | 282 | Sokaogon Chippewa Community of the Mole Lake Band
2 | 283 | Southern Ute Indian Tribe of the Southern Ute Rese
2 | 284 | Spirit Lake Tribe, North Dakota (formerly known as
2 | 285 | Spokane Tribe of the Spokane Reservation, Washingt
2 | 286 | Squaxin Island Tribe of the Squaxin Island Reserva

2 | 287 | St. Croix Chippewa Indians of Wisconsin, St. Croix
2 | 288 | St. Regis Band of Mohawk Indians of New York
2 | 289 | Standing Rock Sioux Tribe of North & South Dakota
2 | 29 | Cahto Indian Tribe of the Laytonville Rancheria, C
2 | 290 | Stockbridge-Munsee Community of Mohican Indians of
2 | 291 | Stillaguamish Tribe of Washington
2 | 292 | Summit Lake Paiute Tribe of Nevada
2 | 293 | Suquamish Indian Tribe of the Port Madison Reserva
2 | 294 | Susanville Indian Rancheria, California
2 | 295 | Swinomish Indians of the Swinomish Reservation, Wa
2 | 296 | Sycuan Band of Diegueno Mission Indians of Califor
2 | 297 | Table Bluff Reservation - Wiyot Tribe, California
2 | 298 | Table Mountain Rancheria of California
2 | 299 | Te-Moak Tribe of Western Shoshone Indians of Nevad
2 | 3 | Ak Chin Indian Community of the Maricopa (Ak Chin)
2 | 30 | California Valley Miwok Tribe, California (formerl
2 | 300 | Thlopthlocco Tribal Town, Oklahoma
2 | 301 | Three Affiliated Tribes of the Fort Berthold Reser
2 | 302 | Tohono O'odham Nation of Arizona
2 | 303 | Tonawanda Band of Seneca Indians of New York
2 | 304 | Tonkawa Tribe of Indians of Oklahoma
2 | 305 | Tonto Apache Tribe of Arizona
2 | 306 | Torres-Martinez Band of Cahuilla Mission Indians o
2 | 307 | Tule River Indian Tribe of the Tule River Reservat
2 | 308 | Tulalip Tribes of the Tulalip Reservation, Washing
2 | 309 | Tunica-Biloxi Indian Tribe of Louisiana
2 | 31 | Campo Band of Diegueno Mission Indians of the Camp
2 | 310 | Tuolumne Band of Me-Wuk Indians of the Tuolumne Ra
2 | 311 | Turtle Mountain Band of Chippewa Indians of North
2 | 312 | Tuscarora Nation of New York
2 | 313 | Twenty-Nine Palms Band of Mission Indians of Calif
2 | 314 | United Auburn Indian Community of the Auburn Ranch
2 | 315 | United Keetoowah Band of Cherokee Indians of Oklah
2 | 316 | Upper Lake Band of Pomo Indians of Upper Lake Ranc
2 | 317 | Upper Sioux Indian Community of the Upper Sioux Re
2 | 318 | Upper Skagit Indian Tribe of Washington
2 | 319 | Ute Indian Tribe of the Uintah & Ouray Reservation
2 | 32 | Capitan Grande Band of Diegueno Mission Indians of
2 | 320 | Ute Mountain Tribe of the Ute Mountain Reservation
2 | 321 | Utu Utu Gwaitu Paiute Tribe of the Benton Paiute R
2 | 322 | Walker River Paiute Tribe of the Walker River Rese
2 | 323 | Wampanoag Tribe of Gay Head (Aquinnah) of Massachu
2 | 324 | Washoe Tribe of Nevada & California (Carson Colony
2 | 325 | White Mountain Apache Tribe of the Fort Apache Res
2 | 326 | Wichita and Affiliated Tribes (Wichita, Keechi, Wa
2 | 327 | Winnebago Tribe of Nebraska
2 | 328 | Winnemucca Indian Colony of Nevada
2 | 329 | Wyandotte Tribe of Oklahoma
2 | 33 | Barona Group of Capitan Grande Band of Mission Ind
2 | 330 | Yankton Sioux Tribe of South Dakota
2 | 331 | Yavapai-Apache Nation of the Camp Verde Indian Res
2 | 332 | Yavapai-Prescott Tribe of the Yavapai Reservation,
2 | 333 | Yerington Paiute Tribe of the Yerington Colony & C
2 | 334 | Yomba Shoshone Tribe of the Yomba Reservation, Nev
2 | 335 | Ysleta Del Sur Pueblo of Texas
2 | 336 | Yurok Tribe of the Yurok Reservation, California
2 | 337 | Zuni Tribe of the Zuni Reservation, New Mexico
2 | 34 | Viejas (Baron Long) Group of Capitan Grande Band o
2 | 35 | Catawba Indian Nation (aka Catawba Tribe of South
2 | 36 | Cayuga Nation of New York
2 | 37 | Cedarville Rancheria, California
2 | 38 | Chemehuevi Indian Tribe of the Chemehuevi Reservat
2 | 39 | Cher-Ae Heights Indian Community of the Trinidad R
2 | 4 | Alabama-Coushatta Tribes of Texas
2 | 40 | Cherokee Nation, Oklahoma
2 | 41 | Cheyenne-Arapaho Tribes of Oklahoma
2 | 42 | Cheyenne River Sioux Tribe of the Cheyenne River R
2 | 43 | Chickasaw Nation, Oklahoma
2 | 44 | Chicken Ranch Rancheria of Me-Wuk Indians of Calif
2 | 45 | Chippewa-Cree Indians of the Rocky Boy's Reservati
2 | 46 | Chitimacha Tribe of Louisiana
2 | 47 | Choctaw Nation of Oklahoma

2 | 48 | Citizen Potawatomi Nation, Oklahoma
2 | 49 | Cloverdale Rancheria of Pomo Indians of California
2 | 5 | Alabama-Quassarte Tribal Town, Oklahoma
2 | 50 | Cocopah Tribe of Arizona
2 | 51 | Coeur D'Alene Tribe of the Coeur D'Alene Reservati
2 | 52 | Cold Springs Rancheria of Mono Indians of Californ
2 | 53 | Colorado River Indian Tribes of the Colorado River
2 | 54 | Comanche Indian Tribe, Oklahoma
2 | 55 | Confederated Salish & Kootenai Tribes of the Flath
2 | 56 | Confederated Tribes of the Chehalis Reservation, W
2 | 57 | Confederated Tribes of the Colville Reservation, W
2 | 58 | Confederated Tribes of the Coos, Lower Umpqua and
2 | 59 | Confederated Tribes of the Goshute Reservation, Ne
2 | 6 | Alturas Indian Rancheria, California
2 | 60 | Confederated Tribes of the Grand Ronde Community o
2 | 61 | Confederated Tribes of the Siletz Reservation, Ore
2 | 62 | Confederated Tribes of the Umatilla Reservation, O
2 | 63 | Confederated Tribes of the Warm Springs Reservatio
2 | 64 | Confederated Tribes and Bands of the Yakama Indian
2 | 65 | Coquille Tribe of Oregon
2 | 66 | Cortina Indian Rancheria of Wintun Indians of Cali
2 | 67 | Coushatta Tribe of Louisiana
2 | 68 | Cow Creek Band of Umpqua Indians of Oregon
2 | 69 | Coyote Valley Band of Pomo Indians of California
2 | 7 | Apache Tribe of Oklahoma
2 | 70 | Crow Tribe of Montana
2 | 71 | Crow Creek Sioux Tribe of the Crow Creek Reservati
2 | 72 | Cuyapaipe Community of Diegueno Mission Indians of
2 | 73 | Death Valley Timbi-Sha Shoshone Band of California
2 | 74 | Delaware Nation, Oklahoma (formerly Delaware Tribe
2 | 75 | Delaware Tribe of Indians, Oklahoma
2 | 76 | Dry Creek Rancheria of Pomo Indians of California
2 | 77 | Duckwater Shoshone Tribe of the Duckwater Reservat
2 | 78 | Eastern Band of Cherokee Indians of North Carolina
2 | 79 | Eastern Shawnee Tribe of Oklahoma
2 | 8 | Arapahoe Tribe of the Wind River Reservation, Wyom
2 | 80 | Elem Indian Colony of Pomo Indians of the Sulphur
2 | 81 | Elk Valley Rancheria, California
2 | 82 | Ely Shoshone Tribe of Nevada
2 | 83 | Enterprise Rancheria of Maidu Indians of Californi
2 | 84 | Flandreau Santee Sioux Tribe of South Dakota
2 | 85 | Forest County Potawatomi Community of Wisconsin Po
2 | 86 | Fort Belknap Indian Community of the Fort Belknap
2 | 87 | Fort Bidwell Indian Community of the Fort Bidwell
2 | 88 | Fort Independence Indian Community of Paiute India
2 | 89 | Fort McDermitt Paiute and Shoshone Tribes of the F
2 | 9 | Aroostook Band of Micmac Indians of Maine
2 | 90 | Fort McDowell Yavapai Nation, Arizona (formerly th
2 | 91 | Fort Mojave Indian Tribe of Arizona, California
2 | 92 | Fort Sill Apache Tribe of Oklahoma
2 | 93 | Gila River Indian Community of the Gila River Indi
2 | 94 | Grand Traverse Band of Ottawa & Chippewa Indians o
2 | 95 | Graton Rancheria, California
2 | 96 | Greenville Rancheria of Maidu Indians of Californi
2 | 97 | Grindstone Indian Rancheria of Wintun-Wailaki Indi
2 | 98 | Guidiville Rancheria of California
2 | 99 | Hannahville Indian Community of Wisconsin Potawato

4.4.2.144 HL7 v3 Code System VaccineManufacturer

URL: https://hl7.org/fhir/R4/v3/VaccineManufacturer/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-VaccineManufacturer) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-VaccineManufacturer
Version: | 2018-08-12
Name: | v3.VaccineManufacturer
Title: | v3 Code System VaccineManufacturer
Definition: | The manufacturer of a vaccine.
OID: | 2.16.840.1.113883.5.144 (for OID based terminology systems)

Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System VaccineManufacturer (The manufacturer of a vaccine.)

4.4.2.144.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | AB | Abbott Laboratories (includes Ross Products Division) | Abbott Laboratories (includes Ross Products Division)
1 | AD | Adams Laboratories | Adams Laboratories
1 | ALP | Alpha Therapeutic Corporation | Alpha Therapeutic Corporation
1 | AR | Armour [Inactive - use CEN] | Armour [Inactive-use CEN]
1 | AVI | Aviron | Aviron
1 | BA | Baxter Healthcare Corporation | Baxter Healthcare Corporation
1 | BAY | Bayer Corporation (includes Miles, Inc. and Cutter Laboratories) | Bayer Corporation (includes Miles, Inc. and Cutter Laboratories)
1 | BP | Berna Products [Inactive - use BPC] | Berna Products [Inactive-use BPC]
1 | BPC | Berna Products Corporation (includes Swiss Serum and Vaccine Institute Berne) | Berna Products Corporation (includes Swiss Serum and Vaccine Institute Berne)
1 | CEN | Centeon L.L.C. (includes Armour Pharmaceutical Company) | Centeon L.L.C. (includes Armour Pharmaceutical Company)
1 | CHI | Chiron Corporation | Chiron Corporation
1 | CON | Connaught [Inactive - use PMC] | Connaught [Inactive-use PMC]
1 | EVN | Evans Medical Limited (an affiliate of Medeva Pharmaceuticals, Inc.) | Evans Medical Limited (an affiliate of Medeva Pharmaceuticals, Inc.)
1 | GRE | Greer Laboratories, Inc. | Greer Laboratories, Inc.
1 | IAG | Immuno International AG | Immuno International AG
1 | IM | Merieux [Inactive - use PMC] | Merieux [Inactive-use PMC]
1 | IUS | Immuno-U.S., Inc. | Immuno-U.S., Inc.
1 | JPN | The Research Foundation for Microbial Diseases of Osaka University (BIKEN) | The Research Foundation for Microbial Diseases of Osaka University (BIKEN)
1 | KGC | Korea Green Cross Corporation | Korea Green Cross Corporation
1 | LED | Lederle [Inactive - use WAL] | Lederle [Inactive-use WAL]
1 | MA | Massachusetts Public Health Biologic Laboratories | Massachusetts Public Health Biologic Laboratories
1 | MED | MedImmune, Inc. | MedImmune, Inc.
1 | MIL | Miles [Inactive - use BAY] | Miles [Inactive-use BAY]
1 | MIP | Bioport Corporation (formerly Michigan Biologic Products Institute) | Bioport Corporation (formerly Michigan Biologic Products Institute)
1 | MSD | Merck and Co., Inc. | Merck & Co., Inc.
1 | NAB | NABI (formerly North American Biologicals, Inc.) | NABI (formerly North American Biologicals, Inc.)
1 | NAV | North American Vaccine, Inc. | North American Vaccine, Inc.
1 | NOV | Novartis Pharmaceutical Corporation (includes Ciba-Geigy Limited and Sandoz Limited) | Novartis Pharmaceutical Corporation (includes Ciba-Geigy Limited and Sandoz Limited)
1 | NYB | New York Blood Center | New York Blood Center
1 | ORT | Ortho Diagnostic Systems, Inc. | Ortho Diagnostic Systems, Inc.
1 | OTC | Organon Teknika Corporation | Organon Teknika Corporation
1 | PD | Parkedale Pharmaceuticals (formerly Parke-Davis) | Parkedale Pharmaceuticals (formerly Parke-Davis)
1 | PMC | Aventis Pasteur Inc. (formerly Pasteur Merieux Connaught; includes Connaught Laboratories and Pasteur Merieux) | Aventis Pasteur Inc. (formerly Pasteur Merieux Connaught; includes Connaught Laboratories and Pasteur Merieux)
1 | PRX | Praxis Biologics [Inactive - use WAL] | Praxis Biologics [Inactive-use WAL]
1 | SCL | Sclavo, Inc. | Sclavo, Inc.
1 | SI | Swiss Serum and Vaccine Inst. [Inactive - use BPC] | Swiss Serum and Vaccine Inst. [Inactive-use BPC]
1 | SKB | SmithKline Beecham | SmithKline Beecham
1 | USA | United States Army Medical Research and Materiel Command | United States Army Medical Research and Materiel Command
1 | WA | Wyeth-Ayerst [Inactive - use WAL] | Wyeth-Ayerst [Inactive-use WAL]
1 | WAL | Wyeth-Ayerst (includes Wyeth-Lederle Vaccines and Pediatrics, Wyeth Laboratories, Lederle Laboratories, and Praxis Biologics) | Wyeth-Ayerst (includes Wyeth-Lederle Vaccines and Pediatrics, Wyeth Laboratories, Lederle Laboratories, and Praxis Biologics)

4.4.2.222 HL7 v3 Code System ObservationCategory

URL: https://hl7.org/fhir/R4/v3/ObservationCategory/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ObservationCategory) is defined as part of HL7 v3. Related FHIR content: Observation Category Codes.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ObservationCategory
Version: | 2018-08-12
Name: | v3.ObservationCategory
Title: | v3 Code System ObservationCategory

Definition: | High level observation categories for the general type of observation being made. URL: http://hl7-fhir.github.io/valueset-observation-category.html This is an inline code system http://hl7.org/fhir/observation-category.
OID: | 2.16.840.1.113883.4.642.1.222 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ObservationCategory (High level observation categories for the general type of observation being made. URL: http://hl7-fhir.github.io/valueset-observation-category.html This is an inline code system http://hl7.org/fhir/observation-category.)

4.4.2.222.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | exam | Exam | Observations generated by physical exam findings including direct observations made by a clinician and use of simple instruments and the result of simple maneuvers performed directly on the patient's body.
1 | imaging | Imaging | Observations generated by imaging. The scope includes observations, plain x-ray, ultrasound, CT, MRI, angiography, echocardiography, nuclear medicine.
1 | laboratory | Laboratory | The results of observations generated by laboratories. Laboratory results are typically generated by laboratories providing analytic services in areas such as chemistry, hematology, serology, histology, cytology, anatomic pathology, microbiology, and/or virology. These observations are based on analysis of specimens obtained from the patient and submitted to the laboratory.
1 | procedure | Procedure | Observations generated by other procedures. This category includes observations resulting from interventional and non-interventional procedures excluding lab and imaging (e.g. cardiology catheterization, endoscopy, electrodiagnostics, etc.). Procedure results are typically generated by a clinician to provide more granular information about component observations made during a procedure, such as where a gastroenterologist reports the size of a polyp observed during a colonoscopy.
1 | social-history | Social History | The Social History Observations define the patient's occupational, personal (e.g. lifestyle), social, and environmental history and health risk factors, as well as administrative data such as marital status, race, ethnicity and religious affiliation.
1 | survey | Survey | Assessment tool/survey instrument observations (e.g. Apgar Scores, Montreal Cognitive Assessment (MoCA))
1 | therapy | Therapy | Observations generated by non-interventional treatment protocols (e.g. occupational, physical, radiation, nutritional and medication therapy)
1 | vital-signs | Vital Signs | Clinical observations measure the body's basic functions such as such as blood pressure, heart rate, respiratory rate, height, weight, body mass index, head circumference, pulse oximetry, temperature, and body surface area.

4.4.2.1001 HL7 v3 Code System ActMood

URL: https://hl7.org/fhir/R4/v3/ActMood/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ActMood) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ActMood
Version: | 2018-08-12
Name: | v3.ActMood
Title: | v3 Code System ActMood
Definition: | OpenIssue: In Ballot 2009May, a strong Negative vote was lodged against several of the concept definitions in the vocabulary used for Act.moodCode. The vote was found "Persuasive With Mod", with the understanding that M and M would undertake a detailed review of these concept definitions for a future release of the RIM.
OID: | 2.16.840.1.113883.5.1001 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ActMood (OpenIssue: In Ballot 2009May, a strong Negative vote was lodged against several of the concept definitions in the vocabulary used for Act.moodCode. The vote was found "Persuasive With Mod", with the understanding that M and M would undertake a detailed review of these concept definitions for a future release of the RIM.)

• ValueSet: V3 Value SetActMoodPredicate (Any of the above service moods (e.g., event, intent, or goal) can be turned into a predicate used as a criterion to express conditionals (or queries.) However, currently we allow only criteria on service events.)

• ValueSet: Example with inactive codes (HL7 v3 ActMood Predicate codes, including inactive codes)

• ValueSet: Example with inactive codes (HL7 v3 ActMood Predicate codes, including inactive codes)

• ValueSet: V3 Value SetActMoodIntent (An intention or plan to perform a service. Historical note: in previous RIM versions, the intent mood was captured as a separate class hierarchy, called Service_intent_or_order.)

4.4.2.1001.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_ActMoodCompletionTrack) Abstract | | These are moods describing activities as they progress in the business cycle, from defined, through planned and ordered to completed.

2 | (_ActMoodPotential) Abstract | | Definition: A possible act.
3 | DEF | definition | Definition: A definition of a kind of act that can occur . OpenIssue: The semantic constructs embodied in DEF and CRT moods seem indistinguishable, and their uses can readily be determined by the context in which these are used. Therefore, this OpenIssue has been created to declare that it is likely that ActMood.DEF will be "retired" in the future in favor of the more general ActMood.CRT.
3 | PERM | permission | Definition: A kind of act that defines a permission that has been granted.
3 | SLOT | resource slot | Definition: A kind of act that may occur during the specified time period.
2 | EVN | event (occurrence) | Definition: An act that actually happens (may be an ongoing act or a documentation of a past act).
2 | INT | intent | Definition: An intention or plan for an act. >UsageNotes: The final outcome of the intent, the act that is intended to occur, is always an event. However the final outcome may be reached indirectly via steps through other intents, such as promise, permission request, or an appointment that may lead to an actual event to occur. Alternatively, the intended act may never occur.
3 | (_ActMoodDesire) Abstract | | Definition: A desire to have an act occur.
4 | (_ActMoodActRequest) Abstract | | Definition: A request (or order) for an act that is part of a defined request/fulfillment cycle. UsageNotes: Use of an HL7 defined request/fulfillment framework is not required to use this mood code.
5 | ARQ | appointment request | Definition: A request act that is specialized for the appointment scheduling request/fulfillment cycle. An appointment request is fulfilled only and completely by an appointment (APT), i.e., all that the appointment request intends is to create an appointment (the actual act may well not happen if that is the professional decision during the appointment).
5 | PERMRQ | permission request | Definition: A request for a permission to perform the act. Typically a payer (or possibly a supervisor) is being requested to give permission to perform the act. As opposed to the RQO, the requestee is not asked to perform or cause to perform the act but only to give the permission.
5 | RQO | request | Definition: A request act that is specialized for an event request/fulfillment cycle. UsageNotes: The fulfillment cycle may involve intermediary fulfilling acts in moods such as PRMS, APT, or even another RQO before being fulfilled by the final event. UsageNotes: The concepts of a "request" and an "order" are viewed as different, because there is an implication of a mandate associated with order. In practice, however, this distinction has no general functional value in the inter-operation of health care computing. "Orders" are commonly refused for a variety of clinical and business reasons, and the notion of a "request" obligates the recipient (the fulfiller) to respond to the sender (the author). Indeed, in many regions, including Australia and Europe, the common term used is "request." Thus, the concept embodies both notions, as there is no useful distinction to be made. If a mandate is to be associated with a request, this will be embodied in the "local" business rules applied to the transactions. Should HL7 desire to provide a distinction between these in the future, the individual concepts could be added as specializations of this concept. The critical distinction here, is the difference between this concept and an "intent", of which it is a specialization. An intent involves decisions by a single party, the author. A request, however, involves decisions by two parties, the author and the fulfiller, with an obligation on the part of the fulfiller to respond to the request indicating that the fulfiller will indeed fulfill the request.
4 | PRP | proposal | Definition: A suggestion that an act might be performed. Not an explicit request, and professional responsibility may or may not be present.
5 | RMD | recommendation | Definition: A suggestion that an act should be performed with an acceptance of some degree of professional responsibility for the resulting act. Not an explicit request. . UsageNotes: Where there is no clear definition or applicable concept of "professional responsibilityâ€?, RMD becomes indistinguishable from PRP. .
3 | PRMS | promise | Definition: A commitment to perform an act (may be either solicited or unsolicited). The committer becomes responsible to the other party for executing the act, and, as a consequence, the other party may rely on the first party to perform or cause to perform the act. UsageNotes: Commitments may be retracted or cancelled.
4 | APT | appointment | Definition: An act that has been scheduled to be performed at a specific place and time.
1 | (_ActMoodPredicate) Abstract | | Definition: An act that expresses condition statements for other acts.
2 | CRT Deprecated | criterion | Deprecation Comment: This concept This codes should no longer be used. Instead, set attribute Act.isCriterionInd to "true" and use the desired mood for your criterion. Definition: A condition that must be true for the source act to be considered.
3 | EVN.CRT Deprecated | event criterion | Deprecation Comment: This concept This codes should no longer be used. Instead, set attribute Act.isCriterionInd to "true" and use the desired mood for your criterion. Definition: A criterion (CRT) that has_match = an event (EVN).
3 | GOL.CRT | goal criterion | A criterion expressed over goals (ActMood.GOL).
3 | INT.CRT | intent criterion | A criterion expressed over intents (ActMood.INT).
4 | PRMS.CRT | promise criterion | A criterion expressed over promises (ActMood.PRMS).
4 | RQO.CRT | request criterion | A criterion expressed over requests or orders (ActMood.RQO).
3 | RSK.CRT | risk criterion | A criterion expressed over risks (ActMood.RSK).
2 | EXPEC | expectation | Definition: An act that is considered to have some noteworthy likelihood of occurring in the future (has_match = event). Examples:Prognosis of a condition, Expected date of discharge from hospital, patient will likely need an emergency decompression of the intracranial pressure by morning. UsageNotes:INT (intent) reflects a plan for the future, which is a declaration to do something. This contrasts with expectation, which is a prediction that something will happen in the future. GOL (goal) reflects a hope rather than a prediction. RSK (risk) reflects a potential negative event that may or may not be expected to happen.
3 | GOL | Goal | Definition: An expectation that is considered to be desirable to occur in the future Examples:Target weight below 80Kg, Stop smoking, Regain ability to walk, goal is to administer thrombolytics to candidate patients presenting with acute myocardial infarction. UsageNotes: INT (intent) reflects a plan for the future, which is a declaration to do something. This contrasts with goal which doesn't represent an intention to act, merely a hope for an eventual result. A goal is distinct from the intended actions to reach that goal. "I will reduce the dose of drug x to 20mg" is an intent. "I hope to be able to get the patient to the point where I can reduce the dose of drug x to 20mg" is a goal. EXPEC (expectation) reflects a prediction rather than a hope. RSK (risk) reflects a potential negative event rather than a hope.
3 | RSK | risk | Definition:An act that may occur in the future and which is regarded as undesirable Examples:Increased risk of DVT, at risk for sub-acute bacterial endocarditis. UsageNotes:Note: An observation in RSK mood expresses the undesirable act, and not the underlying risk factor. A risk factor that is present (e.g. obesity, smoking, etc) should be expressed in event mood. INT (intent) reflects a plan for the future, which is a declaration to do something. This contrasts with RSK (risk), which is the potential that something negative will occur that may or may not ever happen. GOL (goal) reflects a hope to achieve something. EXPEC (expectation) is the prediction of a positive or negative event. This contrasts with RSK (risk), which is the potential that something negative will occur that may or may not ever happen, and may not be expected to happen.

2 | OPT | option | Definition: One of a set of acts that specify an option for the property values that the parent act may have. Typically used in definitions or orders to describe alternatives. An option can only be used as a group, that is, all assigned values must be used together. The actual mood of the act is the same as the parent act, and they must be linked by an actrelationship with type = OPTN.

4.4.2.1002 HL7 v3 Code System ActRelationshipType

URL: https://hl7.org/fhir/R4/v3/ActRelationshipType/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType
Version: | 2018-08-12
Name: | v3.ActRelationshipType
Title: | v3 Code System ActRelationshipType
Definition: | The source is an excerpt from the target.
OID: | 2.16.840.1.113883.5.1002 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ActRelationshipType (The source is an excerpt from the target.)

• ValueSet: V3 Value SetActRelationshipFulfills (The source act fulfills (in whole or in part) the target act. Source act must be in a mood equal or more actual than the target act.)

• ValueSet: V3 Value SetActRelationshipHasComponent (A collection of sub-services as steps or subtasks performed for the source service. Services may be performed sequentially or concurrently.)

• ValueSet: V3 Value SetActRelationshipConditional (Specifies under what circumstances (target Act) the source-Act may, must, must not or has occurred)

• ValueSet: V3 Value SetActRelationshipPertains (This is a very unspecific relationship from one item of clinical information to another. It does not judge about the role the pertinent information plays.)

4.4.2.1002.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | ART | act relationship type | Description: A directed association between a source Act and a target Act. Usage Note: This code should never be transmitted in an instance as the value of ActRelationship.typeCode (attribute)
2 | (_ActClassTemporallyPertains) Abstract
2 | (_ActRelationshipAccounting) Abstract | | Codes that describe the relationship between an Act and a financial instrument such as a financial transaction, account or invoice element.
3 | (_ActRelationshipCostTracking) Abstract | | Expresses values for describing the relationship relationship between an InvoiceElement or InvoiceElementGroup and a billable act.
4 | CHRG | has charge | A relationship that provides an ability to associate a financial transaction (target) as a charge to a clinical act (source). A clinical act may have a charge associated with the execution or delivery of the service. The financial transaction will define the charge (bill) for delivery or performance of the service. Charges and costs are distinct terms. A charge defines what is charged or billed to another organization or entity within an organization. The cost defines what it costs an organization to perform or deliver a service or product.
4 | COST | has cost | A relationship that provides an ability to associate a financial transaction (target) as a cost to a clinical act (source). A clinical act may have an inherit cost associated with the execution or delivery of the service. The financial transaction will define the cost of delivery or performance of the service. Charges and costs are distinct terms. A charge defines what is charged or billed to another organization or entity within an organization. The cost defines what it costs an organization to perform or deliver a service or product.
3 | (_ActRelationshipPosting) Abstract | | Expresses values for describing the relationship between a FinancialTransaction and an Account.
4 | CREDIT | has credit | A credit relationship ties a financial transaction (target) to an account (source). A credit, once applied (posted), may have either a positive or negative effect on the account balance, depending on the type of account. An asset account credit will decrease the account balance. A non-asset account credit will decrease the account balance.
4 | DEBIT | has debit | A debit relationship ties a financial transaction (target) to an account (source). A debit, once applied (posted), may have either a positive or negative effect on the account balance, depending on the type of account. An asset account debit will increase the account balance. A non-asset account debit will decrease the account balance.
2 | (_ActRelationshipConditional) Abstract | | Specifies under what circumstances (target Act) the source-Act may, must, must not or has occurred
3 | CIND | has contra-indication | A contraindication is just a negation of a reason, i.e. it gives a condition under which the action is not to be done. Both, source and target can be any kind of service; target service is in criterion mood. How the strength of a contraindication is expressed (e.g., relative, absolute) is left as an open issue. The priorityNumber attribute could be used.
3 | PRCN | has pre-condition | A requirement to be true before a service is performed. The target can be any service in criterion mood. For multiple pre-conditions a conjunction attribute (AND, OR, XOR) is applicable.
3 | RSON | has reason | Description: The reason or rationale for a service. A reason link is weaker than a trigger, it only suggests that some service may be or might have been a reason for some action, but not that this reason requires/required the action to be taken. Also, as opposed to the trigger, there is no strong timely relation between the reason and the action. As well as providing various types of information about the rationale for a service, the RSON act relationship is routinely used between a SBADM act and an OBS act to describe the indication for use of a medication. Child concepts may be used to describe types of indication.

Discussion: In prior releases, the code "SUGG" (suggests) was expressed as "an inversion of the reason link." That code has been retired in favor of the inversion indicator that is an attribute of ActRelationship.
4 | BLOCK | blocks | Definition: The source act is performed to block the effects of the target act. This act relationship should be used when describing near miss type incidents where potential harm could have occurred, but the action described in the source act blocked the potential harmful effects of the incident actually occurring.
4 | DIAG | diagnoses | Description: The source act is intended to help establish the presence of a (an adverse) situation described by the target act. This is not limited to diseases but can apply to any adverse situation or condition of medical or technical nature.
4 | IMM | immunization against | Description: The source act is intented to provide immunity against the effects of the target act (the target act describes an infectious disease)
5 | ACTIMM | active immunization against | Description: The source act is intended to provide active immunity against the effects of the target act (the target act describes an infectious disease)
5 | PASSIMM | passive immunization against | Description: The source act is intended to provide passive immunity against the effects of the target act (the target act describes an infectious disease).
4 | MITGT | mitigates | The source act removes or lessens the occurrence or effect of the target act.
5 | RCVY | recovers | Definition: The source act is performed to recover from the effects of the target act.
4 | PRYLX | prophylaxis of | Description: The source act is intended to reduce the risk of of an adverse situation to emerge as described by the target act. This is not limited to diseases but can apply to any adverse situation or condition of medical or technical nature.
4 | TREAT | treats | Description: The source act is intended to improve a pre-existing adverse situation described by the target act. This is not limited to diseases but can apply to any adverse situation or condition of medical or technical nature.
5 | ADJUNCT | adjunctive treatment | Description: The source act is intended to offer an additional treatment for the management or cure of a pre-existing adverse situation described by the target act. This is not limited to diseases but can apply to any adverse situation or condition of medical or technical nature. It is not a requirement that the non-adjunctive treatment is explicitly specified.
5 | MTREAT | maintenance treatment | Description: The source act is intended to provide long term maintenance improvement or management of a pre-existing adverse situation described by the target act. This is not limited to diseases but can apply to any adverse situation or condition of medical or technical nature.
5 | PALLTREAT | palliates | Description: The source act is intended to provide palliation for the effects of the target act.
5 | SYMP | symptomatic relief | Description: The source act is intented to provide symptomatic relief for the effects of the target act.
3 | TRIG | has trigger | A pre-condition that if true should result in the source Act being executed. The target is in typically in criterion mood. When reported after the fact (i.e. the criterion has been met) it may be in Event mood. A delay between the trigger and the triggered action can be specified. Discussion: This includes the concept of a required act for a service or financial instrument such as an insurance plan or policy. In such cases, the trigger is the occurrence of a specific condition such as coverage limits being exceeded.
2 | (_ActRelationshipTemporallyPertains) Abstract | | Abstract collector for ActRelationhsip types that relate two acts by their timing.
3 | (_ActRelationshipTemporallyPertainsApproximates) Abstract | | Abstract collector for ActRelationship types that relate two acts by their approximate timing.
4 | ENE | ends near end | A relationship in which the source act's effective time ends near the end of the target act's effective time. Near is defined separately as a time interval. Usage Note: Inverse code is ENS
5 | ECW | ends concurrent with | A relationship in which the source act's effective time ends with the end of the target act's effective time. UsageNote: This code is reflexive. Therefore its inverse code is itself.
6 | CONCURRENT | concurrent with | A relationship in which the source act's effective time is the same as the target act's effective time. UsageNote: This code is reflexive. Therefore its inverse code is itself.
6 | SBSECWE | starts before start of, ends with | The source Act starts before the start of the target Act, and ends with the target Act. UsageNote: Inverse code is SASECWE
4 | ENS | ends near start | A relationship in which the source act's effective time ends near the start of the target act's effective time. Near is defined separately as a time interval. Usage Note: Inverse code is ENE
5 | ECWS | ends concurrent with start of | The source Act ends when the target act starts (i.e. if we say "ActOne ECWS ActTwo", it means that ActOne ends when ActTwo starts, therefore ActOne is the source and ActTwo is the target). UsageNote: Inverse code is SCWE
4 | SNE | starts near end | A relationship in which the source act's effective time starts near the end of the target act's effective time. Near is defined separately as a time interval. Usage Note: Inverse code is SNS
5 | SCWE | starts concurrent with end of | The source Act starts when the target act ends (i.e. if we say "ActOne SCWE ActTwo", it means that ActOne starts when ActTwo ends, therefore ActOne is the source and ActTwo is the target). UsageNote: Inverse code is SBSECWS
4 | SNS | starts near start | A relationship in which the source act's effective time starts near the start of the target act's effective time. Near is defined separately as a time interval. Usage Note: Inverse code is SNE
5 | SCW | starts concurrent with | A relationship in which the source act's effective time starts with the start of the target act's effective time. UsageNote: This code is reflexive. Therefore its inverse code is itself.
6 | CONCURRENT
6 | SCWSEBE | starts with. ends before end of | The source Act starts with.the target Act and ends before the end of the target Act. UsageNote: Inverse code is SCWSEAE
6 | SCWSEAE | starts with, ends after end of | The source Act starts with the target Act, and ends after the end of the target Act.
3 | EAS | ends after start of | A relationship in which the source act ends after the target act starts. UsageNote: Inverse code is SBE
4 | EAE | ends after end of | A relationship in which the source act ends after the target act ends. UsageNote: Inverse code is EBE
5 | SASEAE | starts after start of, ends after end of | The source Act starts after start of the target Act and ends after end of the target Act. UsageNote: Inverse code is SBSEBE
6 | SBEEAE | contains end of | The source Act contains the end of the target Act. UsageNote: Inverse code is EDU
7 | SASSBEEAS | start after start of, contains end of | The source Act start after the start of the target Act, and contains the end of the target Act. UsageNote: Inverse code is SBSEASEBE
7 | SBSEAE | contains time of | The source Act contains the time of the target Act. UsageNote: Inverse code is DURING
7 | SCWSEAE
6 | SASSBEEAS

4 | SAS | starts after start of | The source Act starts after the start of the target Act (i.e. if we say "ActOne SAS ActTwo", it means that ActOne starts after the start of ActTwo, therefore ActOne is the source and ActTwo is the target). UsageNote: Inverse code is SBS
5 | SASEAE
5 | SAE | starts after end of | A relationship in which the source act starts after the target act ends. UsageNote: Inverse code is EBS
5 | DURING | occurs during | A relationship in which the source act's effective time is wholly within the target act's effective time (including end points, as defined in the act's effective times) UsageNote: Inverse code is SBSEAE
5 | SASECWE | starts after start of, ends with | The source Act starts after start of the target Act, and ends with the target Act. UsageNote: Inverse code is SBSECWE
3 | EASORECWS | ends after or concurrent with start of | A relationship in which the source act's effective time ends after or concurrent with the start of the target act's effective time. Usage Note: Inverse code is EBSORECWS
4 | EAS
4 | EAEORECW | ends after or concurrent with end of | A relationship in which the source act's effective time ends after or concurrent with the end of the target act's effective time. Usage Note: Inverse code is EBEORECW
5 | EAE
5 | ECW
4 | ECWS
3 | INDEPENDENT | independent of time of | The source Act is independent of the time of the target Act. UsageNote: This code is reflexive. Therefore its inverse code is itself.
3 | SAEORSCWE | starts after or concurrent with end of | A relationship in which the source act's effective time starts after or concurrent with the end of the target act's effective time. Usage Note: Inverse code is SBEORSCWE
4 | SCWE
4 | SAE
3 | SASORSCW | starts after or concurrent with start of | A relationship in which the source act's effective time starts after or concurrent with the start of the target act's effective time. Usage Note: Inverse code is SBSORSCW
4 | SAS
4 | SCW
3 | SBEORSCWE | starts before or concurrent with end of | A relationship in which the source act's effective time starts before or concurrent with the end of the target act's effective time. Usage Note: Inverse code is SAEORSCWE
4 | OVERLAP | overlaps with | A relationship in which the source act's effective time overlaps the target act's effective time in any way. UsageNote: This code is reflexive. Therefore its inverse code is itself.
5 | ECW
5 | EDU | ends during | A relationship in which the source act ends within the target act's effective time (including end points, as defined in the act's effective times) UsageNote: Inverse code is SBEEAE
6 | SBSEASEBE | contains start of, ends before end of | The source Act contains the start of the target Act, and ends before the end of the target Act. UsageNote: Inverse code is SASSBEEAS
6 | SCWSEBE
5 | SBEEAE
5 | SBSEAS | contains start of | The source Act contains the start of the target Act. UsageNote: Inverse code is SDU
6 | SBSECWE
6 | SBSEASEBE
6 | SBSEAE
5 | SCW
5 | SCWE
5 | SDU | starts during | A relationship in which the source act starts within the target act's effective time (including end points, as defined in the act's effective times) UsageNote: Inverse code is SBSEAS
6 | SASSBEEAS
4 | SBE | starts before end of | The source Act starts before the end of the target Act (i.e. if we say "ActOne SBE ActTwo", it means that ActOne starts before the end of ActTwo, therefore ActOne is the source and ActTwo is the target). UsageNote: Inverse code is EAS
5 | EBE | ends before end of | The source Act ends before the end of the target Act (i.e. if we say "ActOne EBE ActTwo", it means that ActOne ends before the end of ActTwo, therefore ActOne is the source and ActTwo is the target). UsageNote: Inverse code is EAE
6 | SBSEBE | starts before start of, ends before end of | The source Act starts before the start of the target Act, and ends before the end of the target Act. UsageNote: Inverse code is SASEAE
7 | SBSEASEBE
7 | EBSORECWS | ends before or concurrent with start of | A relationship in which the source act's effective time ends before or concurrent with the start of the target act's effective time. Usage Note: Inverse code is EASORECWS
8 | ECWS
8 | EBS | ends before start of | A relationship in which the source act ends before the target act starts. UsageNote: Inverse code is SAE
5 | SBSEASEBE
5 | SCWSEBE
5 | EBEORECW | ends before or concurrent with end of | A relationship in which the source act's effective time ends before or concurrent with the end of the target act's effective time. Usage Note: Inverse code is EAEORECW
6 | ECW
6 | EBE
5 | SBSORSCW | starts before or concurrent with start of | A relationship in which the source act's effective time starts before or concurrent with the start of the target act's effective time. Usage Note: Inverse code is SASORSCW
6 | SCW
6 | SBS | starts before start of | A relationship in which the source act begins before the target act begins. UsageNote: Inverse code is SAS
7 | SBSEBE
7 | SBSEAS

2 | AUTH | authorized by | A relationship in which the target act authorizes or certifies the source act.

2 | CAUS | is etiology for | Description: An assertion that an act was the cause of another act.This is stronger and more specific than the support link. The source (cause) is typically an observation, but may be any act, while the target may be any act. Examples: a growth of Staphylococcus aureus may be considered the cause of an abscess contamination of the infusion bag was deemed to be the cause of the infection that the patient experienced lack of staff on the shift was deemed to be a supporting factor (proximal factor) causing the patient safety incident where the patient fell out of bed because the bed-sides had not been put up which caused the night patient to fall out of bed

2 | COMP | has component | The target act is a component of the source act, with no semantics regarding composition or aggregation implied.

3 | CTRLV | has control variable | A relationship from an Act to a Control Variable. For example, if a Device makes an Observation, this relates the Observation to its Control Variables documenting the device's settings that influenced the observation.

3 | MBR | has member | The target Acts are aggregated by the source Act. Target Acts may have independent existence, participate in multiple ActRelationships, and do not contribute to the meaning of the source. UsageNotes: This explicitly represents the conventional notion of aggregation. The target Act is part of a collection of Acts (no implication is made of cardinality, a source of Acts may contain zero, one, or more member target Acts). It is expected that this will be primarily used with _ActClassRecordOrganizer, BATTERY, and LIST

4 | STEP | has step | A collection of sub-services as steps or subtasks performed for the source service. Services may be performed sequentially or concurrently. UsageNotes: Sequence of steps may be indicated by use of _ActRelationshipTemporallyPertains, as well as via ActRelationship.sequenceNumber, ActRelationship.pauseQuantity, Target.priorityCode. OpenIssue: Need Additional guidelines on when each approach should be used.

5 | ARR | arrival | The relationship that links to a Transportation Act (target) from another Act (source) indicating that the subject of the source Act entered into the source Act by means of the target Transportation act.

5 | DEP | departure | The relationship that links to a Transportation Act (target) from another Act (source) indicating that the subject of the source Act departed from the source Act by means of the target Transportation act.

3 | PART | has part | The source Act is a composite of the target Acts. The target Acts do not have an existence independent of the source Act. UsageNote: In UML 1.1, this is a "composition" defined as: "A form of aggregation with strong ownership and coincident lifetime as part of the whole. Parts with non-fixed multiplicity may be created after the composite itself, but once created they live and die with it (i.e., they share lifetimes). Such parts can also be explicitly removed before the death of the composite. Composition may be recursive."

2 | COVBY | covered by | A relationship in which the source act is covered by or is under the authority of a target act. A financial instrument such as an Invoice Element is covered by one or more specific instances of an Insurance Policy.

2 | DRIV | is derived from | Associates a derived Act with its input parameters. E.G., an anion-gap observation can be associated as being derived from given sodium-, (potassium-,), chloride-, and bicarbonate-observations. The narrative content (Act.text) of a source act is wholly machine-derived from the collection of target acts.

2 | ELNK | episodeLink | Expresses an association that links two instances of the same act over time, indicating that the instance are part of the same episode, e.g. linking two condition nodes for episode of illness; linking two encounters for episode of encounter.

2 | EVID | provides evidence for | Indicates that the target Act provides evidence in support of the action represented by the source Act. The target is not a 'reason' for the source act, but rather gives supporting information on why the source act is an appropriate course of action. Possible targets might be clinical trial results, journal articles, similar successful therapies, etc. Rationale: Provides a mechanism for conveying clinical justification for non-approved or otherwise non-traditional therapies.

2 | EXACBY | exacerbated by | Description:The source act is aggravated by the target act. (Example "chest pain" EXACBY "exercise")

2 | EXPL | has explanation | This is the inversion of support. Used to indicate that a given observation is explained by another observation or condition.

2 | INTF | interfered by | the target act documents a set of circumstances (events, risks) which prevent successful completion, or degradation of quality of, the source Act. UsageNote: This provides the semantics to document barriers to care

2 | ITEMSLOC | items located | Items located

2 | LIMIT | limited by | A relationship that limits or restricts the source act by the elements of the target act. For example, an authorization may be limited by a financial amount (up to $500). Target Act must be in EVN.CRIT mood.

2 | META | has metadata | Definition: Indicates that the attributes and associations of the target act provide metadata (for example, identifiers, authorship, etc.) for the source act. Constraint: Source act must have either a mood code that is not "EVN" (event) or its "isCriterion" attribute must set to "true". Target act must be an Act with a mood code of EVN and with isCriterionInd attribute set to "true".

2 | MFST | is manifestation of | An assertion that a new observation may be the manifestation of another existing observation or action. This assumption is attributed to the same actor who asserts the manifestation. This is stronger and more specific than an inverted support link. For example, an agitated appearance can be asserted to be the manifestation (effect) of a known hyperthyroxia. This expresses that one might not have realized a symptom if it would not be a common manifestation of a known condition. The target (cause) may be any service, while the source (manifestation) must be an observation.

2 | NAME | assigns name | Used to assign a "name" to a condition thread. Source is a condition node, target can be any service.

2 | OUTC | has outcome | An observation that should follow or does actually follow as a result or consequence of a condition or action (sometimes called "post-conditional".) Target must be an observation as a goal, risk or any criterion. For complex outcomes a conjunction attribute (AND, OR, XOR) can be used. An outcome link is often inverted to describe an outcome assessment.

3 | (_ActRelationsipObjective) Abstract | | The target act is a desired outcome of the source act. Source is any act (typically an intervention). Target must be an observation in criterion mood.

4 | OBJC | has continuing objective | A desired state that a service action aims to maintain. E.g., keep systolic blood pressure between 90 and 110 mm Hg. Source is an intervention service. Target must be an observation in criterion mood.

4 | OBJF | has final objective | A desired outcome that a service action aims to meet finally. Source is any service (typically an intervention). Target must be an observation in criterion mood.

3 | GOAL | has goal | A goal that one defines given a patient's health condition. Subsequently planned actions aim to meet that goal. Source is an observation or condition node, target must be an observation in goal mood.

3 | RISK | has risk | A noteworthy undesired outcome of a patient's condition that is either likely enough to become an issue or is less likely but dangerous enough to be addressed.

2 | PERT | has pertinent information | This is a very unspecific relationship from one item of clinical information to another. It does not judge about the role the pertinent information plays.

2 | PREV | has previous instance | A relationship in which the target act is a predecessor instance to the source act. Generally each of these instances is similar, but no identical. In healthcare coverage it is used to link a claim item to a previous claim item that

might have claimed for the same set of services.
2 | REFR | refers to | A relationship in which the target act is referred to by the source act. This permits a simple reference relationship that distinguishes between the referent and the referee.
3 | USE | uses | Indicates that the source act makes use of (or will make use of) the information content of the target act. UsageNotes: A usage relationship only makes sense if the target act is authored and occurs independently of the source act. Otherwise a simpler relationship such as COMP would be appropriate. Rationale: There is a need when defining a clinical trial protocol to indicate that the protocol makes use of other protocol or treatment specifications. This is stronger than the assertion of "references". References may exist without usage, and in a clinical trial protocol is common to assert both: what other specifications does this trial use and what other specifications does it merely reference.
2 | REFV | has reference values | Reference ranges are essentially descriptors of a class of result values assumed to be "normal", "abnormal", or "critical." Those can vary by sex, age, or any other criterion. Source and target are observations, the target is in criterion mood. This link type can act as a trigger in case of alarms being triggered by critical results.
2 | RELVBY | relieved by | Description:The source act is wholly or partially alleviated by the target act. (Example "chest pain" RELVBY "sublingual nitroglycerin administration")
2 | SEQL | is sequel | An act relationship indicating that the source act follows the target act. The source act should in principle represent the same kind of act as the target. Source and target need not have the same mood code (mood will often differ). The target of a sequel is called antecedent. Examples for sequel relationships are: revision, transformation, derivation from a prototype (as a specialization is a derivation of a generalization), followup, realization, instantiation.
3 | APND | is appendage | An addendum (source) to an existing service object (target), containing supplemental information. The addendum is itself an original service object linked to the supplemented service object. The supplemented service object remains in place and its content and status are unaltered.
3 | BSLN | has baseline | Indicates that the target observation(s) provide an initial reference for the source observation or observation group. UsageConstraints: Both source and target must be Observations or specializations thereof.
3 | COMPLY | complies with | Description:The source act complies with, adheres to, conforms to, or is permissible under (in whole or in part) the policy, contract, agreement, law, conformance criteria, certification guidelines or requirement conveyed by the target act. Examples for compliance relationships are: audits of adherence with a security policy, certificate of conformance to system certification requirements, or consent directive in compliance with or permissible under a privacy policy.
3 | DOC | documents | The source act documents the target act.
3 | FLFS | fulfills | The source act fulfills (in whole or in part) the target act. Source act must be in a mood equal or more actual than the target act.
4 | OCCR | occurrence | The source act is a single occurrence of a repeatable target act. The source and target act can be in any mood on the "completion track" but the source act must be as far as or further along the track than the target act (i.e., the occurrence of an intent can be an event but not vice versa).
4 | OREF | references order | Relates either an appointment request or an appointment to the order for the service being scheduled.
4 | SCH | schedules request | Associates a specific time (and associated resources) with a scheduling request or other intent.
3 | GEN | has generalization | The generalization relationship can be used to express categorical knowledge about services (e.g., amilorid, triamterene, and spironolactone have the common generalization potassium sparing diuretic).
3 | GEVL | evaluates (goal) | A goal-evaluation links an observation (intent or actual) to a goal to indicate that the observation evaluates the goal. Given the goal and the observation, a "goal distance" (e.g., goal to observation) can be "calculated" and need not be sent explicitly.
3 | INST | instantiates (master) | Used to capture the link between a potential service ("master" or plan) and an actual service, where the actual service instantiates the potential service. The instantiation may override the master's defaults.
3 | MOD | modifies | Definition: Used to link a newer version or 'snapshot' of a business object (source) to an older version or 'snapshot' of the same business object (target). Usage:The identifier of the Act should be the same for both source and target. If the identifiers are distinct, RPLC should be used instead. Name from source to target = "modifiesPrior" Name from target to source = "modifiesByNew"
3 | MTCH | matches (trigger) | A trigger-match links an actual service (e.g., an observation or procedure that took place) with a service in criterion mood. For example if the trigger is "observation of pain" and pain is actually observed, and if that pain-observation caused the trigger to fire, that pain-observation can be linked with the trigger.
3 | OPTN | has option | A relationship between a source Act that provides more detailed properties to the target Act. The source act thus is a specialization of the target act, but instead of mentioning all the inherited properties it only mentions new property bindings or refinements. The typical use case is to specify certain alternative variants of one kind of Act. The priorityNumber attribute is used to weigh refinements as preferred over other alternative refinements. Example: several routing options for a drug are specified as one SubstanceAdministration for the general treatment with attached refinements for the various routing options.
3 | RCHAL | re-challenge | Description:A relationship in which the target act is carried out to determine whether an effect attributed to the source act can be recreated.
3 | REV | reverses | A relationship between a source Act that seeks to reverse or undo the action of the prior target Act. Example: A posted financial transaction (e.g., a debit transaction) was applied in error and must be reversed (e.g., by a credit transaction) the credit transaction is identified as an undo (or reversal) of the prior target transaction. Constraints: the "completion track" mood of the target Act must be equally or more "actual" than the source act. I.e., when the target act is EVN the source act can be EVN, or any INT. If the target act is INT, the source act can be INT.
3 | RPLC | replaces | A replacement source act replaces an existing target act. The state of the target act being replaced becomes obselete, but the act is typically still retained in the system for historical reference. The source and target must be of the same type.
3 | SUCC | succeeds | Definition: A new act that carries forward the intention of the original act, but does not completely replace it. The status of the predecessor act must be 'completed'. The original act is the target act and the successor is the source act.
3 | UPDT | updates (condition) | A condition thread relationship specifically links condition nodes together to form a condition thread. The source is the new condition node and the target links to the most recent node of the existing condition thread.
3 | XCRPT | Excerpts | The source is an excerpt from the target.
4 | VRXCRPT | Excerpt verbatim | The source is a direct quote from the target.
3 | XFRM | transformation | Used when the target Act is a transformation of the source Act. (For instance, used to show that a CDA document is a transformation of a DICOM SR document.)
2 | SPRT | has support | Used to indicate that an existing service is suggesting evidence for a new observation. The assumption of support is attributed to the same actor who asserts the observation. Source must be an observation, target may be any service (e.g., to indicate a status post).

3 | SPRTBND | has bounded support | A specialization of "has support" (SPRT), used to relate a secondary observation to a Region of Interest on a multidimensional observation, if the ROI specifies the true boundaries of the secondary observation as opposed to only marking the approximate area. For example, if the start and end of an ST elevation episode is visible in an EKG, this relation would indicate the ROI bounds the "ST elevation" observation -- the ROI defines the true beginning and ending of the episode. Conversely, if a ROI simply contains ST elevation, but it does not define the bounds (start and end) of the episode, the more general "has support" relation is used. Likewise, if a ROI on an image defines the true bounds of a "1st degree burn", the relation "has bounded support" is used; but if the ROI only points to the approximate area of the burn, the general "has support" relation is used.
2 | SUBJ | has subject | Relates an Act to its subject Act that the first Act is primarily concerned with. Examples The first Act may be a ControlAct manipulating the subject Act The first act is a region of interest (ROI) that defines a region within the subject Act. The first act is a reporting or notification Act, that echos the subject Act for a specific new purpose. Constraints An Act may have multiple subject acts. Rationale The ActRelationshipType "has subject" is similar to the ParticipationType "subject", Acts that primarily operate on physical subjects use the Participation, those Acts that primarily operate on other Acts (other information) use the ActRelationship.
3 | QUALF | has qualifier | The target observation qualifies (refines) the semantics of the source observation. UsageNote: This is not intended to replace concept refinement and qualification via vocabulary. It is used when there are multiple components which together provide the complete understanding of the source Act.
2 | SUMM | summarized by | An act that contains summary values for a list or set of subordinate acts. For example, a summary of transactions for a particular accounting period.
2 | VALUE | has value | Description:Indicates that the target Act represents the result of the source observation Act. FormalConstraint: Source Act must be an Observation or specialization there-of. Source Act must not have the value attribute specified UsageNote: This relationship allows the result of an observation to be fully expressed as RIM acts as opposed to being embedded in the value attribute. For example, sending a Document act as the result of an imaging observation, sending a list of Procedures and/or other acts as the result of a medical history observation. The valueNegationInd attribute on the source Act has the same semantics of "negated finding" when it applies to the target of a VALUE ActRelationship as it does to the value attribute. On the other hand, if the ActRelationship.negationInd is true for a VALUE ActRelationship, that means the specified observation does not have the indicated value but does not imply a negated finding. Because the semantics are extremely close, it is recommended that Observation.valueNegationInd be used, not ActRelationship.negationInd. OpenIssue: The implications of negationInd on ActRelationship and the valueNegationind on Observation.
1 | CURE | curative indication
1 | CURE.ADJ | adjunct curative indication
1 | MTGT.ADJ | adjunct mitigation
1 | RACT
1 | SUGG

4.4.2.1008 HL7 v3 Code System NullFlavor

URL: https://hl7.org/fhir/R4/v3/NullFlavor/cs.html
Vocabulary Work Group | Maturity Level: N | Normative (from v4.0.0) | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-NullFlavor) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-NullFlavor
Version: | 2018-08-12
Name: | v3.NullFlavor
Title: | v3 Code System NullFlavor
Definition: | A collection of codes specifying why a valid value is not present.
OID: | 2.16.840.1.113883.5.1008 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System NullFlavor (A collection of codes specifying why a valid value is not present.)

• ValueSet: HumanNameAssemblyOrder (A code that represents the preferred display order of the components of a human name.)

• ValueSet: MaritalStatus (This value set defines the set of codes that can be used to indicate the marital status of a person.)

• ValueSet: v2 Expanded yes/no indicator (FHIR Value set/code system definition for HL7 v2 table 0532 (Expanded yes/no indicator))

• ValueSet: HumanNameAssemblyOrder (A code that represents the preferred display order of the components of a human name.)

• ValueSet: Special courtesy (This value set defines a set of codes that can be used to indicate special courtesies provided to the patient.)

• ValueSet: MaritalStatus (This value set defines the set of codes that can be used to indicate the marital status of a person.)

• ValueSet: Special courtesy (This value set defines a set of codes that can be used to indicate special courtesies provided to the patient.)

• ValueSet: v2 CWE statuses (FHIR Value set/code system definition for HL7 v2 table 0353 (CWE statuses))

4.4.2.1008.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | NI | NoInformation | Description:The value is exceptional (missing, omitted, incomplete, improper). No information as to the reason for being an exceptional value is provided. This is the most general exceptional value. It is also the default exceptional value.
2 | INV | invalid | Description:The value as represented in the instance is not a member of the set of permitted data values in the constrained value domain of a variable.
3 | DER | derived | Description:An actual value may exist, but it must be derived from the provided information (usually an EXPR generic data type extension will be used to convey the derivation expressionexpression .
3 | OTH | other | Description:The actual value is not a member of the set of permitted data values in the constrained value domain of a variable. (e.g., concept not provided by required code system). Usage Notes: This flavor and its specializations are most commonly used with the CD datatype and its flavors. However, it may apply to *any* datatype where the constraints of the type are tighter than can be conveyed. For example, a PQ that is for a true measured amount whose units are not supported in UCUM, a need to convey a REAL when the type has been constrained to INT, etc. With coded datatypes, this null flavor may only be used if the vocabulary binding has a coding strength of CNE. By definition, all local codes and original text are part of the value set if the coding strength is CWE.
4 | NINF | negative infinity | Negative infinity of numbers.
4 | PINF | positive infinity | Positive infinity of numbers.
3 | UNC | un-encoded | Description: The actual value has not yet been encoded within the approved value domain. Example: Original text or a local code has been specified but translation or encoding to the approved value set has not yet occurred due to limitations of the sending system. Original text has been captured for a PQ, but not attempt has been made to split the value and unit or to encode the unit in UCUM. Usage Notes: If it is known that it is not possible to encode the concept, OTH should be used instead. However, use of UNC does not necessarily guarantee the concept will be encodable, only that encoding has not been attempted. Data type properties such as original text and translations may be present when this null flavor is included.
2 | MSK | masked | There is information on this item available but it has not been provided by the sender due to security, privacy or other reasons. There may be an alternate mechanism for gaining access to this information. Note: using this null flavor does provide information that may be a breach of confidentiality, even though no detail data is provided. Its primary purpose is for those circumstances where it is necessary to inform the receiver that the information does exist without providing any detail.
2 | NA | not applicable | Known to have no proper value (e.g., last menstrual period for a male).
2 | UNK | unknown | Description:A proper value is applicable, but not known. Usage Notes: This means the actual value is not known. If the only thing that is unknown is how to properly express the value in the necessary constraints (value set, datatype, etc.), then the OTH or UNC flavor should be used. No properties should be included for a datatype with this property unless: Those properties themselves directly translate to a semantic of "unknown". (E.g. a local code sent as a translation that conveys 'unknown') Those properties further qualify the nature of what is unknown. (E.g. specifying a use code of "H" and a URL prefix of "tel:" to convey that it is the home phone number that is unknown.)
3 | ASKU | asked but unknown | Information was sought but not found (e.g., patient was asked but didn't know)
4 | NAV | temporarily unavailable | Information is not available at this time but it is expected that it will be available later.
3 | NASK | not asked | This information has not been sought (e.g., patient was not asked)
3 | NAVU | Not available | Information is not available at this time (with no expectation regarding whether it will or will not be available in the future).
4 | NAV
3 | QS | Sufficient Quantity | Description:The specific quantity is not known, but is known to be non-zero and is not specified because it makes up the bulk of the material. e.g. 'Add 10mg of ingredient X, 50mg of ingredient Y, and sufficient quantity of water to 100mL.' The null flavor would be used to express the quantity of water.
3 | TRC | trace | The content is greater than zero, but too small to be quantified.
1 | NP | not present | Value is not present in a message. This is only defined in messages, never in application data! All values not present in the message must be replaced by the applicable default, or no-information (NI) as the default of all defaults.

4.4.2.1009 HL7 v3 Code System CompressionAlgorithm

URL: https://hl7.org/fhir/R4/v3/CompressionAlgorithm/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-CompressionAlgorithm) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-CompressionAlgorithm
Version: | 2018-08-12
Name: | v3.CompressionAlgorithm
Title: | v3 Code System CompressionAlgorithm
Definition: | **** MISSING DEFINITIONS ****
OID: | 2.16.840.1.113883.5.1009 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System CompressionAlgorithm (**** MISSING DEFINITIONS ****)

4.4.2.1009.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | BZ | bzip | bzip-2 compression format. See [http://www.bzip.org/] for more information.
1 | DF | deflate | The deflate compressed data format as specified in RFC 1951 [http://www.ietf.org/rfc/rfc1951.txt].
1 | GZ | gzip | A compressed data format that is compatible with the widely used GZIP utility as specified in RFC 1952 [http://www.ietf.org/rfc/rfc1952.txt] (uses the deflate algorithm).

1 | Z | compress | Original UNIX compress algorithm and file format using the LZC algorithm (a variant of LZW). Patent encumbered and less efficient than deflate.
1 | Z7 | Z7 | 7z compression file format. See [http://www.7-zip.org/7z.html] for more information.
1 | ZL | zlib | A compressed data format that also uses the deflate algorithm. Specified as RFC 1950 [http://www.ietf.org/rfc/rfc1952.txt]

4.4.2.1010 HL7 v3 Code System IntegrityCheckAlgorithm

URL: https://hl7.org/fhir/R4/v3/IntegrityCheckAlgorithm/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-IntegrityCheckAlgorithm) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-IntegrityCheckAlgorithm
Version: | 2018-08-12
Name: | v3.IntegrityCheckAlgorithm
Title: | v3 Code System IntegrityCheckAlgorithm
Definition: | **** MISSING DEFINITIONS ****
OID: | 2.16.840.1.113883.5.1010 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System IntegrityCheckAlgorithm (**** MISSING DEFINITIONS ****)

4.4.2.1010.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | SHA-1 | secure hash algorithm - 1 | This algorithm is defined in FIPS PUB 180-1: Secure Hash Standard. As of April 17, 1995.
1 | SHA-256 | secure hash algorithm - 256 | This algorithm is defined in FIPS PUB 180-2: Secure Hash Standard.

4.4.2.1017 HL7 v3 Code System CalendarType

URL: https://hl7.org/fhir/R4/v3/CalendarType/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-CalendarType) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-CalendarType
Version: | 2018-08-12
Name: | v3.CalendarType
Title: | v3 Code System CalendarType
Definition: | **** MISSING DEFINITIONS ****
OID: | 2.16.840.1.113883.5.1017 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System CalendarType (**** MISSING DEFINITIONS ****)

4.4.2.1017.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | GREG | Gregorian | The Gregorian calendar is in effect in the most countries of Christian influence since approximately 1582. This calendar superceded the Julian calendar.

4.4.2.1020 HL7 v3 Code System ProbabilityDistributionType

URL: https://hl7.org/fhir/R4/v3/ProbabilityDistributionType/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ProbabilityDistributionType) is defined as part of HL7 v3. Related FHIR content: ProbabilityDistributionType.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ProbabilityDistributionType
Version: | 2018-08-12
Name: | v3.ProbabilityDistributionType
Title: | v3 Code System ProbabilityDistributionType
Definition: | **** MISSING DEFINITIONS ****
OID: | 2.16.840.1.113883.5.1020 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: ProbabilityDistributionType (Codes specifying the type of probability distribution.)

• ValueSet: v3 Code System ProbabilityDistributionType (**** MISSING DEFINITIONS ****)

• ValueSet: ProbabilityDistributionType (Codes specifying the type of probability distribution.)

4.4.2.1020.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition

1 | B | beta | The beta-distribution is used for data that is bounded on both sides and may or may not be skewed (e.g., occurs when probabilities are estimated.) Two parameters a and b are available to adjust the curve. The mean m and variance s2 relate as follows: m = a/ (a + b) and s2 = ab/((a + b)2 (a + b + 1)).

1 | E | exponential | Used for data that describes extinction. The exponential distribution is a special form of g-distribution where a = 1, hence, the relationship to mean m and variance s2 are m = b and s2 = b2.

1 | F | F | Used to describe the quotient of two c2 random variables. The F-distribution has two parameters n1 and n2, which are the numbers of degrees of freedom of the numerator and denominator variable respectively. The relationship to mean m and variance s2 are: m = n2 / (n2 - 2) and s2 = (2 n2 (n2 + n1 - 2)) / (n1 (n2 - 2)2 (n2 - 4)).

1 | G | (gamma) | The gamma-distribution used for data that is skewed and bounded to the right, i.e. where the maximum of the distribution curve is located near the origin. The g-distribution has a two parameters a and b. The relationship to mean m and variance s2 is m = a b and s2 = a b2.

1 | LN | log-normal | The logarithmic normal distribution is used to transform skewed random variable X into a normally distributed random variable U = log X. The log-normal distribution can be specified with the properties mean m and standard deviation s. Note however that mean m and standard deviation s are the parameters of the raw value distribution, not the transformed parameters of the lognormal distribution that are conventionally referred to by the same letters. Those log-normal parameters mlog and slog relate to the mean m and standard deviation s of the data value through slog2 = log (s2/m2 + 1) and mlog = log m - slog2/2.

1 | N | normal (Gaussian) | This is the well-known bell-shaped normal distribution. Because of the central limit theorem, the normal distribution is the distribution of choice for an unbounded random variable that is an outcome of a combination of many stochastic processes. Even for values bounded on a single side (i.e. greater than 0) the normal distribution may be accurate enough if the mean is "far away" from the bound of the scale measured in terms of standard deviations.

1 | T | T | Used to describe the quotient of a normal random variable and the square root of a c2 random variable. The t-distribution has one parameter n, the number of degrees of freedom. The relationship to mean m and variance s2 are: m = 0 and s2 = n / (n - 2)

1 | U | uniform | The uniform distribution assigns a constant probability over the entire interval of possible outcomes, while all outcomes outside this interval are assumed to have zero probability. The width of this interval is 2s sqrt(3). Thus, the uniform distribution assigns the probability densities f(x) = sqrt(2 s sqrt(3)) to values m - s sqrt(3) >= x <= m + s sqrt(3) and f(x) = 0 otherwise.

1 | X2 | chi square | Used to describe the sum of squares of random variables which occurs when a variance is estimated (rather than presumed) from the sample. The only parameter of the c2-distribution is n, so called the number of degrees of freedom (which is the number of independent parts in the sum). The c2-distribution is a special type of g-distribution with parameter a = n /2 and b = 2. Hence, m = n and s2 = 2 n.

4.4.2.1022 HL7 v3 Code System GTSAbbreviation

URL: https://hl7.org/fhir/R4/v3/GTSAbbreviation/cs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-GTSAbbreviation) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-GTSAbbreviation

Version: | 2018-08-12

Name: | v3.GTSAbbreviation

Title: | v3 Code System GTSAbbreviation

Definition: | Open Issue: It appears that the printnames are suboptimal and should be improved for many of the existing codes.

OID: | 2.16.840.1.113883.5.1022 (for OID based terminology systems)

Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: TimingAbbreviation (Code for a known / defined timing pattern.)

• ValueSet: TimingAbbreviation (Code for a known / defined timing pattern.)

• ValueSet: v3 Code System GTSAbbreviation (Open Issue: It appears that the printnames are suboptimal and should be improved for many of the existing codes.)

4.4.2.1022.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition

1 | AM | AM | Every morning at institution specified times.

1 | BED | at bedtime | At bedtime (institution specified time).

1 | BID | BID | Two times a day at institution specified time

1 | JB | JB | Regular business days (Monday to Friday excluding holidays)

1 | JE | JE | Regular weekends (Saturday and Sunday excluding holidays)

1 | JH | GTSAbbreviationHolidays | Holidays
2 | (_GTSAbbreviationHolidaysChristianRoman) Abstract | | Christian Holidays (Roman/Gregorian [Western] Tradition.)
3 | JHCHREAS | JHCHREAS | Easter Sunday. The Easter date is a rather complex calculation based on Astronomical tables describing full moon dates. Details can be found at [http://www.assa.org.au/edm.html, and http://aa.usno.navy.mil/AA/faq/docs/easter.html]. Note that the Christian Orthodox Holidays are based on the Julian calendar.
3 | JHCHRGFR | JHCHRGFR | Good Friday, is the Friday right before Easter Sunday.
3 | JHCHRNEW | JHCHRNEW | New Year's Day (January 1)
3 | JHCHRPEN | JHCHRPEN | Pentecost Sunday, is seven weeks after Easter (the 50th day of Easter).
3 | JHCHRXME | JHCHRXME | Christmas Eve (December 24)
3 | JHCHRXMS | JHCHRXMS | Christmas Day (December 25)
2 | JHNNL | The Netherlands National Holidays | Description:The Netherlands National Holidays.
3 | JHNNLLD | Liberation day (May 5 every five years) | Description:Liberation day (May 5 every five years)
3 | JHNNLQD | Queen's day (April 30) | Description:Queen's day (April 30)
3 | JHNNLSK | Sinterklaas (December 5) | Description:Sinterklaas (December 5)
2 | JHNUS | GTSAbbreviationHolidaysUSNational | United States National Holidays (public holidays for federal employees established by U.S. Federal law 5 U.S.C. 6103).
3 | JHNUSCLM | JHNUSCLM | Columbus Day, the second Monday in October.
3 | JHNUSIND | JHNUSIND | Independence Day (4th of July)
3 | JHNUSIND1 | JHNUSIND1 | Alternative Monday after 4th of July Weekend [5 U.S.C. 6103(b)].
3 | JHNUSIND5 | JHNUSIND5 | Alternative Friday before 4th of July Weekend [5 U.S.C. 6103(b)].
3 | JHNUSLBR | JHNUSLBR | Labor Day, the first Monday in September.
3 | JHNUSMEM | JHNUSMEM | Memorial Day, the last Monday in May.
3 | JHNUSMEM5 | JHNUSMEM5 | Friday before Memorial Day Weekend
3 | JHNUSMEM6 | JHNUSMEM6 | Saturday of Memorial Day Weekend
3 | JHNUSMLK | JHNUSMLK | Dr. Martin Luther King, Jr. Day, the third Monday in January.
3 | JHNUSPRE | JHNUSPRE | Washington's Birthday (Presidential Day) the third Monday in February.
3 | JHNUSTKS | JHNUSTKS | Thanksgiving Day, the fourth Thursday in November.
3 | JHNUSTKS5 | JHNUSTKS5 | Friday after Thanksgiving.
3 | JHNUSVET | JHNUSVET | Veteran's Day, November 11.
1 | MO | monthly | Monthly at institution specified time.
1 | PM | PM | Every afternoon at institution specified times.
1 | Q1H | every hour | Every hour at institution specified times.
1 | Q2H | every 2 hours | Every 2 hours at institution specified times.
1 | Q3H | every 3 hours | Every 3 hours at institution specified times.
1 | Q4H | Q4H | Every 4 hours at institution specified time
1 | Q6H | Q6H | Every 6 hours at institution specified time
1 | Q8H | every 8 hours | Every 8 hours at institution specified times.
1 | QD | QD | Every day at institution specified times.
1 | QID | QID | Four times a day at institution specified time
1 | QOD | QOD | Every other day at institution specified times.
1 | TID | TID | Three times a day at institution specified time
1 | WK | weekly | Weekly at institution specified time.

4.4.2.1050 HL7 v3 Code System AcknowledgementCondition

URL: https://hl7.org/fhir/R4/v3/AcknowledgementCondition/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-AcknowledgementCondition) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-AcknowledgementCondition
Version: | 2018-08-12
Name: | v3.AcknowledgementCondition
Title: | v3 Code System AcknowledgementCondition
Definition: | The codes identify the conditions under which accept acknowledgements are required to be returned in response to this message. Note that accept acknowledgement address two different issues at the same time: reliable transport as well as syntactical correctness
OID: | 2.16.840.1.113883.5.1050 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System AcknowledgementCondition (The codes identify the conditions under which accept acknowledgements are required to be returned in response to this message. Note that accept acknowledgement address two different issues at the same time: reliable transport as well as syntactical correctness)

4.4.2.1050.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | AL | Always | Always send an acknowledgement.

1 | ER | Error/reject only | Send an acknowledgement for error/reject conditions only.
1 | NE | Never | Never send an acknowledgement.
1 | SU | Successful only | Send an acknowledgement for successful completions only.

4.4.2.1051 HL7 v3 Code System ActInvoiceElementModifier

URL: https://hl7.org/fhir/R4/v3/ActInvoiceElementModifier/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ActInvoiceElementModifier) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ActInvoiceElementModifier
Version: | 2018-08-12
Name: | v3.ActInvoiceElementModifier
Title: | v3 Code System ActInvoiceElementModifier
Definition: | Processing consideration and clarification codes.
OID: | 2.16.840.1.113883.5.1051 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ActInvoiceElementModifier (Processing consideration and clarification codes.)

4.4.2.1051.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | EFORM | Electronic Form To Follow | Electronic form with supporting information to follow.
1 | FAX | Fax To Follow | Fax with supporting information to follow.
1 | LINV | Last Invoice | Represents the last invoice from the perspective of the provider.
1 | PAPER | Paper Documentation To Follow | Paper documentation (or other physical format) with supporting information to follow.

4.4.2.1052 HL7 v3 Code System ActSite

URL: https://hl7.org/fhir/R4/v3/ActSite/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ActSite) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ActSite
Version: | 2018-08-12
Name: | v3.ActSite
Title: | v3 Code System ActSite
Definition: | An anatomical location on an organism which can be the focus of an act.
OID: | 2.16.840.1.113883.5.1052 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: Codes for Immunization Site of Administration (The value set to instantiate this attribute should be drawn from a terminologically robust code system that consists of or contains concepts to support describing the body site where the vaccination occurred. This value set is provided as a suggestive example.)

• ValueSet: v3 Code System ActSite (An anatomical location on an organism which can be the focus of an act.)

• ValueSet: Codes for Immunization Site of Administration (The value set to instantiate this attribute should be drawn from a terminologically robust code system that consists of or contains concepts to support describing the body site where the vaccination occurred. This value set is provided as a suggestive example.)

4.4.2.1052.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_HumanActSite) Abstract | | An anatomical location on a human which can be the focus of an act.
2 | (_HumanSubstanceAdministrationSite) Abstract | | The set of body locations to or through which a drug product may be administered.
3 | BE | bilateral ears
3 | BN | bilateral nares
3 | BU | buttock
3 | LA | left arm
3 | LAC | left anterior chest
3 | LACF | left antecubital fossa
3 | LD | left deltoid

3 | LE | left ear
3 | LEJ | left external jugular
3 | LF | left foot
3 | LG | left gluteus medius
3 | LH | left hand
3 | LIJ | left internal jugular
3 | LLAQ | left lower abd quadrant
3 | LLFA | left lower forearm
3 | LMFA | left mid forearm
3 | LN | left naris
3 | LPC | left posterior chest
3 | LSC | left subclavian
3 | LT | left thigh
3 | LUA | left upper arm
3 | LUAQ | left upper abd quadrant
3 | LUFA | left upper forearm
3 | LVG | left ventragluteal
3 | LVL | left vastus lateralis
3 | OD | right eye
3 | OS | left eye
3 | OU | bilateral eyes
3 | PA | perianal
3 | PERIN | perineal
3 | RA | right arm
3 | RAC | right anterior chest
3 | RACF | right antecubital fossa
3 | RD | right deltoid
3 | RE | right ear
3 | REJ | right external jugular
3 | RF | right foot
3 | RG | right gluteus medius
3 | RH | right hand
3 | RIJ | right internal jugular
3 | RLAQ | right lower abd quadrant
3 | RLFA | right lower forearm
3 | RMFA | right mid forearm
3 | RN | right naris
3 | RPC | right posterior chest
3 | RSC | right subclavian
3 | RT | right thigh
3 | RUA | right upper arm
3 | RUAQ | right upper abd quadrant
3 | RUFA | right upper forearm
3 | RVG | right ventragluteal
3 | RVL | right vastus lateralis

4.4.2.1053 HL7 v3 Code System ActUncertainty

URL: https://hl7.org/fhir/R4/v3/ActUncertainty/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ActUncertainty) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ActUncertainty
Version: | 2018-08-12
Name: | v3.ActUncertainty
Title: | v3 Code System ActUncertainty
Definition: | OpenIssue: Missing Description
OID: | 2.16.840.1.113883.5.1053 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ActUncertainty (OpenIssue: Missing Description)

4.4.2.1053.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | N | stated with no assertion of uncertainty | Specifies that the act statement is made without explicit tagging of uncertainty. This is the normal statement, meaning that it is not free of errors and uncertainty may still exist.
1 | U | stated with uncertainty | Specifies that the originator of the Act statement does not have full confidence in the applicability (i.e., in event mood: factual truth) of the statement.

4.4.2.1054 HL7 v3 Code System AmericanIndianAlaskaNativeLanguages

URL: https://hl7.org/fhir/R4/v3/AmericanIndianAlaskaNativeLanguages/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-AmericanIndianAlaskaNativeLanguages) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-AmericanIndianAlaskaNativeLanguages
Version: | 2018-08-12
Name: | v3.AmericanIndianAlaskaNativeLanguages
Title: | v3 Code System AmericanIndianAlaskaNativeLanguages
Definition: | American Indian and Alaska Native languages currently being used in the United States.
OID: | 2.16.840.1.113883.5.1054 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System AmericanIndianAlaskaNativeLanguages (American Indian and Alaska Native languages currently being used in the United States.)

4.4.2.1054.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_Algic) Abstract
2 | (_Algonquian) Abstract
3 | (_Arapahoan) Abstract
4 | (_ArapahoGrosVentre) Abstract
5 | x-ARP | Arapaho
5 | x-ATS | Gros Ventre
3 | (_CreeMontagnais) Abstract
4 | (_Cree) Abstract
5 | x-CRP | Western Cree
3 | (_EasternAlgonquin) Abstract
4 | (_Abenakian) Abstract
5 | x-AAQ | Eastern Abenaki
5 | x-ABE | Western Abenaki
5 | x-MAC | Maliseet-Passamaquoddy
4 | (_Delawaran) Abstract
5 | x-DEL | Unami Delaware
4 | x-MIC | Micmac
3 | (_Ojibwayan) Abstract
4 | x-OJB | Southern Ojibwa
4 | x-POT | Potawatami
3 | (_SaukFoxKickapoo) Abstract
4 | x-KIC | Kickapoo
4 | x-SAC | Mesquakie
4 | x-SJW | Shawnee
3 | x-BLC | Blackfoot
3 | x-CHY | Cheyenne
3 | x-MEZ | Menominee
2 | (_Ritwan) Abstract
3 | x-YUR | Yurok
1 | (_Caddoan) Abstract
2 | (_NorthernCaddoan) Abstract
3 | x-ARI | Arikara
3 | x-PAW | Pawnee
3 | x-WIC | Wichita
2 | (_SouthernCaddoan) Abstract
3 | x-CAD | Caddo
1 | (_Chimakuan) Abstract
2 | x-QUI | Quileute
1 | (_EskimoAleut) Abstract
2 | (_Aleut) Abstract
3 | x-ALW | Aleut
2 | (_Eskimoan) Abstract
3 | (_InuitInupiaq) Abstract
4 | x-ESI | North Alaskan Inuktitut
4 | x-ESK | Northwest Alaska Inuktitut
3 | (_SirenikskiYupik) Abstract

4 | x-EMS | Pacific Yupik Gulf
4 | x-ESS | Central Siberian Yupik
4 | x-ESU | Central Alaskan Yupik
1 | (_Hokan) Abstract
2 | (_CochimiYuman) Abstract
3 | (_Yuman) Abstract
4 | (_DeltaCalifornia) Abstract
5 | (_Diegueno) Abstract
6 | x-DIH | Kumeyaay
5 | x-COC | Cocopa
4 | (_Pai) Abstract
5 | x-YUF | Havasupai
4 | (_River) Abstract
5 | x-MOV | Mohave
5 | x-MRC | Maricopa
5 | x-YUM | Quechan
2 | (_Palaihnihan) Abstract
3 | x-ACH | Achumawi
3 | x-ATW | Atsugewi
2 | (_Pomoan) Abstract
3 | x-KJU | Kashaya
3 | x-PEF | Northeastern Pomo
3 | x-PEO | Southeastern Pomo
3 | x-PEQ | Southern Pomo
3 | x-POO | Central Pomo
2 | (_Shasta) Abstract
3 | x-SHT | Shasta
2 | x-KYH | Karok
2 | x-WAS | Washoe
1 | (_Iroquoian) Abstract
2 | (_NorthernIroquoian) Abstract
3 | x-CAY | Cayuga
3 | x-MOH | Mohawk
3 | x-ONE | Oneida
3 | x-ONO | Onondaga
3 | x-SEE | Seneca
3 | x-TUS | Tuscarora
2 | x-CER | Cherokee
1 | (_Keresan) Abstract
2 | x-KEE | Rio Grande Keresan
2 | x-KJQ | Acoma-Laguna
1 | (_KiowaTanoan) Abstract
2 | (_Tiwa) Abstract
3 | x-TAO | Northern Tiwa
3 | x-TIX | Southern Tiwa
2 | x-KIO | Kiowa
2 | x-TEW | Tewa
2 | x-TOW | Jemez
1 | (_Muskogean) Abstract
2 | (_CentralMuskogean) Abstract
3 | x-AKZ | Alabama
3 | x-CKU | Koasati
3 | x-MIK | Mikasuki
2 | (_WesternMuskogean) Abstract
3 | x-CCT | Choctaw
3 | x-CIC | Chickasaw
2 | x-CRK | Creek
1 | (_Nadene) Abstract
2 | (_AthapaskanEyak) Abstract
3 | (_Athapaskan) Abstract
4 | (_Apachean) Abstract
5 | (_EasternApachean) Abstract
6 | x-APJ | Jicarilla
6 | x-APL | Lipan
5 | (_WesternApachean) Abstract
6 | x-APM | Mescalero-Chiricahua
6 | x-APW | Western Apache
6 | x-NAV | Dine
5 | x-APK | Kiowa Apache
4 | (_CentralAlaskaYukon) Abstract
5 | (_KoyukonIngalik) Abstract
6 | x-HOI | Holikachuk

6 | x-ING | Degexit'an
6 | x-KOY | Koyukon
5 | (_KutchinHan) Abstract
6 | x-HAA | Han
6 | x-KUC | Kutchin
5 | (_TananaTutchone) Abstract
6 | (_Tanana) Abstract
7 | x-TAA | Lower Tanana
7 | x-TAU | Upper Tanana
7 | x-TCB | Tanacross
6 | x-KUU | Upper Kuskokwim
4 | (_PacificCoastAthapaskan) Abstract
5 | (_CaliforniaAthapaskan) Abstract
6 | x-HUP | Hupa
6 | x-KTW | Cahto
5 | (_OregonAthapaskan) Abstract
6 | x-TOL | Tolowa
6 | x-TUU | Tututni
4 | (_SouthernAlaska) Abstract
5 | x-AHT | Ahtna
5 | x-TFN | Tanaina
3 | x-EYA | Eyak
2 | x-TLI | Tlingit
1 | (_Penutian) Abstract
2 | (_Chinookan) Abstract
3 | (_UpperChinook) Abstract
4 | x-WAC | Kiksht
2 | (_Coosan) Abstract
3 | x-COS | Hanis
2 | (_Maiduan) Abstract
3 | x-MAI | Northwest Maidu
3 | x-NMU | Northeast Maidu
3 | x-NSZ | Nisenan
2 | (_PlateauPenutian) Abstract
3 | (_Sahaptian) Abstract
4 | x-UMA | Umatilla
4 | x-WAA | Walla Walla
4 | x-WAR | Tenino
4 | x-YAK | Yakima
3 | x-KLA | Klamath-Modoc
3 | x-NEZ | Nez Perce
2 | (_Takelman) Abstract
3 | (_Kalapuyan) Abstract
4 | x-KAL | Central Kalapuyan
2 | (_Tsimshianic) Abstract
3 | x-TSI | Coast Tsimshain
2 | (_Utian) Abstract
3 | (_Miwokan) Abstract
4 | (_EasternMiwok) Abstract
5 | x-CSM | Central Sierra Miwok
5 | x-NSQ | Northern Sierra Miwok
5 | x-PMW | Plains Miwok
5 | x-SKD | Southern Sierra Miwok
4 | (_WesternMiwok) Abstract
5 | x-CSI | Coast Miwok
5 | x-LMW | Lake Miwok
2 | (_Wintuan) Abstract
3 | x-WIT | Wintu-Nomlaki
2 | (_Yokutsan) Abstract
3 | x-ENH | Kings River
3 | x-GSH | Gashowu
3 | x-PYL | Poso Creek
3 | x-TKH | Tule-Kaweah
1 | (_Pidgin) Abstract
2 | x-CHH | Chinook Wawa
1 | (_Salishan) Abstract
2 | (_CentralSalish) Abstract
3 | x-CLM | Clallam
3 | x-LUT | Lushootseed
3 | x-STR | Northern Straits
2 | (_InteriorSalish) Abstract
3 | x-COL | Columbian

3 | x-CRD | Coeur D'alene
3 | x-FLA | Kalispel
3 | x-OKA | Okanagan
2 | (_Tsamosan) Abstract
3 | x-CEA | Lower Chehalis
3 | x-CJH | Upper Chehalis
3 | x-COW | Cowlitz
3 | x-QUN | Quinault
1 | (_SiouanCatawba) Abstract
2 | (_Siouan) Abstract
3 | (_MississippiValley) Abstract
4 | (_ChiwereWinnebago) Abstract
5 | x-IOW | Chiwere
5 | x-WIN | Hocak
4 | (_Dakotan) Abstract
5 | x-ASB | Assiniboine
5 | x-DHG | Dakota
5 | x-LKT | Lakota
5 | x-NKT | Nakota
4 | (_Dhegiha) Abstract
5 | x-KAA | Kansa
5 | x-OMA | Omaha-Ponca
5 | x-OSA | Osage
5 | x-QUA | Quapaw
3 | (_MissouriRiver) Abstract
4 | x-CRO | Crow
4 | x-HID | Hidatsa
3 | x-MHQ | Mandan
1 | (_UtoAztecan) Abstract
2 | (_Numic) Abstract
3 | (_CentralNumic) Abstract
4 | x-COM | Comanche
4 | x-PAR | Panamint
4 | x-SHH | Shoshone
3 | (_SouthernNumic) Abstract
4 | x-KAW | Kawaiisu
4 | x-UTE | Ute-Southern Paiute
3 | (_WesternNumic) Abstract
4 | x-MON | Mono
4 | x-PAO | Northern Paiute-Bannock
2 | (_Takic) Abstract
3 | (_Cupan) Abstract
4 | x-CHL | Cahuilla
4 | x-CUP | Cupeno
4 | x-LUI | Luiseno
3 | (_SerranoGabrielino) Abstract
4 | x-SER | Serrano
2 | (_Taracahitan) Abstract
3 | (_Cahitan) Abstract
4 | x-YAQ | Yaqui
2 | (_Tepiman) Abstract
3 | x-PAP | Papago-Pima
2 | x-HOP | Hopi
2 | x-TUB | Tubatululabal
1 | (_Wakashan) Abstract
2 | (_Nootkan) Abstract
3 | x-MYH | Makah
1 | (_Yukian) Abstract
2 | x-WAO | Wappo
2 | x-YUK | Yuki
1 | x-HAI | Haida
1 | x-KUN | Kootenai
1 | x-PSD | Plains Indian Sign Language
1 | x-YUC | Yuchi
1 | x-ZUN | Zuni

4.4.2.1055 HL7 v3 Code System Calendar

URL: https://hl7.org/fhir/R4/v3/Calendar/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-Calendar) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-Calendar
Version: | 2018-08-12
Name: | v3.Calendar
Title: | v3 Code System Calendar
Definition: | **** MISSING DEFINITIONS ****
OID: | 2.16.840.1.113883.5.1055 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System Calendar (**** MISSING DEFINITIONS ****)

4.4.2.1055.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | GREG | Gregorian | The Gregorian calendar is the calendar in effect in most countries of Christian influence since approximately 1582. This calendar superceded the Julian calendar.

4.4.2.1056 HL7 v3 Code System CommunicationFunctionType

URL: https://hl7.org/fhir/R4/v3/CommunicationFunctionType/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-CommunicationFunctionType) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-CommunicationFunctionType
Version: | 2018-08-12
Name: | v3.CommunicationFunctionType
Title: | v3 Code System CommunicationFunctionType
Definition: | Describes the type of communication function that the associated entity plays in the associated transmission.
OID: | 2.16.840.1.113883.5.1056 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System CommunicationFunctionType (Describes the type of communication function that the associated entity plays in the associated transmission.)

4.4.2.1056.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | RCV | receiver | The entity is the receiver of the transmission.
1 | RSP | respond to | The entity is the one to which the response or reply to the transmission should be sent.
1 | SND | sender | The entity is the sender of the transmission.

4.4.2.1057 HL7 v3 Code System ContextControl

URL: https://hl7.org/fhir/R4/v3/ContextControl/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ContextControl) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ContextControl
Version: | 2018-08-12
Name: | v3.ContextControl
Title: | v3 Code System ContextControl
Definition: | A code that specifies how an ActRelationship or Participation contributes to the context of an Act, and whether it may be propagated to descendent Acts whose association allows such propagation (see also attributes Participation.contextControlCode, ActRelationship.contextControlCode, ActRelationship.contextConductionInd).
OID: | 2.16.840.1.113883.5.1057 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ContextControl (A code that specifies how an ActRelationship or Participation contributes to the context of an Act, and whether it may be propagated to descendent Acts whose association allows such propagation (see also attributes Participation.contextControlCode, ActRelationship.contextControlCode, ActRelationship.contextConductionInd).)

4.4.2.1057.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_ContextControlAdditive) Abstract | | The association adds to the existing context associated with the Act. Both this association and any associations propagated from ancestor Acts are interpreted as being related to this Act.
2 | AN | additive, non-propagating | The association adds to the existing context associated with the Act, but will not propagate to any descendant Acts reached by conducting ActRelationships (see contextControlCode). Examples: If an 'Author' Participation were marked as "Additive, Non-Propagating" it means that the author will be added to the set of author participations that have propagated from ancestor Acts for the purpose of this Act. However only the previously propagated authors will propagate to any child Acts that allow context to be propagated.
2 | AP | additive, propagating | The association adds to the existing context associated with the Act, and will propagate to any descendant Acts reached by conducting ActRelationships (see contextControlCode). Examples: If an 'Author' Participation were marked as "Additive, Propagating" it means that the author will be added to the set of author participations that have propagated from ancestor Acts, and will itself propagate with the other authors to any child Acts that allow context to be propagated.
1 | (_ContextControlNonPropagating) Abstract | | The association applies only to the current Act and will not propagate to any child Acts that are related via a conducting ActRelationship (refer to contextConductionInd).
2 | AN
2 | ON | overriding, non-propagating | The association is added to the existing context associated with the Act, but overrides an association with the same typeCode. However, this overriding association will not propagate to any descendant Acts reached by conducting ActRelationships (see contextControlCode). Examples: If an 'Author' Participation were marked as "Overriding, Non-Propagating" it means that the author will replace the set of author participations that have propagated from ancestor Acts. Furthermore, no author participations whatsoever will propagate to any child Acts that allow context to be propagated.
1 | (_ContextControlOverriding) Abstract | | The association adds to the existing context associated with the Act, but replaces associations propagated from ancestor Acts whose typeCodes are the same or more specific.
2 | ON
2 | OP | overriding, propagating | The association is added to the existing context associated with the Act, but overrides an association with the same typeCode. This overriding association will propagate to any descendant Acts reached by conducting ActRelationships (see contextControlCode). Examples: If an 'Author' Participation were marked as "Overriding, Propagating" it means that the author will replace the set of author participations that have propagated from ancestor Acts, and will itself be the only author to propagate to any child Acts that allow context to be propagated.
1 | (_ContextControlPropagating) Abstract | | The association propagates to any child Acts that are related via a conducting ActRelationship (refer to contextConductionInd).
2 | AP
2 | OP

4.4.2.1059 HL7 v3 Code System EmployeeJobClass

URL: https://hl7.org/fhir/R4/v3/EmployeeJobClass/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-EmployeeJobClass) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-EmployeeJobClass
Version: | 2018-08-12
Name: | v3.EmployeeJobClass
Title: | v3 Code System EmployeeJobClass
Definition: | **** MISSING DEFINITIONS ****
OID: | 2.16.840.1.113883.5.1059 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System EmployeeJobClass (**** MISSING DEFINITIONS ****)

4.4.2.1059.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | FT | full-time | Employment in which the employee is expected to work at least a standard work week (defined by the US Bureau of Labor Statistics as 35-44 hours per week)
1 | PT | part-time | Employment in which the employee is expected to work less than a standard work week (defined by the US Bureau of Labor Statistics as 35-44 hours per week)

4.4.2.1060 HL7 v3 Code System EntityCode

URL: https://hl7.org/fhir/R4/v3/EntityCode/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-EntityCode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-EntityCode
Version: | 2018-08-12
Name: | v3.EntityCode
Title: | v3 Code System EntityCode
Definition: | OpenIssue: Missing description.

OID: | 2.16.840.1.113883.5.1060 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System EntityCode (OpenIssue: Missing description.)

4.4.2.1060.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_MaterialEntityClassType) Abstract | | Types of Material for EntityClass "MAT"
2 | (_ContainerEntityType) Abstract | | Material intended to hold another material for purpose of storage or transport.
3 | PKG | Package | A material intended to hold other materials for purposes of storage or transportation
4 | (_NonRigidContainerEntityType) Abstract | | A container having dimensions that adjust somewhat based on the amount and shape of the material placed within it.
5 | BAG | Bag | A pouched or pendulous container.
5 | PACKT | Packet | A paper
5 | PCH | Pouch | A small bag or container made of a soft material.
5 | SACH | Sachet | A small bag or packet containing a small portion of a substance.
4 | (_RigidContainerEntityType) Abstract | | A container having a fixed and inflexible dimensions and volume
5 | (_IndividualPackageEntityType) Abstract | | Container intended to contain sufficient material for only one use.
6 | AMP | Ampule | A small sealed glass container that holds a measured amount of a medicinal substance.
6 | MINIM | Minim | Individually dosed ophthalmic solution. One time eye dropper dispenser.
6 | NEBAMP | Nebuamp | Individually dosed inhalation solution.
6 | OVUL | Ovule | A container either glass or plastic and a narrow neck, for storing liquid.
5 | (_MultiUseContainerEntityType) Abstract | | A container intended to contain sufficient material for more than one use. (I.e. Material is intended to be removed from the container at more than one discrete time period.)
6 | BOT | Bottle | A container, typically rounded, either glass or plastic with a narrow neck and capable of storing liquid.
7 | BOTA | Amber Bottle | A bottle of yellow to brown color. Used to store light-sensitive materials
7 | BOTD | Dropper Bottle | A bottle with a cap designed to release the contained liquid in droplets of a specific size.
7 | BOTG | Glass Bottle | A bottle made of glass
7 | BOTP | Plastic Bottle | A bottle made of plastic
8 | BOTPLY | Polyethylene Bottle | A bottle made of polyethylene
6 | BOX | Box | A 6-sided container commonly made from paper or cardboard used for solid forms.
6 | CAN | Can | A metal container in which a material is hermetically sealed to enable storage over long periods.
6 | CART | Cartridge | A sealed container of liquid or powder intended to be loaded into a device.
6 | CNSTR | Canister | A pressurized metal container holding a substance released as a spray or aerosol.
6 | JAR | Jar | A container of glass, earthenware, plastic, etc. Top of the container has a diameter of similar size to the diameter of the container as a whole
6 | JUG | Jug | A deep vessel for holding liquids, with a handle and often with a spout or lip shape for pouring.
6 | TIN | Tin | A lidded container made of thin sheet metal.
6 | TUB | Tub | An open flat bottomed round container.
6 | TUBE | Tube | A long hollow rigid or flexible cylinder. Material is extruded by squeezing the container.
6 | VIAL | Vial | A small cylindrical glass for holding liquid medicines.
5 | BLSTRPK | Blister Pack | A bubblepack. Medications sealed individually, separated into doses.
6 | CARD | Card | A bubble pack card. Multiple individual/separated doses.
4 | COMPPKG | Compliance Package | A container intended to contain sufficient material for more than one use, but grouped or organized to provide individual access to sufficient material for a single use. Often used to ensure that the proper type and amount of material is consumed/expended for each use.
5 | BLSTRPK
5 | DIALPK | Dial Pack | Rotatable dispenser. Eg. Birth control package.
5 | DISK | Disk | Object that is thin, flat, and circular. Doses of medication often contained in bubbles on the disk.
5 | DOSET | Dosette | Special packaging that will help patients take their medications on a regular basis.
5 | STRIP | Strip | A continuous strip of plastic sectioned into individual pouches, each one containing the quantity of 1 or more medications intended to be administered at a specific time
4 | KIT | Kit | A container for a diverse collection of products intended to be used together for some purpose (e.g. Medicinal kits often contain a syringe, a needle and the injectable medication).
5 | SYSTM | System | A kit in which the components are interconnected.
2 | (_MedicalDevice) Abstract | | A device with direct or indirect therapeutic purpose. Values for EntityCode when EntityClass = "DEV"
3 | (_AccessMedicalDevice) Abstract | | A device used to allow access to a part of a body
4 | LINE | Line | A hollow tube used to administer a substance into a vein, artery or body cavity
5 | IALINE | Intra-arterial Line | A line used to administer a substance into an artery
5 | IVLINE | Intraveneous Line | A line used to administer a substance into a vein
3 | (_AdministrationMedicalDevice) Abstract | | A device intended to administer a substance to a subject
4 | (_InjectionMedicalDevice) Abstract | | A device intended to administer liquid into a subject via a
5 | AINJ | AutoInjector | Automatically injects medication.
5 | PEN | Pen | A device which can contain a cartridge for injection purposes. Eg. Insulin pen.
5 | SYR | Syringe | A barrel with a plunger.
4 | APLCTR | Applicator | A device used to apply a liquid or powder to a surface.
4 | INH | Inhaler | A small device used for inhaling medicine in the form of a vapour or gas in order to ease a respiratory condition such as asthma or to relieve nasal congestion.
5 | DSKS | Diskus | The device used to inhale the doses of medication contained in the disk form.

5 | DSKUNH | Diskhaler | The device used to inhale the doses of medication contained in the disk form.
5 | TRBINH | Turbuhaler | Asthma medication delivery device.
4 | PMP | Pump | A device that is used to raise, compress, or transfer liquids or gases and is operated by a piston or similar mechanism.
2 | (_SpecimenAdditiveEntity) Abstract | | Set of codes related to specimen additives
3 | ACDA | ACD Solution A | ACD Solution A of trisodium citrate, 22.0g/L; citric acid, 8.0 g/L; and dextrose 24.5 g/L. Used in Blood banking and histocompatibilty testing
3 | ACDB | ACD Solution B | ACD Solution B of trisodium citrate, 13.2g/L; citric acid, 4.8 g/L; and dextrose 14.7 g/L. Used in Blood banking and histocompatibilty testing.
3 | ACET | Acetic Acid | 50% V/V acetic acid in water. Used as a urine preservative
3 | AMIES | Amies transport medium | Sodium Chloride 3.0g, Potassium Chloride 0.2g, Calcium Chloride 0.1g, Magnesium Chloride 0.1g, Monopotassium Phosphate 0.2g, Disodium Phosphate 1.15g, Sodium Thiogly collate 1.0g, Distilled Water 1 liter
3 | BACTM | Bacterial Transport medium | Any medium used to maintain bacterial viability (e.g. Stuart's, Cary-Blair, Amies)
3 | BF10 | Buffered 10% formalin | Formaldehyde 4% w/v; methyl alcohol 1% w/v; phosphate buffering salts. Tissue preservative
3 | BOR | Boric Acid | Powdered boric acid (usually 10 g) added to 24-hour urine collections as a preservative.
3 | BOUIN | Bouin's solution | Picric acid, saturated aqueous solution (750.0 ml), 37-40% formalin (250.0 ml), glacial acetic acid (50.0 ml). Tissue preservative.
3 | BSKM | Buffered skim milk | 50% skim milk in 0.01 M phosphate-buffered saline. Maintain virus viability
3 | C32 | 3.2% Citrate | A 3.2% solution of Sodium Citrate in water. Used as a blood preservative
3 | C38 | 3.8% Citrate | A 3.8% solution of Sodium Citrate in water. Used as a blood preservative
3 | CARS | Carson's Modified 10% formalin | A modification of buffered 10% formalin used as a general tissue preservative.
3 | CARY | Cary Blair Medium | Sodium Thioglycollate 1.5 g, Disodium Hydrogen Phosphate 1.1 g, Sodium Chloride 5.0 g, Calcium Chloride 0.09 g, Agar 5.0 g, per Liter of Water
3 | CHLTM | Chlamydia transport medium | Any of a number of non-nutritive buffered media used to maintain Chlamydia viability during transportation to the laboratory
3 | CTAD | CTAD | Buffered tri-sodium citrate solution with theophylline, adenosine and dipyridamole
3 | EDTK15 | Potassium/K EDTA 15% | Potassium EDTA 15% solution in water
3 | EDTK75 | Potassium/K EDTA 7.5% | Potassium EDTA 7.5% solution in water
3 | EDTN | Sodium/Na EDTA | Sodium fluoride and Disodium EDTA
3 | ENT | Enteric bacteria transport medium | Any of a number of non-nutritive buffered media used to maintain enteric bacterial viability during transportation to the laboratory
3 | F10 | 10% Formalin | A 10% v/v solution in water of formalin(a 37% solution of formaldehyde and water). Used for tissue preservation.
3 | FDP | Thrombin NIH; soybean trypsin inhibitor | Thrombin plus soybean trypsin inhibitor. For use in identifying fibrn degredation products.
3 | FL10 | Sodium Fluoride, 10mg | Sodium fluoride, 10mg added as a urine preservative.
3 | FL100 | Sodium Fluoride, 100mg | Sodium fluoride, 100mg added as a urine preservative.
3 | HCL6 | 6N HCL | A solution of HCl containing 6moles of hydrogen ion/L. Used as a Urine Preservative.
3 | HEPA | Ammonium heparin | Ammonium heparin
3 | HEPL | Lithium/Li Heparin | Lithium heparin salt
3 | HEPN | Sodium/Na Heparin | Sodium heparin salt
3 | HNO3 | Nitric Acid | 6N Nitric acid used to preserve urine for heavy metal analysis.
3 | JKM | Jones Kendrick Medium | A transport medium formulated to maintain Bordetella pertussis viability.
3 | KARN | Karnovsky's fixative | 5% Glutaraldehyde, 4% Formaldehyde in 0.08M buffer. Tissue preservation
3 | KOX | Potassium Oxalate | Potassium oxalate and sodium fluoride in a 1.25:1 ratio
3 | LIA | Lithium iodoacetate | Iodoacetate lithium salt
3 | M4 | M4 | Modified Hank's balanced salt solution supplemented with bovine serum albumin, gelatin, sucrose and glutamic acid. It is buffered to pH 7.3+ or - 0.2 with HEPES buffer. Phenol red is used to indicate pH. Vancomycin, Amphotericin B and Colistin are used to
3 | M4RT | M4-RT | Modified Hank's balanced salt solution supplemented with bovine serum albumin, gelatin, sucrose and glutamic acid. It is buffered to pH 7.3+ or - 0.2 with Hepes buffer. Phenol red is used to indicate pH. Gentamicin and amphotericin B are used to inhibit c
3 | M5 | M5 | Modified Hank's balanced salt solution supplemented with protein stabilizers, sucrose and glutamic acid. It is buffered to pH 7.3+ or - 0.2 with Hepes buffer. Phenol red is used to indicate pH. Vancomycin, Amphotericin B and Colistin are used to inhibit c
3 | MICHTM | Michel's transport medium | 1M potassium citrate, pH 7.0 2.5 ml, 0.1M magnesium sulfate 5.0 ml, 0.1M N-ethyl malemide 5.0 ml, dH2O 87.5 ml, ammonium sulfate 55gm. Preserve antigens for Immunofluorescence procedures
3 | MMDTM | MMD transport medium | A buffered medium with ammonium sulfate added to preserve antigens for Immunofluorescence procedures
3 | NAF | Sodium Fluoride | Sodium fluoride
3 | NONE | None | No additive. Specifically identifes the specimen as having no additives.
3 | PAGE | Page's Saline | 0.12 g NaCl, 0.004 g MgSO, 0.004 g, CaCl, 0.142 g Na2HPO4 and 0.136 g KH2PO4 per liter of distilled water. Maintain Acanthaoemba viability.
3 | PHENOL | Phenol | Phenol. Urine preservative
3 | PVA | Polyvinylalcohol | Polyvinyl alcohol
3 | RLM | Reagan Lowe Medium | A transport medium formulated to maintain Bordetella pertussis viability.
3 | SILICA | Siliceous earth | Diatomaceous earth. For glucose determination blood samples
3 | SPS | Sodium polyanethol sulfonate 0.35% in 0.85% sodium chloride | Sodium polyanethol sulfonate in saline. Anticomplementary and antiphagocytic properties. Used in blood culture collection.
3 | SST | Serum Separator Tube | Polymer separator gel with clot activator
3 | STUTM | Stuart transport medium | Sodium Glycerophosphate 10.0g, Calcium Chloride 0.1g, Mercaptoacetic Acid 1.0ml, Distilled Water 1 liter
3 | THROM | Thrombin | Thrombin. Accelerates clotting.
3 | THYMOL | Thymol | 2-Isopropyl-5-methyl phenol. A preservative for 24 Hr Urine samples

3 | THYO | Thyoglycolate broth | A nutritive medium with a reducing agent (sodium thioglycolate) which, due to a chemical reaction, removes oxygen from the broth.
3 | TOLU | Toluene | Also known as Methylbenzene; Toluol; Phenylmethane. A preservative for 24 Hr Urine samples
3 | URETM | Ureaplasma transport medium | A buffered salt solution with antifungal agents added for the collection and transport of Ureaplasma specimens.
3 | VIRTM | Viral Transport medium | Sucrose 74.6g, Potassium hydrogenphosphate 0.52g, L-glutamic acid 0.72g, Bovine serum albumin 5.0g, Gentamicin 50mg, Potassium dihydrogenphosphate 1.25g, L-15 medium 9.9L, Water to 10L. Maintain Virus viability.
3 | WEST | Buffered Citrate | 3.8% Citrate buffered to a pH of 5.5 for Westergren Sedimentation Rate
2 | BLDPRD | Blood Product | A manufactured product that is produced from the raw blood oi a donor with the intention of using it in a recipient transfusion.
2 | VCCNE | Vaccine | A Type of medicine that creates an immune protection without the recipient experiencing the disease.
2 | (_DrugEntity) Abstract | | A substance whose therapeutic effect is produced by chemical action within the body.
3 | (_ClinicalDrug) Abstract | | Any substance or mixture of substances manufactured, sold or represented for use in: (a) the diagnosis, treatment, mitigation or prevention of a disease, disorder, abnormal physical state, or its symptoms, in human beings or animals; (b) restoring, correcting or modifying organic functions in human beings or animals.
1 | (_NonDrugAgentEntity) Abstract | | Indicates types of allergy and intolerance agents which are non-drugs. (E.g. foods, latex, etc.)
2 | NDA01 | egg
2 | NDA02 | fish
2 | NDA03 | lactose
2 | NDA04 | peanut
2 | NDA05 | soy
2 | NDA06 | sulfites
2 | NDA07 | wheat or gluten
2 | NDA08 | isocyanates
2 | NDA09 | solvents
2 | NDA10 | oils
2 | NDA11 | venoms
2 | NDA12 | latex
2 | NDA13 | shellfish
2 | NDA14 | strawberries
2 | NDA15 | tomatoes
2 | NDA16 | dust
2 | NDA17 | dust mites
1 | (_OrganizationEntityType) Abstract | | Further classifies entities of classCode ORG.
2 | HHOLD | household | The group of persons who occupy a single housing unit.
2 | NAT | NationEntityType | Codes identifying nation states. Allows for finer grained specification of Entity with classcode <= NAT Example:ISO3166 country codes.
2 | RELIG | religious institution | An organization that provides religious rites of worship.
1 | (_PlaceEntityType) Abstract | | Types of places for EntityClass "PLC"
2 | BED | Bed Location | The location of a bed
2 | BLDG | Building Location | The location of a building
2 | FLOOR | Floor Location | The location of a floor of a building
2 | ROOM | Room Location | The location of a room
2 | WING | Wing Location | The location of a wing of a building (e.g. East Wing). The same room number for the same floor number can be distinguished by wing number in some situations
1 | (_ResourceGroupEntityType) Abstract | | Codes to characterize a Resource Group using categories that typify its membership and/or function . Example: PractitionerGroup
2 | PRAC | PractitionerGroup

4.4.2.1061 HL7 v3 Code System EntityStatus

URL: https://hl7.org/fhir/R4/v3/EntityStatus/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-EntityStatus) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-EntityStatus
Version: | 2018-08-12
Name: | v3.EntityStatus
Title: | v3 Code System EntityStatus
Definition: | Codes representing the defined possible states of an Entity, as defined by the Entity class state machine.
OID: | 2.16.840.1.113883.5.1061 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System EntityStatus (Codes representing the defined possible states of an Entity, as defined by the Entity class state machine.)

4.4.2.1061.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition

1 | normal | normal | The 'typical' state. Excludes "nullified" which represents the termination state of an Entity record instance that was created in error.
2 | active | active | The state representing the fact that the Entity record is currently active.
2 | inactive | inactive | Definition: The state representing the fact that the entity is inactive.
2 | terminated | terminated | The state representing the normal termination of an Entity record.
1 | nullified | nullified | The state representing the termination of an Entity record instance that was created in error.

4.4.2.1062 HL7 v3 Code System ManagedParticipationStatus

URL: https://hl7.org/fhir/R4/v3/ManagedParticipationStatus/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ManagedParticipationStatus) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ManagedParticipationStatus
Version: | 2018-08-12
Name: | v3.ManagedParticipationStatus
Title: | v3 Code System ManagedParticipationStatus
Definition: | Codes representing the defined possible states of a Managed Participation, as defined by the Managed Participation class state machine.
OID: | 2.16.840.1.113883.5.1062 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ManagedParticipationStatus (Codes representing the defined possible states of a Managed Participation, as defined by the Managed Participation class state machine.)

4.4.2.1062.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | normal | normal | The 'typical' state. Excludes "nullified" which represents the termination state of a participation instance that was created in error.
2 | active | active | The state representing the fact that the Participation is in progress.
2 | cancelled | cancelled | The terminal state resulting from cancellation of the Participation prior to activation.
2 | completed | completed | The terminal state representing the successful completion of the Participation.
2 | pending | pending | The state representing that fact that the Participation has not yet become active.
1 | nullified | nullified | The state representing the termination of a Participation instance that was created in error.

4.4.2.1063 HL7 v3 Code System ObservationValue

URL: https://hl7.org/fhir/R4/v3/ObservationValue/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ObservationValue) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ObservationValue
Version: | 2018-08-12
Name: | v3.ObservationValue
Title: | v3 Code System ObservationValue
Definition: | This domain is the root domain to which all HL7-recognized value sets for the Observation.value attribute will be linked when Observation.value has a coded data type. OpenIssue: Description copied from Concept Domain of same name. Must be corrected..
OID: | 2.16.840.1.113883.5.1063 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ObservationValue (This domain is the root domain to which all HL7-recognized value sets for the Observation.value attribute will be linked when Observation.value has a coded data type. OpenIssue: Description copied from Concept Domain of same name. Must be corrected..)

• ValueSet: V3 Value SetWorkScheduleODH (Describes an individual's typical arrangement of working hours for an occupation.)

• ValueSet: V3 Value SetSeverityObservation (Potential values for observations of severity.)

• ValueSet: V3 Value SetemploymentStatusODH (Concepts representing whether a person does or does not currently have a job or is not currently in the labor pool seeking employment.)

• ValueSet: V3 Value SetSecurityIntegrityObservationValue (No Description Provided)

4.4.2.1063.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_ActCoverageAssessmentObservationValue) Abstract | | Codes specify the category of observation, evidence, or document used to assess for services, e.g., discharge planning, or to establish eligibility for coverage under a policy or program. The type of evidence is coded as observation values.
2 | (_ActFinancialStatusObservationValue) Abstract | | Code specifying financial indicators used to assess or establish eligibility for coverage under a policy or program; e.g., pay stub; tax or income document; asset document; living expenses.
3 | ASSET | asset | Codes specifying asset indicators used to assess or establish eligibility for coverage under a policy or program.
4 | ANNUITY | annuity | Indicator of annuity ownership or status as beneficiary.
4 | PROP | real property | Indicator of real property ownership, e.g., deed or real estate contract.
4 | RETACCT | retirement investment account | Indicator of retirement investment account ownership.
4 | TRUST | trust | Indicator of status as trust beneficiary.
3 | INCOME | income | Code specifying income indicators used to assess or establish eligibility for coverage under a policy or program; e.g., pay or pension check, child support payments received or provided, and taxes paid.
4 | CHILD | child support | Indicator of child support payments received or provided.
4 | DISABL | disability pay | Indicator of disability income replacement payment.
4 | INVEST | investment income | Indicator of investment income, e.g., dividend check, annuity payment; real estate rent, investment divestiture proceeds; trust or endowment check.
4 | PAY | paid employment | Indicator of paid employment, e.g., letter of hire, contract, employer letter; copy of pay check or pay stub.
4 | RETIRE | retirement pay | Indicator of retirement payment, e.g., pension check.
4 | SPOUSAL | spousal or partner support | Indicator of spousal or partner support payments received or provided; e.g., alimony payment; support stipulations in a divorce settlement.
4 | SUPPLE | income supplement | Indicator of income supplement, e.g., gifting, parental income support; stipend, or grant.
4 | TAX | tax obligation | Indicator of tax obligation or payment, e.g., statement of taxable income.
3 | LIVEXP | living expense | Codes specifying living expense indicators used to assess or establish eligibility for coverage under a policy or program.
4 | CLOTH | clothing expense | Indicator of clothing expenses.
4 | FOOD | food expense | Indicator of transportation expenses.
4 | HEALTH | health expense | Indicator of health expenses; including medication costs, health service costs, financial participations, and health coverage premiums.
4 | HOUSE | household expense | Indicator of housing expense, e.g., household appliances, fixtures, furnishings, and maintenance and repairs.
4 | LEGAL | legal expense | Indicator of legal expenses.
4 | MORTG | mortgage | Indicator of mortgage amount, interest, and payments.
4 | RENT | rent | Indicator of rental or lease payments.
4 | SUNDRY | sundry expense | Indicator of transportation expenses.
4 | TRANS | transportation expense | Indicator of transportation expenses, e.g., vehicle payments, vehicle insurance, vehicle fuel, and vehicle maintenance and repairs.
4 | UTIL | utility expense | Indicator of transportation expenses.
2 | ELSTAT | eligibility indicator | Code specifying eligibility indicators used to assess or establish eligibility for coverage under a policy or program eligibility status, e.g., certificates of creditable coverage; student enrollment; adoption, marriage or birth certificate.
3 | ADOPT | adoption document | Indicator of adoption.
3 | BTHCERT | birth certificate | Indicator of birth.
3 | CCOC | creditable coverage document | Indicator of creditable coverage.
3 | DRLIC | driver license | Indicator of driving status.
3 | FOSTER | foster child document | Indicator of foster child status.
3 | MEMBER | program or policy member | Indicator of status as covered member under a policy or program, e.g., member id card or coverage document.
3 | MIL | military identification | Indicator of military status.
3 | MRGCERT | marriage certificate | Indicator of marriage status.
3 | PASSPORT | passport | Indicator of citizenship.
3 | STUDENRL | student enrollment | Indicator of student status.
2 | HLSTAT | health status | Code specifying non-clinical indicators related to health status used to assess or establish eligibility for coverage under a policy or program, e.g., pregnancy, disability, drug use, mental health issues.
3 | DISABLE | disabled | Indication of disability.
3 | DRUG | drug use | Indication of drug use.
3 | IVDRG | IV drug use | Indication of IV drug use .
3 | PGNT | pregnant | Non-clinical report of pregnancy.
2 | LIVDEP | living dependency | Code specifying observations related to living dependency, such as dependent upon spouse for activities of daily living.
3 | RELDEP | relative dependent | Continued living in private residence requires functional and health care assistance from one or more relatives.
3 | SPSDEP | spouse dependent | Continued living in private residence requires functional and health care assistance from spouse or life partner.
3 | URELDEP | unrelated person dependent | Continued living in private residence requires functional and health care assistance from one or more unrelated persons.
2 | LIVSIT | living situation | Code specifying observations related to living situation for a person in a private residence.
3 | ALONE | alone | Living alone. Maps to PD1-2 Living arrangement (IS) 00742 [A]
3 | DEPCHD | dependent children | Living with one or more dependent children requiring moderate supervision.
3 | DEPSPS | dependent spouse | Living with disabled spouse requiring functional and health care assistance
3 | DEPYGCHD | dependent young children | Living with one or more dependent children requiring intensive supervision
3 | FAM | live with family | Living with family. Maps to PD1-2 Living arrangement (IS) 00742 [F]
3 | RELAT | relative | Living with one or more relatives. Maps to PD1-2 Living arrangement (IS) 00742 [R]

3 | SPS | spouse only | Living only with spouse or life partner. Maps to PD1-2 Living arrangement (IS) 00742 [S]
3 | UNREL | unrelated person | Living with one or more unrelated persons.
2 | SOECSTAT | socio economic status | Code specifying observations or indicators related to socio-economic status used to assess to assess for services, e.g., discharge planning, or to establish eligibility for coverage under a policy or program.
3 | ABUSE | abuse victim | Indication of abuse victim.
3 | HMLESS | homeless | Indication of status as homeless.
3 | ILGIM | illegal immigrant | Indication of status as illegal immigrant.
3 | INCAR | incarcerated | Indication of status as incarcerated.
3 | PROB | probation | Indication of probation status.
3 | REFUG | refugee | Indication of refugee status.
3 | UNEMPL | unemployed | Indication of unemployed status.
1 | (_AllergyTestValue) Abstract | | Indicates the result of a particular allergy test. E.g. Negative, Mild, Moderate, Severe
2 | A0 | no reaction | Description:Patient exhibits no reaction to the challenge agent.
2 | A1 | minimal reaction | Description:Patient exhibits a minimal reaction to the challenge agent.
2 | A2 | mild reaction | Description:Patient exhibits a mild reaction to the challenge agent.
2 | A3 | moderate reaction | Description:Patient exhibits moderate reaction to the challenge agent.
2 | A4 | severe reaction | Description:Patient exhibits a severe reaction to the challenge agent.
1 | (_CompositeMeasureScoring) Abstract | | Observation values that communicate the method used in a quality measure to combine the component measure results included in an composite measure.
2 | ALLORNONESCR | All-or-nothing Scoring | Code specifying that the measure uses all-or-nothing scoring. All-or-nothing scoring places an individual in the numerator of the composite measure if and only if they are in the numerator of all component measures in which they are in the denominator.
2 | LINEARSCR | Linear Scoring | Code specifying that the measure uses linear scoring. Linear scoring computes the fraction of component measures in which the individual appears in the numerator, giving equal weight to each component measure.
2 | OPPORSCR | Opportunity Scoring | Code specifying that the measure uses opportunity-based scoring. In opportunity-based scoring the measure score is determined by combining the denominator and numerator of each component measure to determine an overall composite score.
2 | WEIGHTSCR | Weighted Scoring | Code specifying that the measure uses weighted scoring. Weighted scoring assigns a factor to each component measure to weight that measure's contribution to the overall score.
1 | (_CoverageLimitObservationValue) Abstract | | Description:Coded observation values for coverage limitations, for e.g., types of claims or types of parties covered under a policy or program.
2 | (_CoverageLevelObservationValue) Abstract | | Description:Coded observation values for types of covered parties under a policy or program based on their personal relationships or employment status.
3 | ADC | adult child | Description:Child over an age as specified by coverage policy or program, e.g., student, differently abled, and income dependent.
3 | CHD | child | Description:Dependent biological, adopted, foster child as specified by coverage policy or program.
3 | DEP | dependent | Description:Person requiring functional and/or financial assistance from another person as specified by coverage policy or program.
3 | DP | domestic partner | Description:Persons registered as a family unit in a domestic partner registry as specified by law and by coverage policy or program.
3 | ECH | employee | Description:An individual employed by an employer who receive remuneration in wages, salary, commission, tips, piece-rates, or pay-in-kind through the employeraTMs payment system (i.e., not a contractor) as specified by coverage policy or program.
3 | FLY | family coverage | Description:As specified by coverage policy or program.
3 | IND | individual | Description:Person as specified by coverage policy or program.
3 | SSP | same sex partner | Description:A pair of people of the same gender who live together as a family as specified by coverage policy or program, e.g., Naomi and Ruth from the Book of Ruth; Socrates and Alcibiades
1 | (_CriticalityObservationValue) Abstract | | A clinical judgment as to the worst case result of a future exposure (including substance administration). When the worst case result is assessed to have a life-threatening or organ system threatening potential, it is considered to be of high criticality.
2 | CRITH | high criticality | Worst case result of a future exposure is assessed to be life-threatening or having high potential for organ system failure.
2 | CRITL | low criticality | Worst case result of a future exposure is not assessed to be life-threatening or having high potential for organ system failure.
2 | CRITU | unable to assess criticality | Unable to assess the worst case result of a future exposure.
1 | (_EmploymentStatus) Abstract | | Concepts representing whether a person does or does not currently have a job or is not currently in the labor pool seeking employment.
2 | Employed | Employed | Individuals who, during the last week: a) did any work for at least 1 hour as paid or unpaid employees of a business or government organization; worked in their own businesses, professions, or on their own farms; or b) were not working, but who have a job or business from which the individual was temporarily absent because of vacation, illness, bad weather, childcare problems, maternity or paternity leave, labor-management dispute, job training, or other family or personal reasons, regardless of whether or not they were paid for the time off or were seeking other jobs.
2 | NotInLaborForce | Not In Labor Force | Persons not classified as employed or unemployed, meaning those who have no job and are not looking for one.
2 | Unemployed | Unemployed | Persons who currently have no employment, but are available for work and have made specific efforts to find employment.
1 | (_GeneticObservationValue) Abstract | | Description: The domain contains genetic analysis specific observation values, e.g. Homozygote, Heterozygote, etc.
2 | Homozygote | HOMO | Description: An individual having different alleles at one or more loci regarding a specific character
1 | (_ObservationMeasureScoring) Abstract | | Observation values used to indicate the type of scoring (e.g. proportion, ratio) used by a health quality measure.
2 | COHORT | cohort measure scoring | A measure in which either short-term cross-section or long-term longitudinal analysis is performed over a group of subjects defined by a set of common properties or defining characteristics (e.g., Male smokers between the ages of 40 and 50 years, exposure to treatment, exposure duration).

2 | CONTVAR | continuous variable measure scoring | A measure score in which each individual value for the measure can fall anywhere along a continuous scale (e.g., mean time to thrombolytics which aggregates the time in minutes from a case presenting with chest pain to the time of administration of thrombolytics).
2 | PROPOR | proportion measure scoring | A score derived by dividing the number of cases that meet a criterion for quality (the numerator) by the number of eligible cases within a given time frame (the denominator) where the numerator cases are a subset of the denominator cases (e.g., percentage of eligible women with a mammogram performed in the last year).
2 | RATIO | ratio measure scoring | A score that may have a value of zero or greater that is derived by dividing a count of one type of data by a count of another type of data (e.g., the number of patients with central lines who develop infection divided by the number of central line days).
1 | (_ObservationMeasureType) Abstract | | Observation values used to indicate what kind of health quality measure is used.
2 | COMPOSITE | composite measure type | A measure that is composed from one or more other measures and indicates an overall summary of those measures.
2 | EFFICIENCY | efficiency measure type | A measure related to the efficiency of medical treatment.
2 | EXPERIENCE | experience measure type | A measure related to the level of patient engagement or patient experience of care.
2 | OUTCOME | outcome measure type | A measure that indicates the result of the performance (or non-performance) of a function or process.
3 | INTERM-OM | intermediate clinical outcome measure | A measure that evaluates the change over time of a physiologic state observable that is associated with a specific long-term health outcome.
3 | PRO-PM | intermediate clinical outcome measure | A measure that is a comparison of patient reported outcomes for a single or multiple patients collected via an instrument specifically designed to obtain input directly from patients.
2 | PROCESS | process measure type | A measure which focuses on a process which leads to a certain outcome, meaning that a scientific basis exists for believing that the process, when executed well, will increase the probability of achieving a desired outcome.
3 | APPROPRIATE | appropriate use process measure | A measure that assesses the use of one or more processes where the expected health benefit exceeds the expected negative consequences.
2 | RESOURCE | resource use measure type | A measure related to the extent of use of clinical resources or cost of care.
2 | STRUCTURE | structure measure type | A measure related to the structure of patient care.
1 | _ObservationPopulationInclusion Deprecated | ObservationPopulationInclusion | Observation values used to assert various populations that a subject falls into.
2 | DENEX Deprecated | denominator exclusions | Patients who should be removed from the eMeasure population and denominator before determining if numerator criteria are met. Denominator exclusions are used in proportion and ratio measures to help narrow the denominator.
2 | DENEXCEP Deprecated | denominator exceptions | Denominator exceptions are those conditions that should remove a patient, procedure or unit of measurement from the denominator only if the numerator criteria are not met. Denominator exceptions allow for adjustment of the calculated score for those providers with higher risk populations. Denominator exceptions are used only in proportion eMeasures. They are not appropriate for ratio or continuous variable eMeasures. Denominator exceptions allow for the exercise of clinical judgment and should be specifically defined where capturing the information in a structured manner fits the clinical workflow. Generic denominator exception reasons used in proportion eMeasures fall into three general categories: Medical reasons Patient reasons System reasons
2 | DENOM Deprecated | denominator | It can be the same as the initial patient population or a subset of the initial patient population to further constrain the population for the purpose of the eMeasure. Different measures within an eMeasure set may have different Denominators. Continuous Variable eMeasures do not have a Denominator, but instead define a Measure Population.
2 | IP Deprecated | initial population | The initial population refers to all entities to be evaluated by a specific quality measure who share a common set of specified characteristics within a specific measurement set to which a given measure belongs.
3 | IPP Deprecated | initial patient population | The initial patient population refers to all patients to be evaluated by a specific quality measure who share a common set of specified characteristics within a specific measurement set to which a given measure belongs. Details often include information based upon specific age groups, diagnoses, diagnostic and procedure codes, and enrollment periods.
2 | MSRPOPL Deprecated | measure population | Measure population is used only in continuous variable eMeasures. It is a narrative description of the eMeasure population. (e.g., all patients seen in the Emergency Department during the measurement period).
2 | NUMER Deprecated | numerator | Numerators are used in proportion and ratio eMeasures. In proportion measures the numerator criteria are the processes or outcomes expected for each patient, procedure, or other unit of measurement defined in the denominator. In ratio measures the numerator is related, but not directly derived from the denominator (e.g., a numerator listing the number of central line blood stream infections and a denominator indicating the days per thousand of central line usage in a specific time period).
2 | NUMEX Deprecated | numerator exclusions | Numerator Exclusions are used only in ratio eMeasures to define instances that should not be included in the numerator data. (e.g., if the number of central line blood stream infections per 1000 catheter days were to exclude infections with a specific bacterium, that bacterium would be listed as a numerator exclusion.)
1 | (_PartialCompletionScale) Abstract
2 | G | Great extent | Value for Act.partialCompletionCode attribute that implies 81-99% completion
2 | LE | Large extent | Value for Act.partialCompletionCode attribute that implies 61-80% completion
2 | ME | Medium extent | Value for Act.partialCompletionCode attribute that implies 41-60% completion
2 | MI | Minimal extent | Value for Act.partialCompletionCode attribute that implies 1-20% completion
2 | N | None | Value for Act.partialCompletionCode attribute that implies 0% completion
2 | S | Some extent | Value for Act.partialCompletionCode attribute that implies 21-40% completion
1 | (_SecurityObservationValue) Abstract | | Observation values used to indicate security observation metadata.
2 | (_SECINTOBV) Abstract | | Abstract security observation values used to indicate security integrity metadata. Examples: Codes conveying integrity status, integrity confidence, and provenance.
3 | (_SECALTINTOBV) Abstract | | Abstract security metadata observation values used to indicate mechanism used for authorized alteration of an IT resource (data, information object, service, or system capability)
4 | ABSTRED | abstracted | Security metadata observation values used to indicate the use of a more abstract version of the content, e.g., replacing exact value of an age or date field with a range, or remove the left digits of a credit card number or SSN.
4 | AGGRED | aggregated | Security metadata observation values used to indicate the use of an algorithmic combination of actual values with the result of an aggregate function, e.g., average, sum, or count in order to limit disclosure of an IT resource (data,

information object, service, or system capability) to the minimum necessary.
4 | ANONYED | anonymized | Security metadata observation value conveying the alteration integrity of an IT resource (data, information object, service, or system capability) by used to indicate the mechanism by which software systems can strip portions of the resource that could allow the identification of the source of the information or the information subject. No key to relink the data is retained.
4 | MAPPED | mapped | Security metadata observation value used to indicate that the IT resource semantic content has been transformed from one encoding to another. Usage Note: "MAP" code does not indicate the semantic fidelity of the transformed content. To indicate semantic fidelity for maps of HL7 to other code systems, this security alteration integrity observation may be further specified using an Act valued with Value Set: MapRelationship (2.16.840.1.113883.1.11.11052). Semantic fidelity of the mapped IT Resource may also be indicated using a SecurityIntegrityConfidenceObservation.
4 | MASKED | masked | Security metadata observation value conveying the alteration integrity of an IT resource (data, information object, service, or system capability) by indicating the mechanism by which software systems can make data unintelligible (that is, as unreadable and unusable by algorithmically transforming plaintext into ciphertext) such that it can only be accessed or used by authorized users. An authorized user may be provided a key to decrypt per license or "shared secret". Usage Note: "MASKED" may be used, per applicable policy, as a flag to indicate to a user or receiver that some portion of an IT resource has been further encrypted, and may be accessed only by an authorized user or receiver to which a decryption key is provided.
4 | PSEUDED | pseudonymized | Security metadata observation value conveying the alteration integrity of an IT resource (data, information object, service, or system capability), by indicating the mechanism by which software systems can strip portions of the resource that could allow the identification of the source of the information or the information subject. Custodian may retain a key to relink data necessary to reidentify the information subject. Rationale: Personal data which has been processed to make it impossible to know whose data it is. Used particularly for secondary use of health data. In some cases, it may be possible for authorized individuals to restore the identity of the individual, e.g.,for public health case management. Based on ISO/TS 25237:2008 Health informaticsâ€”Pseudonymization
4 | REDACTED | redacted | Security metadata observation value used to indicate the mechanism by which software systems can filter an IT resource (data, information object, service, or system capability) to remove any portion of the resource that is not authorized to be access, used, or disclosed. Usage Note: "REDACTED" may be used, per applicable policy, as a flag to indicate to a user or receiver that some portion of an IT resource has filtered and not included in the content accessed or received.
4 | SUBSETTED | subsetted | Metadata observation used to indicate that some information has been removed from the source object when the view this object contains was constructed because of configuration options when the view was created. The content may not be suitable for use as the basis of a record update Usage Note: This is not suitable to be used when information is removed for security reasons - see the code REDACTED for this use.
4 | SYNTAC | syntactic transform | Security metadata observation value used to indicate that the IT resource syntax has been transformed from one syntactical representation to another. Usage Note: "SYNTAC" code does not indicate the syntactical correctness of the syntactically transformed IT resource.
4 | TRSLT | translated | Security metadata observation value used to indicate that the IT resource has been translated from one human language to another. Usage Note: "TRSLT" does not indicate the fidelity of the translation or the languages translated. The fidelity of the IT Resource translation may be indicated using a SecurityIntegrityConfidenceObservation. To indicate languages, use the Value Set:HumanLanguage (2.16.840.1.113883.1.11.11526)
4 | VERSIONED | versioned | Security metadata observation value conveying the alteration integrity of an IT resource (data, information object, service, or system capability) which indicates that the resource only retains versions of an IT resource for access and use per applicable policy Usage Note: When this code is used, expectation is that the system has removed historical versions of the data that falls outside the time period deemed to be the effective time of the applicable version.
3 | (_SECDATINTOBV) Abstract | | Abstract security observation values used to indicate data integrity metadata. Examples: Codes conveying the mechanism used to preserve the accuracy and consistency of an IT resource such as a digital signature and a cryptographic hash function.
4 | CRYTOHASH | cryptographic hash function | Security metadata observation value used to indicate the mechanism by which software systems can establish that data was not modified in transit. Rationale: This definition is intended to align with the ISO 22600-2 3.3.19 definition of cryptographic checkvalue: Information which is derived by performing a cryptographic transformation (see cryptography) on the data unit. The derivation of the checkvalue may be performed in one or more steps and is a result of a mathematical function of the key and a data unit. It is usually used to check the integrity of a data unit. Examples: SHA-1 SHA-2 (Secure Hash Algorithm)
4 | DIGSIG | digital signature | Security metadata observation value used to indicate the mechanism by which software systems use digital signature to establish that data has not been modified. Rationale: This definition is intended to align with the ISO 22600-2 3.3.26 definition of digital signature: Data appended to, or a cryptographic transformation (see cryptography) of, a data unit that allows a recipient of the data unit to prove the source and integrity of the data unit and protect against forgery e.g., by the recipient.
3 | (_SECINTCONOBV) Abstract | | Abstract security observation value used to indicate integrity confidence metadata. Examples: Codes conveying the level of reliability and trustworthiness of an IT resource.
4 | HRELIABLE | highly reliable | Security metadata observation value used to indicate that the veracity or trustworthiness of an IT resource (data, information object, service, or system capability) for a specified purpose of use is perceived to be or deemed by policy to be very high.
4 | RELIABLE | reliable | Security metadata observation value used to indicate that the veracity or trustworthiness of an IT resource (data, information object, service, or system capability) for a specified purpose of use is perceived to be or deemed by policy to be adequate.
4 | UNCERTREL | uncertain reliability | Security metadata observation value used to indicate that the veracity or trustworthiness of an IT resource (data, information object, service, or system capability) for a specified purpose of use is perceived to be or deemed by policy to be uncertain.
4 | UNRELIABLE | unreliable | Security metadata observation value used to indicate that the veracity or trustworthiness of an IT resource (data, information object, service, or system capability) for a specified purpose of use is perceived to be or deemed by policy to be inadequate.
3 | (_SECINTPRVOBV) Abstract | | Abstract security metadata observation value used to indicate the provenance of an IT resource (data, information object, service, or system capability). Examples: Codes conveying the provenance metadata about the entity reporting an IT resource.
4 | (_SECINTPRVABOBV) Abstract | | Abstract security provenance metadata observation value used to indicate the entity that asserted an IT resource (data, information object, service, or system capability). Examples: Codes conveying the provenance metadata about the entity asserting the resource.
5 | CLINAST | clinician asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a clinician.

5 | DEVAST | device asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a device.
5 | HCPAST | healthcare professional asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a healthcare professional.
5 | PACQAST | patient acquaintance asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a patient acquaintance.
5 | PATAST | patient asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a patient.
5 | PAYAST | payer asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a payer.
5 | PROAST | professional asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a professional.
5 | SDMAST | substitute decision maker asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a substitute decision maker.
4 | (_SECINTPRVRBOBV) Abstract | | Abstract security provenance metadata observation value used to indicate the entity that reported the resource (data, information object, service, or system capability). Examples: Codes conveying the provenance metadata about the entity reporting an IT resource.
5 | CLINRPT | clinician reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a clinician.
5 | DEVRPT | device reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a device.
5 | HCPRPT | healthcare professional reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a healthcare professional.
5 | PACQRPT | patient acquaintance reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a patient acquaintance.
5 | PATRPT | patient reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a patient.
5 | PAYRPT | payer reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a payer.
5 | PRORPT | professional reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a professional.
5 | SDMRPT | substitute decision maker reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a substitute decision maker.
2 | (SECTRSTOBV) Abstract | | Observation value used to indicate aspects of trust applicable to an IT resource (data, information object, service, or system capability).
3 | TRSTACCRDOBV | trust accreditation observation | Values for security trust accreditation metadata observation made about the formal declaration by an authority or neutral third party that validates the technical, security, trust, and business practice conformance of Trust Agents to facilitate security, interoperability, and trust among participants within a security domain or trust framework.
3 | TRSTAGREOBV | trust agreement observation | Values for security trust agreement metadata observation made about privacy and security requirements with which a security domain must comply. [ISO IEC 10181-1] [ISO IEC 10181-1]
3 | TRSTCERTOBV | trust certificate observation | Values for security trust certificate metadata observation made about a set of security-relevant data issued by a security authority or trusted third party, together with security information which is used to provide the integrity and data origin authentication services for an IT resource (data, information object, service, or system capability). [Based on ISO IEC 10181-1] For example, a Certificate Policy (CP), which is a named set of rules that indicates the applicability of a certificate to a particular community and/or class of application with common security requirements. A particular Certificate Policy might indicate the applicability of a type of certificate to the authentication of electronic data interchange transactions for the trading of goods within a given price range. Another example is Cross Certification with Federal Bridge.
3 | (TRSTLOAOBV) Abstract | | Values for security trust assurance metadata observation made about the digital quality or reliability of a trust assertion, activity, capability, information exchange, mechanism, process, or protocol.
4 | (LOAAN) Abstract | | The value assigned as the indicator of the digital quality or reliability of the verification and validation process used to verify the claimed identity of an entity by securely associating an identifier and its authenticator. [Based on ISO 7498-2] For example, the degree of confidence in the vetting process used to establish the identity of the individual to whom the credential was issued, and 2) the degree of confidence that the individual who uses the credential is the individual to whom the credential was issued. [OMB M-04-04 E-Authentication Guidance for Federal Agencies]
5 | LOAAN1 | low authentication level of assurance | Indicator of low digital quality or reliability of the digital reliability of the verification and validation process used to verify the claimed identity of an entity by securely associating an identifier and its authenticator. [Based on ISO 7498-2] The degree of confidence in the vetting process used to establish the identity of the individual to whom the credential was issued, and 2) the degree of confidence that the individual who uses the credential is the individual to whom the credential was issued. [OMB M-04-04 E-Authentication Guidance for Federal Agencies] Low authentication level of assurance indicates that the relying party may have little or no confidence in the asserted identity's validity. Level 1 requires little or no confidence in the asserted identity. No identity proofing is required at this level, but the authentication mechanism should provide some assurance that the same claimant is accessing the protected transaction or data. A wide range of available authentication technologies can be employed and any of the token methods of Levels 2, 3, or 4, including Personal Identification Numbers (PINs), may be used. To be authenticated, the claimant must prove control of the token through a secure authentication protocol. At Level 1, long-term shared authentication secrets may be revealed to verifiers. Assertions issued about claimants as a result of a successful authentication are either cryptographically authenticated by relying parties (using approved methods) or are obtained directly from a trusted party via a secure authentication protocol. [Summary of the technical requirements specified in NIST SP 800-63 for the four levels of assurance defined by the December 2003, the Office of Management and Budget (OMB) issued Memorandum M-04-04, E-Authentication Guidance for Federal Agencies.]
5 | LOAAN2 | basic authentication level of assurance | Indicator of basic digital quality or reliability of the digital reliability of the verification and validation process used to verify the claimed identity of an entity by securely associating an identifier and its authenticator. [Based on ISO 7498-2] The degree of confidence in the vetting process used to establish the identity of the individual to whom the credential was issued, and 2) the degree of confidence that the individual who uses the credential is the individual to whom the credential was issued. [OMB M-04-04 E-Authentication Guidance for Federal Agencies] Basic authentication level of assurance indicates that the relying party may have some confidence in the asserted identity's validity. Level 2 requires confidence that the asserted identity is accurate. Level 2 provides for single-factor remote network authentication, including

identity-proofing requirements for presentation of identifying materials or information. A wide range of available authentication technologies can be employed, including any of the token methods of Levels 3 or 4, as well as passwords. Successful authentication requires that the claimant prove through a secure authentication protocol that the claimant controls the token. Eavesdropper, replay, and online guessing attacks are prevented. Long-term shared authentication secrets, if used, are never revealed to any party except the claimant and verifiers operated by the CSP; however, session (temporary) shared secrets may be provided to independent verifiers by the CSP. Approved cryptographic techniques are required. Assertions issued about claimants as a result of a successful authentication are either cryptographically authenticated by relying parties (using approved methods) or are obtained directly from a trusted party via a secure authentication protocol. [Summary of the technical requirements specified in NIST SP 800-63 for the four levels of assurance defined by the December 2003, the Office of Management and Budget (OMB) issued Memorandum M-04-04, E-Authentication Guidance for Federal Agencies.]

5 | LOAAN3 | medium authentication level of assurance | Indicator of medium digital quality or reliability of the digital reliability of verification and validation of the process used to verify the claimed identity of an entity by securely associating an identifier and its authenticator. [Based on ISO 7498-2] The degree of confidence in the vetting process used to establish the identity of the individual to whom the credential was issued, and 2) the degree of confidence that the individual who uses the credential is the individual to whom the credential was issued. [OMB M-04-04 E-Authentication Guidance for Federal Agencies] Medium authentication level of assurance indicates that the relying party may have high confidence in the asserted identity's validity. Level 3 is appropriate for transactions that need high confidence in the accuracy of the asserted identity. Level 3 provides multifactor remote network authentication. At this level, identity-proofing procedures require verification of identifying materials and information. Authentication is based on proof of possession of a key or password through a cryptographic protocol. Cryptographic strength mechanisms should protect the primary authentication token (a cryptographic key) against compromise by the protocol threats, including eavesdropper, replay, online guessing, verifier impersonation, and man-in-the-middle attacks. A minimum of two authentication factors is required. Three kinds of tokens may be used: "soft" cryptographic token, which has the key stored on a general-purpose computer, "hard" cryptographic token, which has the key stored on a special hardware device, and "one-time password" device token, which has symmetric key stored on a personal hardware device that is a cryptographic module validated at FIPS 140-2 Level 1 or higher. Validation testing of cryptographic modules and algorithms for conformance to Federal Information Processing Standard (FIPS) 140-2, Security Requirements for Cryptographic Modules, is managed by NIST. Authentication requires that the claimant prove control of the token through a secure authentication protocol. The token must be unlocked with a password or biometric representation, or a password must be used in a secure authentication protocol, to establish two-factor authentication. Long-term shared authentication secrets, if used, are never revealed to any party except the claimant and verifiers operated directly by the CSP; however, session (temporary) shared secrets may be provided to independent verifiers by the CSP. Approved cryptographic techniques are used for all operations. Assertions issued about claimants as a result of a successful authentication are either cryptographically authenticated by relying parties (using approved methods) or are obtained directly from a trusted party via a secure authentication protocol. [Summary of the technical requirements specified in NIST SP 800-63 for the four levels of assurance defined by the December 2003, the Office of Management and Budget (OMB) issued Memorandum M-04-04, E-Authentication Guidance for Federal Agencies.]

5 | LOAAN4 | high authentication level of assurance | Indicator of high digital quality or reliability of the digital reliability of the verification and validation process used to verify the claimed identity of an entity by securely associating an identifier and its authenticator. [Based on ISO 7498-2] The degree of confidence in the vetting process used to establish the identity of the individual to whom the credential was issued, and 2) the degree of confidence that the individual who uses the credential is the individual to whom the credential was issued. [OMB M-04-04 E-Authentication Guidance for Federal Agencies] High authentication level of assurance indicates that the relying party may have very high confidence in the asserted identity's validity. Level 4 is for transactions that need very high confidence in the accuracy of the asserted identity. Level 4 provides the highest practical assurance of remote network authentication. Authentication is based on proof of possession of a key through a cryptographic protocol. This level is similar to Level 3 except that only â€œhardâ€? cryptographic tokens are allowed, cryptographic module validation requirements are strengthened, and subsequent critical data transfers must be authenticated via a key that is bound to the authentication process. The token should be a hardware cryptographic module validated at FIPS 140-2 Level 2 or higher overall with at least FIPS 140-2 Level 3 physical security. This level requires a physical token, which cannot readily be copied, and operator authentication at Level 2 and higher, and ensures good, two-factor remote authentication. Level 4 requires strong cryptographic authentication of all parties and all sensitive data transfers between the parties. Either public key or symmetric key technology may be used. Authentication requires that the claimant prove through a secure authentication protocol that the claimant controls the token. Eavesdropper, replay, online guessing, verifier impersonation, and man-in-the-middle attacks are prevented. Long-term shared authentication secrets, if used, are never revealed to any party except the claimant and verifiers operated directly by the CSP; however, session (temporary) shared secrets may be provided to independent verifiers by the CSP. Strong approved cryptographic techniques are used for all operations. All sensitive data transfers are cryptographically authenticated using keys bound to the authentication process. [Summary of the technical requirements specified in NIST SP 800-63 for the four levels of assurance defined by the December 2003, the Office of Management and Budget (OMB) issued Memorandum M-04-04, E-Authentication Guidance for Federal Agencies.]

4 | (LOAAP) Abstract | | The value assigned as the indicator of the digital quality or reliability of a defined sequence of messages between a Claimant and a Verifier that demonstrates that the Claimant has possession and control of a valid token to establish his/her identity, and optionally, demonstrates to the Claimant that he or she is communicating with the intended Verifier. [Based on NIST SP 800-63-2]

5 | LOAAP1 | low authentication process level of assurance | Indicator of the low digital quality or reliability of a defined sequence of messages between a Claimant and a Verifier that demonstrates that the Claimant has possession and control of a valid token to establish his/her identity, and optionally, demonstrates to the Claimant that he or she is communicating with the intended Verifier. [Based on NIST SP 800-63-2] Low authentication process level of assurance indicates that (1) long-term shared authentication secrets may be revealed to verifiers; and (2) assertions and assertion references require protection from manufacture/modification and reuse attacks. [Summary of the technical requirements specified in NIST SP 800-63 for the four levels of assurance defined by the December 2003, the Office of Management and Budget (OMB) issued Memorandum M-04-04, E-Authentication Guidance for Federal Agencies.]

5 | LOAAP2 | basic authentication process level of assurance | Indicator of the basic digital quality or reliability of a defined sequence of messages between a Claimant and a Verifier that demonstrates that the Claimant has possession and control of a valid token to establish his/her identity, and optionally, demonstrates to the Claimant that he or she is communicating with the intended Verifier. [Based on NIST SP 800-63-2] Basic authentication process level of assurance indicates that long-term shared authentication secrets are never revealed to any other party except Credential Service Provider (CSP). Sessions (temporary) shared secrets may be provided to independent verifiers by CSP. Long-term shared authentication secrets, if used, are never revealed to any other party except Verifiers operated by the Credential Service Provider (CSP); however, session (temporary) shared secrets may be provided to independent Verifiers by the CSP. In addition to Level 1 requirements, assertions are resistant to disclosure, redirection, capture and substitution attacks. Approved cryptographic techniques are required. [Summary of the technical requirements specified in NIST SP 800-63 for the four levels of assurance defined by the December 2003, the Office of

Management and Budget (OMB) issued Memorandum M-04-04, E-Authentication Guidance for Federal Agencies.]
5 | LOAAP3 | medium authentication process level of assurance | Indicator of the medium digital quality or reliability of a defined sequence of messages between a Claimant and a Verifier that demonstrates that the Claimant has possession and control of a valid token to establish his/her identity, and optionally, demonstrates to the Claimant that he or she is communicating with the intended Verifier. [Based on NIST SP 800-63-2] Medium authentication process level of assurance indicates that the token can be unlocked with password, biometric, or uses a secure multi-token authentication protocol to establish two-factor authentication. Long-term shared authentication secrets are never revealed to any party except the Claimant and Credential Service Provider (CSP). Authentication requires that the Claimant prove, through a secure authentication protocol, that he or she controls the token. The Claimant unlocks the token with a password or biometric, or uses a secure multi-token authentication protocol to establish two-factor authentication (through proof of possession of a physical or software token in combination with some memorized secret knowledge). Long-term shared authentication secrets, if used, are never revealed to any party except the Claimant and Verifiers operated directly by the CSP; however, session (temporary) shared secrets may be provided to independent Verifiers by the CSP. In addition to Level 2 requirements, assertions are protected against repudiation by the Verifier.
5 | LOAAP4 | high authentication process level of assurance | Indicator of the high digital quality or reliability of a defined sequence of messages between a Claimant and a Verifier that demonstrates that the Claimant has possession and control of a valid token to establish his/her identity, and optionally, demonstrates to the Claimant that he or she is communicating with the intended Verifier. [Based on NIST SP 800-63-2] High authentication process level of assurance indicates all sensitive data transfer are cryptographically authenticated using keys bound to the authentication process. Level 4 requires strong cryptographic authentication of all communicating parties and all sensitive data transfers between the parties. Either public key or symmetric key technology may be used. Authentication requires that the Claimant prove through a secure authentication protocol that he or she controls the token. All protocol threats at Level 3 are required to be prevented at Level 4. Protocols shall also be strongly resistant to man-in-the-middle attacks. Long-term shared authentication secrets, if used, are never revealed to any party except the Claimant and Verifiers operated directly by the CSP; however, session (temporary) shared secrets may be provided to independent Verifiers by the CSP. Approved cryptographic techniques are used for all operations. All sensitive data transfers are cryptographically authenticated using keys bound to the authentication process. [Summary of the technical requirements specified in NIST SP 800-63 for the four levels of assurance defined by the December 2003, the Office of Management and Budget (OMB) issued Memorandum M-04-04, E-Authentication Guidance for Federal Agencies.]
4 | (LOAAS) Abstract | | The value assigned as the indicator of the high quality or reliability of the statement from a Verifier to a Relying Party (RP) that contains identity information about a Subscriber. Assertions may also contain verified attributes.
5 | LOAAS1 | low assertion level of assurance | Indicator of the low quality or reliability of the statement from a Verifier to a Relying Party (RP) that contains identity information about a Subscriber. Assertions may also contain verified attributes. Assertions and assertion references require protection from modification and reuse attacks. [Summary of the technical requirements specified in NIST SP 800-63 for the four levels of assurance defined by the December 2003, the Office of Management and Budget (OMB) issued Memorandum M-04-04, E-Authentication Guidance for Federal Agencies.]
5 | LOAAS2 | basic assertion level of assurance | Indicator of the basic quality or reliability of the statement from a Verifier to a Relying Party (RP) that contains identity information about a Subscriber. Assertions may also contain verified attributes. Assertions are resistant to disclosure, redirection, capture and substitution attacks. Approved cryptographic techniques are required for all assertion protocols. [Summary of the technical requirements specified in NIST SP 800-63 for the four levels of assurance defined by the December 2003, the Office of Management and Budget (OMB) issued Memorandum M-04-04, E-Authentication Guidance for Federal Agencies.]
5 | LOAAS3 | medium assertion level of assurance | Indicator of the medium quality or reliability of the statement from a Verifier to a Relying Party (RP) that contains identity information about a Subscriber. Assertions may also contain verified attributes. Assertions are protected against repudiation by the verifier. [Summary of the technical requirements specified in NIST SP 800-63 for the four levels of assurance defined by the December 2003, the Office of Management and Budget (OMB) issued Memorandum M-04-04, E-Authentication Guidance for Federal Agencies.]
5 | LOAAS4 | high assertion level of assurance | Indicator of the high quality or reliability of the statement from a Verifier to a Relying Party (RP) that contains identity information about a Subscriber. Assertions may also contain verified attributes. Strongly resistant to man-in-the-middle attacks. "Bearer" assertions are not used. "Holder-of-key" assertions may be used. RP maintains records of the assertions. [Summary of the technical requirements specified in NIST SP 800-63 for the four levels of assurance defined by the December 2003, the Office of Management and Budget (OMB) issued Memorandum M-04-04, E-Authentication Guidance for Federal Agencies.]
4 | (LOACM) Abstract | | Indicator of the digital quality or reliability of the activities performed by the Credential Service Provider (CSP) subsequent to electronic authentication registration, identity proofing and issuance activities to manage and safeguard the integrity of an issued credential and its binding to an identity. [Electronic Authentication Guideline - Recommendations of the National Institute of Standards and Technology, NIST Special Publication 800-63-1, Dec 2011]
5 | LOACM1 | low token and credential management level of assurance | Indicator of the low digital quality or reliability of the activities performed by the Credential Service Provider (CSP) subsequent to electronic authentication registration, identity proofing and issuance activities to manage and safeguard the integrity of an issued credential and its binding to an identity. Little or no confidence that an individual has maintained control over a token that has been entrusted to him or her and that that token has not been compromised. Characteristics include weak identity binding to tokens and plaintext passwords or secrets not transmitted across a network. [Electronic Authentication Guideline - Recommendations of the National Institute of Standards and Technology, NIST Special Publication 800-63-1, Dec 2011]
5 | LOACM2 | basic token and credential management level of assurance | Indicator of the basic digital quality or reliability of the activities performed by the Credential Service Provider (CSP) subsequent to electronic authentication registration, identity proofing and issuance activities to manage and safeguard the integrity of an issued credential and its binding to an identity. Some confidence that an individual has maintained control over a token that has been entrusted to him or her and that that token has not been compromised. Characteristics include: Verification must prove claimant controls the token; token resists online guessing, replay, session hijacking, and eavesdropping attacks; and token is at least weakly resistant to man-in-the middle attacks. [Electronic Authentication Guideline - Recommendations of the National Institute of Standards and Technology, NIST Special Publication 800-63-1, Dec 2011]
5 | LOACM3 | medium token and credential management level of assurance | Indicator of the medium digital quality or reliability of the activities performed by the Credential Service Provider (CSP) subsequent to electronic authentication registration, identity proofing and issuance activities to manage and safeguard the integrity of an issued credential and itâ€™s binding to an identity. High confidence that an individual has maintained control over a token that has been entrusted to him or her and that that token has not been compromised. Characteristics include: Ownership of token verifiable through security authentication protocol and credential management protects against verifier impersonation attacks. [Electronic Authentication Guideline - Recommendations of the National Institute of Standards and Technology, NIST Special Publication 800-63-1, Dec 2011]

5 | LOACM4 | high token and credential management level of assurance | Indicator of the high digital quality or reliability of the activities performed by the Credential Service Provider (CSP) subsequent to electronic authentication registration, identity proofing and issuance activities to manage and safeguard the integrity of an issued credential and itâ€™s binding to an identity. Very high confidence that an individual has maintained control over a token that has been entrusted to him or her and that that token has not been compromised. Characteristics include: Verifier can prove control of token through a secure protocol; credential management supports strong cryptographic authentication of all communication parties. [Electronic Authentication Guideline - Recommendations of the National Institute of Standards and Technology, NIST Special Publication 800-63-1, Dec 2011]

4 | (LOAID) Abstract | | Indicator of the quality or reliability in the process of ascertaining that an individual is who he or she claims to be.

5 | LOAID1 | low identity proofing level of assurance | Indicator of low digital quality or reliability in the process of ascertaining that an individual is who he or she claims to be. Requires that a continuity of identity be maintained but does not require identity proofing. [Based on Electronic Authentication Guideline - Recommendations of the National Institute of Standards and Technology, NIST Special Publication 800-63-1, Dec 2011]

5 | LOAID2 | basic identity proofing level of assurance | Indicator of some digital quality or reliability in the process of ascertaining that that an individual is who he or she claims to be. Requires identity proofing via presentation of identifying material or information. [Based on Electronic Authentication Guideline - Recommendations of the National Institute of Standards and Technology, NIST Special Publication 800-63-1, Dec 2011]

5 | LOAID3 | medium identity proofing level of assurance | Indicator of high digital quality or reliability in the process of ascertaining that an individual is who he or she claims to be. Requires identity proofing procedures for verification of identifying materials and information. [Based on Electronic Authentication Guideline - Recommendations of the National Institute of Standards and Technology, NIST Special Publication 800-63-1, Dec 2011]

5 | LOAID4 | high identity proofing level of assurance | Indicator of high digital quality or reliability in the process of ascertaining that an individual is who he or she claims to be. Requires identity proofing procedures for verification of identifying materials and information. [Based on Electronic Authentication Guideline - Recommendations of the National Institute of Standards and Technology, NIST Special Publication 800-63-1, Dec 2011]

4 | (LOANR) Abstract | | Indicator of the digital quality or reliability in the process of establishing proof of delivery and proof of origin. [Based on ISO 7498-2]

5 | LOANR1 | low non-repudiation level of assurance | Indicator of low digital quality or reliability in the process of establishing proof of delivery and proof of origin. [Based on ISO 7498-2]

5 | LOANR2 | basic non-repudiation level of assurance | Indicator of basic digital quality or reliability in the process of establishing proof of delivery and proof of origin. [Based on ISO 7498-2]

5 | LOANR3 | medium non-repudiation level of assurance | Indicator of medium digital quality or reliability in the process of establishing proof of delivery and proof of origin. [Based on ISO 7498-2]

5 | LOANR4 | high non-repudiation level of assurance | Indicator of high digital quality or reliability in the process of establishing proof of delivery and proof of origin. [Based on ISO 7498-2]

4 | (LOARA) Abstract | | Indicator of the digital quality or reliability of the information exchange between network-connected devices where the information cannot be reliably protected end-to-end by a single organizationâ€™s security controls. [Based on NIST SP 800-63-2]

5 | LOARA1 | low remote access level of assurance | Indicator of low digital quality or reliability of the information exchange between network-connected devices where the information cannot be reliably protected end-to-end by a single organizationâ€™s security controls. [Based on NIST SP 800-63-2]

5 | LOARA2 | basic remote access level of assurance | Indicator of basic digital quality or reliability of the information exchange between network-connected devices where the information cannot be reliably protected end-to-end by a single organizationâ€™s security controls. [Based on NIST SP 800-63-2]

5 | LOARA3 | medium remote access level of assurance | Indicator of medium digital quality or reliability of the information exchange between network-connected devices where the information cannot be reliably protected end-to-end by a single organizationâ€™s security controls. [Based on NIST SP 800-63-2]

5 | LOARA4 | high remote access level of assurance | Indicator of high digital quality or reliability of the information exchange between network-connected devices where the information cannot be reliably protected end-to-end by a single organization's security controls. [Based on NIST SP 800-63-2]

4 | (LOATK) Abstract | | Indicator of the digital quality or reliability of single and multi-token authentication. [Electronic Authentication Guideline - Recommendations of the National Institute of Standards and Technology, NIST Special Publication 800-63-1, Dec 2011]

5 | LOATK1 | low token level of assurance | Indicator of the low digital quality or reliability of single and multi-token authentication. Permits the use of any of the token methods of Levels 2, 3, or 4. [Electronic Authentication Guideline - Recommendations of the National Institute of Standards and Technology, NIST Special Publication 800-63-1, Dec 2011]

5 | LOATK2 | basic token level of assurance | Indicator of the basic digital quality or reliability of single and multi-token authentication. Requires single factor authentication using memorized secret tokens, pre-registered knowledge tokens, look-up secret tokens, out of band tokens, or single factor one-time password devices. [Electronic Authentication Guideline - Recommendations of the National Institute of Standards and Technology, NIST Special Publication 800-63-1, Dec 2011]

5 | LOATK3 | medium token level of assurance | Indicator of the medium digital quality or reliability of single and multi-token authentication. Requires two authentication factors. Provides multi-factor remote network authentication. Permits multi-factor software cryptographic token. [Electronic Authentication Guideline - Recommendations of the National Institute of Standards and Technology, NIST Special Publication 800-63-1, Dec 2011]

5 | LOATK4 | high token level of assurance | Indicator of the high digital quality or reliability of single and multi-token authentication. Requires token that is a hardware cryptographic module validated at validated at Federal Information Processing Standard (FIPS) 140-2 Level 2 or higher overall with at least FIPS 140-2 Level 3 physical security. Level 4 token requirements can be met by using the PIV authentication key of a FIPS 201 compliant Personal Identity Verification (PIV) Card. [Electronic Authentication Guideline - Recommendations of the National Institute of Standards and Technology, NIST Special Publication 800-63-1, Dec 2011]

3 | TRSTMECOBV | none supplied 6 | Values for security trust mechanism metadata observation made about a security architecture system component that supports enforcement of security policies.

1 | (_SeverityObservation) Abstract | | Potential values for observations of severity.

2 | H | High | Indicates the condition may be life-threatening or has the potential to cause permanent injury.

2 | L | Low | Indicates the condition may result in some adverse consequences but is unlikely to substantially affect the situation of the subject.

2 | M | Moderate | Indicates the condition may result in noticable adverse adverse consequences but is unlikely to be life-threatening or cause permanent injury.
1 | (_SubjectBodyPosition) Abstract | | Contains codes for defining the observed, physical position of a subject, such as during an observation, assessment, collection of a specimen, etc. ECG waveforms and vital signs, such as blood pressure, are two examples where a general, observed position typically needs to be noted.
2 | LLD | left lateral decubitus | Lying on the left side.
2 | PRN | prone | Lying with the front or ventral surface downward; lying face down.
2 | RLD | right lateral decubitus | Lying on the right side.
2 | SFWL | Semi-Fowler's | A semi-sitting position in bed with the head of the bed elevated approximately 45 degrees.
2 | SIT | sitting | Resting the body on the buttocks, typically with upper torso erect or semi erect.
2 | STN | standing | To be stationary, upright, vertical, on one's legs.
2 | SUP | supine
3 | RTRD | reverse trendelenburg | Lying on the back, on an inclined plane, typically about 30-45 degrees with head raised and feet lowered.
3 | TRD | trendelenburg | Lying on the back, on an inclined plane, typically about 30-45 degrees, with head lowered and feet raised.
1 | (_VerificationOutcomeValue) Abstract | | Values for observations of verification act results Examples: Verified, not verified, verified with warning.
2 | ACT | active coverage | Definition: Coverage is in effect for healthcare service(s) and/or product(s).
2 | ACTPEND | active - pending investigation | Definition: Coverage is in effect for healthcare service(s) and/or product(s) - Pending Investigation
2 | ELG | eligible | Definition: Coverage is in effect for healthcare service(s) and/or product(s).
2 | INACT | inactive | Definition: Coverage is not in effect for healthcare service(s) and/or product(s).
2 | INPNDINV | inactive - pending investigation | Definition: Coverage is not in effect for healthcare service(s) and/or product(s) - Pending Investigation.
2 | INPNDUPD | inactive - pending eligibility update | Definition: Coverage is not in effect for healthcare service(s) and/or product(s) - Pending Eligibility Update.
2 | NELG | not eligible | Definition: Coverage is not in effect for healthcare service(s) and/or product(s). May optionally include reasons for the ineligibility.
1 | (_WorkSchedule) Abstract | | Concepts that describe an individual's typical arrangement of working hours for an occupation.
2 | DS | daytime shift | A person who is scheduled for work during daytime hours (for example between 6am and 6pm) on a regular basis.
2 | EMS | early morning shift | Consistent Early morning schedule of 13 hours or less per shift (between 2 am and 2 pm)
2 | ES | evening shift | A person who is scheduled for work during evening hours (for example between 2pm and midnight) on a regular basis.
2 | NS | night shift | Scheduled for work during nighttime hours (for example between 9pm and 8am) on a regular basis.
2 | RSWN | rotating shift with nights | Scheduled for work times that change periodically between days, and/or evenings, and includes some night shifts.
2 | RSWON | rotating shift without nights | Scheduled for work days/times that change periodically between days, but does not include night or evening work.
2 | SS | split shift | Shift consisting of two distinct work periods each day that are separated by a break of a few hours (for example 2 to 4 hours)
2 | VLS | very long shift | Shifts of 17 or more hours.
2 | VS | variable shift | Irregular, unpredictable hours scheduled on a short notice (for example, less than 2 day notice): inconsistent schedule, on-call, as needed, as available.
1 | (_AnnotationValue) Abstract
1 | (_CommonClinicalObservationValue) Abstract | | Description:Used in a patient care message to value simple clinical (non-lab) observations.
1 | (_IndividualCaseSafetyReportValueDomains) Abstract | | This domain is established as a parent to a variety of value domains being defined to support the communication of Individual Case Safety Reports to regulatory bodies. Arguably, this aggregation is not taxonomically pure, but the grouping will facilitate the management of these domains.
1 | (_IndicationValue) Abstract | | Indicates the specific observation result which is the reason for the action (prescription, lab test, etc.). E.g. Headache, Ear infection, planned diagnostic image (requiring contrast agent), etc.

4.4.2.1064 HL7 v3 Code System ParticipationMode

URL: https://hl7.org/fhir/R4/v3/ParticipationMode/cs.html
Vocabulary Work Group | Maturity Level: 1 | Draft | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ParticipationMode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ParticipationMode
Version: | 2018-08-12
Name: | v3.ParticipationMode
Title: | v3 Code System ParticipationMode
Definition: | A set of codes specifying the modality by which the Entity playing the Role is participating in the Act. Examples: Physically present, over the telephone, written communication. Rationale: Particularly for author (originator) participants this is used to specify whether the information represented by the act was initially provided verbally, (hand-)written, or electronically. Open Issue: There needs to be a reexamination of the hierarchies as there seems to be some muddling between ELECTRONIC and other concepts that involve electronic communication that are in other hierarchies.
OID: | 2.16.840.1.113883.5.1064 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: Questionnaire Response Mode (Codes describing how the QuestionnaireResponse was populated)

• ValueSet: v3 Code System ParticipationMode (A set of codes specifying the modality by which the Entity playing the Role is participating in the Act. Examples: Physically present, over the telephone, written communication. Rationale: Particularly for author (originator) participants this is used to specify whether the information represented by the act was initially provided verbally, (hand-)written, or electronically. Open Issue: There needs to be a reexamination of the hierarchies as there seems to be some muddling between ELECTRONIC and other concepts that involve electronic communication that are in other hierarchies.)

• ValueSet: Questionnaire Response Mode (Codes describing how the QuestionnaireResponse was populated)

4.4.2.1064.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | ELECTRONIC | electronic data | Participation by non-human-languaged based electronic signal
1 | PHYSICAL | physical presence | Participation by direct action where subject and actor are in the same location. (The participation involves more than communication.)
1 | REMOTE | remote presence | Participation by direct action where subject and actor are in separate locations, and the actions of the actor are transmitted by electronic or mechanical means. (The participation involves more than communication.)
1 | VERBAL | verbal | Participation by voice communication
2 | DICTATE | dictated | Participation by pre-recorded voice. Communication is limited to one direction (from the recorder to recipient).
2 | FACE | face-to-face | Participation by voice communication where parties speak to each other directly.
2 | PHONE | telephone | Participation by voice communication where the voices of the communicating parties are transported over an electronic medium
2 | VIDEOCONF | videoconferencing | Participation by voice and visual communication where the voices and images of the communicating parties are transported over an electronic medium
1 | WRITTEN | written | Participation by human language recorded on a physical material
2 | FAXWRIT | telefax | Participation by text or diagrams printed on paper that have been transmitted over a fax device
2 | HANDWRIT | handwritten | Participation by text or diagrams printed on paper or other recording medium
2 | MAILWRIT | mail | Participation by text or diagrams printed on paper transmitted physically (e.g. by courier service, postal service).
2 | ONLINEWRIT | online written | Participation by text or diagrams submitted by computer network, e.g. online survey.
3 | EMAILWRIT | email | Participation by text or diagrams transmitted over an electronic mail system.
2 | TYPEWRIT | typewritten | Participation by text or diagrams printed on paper or other recording medium where the recording was performed using a typewriter, typesetter, computer or similar mechanism.

4.4.2.1067 HL7 v3 Code System QueryResponse

URL: https://hl7.org/fhir/R4/v3/QueryResponse/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-QueryResponse) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-QueryResponse
Version: | 2018-08-12
Name: | v3.QueryResponse
Title: | v3 Code System QueryResponse
Definition: | A code classifying the general nature of the response to a given query. Includes whether or not data was found, or whether an error occurred.
OID: | 2.16.840.1.113883.5.1067 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System QueryResponse (A code classifying the general nature of the response to a given query. Includes whether or not data was found, or whether an error occurred.)

4.4.2.1067.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | AE | ApplicationError | Query Error. Application Error.
1 | NF | No data found | No errors, but no data was found matching the query request specification.
1 | OK | Data found | Query reponse data found for 1 or more result sets matching the query request specification.
1 | QE | QueryParameterError | QueryError. Problem with input ParmetersError

4.4.2.1068 HL7 v3 Code System RoleStatus

URL: https://hl7.org/fhir/R4/v3/RoleStatus/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-RoleStatus) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-RoleStatus
Version: | 2018-08-12
Name: | v3.RoleStatus
Title: | v3 Code System RoleStatus
Definition: | Codes representing the defined possible states of an Role, as defined by the Role class state machine.
OID: | 2.16.840.1.113883.5.1068 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System RoleStatus (Codes representing the defined possible states of an Role, as defined by the Role class state machine.)

4.4.2.1068.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | normal | normal | The 'typical' state. Excludes "nullified" which represents the termination state of a Role instance that was created in error.
2 | active | active | The state representing the fact that the Entity is currently active in the Role.
2 | cancelled | cancelled | The terminal state resulting from cancellation of the role prior to activation.
2 | pending | pending | The state representing that fact that the role has not yet become active.
2 | suspended | suspended | The state that represents a suspension of the Entity playing the Role. This state is arrived at from the "active" state.
2 | terminated | terminated | The state representing the successful termination of the Role.
1 | nullified | nullified | The state representing the termination of a Role instance that was created in error.

4.4.2.1069 HL7 v3 Code System SetOperator

URL: https://hl7.org/fhir/R4/v3/SetOperator/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-SetOperator) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-SetOperator
Version: | 2018-08-12
Name: | v3.SetOperator
Title: | v3 Code System SetOperator
Definition: | **** MISSING DEFINITIONS ****
OID: | 2.16.840.1.113883.5.1069 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System SetOperator (**** MISSING DEFINITIONS ****)

4.4.2.1069.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_ValueSetOperator) Abstract | | Operations that can be used to associate concepts in a terminology.
2 | E | exclude | Form the set-difference with this value, i.e., exclude this element or set from the resulting set.
2 | I | include | Form the union with this value, i.e., include this element or set in the resulting set.
1 | A | intersect | Form the intersection with the value.
1 | H | convex hull | Form the convex hull with the value. The convex hull is defined over ordered domains and is the smallest contiguous superset (interval) that of all the operand sets.
1 | P | periodic hull | Form the periodic hull with the value. The periodic hull is defined over ordered domains and is the periodic set that contains all contiguous supersets of pairs of intervals generated by the operand periodic intervals.

4.4.2.1070 HL7 v3 Code System substanceAdminSubstitution

URL: https://hl7.org/fhir/R4/v3/substanceAdminSubstitution/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-substanceAdminSubstitution) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-substanceAdminSubstitution
Version: | 2018-08-12
Name: | v3.substanceAdminSubstitution
Title: | v3 Code System substanceAdminSubstitution
Definition: | Identifies what sort of change is permitted or has occurred between the therapy that was ordered and the therapy that was/will be provided.

OID: | 2.16.840.1.113883.5.1070 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System substanceAdminSubstitution (Identifies what sort of change is permitted or has occurred between the therapy that was ordered and the therapy that was/will be provided.)

• ValueSet: V3 Value SetActSubstanceAdminSubstitutionCode (No Description Provided)

4.4.2.1070.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_ActSubstanceAdminSubstitutionCode) Abstract | | Description: Substitution occurred or is permitted with another product that may potentially have different ingredients, but having the same biological and therapeutic effects.
2 | E | equivalent | Description: Substitution occurred or is permitted with another bioequivalent and therapeutically equivalent product.
3 | EC | equivalent composition | Description: Substitution occurred or is permitted with another product that is a: pharmaceutical alternative containing the same active ingredient but is formulated with different salt, ester pharmaceutical equivalent that has the same active ingredient, strength, dosage form and route of administration Examples: Pharmaceutical alternative: Erythromycin Ethylsuccinate for Erythromycin Stearate Pharmaceutical equivalent: Lisonpril for Zestril
4 | BC | brand composition | Description: Substitution occurred or is permitted between equivalent Brands but not Generics Examples: Zestril for Prinivil Coumadin for Jantoven
4 | G | generic composition | Description: Substitution occurred or is permitted between equivalent Generics but not Brands Examples: Lisnopril (Lupin Corp) for Lisnopril (Wockhardt Corp)
3 | TE | therapeutic alternative | Description: Substitution occurred or is permitted with another product having the same therapeutic objective and safety profile. Examples: ranitidine for Tagamet
4 | TB | therapeutic brand | Description: Substitution occurred or is permitted between therapeutically equivalent Brands but not Generics > Examples: Zantac for Tagamet
4 | TG | therapeutic generic | Description: Substitution occurred or is permitted between therapeutically equivalent Generics but not Brands > Examples: Ranitidine for cimetidine
2 | F | formulary | Description: This substitution was performed or is permitted based on formulary guidelines.
2 | N | none | No substitution occurred or is permitted.

4.4.2.1071 HL7 v3 Code System SubstitutionCondition

URL: https://hl7.org/fhir/R4/v3/SubstitutionCondition/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-SubstitutionCondition) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-SubstitutionCondition
Version: | 2018-08-12
Name: | v3.SubstitutionCondition
Title: | v3 Code System SubstitutionCondition
Definition: | Identifies what sort of change is permitted or has occurred between the item that was ordered/requested and the one that was/will be provided.
OID: | 2.16.840.1.113883.5.1071 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System SubstitutionCondition (Identifies what sort of change is permitted or has occurred between the item that was ordered/requested and the one that was/will be provided.)

4.4.2.1071.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_Conditional) Abstract | | Some conditions may be attached to an allowable substitution. An allowable substitution is based on a match to any other attributes that may be specified.
2 | CONFIRM | Confirm first | Confirmation with Contact Person prior to making any substitutions has or will occur.
2 | NOTIFY | Notify first | Notification to the Contact Person, prior to substitution and through normal institutional procedures, has or will be made.
1 | NOSUB | No substitution | Substitution is not permitted.
1 | UNCOND | Unconditional | No conditions are required.

4.4.2.1074 HL7 v3 Code System CodingRationale

URL: https://hl7.org/fhir/R4/v3/CodingRationale/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-CodingRationale) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-CodingRationale
Version: | 2018-08-12
Name: | v3.CodingRationale
Title: | v3 Code System CodingRationale
Definition: | Identifies how to interpret the instance of the code, codeSystem value in a set of translations. Since HL7 (or a government body) may mandate that codes from certain code systems be sent in conformant messages, other synonyms that are sent in the translation set need to be distinguished among the originally captured source, the HL7 specified code, or some future role. When this code is NULL, it indicates that the translation is an undefined type. When valued, this property must contain one of the following values: SRC - Source (or original) code HL7 - HL7 Specified or Mandated SH - both HL7 mandated and the original code (precoordination) There may be additional values added to this value set as we work through the use of codes in messages and determine other Use Cases requiring special interpretation of the translations.
OID: | 2.16.840.1.113883.5.1074 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System CodingRationale (Identifies how to interpret the instance of the code, codeSystem value in a set of translations. Since HL7 (or a government body) may mandate that codes from certain code systems be sent in conformant messages, other synonyms that are sent in the translation set need to be distinguished among the originally captured source, the HL7 specified code, or some future role. When this code is NULL, it indicates that the translation is an undefined type. When valued, this property must contain one of the following values: SRC - Source (or original) code HL7 - HL7 Specified or Mandated SH - both HL7 mandated and the original code (precoordination) There may be additional values added to this value set as we work through the use of codes in messages and determine other Use Cases requiring special interpretation of the translations.)

4.4.2.1074.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | O | originally produced code | Description: Originally produced code.
1 | OR | original and required | Originally produced code, required by the specification describing the use of the coded concept.
1 | P | post-coded | Description: Post-coded from free text source</description>
1 | PR | post-coded and required | Post-coded from free text source, required by the specification describing the use of the coded concept.
1 | R | required | Description: Required standard code for HL7.
1 | HL7 | HL7 Specified or Mandated | HL7 Specified or Mandated
1 | SH | Both HL7 mandated and the original code | Both HL7 mandated and the original code (precoordination)
1 | SRC | Source (or original) code | Source (or original) code

4.4.2.1075 HL7 v3 Code System PatientImportance

URL: https://hl7.org/fhir/R4/v3/PatientImportance/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-PatientImportance) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-PatientImportance
Version: | 2018-08-12
Name: | v3.PatientImportance
Title: | v3 Code System PatientImportance
Definition: | Patient VIP code
OID: | 2.16.840.1.113883.5.1075 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System PatientImportance (Patient VIP code)

4.4.2.1075.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | BM | Board Member | Board member of health care organization
1 | DFM | Physician Family Member | Family member of staff physician
1 | DR | Staff Physician | Member of the health care organization physician staff
1 | FD | Financial Donor | Financial donor to the health care organization
1 | FOR | Foreign Dignitary | Foreign citizen dignitary of interest to the health care organization
1 | GOVT | Government Dignitary | Government dignitary of interest to the organization
1 | SFM | Staff Family Member | Family member of staff member
1 | STF | Staff Member | Staff member of the health care organization
1 | VIP | Very Important Person | Very important person of interest to the health care organization

4.4.2.1076 HL7 v3 Code System ReligiousAffiliation

URL: https://hl7.org/fhir/R4/v3/ReligiousAffiliation/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ReligiousAffiliation) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ReligiousAffiliation
Version: | 2018-08-12
Name: | v3.ReligiousAffiliation
Title: | v3 Code System ReligiousAffiliation
Definition: | Assigment of spiritual faith affiliation
OID: | 2.16.840.1.113883.5.1076 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ReligiousAffiliation (Assigment of spiritual faith affiliation)

4.4.2.1076.1 Code System Content

Release Date: 2018-08-12

Level	Code	Display	Definition
1	1001	Adventist	
1	1002	African Religions	
1	1003	Afro-Caribbean Religions	
1	1004	Agnosticism	
1	1005	Anglican	
1	1006	Animism	
1	1007	Atheism	
1	1008	Babi & Baha'I faiths	
1	1009	Baptist	
1	1010	Bon	
1	1011	Cao Dai	
1	1012	Celticism	
1	1013	Christian (non-Catholic, non-specific)	
1	1014	Confucianism	
1	1015	Cyberculture Religions	
1	1016	Divination	
1	1017	Fourth Way	
1	1018	Free Daism	
1	1019	Gnosis	
1	1020	Hinduism	
1	1021	Humanism	
1	1022	Independent	
1	1023	Islam	
1	1024	Jainism	
1	1025	Jehovah's Witnesses	
1	1026	Judaism	
1	1027	Latter Day Saints	
1	1028	Lutheran	
1	1029	Mahayana	
1	1030	Meditation	
1	1031	Messianic Judaism	
1	1032	Mitraism	
1	1033	New Age	
1	1034	non-Roman Catholic	
1	1035	Occult	
1	1036	Orthodox	
1	1037	Paganism	
1	1038	Pentecostal	
1	1039	Process, The	
1	1040	Reformed/Presbyterian	
1	1041	Roman Catholic Church	
1	1042	Satanism	
1	1043	Scientology	
1	1044	Shamanism	
1	1045	Shiite (Islam)	
1	1046	Shinto	
1	1047	Sikism	
1	1048	Spiritualism	
1	1049	Sunni (Islam)	
1	1050	Taoism	

1 | 1051 | Theravada
1 | 1052 | Unitarian-Universalism
1 | 1053 | Universal Life Church
1 | 1054 | Vajrayana (Tibetan)
1 | 1055 | Veda
1 | 1056 | Voodoo
1 | 1057 | Wicca
1 | 1058 | Yaohushua
1 | 1059 | Zen Buddhism
1 | 1060 | Zoroastrianism
1 | 1061 | Assembly of God
1 | 1062 | Brethren
1 | 1063 | Christian Scientist
1 | 1064 | Church of Christ
1 | 1065 | Church of God
1 | 1066 | Congregational
1 | 1067 | Disciples of Christ
1 | 1068 | Eastern Orthodox
1 | 1069 | Episcopalian
1 | 1070 | Evangelical Covenant
1 | 1071 | Friends
1 | 1072 | Full Gospel
1 | 1073 | Methodist
1 | 1074 | Native American
1 | 1075 | Nazarene
1 | 1076 | Presbyterian
1 | 1077 | Protestant
1 | 1078 | Protestant, No Denomination
1 | 1079 | Reformed
1 | 1080 | Salvation Army
1 | 1081 | Unitarian Universalist
1 | 1082 | United Church of Christ

4.4.2.1077 HL7 v3 Code System EducationLevel

URL: https://hl7.org/fhir/R4/v3/EducationLevel/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-EducationLevel) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-EducationLevel
Version: | 2018-08-12
Name: | v3.EducationLevel
Title: | v3 Code System EducationLevel
Definition: | Years of education that a person has completed
OID: | 2.16.840.1.113883.5.1077 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System EducationLevel (Years of education that a person has completed)

4.4.2.1077.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | ASSOC | Associate's or technical degree complete | Associate's or technical degree complete
1 | BD | College or baccalaureate degree complete | College or baccalaureate degree complete
1 | ELEM | Elementary School | Elementary School
1 | GD | Graduate or professional Degree complete | Graduate or professional Degree complete
1 | HS | High School or secondary school degree complete | High School or secondary school degree complete
1 | PB | Some post-baccalaureate education | Some post-baccalaureate education
1 | POSTG | Doctoral or post graduate education | Doctoral or post graduate education
1 | SCOL | Some College education | Some College education
1 | SEC | Some secondary or high school education | Some secondary or high school education

4.4.2.1080 HL7 v3 Code System Dentition

URL: https://hl7.org/fhir/R4/v3/Dentition/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-Dentition) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-Dentition
Version: | 2018-08-12
Name: | v3.Dentition
Title: | v3 Code System Dentition
Definition: | **** MISSING DEFINITIONS ****
OID: | 2.16.840.1.113883.5.1080 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System Dentition (**** MISSING DEFINITIONS ****)

4.4.2.1080.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | ArtificialDentition | Artificial dentition | Artificial dentition, artificial subsitutes for the natural dentition
2 | TID10a | 10a | Maxillary left lateral incisor abutment
2 | TID10i | 10i | Maxillary left lateral incisor implant
2 | TID10p | 10p | Maxillary left lateral incisor prosthesis
2 | TID10pd | 10pd | Maxillary left lateral incisor distal prosthesis
2 | TID10pm | 10pm | Maxillary left lateral incisor mesial prosthesis
2 | TID11a | 11a | Maxillary left canine abutment
2 | TID11i | 11i | Maxillary left canine implant
2 | TID11p | 11p | Maxillary left canine prosthesis
2 | TID11pd | 11pd | Maxillary left canine distal prosthesis
2 | TID11pm | 11pm | Maxillary left canine mesial prosthesis
2 | TID12a | 12a | Maxillary left first premolar abutment
2 | TID12i | 12i | Maxillary left first premolar implant
2 | TID12p | 12p | Maxillary left first premolar prosthesis
2 | TID12pd | 12pd | Maxillary left first premolar distal prosthesis
2 | TID12pm | 12pm | Maxillary left first premolar mesial prosthesis
2 | TID13a | 13a | Maxillary left second premolar abutment
2 | TID13i | 13i | Maxillary left second premolar implant
2 | TID13p | 13p | Maxillary left second premolar prosthesis
2 | TID13pd | 13pd | Maxillary left second premolar distal prosthesis
2 | TID13pm | 13pm | Maxillary left second premolar mesial prosthesis
2 | TID14a | 14a | Maxillary left first molar abutment
2 | TID14i | 14i | Maxillary left first molar implant
2 | TID14p | 14p | Maxillary left first molar prosthesis
2 | TID14pd | 14pd | Maxillary left first molar distal prosthesis
2 | TID14pm | 14pm | Maxillary left first molar mesial prosthesis
2 | TID15a | 15a | Maxillary left second molar abutment
2 | TID15i | 15i | Maxillary left second molar implant
2 | TID15p | 15p | Maxillary left second molar prosthesis
2 | TID15pd | 15pd | Maxillary left second molar distal prosthesis
2 | TID15pm | 15pm | Maxillary left second molar mesial prosthesis
2 | TID16a | 16a | Maxillary left third molar abutment
2 | TID16i | 16i | Maxillary left third molar implant
2 | TID16p | 16p | Maxillary left third molar prosthesis
2 | TID16pd | 16pd | Maxillary left third molar distal prosthesis
2 | TID16pm | 16pm | Maxillary left third molar mesial prosthesis
2 | TID17a | 17a | Mandibular left third molar abutment
2 | TID17ad | 17ad | Mandibular left third molar abutment distal hemisection
2 | TID17am | 17am | Mandibular left third molar abutment mesial hemisection
2 | TID17i | 17i | Mandibular left third molar implant
2 | TID17id | 17id | Mandibular left third molar implant distal hemisection
2 | TID17im | 17im | Mandibular left third molar implant mesial hemisection
2 | TID17p | 17p | Mandibular left third molar prosthesis
2 | TID17pd | 17pd | Mandibular left third molar distal prosthesis
2 | TID17pm | 17pm | Mandibular left third molar mesial prosthesis
2 | TID18a | 18a | Mandibular left second molar abutment
2 | TID18ad | 18ad | Mandibular left second molar abutment distal hemisection
2 | TID18am | 18am | Mandibular left second molar abutment mesial hemisection
2 | TID18i | 18i | Mandibular left second molar implant
2 | TID18id | 18id | Mandibular left second molar implant distal hemisection
2 | TID18im | 18im | Mandibular left second molar implant mesial hemisection
2 | TID18p | 18p | Mandibular left second molar prosthesis
2 | TID18pd | 18pd | Mandibular left second molar distal prosthesis
2 | TID18pm | 18pm | Mandibular left second molar mesial prosthesis
2 | TID19a | 19a | Mandibular left first molar abutment
2 | TID19ad | 19ad | Mandibular left first molar abutment distal hemisection
2 | TID19am | 19am | Mandibular left first molar abutment mesial hemisection
2 | TID19i | 19i | Mandibular left first molar implant

2 | TID19id | 19id | Mandibular left first molar implant distal hemisection
2 | TID19im | 19im | Mandibular left first molar implant mesial hemisection
2 | TID19p | 19p | Mandibular left first molar prosthesis
2 | TID19pd | 19pd | Mandibular left first molar distal prosthesis
2 | TID19pm | 19pm | Mandibular left first molar mesial prosthesis
2 | TID1a | 1a | Maxillary right third molar abutment
2 | TID1i | 1i | Maxillary right third molar implant
2 | TID1p | 1p | Maxillary right third molar prosthesis
2 | TID1pd | 1pd | Maxillary right third molar distal prosthesis
2 | TID1pm | 1pm | Maxillary right third molar mesial prosthesis
2 | TID20a | 20a | Mandibular left second premolar abutment
2 | TID20i | 20i | Mandibular left second premolar implant
2 | TID20p | 20p | Mandibular left second premolar prosthesis
2 | TID20pd | 20pd | Mandibular left second premolar distal prosthesis
2 | TID20pm | 20pm | Mandibular left second premolar mesial prosthesis
2 | TID21a | 21a | Mandibular left first premolar abutment
2 | TID21i | 21i | Mandibular left first premolar implant
2 | TID21p | 21p | Mandibular left first premolar prosthesis
2 | TID21pd | 21pd | Mandibular left first premolar distal prosthesis
2 | TID21pm | 21pm | Mandibular left first premolar mesial prosthesis
2 | TID22a | 22a | Mandibular left canine abutment
2 | TID22i | 22i | Mandibular left canine implant
2 | TID22p | 22p | Mandibular left canine prosthesis
2 | TID22pd | 22pd | Mandibular left canine distal prosthesis
2 | TID22pm | 22pm | Mandibular left canine mesial prosthesis
2 | TID23a | 23a | Mandibular left lateral incisor abutment
2 | TID23i | 23i | Mandibular left lateral incisor implant
2 | TID23p | 23p | Mandibular left lateral incisor prosthesis
2 | TID23pd | 23pd | Mandibular left lateral incisor distal prosthesis
2 | TID23pm | 23pm | Mandibular left lateral incisor mesial prosthesis
2 | TID24a | 24a | Mandibular left central incisor abutment
2 | TID24i | 24i | Mandibular left central incisor implant
2 | TID24p | 24p | Mandibular left central incisor prosthesis
2 | TID24pd | 24pd | Mandibular left central incisor distal prosthesis
2 | TID24pm | 24pm | Mandibular left central incisor mesial prosthesis
2 | TID25a | 25a | Mandibular right central incisor abutment
2 | TID25i | 25i | Mandibular right central incisor implant
2 | TID25p | 25p | Mandibular right central incisor prosthesis
2 | TID25pd | 25pd | Mandibular right central incisor distal prosthesis
2 | TID25pm | 25pm | Mandibular right central incisor mesial prosthesis
2 | TID26a | 26a | Mandibular right lateral incisor abutment
2 | TID26i | 26i | Mandibular right lateral incisor implant
2 | TID26p | 26p | Mandibular right lateral incisor prosthesis
2 | TID26pd | 26pd | Mandibular right lateral incisor distal prosthesis
2 | TID26pm | 26pm | Mandibular right lateral incisor mesial prosthesis
2 | TID27a | 27a | Mandibular right canine abutment
2 | TID27i | 27i | Mandibular right canine implant
2 | TID27p | 27p | Mandibular right canine prosthesis
2 | TID27pd | 27pd | Mandibular right canine distal prosthesis
2 | TID27pm | 27pm | Mandibular right canine mesial prosthesis
2 | TID28a | 28a | Mandibular right first premolar abutment
2 | TID28i | 28i | Mandibular right first premolar implant
2 | TID28p | 28p | Mandibular right first premolar prosthesis
2 | TID28pd | 28pd | Mandibular right first premolar distal prosthesis
2 | TID28pm | 28pm | Mandibular right first premolar mesial prosthesis
2 | TID29a | 29a | Mandibular right second premolar abutment
2 | TID29i | 29i | Mandibular right second premolar implant
2 | TID29p | 29p | Mandibular right second premolar prosthesis
2 | TID29pd | 29pd | Mandibular right second premolar distal prosthesis
2 | TID29pm | 29pm | Mandibular right second premolar mesial prosthesis
2 | TID2a | 2a | Maxillary right second molar abutment
2 | TID2i | 2i | Maxillary right second molar implant
2 | TID2p | 2p | Maxillary right second molar prosthesis
2 | TID2pd | 2pd | Maxillary right second molar distal prosthesis
2 | TID2pm | 2pm | Maxillary right second molar mesial prosthesis
2 | TID30a | 30a | Mandibular right first molar abutment
2 | TID30ad | 30ad | Mandibular right first molar abutment distal hemisection
2 | TID30am | 30am | Mandibular right first molar abutment mesial hemisection
2 | TID30i | 30i | Mandibular right first molar implant
2 | TID30id | 30id | Mandibular right first molar implant distal hemisection
2 | TID30im | 30im | Mandibular right first molar implant mesial hemisection
2 | TID30p | 30p | Mandibular right first molar prosthesis

2 | TID30pd | 30pd | Mandibular right first molar distal prosthesis
2 | TID30pm | 30pm | Mandibular right first molar mesial prosthesis
2 | TID31a | 31a | Mandibular right second molar abutment
2 | TID31ad | 31ad | Mandibular right second molar abutment distal hemisection
2 | TID31am | 31am | Mandibular right second molar abutment mesial hemisection
2 | TID31i | 31i | Mandibular right second molar implant
2 | TID31id | 31id | Mandibular right second molar implant distal hemisection
2 | TID31im | 31im | Mandibular right second molar implant mesial hemisection
2 | TID31p | 31p | Mandibular right second molar prosthesis
2 | TID31pd | 31pd | Mandibular right second molar distal prosthesis
2 | TID31pm | 31pm | Mandibular right second molar mesial prosthesis
2 | TID32a | 32a | Mandibular right third molar abutment
2 | TID32ad | 32ad | Mandibular right third molar abutment distal hemisection
2 | TID32am | 32am | Mandibular right third molar abutment mesial hemisection
2 | TID32i | 32i | Mandibular right third molar implant
2 | TID32id | 32id | Mandibular right third molar implant distal hemisection
2 | TID32im | 32im | Mandibular right third molar implant mesial hemisection
2 | TID32p | 32p | Mandibular right third molar prosthesis
2 | TID32pd | 32pd | Mandibular right third molar distal prosthesis
2 | TID32pm | 32pm | Mandibular right third molar mesial prosthesis
2 | TID3a | 3a | Maxillary right first molar abutment
2 | TID3i | 3i | Maxillary right first molar implant
2 | TID3p | 3p | Maxillary right first molar prosthesis
2 | TID3pd | 3pd | Maxillary right first molar distal prosthesis
2 | TID3pm | 3pm | Maxillary right first molar mesial prosthesis
2 | TID4a | 4a | Maxillary right second premolar abutment
2 | TID4i | 4i | Maxillary right second premolar implant
2 | TID4p | 4p | Maxillary right second premolar prosthesis
2 | TID4pd | 4pd | Maxillary right second premolar distal prosthesis
2 | TID4pm | 4pm | Maxillary right second premolar mesial prosthesis
2 | TID5a | 5a | Maxillary right first premolar abutment
2 | TID5i | 5i | Maxillary right first premolar implant
2 | TID5p | 5p | Maxillary right first premolar prosthesis
2 | TID5pd | 5pd | Maxillary right first premolar distal prosthesis
2 | TID5pm | 5pm | Maxillary right first premolar mesial prosthesis
2 | TID6a | 6a | Maxillary right canine abutment
2 | TID6i | 6i | Maxillary right canine implant
2 | TID6p | 6p | Maxillary right canine prosthesis
2 | TID6pd | 6pd | Maxillary right canine distal prosthesis
2 | TID6pm | 6pm | Maxillary right canine mesial prosthesis
2 | TID7a | 7a | Maxillary right lateral incisor abutment
2 | TID7i | 7i | Maxillary right lateral incisor implant
2 | TID7p | 7p | Maxillary right lateral incisor prosthesis
2 | TID7pd | 7pd | Maxillary right lateral incisor distal prosthesis
2 | TID7pm | 7pm | Maxillary right lateral incisor mesial prosthesis
2 | TID8a | 8a | Maxillary right central incisor abutment
2 | TID8i | 8i | Maxillary right central incisor implant
2 | TID8p | 8p | Maxillary right central incisor prosthesis
2 | TID8pd | 8pd | Maxillary right central incisor distal prosthesis
2 | TID8pm | 8pm | Maxillary right central incisor mesial prosthesis
2 | TID9a | 9a | Maxillary left central incisor abutment
2 | TID9i | 9i | Maxillary left central incisor implant
2 | TID9p | 9p | Maxillary left central incisor prosthesis
2 | TID9pd | 9pd | Maxillary left central incisor distal prosthesis
2 | TID9pm | 9pm | Maxillary left central incisor mesial prosthesis
1 | PermanentDentition | Permanent dentition | Permanent dentition, the natural teeth of adulthood that replace or are added to the deciduous teeth
2 | TID1 | 1 | Maxillary right third molar
2 | TID10 | 10 | Maxillary left lateral incisor
2 | TID11 | 11 | Maxillary left canine
2 | TID12 | 12 | Maxillary left first premolar
2 | TID13 | 13 | Maxillary left second premolar
2 | TID14 | 14 | Maxillary left first molar
2 | TID15 | 15 | Maxillary left second molar
2 | TID16 | 16 | Maxillary left third molar
2 | TID17 | 17 | Mandibular left third molar
2 | TID17d | 17d | Mandibular left third molar distal hemisection
2 | TID17m | 17m | Mandibular left third molar mesial hemisection
2 | TID18 | 18 | Mandibular left second molar
2 | TID18d | 18d | Mandibular left second molar distal hemisection
2 | TID18m | 18m | Mandibular left second molar mesial hemisection
2 | TID19 | 19 | Mandibular left first molar

2 | TID19d | 19d | Mandibular left first molar distal hemisection
2 | TID19m | 19m | Mandibular left first molar mesial hemisection
2 | TID2 | 2 | Maxillary right second molar
2 | TID20 | 20 | Mandibular left second premolar
2 | TID21 | 21 | Mandibular left first premolar
2 | TID22 | 22 | Mandibular left canine
2 | TID23 | 23 | Mandibular left lateral incisor
2 | TID24 | 24 | Mandibular left central incisor
2 | TID25 | 25 | Mandibular right central incisor
2 | TID26 | 26 | Mandibular right lateral incisor
2 | TID27 | 27 | Mandibular right canine
2 | TID28 | 28 | Mandibular right first premolar
2 | TID29 | 29 | Mandibular right second premolar
2 | TID3 | 3 | Maxillary right first molar
2 | TID30 | 30 | Mandibular right first molar
2 | TID30d | 30d | Mandibular right first molar distal hemisection
2 | TID30m | 30m | Mandibular right first molar mesial hemisection
2 | TID31 | 31 | Mandibular right second molar
2 | TID31d | 31d | Mandibular right second molar distal hemisection
2 | TID31m | 31m | Mandibular right second molar mesial hemisection
2 | TID32 | 32 | Mandibular right third molar
2 | TID32d | 32d | Mandibular right third molar distal hemisection
2 | TID32m | 32m | Mandibular right third molar mesial hemisection
2 | TID4 | 4 | Maxillary right second premolar
2 | TID5 | 5 | Maxillary right first premolar
2 | TID6 | 6 | Maxillary right canine
2 | TID7 | 7 | Maxillary right lateral incisor
2 | TID8 | 8 | Maxillary right central incisor
2 | TID9 | 9 | Maxillary left central incisor
1 | PrimaryDentition | Primary dentition | Primary dentition, the first teeth to errupt and usually replaced with permanent dentition
2 | TIDA | A | Maxillary right second primary molar
2 | TIDB | B | Maxillary right first primary molar
2 | TIDC | C | Maxillary right primary canine
2 | TIDD | D | Maxillary right lateral primary incisor
2 | TIDE | E | Maxillary right central primary incisor
2 | TIDF | F | Maxillary left central primary incisor
2 | TIDG | G | Maxillary left lateral primary incisor
2 | TIDH | H | Maxillary left primary canine
2 | TIDI | I | Maxillary left first primary molar
2 | TIDJ | J | Maxillary left second primary molar
2 | TIDK | K | Mandibular left second primary molar
2 | TIDL | L | Mandibular left first primary molar
2 | TIDM | M | Mandibular left primary canine
2 | TIDN | N | Mandibular left lateral primary incisor
2 | TIDO | O | Mandibular left central primary incisor
2 | TIDP | P | Mandibular right central primary incisor
2 | TIDQ | Q | Mandibular right lateral primary incisor
2 | TIDR | R | Mandibular right primary canine
2 | TIDS | S | Mandibular right first primary molar
2 | TIDT | T | Mandibular right second primary molar
1 | SupernumeraryTooth | Supernumerary Tooth | Supernumerary tooth, any tooth in addition to the normal permanent and primary dentition
2 | TID10s | 10s | Supernumerary maxillary left lateral incisor
2 | TID11s | 11s | Supernumerary maxillary left canine
2 | TID12s | 12s | Supernumerary maxillary left first premolar
2 | TID13s | 13s | Supernumerary maxillary left second premolar
2 | TID14s | 14s | Supernumerary maxillary left first molar
2 | TID15s | 15s | Supernumerary maxillary left second molar
2 | TID16s | 16s | Supernumerary maxillary left third molar
2 | TID17s | 17s | Supernumerary mandibular left third molar
2 | TID18s | 18s | Supernumerary mandibular left second molar
2 | TID19s | 19s | Supernumerary mandibular left first molar
2 | TID1s | 1s | Supernumerary maxillary right third molar
2 | TID20s | 20s | Supernumerary mandibular left second premolar
2 | TID21s | 21s | Supernumerary mandibular left first premolar
2 | TID22s | 22s | Supernumerary mandibular left canine
2 | TID23s | 23s | Supernumerary mandibular left lateral incisor
2 | TID24s | 24s | Supernumerary mandibular left central incisor
2 | TID25s | 25s | Supernumerary mandibular right central incisor
2 | TID26s | 26s | Supernumerary mandibular right lateral incisor
2 | TID27s | 27s | Supernumerary mandibular right canine
2 | TID28s | 28s | Supernumerary mandibular right first premolar

2 | TID29s | 29s | Supernumerary mandibular right second premolar
2 | TID2s | 2s | Supernumerary maxillary right second molar
2 | TID30s | 30s | Supernumerary mandibular right first molar
2 | TID31s | 31s | Supernumerary mandibular right second molar
2 | TID32s | 32s | Supernumerary mandibular right third molar
2 | TID3s | 3s | Supernumerary maxillary right first molar
2 | TID4s | 4s | Supernumerary maxillary right second premolar
2 | TID5s | 5s | Supernumerary maxillary right first premolar
2 | TID6s | 6s | Supernumerary maxillary right canine
2 | TID7s | 7s | Supernumerary maxillary right lateral incisor
2 | TID8s | 8s | Supernumerary maxillary right central incisor
2 | TID9s | 9s | Supernumerary maxillary left central incisor
2 | TIDAs | As | Supernumerary maxillary right second primary molar
2 | TIDBs | Bs | Supernumerary maxillary right first primary molar
2 | TIDCs | Cs | Supernumerary maxillary right primary canine
2 | TIDDs | Ds | Supernumerary maxillary right lateral primary incisor
2 | TIDEs | Es | Supernumerary maxillary right central primary incisor
2 | TIDFs | Fs | Supernumerary maxillary left central primary incisor
2 | TIDGs | Gs | Supernumerary maxillary left lateral primary incisor
2 | TIDHs | Hs | Supernumerary maxillary left primary canine
2 | TIDIs | Is | Supernumerary maxillary left first primary molar
2 | TIDJs | Js | Supernumerary maxillary left second primary molar
2 | TIDKs | Ks | Supernumerary mandibular left second primary molar
2 | TIDLs | Ls | Supernumerary mandibular left first primary molar
2 | TIDMs | Ms | Supernumerary mandibular left primary canine
2 | TIDNs | Ns | Supernumerary mandibular left lateral primary incisor
2 | TIDOs | Os | Supernumerary mandibular left central primary incisor
2 | TIDPs | Ps | Supernumerary mandibular right central primary incisor
2 | TIDQs | Qs | Supernumerary mandibular right lateral primary incisor
2 | TIDRs | Rs | Supernumerary mandibular right primary canine
2 | TIDSs | Ss | Supernumerary mandibular right first primary molar
2 | TIDTs | Ts | Supernumerary mandibular right second primary molar

4.4.2.1082 HL7 v3 Code System AcknowledgementDetailType

URL: https://hl7.org/fhir/R4/v3/AcknowledgementDetailType/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-AcknowledgementDetailType) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-AcknowledgementDetailType
Version: | 2018-08-12
Name: | v3.AcknowledgementDetailType
Title: | v3 Code System AcknowledgementDetailType
Definition: | A code identifying the specific message to be provided. Discussion: A textual value may be specified as the print name, or for non-coded messages, as the original text. Examples: 'Required attribute xxx is missing', 'System will be unavailable March 19 from 0100 to 0300'
OID: | 2.16.840.1.113883.5.1082 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System AcknowledgementDetailType (A code identifying the specific message to be provided. Discussion: A textual value may be specified as the print name, or for non-coded messages, as the original text. Examples: 'Required attribute xxx is missing', 'System will be unavailable March 19 from 0100 to 0300')

4.4.2.1082.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | E | Error | Definition:An issue which has prevented, or will prevent (unless a management is provided for the issue by the sender), the successful processing of an interaction. Response interactions which include an issue which is an Error are a 'rejection', indicating that the request was not successfully processed. Example:Unable to find specified patient.
1 | I | Information | Definition: The message relates to an issue which has no bearing on the successful processing of the request. Information issues cannot be overridden by specifying a management. Example: A Patient's coverage will expire in 5 days.
1 | W | Warning | Definition: The message relates to an issue which cannot prevent the successful processing of a request, but which could result in the processing not having the ideal or intended effect. Managing a warning issue is not required for successful processing, but will suppress the warning from being raised. Example: Unexpected additional repetitions of phone number have been ignored.
1 | ERR
1 | INFO

1 | WARN

4.4.2.1083 HL7 v3 Code System MessageWaitingPriority

URL: https://hl7.org/fhir/R4/v3/MessageWaitingPriority/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-MessageWaitingPriority) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-MessageWaitingPriority
Version: | 2018-08-12
Name: | v3.MessageWaitingPriority
Title: | v3 Code System MessageWaitingPriority
Definition: | Indicates that the receiver has messages for the sender OpenIssue: Description does not make sense relative to name of coding system. Must be reviewed and improved.
OID: | 2.16.840.1.113883.5.1083 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System MessageWaitingPriority (Indicates that the receiver has messages for the sender OpenIssue: Description does not make sense relative to name of coding system. Must be reviewed and improved.)

4.4.2.1083.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | H | High | High priority messages are available
1 | L | Low | Low priority messages are available
1 | M | Medium | Medium priority messages are available

4.4.2.1092 HL7 v3 Code System hl7ITSVersionCode

URL: https://hl7.org/fhir/R4/v3/hl7ITSVersionCode/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-hl7ITSVersionCode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-hl7ITSVersionCode
Version: | 2018-08-12
Name: | v3.hl7ITSVersionCode
Title: | v3 Code System hl7ITSVersionCode
Definition: | HL7 implementation technology specification versions. These codes will document the ITS type and version for message encoding. The code will appear in the instances based upon rules expressed in the ITS, and do not appear in the abstract message, either as it is presented to received from the ITS.
OID: | 2.16.840.1.113883.5.1092 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System hl7ITSVersionCode (HL7 implementation technology specification versions. These codes will document the ITS type and version for message encoding. The code will appear in the instances based upon rules expressed in the ITS, and do not appear in the abstract message, either as it is presented to received from the ITS.)

4.4.2.1092.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | XMLV1PR1 | XMLV1PR1 | XML ITS version 1 pre-release 1.

4.4.2.1095 HL7 v3 Code System styleType

URL: https://hl7.org/fhir/R4/v3/styleType/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-styleType) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-styleType
Version: | 2018-08-12
Name: | v3.styleType

Title: | v3 Code System styleType
Definition: | <ns1:p>The style code is used within the CDA/SPL narrative block to give the instance author some control over various aspects of style</ns1:p>
OID: | 2.16.840.1.113883.5.1095 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System styleType (<ns1:p>The style code is used within the CDA/SPL narrative block to give the instance author some control over various aspects of style</ns1:p>)

4.4.2.1095.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_FontStyle) Abstract | | Defines font rendering characteristics
2 | bold | Bold Font | Render with a bold font
2 | emphasis | Emphasised Font | Render with with some type of emphasis
2 | italics | Italics Font | Render italicized
2 | underline | Underline Font | Render with an underline font
1 | (_ListStyle) Abstract | | Defines list rendering characteristics
2 | (_OrderedListStyle) Abstract | | Defines rendering characteristics for ordered lists
3 | Arabic | Arabic | List is ordered using Arabic numerals: 1, 2, 3
3 | BigAlpha | Big Alpha | List is ordered using big alpha characters: A, B, C
3 | BigRoman | Big Roman | List is ordered using big Roman numerals: I, II, III
3 | LittleAlpha | Little Alpha | List is order using little alpha characters: a, b, c
3 | LittleRoman | Little Roman | List is ordered using little Roman numerals: i, ii, iii
2 | (_UnorderedListStyle) Abstract | | Defines rendering characteristics for unordered lists
3 | Circle | Circle | List bullets are hollow discs
3 | Disc | Disc | List bullets are simple solid discs
3 | Square | Square | List bullets are solid squares
1 | (_TableRuleStyle) Abstract | | Defines table cell rendering characteristics
2 | Botrule | Bottom Rule | Render cell with rule on bottom
2 | Lrule | Left-sided rule | Render cell with left-sided rule
2 | Rrule | Right-sided rule | Render cell with right-sided rule
2 | Toprule | Top Rule | Render cell with rule on top

4.4.2.1096 HL7 v3 Code System QueryParameterValue

URL: https://hl7.org/fhir/R4/v3/QueryParameterValue/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-QueryParameterValue) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-QueryParameterValue
Version: | 2018-08-12
Name: | v3.QueryParameterValue
Title: | v3 Code System QueryParameterValue
Definition: | The domain of coded values used as parameters within QueryByParameter queries.
OID: | 2.16.840.1.113883.5.1096 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System QueryParameterValue (The domain of coded values used as parameters within QueryByParameter queries.)

4.4.2.1096.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_DispenseQueryFilterCode) Abstract | | Description:Filter codes used to manage volume of dispenses returned by a parameter-based queries.
2 | ALLDISP | all dispenses | Description:Returns all dispenses to date for a prescription.
2 | LASTDISP | last dispense | Description:Returns the most recent dispense for a prescription.
2 | NODISP | no dispense | Description:Returns no dispense for a prescription.
1 | (_OrderFilterCode) Abstract | | Filter codes used to manage types of orders being returned by a parameter-based query.
2 | AO | all orders | Return all orders.
2 | ONR | orders without results | Return only those orders that do not have results.
2 | OWR | orders with results | Return only those orders that have results.
1 | (_PrescriptionDispenseFilterCode) Abstract | | A "helper" vocabulary used to construct complex query filters based on how and whether a prescription has been dispensed.
2 | C | Completely dispensed | Filter to only include SubstanceAdministration orders which have no remaining quantity authorized to be dispensed.

2 | N | Never Dispensed | Filter to only include SubstanceAdministration orders which have no fulfilling supply events performed.
2 | R | Dispensed with remaining fills | Filter to only include SubstanceAdministration orders which have had at least one fulfilling supply event, but which still have outstanding quantity remaining to be authorized.
1 | (_QueryParameterValue) Abstract | | Description:Indicates how result sets should be filtered based on whether they have associated issues.
2 | ISSFA | all | Description:Result set should not be filtered based on the presence of issues.
2 | ISSFI | with issues | Description:Result set should be filtered to only include records with associated issues.
2 | ISSFU | with unmanaged issues | Description:Result set should be filtered to only include records with associated unmanaged issues.

4.4.2.1097 HL7 v3 Code System HL7StandardVersionCode

URL: https://hl7.org/fhir/R4/v3/HL7StandardVersionCode/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-HL7StandardVersionCode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-HL7StandardVersionCode
Version: | 2018-08-12
Name: | v3.HL7StandardVersionCode
Title: | v3 Code System HL7StandardVersionCode
Definition: | This code system holds version codes for the Version 3 standards. Values are to be determined by HL7 and added with each new version of the HL7 Standard.
OID: | 2.16.840.1.113883.5.1097 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System HL7StandardVersionCode (This code system holds version codes for the Version 3 standards. Values are to be determined by HL7 and added with each new version of the HL7 Standard.)

4.4.2.1097.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | Ballot2008Jan | Ballot 2008 January | The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in January 2008.
1 | Ballot2008May | Ballot 2008 May | The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in May 2008.
1 | Ballot2008Sep | Ballot 2008 September | The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in September 2008.
1 | Ballot2009Jan | Ballot 2009 January | The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in January 2009.
1 | Ballot2009May | Ballot 2009 May | The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in May 2009.
1 | Ballot2009Sep | Ballot 2009 September | The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in September 2009.
1 | Ballot2010Jan | Ballot 2010 Jan | Definition: The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in January 2010.
1 | Ballot2010May | Ballot 2010 May | Definition: The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in May 2010.
1 | Ballot2010Sep | Ballot 2010 Sep | Definition: The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in September 2010.
1 | Ballot2011Jan | Ballot 2011 Jan | Definition: The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in January 2011.
1 | Ballot2011May | Ballot 2011 May | Definition: The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in May 2011.
1 | Ballot2011Sep | Ballot 2011 Sep | Definition: The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in September 2011.
1 | Ballot2012Jan | Ballot 2012 Jan | Definition: The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in January 2012.
1 | Ballot2012May | Ballot 2012 May | Definition: The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in May 2012.
1 | Ballot2012Sep | Ballot 2012 Sep | Definition: The complete set of normative, DSTU, proposed (under ballot) and draft artifacts as published in the ballot whose ballot cycle ended in September 2012.
1 | V3-2003-12 | HL7 Version V3-2003-12 | The consistent set of messaging artefacts as published or contained in repositories in December of 2003, based on the latest version of any V3 models or artefacts (RIM, Datatypes, CMETS, Common Messages, Vocabularies) as available in December of 2003. Note: This versioncode does not cover the version of the XML ITS.
1 | V3-2005N | 2005 Normative Edition | Description:The consistent set of normative and DSTU messaging artifacts as published or contained in repositories in December of 2004, based on the latest version of any V3 models or artifacts (RIM, Datatypes, CMETS, Common Messages, Vocabularies) as published under the title of Normative Edition 2005. Note: This versioncode does not cover the version of the XML ITS.
1 | V3-2006N | 2006 Normative Edition | Description:The consistent set of normative and DSTU messaging artifacts as published or contained in repositories in December of 2005, based on the latest version of any V3 models or artifacts (RIM, Datatypes, CMETS,

Common Messages, Vocabularies) as published under the title of Normative Edition 2006. Note: This versioncode does not cover the version of the XML ITS.

1 | V3-2008N | 2008 Normative Edition | Definition: The consistent set of normative and DSTU messaging artifacts as published or contained in repositories in December of 2007, based on the latest version of any V3 models or artifacts (RIM, Data Types, CMETS, Common Messages, Vocabularies) published under the title of Normative Edition 2008. Note: This version code does not cover the version of the XML ITS.

1 | V3-2009N | 2009 Normative Edition | Description: The consistent set of normative and DSTU messaging artifacts as published or contained in repositories in December of 2008, based on the latest version of any V3 models or artifacts (RIM, Data Types, CMETS, Common Messages, Vocabularies) published under the title of Normative Edition 2009. Note: This version code does not cover the version of the XML ITS.

1 | V3-2010N | 2010 Normative Edition | Definition: The consistent set of normative and DSTU messaging artifacts as published or contained in repositories in December of 2009, based on the latest version of any V3 models or artifacts (RIM, Data Types, CMETS, Common Messages, Vocabularies) published under the title of Normative Edition 2010. Note: This version code does not cover the version of the XML ITS.

1 | V3-2011N | 2011 Normative Edition | Definition: The consistent set of normative and DSTU messaging artifacts as published or contained in repositories in December of 2010, based on the latest version of any V3 models or artifacts (RIM, Data Types, CMETS, Common Messages, Vocabularies) published under the title of Normative Edition 2011. Note: This version code does not cover the version of the XML ITS.

1 | V3-2012N | 2012 Normative Edition | Definition: The consistent set of normative and DSTU messaging artifacts as published or contained in repositories in December of 2011, based on the latest version of any V3 models or artifacts (RIM, Data Types, CMETS, Common Messages, Vocabularies) published under the title of Normative Edition 2012. Note: This version code does not cover the version of the XML ITS.

1 | V3PR1 | Version3 Pre-release #1 | Includes all material published as part of the ballot package released for vote in July-August 2003.

1 | V3-2007N | 2007 Normative Edition | Description:The consistent set of normative and DSTU messaging artifacts as published or contained in repositories in December of 2006, based on the latest version of any V3 models or artifacts (RIM, Datatypes, CMETS, Common Messages, Vocabularies) published under the title of Normative Edition 2007. Note: This versioncode does not cover the version of the XML ITS.

4.4.2.1099 HL7 v3 Code System ActRelationshipSubset

URL: https://hl7.org/fhir/R4/v3/ActRelationshipSubset/cs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ActRelationshipSubset) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ActRelationshipSubset

Version: | 2018-08-12

Name: | v3.ActRelationshipSubset

Title: | v3 Code System ActRelationshipSubset

Definition: | <ns1:p>Used to indicate that the target of the relationship will be a filtered subset of the total related set of targets.</ns1:p><ns1:p>Used when there is a need to limit the number of components to the first, the last, the next, the total, the average or some other filtered or calculated subset.</ns1:p>

OID: | 2.16.840.1.113883.5.1099 (for OID based terminology systems)

Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ActRelationshipSubset (<ns1:p>Used to indicate that the target of the relationship will be a filtered subset of the total related set of targets.</ns1:p><ns1:p>Used when there is a need to limit the number of components to the first, the last, the next, the total, the average or some other filtered or calculated subset.</ns1:p>)

4.4.2.1099.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition

1 | (_ParticipationSubset) Abstract | | Used to indicate that the participation is a filtered subset of the total participations of the same type owned by the Act. Used when there is a need to limit the participations to the first, the last, the next or some other filtered subset.

2 | FUTURE | expected future | An occurrence that is scheduled to occur in the future. An Act whose effective time is greater than 'now', where 'now' is the time the instance is authored.

3 | FUTSUM | future summary | Represents a 'summary' of all acts that are scheduled to occur in the future (whose effective time is greater than 'now' where is the time the instance is authored.). The effectiveTime represents the outer boundary of all occurrences, repeatNumber represents the total number of repetitions, etc.

3 | LAST | expected last | Restricted to the latest known occurrence that is scheduled to occur. The Act with the highest known effective time.

3 | NEXT | expected next | Restricted to the nearest recent known occurrence scheduled to occur in the future. The Act with the lowest effective time, still greater than 'now'. ('now' is the time the instance is authored.)

2 | PAST | previous | An occurrence that occurred or was scheduled to occur in the past. An Act whose effective time is less than 'now'. ('now' is the time the instance is authored.)

3 | FIRST | first known | Restricted to the earliest known occurrence that occurred or was scheduled to occur in the past. The Act with the lowest effective time. ('now' is the time the instance is authored.)

3 | PREVSUM | previous summary | Represents a 'summary' of all acts that previously occurred or were scheduled to occur. The effectiveTime represents the outer boundary of all occurrences, repeatNumber represents the total number of repetitions, etc.

('now' is the time the instance is authored.)
3 | RECENT | most recent | Restricted to the most recent known occurrence that occurred or was scheduled to occur in the past. The Act with the most recent effective time, still less than 'now'. ('now' is the time the instance is authored.)
2 | SUM | summary | Represents a 'summary' of all acts that have occurred or were scheduled to occur and which are scheduled to occur in the future. The effectiveTime represents the outer boundary of all occurrences, repeatNumber represents the total number of repetitions, etc.
1 | ActRelationshipExpectedSubset | ActRelationshipExpectedSubset
1 | ActRelationshipPastSubset | ActRelationshipPastSubset
1 | MAX | maximum | The occurrence whose value attribute is greater than all other occurrences at the time the instance is created.
1 | MIN | minimum | The occurrence whose value attribute is less than all other occurrences at the time the instance is created.

4.4.2.1100 HL7 v3 Code System AcknowledgementDetailCode

URL: https://hl7.org/fhir/R4/v3/AcknowledgementDetailCode/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-AcknowledgementDetailCode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-AcknowledgementDetailCode
Version: | 2018-08-12
Name: | v3.AcknowledgementDetailCode
Title: | v3 Code System AcknowledgementDetailCode
Definition: | OpenIssue: Missing description.
OID: | 2.16.840.1.113883.5.1100 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System AcknowledgementDetailCode (OpenIssue: Missing description.)

4.4.2.1100.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_AcknowledgementDetailNotSupportedCode) Abstract | | Refelects rejections because elements of the communication are not supported in the current context.
2 | NS200 | Unsupported interaction | The interaction (or: this version of the interaction) is not supported.
2 | NS202 | Unsupported processing id | The Processing ID is not supported.
2 | NS203 | Unsupported version id | The Version ID is not supported.
2 | NS250 | Unsupported processing Mode | The processing mode is not supported.
2 | NS260 | Unknown sender | The Device.id of the sender is unknown.
2 | NS261 | Unrecognized attentionline | The receiver requires information in the attentionLine classes for routing purposes.
1 | INTERR | Internal system error | An internal software component (database, application, queue mechanism, etc.) has failed, leading to inability to process the message.
1 | NOSTORE | No storage space for message. | Rejection: The message can't be stored by the receiver due to an unspecified internal application issue. The message was neither processed nor stored by the receiving application.
1 | RTEDEST | Message routing error, destination unreachable. | Error: The destination of this message is known to the receiving application. Messages have been successfully routed to that destination in the past. The link to the destination application or an intermediate application is unavailable.
1 | RTUDEST | Error: Message routing error, unknown destination. | The destination of this message is unknown to the receiving application. The receiving application in the message does not match the application which received the message. The message was neither routed, processed nor stored by the receiving application.
1 | RTWDEST | Message routing warning, destination unreachable. | Warning: The destination of this message is known to the receiving application. Messages have been successfully routed to that destination in the past. The link to the destination application or an intermediate application is (temporarily) unavailable. The receiving application will forward the message as soon as the destination can be reached again.
1 | SYN | Syntax error | Reflects errors in the syntax or structure of the communication.
2 | SYN102 | Data type error | The attribute contained data of the wrong data type, e.g. a numeric attribute contained "FOO".
2 | SYN105 | Required element missing | Description: Required association or attribute missing in message; or the sequence of the classes is different than required by the standard or one of the conformance profiles identified in the message.
3 | SYN100 | Required association missing | Required association missing in message; or the sequence of the classes is different than required by the standard or one of the conformance profiles identified in the message.
3 | SYN101 | Required attribute missing | A required attribute is missing in a class.
3 | SYN114 | Insufficient repetitions | Description: The number of repetitions of a group of association or attributes is less than the required minimum for the standard or of one of the conformance profiles or templates identified in the message.
2 | SYN106 | Terminology error | Description: A coded attribute or datatype property violates one of the terminology constraints specified in the standard or one of the conformance profiles or templates declared by the instance.
3 | SYN103 | Value not found in code system | An attribute value was compared against the corresponding code system, and no match was found.
3 | SYN104 | Invalid code system in CNE | An attribute value referenced a code system that is not valid for an attribute constrained to CNE.
3 | SYN107 | Deprecated code | Description: A coded attribute is referencing a code that has been deprecated by the owning code system.

2 | SYN108 | Number of repetitions exceeds limit | Description: The number of repetitions of a (group of) association(s) or attribute(s) exceeds the limits of the standard or of one of the conformance profiles or templates identified in the message.
3 | SYN110 | Number of association repetitions exceeds limit | The number of repetitions of a (group of) association(s) exceeds the limits of the standard or of one of the conformance profiles identified in the message.
3 | SYN112 | Number of attribute repetitions exceeds limit | The number of repetitions of an attribute exceeds the limits of the standard or of one of the conformance profiles identified in the message.
2 | SYN109 | Mandatory element with null value | Description: An attribute or association identified as mandatory in a specification or declared conformance profile or template has been specified with a null flavor.
2 | SYN111 | Value does not match fixed value | Description: The value of an attribute or property differs from the fixed value asserted in the standard or one of the conformance profiles or templates declared in the message.
2 | SYN113 | Formal constraint violation | Description: A formal constraint asserted in the standard or one of the conformance profiles or templates declared in the message has been violated.

4.4.2.1110 HL7 v3 Code System ContentProcessingMode

URL: https://hl7.org/fhir/R4/v3/ContentProcessingMode/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ContentProcessingMode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ContentProcessingMode
Version: | 2018-08-12
Name: | v3.ContentProcessingMode
Title: | v3 Code System ContentProcessingMode
Definition: | Description: Identifies the order in which content should be processed.
OID: | 2.16.840.1.113883.5.1110 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ContentProcessingMode (Description: Identifies the order in which content should be processed.)

4.4.2.1110.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | SEQL | Sequential | Description:The content should be processed in a sequential fashion.
1 | UNOR | Unordered | Description:The content may be processed in any order.

4.4.2.1111 HL7 v3 Code System TransmissionRelationshipTypeCode

URL: https://hl7.org/fhir/R4/v3/TransmissionRelationshipTypeCode/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-TransmissionRelationshipTypeCode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-TransmissionRelationshipTypeCode
Version: | 2018-08-12
Name: | v3.TransmissionRelationshipTypeCode
Title: | v3 Code System TransmissionRelationshipTypeCode
Definition: | Description: A code specifying the meaning and purpose of every TransmissionRelationship instance. Each of its values implies specific constraints to what kinds of Transmission objects can be related and in which way.
OID: | 2.16.840.1.113883.5.1111 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System TransmissionRelationshipTypeCode (Description: A code specifying the meaning and purpose of every TransmissionRelationship instance. Each of its values implies specific constraints to what kinds of Transmission objects can be related and in which way.)

4.4.2.1111.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | SEQL | sequence | Description:A transmission relationship indicating that the source transmission follows the target transmission.

4.4.2.1112 HL7 v3 Code System QueryRequestLimit

URL: https://hl7.org/fhir/R4/v3/QueryRequestLimit/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-QueryRequestLimit) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-QueryRequestLimit
Version: | 2018-08-12
Name: | v3.QueryRequestLimit
Title: | v3 Code System QueryRequestLimit
Definition: | Definition: Defines the units associated with the magnitude of the maximum size limit of a query response that can be accepted by the requesting application.
OID: | 2.16.840.1.113883.5.1112 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System QueryRequestLimit (Definition: Defines the units associated with the magnitude of the maximum size limit of a query response that can be accepted by the requesting application.)

4.4.2.1112.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_QueryRequestLimit) Abstract | | Definition: The number of matching instances (number of focal classes). The document header class is the focal class of a document, a record would therefore be equal to a document.
2 | RD | record | Definition: The number of matching instances (number of focal classes). The document header class is the focal class of a document, a record would therefore be equal to a document.

4.4.2.1113 HL7 v3 Code System ExposureMode

URL: https://hl7.org/fhir/R4/v3/ExposureMode/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ExposureMode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ExposureMode
Version: | 2018-08-12
Name: | v3.ExposureMode
Title: | v3 Code System ExposureMode
Definition: | Code for the mechanism by which the exposure agent was exchanged or potentially exchanged by the participants involved in the exposure.
OID: | 2.16.840.1.113883.5.1113 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ExposureMode (Code for the mechanism by which the exposure agent was exchanged or potentially exchanged by the participants involved in the exposure.)

4.4.2.1113.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_ExposureMode) Abstract | | Code for the mechanism by which the exposure agent was exchanged or potentially exchanged by the participants involved in the exposure.
2 | AIRBORNE | airborne | Description: Communication of an agent from a living subject or environmental source to a living subject through indirect contact via oral or nasal inhalation.
2 | CONTACT | contact | Description: Communication of an agent from a living subject or environmental source to a living subject through direct physical contact.
2 | FOODBORNE | foodborne | Description: Communication of an agent from a food source to a living subject via oral consumption.
2 | WATERBORNE | waterborne | Description: Communication of an agent to a living subject by contact and/or consumption via an aqueous medium

4.4.2.1114 HL7 v3 Code System ActExposureLevelCode

URL: https://hl7.org/fhir/R4/v3/ActExposureLevelCode/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ActExposureLevelCode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ActExposureLevelCode
Version: | 2018-08-12
Name: | v3.ActExposureLevelCode

Title: | v3 Code System ActExposureLevelCode
Definition: | A qualitative measure of the degree of exposure to the causative agent. This includes concepts such as "low", "medium" and "high". This quantifies how the quantity that was available to be administered to the target differs from typical or background levels of the substance.
OID: | 2.16.840.1.113883.5.1114 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ActExposureLevelCode (A qualitative measure of the degree of exposure to the causative agent. This includes concepts such as "low", "medium" and "high". This quantifies how the quantity that was available to be administered to the target differs from typical or background levels of the substance.)

4.4.2.1114.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_ActExposureLevelCode) Abstract | | A qualitative measure of the degree of exposure to the causative agent. This includes concepts such as "low", "medium" and "high". This quantifies how the quantity that was available to be administered to the target differs from typical or background levels of the substance.
2 | HIGH | high | Description: Exposure to an agent at a relatively high level above background.
2 | LOW | low | Description: Exposure to an agent at a relatively low level above background.
2 | MEDIUM | medium | Description: Exposure to an agent at a relatively moderate level above background.A

4.4.2.1116 HL7 v3 Code System IdentifierScope

URL: https://hl7.org/fhir/R4/v3/IdentifierScope/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-IdentifierScope) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-IdentifierScope
Version: | 2018-08-12
Name: | v3.IdentifierScope
Title: | v3 Code System IdentifierScope
Definition: | Description: Codes to specify the scope in which the identifier applies to the object with which it is associated, and used in the datatype property II.
OID: | 2.16.840.1.113883.5.1116 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System IdentifierScope (Description: Codes to specify the scope in which the identifier applies to the object with which it is associated, and used in the datatype property II.)

4.4.2.1116.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | BUSN | Business Identifier | Description: An identifier whose scope is defined by the business practices associated with the object. In contrast to the other scope identifiers, the scope of the use of the id is not necessarily restricted to a single object, but may be reused for other objects closely associated with the object due to business practice.
1 | OBJ | Object Identifier | Description: The identifier associated with a particular object. It remains consistent as the object undergoes state transitions.
1 | VER | Version Identifier | Description: An identifier that references a particular object as it existed at a given point in time. The identifier SHALL change with each state transition on the object. I.e. The version identifier of an object prior to a 'suspend' state transition is distinct from the identifier of the object after the state transition. Each version identifier can be tied to exactly one ControlAct event which brought that version into being (though the control act may never be instantiated). NOTE: Applications that do not support versioning of objects must ignore and not persist these ids to avoid confusion resulting from leaving the same identifier on an object that undergoes changes.
1 | VW | View Specific Identifier | Description: An identifier that references a particular object as it existed at a given point in time. The identifier SHALL change with each state transition on the object. Example The version identifier of an object prior to a 'suspend' state transition is distinct from the identifier of the object after the state transition. Each version identifier can be tied to exactly one ControlAct event which brought that version into being (though the control act may never be instantiated). NOTE: Applications that do not support versioning of objects must ignore and not persist these ids to avoid confusion resulting from leaving the same identifier on an object that undergoes changes.

4.4.2.1117 HL7 v3 Code System IdentifierReliability

URL: https://hl7.org/fhir/R4/v3/IdentifierReliability/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-IdentifierReliability) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-IdentifierReliability

Version: | 2018-08-12
Name: | v3.IdentifierReliability
Title: | v3 Code System IdentifierReliability
Definition: | Specifies the reliability with which the identifier is known. This attribute MAY be used to assist with identifier matching algorithms.
OID: | 2.16.840.1.113883.5.1117 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System IdentifierReliability (Specifies the reliability with which the identifier is known. This attribute MAY be used to assist with identifier matching algorithms.)

4.4.2.1117.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | ISS | Issued by System | Description: The identifier was issued by the system responsible for constructing the instance.
1 | UNV | Unverified by system | Description: The identifier was provided to the system that constructed the instance, but has not been verified. e.g. a Drivers license entered manually into a system by a user.
1 | VRF | Verified by system | Description: The identifier was not issued by the system responsible for constructing the instance, but the system that captured the id has verified the identifier with the issuing authority, or with another system that has verified the identifier.

4.4.2.1118 HL7 v3 Code System TelecommunicationCapabilities

URL: https://hl7.org/fhir/R4/v3/TelecommunicationCapabilities/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-TelecommunicationCapabilities) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-TelecommunicationCapabilities
Version: | 2018-08-12
Name: | v3.TelecommunicationCapabilities
Title: | v3 Code System TelecommunicationCapabilities
Definition: | Description: Concepts that define the telecommunication capabilities of a particular device. Used to identify the expected capabilities to be found at a particular telecommunication address.
OID: | 2.16.840.1.113883.5.1118 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System TelecommunicationCapabilities (Description: Concepts that define the telecommunication capabilities of a particular device. Used to identify the expected capabilities to be found at a particular telecommunication address.)

4.4.2.1118.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | data | data | Description: This device can receive data calls (i.e. modem).
1 | fax | fax | Description: This device can receive faxes.
1 | sms | sms | Description: This device can receive SMS messages.
1 | tty | text | Description: This device is a text telephone.
1 | voice | voice | Description: This device can receive voice calls (i.e. talking to another person, or a recording device, or a voice activated computer).

4.4.2.1119 HL7 v3 Code System AddressUse

URL: https://hl7.org/fhir/R4/v3/AddressUse/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-AddressUse) is defined as part of HL7 v3. Related FHIR content: AddressUse.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-AddressUse
Version: | 2018-08-12
Name: | v3.AddressUse
Title: | v3 Code System AddressUse
Definition: | Codes that provide guidance around the circumstances in which a given address should be used.
OID: | 2.16.840.1.113883.5.1119 (for OID based terminology systems)

Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: PostalAddressUse (Uses of an address not included in Address.use.)

• ValueSet: PostalAddressUse (Uses of an address not included in Address.use.)

• ValueSet: v3 Code System AddressUse (Codes that provide guidance around the circumstances in which a given address should be used.)

4.4.2.1119.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (_GeneralAddressUse) Abstract | | Description: Address uses that can apply to both postal and telecommunication addresses.
2 | BAD | bad address | Description: A flag indicating that the address is bad, in fact, useless.
2 | CONF | confidential address | Description: Indicates that the address is considered sensitive and should only be shared or published in accordance with organizational controls governing patient demographic information with increased sensitivity. Uses of Addresses. Lloyd to supply more complete description.
2 | H | home address | Description: A communication address at a home, attempted contacts for business purposes might intrude privacy and chances are one will contact family or other household members instead of the person one wishes to call. Typically used with urgent cases, or if no other contacts are available.
3 | HP | primary home | Description: The primary home, to reach a person after business hours.
3 | HV | vacation home | Description: A vacation home, to reach a person while on vacation.
2 | OLD | no longer in use | This address is no longer in use. Usage Note: Address may also carry valid time ranges. This code is used to cover the situations where it is known that the address is no longer valid, but no particular time range for its use is known.
2 | TMP | temporary address | Description: A temporary address, may be good for visit or mailing. Note that an address history can provide more detailed information.
2 | WP | work place | Description: An office address. First choice for business related contacts during business hours.
3 | DIR | direct | Description: Indicates a work place address or telecommunication address that reaches the individual or organization directly without intermediaries. For phones, often referred to as a 'private line'.
3 | PUB | public | Description: Indicates a work place address or telecommunication address that is a 'standard' address which may reach a reception service, mail-room, or other intermediary prior to the target entity.
1 | (_PostalAddressUse) Abstract | | Description: Address uses that only apply to postal addresses, not telecommunication addresses.
2 | PHYS | physical visit address | Description: Used primarily to visit an address.
2 | PST | postal address | Description: Used to send mail.
1 | (_TelecommunicationAddressUse) Abstract | | Description: Address uses that only apply to telecommunication addresses, not postal addresses.
2 | AS | answering service | Description: An automated answering machine used for less urgent cases and if the main purpose of contact is to leave a message or access an automated announcement.
2 | EC | emergency contact | Description: A contact specifically designated to be used for emergencies. This is the first choice in emergencies, independent of any other use codes.
2 | MC | mobile contact) | Description: A telecommunication device that moves and stays with its owner. May have characteristics of all other use codes, suitable for urgent matters, not the first choice for routine business.
2 | PG | pager | Description: A paging device suitable to solicit a callback or to leave a very short message.

4.4.2.1120 HL7 v3 Code System EntityNameUseR2

URL: https://hl7.org/fhir/R4/v3/EntityNameUseR2/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-EntityNameUseR2) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-EntityNameUseR2
Version: | 2018-08-12
Name: | v3.EntityNameUseR2
Title: | v3 Code System EntityNameUseR2
Definition: | Description: A set of codes advising a system or user which name in a set of names to select for a given purpose.
OID: | 2.16.840.1.113883.5.1120 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System EntityNameUseR2 (Description: A set of codes advising a system or user which name in a set of names to select for a given purpose.)

4.4.2.1120.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (Assumed) Abstract | | Description:A name that a person has assumed or has been assumed to them.
2 | A | business name | Description:A name used in a Professional or Business context . Examples: Continuing to use a maiden name in a professional context, or using a stage performing name (some of these names are also pseudonyms)
2 | ANON | Anonymous | Description:Anonymous assigned name (used to protect a persons identity for privacy reasons)
2 | I | Indigenous/Tribal | Description:e.g . Chief Red Cloud

2 | P | Other/Pseudonym/Alias | Description:A non-official name by which the person is sometimes known. (This may also be used to record informal names such as a nickname)
2 | R | religious | Description:A name assumed as part of a religious vocation . e.g . Sister Mary Francis, Brother John
1 | C | customary | Description:Known as/conventional/the one you normally use
1 | M | maiden name | Description:A name used prior to marriage.Note that marriage naming customs vary greatly around the world. This name use is for use by applications that collect and store maiden names. Though the concept of maiden name is often gender specific, the use of this term is not gender specific. The use of this term does not imply any particular history for a person's name, nor should the maiden name be determined algorithmically
1 | (NameRepresentationUse) Abstract | | Description:Identifies the different representations of a name . The representation may affect how the name is used . (E.g . use of Ideographic for formal communications.)
2 | ABC | alphabetic | Description:Alphabetic transcription of name in latin alphabet (Japanese: romaji)
2 | IDE | ideographic | Description:Ideographic representation of name (e.g., Japanese kanji, Chinese characters)
2 | SYL | syllabic | Description:Syllabic transcription of name (e.g., Japanese kana, Korean hangul)
1 | OLD | no longer in use | Description:This name is no longer in use (note: Names may also carry valid time ranges . This code is used to cover the situations where it is known that the name is no longer valid, but no particular time range for its use is known)
2 | DN | do not use | Description:This name should no longer be used when interacting with the person (i.e . in addition to no longer being used, the name should not be even mentioned when interacting with the person)Note: applications are not required to compare names labeled "Do Not Use" and other names in order to eliminate name parts that are common between the other name and a name labeled "Do Not Use".
1 | OR | official registry name | Description:The formal name as registered in an official (government) registry, but which name might not be commonly used . May correspond to the concept of legal name
1 | PHON | phonetic | Description:The name as understood by the data enterer, i.e. a close approximation of a phonetic spelling of the name, not based on a phonetic algorithm.
1 | SRCH | search | Description:A name intended for use in searching or matching. This is used when the name is incomplete and contains enough details for search matching, but not enough for other uses.
1 | T | temporary | Description:A temporary name. Note that a name valid time can provide more detailed information. This may also be used for temporary names assigned at birth or in emergency situations.

4.4.2.1121 HL7 v3 Code System EntityNamePartTypeR2

URL: https://hl7.org/fhir/R4/v3/EntityNamePartTypeR2/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-EntityNamePartTypeR2) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-EntityNamePartTypeR2
Version: | 2018-08-12
Name: | v3.EntityNamePartTypeR2
Title: | v3 Code System EntityNamePartTypeR2
Definition: | Description: Indicates whether the name part is a given name, family name, prefix, suffix, etc.
OID: | 2.16.840.1.113883.5.1121 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System EntityNamePartTypeR2 (Description: Indicates whether the name part is a given name, family name, prefix, suffix, etc.)

4.4.2.1121.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | DEL | delimiter | Description:A delimiter has no meaning other than being literally printed in this name representation. A delimiter has no implicit leading and trailing white space.
1 | FAM | family | Description:Family name, this is the name that links to the genealogy. In some cultures (e.g. Eritrea) the family name of a son is the first name of his father.
1 | GIV | given | Description:Given name. Note: don't call it "first name" since the given names do not always come first.
1 | TITLE | title | Description:Part of the name that is acquired as a title due to academic, legal, employment or nobility status etc. Note:Title name parts include name parts that come after the name such as qualifications, even if they are not always considered to be titles.

4.4.2.1122 HL7 v3 Code System EntityNamePartQualifierR2

URL: https://hl7.org/fhir/R4/v3/EntityNamePartQualifierR2/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-EntityNamePartQualifierR2) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-EntityNamePartQualifierR2
Version: | 2018-08-12
Name: | v3.EntityNamePartQualifierR2
Title: | v3 Code System EntityNamePartQualifierR2

Definition: | Description: The qualifier is a set of codes each of which specifies a certain subcategory of the name part in addition to the main name part type. For example, a given name may be flagged as a nickname, a family name may be a pseudonym or a name of public records.
OID: | 2.16.840.1.113883.5.1122 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: EntityNamePartQualifier (A set of codes each of which specifies a certain subcategory of the name part in addition to the main name part type.)

• ValueSet: v3 Code System EntityNamePartQualifierR2 (Description: The qualifier is a set of codes each of which specifies a certain subcategory of the name part in addition to the main name part type. For example, a given name may be flagged as a nickname, a family name may be a pseudonym or a name of public records.)

• ValueSet: EntityNamePartQualifier (A set of codes each of which specifies a certain subcategory of the name part in addition to the main name part type.)

4.4.2.1122.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | AD | acquired | Description:A name part a person acquired. The name part may be acquired by adoption, or the person may have chosen to use the name part for some other reason. Note: this differs from an Other/Psuedonym/Alias in that an acquired name part is acquired on a formal basis rather than an informal one (e.g. registered as part of the official name).
2 | SP | spouse | Description:The name assumed from the partner in a marital relationship. Usually the spouse's family name. Note that no inference about gender can be made from the existence of spouse names.
1 | BR | birth | Description:A name that a person was given at birth or established as a consequence of adoption. Note: This is not used for temporary names assigned at birth such as "Baby of Smith" a" which is just a name with a use code of "TEMP".
1 | CL | callme | Description:Used to indicate which of the various name parts is used when interacting with the person.
1 | IN | initial | Description:Indicates that a name part is just an initial. Initials do not imply a trailing period since this would not work with non-Latin scripts. In some languages, initials may consist of more than one letter, e.g., "Ph" could stand for "Philippe" or "Th" For "Thomas".
1 | LS | legal status | Description:For organizations a suffix indicating the legal status, e.g., "Inc.", "Co.", "AG", "GmbH", "B.V." "S.A.", "Ltd." etc.
1 | MID | middle name | Description:Indicates that the name part is a middle name. Usage Notes: In general, the english "middle name" concept is all of the given names after the first. This qualifier may be used to explicitly indicate which given names are considered to be middle names. The middle name qualifier may also be used with family names. This is a Scandinavian use case, matching the concept of "mellomnavn","mellannamn". Note that there are specific rules that indicate what names may be taken as a mellannamn in different Scandinavian countries.
1 | PFX | prefix | Description:A prefix has a strong association to the immediately following name part. A prefix has no implicit trailing white space (it has implicit leading white space though).
1 | (PharmaceuticalEntityNamePartQualifiers) Abstract | | Description: Medication Name Parts are a means of specifying a range of acceptable "official" forms of the name of a product. They are used as patterns against which input name strings may be matched for automatic identification of products from input text reports. While they cover the concepts held under "doseForm" or "route" or "strength" the name parts are not the same and do not fit into a controlled vocabulary in the same way. By specifying up to 8 name parts a much larger range of possible names can be generated.
2 | CON | container name | Description: This refers to the container if present in the medicinal product name. EXAMPLES: For Optaflu suspension for injection in pre-filled syringe Influenza vaccine (surface antigen, inactivated, prepared in cell culture) (2007/2008 season): pre-filled syringe
2 | DEV | device name | Description: This refers to the qualifiers in the name for devices and is at the moment mainly applicable to insulins and inhalation products. EXAMPLES: For the medicinal product Actrapid FlexPen 100 IU/ml Solution for injection Subcutaneous use: FlexPen.
2 | FLAV | FlavorName | Description: This refers to a flavor of the medicinal product if present in the medicinal product name. Examples: For 'CoughCure Linctus Orange Flavor', the flavor part is "Orange" For 'Wonderdrug Syrup Cherry Flavor', the flavor part is "Cherry"
2 | FORMUL | FormulationPartName | Description: This refers to the formulation of the medicinal product if present in the medicinal product name. Examples: For 'SpecialMed Sugar Free Cough Syrup', the formulation name part is "Sugar Free" For 'QuickCure Gluten-free Bulk Fibre', the formulation name part is "gluten-free"
2 | FRM | form name | Description: This refers to the pharmaceutical form/ if present in the medicinal product name. EXAMPLES: For Agenerase 50 mg soft capsules: Soft Capsules For Ludiomil 25mg-Filmtabletten: Filmtabletten For Optaflu suspension for injection in pre-filled syringe Influenza vaccine (surface antigen, inactivated, prepared in cell culture) (2007/2008 season): suspension for injection
2 | INV | invented name | Description: This refers to the product name without the trademark or the name of the marketing authorization holder or any other descriptor reflected in the product name and, if appropriate, whether it is intended e.g. for babies, children or adults. EXAMPLES: Agenerase Optaflu Ludiomil
2 | POPUL | TargetPopulationName | Description: This refers to the target population for the medicinal product if present in the medicinal product name Examples: For 'Broncho-Drug 3.5 mg-capsules for children', the target population part is "children" For 'Adult Chesty Cough Syrup', the target population part is "adult"
2 | SCI | scientific name | Description: This refers to the product common or scientific name without the trademark or the name of the marketing authorization holder or any other descriptor reflected in the product name. EXAMPLES: For Agenerase: N/A For Optaflu: Influenza vaccine (surface antigen, inactivated, prepared in cell culture) (2007/2008 season) For Ludiomil: N/A
2 | STR | strength name | Description: This refers to the strength if present in the medicinal product name. The use of decimal points should be accommodated if required. EXAMPLES: For Agenerase 50 mg soft capsules: 50mg For Ludiomil 25mg-Filmtabletten: 25 mg For Optaflu suspension for injection in pre-filled syringe Influenza vaccine (surface antigen, inactivated, prepared in cell culture) (2007/2008 season): N/A
2 | TIME | TimeOrPeriodName | Description: This refers to a time or time period that may be specified in the text of the medicinal product name Example: For an influenza vaccine 'Drug-FLU season 2008/2009', the time/period part is "2008/2009 season"

2 | TMK | trademark name | Description: This refers to trademark/company element if present in the medicinal product name. EXAMPLES: for Insulin Human Winthrop Comb 15: Winthrop

2 | USE | intended use name | Description: This refers to the intended use if present in the medicinal product name without the trademark or the name of the marketing authorization holder or any other descriptor reflected in the product name. Examples: For 'Drug-BI Caplets - Heartburn Relief', the intended use part is: "Heartburn Relief" For 'Medicine Honey Syrup for Soothing Coughs' the intended use part is "Soothing Coughs"

1 | SFX | suffix | Description:A suffix has a strong association to the immediately preceding name part. A suffix has no implicit leading white space (it has implicit trailing white space though).

1 | (TitleStyles) Abstract | | Description:Extra information about the style of a title

2 | AC | academic | Description:Indicates that a title like "Dr.", "M.D." or "Ph.D." is an academic title.

2 | HON | honorific | Description:A honorific such as "The Right Honourable" or "Weledelgeleerde Heer".

2 | NB | nobility | Description:A nobility title such as Sir, Count, Grafin.

2 | PR | professional | Description:Primarily in the British Imperial culture people tend to have an abbreviation of their professional organization as part of their credential titles.

4.4.2.1123 HL7 v3 Code System DataOperation

URL: https://hl7.org/fhir/R4/v3/DataOperation/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-DataOperation) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-DataOperation
Version: | 2018-08-12
Name: | v3.DataOperation
Title: | v3 Code System DataOperation
Definition: | **** MISSING DEFINITIONS ****
OID: | 2.16.840.1.113883.5.1123 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: Provenance activity type (This value set contains representative Activity Type codes, which includes codes from the HL7 DocumentCompletion, ActStatus, and DataOperations code system, W3C PROV-DM and PROV-N concepts and display names, several HL7 Lifecycle Event codes for which there are agreed upon definitions, and non-duplicated codes from the HL7 Security and Privacy Ontology Operations codes.)

• ValueSet: Provenance History Record Activity Codes (Codes for Provenance activities that are relevant when capturing event history for resources.)

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• ValueSet: v3 Code System DataOperation (**** MISSING DEFINITIONS ****)

4.4.2.1123.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition

1 | OPERATE | operate | Description:Act on an object or objects.

2 | CREATE | create | Description:Fundamental operation in an Information System (IS) that results only in the act of bringing an object into existence. Note: The preceding definition is taken from the HL7 RBAC specification. There is no restriction on how the operation is invoked, e.g., via a user interface. For an HL7 Act, the state transitions per the HL7 Reference Information Model.

2 | DELETE | delete | Description:Fundamental operation in an Information System (IS) that results only in the removal of information about an object from memory or storage. Note: The preceding definition is taken from the HL7 RBAC specification. There is no restriction on how the operation is invoked, e.g., via a user interface.

2 | EXECUTE | execute | Description:Fundamental operation in an IS that results only in initiating performance of a single or set of programs (i.e., software objects). Note: The preceding definition is taken from the HL7 RBAC specification. There is no restriction on how the operation is invoked, e.g., via a user interface.

2 | READ | read | Description:Fundamental operation in an Information System (IS) that results only in the flow of information about an object to a subject. Note: The preceding definition is taken from the HL7 RBAC specification. There is no restriction on how the operation is invoked, e.g., via a user interface.

2 | UPDATE | revise | Definition:Fundamental operation in an Information System (IS) that results only in the revision or alteration of an object. Note: The preceding definition is taken from the HL7 RBAC specification. There is no restriction on how the operation is invoked, e.g., via a user interface.

3 | APPEND | append | Description:Fundamental operation in an Information System (IS) that results only in the addition of information to an object already in existence. Note: The preceding definition is taken from the HL7 RBAC specification. There is no restriction on how the operation is invoked, e.g., via a user interface.

3 | MODIFYSTATUS | modify status | Description:Change the status of an object representing an Act.

4 | ABORT | abort | Description:Change the status of an object representing an Act to "aborted", i.e., terminated prior to the originally intended completion. For an HL7 Act, the state transitions per the HL7 Reference Information Model.

4 | ACTIVATE | activate | Description:Change the status of an object representing an Act to "active", i.e., so it can be performed or is being performed, for the first time. (Contrast with REACTIVATE.) For an HL7 Act, the state transitions per the HL7 Reference Information Model.

4 | CANCEL | cancel | Description:Change the status of an object representing an Act to "cancelled", i.e., abandoned before activation. For an HL7 Act, the state transitions per the HL7 Reference Information Model.

4 | COMPLETE | complete | Description:Change the status of an object representing an Act to "completed", i.e., terminated normally after all of its constituents have been performed. For an HL7 Act, the state transitions per the HL7 Reference Information Model.

4 | HOLD | hold | Description:Change the status of an object representing an Act to "held", i.e., put aside an Act that is still in preparatory stages. No action can occur until the Act is released. For an HL7 Act, the state transitions per the HL7 Reference Information Model.

4 | JUMP | jump | Description:Change the status of an object representing an Act to a normal state. For an HL7 Act, the state transitions per the HL7 Reference Information Model.

4 | NULLIFY | nullify | Description:Change the status of an object representing an Act to "nullified", i.e., treat as though it never existed. For an HL7 Act, the state transitions per the HL7 Reference Information Model.

4 | OBSOLETE | obsolete | Description:Change the status of an object representing an Act to "obsolete" when it has been replaced by a new instance. For an HL7 Act, the state transitions per the HL7 Reference Information Model.

4 | REACTIVATE | reactivate | Description:Change the status of a formerly active object representing an Act to "active", i.e., so it can again be performed or is being performed. (Contrast with ACTIVATE.) For an HL7 Act, the state transitions per the HL7 Reference Information Model.

4 | RELEASE | release | Description:Change the status of an object representing an Act so it is no longer "held", i.e., allow action to occur. For an HL7 Act, the state transitions per the HL7 Reference Information Model.

4 | RESUME | resume | Description:Change the status of a suspended object representing an Act to "active", i.e., so it can be performed or is being performed. For an HL7 Act, the state transitions per the HL7 Reference Information Model.

4 | SUSPEND | suspend | Definition:Change the status of an object representing an Act to suspended, i.e., so it is temporarily not in service.

4.4.2.1124 HL7 v3 Code System hl7Realm

URL: https://hl7.org/fhir/R4/v3/hl7Realm/cs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-hl7Realm) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-hl7Realm

Version: | 2018-08-12

Name: | v3.hl7Realm

Title: | v3 Code System hl7Realm

Definition: | Description: Coded concepts representing Binding Realms (used for Context Binding of terminology in HL7 models) and/or Namespace Realms (used to help ensure unique identification of HL7 artifacts). This code system is partitioned into three sections: Affiliate realms, Binding realms and Namespace realms. All affiliate realm codes may automatically be used as both binding realms and namespace realms. Furthermore, affiliate realms are the only realms that have authority over the creation of binding realms. (Note that 'affiliate' includes the idea of both international affiliates and the HL7 International organization.) All other codes must be associated with an owning affiliate realm and must appear as a specialization of _BindingRealm or _NamespaceRealm. For affiliates whose concepts align with nations, the country codes from ISO 3166-1 2-character alpha are used for the code when possible so these codes should not be used for other realm types. It is recommended that binding realm and namespace codes submitted by affiliates use the realm code as a prefix to avoid possible collisions with ISO codes. However, tooling does not currently support namepace realm codes greater than 2 characters. Open Issue: The name of the concept property "owningAffiliate" should be changed to better reflect that the property value is the human readable name of the organizational entity that manages the Realm identified by the Realm Code. Open Issue: In spite of the inability of tooling to process codes longer than 2 characters, there is at least one realm codes ('SOA') that was added that is 3 characters in length.

OID: | 2.16.840.1.113883.5.1124 (for OID based terminology systems)

Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System hl7Realm (Description: Coded concepts representing Binding Realms (used for Context Binding of terminology in HL7 models) and/or Namespace Realms (used to help ensure unique identification of HL7 artifacts). This code system is partitioned into three sections: Affiliate realms, Binding realms and Namespace realms. All affiliate realm codes may automatically be used as both binding realms and namespace realms. Furthermore, affiliate realms are the only realms that have authority over the creation of binding realms. (Note that 'affiliate' includes the idea of both international affiliates and the HL7 International organization.) All other codes must be associated with an owning affiliate realm and must appear as a specialization of _BindingRealm or _NamespaceRealm. For affiliates whose concepts align with nations, the country codes from ISO 3166-1 2-character alpha are used for the code when possible so these codes should not be used for other realm types. It is recommended that binding realm and namespace codes submitted by affiliates use the realm code as a prefix to avoid possible collisions with ISO codes. However, tooling does not currently support namepace realm codes greater than 2 characters. Open Issue: The name of the concept property "owningAffiliate" should be changed to better reflect that the property value is the human readable name of the organizational entity that manages the Realm identified by the Realm Code. Open Issue: In spite of the inability of tooling to process codes longer than 2 characters, there is at least one realm codes ('SOA') that was added that is 3 characters in length.)

4.4.2.1124.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (BindingRealms) Abstract | | Description: Concepts that can be used as Binding Realms when creating Binding Statements. These codes are permitted to appear in the InfrastructureRoot.realmCode attribute.
2 | (AffiliateRealms) Abstract | | Description: Realm codes for official HL7 organizational bodies. This includes both the HL7 International organization as well as all recognized international affiliates (past and present). These groups have the ability to bind vocabulary and develop artifacts. As well, they have the ability to have "ownership" over other binding realms and namespace realms via the owningAffiliate property of those other realm codes.
3 | AR | Argentina | Description: Realm code for use of Argentina
3 | AT | Austria | Description: Realm code for use of Austria
3 | AU | Australia | Description: Realm code for use of Australia
3 | BR | Brazil | Description: Realm code for use of Brazil
3 | CA | Canada | Description: Realm code for use of Canada
3 | CH | Switzerland | Description: Realm code for use of Switzerland
3 | CL | Chile | Description: Realm code for use of Chile
3 | CN | China | Description: Realm code for use of China
3 | CO | Columbia | Description: Realm code for use of Localized Version
3 | CZ | Czech Republic | Description: Realm code for use of Czech Republic
3 | DE | Germany | Description: Realm code for use of Germany
3 | DK | Denmark | Description: Realm code for use of Denmark
3 | ES | Spain | Description: Realm code for use of Spain
3 | FI | Finland | Description: Realm code for use of Finland
3 | FR | France | Description: Realm code for use of France
3 | GR | Greece | Description: Realm code for use of Greece
3 | HR | Croatia | Description: Realm code for use of Croatia
3 | IE | Ireland | Description: Realm code for use of Ireland
3 | IN | India | Description: Realm code for use of India
3 | IT | Italy | Description: Realm code for use of Italy
3 | JP | Japan | Description: Realm code for use of Japan
3 | KR | Korea | Description: Realm code for use of Korea
3 | LT | Lithuania | Description: Realm code for use of Lithuania
3 | MX | Mexico | Description: Realm code for use of Mexico
3 | NL | The Netherlands | Description: Realm code for use of The Netherlands
3 | NZ | New Zealand | Description: Realm code for use of New Zealand
3 | RO | Romania | Description: Realm code for use of Romania
3 | RU | Russian Federation | Description: Realm code for use of Russian Federation
3 | SE | Sweden | Description: Realm code for use of Sweden
3 | SG | Singapore | Description: Realm code for use of Localized Version
3 | SOA | Southern Africa | Description: Realm code for use of Southern Africa
3 | TR | Turkey | Description: Realm code for use of Turkey
3 | TW | Taiwan | Description: Realm code for use of Taiwan
3 | UK | United Kingdom | Description: Realm code for use of United Kingdom
3 | US | United States of America | Description: Realm code for use of United States of America
3 | UV | Universal | Description: Realm code for use of Universal realm or context, used in every instance
3 | UY | Uruguay | Description: Realm code for use of Uruguay
2 | C1 | Unclassified Realm | Description: Realm code for use of Unclassified Realm
2 | GB | Great Britain | Description: Realm code for use of Great Britain
2 | R1 | Representative Realm | Description: Realm code for use of Representative Realm
2 | X1 | Example Realm | Description: Realm code for use of Example Realm
1 | (NamespaceRealms) Abstract | | Description: Codes that can be used in the "realm" portion of HL7 v3 artifact identifiers.
2 | AffiliateRealms
2 | GB
2 | ZZ | Localized Version | Description: An artifact created for local use only. This realm namespace has no owning affiliate. Its use is uncontrolled, i.e. anyone can create artifacts using this realm namespace. Because of this, there is a significant likelihood of artifact identifier collisions. Implementers are encouraged to register their artifacts under an affiliate owned and controlled namespace to avoid such collision problems where possible.

4.4.2.1125 HL7 v3 Code System hl7V3Conformance

URL: https://hl7.org/fhir/R4/v3/hl7V3Conformance/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-hl7V3Conformance) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-hl7V3Conformance
Version: | 2018-08-12
Name: | v3.hl7V3Conformance
Title: | v3 Code System hl7V3Conformance
Definition: | Description: Identifies allowed codes for HL7aTMs v3 conformance property.
OID: | 2.16.840.1.113883.5.1125 (for OID based terminology systems)

Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System hl7V3Conformance (Description: Identifies allowed codes for HL7aTMs v3 conformance property.)

4.4.2.1125.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | I | ignored | Description: Implementers receiving this property must not raise an error if the data is received, but will not perform any useful function with the data. This conformance level is not used in profiles or other artifacts that are specific to the "sender" or "initiator" of a communication.
1 | NP | not permitted | Description: All implementers are prohibited from transmitting this content, and may raise an error if they receive it.
1 | R | required | Description: All implementers must support this property. I.e. they must be able to transmit, or to receive and usefully handle the concept.
1 | RC | required for consumer | Description: The element is considered "required" (i.e. must be supported) from the perspective of systems that consume instances, but is "undetermined" for systems that generate instances. Used only as part of specifications that define both initiator and consumer expectations.
1 | RI | required for initiator | Description: The element is considered "required" (i.e. must be supported) from the perspective of systems that generate instances, but is "undetermined" for systems that consume instances. Used only as part of specifications that define both initiator and consumer expectations.
1 | U | undetermined | Description: The conformance expectations for this element have not yet been determined.

4.4.2.1126 HL7 v3 Code System ResponseMode

URL: https://hl7.org/fhir/R4/v3/ResponseMode/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ResponseMode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ResponseMode
Version: | 2018-08-12
Name: | v3.ResponseMode
Title: | v3 Code System ResponseMode
Definition: | Specifies the mode, immediate versus deferred or queued, by which a receiver should communicate its receiver responsibilities.
OID: | 2.16.840.1.113883.5.1126 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ResponseMode (Specifies the mode, immediate versus deferred or queued, by which a receiver should communicate its receiver responsibilities.)

4.4.2.1126.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | D | deferred | The receiver may respond in a non-immediate manner. Note: this will be the default value.
1 | I | immediate | The receiver is required to assume that the sender is blocking and behave appropriately by sending an immediate response.
1 | Q | queue | The receiver shall keep any application responses in a queue until such time as the queue is polled.

4.4.2.1128 HL7 v3 Code System policyHolderRole

URL: https://hl7.org/fhir/R4/v3/policyHolderRole/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-policyHolderRole) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-policyHolderRole
Version: | 2018-08-12
Name: | v3.policyHolderRole
Title: | v3 Code System policyHolderRole
Definition: | This vocabulary is defined by Implementation Guide for CDA Release 2 - Level 1 - Care Record Summary (US realm). It describes roles recognized through the issuance of an insurance policy to a policyholder who a relationship with the covered party, such as spouse, child, etc. This vocabulary is essentially an inversion of the role relations of the HL7 CoverageRoleType vocabulary. It provides more detailed roles with respect to the underwriter (the scoping organization) for those participants in the policyholder role for a patient. Open Issue: The code values for this coding system must be extracted from the CDA documentation and brought forward through Harmonization for instantiation in this repository.
OID: | 2.16.840.1.113883.5.1128 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System policyHolderRole (This vocabulary is defined by Implementation Guide for CDA Release 2 - Level 1 - Care Record Summary (US realm). It describes roles recognized through the issuance of an insurance policy to a policyholder who a relationship with the covered party, such as spouse, child, etc. This vocabulary is essentially an inversion of the role relations of the HL7 CoverageRoleType vocabulary. It provides more detailed roles with respect to the underwriter (the scoping organization) for those participants in the policyholder role for a patient. Open Issue: The code values for this coding system must be extracted from the CDA documentation and brought forward through Harmonization for instantiation in this repository.)

4.4.2.1128.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition

4.4.2.1129 HL7 v3 Code System HL7ContextConductionStyle

URL: https://hl7.org/fhir/R4/v3/HL7ContextConductionStyle/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-HL7ContextConductionStyle) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-HL7ContextConductionStyle
Version: | 2018-08-12
Name: | v3.HL7ContextConductionStyle
Title: | v3 Code System HL7ContextConductionStyle
Definition: | The styles of context conduction usable by relationships within a static model derived from tyhe HL7 Reference Information Model.
OID: | 2.16.840.1.113883.5.1129 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System HL7ContextConductionStyle (The styles of context conduction usable by relationships within a static model derived from tyhe HL7 Reference Information Model.)

4.4.2.1129.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | C | conduction-indicator-based | Definition: Context conduction is defined using the contextConductionCode and contextConductionInd attributes on ActRelationship and Participation. UsageNotes: This approach is deprecated as of March, 2010.
1 | I | inferred | Definition: Context conduction is not explicitly defined. The recipient of an instance must infer conduction based on the semantics of the model and what is deemed "reasonable". UsageNotes: Because this approach can lead to variation in instance interpretation, its use is discouraged.
1 | V | vocabulary-based | Definition: Context conduction is defined using the ActRelationship.blockedContextActRelationshipType and blockedContextParticipationType attributes and the "conductible" property on the ActRelationshipType and ParticipationType code systems.

4.4.2.1130 HL7 v3 Code System hl7ApprovalStatus

URL: https://hl7.org/fhir/R4/v3/hl7ApprovalStatus/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-hl7ApprovalStatus) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-hl7ApprovalStatus
Version: | 2018-08-12
Name: | v3.hl7ApprovalStatus
Title: | v3 Code System hl7ApprovalStatus
Definition: | Description: Codes for concepts describing the approval level of HL7 artifacts. This code system reflects the concepts expressed in HL7's Governance & Operations Manual (GOM) past and present.
OID: | 2.16.840.1.113883.5.1130 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System hl7ApprovalStatus (Description: Codes for concepts describing the approval level of HL7 artifacts. This code system reflects the concepts expressed in HL7's Governance & Operations Manual (GOM) past and present.)

4.4.2.1130.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | affd | affiliate ballot - DSTU | Description: Content that is being presented to an international affiliate for consideration as a realm-specific draft standard for trial use.
1 | affi | affiliate ballot - informative | Description: Content that is being presented to an international affiliate for consideration as a realm-specific informative standard.
1 | affn | affiliate ballot - normative | Description: Content that is being presented to an international affiliate for consideration as a realm-specific normative standard.
1 | appad | approved affiliate DSTU | Description: Content that has passed ballot as a realm-specific draft standard for trial use.
1 | appai | approved affiliate informative | Description: Content that has passed ballot as a realm-specific informative standard.
1 | appan | approved affiliate normative | Description: Content that has passed ballot as a realm-specific normative standard
1 | appd | approved DSTU | Description: Content that has passed ballot as a draft standard for trial use.
1 | appi | approved informative | Description: Content that has passed ballot as a normative standard.
1 | appn | approved normative | Description: Content that has passed ballot as a normative standard.
1 | comi Deprecated | committee ballot - informative | Description: Content prepared by a committee and submitted for internal consideration as an informative standard. Deprecation Comment No longer supported as ballot statuses within the HL7 Governance and Operations Manual. Use normative or informative variants instead.
1 | comn Deprecated | committee ballot - normative | Description: Content prepared by a committee and submitted for internal consideration as an informative standard. Deprecation Comment No longer supported as ballot statuses within the HL7 Governance and Operations Manual. Use normative or informative variants instead.
1 | draft | draft | Description: Content that is under development and is not intended to be used.
1 | loc | localized adaptation | Description: Content that represents an adaption of a implementable balloted material to represent the needs or capabilities of a particular installation.
1 | memd | membership ballot - DSTU | Description: Content prepared by a committee and submitted for membership consideration as a draft standard for trial use.
1 | memi | membership ballot - informative | Description: Content prepared by a committee and submitted for membership consideration as an informative standard.
1 | memn | membership ballot - normative | Description: Content prepared by a committee and submitted for membership consideration as a normative standard.
1 | ns | non-standard - available for use | Description: Content developed independently by an organization or individual that is declared to be 'usable' but for which there is no present intention to submit through the standards submission and review process.
1 | prop | proposal | Description: Content submitted to a committee for consideration for future inclusion in the standard.
1 | ref | reference | Description: Content intended to support other content that is subject to approval, but which is not itself subject to formal approval.
1 | wd | withdrawn | Description: Content that represents an item that was at one point a normative or informative standard, but was subsequently withdrawn.

4.4.2.1131 HL7 v3 Code System hl7VoteResolution

URL: https://hl7.org/fhir/R4/v3/hl7VoteResolution/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-hl7VoteResolution) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-hl7VoteResolution
Version: | 2018-08-12
Name: | v3.hl7VoteResolution
Title: | v3 Code System hl7VoteResolution
Definition: | Description: Based on concepts for resolutions from HL7 ballot spreadsheet according to HL7's Governance & Operations Manual (GOM).
OID: | 2.16.840.1.113883.5.1131 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System hl7VoteResolution (Description: Based on concepts for resolutions from HL7 ballot spreadsheet according to HL7's Governance & Operations Manual (GOM).)

4.4.2.1131.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | (affirmativeResolution) Abstract | | Description: An abstract concept grouping resolutions that can be applied to affirmative ballot comments.
2 | affdef | affirmative-deferred | Description: The recommended change has been deferred to consideration for a future release.
2 | affi | affirmative-incorporated | Description: The recommended change has been incorporated or identified issue has been answered.
2 | affr | affirmative-rejected | Description: The recommended change has been refused and is not expected to be incorporated.
1 | (negativeResolution) Abstract | | Description: An abstract concept grouping resolutions that can be applied to negative ballot comments.
2 | nonsubp | non-substantive proposed | Description: Responsible group has recommended that the negative vote be considered non-substantive. (Issue raised does not provide sufficiently convincing reason to make changes to the item under ballot, or otherwise impede its adoption.)
2 | nonsubv | non-substantive voted | Description: Ballot group has voted and declared the negative vote non-substantive.
2 | notrelp | not related proposed | Description: Responsible group has recommended that the negative vote be considered not-related. (Issue raised is not related to the current scope of the item under ballot, or does not prevent the item under ballot for

being used for its defined intent. Recommended changes may be considered as part of future versions.) (Perhaps after further reading or explanation).

2 | notrelv | not related voted | Description: Ballot group has voted and declared the negative vote non-related.

2 | prevcons | previously considered | Description: Committee identifies that the same issue has been raised as part of a previous ballot on the same element version and was found by the ballot group to be non-substantive or not related.)

2 | retract | retracted | Description: Voter has formally withdrawn their vote or comment as having been in error. (Perhaps after further reading or explanation).

2 | unresolved | unresolved | Description: Vote has not yet gone through resolution.

2 | withdraw | withdrawn | Description: Voter has formally withdrawn their vote or comment on the basis of agreed changes or proposed future changes.

4.4.2.1132 HL7 v3 Code System hl7CMETAttribution

URL: https://hl7.org/fhir/R4/v3/hl7CMETAttribution/cs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-hl7CMETAttribution) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-hl7CMETAttribution

Version: | 2018-08-12

Name: | v3.hl7CMETAttribution

Title: | v3 Code System hl7CMETAttribution

Definition: | **** MISSING DEFINITIONS ****

OID: | 2.16.840.1.113883.5.1132 (for OID based terminology systems)

Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System hl7CMETAttribution (**** MISSING DEFINITIONS ****)

4.4.2.1132.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition

1 | contact | contact | Description: Provides sufficient information to allow the object identified to be contacted. This is likely to have the content of identified and confirmable plus telephone number.

1 | identified | identified | Description: This variant is a proper subset of universal and is intended to provide sufficient information to identify the object(s) modeled by the CMET. This variant is only suitable for use within TIGHTLY COUPLED SYSTEMS ONLY. This variant provides ONLY the ID (and code where applicable) and Name. Other variants may not be substituted at runtime.

1 | identified-confirmable | identified-confirmable | Description: This extends the identified variant by adding just sufficient additional information to allow the identity of object modeled to be confirmed by a number of corroborating items of data; for instance a patient's date of birth and current address. However, specific contact information, such as telephone number, are not viewed as confirming information.

1 | identified-informational | identified-informational | Description: Generally the same information content as "contactable" but using new "informational" CMETs as dependant CMETs. This flavor allows expression of the CMET when non-focal class information is not known.

1 | informational | informational | Description: Generally the same information content as "contactable", but with required (not mandatory) ids on entry point class. This flavor allows expression of the CMET even when mandatory information is not known.

1 | minimal | minimal | Description: Provides more than identified, but not as much as universal. There are not expected to be many of these.

1 | universal | universal | Description: This variant includes all attributes and associations present in the R-MIM. Any of non-mandatory and non-required attributes and/or associations may be present or absent, as permitted in the cardinality constraints.

4.4.2.1133 HL7 v3 Code System hl7PublishingSection

URL: https://hl7.org/fhir/R4/v3/hl7PublishingSection/cs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-hl7PublishingSection) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-hl7PublishingSection

Version: | 2018-08-12

Name: | v3.hl7PublishingSection

Title: | v3 Code System hl7PublishingSection

Definition: | Description: Codes for HL7 publishing sections (major business categories)

OID: | 2.16.840.1.113883.5.1133 (for OID based terminology systems)

Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System hl7PublishingSection (Description: Codes for HL7 publishing sections (major business categories))

4.4.2.1133.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition

1 | AM | administrative management | Description: Represents the HL7 V3 publishing section that deals with the administration and management of health care activities and organizations. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications. For publishing purposes, these domains are aggregated into sub-sections of related health care areas and these sub-sections are further aggregated into three major sets.

1 | HM | health and clinical management | Description: Represents the HL7 V3 publishing section that deals with the health care provision and clinical management. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications. For publishing purposes, these domains are aggregated into sub-sections of related health care areas and these sub-sections are further aggregated into three major sets.

1 | IM | infrastructure management | Description: Represents the HL7 V3 publishing section that deals with the definition and management of the computing and communication infrastructure necessary to support health care. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications. For publishing purposes, these domains are aggregated into sub-sections of related health care areas and these sub-sections are further aggregated into three major sets.

1 | UU | unknown | Description: Represents the HL7 V3 publishing section that holds specifications that are unassigned - that have not yet been assigned to one of the formal publishing sections. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications. For publishing purposes, these domains are aggregated into sub-sections of related health care areas and these sub-sections are further aggregated into three major sets.

4.4.2.1134 HL7 v3 Code System hl7PublishingSubSection

URL: https://hl7.org/fhir/R4/v3/hl7PublishingSubSection/cs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-hl7PublishingSubSection) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-hl7PublishingSubSection

Version: | 2018-08-12

Name: | v3.hl7PublishingSubSection

Title: | v3 Code System hl7PublishingSubSection

Definition: | Description: Codes for HL7 publishing sub-sections (business sub-categories)

OID: | 2.16.840.1.113883.5.1134 (for OID based terminology systems)

Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System hl7PublishingSubSection (Description: Codes for HL7 publishing sub-sections (business sub-categories))

4.4.2.1134.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition

1 | CO | common | Description: Represents the HL7 V3 publishing sub-section that holds common or shared specifications within the Infrastructure Management (IM) section. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications. For publishing purposes, these domains are aggregated into sub-sections of related health care areas and these sub-sections are further aggregated into three major sets.

1 | FI | financial information | Description: Represents the HL7 V3 publishing sub-section that holds specifications related to the management of financial information within the Administrative Management (AM) section. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications. For publishing purposes, these domains are aggregated into sub-sections of related health care areas and these sub-sections are further aggregated into three major sets.

1 | MC | message control | Description: Represents the HL7 V3 publishing sub-section that holds specifications related to the definition and control of interoperability messages within the Infrastructure Management (IM) section. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications. For publishing purposes, these domains are aggregated into sub-sections of related health care areas and these sub-sections are further aggregated into three major sets.

1 | MF | master file | Description: Represents the HL7 V3 publishing sub-section that holds specifications related to master file and registry management activities within the Infrastructure Management (IM) section. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications. For publishing purposes, these domains are aggregated into sub-sections of related health care areas and these sub-sections are further aggregated into three major sets.

1 | PO | operations | Description: Represents the HL7 V3 publishing sub-section that holds specifications related to managing clinical operations within the Health and Clinical Management (HM) section. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications. For publishing purposes, these domains are aggregated into sub-sections of related health care areas and these sub-sections are further aggregated into three major sets.

1 | PR | practice | Description: Represents the HL7 V3 publishing sub-section that holds specifications related to the management of practice settings within the Administrative Management (AM) section. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications. For publishing purposes, these domains are aggregated into sub-sections of related health care areas and these sub-sections are further aggregated into three major sets.
1 | QU | query | Description: Represents the HL7 V3 publishing sub-section that holds specifications related to query/response activities within the Infrastructure Management (IM) section. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications. For publishing purposes, these domains are aggregated into sub-sections of related health care areas and these sub-sections are further aggregated into three major sets.
1 | RC | records | Description: Represents the HL7 V3 publishing sub-section that holds specifications related to the definition and communication of records of clinical care within the Health and Clinical Management (HM) section. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications. For publishing purposes, these domains are aggregated into sub-sections of related health care areas and these sub-sections are further aggregated into three major sets.
1 | RE | reasoning | Description: Represents the HL7 V3 publishing sub-section that holds specifications related to the definition and communication of reasoning (knowledge) within the Health and Clinical Management (HM) section. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications. For publishing purposes, these domains are aggregated into sub-sections of related health care areas and these sub-sections are further aggregated into three major sets.
1 | UU | unknown | Description: Represents the HL7 V3 publishing sub-section that holds specifications that are unassigned - that have not yet been assigned to one of the formal publishing sections. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications. For publishing purposes, these domains are aggregated into sub-sections of related health care areas and these sub-sections are further aggregated into three major sets.

4.4.2.1135 HL7 v3 Code System hl7ITSType

URL: https://hl7.org/fhir/R4/v3/hl7ITSType/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-hl7ITSType) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-hl7ITSType
Version: | 2018-08-12
Name: | v3.hl7ITSType
Title: | v3 Code System hl7ITSType
Definition: | Description: Codes identifying types of HL7 Implementation Technology Specifications
OID: | 2.16.840.1.113883.5.1135 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System hl7ITSType (Description: Codes identifying types of HL7 Implementation Technology Specifications)

4.4.2.1135.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | UML | universal modeling language | Description: ITS based on the Universal Modeling Language.
1 | XML | extensible markup language | Description: ITS based on the Extensible Markup Language.

4.4.2.1136 HL7 v3 Code System hl7PublishingDomain

URL: https://hl7.org/fhir/R4/v3/hl7PublishingDomain/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-hl7PublishingDomain) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-hl7PublishingDomain
Version: | 2018-08-12
Name: | v3.hl7PublishingDomain
Title: | v3 Code System hl7PublishingDomain
Definition: | Description: Codes for HL7 publishing domains (specific content area)
OID: | 2.16.840.1.113883.5.1136 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System hl7PublishingDomain (Description: Codes for HL7 publishing domains (specific content area))

4.4.2.1136.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition

1 | AB | accounting & billing | Description: Represents the HL7 content "domain" that supports accounting and billing functions - and "provides support for the creation and management of patient billing accounts and the post of financial transactions against patient billing accounts for the purpose of aggregating financial transactions that will be submitted as claims or invoices for reimbursemen" UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | AI | trigger event control act infrastructure | Description: Represents the HL7 content "domain" that supports trigger event control act infrastructure - and "covers the alternate structures of the message Trigger Event Control Acts in the HL7 Composite Message." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | AL | artificial listing for test purposes - faux Domain for testing | Description: Represents the HL7 content "domain" that was defined as an "artificial listing" domain to support publication testing.

1 | BB | blood tissue and organ | Description: Represents the HL7 content "domain" that supports blood tissue and organ domain - and "comprises the models, messages, and other artIfacts that are needed to support messaging related to the process of blood, tissue, and organ banking operations such as donations, eligibility, storage, dispense, administration/transfusion, explantation, and implantation. " UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | CD | clinical document architecture | Description: Represents the HL7 content "domain" that supports the clinical document architecture. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | CG | clinical genomics | Description: Represents the HL7 content "domain" that supports clinical genomics - and includes " standards to enable the exchange of interrelated clinical and personalized genomic data between interested parties." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | CI | transmission infrastructure | Description: Represents the HL7 content "domain" that supports transmission infrastructure - and " is primarily concerned with the data content of exchanges between healthcare applications, the sequence or interrelationships in the flow of messages and the communication of significant application level exceptions or error conditions." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | CO | coverage | Description: Represents the HL7 content "domain" that supports Coverage - and provides support for managing health care coverage in the reimbursement system(s). UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | CP | common product model | Description: Represents the HL7 content "domain" that supports the common product model - which "is used to improve the alignment between the different representations of products used within the body of HL7 Version 3 models." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | CR | claims and reimbursement | Description: Represents the HL7 content "domain" that supports Claims and Reimbursement - and "provides support for Generic, Pharmacy, Preferred Accommodation, Physician, Oral Health Vision Care and Hospital claims for eligibility, authorization, coverage extension, pre-determination, invoice adjudication, payment advice and Statement of Financial Activity (SOFA) Release 3 of this document adds claims messaging support for Physician, Oral Health Vision Care and Hospital claims." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | CS | clinical statement | Description: Represents the HL7 content "domain" that supports a common clinical statement pattern - and "is a 'pattern' designed to be used within multiple HL7 Version 3 domain models. This pattern is intended to facilitate the consistent design of communications that convey clinical information to meet specific use cases." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | CT | common types | Description: Represents the HL7 content "domain" that supports common model types - and "are a work product produced by a particular committee for expressing a common, useful and reusable concept." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | DD | dummy domain | Description: Represents the HL7 content "domain" that was created to support testing and initial set-up functions.

1 | DI | diagnostic imaging | Description: This domain has been retired in favor of "imaging integration" (II).

1 | DS | decision support | Description: Represents the HL7 content "domain" that provides decision support. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | EM | emergency medical services | Description: Represents the HL7 content "domain" that supports Emergency Medical Services. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | II | imaging integration | Description: Represents the HL7 content "domain" that supports imaging integration - and is "comprises the models, implementation guides, sample documents and images that are needed to illustrate the transformation of DICOM structured reports to CDA Release 2 as well as the creation of CDA diagnostic imaging reports." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | IZ | immunization | Description: Represents the HL7 content "domain" that supports immunization - and "describes communication of information about immunization: the administration of vaccines (and/or antisera) to individuals to prevent infectious disease." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | LB | laboratory | Description: Represents the HL7 content "domain" that supports clinical laboratory functions - and is "comprises the models, messages, and other artifacts that are needed to support messaging related to laboratory tests or observations. " UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | ME | medication | Description: Represents the HL7 content "domain" that supports medication - and "deals with the description of a medicine for the purposes of messaging information about medicines" and the applications of these descriptions. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | MI | masterfile infrastructure | Description: Represents the HL7 content "domain" that supports master file infrastructure - and is "comprises the classes and attributes needed to support Master Files and Registries." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | MM | materials management | Description: Represents the HL7 content "domain" that supports Materials Management - and is "supports the simple scenario of a Materials Management application sending requests, notifications and queries to an auxiliary application. The intent is to establish a standard for the minimum functionality that is useful and comprehensive enough to explore the important concepts relative to inventory management." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | MR | medical records | Description: Represents the HL7 content "domain" that supports medical records - and is "supports clinical document management, and document querying." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | MT | shared messages | Description: Represents the HL7 content "domain" that supports shared messages - and "are a work product produced for expressing common, useful and reusable message types." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | OB | observations | Description: Represents the HL7 content "domain" that supports observations - and is "comprises the models, messages, and other artifacts that are needed to support messaging related to resulting basic healthcare diagnostic services. " UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | OO | orders & observations | Description: Represents the HL7 content "domain" that supports orders and observations - and will provide over-arching support information for the "Orders" (OR) and "Observations" (OB) domains. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | OR | orders | Description: Represents the HL7 content "domain" that supports orders - and "comprises the models, messages, and other artifacts that are needed to support messaging related to ordering basic healthcare services." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | PA | patient administration | Description: Represents the HL7 content "domain" that supports Patient Administration - and "defines person and patient demographics and visit information about patients" UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | PC | care provision | Description: Represents the HL7 content "domain" that supports Care Provision - and "addresses the information that is needed for the ongoing care of individuals, populations, and other targets of care." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | PH | public health | Description: Represents the HL7 content "domain" that supports public health - and is "the source of a number of Common Model Element Types (CMET) designed to meet the needs of public health data exchange." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | PM | personnel management | Description: Represents the HL7 content "domain" that supports Personnel Management - and "spans a variety of clinical-administrative information functions associated with the organizations, individuals, animals and devices involved in the delivery and support of healthcare services." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | QI | query infrastructure | Description: Represents the HL7 content "domain" that supports query infrastructure - and "specifies the formation of information queries and the responses to these queries to meet the needs of healthcare applications using the HL7 version 3 messaging standard." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | QM | quality measures | Description: Represents the HL7 content "domain" that supports Quality Measures - and "is a standard for representing a health quality measure as an electronic document." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | RG | registries | Description: Represents the HL7 content "domain" that supports Registries - and "collects HL7 artifacts for administrative registries." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | RI | informative public health | Description: Represents the HL7 content "domain" that supports Informative Public Health. UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | RP | regulated products | Description: Represents the HL7 content "domain" that supports Regulated Products - and "includes standards developed as part of the family of messages targeted for the exchange of information about regulated products and the exchange of the data needed to provide approval for such products." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

1 | RR | public health reporting | Description: Represents the HL7 content "domain" that supports Public Health Reporting - and "includes messages and documents that are specifically designed to support managment, reporting and investigation in the public health context." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.
1 | RT | regulated studies | Description: Represents the HL7 content "domain" that supports Regulated Studies - and is "includes standards developed as part of the family of messages targeted for the exchange of information about the conduct of regulated studies, and the exchange of the data collected during those studies." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.
1 | RX | pharmacy | Description: Represents the HL7 content "domain" that supports pharmacy - and is a "model used to derive message patterns to describe and communicate processes related to medication." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.
1 | SC | scheduling | Description: Represents the HL7 content "domain" that supports Scheduling - and "offers a generic set of messages and behavior to implement any number of Scheduling scenarios." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.
1 | SP | specimen | Description: Represents the HL7 content "domain" that supports Specimen - and "comprises the models and artifacts that are needed to support the creation of messaging related to specimen." UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.
1 | TD | therapeutic devices | Description: Represents the HL7 content "domain" that supports Therapeutic Devices - and is "comprises the models, messages, and other artifacts that are needed to support messaging related to therapy delivery and observations made by a medical device. " UsageNote: V3 Specifications are published in a set of "domains", which contain interactions and related specifications for a single area of health care within which can be supported by a single, coherent set of interoperability specifications.

4.4.2.1137 HL7 v3 Code System RoleLinkStatus

URL: https://hl7.org/fhir/R4/v3/RoleLinkStatus/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-RoleLinkStatus) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-RoleLinkStatus
Version: | 2018-08-12
Name: | v3.RoleLinkStatus
Title: | v3 Code System RoleLinkStatus
Definition: | Description: Codes representing possible states of a RoleLink, as defined by the RoleLink class state machine.
OID: | 2.16.840.1.113883.5.1137 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System RoleLinkStatus (Description: Codes representing possible states of a RoleLink, as defined by the RoleLink class state machine.)

4.4.2.1137.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | NORMAL | normal | Description: The 'typical' state. Excludes "nullified" which represents the termination state of a RoleLink instance that was created in error.
2 | ACTIVE | active | Description: The state indicates the RoleLink is in progress.
2 | CANCELLED | cancelled | Description: The terminal state resulting from abandoning the RoleLink prior to or after activation.
2 | COMPLETED | completed | Description: The terminal state representing the successful completion of the RoleLink.
2 | PENDING | pending | Description: The state indicates the RoleLink has not yet become active.
1 | NULLIFIED | nullified | Description: The state representing the termination of a RoleLink instance that was created in error.

4.4.2.1138 HL7 v3 Code System ActUSPrivacyLaw

URL: https://hl7.org/fhir/R4/v3/ActUSPrivacyLaw/cs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-ActUSPrivacyLaw) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-ActUSPrivacyLaw
Version: | 2018-08-12
Name: | v3.ActUSPrivacyLaw
Title: | v3 Code System ActUSPrivacyLaw
Definition: | A jurisdictional mandate in the US relating to privacy. Deprecation Comment: Content moved to ActCode under _ActPrivacyLaw; use that instead.
OID: | 2.16.840.1.113883.5.1138 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System ActUSPrivacyLaw (A jurisdictional mandate in the US relating to privacy. Deprecation Comment: Content moved to ActCode under _ActPrivacyLaw; use that instead.)

4.4.2.1138.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition

1 | _ActUSPrivacyLaw Deprecated | ActUSPrivacyLaw | Definition: A jurisdictional mandate in the U.S. relating to privacy. Usage Note: ActPrivacyLaw codes may be associated with an Act or a Role to indicate the legal provision to which the assignment of an Act.confidentialityCode or Role.confidentialtyCode complies. May be used to further specify rationale for assignment of other ActPrivacyPolicy codes in the US realm, e.g., ETH and 42CFRPart2 can be differentiated from ETH and Title38Part1.

2 | 42CFRPart2 Deprecated | 42 CFR Part2 | 42 CFR Part 2 stipulates the right of an individual who has applied for or been given diagnosis or treatment for alcohol or drug abuse at a federally assisted program. Definition: Non-disclosure of health information relating to health care paid for by a federally assisted substance abuse program without patient consent. Usage Note: May be associated with an Act or a Role to indicate the legal provision to which the assignment of an Act.confidentialityCode or Role.confidentialityCode complies.

2 | CommonRule Deprecated | Common Rule | U.S. Federal regulations governing the protection of human subjects in research (codified at Subpart A of 45 CFR part 46) that has been adopted by 15 U.S. Federal departments and agencies in an effort to promote uniformity, understanding, and compliance with human subject protections. Existing regulations governing the protection of human subjects in Food and Drug Administration (FDA)-regulated research (21 CFR parts 50, 56, 312, and 812) are separate from the Common Rule but include similar requirements. Definition: U.S. federal laws governing research-related privacy policies. Usage Note: May be associated with an Act or a Role to indicate the legal provision to which the assignment of an Act.confidentialityCode or Role.confidentialtyCode complies.

2 | HIPAANOPP Deprecated | HIPAA notice of privacy practices | The U.S. Public Law 104-191 Health Insurance Portability and Accountability Act (HIPAA) Privacy Rule (45 CFR Part 164 Subpart E) permits access, use and disclosure of certain personal health information (PHI as defined under the law) for purposes of Treatment, Payment, and Operations, and requires that the provider ask that patients acknowledge the Provider's Notice of Privacy Practices as permitted conduct under the law. Definition: Notification of HIPAA Privacy Practices. Usage Note: May be associated with an Act or a Role to indicate the legal provision to which the assignment of an Act.confidentialityCode or Role.confidentialtyCode complies.

2 | HIPAAPsyNotes Deprecated | HIPAA psychotherapy notes | The U.S. Public Law 104-191 Health Insurance Portability and Accountability Act (HIPAA) Privacy Rule (45 CFR Part 164 Section 164.508) requires authorization for certain uses and disclosure of psychotherapy notes. Definition: Authorization that must be obtained for disclosure of psychotherapy notes. Usage Note: May be associated with an Act or a Role to indicate the legal provision to which the assignment of an Act.confidentialityCode or Role.confidentialityCode complies.

2 | HIPAASelfPay Deprecated | HIPAA self-pay | Section 13405(a) of the Health Information Technology for Economic and Clinical Health Act (HITECH) stipulates the right of an individual to have disclosures regarding certain health care items or services for which the individual pays out of pocket in full restricted from a health plan. Definition: Non-disclosure of health information to a health plan relating to health care items or services for which an individual pays out of pocket in full. Usage Note: May be associated with an Act or a Role to indicate the legal provision to which the assignment of an Act.confidentialityCode or Role.confidentialityCode complies.

2 | Title38Section7332 Deprecated | Title 38 Section 7332 | Title 38 Part 1-protected information may only be disclosed to a third party with the special written consent of the patient except where expressly authorized by 38 USC 7332. VA may disclose this information for specific purposes to: VA employees on a need to know basis - more restrictive than Privacy Act need to know; contractors who need the information in order to perform or fulfill the duties of the contract; and researchers who provide assurances that the information will not be identified in any report. This information may also be disclosed without consent where patient lacks decision-making capacity; in a medical emergency for the purpose of treating a condition which poses an immediate threat to the health of any individual and which requires immediate medical intervention; for eye, tissue, or organ donation purposes; and disclosure of HIV information for public health purposes. Definition: Title 38 Part 1 - Â§1.462 Confidentiality restrictions. (a) General. The patient records to which Â§Â§1.460 through 1.499 of this part apply may be disclosed or used only as permitted by these regulations and may not otherwise be disclosed or used in any civil, criminal, administrative, or legislative proceedings conducted by any Federal, State, or local authority. Any disclosure made under these regulations must be limited to that information which is necessary to carry out the purpose of the disclosure. SUBCHAPTER III--PROTECTION OF PATIENT RIGHTS Sec. 7332. Confidentiality of certain medical records (a)(1) Records of the identity, diagnosis, prognosis, or treatment of any patient or subject which are maintained in connection with the performance of any program or activity (including education, training, treatment, rehabilitation, or research) relating to drug abuse, alcoholism or alcohol abuse, infection with the human immunodeficiency virus, or sickle cell anemia which is carried out by or for the Department under this title shall, except as provided in subsections (e) and (f), be confidential, and (section 5701 of this title to the contrary notwithstanding) such records may be disclosed only for the purposes and under the circumstances expressly authorized under subsection (b). Usage Note: May be associated with an Act or a Role to indicate the legal provision to which the assignment of an Act.confidentialityCode or Role.confidentialityCode complies.

4.4.2.1139 HL7 v3 Code System WorkClassificationODH

URL: https://hl7.org/fhir/R4/v3/WorkClassificationODH/cs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This code system (http://terminology.hl7.org/CodeSystem/v3-WorkClassificationODH) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/CodeSystem/v3-WorkClassificationODH

Version: | 2018-08-12

Name: | v3.WorkClassificationODH

Title: | v3 Code System WorkClassificationODH

Definition: | Code system of concepts representing a person's job type as defined by compensation and sector (e.g. paid vs. unpaid, self-employed vs. not self-employed, government vs. private, etc.).
OID: | 2.16.840.1.113883.5.1139 (for OID based terminology systems)
Source Resource | XML / JSON

This Code system is used in the following value sets:

• ValueSet: v3 Code System WorkClassificationODH (Code system of concepts representing a person's job type as defined by compensation and sector (e.g. paid vs. unpaid, self-employed vs. not self-employed, government vs. private, etc.).)

4.4.2.1139.1 Code System Content

Release Date: 2018-08-12

Level | Code | Display | Definition
1 | PWAF | Paid work, Armed Forces | A situation in which an individual serves in a government-sponsored military force.
1 | PWFG | Paid work, national government, not armed forces | A situation in which an individual works for a national government organization, not including armed forces, and receives a paid salary or wage.
1 | PWLG | Paid work, local government, not armed forces | A situation in which an individual works for a government organization with jurisdiction below the level of state/provincial/territorial/tribal government (e.g., city, town, township), not armed forces, and receives a paid salary or wage.
1 | PWNSE | Paid non-governmental work, not self-employed | A situation in which an individual works for a business (not government) that they do not own and receives a paid salary or wage.
1 | PWSE | Paid work, self-employed | A situation in which an individual earns a salary or wage working for himself or herself instead of working for an employer.
1 | PWSG | Paid work, state government, not armed forces | A situation in which an individual works for a government organization with jurisdiction immediately below the level of national government (between national government and local government), not armed forces and receives a paid salary or wage. Often called a state, provincial, territorial, or tribal government.
1 | UWNSE | Unpaid non-governmental work, not self-employed | A situation in which an individual works for a business (not government) that they do not own without receiving a paid salary or wage.
1 | UWSE | Unpaid work, self-employed | A situation in which an individual works for himself or herself without receiving a paid salary or wage.
1 | VW | Voluntary work | A situation in which an individual chooses to do something, especially for other people or for an organization, willingly and without being forced or compensated to do it. This can include formal activity undertaken through public, private and voluntary organizations as well as informal community participation.

4.4.2.10199 HL7 v3 Value Set ActMoodIntent

URL: https://hl7.org/fhir/R4/v3/ActMoodIntent/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActMoodIntent) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActMoodIntent
Version: | 2014-03-26
Name: | v3.ActMoodIntent
Title: | V3 Value SetActMoodIntent
Definition: | An intention or plan to perform a service. Historical note: in previous RIM versions, the intent mood was captured as a separate class hierarchy, called Service_intent_or_order.
OID: | 2.16.840.1.113883.1.11.10199 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.10199.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActMood where concept is-a INT

4.4.2.10199.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 8 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActMood version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActMood

Lvl | Code | Display | Definition
0 | INT | intent | Definition: An intention or plan for an act. >UsageNotes: The final outcome of the intent, the act that is intended to occur, is always an event. However the final outcome may be reached indirectly via steps through other intents, such as promise, permission request, or an appointment that may lead to an actual event to occur. Alternatively, the intended act may never occur.
1 | _ActMoodDesire | desire | Definition: A desire to have an act occur.
2 | _ActMoodActRequest | act request | Definition: A request (or order) for an act that is part of a defined request/fulfillment cycle. UsageNotes: Use of an HL7 defined request/fulfillment framework is not required to use this mood code.

3 | ARQ | appointment request | Definition: A request act that is specialized for the appointment scheduling request/fulfillment cycle. An appointment request is fulfilled only and completely by an appointment (APT), i.e., all that the appointment request intends is to create an appointment (the actual act may well not happen if that is the professional decision during the appointment).

3 | PERMRQ | permission request | Definition: A request for a permission to perform the act. Typically a payer (or possibly a supervisor) is being requested to give permission to perform the act. As opposed to the RQO, the requestee is not asked to perform or cause to perform the act but only to give the permission.

3 | RQO | request | Definition: A request act that is specialized for an event request/fulfillment cycle. UsageNotes: The fulfillment cycle may involve intermediary fulfilling acts in moods such as PRMS, APT, or even another RQO before being fulfilled by the final event. UsageNotes: The concepts of a "request" and an "order" are viewed as different, because there is an implication of a mandate associated with order. In practice, however, this distinction has no general functional value in the inter-operation of health care computing. "Orders" are commonly refused for a variety of clinical and business reasons, and the notion of a "request" obligates the recipient (the fulfiller) to respond to the sender (the author). Indeed, in many regions, including Australia and Europe, the common term used is "request." Thus, the concept embodies both notions, as there is no useful distinction to be made. If a mandate is to be associated with a request, this will be embodied in the "local" business rules applied to the transactions. Should HL7 desire to provide a distinction between these in the future, the individual concepts could be added as specializations of this concept. The critical distinction here, is the difference between this concept and an "intent", of which it is a specialization. An intent involves decisions by a single party, the author. A request, however, involves decisions by two parties, the author and the fulfiller, with an obligation on the part of the fulfiller to respond to the request indicating that the fulfiller will indeed fulfill the request.

2 | PRP | proposal | Definition: A suggestion that an act might be performed. Not an explicit request, and professional responsibility may or may not be present.

3 | RMD | recommendation | Definition: A suggestion that an act should be performed with an acceptance of some degree of professional responsibility for the resulting act. Not an explicit request. . UsageNotes: Where there is no clear definition or applicable concept of "professional responsibilityâ€?, RMD becomes indistinguishable from PRP. .

1 | PRMS | promise | Definition: A commitment to perform an act (may be either solicited or unsolicited). The committer becomes responsible to the other party for executing the act, and, as a consequence, the other party may rely on the first party to perform or cause to perform the act. UsageNotes: Commitments may be retracted or cancelled.

2 | APT | appointment | Definition: An act that has been scheduled to be performed at a specific place and time.

4.4.2.10202 HL7 v3 Value Set ActMoodPredicate

URL: https://hl7.org/fhir/R4/v3/ActMoodPredicate/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActMoodPredicate) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActMoodPredicate

Version: | 2014-03-26

Name: | v3.ActMoodPredicate

Title: | V3 Value SetActMoodPredicate

Definition: | Any of the above service moods (e.g., event, intent, or goal) can be turned into a predicate used as a criterion to express conditionals (or queries.) However, currently we allow only criteria on service events.

OID: | 2.16.840.1.113883.1.11.10202 (for OID based terminology systems)

Source Resource | XML / JSON

This value set is not currently used

4.4.2.10202.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActMood where concept is-a _ActMoodPredicate

http://terminology.hl7.org/CodeSystem/v3-ActMood

Code | Display

_ActMoodPredicate | ActMoodPredicate | Definition: An act that expresses condition statements for other acts.

4.4.2.10202.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 9 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActMood version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActMood

Code | Display | Definition

GOL.CRT | goal criterion | A criterion expressed over goals (ActMood.GOL).

INT.CRT | intent criterion | A criterion expressed over intents (ActMood.INT).

PRMS.CRT | promise criterion | A criterion expressed over promises (ActMood.PRMS).

RQO.CRT | request criterion | A criterion expressed over requests or orders (ActMood.RQO).

RSK.CRT | risk criterion | A criterion expressed over risks (ActMood.RSK).

EXPEC | expectation | Definition: An act that is considered to have some noteworthy likelihood of occurring in the future (has_match = event). Examples:Prognosis of a condition, Expected date of discharge from hospital, patient will likely need an emergency decompression of the intracranial pressure by morning. UsageNotes:INT (intent) reflects a plan for the future, which is a declaration to do something. This contrasts with expectation, which is a prediction that something will happen in the future. GOL

(goal) reflects a hope rather than a prediction. RSK (risk) reflects a potential negative event that may or may not be expected to happen.

GOL | Goal | Definition: An expectation that is considered to be desirable to occur in the future Examples:Target weight below 80Kg, Stop smoking, Regain ability to walk, goal is to administer thrombolytics to candidate patients presenting with acute myocardial infarction. UsageNotes: INT (intent) reflects a plan for the future, which is a declaration to do something. This contrasts with goal which doesn't represent an intention to act, merely a hope for an eventual result. A goal is distinct from the intended actions to reach that goal. "I will reduce the dose of drug x to 20mg" is an intent. "I hope to be able to get the patient to the point where I can reduce the dose of drug x to 20mg" is a goal. EXPEC (expectation) reflects a prediction rather than a hope. RSK (risk) reflects a potential negative event rather than a hope.

RSK | risk | Definition:An act that may occur in the future and which is regarded as undesirable Examples:Increased risk of DVT, at risk for sub-acute bacterial endocarditis. UsageNotes:Note: An observation in RSK mood expresses the undesirable act, and not the underlying risk factor. A risk factor that is present (e.g. obesity, smoking, etc) should be expressed in event mood. INT (intent) reflects a plan for the future, which is a declaration to do something. This contrasts with RSK (risk), which is the potential that something negative will occur that may or may not ever happen. GOL (goal) reflects a hope to achieve something. EXPEC (expectation) is the prediction of a positive or negative event. This contrasts with RSK (risk), which is the potential that something negative will occur that may or may not ever happen, and may not be expected to happen.

OPT | option | Definition: One of a set of acts that specify an option for the property values that the parent act may have. Typically used in definitions or orders to describe alternatives. An option can only be used as a group, that is, all assigned values must be used together. The actual mood of the act is the same as the parent act, and they must be linked by an actrelationship with type = OPTN.

4.4.2.10228 HL7 v3 Value Set ConfidentialityClassification

URL: https://hl7.org/fhir/R4/v3/ConfidentialityClassification/vs.html

Vocabulary Work Group | Maturity Level: 2 | Trial Use | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ConfidentialityClassification) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ConfidentialityClassification

Version: | 2014-03-26

Name: | v3.ConfidentialityClassification

Title: | V3 Value SetConfidentialityClassification

Definition: | Set of codes used to value Act.Confidentiality and Role.Confidentiality attribute in accordance with the definition for concept domain "Confidentiality".

OID: | 2.16.840.1.113883.1.11.10228 (for OID based terminology systems)

Source Resource | XML / JSON

This value set is used in the following places:

• Resource: Composition.confidentiality (code / Required)

4.4.2.10228.1 Content Logical Definition

This value set includes codes from the following code systems:

http://terminology.hl7.org/CodeSystem/v3-Confidentiality

Code | Display

U | unrestricted | Definition: Privacy metadata indicating that the information is not classified as sensitive. Examples: Includes publicly available information, e.g., business name, phone, email or physical address. Usage Note: This metadata indicates that the receiver has no obligation to consider additional policies when making access control decisions. Note that in some jurisdictions, personally identifiable information must be protected as confidential, so it would not be appropriate to assign a confidentiality code of "unrestricted" to that information even if it is publicly available.

L | low | Definition: Privacy metadata indicating that the information has been de-identified, and there are mitigating circumstances that prevent re-identification, which minimize risk of harm from unauthorized disclosure. The information requires protection to maintain low sensitivity. Examples: Includes anonymized, pseudonymized, or non-personally identifiable information such as HIPAA limited data sets. Map: No clear map to ISO 13606-4 Sensitivity Level (1) Care Management: RECORD_COMPONENTs that might need to be accessed by a wide range of administrative staff to manage the subject of care's access to health services. Usage Note: This metadata indicates the receiver may have an obligation to comply with a data use agreement.

M | moderate | Definition: Privacy metadata indicating moderately sensitive information, which presents moderate risk of harm if disclosed without authorization. Examples: Includes allergies of non-sensitive nature used inform food service; health information a patient authorizes to be used for marketing, released to a bank for a health credit card or savings account; or information in personal health record systems that are not governed under health privacy laws. Map: Partial Map to ISO 13606-4 Sensitivity Level (2) Clinical Management: Less sensitive RECORD_COMPONENTs that might need to be accessed by a wider range of personnel not all of whom are actively caring for the patient (e.g. radiology staff). Usage Note: This metadata indicates that the receiver may be obligated to comply with the receiver's terms of use or privacy policies.

N | normal | Definition: Privacy metadata indicating that the information is typical, non-stigmatizing health information, which presents typical risk of harm if disclosed without authorization. Examples: In the US, this includes what HIPAA identifies as the minimum necessary protected health information (PHI) given a covered purpose of use (treatment, payment, or operations). Includes typical, non-stigmatizing health information disclosed in an application for health, workers compensation, disability, or life insurance. Map: Partial Map to ISO 13606-4 Sensitivity Level (3) Clinical Care: Default for normal clinical care access (i.e. most clinical staff directly caring for the patient should be able to access nearly all of the EHR). Maps to normal confidentiality for treatment information but not to ancillary care, payment and operations. Usage Note: This metadata indicates that the receiver may be obligated to comply with applicable jurisdictional privacy law or disclosure authorization.

R | restricted | Privacy metadata indicating highly sensitive, potentially stigmatizing information, which presents a high risk to the information subject if disclosed without authorization. May be pre-empted by jurisdictional law, e.g., for public health reporting or emergency treatment. Examples: Includes information that is additionally protected such as sensitive conditions mental health,

HIV, substance abuse, domestic violence, child abuse, genetic disease, and reproductive health; or sensitive demographic information such as a patient's standing as an employee or a celebrity. May be used to indicate proprietary or classified information that is not related to an individual, e.g., secret ingredients in a therapeutic substance; or the name of a manufacturer. Map: Partial Map to ISO 13606-4 Sensitivity Level (3) Clinical Care: Default for normal clinical care access (i.e. most clinical staff directly caring for the patient should be able to access nearly all of the EHR). Maps to normal confidentiality for treatment information but not to ancillary care, payment and operations.. Usage Note: This metadata indicates that the receiver may be obligated to comply with applicable, prevailing (default) jurisdictional privacy law or disclosure authorization..

V | very restricted | . Privacy metadata indicating that the information is extremely sensitive and likely stigmatizing health information that presents a very high risk if disclosed without authorization. This information must be kept in the highest confidence. Examples: Includes information about a victim of abuse, patient requested information sensitivity, and taboo subjects relating to health status that must be discussed with the patient by an attending provider before sharing with the patient. May also include information held under â€œlegal lockâ€? or attorney-client privilege Map: This metadata indicates that the receiver may not disclose this information except as directed by the information custodian, who may be the information subject. Usage Note: This metadata indicates that the receiver may not disclose this information except as directed by the information custodian, who may be the information subject.

4.4.2.10248 HL7 v3 Value Set ParticipationPhysicalPerformer

URL: https://hl7.org/fhir/R4/v3/ParticipationPhysicalPerformer/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ParticipationPhysicalPerformer) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ParticipationPhysicalPerformer
Version: | 2014-03-26
Name: | v3.ParticipationPhysicalPerformer
Title: | V3 Value SetParticipationPhysicalPerformer
Definition: | A person who actually and principally carries out the action. Need not be the principal responsible actor, e.g. a surgery resident operating under supervision of attending surgeon, and may be the patient in self-care, e.g. fingerstick blood sugar. The traditional order filler is a performer. This information should accompany every service event.
OID: | 2.16.840.1.113883.1.11.10248 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.10248.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ParticipationType where concept is-a PRF

4.4.2.10248.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 4 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ParticipationType version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ParticipationType

Lvl | Code | Display | Definition

0 | PRF | performer | Definition: A person, non-person living subject, organization or device that who actually and principally carries out the action. Device should only be assigned as a performer in circumstances where the device is performing independent of human intervention. Need not be the principal responsible actor. Exampe: A surgery resident operating under supervision of attending surgeon, a search and rescue dog locating survivors, an electronic laboratory analyzer or the laboratory discipline requested to perform a laboratory test. The performer may also be the patient in self-care, e.g. fingerstick blood sugar. The traditional order filler is a performer. This information should accompany every service event. Note: that existing HL7 designs assign an organization as the playing entity of the Role that is the performer. These designs should be revised in subsequent releases to make this the scooping entity for the role involved.

1 | DIST | distributor | Distributes material used in or generated during the act.

1 | PPRF | primary performer | The principal or primary performer of the act.

1 | SPRF | secondary performer | A person assisting in an act through his substantial presence and involvement This includes: assistants, technicians, associates, or whatever the job titles may be.

4.4.2.10251 HL7 v3 Value Set ParticipationInformationGenerator

URL: https://hl7.org/fhir/R4/v3/ParticipationInformationGenerator/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ParticipationInformationGenerator) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ParticipationInformationGenerator
Version: | 2014-03-26

Name: | v3.ParticipationInformationGenerator
Title: | V3 Value SetParticipationInformationGenerator
Definition: | Parties that may or should contribute or have contributed information to the Act. Such information includes information leading to the decision to perform the Act and how to perform the Act (e.g., consultant), information that the Act itself seeks to reveal (e.g., informant of clinical history), or information about what Act was performed (e.g., informant witness).
OID: | 2.16.840.1.113883.1.11.10251 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.10251.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ParticipationType where concept is-a _ParticipationInformationGenerator

http://terminology.hl7.org/CodeSystem/v3-ParticipationType

Code | Display
_ParticipationInformationGenerator | ParticipationInformationGenerator | Parties that may or should contribute or have contributed information to the Act. Such information includes information leading to the decision to perform the Act and how to perform the Act (e.g., consultant), information that the Act itself seeks to reveal (e.g., informant of clinical history), or information about what Act was performed (e.g., informant witness).

4.4.2.10251.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 5 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ParticipationType version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ParticipationType

Code | Display | Definition
AUT | author (originator) | Definition: A party that originates the Act and therefore has responsibility for the information given in the Act and ownership of this Act. Example: the report writer, the person writing the act definition, the guideline author, the placer of an order, the EKG cart (device) creating a report etc. Every Act should have an author. Authorship is regardless of mood always actual authorship. Examples of such policies might include: The author and anyone they explicitly delegate may update the report; All administrators within the same clinic may cancel and reschedule appointments created by other administrators within that clinic; A party that is neither an author nor a party who is extended authorship maintenance rights by policy, may only amend, reverse, override, replace, or follow up in other ways on this Act, whereby the Act remains intact and is linked to another Act authored by that other party.
INF | informant | A source of reported information (e.g., a next of kin who answers questions about the patient's history). For history questions, the patient is logically an informant, yet the informant of history questions is implicitly the subject.
TRANS | Transcriber | An entity entering the data into the originating system. The data entry entity is collected optionally for internal quality control purposes. This includes the transcriptionist for dictated text transcribed into electronic form.
ENT | data entry person | A person entering the data into the originating system. The data entry person is collected optionally for internal quality control purposes. This includes the transcriptionist for dictated text.
WIT | witness | Only with service events. A person witnessing the action happening without doing anything. A witness is not necessarily aware, much less approves of anything stated in the service event. Example for a witness is students watching an operation or an advanced directive witness.

4.4.2.10259 HL7 v3 Value Set ParticipationVerifier

URL: https://hl7.org/fhir/R4/v3/ParticipationVerifier/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ParticipationVerifier) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ParticipationVerifier
Version: | 2014-03-26
Name: | v3.ParticipationVerifier
Title: | V3 Value SetParticipationVerifier
Definition: | A person who verifies the correctness and appropriateness of the service (plan, order, event, etc.) and hence takes on accountability.
OID: | 2.16.840.1.113883.1.11.10259 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.10259.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ParticipationType where concept is-a VRF

4.4.2.10259.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 3 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ParticipationType version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ParticipationType

Lvl | Code | Display | Definition

0 | VRF | verifier | A person who verifies the correctness and appropriateness of the service (plan, order, event, etc.) and hence takes on accountability.

1 | AUTHEN | authenticator | A verifier who attests to the accuracy of an act, but who does not have privileges to legally authenticate the act. An example would be a resident physician who sees a patient and dictates a note, then later signs it. Their signature constitutes an authentication.

1 | LA | legal authenticator | A verifier who legally authenticates the accuracy of an act. An example would be a staff physician who sees a patient and dictates a note, then later signs it. Their signature constitutes a legal authentication.

4.4.2.10286 HL7 v3 Value Set ParticipationTargetDirect

URL: https://hl7.org/fhir/R4/v3/ParticipationTargetDirect/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ParticipationTargetDirect) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ParticipationTargetDirect

Version: | 2014-03-26

Name: | v3.ParticipationTargetDirect

Title: | V3 Value SetParticipationTargetDirect

Definition: | Target that is substantially present in the service and which is directly affected by the service action (includes consumed material, devices, etc.).

OID: | 2.16.840.1.113883.1.11.10286 (for OID based terminology systems)

Source Resource | XML / JSON

This value set is not currently used

4.4.2.10286.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ParticipationType where concept is-a DIR

4.4.2.10286.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 17 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ParticipationType version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ParticipationType

Lvl | Code | Display | Definition

0 | DIR | direct target | Target participant that is substantially present in the act and which is directly involved in the action (includes consumed material, devices, etc.).

1 | ALY | analyte | The target of an Observation action. Links an observation to a Role whose player is the substance or most specific component entity (material, micro-organism, etc.) being measured within the subject. Examples: A "plasma porcelain substance concentration" has analyte a Role with player substance Entity "porcelain". UsageNotes: The Role that this participation connects to may be any Role whose player is that substance measured. Very often, the scoper may indicate the system in which the component is being measured. E.g., for "plasma porcelain" the scoper could be "Plasma".

1 | BBY | baby | In an obstetric service, the baby.

1 | CAT | catalyst | The catalyst of a chemical reaction, such as an enzyme or a platinum surface. In biochemical reactions, connects the enzyme with the molecular interaction

1 | CSM | consumable | Participant material that is taken up, diminished, altered, or disappears in the act.

2 | TPA | therapeutic agent | Something incorporated in the subject of a therapy service to achieve a physiologic effect (e.g., heal, relieve, provoke a condition, etc.) on the subject. In an administration service the therapeutic agent is a consumable, in a preparation or dispense service, it is a product. Thus, consumable or product must be specified in accordance with the kind of service.

1 | DEV | device | Participant used in performing the act without being substantially affected by the act (i.e. durable or inert with respect to that particular service). Examples: monitoring equipment, tools, but also access/drainage lines, prostheses, pace maker, etc.

2 | NRD | non-reuseable device | A device that changes ownership due to the service, e.g., a pacemaker, a prosthesis, an insulin injection equipment (pen), etc. Such material may need to be restocked after he service.

2 | RDV | reusable device | A device that does not change ownership due to the service, i.e., a surgical instrument or tool or an endoscope. The distinction between reuseable and non-reuseable must be made in order to know whether material must be re-stocked.

1 | DON | donor | In some organ transplantation services and rarely in transfusion services a donor will be a target participant in the service. However, in most cases transplantation is decomposed in three services: explantation, transport, and implantation. The identity of the donor (recipient) is often irrelevant for the explantation (implantation) service.

1 | EXPAGNT | ExposureAgent | Description: The entity playing the associated role is the physical (including energy), chemical or biological substance that is participating in the exposure. For example in communicable diseases, the associated playing entity is the disease causing pathogen.

1 | EXPART | ExposureParticipation | Description:Direct participation in an exposure act where it is unknown that the participant is the source or subject of the exposure. If the participant is known to be the contact of an exposure then the SBJ participation type should be used. If the participant is known to be the source then the EXSRC participation type should be used.
2 | EXPTRGT | ExposureTarget | Description: The entity playing the associated role is the target (contact) of exposure.
2 | EXSRC | ExposureSource | Description:The entity playing the associated role is the source of exposure.
1 | PRD | product | Participant material that is brought forth (produced) in the act (e.g., specimen in a specimen collection, access or drainage in a placement service, medication package in a dispense service). It does not matter whether the material produced had existence prior to the service, or whether it is created in the service (e.g., in supply services the product is taken from a stock).
1 | SBJ | subject | The principle target on which the action happens. Examples: The patient in physical examination, a specimen in a lab observation. May also be a patient's family member (teaching) or a device or room (cleaning, disinfecting, housekeeping). UsageNotes: Not all direct targets are subjects. Consumables and devices used as tools for an act are not subjects. However, a device may be a subject of a maintenance action.
2 | SPC | specimen | The subject of non-clinical (e.g. laboratory) observation services is a specimen.

4.4.2.10302 HL7 v3 Value Set ParticipationTargetLocation

URL: https://hl7.org/fhir/R4/v3/ParticipationTargetLocation/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ParticipationTargetLocation) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ParticipationTargetLocation
Version: | 2014-03-26
Name: | v3.ParticipationTargetLocation
Title: | V3 Value SetParticipationTargetLocation
Definition: | The facility where the service is done. May be a static building (or room therein) or a moving location (e.g., ambulance, helicopter, aircraft, train, truck, ship, etc.)
OID: | 2.16.840.1.113883.1.11.10302 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.10302.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ParticipationType where concept is-a LOC

4.4.2.10302.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 6 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ParticipationType version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ParticipationType

Lvl | Code | Display | Definition
0 | LOC | location | The facility where the service is done. May be a static building (or room therein) or a moving location (e.g., ambulance, helicopter, aircraft, train, truck, ship, etc.)
1 | DST | destination | The destination for services. May be a static building (or room therein) or a movable facility (e.g., ship).
1 | ELOC | entry location | A location where data about an Act was entered.
1 | ORG | origin | The location of origin for services. May be a static building (or room therein) or a movable facility (e.g., ship).
1 | RML | remote | Some services take place at multiple concurrent locations (e.g., telemedicine, telephone consultation). The location where the principal performing actor is located is taken as the primary location (LOC) while the other location(s) are considered "remote."
1 | VIA | via | For services, an intermediate location that specifies a path between origin an destination.

4.4.2.10318 HL7 v3 Value Set ActRelationshipHasComponent

URL: https://hl7.org/fhir/R4/v3/ActRelationshipHasComponent/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActRelationshipHasComponent) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActRelationshipHasComponent
Version: | 2014-03-26
Name: | v3.ActRelationshipHasComponent
Title: | V3 Value SetActRelationshipHasComponent
Definition: | A collection of sub-services as steps or subtasks performed for the source service. Services may be performed sequentially or concurrently.
OID: | 2.16.840.1.113883.1.11.10318 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.10318.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType where concept is-a COMP

4.4.2.10318.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 7 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType

Lvl | Code | Display | Definition
0 | COMP | has component | The target act is a component of the source act, with no semantics regarding composition or aggregation implied.
1 | CTRLV | has control variable | A relationship from an Act to a Control Variable. For example, if a Device makes an Observation, this relates the Observation to its Control Variables documenting the device's settings that influenced the observation.
1 | MBR | has member | The target Acts are aggregated by the source Act. Target Acts may have independent existence, participate in multiple ActRelationships, and do not contribute to the meaning of the source. UsageNotes: This explicitly represents the conventional notion of aggregation. The target Act is part of a collection of Acts (no implication is made of cardinality, a source of Acts may contain zero, one, or more member target Acts). It is expected that this will be primarily used with _ActClassRecordOrganizer, BATTERY, and LIST
2 | STEP | has step | A collection of sub-services as steps or subtasks performed for the source service. Services may be performed sequentially or concurrently. UsageNotes: Sequence of steps may be indicated by use of _ActRelationshipTemporallyPertains, as well as via ActRelationship.sequenceNumber, ActRelationship.pauseQuantity, Target.priorityCode. OpenIssue: Need Additional guidelines on when each approach should be used.
3 | ARR | arrival | The relationship that links to a Transportation Act (target) from another Act (source) indicating that the subject of the source Act entered into the source Act by means of the target Transportation act.
3 | DEP | departure | The relationship that links to a Transportation Act (target) from another Act (source) indicating that the subject of the source Act departed from the source Act by means of the target Transportation act.
1 | PART | has part | The source Act is a composite of the target Acts. The target Acts do not have an existence independent of the source Act. UsageNote: In UML 1.1, this is a "composition" defined as: "A form of aggregation with strong ownership and coincident lifetime as part of the whole. Parts with non-fixed multiplicity may be created after the composite itself, but once created they live and die with it (i.e., they share lifetimes). Such parts can also be explicitly removed before the death of the composite. Composition may be recursive."

4.4.2.10329 HL7 v3 Value Set ActRelationshipPertains

URL: https://hl7.org/fhir/R4/v3/ActRelationshipPertains/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActRelationshipPertains) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActRelationshipPertains
Version: | 2014-03-26
Name: | v3.ActRelationshipPertains
Title: | V3 Value SetActRelationshipPertains
Definition: | This is a very unspecific relationship from one item of clinical information to another. It does not judge about the role the pertinent information plays.
OID: | 2.16.840.1.113883.1.11.10329 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.10329.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType where concept is-a PERT

4.4.2.10329.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 1 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType

Code | Display | Definition
PERT | has pertinent information | This is a very unspecific relationship from one item of clinical information to another. It does not judge about the role the pertinent information plays.

4.4.2.10342 HL7 v3 Value Set ActRelationshipFulfills

URL: https://hl7.org/fhir/R4/v3/ActRelationshipFulfills/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActRelationshipFulfills) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActRelationshipFulfills
Version: | 2014-03-26
Name: | v3.ActRelationshipFulfills
Title: | V3 Value SetActRelationshipFulfills
Definition: | The source act fulfills (in whole or in part) the target act. Source act must be in a mood equal or more actual than the target act.
OID: | 2.16.840.1.113883.1.11.10342 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.10342.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType where concept is-a FLFS

4.4.2.10342.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 4 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType

Lvl | Code | Display | Definition
0 | FLFS | fulfills | The source act fulfills (in whole or in part) the target act. Source act must be in a mood equal or more actual than the target act.
1 | OCCR | occurrence | The source act is a single occurrence of a repeatable target act. The source and target act can be in any mood on the "completion track" but the source act must be as far as or further along the track than the target act (i.e., the occurrence of an intent can be an event but not vice versa).
1 | OREF | references order | Relates either an appointment request or an appointment to the order for the service being scheduled.
1 | SCH | schedules request | Associates a specific time (and associated resources) with a scheduling request or other intent.

4.4.2.10416 HL7 v3 Value Set RoleClassRelationshipFormal

URL: https://hl7.org/fhir/R4/v3/RoleClassRelationshipFormal/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-RoleClassRelationshipFormal) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-RoleClassRelationshipFormal
Version: | 2014-03-26
Name: | v3.RoleClassRelationshipFormal
Title: | V3 Value SetRoleClassRelationshipFormal
Definition: | A relationship between two entities that is formally recognized, frequently by a contract or similar agreement.
OID: | 2.16.840.1.113883.1.11.10416 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.10416.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-RoleClass where concept is-a _RoleClassRelationshipFormal

http://terminology.hl7.org/CodeSystem/v3-RoleClass

Code | Display
_RoleClassRelationshipFormal | RoleClassRelationshipFormal | A relationship between two entities that is formally recognized, frequently by a contract or similar agreement.

4.4.2.10416.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 36 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-RoleClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-RoleClass

Code | Display | Definition

AFFL | affiliate | Player of the Affiliate role has a business/professional relationship with scoper. Player and scoper may be persons or organization. The Affiliate relationship does not imply membership in a group, nor does it exist for resource scheduling purposes. Example: A healthcare provider is affiliated with another provider as a business associate.

AGNT | agent | An entity (player) that acts or is authorized to act on behalf of another entity (scoper).

ASSIGNED | assigned entity | An agent role in which the agent is an Entity acting in the employ of an organization. The focus is on functional role on behalf of the organization, unlike the Employee role where the focus is on the 'Human Resources' relationship between the employee and the organization.

COMPAR | commissioning party | An Entity that is authorized to issue or instantiate permissions, privileges, credentials or other formal/legal authorizations.

SGNOFF | signing authority or officer | The role of a person (player) who is the officer or signature authority for of a scoping entity, usually an organization (scoper).

CON | contact | A person or an organization (player) which provides or receives information regarding another entity (scoper). Examples; patient NOK and emergency contacts; guarantor contact; employer contact.

ECON | emergency contact | An entity to be contacted in the event of an emergency.

NOK | next of kin | An individual designated for notification as the next of kin for a given entity.

GUARD | guardian | Guardian of a ward

CIT | citizen | Citizen of apolitical entity

COVPTY | covered party | A role class played by a person who receives benefit coverage under the terms of a particular insurance policy. The underwriter of that policy is the scoping entity. The covered party receives coverage because of some contractual or other relationship with the holder of that policy. Discussion:This reason for coverage is captured in 'Role.code' and a relationship link with type code of indirect authority should be included using the policy holder role as the source, and the covered party role as the target. Note that a particular policy may cover several individuals one of whom may be, but need not be, the policy holder. Thus the notion of covered party is a role that is distinct from that of the policy holder.

CLAIM | claimant | Description: A role played by a party making a claim for coverage under a policy or program. A claimant must be either a person or organization, or a group of persons or organizations. A claimant is not a named insured or a program eligible. Discussion: With respect to liability insurance such as property and casualty insurance, a claimant must file a claim requesting indemnification for a loss that the claimant considers covered under the policy of a named insured. The claims adjuster for the policy underwriter will review the claim to determine whether the loss meets the benefit coverage criteria under a policy, and base any indemnification or coverage payment on that review. If a third party is liable in whole or part for the loss, the underwriter may pursue third party liability recovery. A claimant may be involved in civil or criminal legal proceedings involving claims against a defendant party that is indemnified by an insurance policy or to protest the finding of a claims adjustor. With respect to life insurance, a beneficiary designated by a named insured becomes a claimant of the proceeds of coverage, as in the case of a life insurance policy. However, a claimant for coverage under life insurance is not necessarily a designated beneficiary. Note: A claimant is not a named insured. However, a named insured may make a claim under a policy, e.g., an insured driver may make a claim for an injury under his or her comprehensive automobile insurance policy. Similarly, a program eligible may make a claim under program, e.g., an unemployed worker may claim benefits under an unemployment insurance program, but parties playing these covered party role classes are not, for purposes of this vocabulary and in an effort to clearly distinguish role classes, considered claimants. In the case of a named insured making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that either a named insured or an individual insured has filed a claim for a loss. In the case of a program eligible, a role type code INJWKR (injured worker) subtypes the class to indicate that the covered party in a workers compensation program is an injured worker, and as such, has filed a "claim" under the program for benefits. Likewise, a covered role type code UNEMP (unemployed worker) subtypes the program eligible class to indicate that the covered party in an unemployment insurance program has filed a claim for unemployment benefits. Example: A claimant under automobile policy that is not the named insured.

NAMED | named insured | Description: A role played by a party to an insurance policy to which the insurer agrees to indemnify for losses, provides benefits for, or renders services. A named insured may be either a person, non-person living subject, or an organization, or a group of persons, non-person living subjects, or organizations. Discussion: The coded concept NAMED should not be used where a more specific child concept in this Specializable value set applies. In some cases, the named insured may not be the policy holder, e.g., where a policy holder purchases life insurance policy in which another party is the named insured and the policy holder is the beneficiary of the policy. Note: The party playing the role of a named insured is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a named insured may make a claim under a policy, e.g., e.g., a party that is the named insured and policy holder under a comprehensive automobile insurance policy may become the claimant for coverage under that policy e.g., if injured in an automobile accident and there is no liable third party. In the case of a named insured making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that a named insured has filed a claim for a loss. Example: The named insured under a comprehensive automobile, disability, or property and casualty policy that is the named insured and may or may not be the policy holder.

DEPEN | dependent | Description: A role played by a person covered under a policy or program based on an association with a subscriber, which is recognized by the policy holder. Note: The party playing the role of a dependent is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a dependent may make a claim under a policy, e.g., a dependent under a health insurance policy may become the claimant for coverage under that policy for wellness examines or if injured and there is no liable third party. In the case of a dependent making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that the dependent has filed a claim for services covered under the health insurance policy. Example: The dependent has an association with the subscriber such as a financial dependency or personal relationship such as that of a spouse, or a natural or adopted child. The policy holder may be required by law to recognize certain associations or may have discretion about the associations. For example, a policy holder may dictate the criteria for the dependent status of adult children who are students, such as requiring full time enrollment, or may recognize domestic partners as dependents. Under certain circumstances, the dependent may be under the indirect authority of a responsible party acting as a surrogate for the subscriber, for example, if the subscriber is differently abled or deceased, a guardian ad Lidem or estate executor may be appointed to assume the subscriberaTMs legal standing in the relationship with the dependent.

INDIV | individual | Description: A role played by a party covered under a policy as the policy holder. An individual may be either a person or an organization. Note: The party playing the role of an individual insured is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a named insured may make a claim under a policy, e.g., a party that is the named insured and policy holder under a comprehensive automobile insurance policy may become the claimant for coverage under that policy if injured in an automobile accident and there is no liable third party. In the case of an individual insured making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that an individual insured has filed a claim for a

loss. Example: The individual insured under a comprehensive automobile, disability, or property and casualty policy that is the policy holder.

SUBSCR | subscriber | Description: A role played by a person covered under a policy based on association with a sponsor who is the policy holder, and whose association may provide for the eligibility of dependents for coverage. Discussion: The policy holder holds the contract with the policy or program underwriter. The subscriber holds a certificate of coverage under the contract. In legal proceedings concerning the policy or program, the terms of the contract takes precedence over the terms of the certificate of coverage if there are any inconsistencies. Note: The party playing the role of a subscriber is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a subscriber may make a claim under a policy, e.g., a subscriber under a health insurance policy may become the claimant for coverage under that policy for wellness examines or if injured and there is no liable third party. In the case of a subscriber making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that the subscriber has filed a claim for services covered under the health insurance policy. Example: An employee or a member of an association.

PROG | program eligible | Description: A role played by a party that meets the eligibility criteria for coverage under a program. A program eligible may be either a person, non-person living subject, or an organization, or a group of persons, non-person living subjects, or organizations. Discussion: A program as typically government administered coverage for parties determined eligible under the terms of the program. Note: The party playing a program eligible is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However a program eligible may make a claim under program, e.g., an unemployed worker may claim benefits under an unemployment insurance program, but parties playing these covered party role classes are not, for purposes of this vocabulary and in an effort to clearly distinguish role classes, considered claimants. In the case of a program eligible, a role type code INJWKR (injured worker) subtypes the class to indicate that the covered party in a workers compensation program is an injured worker, and as such, has filed a "claim" under the program for benefits. Likewise, a covered role type code UNEMP (unemployed worker) subtypes the program eligible class to indicate that the covered party in an unemployment insurance program has filed a claim for unemployment benefits. Example: A party meeting eligibility criteria related to health or financial status, e.g., in the U.S., persons meeting health, demographic, or financial criteria established by state and federal law are eligible for Medicaid.

CRINV | clinical research investigator | A role played by a provider, always a person, who has agency authority from a Clinical Research Sponsor to direct the conduct of a clinical research trial or study on behalf of the sponsor.

CRSPNSR | clinical research sponsor | A role played by an entity, usually an organization, that is the sponsor of a clinical research trial or study. The sponsor commissions the study, bears the expenses, is responsible for satisfying all legal requirements concerning subject safety and privacy, and is generally responsible for collection, storage and analysis of the data generated during the trial. No scoper is necessary, as a clinical research sponsor undertakes the role on its own authority and declaration. Clinical research sponsors are usually educational or other research organizations, government agencies or biopharmaceutical companies.

EMP | employee | A relationship between a person or organization and a person or organization formed for the purpose of exchanging work for compensation. The purpose of the role is to identify the type of relationship the employee has to the employer, rather than the nature of the work actually performed. (Contrast with AssignedEntity.)

MIL | military person | A role played by a member of a military service. Scoper is the military service (e.g. Army, Navy, Air Force, etc.) or, more specifically, the unit (e.g. Company C, 3rd Battalion, 4th Division, etc.)

GUAR | guarantor | A person or organization (player) that serves as a financial guarantor for another person or organization (scoper).

INVSBJ | Investigation Subject | An entity that is the subject of an investigation. This role is scoped by the party responsible for the investigation.

CASEBJ | Case Subject | A person, non-person living subject, or place that is the subject of an investigation related to a notifiable condition (health circumstance that is reportable within the applicable public health jurisdiction)

RESBJ | research subject | Definition:Specifies the administrative functionality within a formal experimental design for which the ResearchSubject role was established. Examples: Screening - role is used for pre-enrollment evaluation portion of the design; enrolled - role is used for subjects admitted to the experimental portion of the design.

LIC | licensed entity | A relationship in which the scoper certifies the player (e. g. a medical care giver, a medical device or a provider organization) to perform certain activities that fall under the jurisdiction of the scoper (e.g., a health authority licensing healthcare providers, a medical quality authority certifying healthcare professionals).

NOT | notary public | notary public

PROV | healthcare provider | An Entity (player) that is authorized to provide health care services by some authorizing agency (scoper).

PAT | patient | A Role of a LivingSubject (player) as an actual or potential recipient of health care services from a healthcare provider organization (scoper). Usage Note: Communication about relationships between patients and specific healthcare practitioners (people) is not done via scoper. Instead this is generally done using the CareProvision act. This allows linkage between patient and a particular healthcare practitioner role and also allows description of the type of care involved in the relationship.

PAYEE | payee | The role of an organization or individual designated to receive payment for a claim against a particular coverage. The scoping entity is the organization that is the submitter of the invoice in question.

PAYOR | invoice payor | The role of an organization that undertakes to accept claims invoices, assess the coverage or payments due for those invoices and pay to the designated payees for those invoices. This role may be either the underwriter or a third-party organization authorized by the underwriter. The scoping entity is the organization that underwrites the claimed coverage.

POLHOLD | policy holder | A role played by a person or organization that holds an insurance policy. The underwriter of that policy is the scoping entity. Discussion:The identifier of the policy is captured in 'Role.id' when the Role is a policy holder. A particular policy may cover several individuals one of whom may be, but need not be, the policy holder. Thus the notion of covered party is a role that is distinct from that of the policy holder.

QUAL | qualified entity | An entity (player) that has been recognized as having certain training/experience or other characteristics that would make said entity an appropriate performer for a certain activity. The scoper is an organization that educates or qualifies entities.

SPNSR | coverage sponsor | A role played by an entity, usually an organization that is the sponsor of an insurance plan or a health program. A sponsor is the party that is ultimately accountable for the coverage by employment contract or by law. A sponsor can be an employer, union, government agency, or association. Fully insured sponsors establish the terms of the plan and contract with health insurance plans to assume the risk and to administer the plan. Self-insured sponsors delegate coverage administration, but not risk, to third-party administrators. Program sponsors designate services to be covered in accordance with statute. Program sponsors may administer the coverage themselves, delegate coverage administration, but not risk to third-party administrators, or contract with health insurance plans to assume the risk and administrator a program. Sponsors qualify

individuals who may become a policy holder of the plan; where the sponsor is the policy holder, who may become a subscriber or a dependent to a policy under the plan; or where the sponsor is a government agency, who may become program eligibles under a program. The sponsor role may be further qualified by the SponsorRole.code. Entities playing the sponsor role may also play the role of a Coverage Administrator. Example: An employer, union, government agency, or association.

STD | student | A role played by an individual who is a student of a school, which is the scoping entity.

UNDWRT | underwriter | A role played by a person or an organization. It is the party that accepts fiscal responsibility for insurance plans and the policies created under those plans; administers and accepts fiscal responsibility for a program that provides coverage for services to eligible individuals; and/or has the responsibility to assess the merits of each risk and decide a suitable premium for accepting all or part of the risk. If played by an organization, this role may be further specified by an appropriate RoleCode. Example: A health insurer; Medicaid Program; Lloyd's of London

4.4.2.10429 HL7 v3 Value Set RoleClassPartitive

URL: https://hl7.org/fhir/R4/v3/RoleClassPartitive/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-RoleClassPartitive) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-RoleClassPartitive

Version: | 2014-03-26

Name: | v3.RoleClassPartitive

Title: | V3 Value SetRoleClassPartitive

Definition: | An association between two Entities where the playing Entity is considered in some way "part" of the scoping Entity, e.g., as a member, component, ingredient, or content. Being "part" in the broadest sense of the word can mean anything from being an integral structural component to a mere incidental temporary association of a playing Entity with a (generally larger) scoping Entity.

OID: | 2.16.840.1.113883.1.11.10429 (for OID based terminology systems)

Source Resource | XML / JSON

This value set is not currently used

4.4.2.10429.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-RoleClass where concept is-a _RoleClassPartitive

http://terminology.hl7.org/CodeSystem/v3-RoleClass

Code | Display

_RoleClassPartitive | RoleClassPartitive | An association between two Entities where the playing Entity is considered in some way "part" of the scoping Entity, e.g., as a member, component, ingredient, or content. Being "part" in the broadest sense of the word can mean anything from being an integral structural component to a mere incidental temporary association of a playing Entity with a (generally larger) scoping Entity.

4.4.2.10429.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 27 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-RoleClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-RoleClass

Code | Display | Definition

CONT | content | Relates a material as the content (player) to a container (scoper). Unlike ingredients, the content and a container remain separate (not mixed) and the content can be removed from the container. A content is not part of an empty container.

EXPAGTCAR | exposure agent carrier | An exposure agent carrier is an entity that is capable of conveying an exposure agent from one entity to another. The scoper of the role must be the exposure agent (e.g., pathogen).

EXPVECTOR | exposure vector | Description: A vector is a living subject that carries an exposure agent. The vector does not cause the disease itself, but exposes targets to the exposure agent. A mosquito carrying malaria is an example of a vector. The scoper of the role must be the exposure agent (e.g., pathogen).

FOMITE | fomite | Description: A fomite is a non-living entity that is capable of conveying exposure agent from one entity to another. A doorknob contaminated with a Norovirus is an example of a fomite. Anyone touching the doorknob would be exposed to the virus. The scoper of the role must be the exposure agent (e.g., pathogen).

INGR | ingredient | Relates a component (player) to a mixture (scoper). E.g., Glucose and Water are ingredients of D5W, latex may be an ingredient in a tracheal tube.

ACTI | active ingredient | Definition: a therapeutically active ingredient (player) in a mixture (scoper), where the mixture is typically a manufactured pharmaceutical. It is unknown if the quantity of such an ingredient is expressed precisely in terms of the playing ingredient substance, or, if it is specified in terms of a closely related substance (active moiety or reference substance).

ACTIB | active ingredient - basis of strength | Description: Active ingredient, where the ingredient substance (player) is itself the "basis of strength", i.e., where the Role.quantity specifies exactly the quantity of the player substance in the medicine formulation. Examples: Lopressor 50 mg actually contains 50 mg of metoprolol succinate, even though the active moiety is metoprolol, but also: Tenormin 50 mg contain 50 mg of atenolol, as free base, i.e., where the active ingredient atenolol is also the active moiety.

ACTIM | active ingredient - moiety is basis of strength | Description: Active ingredient, where not the ingredient substance (player), but itaTMs active moiety is the "basis of strength", i.e., where the Role.quantity specifies the quantity of the player substance's active moiety in the medicine formulation. Examples: 1 mL of Betopic 5mg/mL eye drops contains 5.6 mg betaxolol hydrochloride equivalent to betaxolol base 5 mg.

ACTIR | active ingredient - reference substance is basis of strength | Description: Active ingredient, where not the ingredient substance (player) but another reference substance with the same active moiety, is the "basis of strength", i.e., where the Role.quantity specifies the quantity of a reference substance, similar but different from the player substance's in the medicine formulation. Examples: Toprol-XL 50 mg contains 47.5 mg of metoprolol succinate equivalent to 50 mg of metoprolol tartrate.
ADJV | adjuvant | A component (player) added to enhance the action of an active ingredient (scoper) (in the manner of a catalyst) but which has no active effect in and of itself. Such ingredients are significant in defining equivalence of products in a way that inactive ingredients are not.
ADTV | additive | An ingredient (player) that is added to a base (scoper), that amounts to a minor part of the overall mixture.
BASE | base | A base ingredient (player) is what comprises the major part of a mixture (scoper). E.g., Water in most i.v. solutions, or Vaseline in salves. Among all ingredients of a material, there should be only one base. A base substance can, in turn, be a mixture.
CNTM | contaminant ingredient | An ingredient whose presence is not intended but may not be reasonably avoided given the circumstances of the mixture's nature or origin.
IACT | inactive ingredient | An ingredient which is not considered therapeutically active, e.g., colors, flavors, stabilizers, or preservatives, fillers, or structural components added to an active ingredient in order to facilitate administration of the active ingredient but without being considered therapeutically active. An inactive ingredient need not be biologically inert, e.g., might be active as an allergen or might have a pleasant taste, but is not an essential constituent delivering the therapeutic effect.
COLR | color additive | A substance (player) influencing the optical aspect of material (scoper).
FLVR | flavor additive | A substance (player) added to a mixture (scoper) to make it taste a certain way. In food the use is obvious, in pharmaceuticals flavors can hide disgusting taste of the active ingredient (important in pediatric treatments).
PRSV | preservative | A substance (player) added to a mixture (scoper) to prevent microorganisms (fungi, bacteria) to spoil the mixture.
STBL | stabilizer | A stabilizer (player) added to a mixture (scoper) in order to prevent the molecular disintegration of the main substance.
MECH | mechanical ingredient | An ingredient (player) of a medication (scoper) that is inseparable from the active ingredients, but has no intended chemical or pharmaceutical effect itself, but which may have some systemic effect on the patient. An example is a collagen matrix used as a base for transplanting skin cells. The collagen matrix can be left permanently in the graft site. Because it is of bovine origin, the patient may exhibit allergies or may have cultural objections to its use.
LOCE | located entity | Relates an entity (player) to a location (scoper) at which it is present in some way. This presence may be limited in time.
STOR | stored entity | Relates an entity (player) (e.g. a device) to a location (scoper) at which it is normally found or stored when not used.
MBR | member | A role played by an entity that is a member of a group. The group provides the scope for this role. Among other uses, groups as used in insurance (groups of covered individuals) and in scheduling where resources may be grouped for scheduling and logistical purposes.
PART | part | Definition: an association between two Entities where the playing Entity (the part) is a component of the whole (scoper) in the sense of an integral structural component, that is distinct from other parts in the same whole, has a distinct function in the whole, and, as an effect, the full integrity of the whole depends (to some degree) on the presence of this part, even though the part may often be separable from the whole. Discussion: Part is defined in opposition to (a) ingredient (not separable), (b) content (not a functional component), and (c) member (not functionally distinct from other members).
ACTM | active moiety | The molecule or ion that is responsible for the intended pharmacological action of the drug substance, excluding those appended or associated parts of the molecule that make the molecule an ester, salt (including a salt with hydrogen or coordination bonds), or other noncovalent derivative (such as a complex, chelate, or clathrate). Examples: heparin-sodium and heparin-potassium have the same active moiety, heparin; the active moiety of morphine-hydrochloride is morphine.
SPEC | specimen | A role played by a material entity that is a specimen for an act. It is scoped by the source of the specimen.
ALQT | aliquot | A portion (player) of an original or source specimen (scoper) used for testing or transportation.
ISLT | isolate | A microorganism that has been isolated from other microorganisms or a source matrix.

4.4.2.10871 HL7 v3 Value Set DocumentSectionType

URL: https://hl7.org/fhir/R4/v3/DocumentSectionType/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-DocumentSectionType) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-DocumentSectionType
Version: | 2014-03-26
Name: | v3.DocumentSectionType
Title: | V3 Value SetDocumentSectionType
Definition: | The type of document section. Possible values: review of systems, medical history, family history, microscopic findings, etc.
OID: | 2.16.840.1.113883.1.11.10871 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.10871.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include all codes defined in http://loinc.org

4.4.2.10871.2 Expansion

This expansion generated 01 Nov 2019

This value set has >1000 codes in it. In order to keep the publication size manageable, only a selection (1000 codes) of the whole set of codes is shown

All codes from system http://loinc.org

Code	Display
1-8	Acyclovir [Susceptibility]
10-9	Amdinocillin [Susceptibility] by Serum bactericidal titer
100-8	Cefoperazone [Susceptibility] by Minimum inhibitory concentration (MIC)
1000-9	DBG Ab [Presence] in Serum or Plasma from Blood product unit
10000-8	R wave duration in lead AVR
10001-6	R wave duration in lead I
10002-4	R wave duration in lead II
10003-2	R wave duration in lead III
10004-0	R wave duration in lead V1
10005-7	R wave duration in lead V2
10006-5	R wave duration in lead V3
10007-3	R wave duration in lead V4
10008-1	R wave duration in lead V5
10009-9	R wave duration in lead V6
1001-7	DBG Ab [Presence] in Serum or Plasma from Donor
10010-7	R' wave amplitude in lead AVF
10011-5	R' wave amplitude in lead AVL
10012-3	R' wave amplitude in lead AVR
10013-1	R' wave amplitude in lead I
10014-9	R' wave amplitude in lead II
10015-6	R' wave amplitude in lead III
10016-4	R' wave amplitude in lead V1
10017-2	R' wave amplitude in lead V2
10018-0	R' wave amplitude in lead V3
10019-8	R' wave amplitude in lead V4
1002-5	DBG Ab [Presence] in Serum or Plasma
10020-6	R' wave amplitude in lead V5
10021-4	R' wave amplitude in lead V6
10022-2	R' wave duration in lead AVF
10023-0	R' wave duration in lead AVL
10024-8	R' wave duration in lead AVR
10025-5	R' wave duration in lead I
10026-3	R' wave duration in lead II
10027-1	R' wave duration in lead III
10028-9	R' wave duration in lead V1
10029-7	R' wave duration in lead V2
1003-3	Indirect antiglobulin test.complement specific reagent [Presence] in Serum or Plasma
10030-5	R' wave duration in lead V3
10031-3	R' wave duration in lead V4
10032-1	R' wave duration in lead V5
10033-9	R' wave duration in lead V6
10034-7	S wave amplitude in lead AVF
10035-4	S wave amplitude in lead AVL
10036-2	S wave amplitude in lead AVR
10037-0	S wave amplitude in lead I
10038-8	S wave amplitude in lead II
10039-6	S wave amplitude in lead III
1004-1	Direct antiglobulin test.complement specific reagent [Presence] on Red Blood Cells
10040-4	S wave amplitude in lead V1
10041-2	S wave amplitude in lead V2
10042-0	S wave amplitude in lead V3
10043-8	S wave amplitude in lead V4
10044-6	S wave amplitude in lead V5
10045-3	S wave amplitude in lead V6
10046-1	S wave duration in lead AVF
10047-9	S wave duration in lead AVL
10048-7	S wave duration in lead AVR
10049-5	S wave duration in lead I
1005-8	Indirect antiglobulin test.IgG specific reagent [Presence] in Serum or Plasma
10050-3	S wave duration in lead II
10051-1	S wave duration in lead III
10052-9	S wave duration in lead V1
10053-7	S wave duration in lead V2
10054-5	S wave duration in lead V3
10055-2	S wave duration in lead V4
10056-0	S wave duration in lead V5
10057-8	S wave duration in lead V6

10058-6 | S' wave amplitude in lead AVF
10059-4 | S' wave amplitude in lead AVL
1006-6 | Direct antiglobulin test.IgG specific reagent [Interpretation] on Red Blood Cells
10060-2 | S' wave amplitude in lead AVR
10061-0 | S' wave amplitude in lead I
10062-8 | S' wave amplitude in lead II
10063-6 | S' wave amplitude in lead III
10064-4 | S' wave amplitude in lead V1
10065-1 | S' wave amplitude in lead V2
10066-9 | S' wave amplitude in lead V3
10067-7 | S' wave amplitude in lead V4
10068-5 | S' wave amplitude in lead V5
10069-3 | S' wave amplitude in lead V6
1007-4 | Direct antiglobulin test.poly specific reagent [Presence] on Red Blood Cells
10070-1 | S' wave duration in lead AVF
10071-9 | S' wave duration in lead AVL
10072-7 | S' wave duration in lead AVR
10073-5 | S' wave duration in lead I
10074-3 | S' wave duration in lead II
10075-0 | S' wave duration in lead III
10076-8 | S' wave duration in lead V1
10077-6 | S' wave duration in lead V2
10078-4 | S' wave duration in lead V3
10079-2 | S' wave duration in lead V4
1008-2 | Indirect antiglobulin test.poly specific reagent [Presence] in Serum or Plasma
10080-0 | S' wave duration in lead V5
10081-8 | S' wave duration in lead V6
10082-6 | ST initial amplitude 6 ms in lead AVF
10083-4 | ST initial amplitude 6 ms in lead AVL
10084-2 | ST initial amplitude 6 ms in lead AVR
10085-9 | ST initial amplitude 6 ms in lead I
10086-7 | ST initial amplitude 6 ms in lead II
10087-5 | ST initial amplitude 6 ms in lead III
10088-3 | ST initial amplitude 6 ms in lead V1
10089-1 | ST initial amplitude 6 ms in lead V2
1009-0 | Deprecated Direct antiglobulin test.poly specific reagent [Presence] on Red Blood Cells
10090-9 | ST initial amplitude 6 ms in lead V3
10091-7 | ST initial amplitude 6 ms in lead V4
10092-5 | ST initial amplitude 6 ms in lead V5
10093-3 | ST initial amplitude 6 ms in lead V6
10094-1 | ST slope in lead AVF
10095-8 | ST slope in lead AVL
10096-6 | ST slope in lead AVR
10097-4 | ST slope in lead I
10098-2 | ST slope in lead II
10099-0 | ST slope in lead III
101-6 | Cefoperazone [Susceptibility] by Disk diffusion (KB)
1010-8 | E sup(w) Ab [Presence] in Serum or Plasma from Blood product unit
10100-6 | ST slope in lead V1
10101-4 | ST slope in lead V2
10102-2 | ST slope in lead V3
10103-0 | ST slope in lead V4
10104-8 | ST slope in lead V5
10105-5 | ST slope in lead V6
10106-3 | ST wave end displacement in lead AVF
10107-1 | ST wave end displacement in lead AVL
10108-9 | ST wave end displacement in lead AVR
10109-7 | ST wave end displacement in lead I
1011-6 | E sup(w) Ab [Presence] in Serum or Plasma from Donor
10110-5 | ST wave end displacement in lead II
10111-3 | ST wave end displacement in lead III
10112-1 | ST wave end displacement in lead V1
10113-9 | ST wave end displacement in lead V2
10114-7 | ST wave end displacement in lead V3
10115-4 | ST wave end displacement in lead V4
10116-2 | ST wave end displacement in lead V5
10117-0 | ST wave end displacement in lead V6
10118-8 | ST wave mid displacement in lead AVF
10119-6 | ST wave mid displacement in lead AVL
1012-4 | E sup(w) Ab [Presence] in Serum or Plasma
10120-4 | ST wave mid displacement in lead AVR
10121-2 | ST wave mid displacement in lead I

10122-0 | ST wave mid displacement in lead II
10123-8 | ST wave mid displacement in lead III
10124-6 | ST wave mid displacement in lead V1
10125-3 | ST wave mid displacement in lead V2
10126-1 | ST wave mid displacement in lead V3
10127-9 | ST wave mid displacement in lead V4
10128-7 | ST wave mid displacement in lead V5
10129-5 | ST wave mid displacement in lead V6
1013-2 | E sup(w) Ag [Presence] on Red Blood Cells from Blood product unit
10130-3 | T' wave amplitude in lead AVF
10131-1 | T' wave amplitude in lead AVL
10132-9 | T' wave amplitude in lead AVR
10133-7 | T' wave amplitude in lead I
10134-5 | T' wave amplitude in lead II
10135-2 | T' wave amplitude in lead III
10136-0 | T' wave amplitude in lead V1
10137-8 | T' wave amplitude in lead V2
10138-6 | T' wave amplitude in lead V3
10139-4 | T' wave amplitude in lead V4
1014-0 | E sup(w) Ag [Presence] on Red Blood Cells from Donor
10140-2 | T' wave amplitude in lead V5
10141-0 | T' wave amplitude in lead V6
10142-8 | T wave amplitude in lead AVF
10143-6 | T wave amplitude in lead AVL
10144-4 | T wave amplitude in lead AVR
10145-1 | T wave amplitude in lead I
10146-9 | T wave amplitude in lead II
10147-7 | T wave amplitude in lead III
10148-5 | T wave amplitude in lead V1
10149-3 | T wave amplitude in lead V2
1015-7 | E sup(w) Ag [Presence] on Red Blood Cells
10150-1 | T wave amplitude in lead V3
10151-9 | T wave amplitude in lead V4
10152-7 | T wave amplitude in lead V5
10153-5 | T wave amplitude in lead V6
10154-3 | Chief complaint Narrative - Reported
10155-0 | History of allergies, reported
10156-8 | History of Childhood diseases Narrative
10157-6 | History of family member diseases Narrative
10158-4 | History of Functional status Narrative
10159-2 | History of Industrial exposure Narrative
1016-5 | E Ab [Presence] in Serum or Plasma from Blood product unit
10160-0 | History of Medication use Narrative
10161-8 | History of Occupational exposure Narrative
10162-6 | History of pregnancies Narrative
10163-4 | History of pregnancies
10164-2 | History of Present illness Narrative
10165-9 | Deprecated History of psychiatric symptoms and diseases Narrative
10166-7 | History of Social function Narrative
10167-5 | History of Surgical procedures Narrative
10168-3 | History of Cardiovascular system disorders Narrative
10169-1 | History of Ear disorders Narrative
1017-3 | E Ab [Presence] in Serum or Plasma from Donor
10170-9 | History of Endocrine system disorders Narrative
10171-7 | History of Eyes disorders Narrative
10172-5 | History of Hematologic system disorders Narrative
10173-3 | History of Musculoskeletal system disorders Narrative
10174-1 | History of Nose disorders Narrative
10175-8 | History of Oral cavity disorders Narrative
10176-6 | History of Reproductive system disorders Narrative
10177-4 | History of Respiratory system disorders Narrative
10178-2 | History of Skin disorders Narrative
10179-0 | History of Throat and Neck disorders Narrative
1018-1 | E Ab [Presence] in Serum or Plasma
10180-8 | History of Throat and Neck disorders
10181-6 | History of Urinary tract disorders Narrative
10182-4 | History of Travel Narrative
10183-2 | Hospital discharge medications Narrative
10184-0 | Hospital discharge physical findings Narrative
10185-7 | Hospital discharge procedures Narrative
10186-5 | Identifying information Narrative Observed
10187-3 | Review of systems Narrative - Reported

10188-1 | Review of systems overview Narrative - Reported
10189-9 | Review of systems overview - Reported
1019-9 | E Ag [Presence] on Red Blood Cells from Blood product unit
10190-7 | Mental status Narrative
10191-5 | Physical findings of Abdomen Narrative
10192-3 | Physical findings of Back Narrative
10193-1 | Physical findings of Breasts Narrative
10194-9 | Physical findings of Neurologic deep tendon reflexes Narrative
10195-6 | Physical findings of Ear Narrative
10196-4 | Physical findings of Extremities Narrative
10197-2 | Physical findings of Eye Narrative
10198-0 | Physical findings of Genitourinary tract Narrative
10199-8 | Physical findings of Head Narrative
102-4 | Cefoperazone [Susceptibility] by Serum bactericidal titer
1020-7 | E Ag [Presence] on Red Blood Cells from Donor
10200-4 | Physical findings of Heart Narrative
10201-2 | Physical findings of Mouth and Throat and Teeth Narrative
10202-0 | Physical findings of Nervous system Narrative
10203-8 | Physical findings of Nose Narrative
10204-6 | Physical findings of Pelvis Narrative
10205-3 | Physical findings of Rectum Narrative
10206-1 | Physical findings of Skin Narrative
10207-9 | Physical findings of Thorax and Lungs Narrative
10208-7 | Physical findings of Vessels Narrative
10209-5 | Physical findings of Neurologic balance and Coordination Narrative
1021-5 | E Ag [Presence] on Red Blood Cells
10210-3 | Physical findings of General status Narrative
10211-1 | Physical findings of Sense of touch Narrative
10212-9 | Physical findings of Strength Narrative
10213-7 | Surgical operation note anesthesia Narrative
10214-5 | Surgical operation note anesthesia duration
10215-2 | Surgical operation note findings Narrative
10216-0 | Surgical operation note fluids Narrative
10217-8 | Surgical operation note indications [Interpretation] Narrative
10218-6 | Surgical operation note postoperative diagnosis Narrative
10219-4 | Surgical operation note preoperative diagnosis Narrative
1022-3 | Fy sup(a) Ab [Presence] in Serum or Plasma from Blood product unit
10220-2 | Surgical operation note prep time duration
10221-0 | Surgical operation note specimens taken Narrative
10222-8 | Surgical operation note surgical complications [Interpretation] Narrative
10223-6 | Surgical operation note surgical procedure Narrative
10224-4 | Hemodynamic method special circumstances
10225-1 | Cardiac measurement device Institution inventory number
10226-9 | Oxygen content in Intravascular space
10227-7 | Cardiac measurement device Vendor model number
10228-5 | Cardiac measurement device Vendor serial number
10229-3 | Hemodynamic method Type of
1023-1 | Fy sup(a) Ab [Presence] in Serum or Plasma from Donor
10230-1 | Left ventricular Ejection fraction
10231-9 | Right ventricular Ejection fraction
10232-7 | Oxygen content in Aorta root
10233-5 | Oxygen content in Left atrium
10234-3 | Oxygen content in Right atrium
10235-0 | Oxygen content in High right atrium
10236-8 | Oxygen content in Low right atrium
10237-6 | Oxygen content in Mid right atrium
10238-4 | Oxygen content in Left ventricle
10239-2 | Oxygen content in Right ventricular outflow tract
1024-9 | Fy sup(a) Ab [Presence] in Serum or Plasma
10240-0 | Oxygen content in Right ventricle
10241-8 | Oxygen content in Coronary sinus
10242-6 | Oxygen content in Ductus arteriosus
10243-4 | Oxygen content in Inferior vena cava
10244-2 | Oxygen content in Left pulmonary artery
10245-9 | Oxygen content in Main pulmonary artery
10246-7 | Oxygen content in Right pulmonary artery
10247-5 | Oxygen content in Pulmonary wedge
10248-3 | Oxygen content in Superior vena cava
10249-1 | Heart rate device Institution inventory number
1025-6 | Fy sup(a) Ag [Presence] on Red Blood Cells from Blood product unit
10250-9 | Heart rate device Vendor model number
10251-7 | Heart rate device Vendor serial number

10252-5 | Biliary drain site
10253-3 | Type of Biliary drain
10254-1 | Type of Peritoneal drain
10255-8 | Type of Subarachnoid drain
10256-6 | Biliary drain Institution inventory number
10257-4 | Chest tube Institution inventory number
10258-2 | Enteral tube Institution inventory number
10259-0 | Gastrostomy tube Institution inventory number
1026-4 | Fy sup(a) Ag [Presence] on Red Blood Cells from Donor
10260-8 | GI tube Institution inventory number
10261-6 | Intravascular tube Institution inventory number
10262-4 | IO tube Institution inventory number
10263-2 | Nasogastric tube Institution inventory number
10264-0 | Oral tube Institution inventory number
10265-7 | Peritoneal drain Institution inventory number
10266-5 | Peritoneal tube Institution inventory number
10267-3 | Rate control device Institution inventory number
10268-1 | Stool collection device Institution inventory number
10269-9 | Subarachnoid drain Institution inventory number
1027-2 | Fy sup(a) Ag [Presence] on Red Blood Cells
10270-7 | Synovial drain Institution inventory number
10271-5 | Tube or drain Institution inventory number
10272-3 | Upper GI tube Institution inventory number
10273-1 | Bladder irrigation tube Institution inventory number
10274-9 | Urine tube Institution inventory number
10275-6 | Wound drain device Institution inventory number
10276-4 | Peritoneal drain site
10277-2 | Stool collection site
10278-0 | Subarachnoid drain site
10279-8 | Biliary drain Vendor model number
1028-0 | Fy sup(b) Ab [Presence] in Serum or Plasma from Blood product unit
10280-6 | Chest tube Vendor model number
10281-4 | Enteral tube Vendor model number
10282-2 | Gastrostomy tube Vendor model number
10283-0 | GI tube Vendor model number
10284-8 | Intravascular tube Vendor model number
10285-5 | IO tube Vendor model number
10286-3 | Nasogastric tube Vendor model number
10287-1 | Oral tube Vendor model number
10288-9 | Peritoneal drain Vendor model number
10289-7 | Peritoneal tube Vendor model number
1029-8 | Fy sup(b) Ab [Presence] in Serum or Plasma from Donor
10290-5 | Rate control device Vendor model number
10291-3 | Stool collection device Vendor model number
10292-1 | Subarachnoid drain Vendor model number
10293-9 | Synovial drain Vendor model number
10294-7 | Tube or drain Vendor model number
10295-4 | Upper GI tube Vendor model number
10296-2 | Bladder irrigation tube Vendor model number
10297-0 | Urine tube Vendor model number
10298-8 | Wound drain device Vendor model number
10299-6 | Biliary drain Vendor serial number
103-2 | Ceforanide [Susceptibility] by Minimum lethal concentration (MLC)
1030-6 | Fy sup(b) Ab [Presence] in Serum or Plasma
10300-2 | Chest tube Vendor serial number
10301-0 | Enteral tube Vendor serial number
10302-8 | Gastrostomy tube Vendor serial number
10303-6 | GI tube Vendor serial number
10304-4 | Intravascular tube Vendor serial number
10305-1 | IO tube Vendor serial number
10306-9 | Nasogastric tube Vendor serial number
10307-7 | Oral tube Vendor serial number
10308-5 | Peritoneal drain Vendor serial number
10309-3 | Peritoneal tube Vendor serial number
1031-4 | Fy sup(b) Ag [Presence] on Red Blood Cells from Blood product unit
10310-1 | Rate control device Vendor serial number
10311-9 | Stool collection device Vendor serial number
10312-7 | Subarachnoid drain Vendor serial number
10313-5 | Synovial drain Vendor serial number
10314-3 | Tube or drain Vendor serial number
10315-0 | Upper GI tube Vendor serial number
10316-8 | Bladder irrigation tube Vendor serial number

10317-6 | Urine tube Vendor serial number
10318-4 | Wound drain device Vendor serial number
10319-2 | Enema device Institution inventory number
1032-2 | Fy sup(b) Ag [Presence] on Red Blood Cells from Donor
10320-0 | Enema device Vendor model number
10321-8 | Enema device Vendor serial number
10322-6 | Potassium intake 24 hour
10323-4 | Wound drain fluid [Appearance] Lower GI tract
10324-2 | Breath rate device Institution inventory number
10325-9 | Breath rate device Vendor model number
10326-7 | Breath rate device Vendor serial number
10327-5 | Eosinophils/100 leukocytes in Sputum by Manual count
10328-3 | Lymphocytes/100 leukocytes in Cerebral spinal fluid by Manual count
10329-1 | Monocytes/100 leukocytes in Cerebral spinal fluid by Manual count
1033-0 | Fy sup(b) Ag [Presence] on Red Blood Cells
10330-9 | Monocytes/100 leukocytes in Body fluid by Manual count
10331-7 | Rh [Type] in Blood
10332-5 | Cortisol [Mass/volume] in Serum or Plasma --pre 250 ug corticotropin IM
10333-3 | Appearance of Cerebral spinal fluid
10334-1 | Cancer Ag 125 [Units/volume] in Serum or Plasma
10335-8 | Color of Cerebral spinal fluid
10336-6 | Gonadotropin peptide [Mass/volume] in Urine
10337-4 | Procollagen type I [Mass/volume] in Serum
10338-2 | Barbiturates [Mass/volume] in Serum or Plasma
10339-0 | Fluoxetine+Norfluoxetine [Mass/volume] in Serum or Plasma
1034-8 | Fetal cell screen [Interpretation] in Blood
10340-8 | Molindone [Mass/volume] in Serum or Plasma
10341-6 | Norpropoxyphene [Mass/volume] in Serum or Plasma
10342-4 | Sulfamethoxazole [Mass/volume] in Serum or Plasma
10343-2 | Temazepam [Mass/volume] in Serum or Plasma
10344-0 | Tranylcypromine [Mass/volume] in Serum or Plasma
10345-7 | Trihexyphenidyl [Mass/volume] in Urine
10346-5 | Hemoglobin A [Units/volume] in Blood by Electrophoresis
10347-3 | Babesia microti identified in Blood by Light microscopy
10348-1 | Bordetella parapertussis Ab [Presence] in Serum
10349-9 | Brucella sp Ab [Units/volume] in Serum
1035-5 | Fresh frozen plasma given [Volume]
10350-7 | Herpes simplex virus IgM Ab [Titer] in Serum by Immunoassay
10351-5 | HIV 1 RNA [Units/volume] (viral load) in Serum or Plasma by Probe with amplification
10352-3 | Bacteria identified in Genital specimen by Aerobe culture
10353-1 | Bacteria identified in Nose by Aerobe culture
10354-9 | Bacteria identified in Urethra by Culture
10355-6 | Microscopic observation [Identifier] in Bone marrow by Wright Giemsa stain
10356-4 | Deprecated Microscopic observation [Identifier] in Stool by Trichrome stain
10357-2 | Microscopic observation [Identifier] in Wound by Gram stain
10358-0 | Teichoate Ab [Titer] in Serum by Immune diffusion (ID)
10359-8 | Asialoganglioside GM1 IgM Ab [Titer] in Serum
1036-3 | G Ab [Presence] in Serum or Plasma from Blood product unit
10360-6 | DNA single strand IgG Ab [Units/volume] in Serum
10361-4 | DNA single strand IgM Ab [Units/volume] in Serum
10362-2 | Endomysium IgA Ab [Presence] in Serum
10363-0 | Barbiturates [Presence] in Unspecified specimen
10364-8 | Cotinine [Mass/mass] in Hair
10365-5 | Cotinine [Mass/volume] in Serum or Plasma
10366-3 | Cotinine [Mass/volume] in Urine
10367-1 | Ethanol [Mass/volume] in Gastric fluid
10368-9 | Lead [Mass/volume] in Capillary blood
10369-7 | Opiates [Mass/mass] in Hair
1037-1 | G Ab [Presence] in Serum or Plasma from Donor
10370-5 | Phencyclidine [Mass/mass] in Hair
10371-3 | Bite cells [Presence] in Blood by Light microscopy
10372-1 | Blister cells [Presence] in Blood by Light microscopy
10373-9 | Fragments [Presence] in Blood by Light microscopy
10374-7 | Helmet cells [Presence] in Blood by Light microscopy
10375-4 | Irregularly contracted cells [Presence] in Blood by Light microscopy
10376-2 | Oval macrocytes [Presence] in Blood by Light microscopy
10377-0 | Pencil cells [Presence] in Blood by Light microscopy
10378-8 | Polychromasia [Presence] in Blood by Light microscopy
10379-6 | Erythrocytes.dual population [Presence] in Blood by Light microscopy
1038-9 | G Ab [Presence] in Serum or Plasma
10380-4 | Stomatocytes [Presence] in Blood by Light microscopy
10381-2 | Target cells [Presence] in Blood by Light microscopy

10382-0 | A,B variant NOS Ag [Presence] on Red Blood Cells from Blood product unit
10383-8 | A,B variant NOS Ag [Presence] on Red Blood Cells from Donor
10384-6 | A,B variant NOS Ag [Presence] on Red Blood Cells
10385-3 | Albumin concentration given
10386-1 | Albumin given [Volume]
10387-9 | Autologous erythrocytes given [Volume]
10388-7 | Autologous whole blood given [Volume]
10389-5 | Blood product.other [Type]
1039-7 | G Ag [Presence] on Red Blood Cells from Blood product unit
10390-3 | Blood product special preparation [Type]
10391-1 | Cytomegalovirus immune globulin given [Volume]
10392-9 | Cryoprecipitate given [Volume]
10393-7 | Factor IX given [Type]
10394-5 | Factor IX given [Volume]
10395-2 | Factor VIII given [Type]
10396-0 | Factor VIII given [Volume]
10397-8 | Hepatitis B immune globulin given [Volume]
10398-6 | I Ag [Presence] on Red Blood Cells from Blood product unit
10399-4 | I Ag [Presence] on Red Blood Cells from Donor
104-0 | Ceforanide [Susceptibility] by Minimum inhibitory concentration (MIC)
1040-5 | G Ag [Presence] on Red Blood Cells from Donor
10400-0 | I Ag [Presence] on Red Blood Cells
10401-8 | Immune serum globulin given [Type]
10402-6 | Immune serum globulin given [Volume]
10403-4 | Inject immune serum globulin [Volume]
10404-2 | Inject Rh immune globulin [Volume]
10405-9 | Inject varicella zoster virus immune globulin [Volume]
10406-7 | little i Ag [Presence] on Red Blood Cells from Blood product unit
10407-5 | little i Ag [Presence] on Red Blood Cells from Donor
10408-3 | little i Ag [Presence] on Red Blood Cells
10409-1 | Pentaspan given [Volume]
1041-3 | G Ag [Presence] on Red Blood Cells
10410-9 | Plasma given [Type]
10411-7 | Plasma given [Volume]
10412-5 | Platelets given [Type]
10413-3 | Rh immune globulin given [Volume]
10414-1 | Transfuse albumin [Volume]
10415-8 | Transfuse blood exchange transfusion [Volume]
10416-6 | Transfuse blood product other [Volume]
10417-4 | Transfuse cryoprecipitate [Volume]
10418-2 | Transfuse factor IX [Volume]
10419-0 | Transfuse factor VIII [Volume]
1042-1 | H Ab [Presence] in Serum or Plasma from Blood product unit
10420-8 | Transfuse immune serum globulin [Volume]
10421-6 | Transfuse Pentaspan [Volume]
10422-4 | Transfuse plasma [Volume]
10423-2 | Transfuse platelets [Volume]
10424-0 | Transfuse erythrocytes [Volume]
10425-7 | Transfuse Rh immune globulin [Volume]
10426-5 | Transfuse whole blood [Volume]
10427-3 | Transfuse whole blood autologous [Volume]
10428-1 | Varicella zoster virus immune globulin given [Volume]
10429-9 | AE 1 Ag [Presence] in Tissue by Immune stain
1043-9 | H Ab [Presence] in Serum or Plasma from Donor
10430-7 | AE 3 Ag [Presence] in Tissue by Immune stain
10431-5 | B-cell Ag [Presence] in Tissue by Immune stain
10432-3 | CD30 Ag [Presence] in Tissue by Immune stain
10433-1 | BR-2 Ag [Presence] in Tissue by Immune stain
10434-9 | Complement C3 Ag [Presence] in Tissue by Immune stain
10435-6 | C5B-9 Ag [Presence] in Tissue by Immune stain
10436-4 | CD15 Ag [Presence] in Tissue by Immune stain
10437-2 | CD16 Ag [Presence] in Tissue by Immune stain
10438-0 | CD20 Ag [Presence] in Tissue by Immune stain
10439-8 | CD3 Ag [Presence] in Tissue by Immune stain
1044-7 | H Ab [Presence] in Serum
10440-6 | Deprecated CD30 Ag [Presence] in Tissue by Immune stain
10441-4 | CD34 Ag [Presence] in Tissue by Immune stain
10442-2 | CD56 Ag [Presence] in Tissue by Immune stain
10443-0 | CD43 T-cell monocyte+myeloid cell Ag [Presence] in Tissue by Immune stain
10444-8 | CD57 Ag [Presence] in Tissue by Immune stain
10445-5 | CD11c Ag [Presence] in Tissue by Immune stain
10446-3 | Leukocyte common Ag [Presence] in Tissue by Immune stain

10447-1 | M-5 Ag [Presence] in Tissue by Immune stain
10448-9 | T-cell Ag [Presence] in Tissue by Immune stain
10449-7 | Glucose [Mass/volume] in Serum or Plasma --1 hour post meal
1045-4 | H NOS Ab [Presence] in Serum or Plasma from Blood product unit
10450-5 | Glucose [Mass/volume] in Serum or Plasma --10 hours fasting
10451-3 | Proinsulin [Moles/volume] in Serum or Plasma --12 hours fasting
10452-1 | Xylose [Mass/volume] in Serum or Plasma --1 hour post 25 g xylose PO
10453-9 | Xylose [Mass/volume] in Serum or Plasma --1 hour post dose xylose PO
10454-7 | Xylose [Mass/volume] in Serum or Plasma --2 hours post 25 g xylose PO
10455-4 | Xylose [Mass/volume] in Serum or Plasma --30 minutes post 25 g xylose PO
10456-2 | Xylose [Mass/volume] in Serum or Plasma --6 hours fasting
10457-0 | Actin Ag [Presence] in Tissue by Immune stain
10458-8 | Alkaline phosphatase.placental Ag [Presence] in Tissue by Immune stain
10459-6 | Alpha-1-Fetoprotein Ag [Presence] in Tissue by Immune stain
1046-2 | H NOS Ab [Presence] in Serum or Plasma from Donor
10460-4 | Lactalbumin alpha Ag [Presence] in Tissue by Immune stain
10461-2 | Alpha-1-Antichymotrypsin Ag [Presence] in Tissue by Immune stain
10462-0 | Alpha 1 antitrypsin Ag [Presence] in Tissue by Immune stain
10463-8 | Amyloid A component Ag [Presence] in Tissue by Immune stain
10464-6 | Amyloid P component Ag [Presence] in Tissue by Immune stain
10465-3 | Amyloid.prealbumin Ag [Presence] in Tissue by Immune stain
10466-1 | Anion gap 3 in Serum or Plasma
10467-9 | Beta-2-Microglobulin amyloid Ag [Presence] in Tissue by Immune stain
10468-7 | Calcitonin Ag [Presence] in Tissue by Immune stain
10469-5 | Carcinoembryonic Ag [Presence] in Tissue by Immune stain
1047-0 | H NOS Ab [Presence] in Serum or Plasma
10470-3 | Choriogonadotropin Ag [Presence] in Tissue by Immune stain
10471-1 | Chromogranin A Ag [Presence] in Tissue by Immune stain
10472-9 | Chromogranin Ag [Presence] in Tissue by Immune stain
10473-7 | Chymotrypsin Ag [Presence] in Tissue by Immune stain
10474-5 | Collagen type 4 Ag [Presence] in Tissue by Immune stain
10475-2 | Corticotropin Ag [Presence] in Tissue by Immune stain
10476-0 | Desmin Ag [Presence] in Tissue by Immune stain
10477-8 | Enolase.neuron specific Ag [Presence] in Tissue by Immune stain
10478-6 | Eosinophil major basic protein Ag [Presence] in Tissue by Immune stain
10479-4 | CD227 Ag [Presence] in Tissue by Immune stain
1048-8 | H NOS Ag [Presence] on Red Blood Cells from Blood product unit
10480-2 | Estrogen+Progesterone receptor Ag [Presence] in Tissue by Immune stain
10481-0 | Follitropin.alpha subunit Ag [Presence] in Tissue by Immune stain
10482-8 | Follitropin.beta subunit Ag [Presence] in Tissue by Immune stain
10483-6 | Gastrin Ag [Presence] in Tissue by Immune stain
10484-4 | Glial fibrillary acidic protein Ag [Presence] in Tissue by Immune stain
10485-1 | Glucagon Ag [Presence] in Tissue by Immune stain
10486-9 | Hemoglobin Ag [Presence] in Tissue by Immune stain
10487-7 | HMB-45 Ag [Presence] in Tissue by Immune stain
10488-5 | IgA Ag [Presence] in Tissue by Immune stain
10489-3 | IgA.heavy chain Ag [Presence] in Tissue by Immune stain
1049-6 | H NOS Ag [Presence] on Red Blood Cells from Donor
10490-1 | IgE Ag [Presence] in Tissue by Immune stain
10491-9 | IgG Ag [Presence] in Tissue by Immune stain
10492-7 | IgG.heavy chain Ag [Presence] in Tissue by Immune stain
10493-5 | IgM Ag [Presence] in Tissue by Immune stain
10494-3 | IgM.heavy chain Ag [Presence] in Tissue by Immune stain
10495-0 | Insulin Ag [Presence] in Tissue by Immune stain
10496-8 | Kappa light chains Ag [Presence] in Tissue by Immune stain
10497-6 | Immunoglobulin light chains.kappa amyloid Ag [Presence] in Tissue by Immune stain
10498-4 | Keratin Ag [Presence] in Tissue by Immune stain
10499-2 | Lambda light chains Ag [Presence] in Tissue by Immune stain
105-7 | Ceforanide [Susceptibility] by Disk diffusion (KB)
1050-4 | H NOS Ag [Presence] on Red Blood Cells
10500-7 | Immunoglobulin light chains.lambda amyloid Ag [Presence] in Tissue by Immune stain
10501-5 | Lutropin [Units/volume] in Serum or Plasma
10502-3 | Lutropin Ag [Presence] in Tissue by Immune stain
10503-1 | Lysozyme Ag [Presence] in Tissue by Immune stain
10504-9 | Myelin basic protein Ag [Presence] in Tissue by Immune stain
10505-6 | Myoglobin Ag [Presence] in Tissue by Immune stain
10506-4 | Peanut agglutinin Ag [Presence] in Tissue by Immune stain
10507-2 | Prolactin Ag [Presence] in Tissue by Immune stain
10508-0 | Prostate specific Ag [Presence] in Tissue by Immune stain
10509-8 | Prostatic acid phosphatase Ag [Presence] in Tissue by Immune stain
1051-2 | Hemolytic disease of newborn screen [Interpretation] in Blood
10510-6 | S-100 Ag Ag [Presence] in Tissue by Immune stain

10511-4 | Serotonin Ag [Presence] in Tissue by Immune stain
10512-2 | Somatostatin Ag [Presence] in Tissue by Immune stain
10513-0 | Somatotropin Ag [Presence] in Tissue by Immune stain
10514-8 | Synaptophysin Ag [Presence] in Tissue by Immune stain
10515-5 | Thyroglobulin Ag [Presence] in Tissue by Immune stain
10516-3 | Thyrotropin Ag [Presence] in Tissue by Immune stain
10517-1 | Trypsin Ag [Presence] in Tissue by Immune stain
10518-9 | Ulex europaeus I lectin Ag [Presence] in Tissue by Immune stain
10519-7 | Vimentin Ag [Presence] in Tissue by Immune stain
1052-0 | Deprecated I (intermediate) subtype [Presence] in Serum or Plasma from Blood product unit
10520-5 | Coagulation factor VI Ag [Presence] in Tissue by Immune stain
10521-3 | Coagulation factor VIII Ag [Presence] in Tissue by Immune stain
10522-1 | Coagulation factor XIII Ag [Presence] in Tissue by Immune stain
10523-9 | Fibrinogen Ag [Presence] in Tissue by Immune stain
10524-7 | Microscopic observation [Identifier] in Cervix by Cyto stain
10525-4 | Microscopic observation [Identifier] in Unspecified specimen by Cyto stain
10526-2 | Microscopic observation [Identifier] in Sputum by Cyto stain
10527-0 | Microscopic observation [Identifier] in Tissue by Cyto stain
10528-8 | Acetophenazine [Mass/volume] in Serum or Plasma
10529-6 | Amoxapine metabolite [Mass/volume] in Serum or Plasma
1053-8 | Deprecated I (intermediate) subtype [Presence] in serum or plasma from donor
10530-4 | Aprobarbital [Mass/volume] in Serum or Plasma
10531-2 | Bretylium [Mass/volume] in Serum or Plasma
10532-0 | Deprecated Desethylamiodarone [Mass/volume] in Serum or Plasma
10533-8 | Propoxyphene+Acetaminophen [Mass/volume] in Serum or Plasma
10534-6 | Diazoxide [Mass/volume] in Serum or Plasma
10535-3 | Digoxin [Mass/volume] in Serum or Plasma
10536-1 | Dipyridamole [Mass/volume] in Serum or Plasma
10537-9 | Deprecated Doxepin+Nordoxepin [Mass/volume] in Serum or Plasma
10538-7 | Deprecated Fluoxetine+Norfluoxetine [Mass/volume] in Serum or Plasma
10539-5 | Glipizide [Mass/volume] in Serum or Plasma
1054-6 | Deprecated I (intermediate) subtype [Presence] in srum
10540-3 | Glyburide [Mass/volume] in Serum or Plasma
10541-1 | Mepivacaine [Mass/volume] in Serum or Plasma
10542-9 | Metharbital [Mass/volume] in Serum or Plasma
10543-7 | Methsuximide+Normethsuximide [Mass/volume] in Serum or Plasma
10544-5 | Normeperidine [Presence] in Serum or Plasma
10545-2 | Normephenytoin [Mass/volume] in Serum or Plasma
10546-0 | Normethsuximide [Mass/volume] in Serum or Plasma
10547-8 | Primidone+Phenobarbital [Mass/volume] in Serum or Plasma
10548-6 | Phenytoin Free/Phenytoin.total in Serum or Plasma
10549-4 | Pirmenol [Mass/volume] in Serum or Plasma
1055-3 | Deprecated I (intermediate) subtype [Presence] on Red Blood Cells from Blood product unit
10550-2 | Deprecated Temazepam [Mass/volume] in Serum or Plasma
10551-0 | Triamterene [Mass/volume] in Serum or Plasma
10552-8 | Tricyclic antidepressants [Mass/volume] in Serum or Plasma
10553-6 | Prostatic acid phosphatase [Enzymatic activity/volume] in Genital fluid
10554-4 | Deprecated Prostatic acid phosphatase [Enzymatic activity/volume] in Seminal plasma
10555-1 | Acrosin [Entitic Catalytic Activity] in Spermatozoa
10556-9 | Deprecated Adenosine triphosphatase [Enzymatic activity/volume] in Seminal plasma
10557-7 | Adenosine triphosphate [Moles/volume] in Semen
10558-5 | Albumin [Moles/volume] in Semen
10559-3 | Deprecated Calcium [Molecules/volume] in Semen
1056-1 | Deprecated I (intermediate) subtype [Presence] on Red Blood Cells from donor
10560-1 | Carcinoembryonic Ag [Units/volume] in Semen
10561-9 | Carnitine [Moles/volume] in Semen
10562-7 | Cells [#/volume] in Cervical mucus
10563-5 | Cells other than spermatozoa [#/volume] in Semen
10564-3 | Cervical mucus [Volume]
10565-0 | Choriogonadotropin [Units/volume] in Semen
10566-8 | Deprecated Choriogonadotropin [Molecules/volume] in Semen
10567-6 | Citrate [Moles/volume] in Semen
10568-4 | Clarity of Semen
10569-2 | Color of Semen
1057-9 | Deprecated I (intermediate) subtype [Presence] on Red Blood Cells
10570-0 | Consistency of Cervical mucus
10571-8 | Deprecated Consistency of Semen
10572-6 | Deprecated Duration^post ejaculation
10573-4 | Ferning [Type] in Cervical mucus
10574-2 | Fructose [Moles/volume] in Semen
10575-9 | Deprecated Gamma glutamyl transferase [Enzymatic activity/volume] in Semen
10576-7 | Germ cells.immature [#/volume] in Semen

10577-5 | Glucosidase [Enzymatic activity/volume] in Seminal plasma
10578-3 | Glycerophosphocholine [Moles/volume] in Seminal plasma
10579-1 | Leukocytes [#/volume] in Semen
1058-7 | I Ab [Presence] in Serum or Plasma from Blood product unit
10580-9 | Liquefaction [Time] in Semen
10581-7 | Deprecated Number of entities [#] in Spermatozoa
10582-5 | pH of Cervical mucus
10583-3 | Prostaglandin F1 alpha [Moles/volume] in Semen
10584-1 | Deprecated Protein [Mass/volume] in Semen
10585-8 | Round cells [#/volume] in Semen
10586-6 | Deprecated Volume of Semen
10587-4 | Sexual abstinence duration
10588-2 | Spermatogonia [#/volume] in Semen
10589-0 | Spermatids [#/volume] in Semen by Streptoavidin-biotin method (SAB)
1059-5 | I Ab [Presence] in Serum or Plasma from Donor
10590-8 | Spermatids [#/volume] in Semen by Sertoli cell barrier method (SCB)
10591-6 | Primary Spermatocytes [#/volume] in Semen
10592-4 | Secondary Spermatocytes [#/volume] in Semen
10593-2 | Spermatozoa Pyriform Head/100 spermatozoa in Semen
10594-0 | Spermatozoa Tapering Head/100 spermatozoa in Semen
10595-7 | Deprecated Spermatozoa [#/volume] in Semen
10596-5 | Deprecated Spermatozoa Ab in cervical mucosa
10597-3 | Deprecated Spermatozoa Ab in semen
10598-1 | Deprecated Spermatozoa Ab in serum
10599-9 | Spermatozoa penetration [Presence] in Seminal fluid and Cervical mucosa by Kremer test
106-5 | Ceforanide [Susceptibility] by Serum bactericidal titer
1060-3 | I Ab [Presence] in Serum or Plasma
10600-5 | Spermatozoa penetration [Presence] in Seminal fluid and Cervical mucosa
10601-3 | Spermatozoa penetration [Presence] in Seminal fluid and Cervical mucosa --post coitus
10602-1 | Spermatozoa Abnormal Head/100 spermatozoa in Semen
10603-9 | Spermatozoa Abnormal Midpiece/100 spermatozoa in Semen
10604-7 | Spermatozoa Abnormal Tail/100 spermatozoa in Semen
10605-4 | Spermatozoa Agglutinated/100 spermatozoa in Semen
10606-2 | Spermatozoa Amorphous Head/100 spermatozoa in Semen
10607-0 | Spermatozoa Coiled Tail/100 spermatozoa in Semen
10608-8 | Spermatozoa Cytoplasmic Droplet/100 spermatozoa in Semen
10609-6 | Spermatozoa Duplicate Head/100 spermatozoa in Semen
1061-1 | I NOS Ab [Presence] in Serum or Plasma from Blood product unit
10610-4 | Spermatozoa Duplicate Tail/100 spermatozoa in Semen
10611-2 | Spermatozoa Immotile/100 spermatozoa in Semen
10612-0 | Spermatozoa Large Oval Head/100 spermatozoa in Semen
10613-8 | Spermatozoa Viable/100 spermatozoa in Semen
10614-6 | Spermatozoa Motile w IgA/100 spermatozoa in Semen by Immunobead
10615-3 | Spermatozoa Motile w IgA/100 spermatozoa in Semen by Mixed antiglobulin reaction
10616-1 | Spermatozoa Motile w IgG/100 spermatozoa in Semen by Immunobead
10617-9 | Spermatozoa Motile w IgG/100 spermatozoa in Semen by Mixed antiglobulin reaction
10618-7 | Spermatozoa Motile w IgM/100 spermatozoa in Semen by Immunobead
10619-5 | Spermatozoa Motile w IgM/100 spermatozoa in Semen by Mixed antiglobulin reaction
1062-9 | I NOS Ab [Presence] in Serum or Plasma from Donor
10620-3 | Spermatozoa Nonprogressive/100 spermatozoa in Semen
10621-1 | Spermatozoa Normal Head/100 spermatozoa in Semen
10622-9 | Spermatozoa Normal/100 spermatozoa in Semen
10623-7 | Spermatozoa Pin Head/100 spermatozoa in Semen
10624-5 | Spermatozoa Rapid/100 spermatozoa in Semen
10625-2 | Spermatozoa Round Head/100 spermatozoa in Semen
10626-0 | Spermatozoa Slow/100 spermatozoa in Semen
10627-8 | Spermatozoa Small Oval Head/100 spermatozoa in Semen
10628-6 | Spermatozoa Tail Swelling/100 spermatozoa in Semen
10629-4 | Spermatozoa Vacuolated Head/100 spermatozoa in Semen
1063-7 | I NOS Ab [Presence] in Serum or Plasma
10630-2 | Spinnbarkeit [Length] in Cervical mucus
10631-0 | Testosterone [Moles/volume] in Semen
10632-8 | Time until next menstrual period
10633-6 | Zinc [Moles/volume] in Semen
10634-4 | Complement C1 esterase inhibitor.functional/Complement C1 esterase inhibitor.total in Serum or Plasma
10635-1 | Acanthamoeba sp identified in Eye by Organism specific culture
10636-9 | Acanthamoeba sp identified in Eye by Wet preparation
10637-7 | Microscopic observation [Identifier] in Tissue by Night blue stain
10638-5 | Thermophilic Actinomycetes colony count [#/volume] in Unspecified specimen by Organism specific culture
10639-3 | Thermophilic Actinomycetes identified in Unspecified specimen by Organism specific culture
1064-5 | I NOS Ag [Presence] on Red Blood Cells from Blood product unit
10640-1 | Deprecated Adenovirus 40+41 [Identifier] in Stool by Electron microscopy

10641-9 | Amoeba identified in Aspirate by Immune stain
10642-7 | Amoeba identified in Aspirate by Wet preparation
10643-5 | Amoeba identified in Stool by Organism specific culture
10644-3 | Arthropod identified in Unspecified specimen
10645-0 | Aspergillus fumigatus Ag [Presence] in Tissue by Immunofluorescence
10646-8 | Astrovirus [Identifier] in Stool by Electron microscopy
10647-6 | Babesia sp identified in Blood by Thick film
10648-4 | Babesia sp identified in Blood by Thin film
10649-2 | Calicivirus [Identifier] in Stool by Electron microscopy
1065-2 | I NOS Ag [Presence] on Red Blood Cells from Donor
10650-0 | Candida sp DNA [Presence] in Blood by Probe with amplification
10651-8 | Chlamydophila pneumoniae Ag [Presence] in Sputum or Bronchial
10652-6 | Chlamydophila pneumoniae DNA [Presence] in Sputum or Bronchial by Probe with amplification
10653-4 | Clotrimazole [Susceptibility] by Minimum inhibitory concentration (MIC)
10654-2 | Clotrimazole [Susceptibility] by Minimum lethal concentration (MLC)
10655-9 | Coccidia identified in Duodenal fluid by Acid fast stain
10656-7 | Coccidia identified in Stool by Acid fast stain
10657-5 | Deprecated Cryptococcus neoformans Ag [Presence] in Tissue by Immunofluorescence
10658-3 | Cyanobacterium identified in Water by Light microscopy
10659-1 | Cyclospora sp identified in Stool by Acid fast stain
1066-0 | I NOS Ag [Presence] on Red Blood Cells
10660-9 | Cytomegalovirus Ag [Presence] in Tissue by Immune stain
10661-7 | Dinoflagellate identified in Water by Light microscopy
10662-5 | Filaria identified in Blood by Concentration
10663-3 | Filaria identified in Blood by Thick film
10664-1 | Filaria identified in Blood by Thin film
10665-8 | Fungus colony count [#/volume] in Unspecified specimen by Environmental culture
10666-6 | Fungus identified in Tissue by Fontana-Masson stain
10667-4 | Fungus identified in Unspecified specimen by Animal inoculation
10668-2 | Fungus identified in Unspecified specimen by Environmental culture
10669-0 | Fungus identified in Unspecified specimen by Sticky tape for environmental fungus
1067-8 | Jk sup(a) Ab [Presence] in Serum or Plasma from Blood product unit
10670-8 | Giardia lamblia [Presence] in Stool by Organism specific culture
10671-6 | Helminth identified in Unspecified specimen
10672-4 | Helminth+Arthropod identified in Unspecified specimen
10673-2 | Hepatitis B virus core Ag [Presence] in Tissue by Immune stain
10674-0 | Hepatitis B virus surface Ag [Presence] in Tissue by Immune stain
10675-7 | Hepatitis B virus surface Ag [Presence] in Tissue by Orcein stain
10676-5 | Hepatitis C virus RNA [Units/volume] (viral load) in Serum or Plasma by Probe with amplification
10677-3 | Herpes simplex virus 1 Ag [Presence] in Tissue by Immune stain
10678-1 | Herpes simplex virus 1+2 Ag [Presence] in Tissue by Immune stain
10679-9 | Herpes simplex virus 2 Ag [Presence] in Tissue by Immune stain
1068-6 | Jk sup(a) Ab [Presence] in Serum or Plasma from Donor
10680-7 | Herpes simplex virus identified in Cerebral spinal fluid by Electron microscopy
10681-5 | Herpes simplex virus identified in Tissue by Electron microscopy
10682-3 | Deprecated HIV 1 RNA [Units/volume] in Serum or Plasma by Probe with amplification
10683-1 | Hydatid cyst identified in Aspirate by Immune stain
10684-9 | Hydatid cyst identified in Liver by Wet preparation
10685-6 | Hydatid cyst identified in Lung tissue by Wet preparation
10686-4 | Leishmania sp identified in Tissue by Giemsa stain
10687-2 | Leishmania sp identified in Tissue by Organism specific culture
10688-0 | Microscopic observation [Identifier] in Hair by KOH preparation
10689-8 | Microscopic observation [Identifier] in Nail by KOH preparation
1069-4 | Jk sup(a) Ab [Presence] in Serum or Plasma
10690-6 | Microsporidia identified in Duodenal fluid by Trichrome stain modified
10691-4 | Mushroom.toxic identified by Inspection
10692-2 | Mushroom.toxic identified in Food by Light microscopy
10693-0 | Mushroom.toxic identified in Vomitus by Light microscopy
10694-8 | Naegleria sp identified in Tissue by Organism specific culture
10695-5 | Naegleria sp identified in Tissue by Wet preparation
10696-3 | Norovirus [Identifier] in Stool by Electron microscopy
10697-1 | Nystatin [Susceptibility] by Minimum inhibitory concentration (MIC)
10698-9 | Nystatin [Susceptibility] by Minimum lethal concentration (MLC)
10699-7 | Onchocerca sp identified in Tissue by Wet preparation
107-3 | Cefotaxime [Susceptibility] by Minimum lethal concentration (MLC)
1070-2 | Jk sup(a) Ag [Presence] on Red Blood Cells from Blood product unit
10700-3 | Orthopoxvirus [Identifier] in Skin by Electron microscopy
10701-1 | Ova and parasites identified in Stool by Concentration
10702-9 | Ova and parasites identified in Stool by Immune stain
10703-7 | Ova and parasites identified in Stool by Kinyoun iron hematoxylin stain
10704-5 | Ova and parasites identified in Stool by Light microscopy
10705-2 | Human papilloma virus Ag [Presence] in Tissue by Immune stain

10706-0 | Picornavirus [Identifier] in Stool by Electron microscopy
10707-8 | Toxic Plant identified in Plant specimen by Inspection
10708-6 | Toxic Plant identified in Vomitus by Light microscopy
10709-4 | Plasmodium falciparum Ag [Units/volume] in Blood by Immunofluorescence
1071-0 | Jk sup(a) Ag [Presence] on Red Blood Cells from Donor
10710-2 | Plasmodium sp identified in Blood by Thin film
10711-0 | Plasmodium vivax Ag [Units/volume] in Blood by Immunofluorescence
10712-8 | Pneumocystis sp identified in Lung tissue
10713-6 | Prototheca identified in Unspecified specimen by Culture
10714-4 | Deprecated Rotavirus [Identifier] in Stool by Electron microscopy
10715-1 | Schistosoma sp identified in Urine sediment by Light microscopy
10716-9 | Schistosoma sp identified in Unspecified specimen by Organism specific culture
10717-7 | Streptococcus pneumoniae Ab [Units/volume] in Serum by Latex agglutination
10718-5 | Strongyloides sp Ab [Units/volume] in Serum by Immunoassay
10719-3 | Taenia solium adult Ab [Units/volume] in Serum by Immunoblot
1072-8 | Jk sup(a) Ag [Presence] on Red Blood Cells
10720-1 | Terbinafine [Susceptibility] by Minimum inhibitory concentration (MIC)
10721-9 | Terbinafine [Susceptibility] by Minimum lethal concentration (MLC)
10722-7 | Torovirus [Identifier] in Stool by Electron microscopy
10723-5 | Toxoplasma gondii IgA Ab [Units/volume] in Serum by Immunoassay
10724-3 | Toxoplasma gondii IgE Ab [Units/volume] in Serum by Immunoassay
10725-0 | Toxoplasma gondii DNA [Units/volume] in Body fluid by Probe with amplification
10726-8 | Toxoplasma gondii [Presence] in Tissue by Giemsa stain
10727-6 | Toxoplasma gondii identified in Tissue
10728-4 | Trichomonas sp identified in Genital specimen by Organism specific culture
10729-2 | Trypanosoma sp identified in Blood by Light microscopy
1073-6 | Jk sup(b) Ab [Presence] in Serum or Plasma from Blood product unit
10730-0 | Trypanosoma sp identified in Blood by Organism specific culture
10731-8 | Trypanosoma sp identified in Blood by Thick film
10732-6 | Trypanosoma sp identified in Blood by Thin film
10733-4 | Trypanosoma sp identified in Blood by Wet preparation
10734-2 | Varicella zoster virus identified in Skin by Electron microscopy
10735-9 | Viral sequencing [Identifier] in Serum by Sequencing
10736-7 | Virus identified in Cerebral spinal fluid by Electron microscopy
10737-5 | Virus identified in Stool by Electron microscopy
10738-3 | Virus identified in Tissue by Electron microscopy
10739-1 | Virus identified in Unspecified specimen by Electron microscopy
1074-4 | Jk sup(b) Ab [Presence] in Serum or Plasma from Donor
10740-9 | Aluminum.microscopic observation [Identifier] in Bone by Histomorphometry stain
10741-7 | Amyloid.microscopic observation [Identifier] in Brain by Thioflavine-S stain
10742-5 | Amyloid.microscopic observation [Identifier] in Tissue by Bennhold stain.Putchler modified
10743-3 | Amyloid.microscopic observation [Identifier] in Tissue by Highman stain
10744-1 | Amyloid.microscopic observation [Identifier] in Tissue by Vassar-culling stain
10745-8 | Bile.microscopic observation [Identifier] in Tissue by Fouchet stain
10746-6 | Calcium.microscopic observation [Identifier] in Tissue by Von Kossa stain
10747-4 | Collagen fibers+Elastic fibers.microscopic observation [Identifier] in Tissue by Lawson-Van Gieson stain
10748-2 | Collagen fibers+Elastic fibers.microscopic observation [Identifier] in Tissue by Verhoeff-Van Gieson stain
10749-0 | Collagen fibers.microscopic observation [Identifier] in Tissue by Van Gieson stain
1075-1 | Jk sup(b) Ab [Presence] in Serum or Plasma
10750-8 | Connective tissue.microscopic observation [Identifier] in Tissue by Trichrome stain.Masson
10751-6 | Copper.microscopic observation [Identifier] in Tissue by Rhodamine stain
10752-4 | Fat.microscopic observation [Identifier] in Milk by Sudan IV stain
10753-2 | Fat.microscopic observation [Identifier] in Stool by Sudan IV stain
10754-0 | Fat.microscopic observation [Identifier] in Tissue by Sudan IV stain
10755-7 | Fungus.microscopic observation [Identifier] in Tissue by Methenamine silver stain.Grocott
10756-5 | Glial fibers.microscopic observation [Identifier] in Tissue by Holzer stain
10757-3 | Hematologic+Nuclear elements.microscopic observation [Identifier] in Tissue by Giemsa stain.May-Grunwald
10758-1 | Iron.microscopic observation [Identifier] in Bone by Histomorphometry stain
10759-9 | Iron.microscopic observation [Identifier] in Sputum by Gomori stain
1076-9 | Jk sup(b) Ag [Presence] on Red Blood Cells from Blood product unit
10760-7 | Iron.microscopic observation [Identifier] in Tissue by Gomori stain
10761-5 | Iron.microscopic observation [Identifier] in Tissue by Other stain
10762-3 | Microscopic observation [Identifier] in Blood by Hemosiderin stain
10763-1 | Microscopic observation [Identifier] in Body fluid by Sudan black stain
10764-9 | Microscopic observation [Identifier] in Sputum by Silver stain
10765-6 | Microscopic observation [Identifier] in Tissue by Acetate esterase stain
10766-4 | Microscopic observation [Identifier] in Tissue by Alcian blue stain
10767-2 | Microscopic observation [Identifier] in Tissue by Alcian blue stain.sulfated
10768-0 | Microscopic observation [Identifier] in Tissue by Alcian blue stain.with periodic acid-Schiff
10769-8 | Microscopic observation [Identifier] in Tissue by Alizarin red S stain
1077-7 | Jk sup(b) Ag [Presence] on Red Blood Cells from Donor
10770-6 | Microscopic observation [Identifier] in Tissue by Argentaffin stain

10771-4 | Deprecated Microscopic observation [Identifier] in Tissue by Rhodamine-auramine fluorochrome stain
10772-2 | Microscopic observation [Identifier] in Tissue by Azure-eosin stain
10773-0 | Microscopic observation [Identifier] in Tissue by Basic fuchsin stain
10774-8 | Microscopic observation [Identifier] in Tissue by Bielschowsky stain
10775-5 | Microscopic observation [Identifier] in Tissue by Bleach stain
10776-3 | Microscopic observation [Identifier] in Tissue by Bodian stain
10777-1 | Microscopic observation [Identifier] in Tissue by Brown and Brenn stain
10778-9 | Microscopic observation [Identifier] in Tissue by Butyrate esterase stain
10779-7 | Microscopic observation [Identifier] in Tissue by Carmine stain.Best
1078-5 | Jk sup(b) Ag [Presence] on Red Blood Cells
10780-5 | Microscopic observation [Identifier] in Tissue by Chloracetate esterase stain
10781-3 | Microscopic observation [Identifier] in Tissue by Churukian-Schenk stain
10782-1 | Microscopic observation [Identifier] in Tissue by Congo red stain
10783-9 | Microscopic observation [Identifier] in Tissue by Crystal violet stain
10784-7 | Microscopic observation [Identifier] in Tissue by Esterase stain.non-specific
10785-4 | Microscopic observation [Identifier] in Tissue by Fite-Faraco stain
10786-2 | Deprecated Microscopic observation [Identifier] in Tissue by Giemsa stain.3 micron
10787-0 | Microscopic observation [Identifier] in Tissue by Gridley stain
10788-8 | Microscopic observation [Identifier] in Tissue by Hansel stain
10789-6 | Microscopic observation [Identifier] in Tissue by Hematoxylin-eosin-Harris regressive stain
1079-3 | Js sup(a) Ab [Presence] in Serum or Plasma from Blood product unit
10790-4 | Microscopic observation [Identifier] in Tissue by Hematoxylin-eosin-Mayers progressive stain
10791-2 | Microscopic observation [Identifier] in Tissue by Mallory-Heidenhain stain
10792-0 | Microscopic observation [Identifier] in Tissue by Methenamine silver stain.Jones
10793-8 | Microscopic observation [Identifier] in Tissue by Methyl green stain
10794-6 | Microscopic observation [Identifier] in Tissue by Methyl green-pyronine Y stain
10795-3 | Microscopic observation [Identifier] in Tissue by Methyl violet stain
10796-1 | Microscopic observation [Identifier] in Tissue by Mucicarmine stain
10797-9 | Microscopic observation [Identifier] in Tissue by Neutral red stain
10798-7 | Microscopic observation [Identifier] in Tissue by Oil red O stain
10799-5 | Microscopic observation [Identifier] in Tissue by Pentachrome stain.Movat
108-1 | Cefotaxime [Susceptibility] by Minimum inhibitory concentration (MIC)
1080-1 | Js sup(a) Ab [Presence] in Serum or Plasma from Donor
10800-1 | Microscopic observation [Identifier] in Tissue by Periodic acid-Schiff stain with diatase digestion
10801-9 | Microscopic observation [Identifier] in Tissue by Phosphotungstic acid Hematoxylin (PTAH) Stain
10802-7 | Microscopic observation [Identifier] in Tissue by Prussian blue stain
10803-5 | Microscopic observation [Identifier] in Tissue by Quinacrine fluorescent stain
10804-3 | Microscopic observation [Identifier] in Tissue by Reticulin stain
10805-0 | Microscopic observation [Identifier] in Tissue by Safranin stain
10806-8 | Microscopic observation [Identifier] in Tissue by Schmorl stain
10807-6 | Microscopic observation [Identifier] in Tissue by Sevier-Munger stain
10808-4 | Microscopic observation [Identifier] in Tissue by Silver impregnation stain.Dieterle
10809-2 | Microscopic observation [Identifier] in Tissue by Silver nitrate stain
1081-9 | Js sup(a) Ab [Presence] in Serum or Plasma
10810-0 | Microscopic observation [Identifier] in Tissue by Silver stain.Fontana-Masson
10811-8 | Microscopic observation [Identifier] in Tissue by Silver stain.Grimelius
10812-6 | Microscopic observation [Identifier] in Tissue by Steiner stain
10813-4 | Microscopic observation [Identifier] in Tissue by Sudan black stain
10814-2 | Microscopic observation [Identifier] in Tissue by Supravital stain
10815-9 | Microscopic observation [Identifier] in Tissue by Tetrachrome stain
10816-7 | Deprecated Microscopic observation [Identifier] in Tissue by Toluidine blue O stain
10817-5 | Microscopic observation [Identifier] in Tissue by Trichrome stain.Gomori-Wheatley
10818-3 | Microscopic observation [Identifier] in Tissue by Trichrome stain.Masson modified
10819-1 | Microscopic observation [Identifier] in Tissue by Wade stain
1082-7 | Js sup(a) Ag [Presence] on Red Blood Cells from Blood product unit
10820-9 | Deprecated Microscopic observation [Identifier] in Tissue by Warthin-Starry stain
10821-7 | Deprecated Microscopic observation [Identifier] in Tissue by Wright Giemsa stain
10822-5 | Mucin.microscopic observation [Identifier] in Tissue by Mucicarmine stain.Mayer
10823-3 | Mucopolysaccharides.microscopic observation [Identifier] in Tissue by Colloidal ferric oxide stain.Hale
10824-1 | Myelin+Myelin breakdown products.microscopic observation [Identifier] in Tissue by Luxol fast blue/Periodic acid-Schiff stain
10825-8 | Myelin+Nerve cells.microscopic observation [Identifier] in Tissue by Luxol fast blue/Cresyl violet stain
10826-6 | Nissel.microscopic observation [Identifier] in Tissue by Cresyl echt violet stain
10827-4 | Reticulum.microscopic observation [Identifier] in Tissue by Gomori stain
10828-2 | Urate crystals.microscopic observation [type] in Tissue by De Galantha stain
10829-0 | Silicon [Mass/volume] in Serum or Plasma
1083-5 | Js sup(a) Ag [Presence] on Red Blood Cells from Donor
10830-8 | Deprecated Surgical operation note complications
10831-6 | Surgical operation note complications [Interpretation]
10832-4 | Glucose [Mass/volume] in Serum or Plasma --15 minutes post 50 g lactose PO
10833-2 | Insulin [Mass/volume] in Serum or Plasma --7 hours post 75 g glucose PO
10834-0 | Globulin [Mass/volume] in Serum by calculation

10835-7 | Lipoprotein a [Mass/volume] in Serum or Plasma
10836-5 | Niacin [Mass/volume] in Blood
10837-3 | Organic acids [Presence] in Serum or Plasma
10838-1 | Phosphoserine [Mass/volume] in Serum or Plasma
10839-9 | Troponin I.cardiac [Mass/volume] in Serum or Plasma
1084-3 | Js sup(a) Ag [Presence] on Red Blood Cells
10840-7 | Atropine [Mass/volume] in Urine
10841-5 | Methazolamide [Mass/volume] in Serum or Plasma
10842-3 | Deprecated HLA-DQ1 [Presence]
10843-1 | Deprecated HLA-DQ2 [Presence]
10844-9 | Deprecated HLA-DQ3 [Presence]
10845-6 | Deprecated HLA-DQ4 [Presence]
10846-4 | Borrelia burgdorferi DNA [Presence] in Blood by Probe with amplification
10847-2 | Borrelia burgdorferi DNA [Presence] in Body fluid by Probe with amplification
10848-0 | Chlamydia sp IgG Ab [Titer] in Serum by Immunofluorescence
10849-8 | Chlamydia sp IgM Ab [Titer] in Serum by Immunofluorescence
1085-0 | Js sup(b) Ab [Presence] in Serum or Plasma from Blood product unit
10850-6 | Cyclospora cayetanensis [Presence] in Unspecified specimen
10851-4 | Escherichia coli O157:H7 [Presence] in Stool by Organism specific culture
10852-2 | Deprecated Fungus identified in Blood by Culture
10853-0 | Isospora belli [Presence] in Unspecified specimen by Acid fast stain.Kinyoun modified
10854-8 | Deprecated Midrofilaria identified in Blood by Concentration
10855-5 | Ova and parasites identified in Duodenal fluid or Gastric fluid by Light microscopy
10856-3 | Microscopic observation [Identifier] in Genital mucus by Gram stain
10857-1 | Microsporidia identified in Unspecified specimen by Light microscopy
10858-9 | Teichoate Ab [Presence] in Serum
10859-7 | Trypanosoma sp [Identifier] in Blood by Acridine Orange + Giemsa Stain
1086-8 | Js sup(b) Ab [Presence] in Serum or Plasma from Donor
10860-5 | Varicella zoster virus [Presence] in Unspecified specimen by Organism specific culture
10861-3 | Progesterone receptor [Mass/mass] in Tissue
10862-1 | Basement membrane Ab [Titer] in Serum
10863-9 | Endomysium IgA Ab [Titer] in Serum
10864-7 | Immune complex [Units/volume] in Serum or Plasma by Raji cell assay
10865-4 | Intercellular substance Ab [Presence] in Serum
10866-2 | Sulfatide IgG Ab [Titer] in Serum
10867-0 | Sulfatide IgM Ab [Titer] in Serum
10868-8 | Bacitracin [Susceptibility] by Disk diffusion (KB)
10869-6 | Xylose [Mass/volume] in Blood --3 hours post 25 g xylose PO
1087-6 | Js sup(b) Ab [Presence] in Serum or Plasma
10870-4 | Xylose [Mass/volume] in Blood --4 hours post 25 g xylose PO
10871-2 | Xylose [Mass/volume] in Blood --5 hours post 25 g xylose PO
10872-0 | Xylose [Mass/volume] in Blood --baseline
10873-8 | Beta-2-Microglobulin [Mass/time] in 24 hour Urine
10874-6 | Bombesin [Mass/volume] in Plasma
10875-3 | Carnitine esters [Mass/volume] in Urine
10876-1 | Carnitine esters [Mass/volume] in Serum or Plasma
10877-9 | Carnitine free (C0) [Mass/volume] in Urine
10878-7 | Iodine.protein bound [Mass/volume] in Serum or Plasma
10879-5 | Isovalerylglycine [Mass/volume] in Urine
1088-4 | Js sup(b) Ag [Presence] on Red Blood Cells from Blood product unit
10880-3 | Magnesium [Mass/mass] in Stool
10881-1 | Pentacarboxylporphyrins [Mass/volume] in Red Blood Cells
10882-9 | Pentacarboxylporphyrins [Mass/volume] in Serum or Plasma
10883-7 | Phenolphthalein [Mass/mass] in Stool
10884-5 | Phosphate [Mass/mass] in Stool
10885-2 | Porphyrins [Mass/time] in 24 hour Urine
10886-0 | Prostate Specific Ag Free [Mass/volume] in Serum or Plasma
10887-8 | Pyridinoline [Mass/time] in 24 hour Urine
10888-6 | Lipase [Enzymatic activity/volume] in Urine
10889-4 | Bisacodyl [Mass/mass] in Stool
1089-2 | Js sup(b) Ag [Presence] on Red Blood Cells from Donor
10890-2 | Deprecated Normethsuximide [Mass/volume] in Serum or Plasma
10891-0 | Oxyphenisatin [Mass/mass] in Stool
10892-8 | Pentoxifylline [Mass/volume] in Serum or Plasma
10893-6 | Trenbolone [Mass/volume] in Urine
10894-4 | Aspergillus niger Ab [Presence] in Serum by Immune diffusion (ID)
10895-1 | Clostridium difficile toxin B [Presence] in Stool
10896-9 | Eastern equine encephalitis virus IgG Ab [Titer] in Serum by Immunofluorescence
10897-7 | Eastern equine encephalitis virus IgG Ab [Titer] in Cerebral spinal fluid by Immunofluorescence
10898-5 | Eastern equine encephalitis virus IgM Ab [Titer] in Serum by Immunofluorescence

4.4.2.10879 HL7 v3 Value Set EntityDeterminerDetermined

URL: https://hl7.org/fhir/R4/v3/EntityDeterminerDetermined/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-EntityDeterminerDetermined) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-EntityDeterminerDetermined
Version: | 2014-03-26
Name: | v3.EntityDeterminerDetermined
Title: | V3 Value SetEntityDeterminerDetermined
Definition: | The described determiner is used to indicate that the given Entity is taken as a general description of a kind of thing that can be taken in whole, in part, or in multiples.
OID: | 2.16.840.1.113883.1.11.10879 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.10879.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-EntityDeterminer where concept is-a KIND

4.4.2.10879.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 2 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-EntityDeterminer version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-EntityDeterminer

Lvl | Code | Display | Definition
0 | KIND | described | Description:A determiner that specifies that the Entity object represents a universal, kind or class of physical thing (as opposed to a particular thing).
1 | GROUPKIND | described group | A determiner that specifies that the Entity object represents a universal, kind or class of collections physical things. While the collection may resolve to having only a single individual (or even no individuals), the potential should exist for it to cover multiple individuals.

4.4.2.10884 HL7 v3 Value Set EntityClassLivingSubject

URL: https://hl7.org/fhir/R4/v3/EntityClassLivingSubject/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-EntityClassLivingSubject) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-EntityClassLivingSubject
Version: | 2014-03-26
Name: | v3.EntityClassLivingSubject
Title: | V3 Value SetEntityClassLivingSubject
Definition: | Anything that essentially has the property of life, independent of current state (a dead human corpse is still essentially a living subject.)
OID: | 2.16.840.1.113883.1.11.10884 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.10884.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-EntityClass where concept is-a LIV

4.4.2.10884.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 6 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-EntityClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-EntityClass

Lvl | Code | Display | Definition
0 | LIV | living subject | Anything that essentially has the property of life, independent of current state (a dead human corpse is still essentially a living subject).
1 | NLIV | non-person living subject | A subtype of living subject that includes all living things except the species Homo Sapiens.
2 | ANM | animal | A living subject from the animal kingdom.

2 | MIC | microorganism | All single celled living organisms including protozoa, bacteria, yeast, viruses, etc.
2 | PLNT | plant | A living subject from the order of plants.
1 | PSN | person | A living subject of the species homo sapiens.

4.4.2.10889 HL7 v3 Value Set EntityClassOrganization

URL: https://hl7.org/fhir/R4/v3/EntityClassOrganization/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-EntityClassOrganization) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-EntityClassOrganization
Version: | 2014-03-26
Name: | v3.EntityClassOrganization
Title: | V3 Value SetEntityClassOrganization
Definition: | A social or legal structure formed by human beings.
OID: | 2.16.840.1.113883.1.11.10889 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.10889.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-EntityClass where concept is-a ORG

4.4.2.10889.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 4 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-EntityClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-EntityClass

Lvl | Code | Display | Definition
0 | ORG | organization | A social or legal structure formed by human beings.
1 | PUB | public institution | An agency of the people of a state often assuming some authority over a certain matter. Includes government, governmental agencies, associations.
1 | STATE | state | A politically organized body of people bonded by territory, culture, or ethnicity, having sovereignty (to a certain extent) granted by other states (enclosing or neighboring states). This includes countries (nations), provinces (e.g., one of the United States of America or a French departement), counties or municipalities. Refine using, e.g., ISO country codes, FIPS-PUB state codes, etc.
2 | NAT | Nation | A politically organized body of people bonded by territory and known as a nation.

4.4.2.10892 HL7 v3 Value Set EntityClassPlace

URL: https://hl7.org/fhir/R4/v3/EntityClassPlace/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-EntityClassPlace) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-EntityClassPlace
Version: | 2014-03-26
Name: | v3.EntityClassPlace
Title: | V3 Value SetEntityClassPlace
Definition: | A physicial place or site with its containing structure. May be natural or man-made. The geographic position of a place may or may not be constant.
OID: | 2.16.840.1.113883.1.11.10892 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.10892.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-EntityClass where concept is-a PLC

4.4.2.10892.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 5 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-EntityClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-EntityClass

Lvl | Code | Display | Definition
0 | PLC | place | A physical place or site with its containing structure. May be natural or man-made. The geographic position of a place may or may not be constant.
1 | CITY | city or town | The territory of a city, town or other municipality.
1 | COUNTRY | country | The territory of a sovereign nation.
1 | COUNTY | county or parish | The territory of a county, parish or other division of a state or province.
1 | PROVINCE | state or province | The territory of a state, province, department or other division of a sovereign country.

4.4.2.11526 HL7 v3 Value Set HumanLanguage

URL: https://hl7.org/fhir/R4/v3/HumanLanguage/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-HumanLanguage) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-HumanLanguage
Version: | 2014-03-26
Name: | v3.HumanLanguage
Title: | V3 Value SetHumanLanguage
Definition: | Codes for the representation of the names of human languages.
OID: | 2.16.840.1.113883.1.11.11526 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.11526.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include all codes defined in urn:iso:std:iso:3166

4.4.2.11526.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 747 concepts

All codes from system urn:iso:std:iso:3166

Code | Display
AD | Andorra
AE | United Arab Emirates
AF | Afghanistan
AG | Antigua and Barbuda
AI | Anguilla
AL | Albania
AM | Armenia
AO | Angola
AQ | Antarctica
AR | Argentina
AS | American Samoa
AT | Austria
AU | Australia
AW | Aruba
AX | Åland Islands
AZ | Azerbaijan
BA | Bosnia and Herzegovina
BB | Barbados
BD | Bangladesh
BE | Belgium
BF | Burkina Faso
BG | Bulgaria
BH | Bahrain
BI | Burundi
BJ | Benin
BL | Saint Barthélemy
BM | Bermuda
BN | Brunei Darussalam
BO | Bolivia, Plurinational State of
BQ | Bonaire, Sint Eustatius and Saba
BR | Brazil
BS | Bahamas
BT | Bhutan
BV | Bouvet Island
BW | Botswana
BY | Belarus

BZ | Belize
CA | Canada
CC | Cocos (Keeling) Islands
CD | Congo, the Democratic Republic of the
CF | Central African Republic
CG | Congo
CH | Switzerland
CI | Côte d'Ivoire
CK | Cook Islands
CL | Chile
CM | Cameroon
CN | China
CO | Colombia
CR | Costa Rica
CU | Cuba
CV | Cabo Verde
CW | Curaçao
CX | Christmas Island
CY | Cyprus
CZ | Czechia
DE | Germany
DJ | Djibouti
DK | Denmark
DM | Dominica
DO | Dominican Republic
DZ | Algeria
EC | Ecuador
EE | Estonia
EG | Egypt
EH | Western Sahara
ER | Eritrea
ES | Spain
ET | Ethiopia
FI | Finland
FJ | Fiji
FK | Falkland Islands (Malvinas)
FM | Micronesia, Federated States of
FO | Faroe Islands
FR | France
GA | Gabon
GB | United Kingdom of Great Britain and Northern Ireland
GD | Grenada
GE | Georgia
GF | French Guiana
GG | Guernsey
GH | Ghana
GI | Gibraltar
GL | Greenland
GM | Gambia
GN | Guinea
GP | Guadeloupe
GQ | Equatorial Guinea
GR | Greece
GS | South Georgia and the South Sandwich Islands
GT | Guatemala
GU | Guam
GW | Guinea-Bissau
GY | Guyana
HK | Hong Kong
HM | Heard Island and McDonald Islands
HN | Honduras
HR | Croatia
HT | Haiti
HU | Hungary
ID | Indonesia
IE | Ireland
IL | Israel
IM | Isle of Man
IN | India
IO | British Indian Ocean Territory
IQ | Iraq
IR | Iran, Islamic Republic of

IS | Iceland
IT | Italy
JE | Jersey
JM | Jamaica
JO | Jordan
JP | Japan
KE | Kenya
KG | Kyrgyzstan
KH | Cambodia
KI | Kiribati
KM | Comoros
KN | Saint Kitts and Nevis
KP | Korea, Democratic People's Republic of
KR | Korea, Republic of
KW | Kuwait
KY | Cayman Islands
KZ | Kazakhstan
LA | Lao People's Democratic Republic
LB | Lebanon
LC | Saint Lucia
LI | Liechtenstein
LK | Sri Lanka
LR | Liberia
LS | Lesotho
LT | Lithuania
LU | Luxembourg
LV | Latvia
LY | Libya
MA | Morocco
MC | Monaco
MD | Moldova, Republic of
ME | Montenegro
MF | Saint Martin (French part)
MG | Madagascar
MH | Marshall Islands
MK | Macedonia, the former Yugoslav Republic of
ML | Mali
MM | Myanmar
MN | Mongolia
MO | Macao
MP | Northern Mariana Islands
MQ | Martinique
MR | Mauritania
MS | Montserrat
MT | Malta
MU | Mauritius
MV | Maldives
MW | Malawi
MX | Mexico
MY | Malaysia
MZ | Mozambique
NA | Namibia
NC | New Caledonia
NE | Niger
NF | Norfolk Island
NG | Nigeria
NI | Nicaragua
NL | Netherlands
NO | Norway
NP | Nepal
NR | Nauru
NU | Niue
NZ | New Zealand
OM | Oman
PA | Panama
PE | Peru
PF | French Polynesia
PG | Papua New Guinea
PH | Philippines
PK | Pakistan
PL | Poland
PM | Saint Pierre and Miquelon

PN | Pitcairn
PR | Puerto Rico
PS | Palestine, State of
PT | Portugal
PW | Palau
PY | Paraguay
QA | Qatar
RE | Réunion
RO | Romania
RS | Serbia
RU | Russian Federation
RW | Rwanda
SA | Saudi Arabia
SB | Solomon Islands
SC | Seychelles
SD | Sudan
SE | Sweden
SG | Singapore
SH | Saint Helena, Ascension and Tristan da Cunha
SI | Slovenia
SJ | Svalbard and Jan Mayen
SK | Slovakia
SL | Sierra Leone
SM | San Marino
SN | Senegal
SO | Somalia
SR | Suriname
SS | South Sudan
ST | Sao Tome and Principe
SV | El Salvador
SX | Sint Maarten (Dutch part)
SY | Syrian Arab Republic
SZ | Swaziland
TC | Turks and Caicos Islands
TD | Chad
TF | French Southern Territories
TG | Togo
TH | Thailand
TJ | Tajikistan
TK | Tokelau
TL | Timor-Leste
TM | Turkmenistan
TN | Tunisia
TO | Tonga
TR | Turkey
TT | Trinidad and Tobago
TV | Tuvalu
TW | Taiwan, Province of China
TZ | Tanzania, United Republic of
UA | Ukraine
UG | Uganda
UM | United States Minor Outlying Islands
US | United States of America
UY | Uruguay
UZ | Uzbekistan
VA | Holy See
VC | Saint Vincent and the Grenadines
VE | Venezuela, Bolivarian Republic of
VG | Virgin Islands, British
VI | Virgin Islands,
VN | Viet Nam
VU | Vanuatu
WF | Wallis and Futuna
WS | Samoa
YE | Yemen
YT | Mayotte
ZA | South Africa
ZM | Zambia
ZW | Zimbabwe
ABW | Aruba
AFG | Afghanistan
AGO | Angola

AIA | Anguilla
ALA | Åland Islands
ALB | Albania
AND | Andorra
ARE | United Arab Emirates
ARG | Argentina
ARM | Armenia
ASM | American Samoa
ATA | Antarctica
ATF | French Southern Territories
ATG | Antigua and Barbuda
AUS | Australia
AUT | Austria
AZE | Azerbaijan
BDI | Burundi
BEL | Belgium
BEN | Benin
BES | Bonaire, Sint Eustatius and Saba
BFA | Burkina Faso
BGD | Bangladesh
BGR | Bulgaria
BHR | Bahrain
BHS | Bahamas
BIH | Bosnia and Herzegovina
BLM | Saint Barthélemy
BLR | Belarus
BLZ | Belize
BMU | Bermuda
BOL | Bolivia, Plurinational State of
BRA | Brazil
BRB | Barbados
BRN | Brunei Darussalam
BTN | Bhutan
BVT | Bouvet Island
BWA | Botswana
CAF | Central African Republic
CAN | Canada
CCK | Cocos (Keeling) Islands
CHE | Switzerland
CHL | Chile
CHN | China
CIV | Côte d'Ivoire
CMR | Cameroon
COD | Congo, the Democratic Republic of the
COG | Congo
COK | Cook Islands
COL | Colombia
COM | Comoros
CPV | Cabo Verde
CRI | Costa Rica
CUB | Cuba
CUW | Curaçao
CXR | Christmas Island
CYM | Cayman Islands
CYP | Cyprus
CZE | Czechia
DEU | Germany
DJI | Djibouti
DMA | Dominica
DNK | Denmark
DOM | Dominican Republic
DZA | Algeria
ECU | Ecuador
EGY | Egypt
ERI | Eritrea
ESH | Western Sahara
ESP | Spain
EST | Estonia
ETH | Ethiopia
FIN | Finland
FJI | Fiji
FLK | Falkland Islands (Malvinas)

FRA | France
FRO | Faroe Islands
FSM | Micronesia, Federated States of
GAB | Gabon
GBR | United Kingdom
GEO | Georgia
GGY | Guernsey
GHA | Ghana
GIB | Gibraltar
GIN | Guinea
GLP | Guadeloupe
GMB | Gambia
GNB | Guinea-Bissau
GNQ | Equatorial Guinea
GRC | Greece
GRD | Grenada
GRL | Greenland
GTM | Guatemala
GUF | French Guiana
GUM | Guam
GUY | Guyana
HKG | Hong Kong
HMD | Heard Island and McDonald Islands
HND | Honduras
HRV | Croatia
HTI | Haiti
HUN | Hungary
IDN | Indonesia
IMN | Isle of Man
IND | India
IOT | British Indian Ocean Territory
IRL | Ireland
IRN | Iran, Islamic Republic of
IRQ | Iraq
ISL | Iceland
ISR | Israel
ITA | Italy
JAM | Jamaica
JEY | Jersey
JOR | Jordan
JPN | Japan
KAZ | Kazakhstan
KEN | Kenya
KGZ | Kyrgyzstan
KHM | Cambodia
KIR | Kiribati
KNA | Saint Kitts and Nevis
KOR | Korea, Republic of
KWT | Kuwait
LAO | Lao People's Democratic Republic
LBN | Lebanon
LBR | Liberia
LBY | Libya
LCA | Saint Lucia
LIE | Liechtenstein
LKA | Sri Lanka
LSO | Lesotho
LTU | Lithuania
LUX | Luxembourg
LVA | Latvia
MAC | Macao
MAF | Saint Martin (French part)
MAR | Morocco
MCO | Monaco
MDA | Moldova, Republic of
MDG | Madagascar
MDV | Maldives
MEX | Mexico
MHL | Marshall Islands
MKD | Macedonia, the former Yugoslav Republic of
MLI | Mali
MLT | Malta

MMR | Myanmar
MNE | Montenegro
MNG | Mongolia
MNP | Northern Mariana Islands
MOZ | Mozambique
MRT | Mauritania
MSR | Montserrat
MTQ | Martinique
MUS | Mauritius
MWI | Malawi
MYS | Malaysia
MYT | Mayotte
NAM | Namibia
NCL | New Caledonia
NER | Niger
NFK | Norfolk Island
NGA | Nigeria
NIC | Nicaragua
NIU | Niue
NLD | Netherlands
NOR | Norway
NPL | Nepal
NRU | Nauru
NZL | New Zealand
OMN | Oman
PAK | Pakistan
PAN | Panama
PCN | Pitcairn
PER | Peru
PHL | Philippines
PLW | Palau
PNG | Papua New Guinea
POL | Poland
PRI | Puerto Rico
PRK | Korea, Democratic People's Republic of
PRT | Portugal
PRY | Paraguay
PSE | Palestine, State of
PYF | French Polynesia
QAT | Qatar
REU | Réunion
ROU | Romania
RUS | Russian Federation
RWA | Rwanda
SAU | Saudi Arabia
SDN | Sudan
SEN | Senegal
SGP | Singapore
SGS | South Georgia and the South Sandwich Islands
SHN | Saint Helena, Ascension and Tristan da Cunha
SJM | Svalbard and Jan Mayen
SLB | Solomon Islands
SLE | Sierra Leone
SLV | El Salvador
SMR | San Marino
SOM | Somalia
SPM | Saint Pierre and Miquelon
SRB | Serbia
SSD | South Sudan
STP | Sao Tome and Principe
SUR | Suriname
SVK | Slovakia
SVN | Slovenia
SWE | Sweden
SWZ | Swaziland
SXM | Sint Maarten (Dutch part)
SYC | Seychelles
SYR | Syrian Arab Republic
TCA | Turks and Caicos Islands
TCD | Chad
TGO | Togo
THA | Thailand

TJK | Tajikistan
TKL | Tokelau
TKM | Turkmenistan
TLS | Timor-Leste
TON | Tonga
TTO | Trinidad and Tobago
TUN | Tunisia
TUR | Turkey
TUV | Tuvalu
TWN | Taiwan, Province of China
TZA | Tanzania, United Republic of
UGA | Uganda
UKR | Ukraine
UMI | United States Minor Outlying Islands
URY | Uruguay
USA | United States of America
UZB | Uzbekistan
VAT | Holy See
VCT | Saint Vincent and the Grenadines
VEN | Venezuela, Bolivarian Republic of
VGB | Virgin Islands, British
VIR | Virgin Islands, U.S.
VNM | Viet Nam
VUT | Vanuatu
WLF | Wallis and Futuna
WSM | Samoa
YEM | Yemen
ZAF | South Africa
ZMB | Zambia
ZWE | Zimbabwe
004 | Afghanistan
008 | Albania
010 | Antarctica
012 | Algeria
016 | American Samoa
020 | Andorra
024 | Angola
028 | Antigua and Barbuda
031 | Azerbaijan
032 | Argentina
036 | Australia
040 | Austria
044 | Bahamas
048 | Bahrain
050 | Bangladesh
051 | Armenia
052 | Barbados
056 | Belgium
060 | Bermuda
064 | Bhutan
068 | Bolivia, Plurinational State of
070 | Bosnia and Herzegovina
072 | Botswana
074 | Bouvet Island
076 | Brazil
084 | Belize
086 | British Indian Ocean Territory
090 | Solomon Islands
092 | Virgin Islands, British
096 | Brunei Darussalam
100 | Bulgaria
104 | Myanmar
108 | Burundi
112 | Belarus
116 | Cambodia
120 | Cameroon
124 | Canada
132 | Cabo Verde
136 | Cayman Islands
140 | Central African Republic
144 | Sri Lanka
148 | Chad

152 | Chile
156 | China
158 | Taiwan, Province of China
162 | Christmas Island
166 | Cocos (Keeling) Islands
170 | Colombia
174 | Comoros
175 | Mayotte
178 | Congo
180 | Congo, the Democratic Republic of the
184 | Cook Islands
188 | Costa Rica
191 | Croatia
192 | Cuba
196 | Cyprus
203 | Czechia
204 | Benin
208 | Denmark
212 | Dominica
214 | Dominican Republic
218 | Ecuador
222 | El Salvador
226 | Equatorial Guinea
231 | Ethiopia
232 | Eritrea
233 | Estonia
234 | Faroe Islands
238 | Falkland Islands (Malvinas)
239 | South Georgia and the South Sandwich Islands
242 | Fiji
246 | Finland
248 | Åland Islands
250 | France
254 | French Guiana
258 | French Polynesia
260 | French Southern Territories
262 | Djibouti
266 | Gabon
268 | Georgia
270 | Gambia
275 | Palestine, State of
276 | Germany
288 | Ghana
292 | Gibraltar
296 | Kiribati
300 | Greece
304 | Greenland
308 | Grenada
312 | Guadeloupe
316 | Guam
320 | Guatemala
324 | Guinea
328 | Guyana
332 | Haiti
334 | Heard Island and McDonald Islands
336 | Holy See
340 | Honduras
344 | Hong Kong
348 | Hungary
352 | Iceland
356 | India
360 | Indonesia
364 | Iran, Islamic Republic of
368 | Iraq
372 | Ireland
376 | Israel
380 | Italy
384 | Côte d'Ivoire
388 | Jamaica
392 | Japan
398 | Kazakhstan
400 | Jordan

404 | Kenya
408 | Korea, Democratic People's Republic of
410 | Korea, Republic of
414 | Kuwait
417 | Kyrgyzstan
418 | Lao People's Democratic Republic
422 | Lebanon
426 | Lesotho
428 | Latvia
430 | Liberia
434 | Libya
438 | Liechtenstein
440 | Lithuania
442 | Luxembourg
446 | Macao
450 | Madagascar
454 | Malawi
458 | Malaysia
462 | Maldives
466 | Mali
470 | Malta
474 | Martinique
478 | Mauritania
480 | Mauritius
484 | Mexico
492 | Monaco
496 | Mongolia
498 | Moldova, Republic of
499 | Montenegro
500 | Montserrat
504 | Morocco
508 | Mozambique
512 | Oman
516 | Namibia
520 | Nauru
524 | Nepal
528 | Netherlands
531 | Curaçao
533 | Aruba
534 | Sint Maarten (Dutch part)
535 | Bonaire, Sint Eustatius and Saba
540 | New Caledonia
548 | Vanuatu
554 | New Zealand
558 | Nicaragua
562 | Niger
566 | Nigeria
570 | Niue
574 | Norfolk Island
578 | Norway
580 | Northern Mariana Islands
581 | United States Minor Outlying Islands
583 | Micronesia, Federated States of
584 | Marshall Islands
585 | Palau
586 | Pakistan
591 | Panama
598 | Papua New Guinea
600 | Paraguay
604 | Peru
608 | Philippines
612 | Pitcairn
616 | Poland
620 | Portugal
624 | Guinea-Bissau
626 | Timor-Leste
630 | Puerto Rico
634 | Qatar
638 | Réunion
642 | Romania
643 | Russian Federation
646 | Rwanda

652 | Saint Barthélemy
654 | Saint Helena, Ascension and Tristan da Cunha
659 | Saint Kitts and Nevis
660 | Anguilla
662 | Saint Lucia
663 | Saint Martin (French part)
666 | Saint Pierre and Miquelon
670 | Saint Vincent and the Grenadines
674 | San Marino
678 | Sao Tome and Principe
682 | Saudi Arabia
686 | Senegal
688 | Serbia
690 | Seychelles
694 | Sierra Leone
702 | Singapore
703 | Slovakia
704 | Viet Nam
705 | Slovenia
706 | Somalia
710 | South Africa
716 | Zimbabwe
724 | Spain
728 | South Sudan
729 | Sudan
732 | Western Sahara
740 | Suriname
744 | Svalbard and Jan Mayen
748 | Swaziland
752 | Sweden
756 | Switzerland
760 | Syrian Arab Republic
762 | Tajikistan
764 | Thailand
768 | Togo
772 | Tokelau
776 | Tonga
780 | Trinidad and Tobago
784 | United Arab Emirates
788 | Tunisia
792 | Turkey
795 | Turkmenistan
796 | Turks and Caicos Islands
798 | Tuvalu
800 | Uganda
804 | Ukraine
807 | Macedonia, the former Yugoslav Republic of
818 | Egypt
826 | United Kingdom
831 | Guernsey
832 | Jersey
833 | Isle of Man
834 | Tanzania, United Republic of
840 | United States of America
850 | Virgin Islands, U.S.
854 | Burkina Faso
858 | Uruguay
860 | Uzbekistan
862 | Venezuela, Bolivarian Republic of
876 | Wallis and Futuna
882 | Samoa
887 | Yemen
894 | Zambia

4.4.2.11529 HL7 v3 Value Set ActClassObservation

URL: https://hl7.org/fhir/R4/v3/ActClassObservation/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActClassObservation) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActClassObservation

Version: | 2014-03-26
Name: | v3.ActClassObservation
Title: | V3 Value SetActClassObservation
Definition: | Description: An act that is intended to result in new information about a subject. The main difference between Observations and other Acts is that Observations have a value attribute. Thecode attribute of Observation and thevalue attribute of Observation must be considered in combination to determine the semantics of the observation. Discussion: Structurally, many observations are name-value-pairs, where the Observation.code (inherited from Act) is the name and the Observation.value is the value of the property. Such a construct is also known as a variable (a named feature that can assume a value) hence, the Observation class is always used to hold generic name-value-pairs or variables, even though the variable valuation may not be the result of an elaborate observation method. It may be a simple answer to a question or it may be an assertion or setting of a parameter. As with all Act statements, Observation statements describe what was done, and in the case of Observations, this includes a description of what was actually observed (results or answers); and those results or answers are part of the observation and not split off into other objects. The method of action is asserted by the Observation classCode or its subclasses at the least granular level, by the Observation.code attribute value at the medium level of granularity, and by the attribute value of observation.methodCode when a finer level of granularity is required. The method in whole or in part may also appear in the attribute value of Observation.value when using coded data types to express the value of the attribute. Relevant aspects of methodology may also be restated in value when the results themselves imply or state a methodology. An observation may consist of component observations each having their own Observation.code and Observation.value. In this case, the composite observation may not have an Observation.value for itself. For instance, a white blood cell count consists of the sub-observations for the counts of the various granulocytes, lymphocytes and other normal or abnormal blood cells (e.g., blasts). The overall white blood cell count Observation itself may therefore not have a value by itself (even though it could have one, e.g., the sum total of white blood cells). Thus, as long as an Act is essentially an Act of recognizing and noting information about a subject, it is an Observation, regardless of whether it has a simple value by itself or whether it has sub-observations. Even though observations are professional acts (see Act) and as such are intentional actions, this does not require that every possible outcome of an observation be pondered in advance of it being actually made. For instance, differential white blood cell counts (WBC) rarely show blasts, but if they do, this is part of the WBC observation even though blasts might not be predefined in the structure of a normal WBC. Clinical documents commonly have Subjective and Objective findings, both of which are kinds of Observations. In addition, clinical documents commonly contain Assessments, which are also kinds of Observations. Thus, the establishment of a diagnosis is an Observation. Examples: Recording the results of a Family History Assessment Laboratory test and associated result Physical exam test and associated result Device temperature Soil lead level
OID: | 2.16.840.1.113883.1.11.11529 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.11529.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActClass where concept is-a OBS

4.4.2.11529.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 23 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActClass

Lvl | Code | Display | Definition
0 | OBS | observation | Description:An act that is intended to result in new information about a subject. The main difference between Observations and other Acts is that Observations have a value attribute. The code attribute of Observation and the value attribute of Observation must be considered in combination to determine the semantics of the observation. Discussion: Structurally, many observations are name-value-pairs, where the Observation.code (inherited from Act) is the name and the Observation.value is the value of the property. Such a construct is also known as a variable (a named feature that can assume a value) hence, the Observation class is always used to hold generic name-value-pairs or variables, even though the variable valuation may not be the result of an elaborate observation method. It may be a simple answer to a question or it may be an assertion or setting of a parameter. As with all Act statements, Observation statements describe what was done, and in the case of Observations, this includes a description of what was actually observed (results or answers); and those results or answers are part of the observation and not split off into other objects. The method of action is asserted by the Observation classCode or its subclasses at the least granular level, by the Observation.code attribute value at the medium level of granularity, and by the attribute value of observation.methodCode when a finer level of granularity is required. The method in whole or in part may also appear in the attribute value of Observation.value when using coded data types to express the value of the attribute. Relevant aspects of methodology may also be restated in value when the results themselves imply or state a methodology. An observation may consist of component observations each having their own Observation.code and Observation.value. In this case, the composite observation may not have an Observation.value for itself. For instance, a white blood cell count consists of the sub-observations for the counts of the various granulocytes, lymphocytes and other normal or abnormal blood cells (e.g., blasts). The overall white blood cell count Observation itself may therefore not have a value by itself (even though it could have one, e.g., the sum total of white blood cells). Thus, as long as an Act is essentially an Act of recognizing and noting information about a subject, it is an Observation, regardless of whether it has a simple value by itself or whether it has sub-observations. Even though observations are professional acts (see Act) and as such are intentional actions, this does not require that every possible outcome of an observation be pondered in advance of it being actually made. For instance, differential white blood cell counts (WBC) rarely show blasts, but if they do, this is part of the WBC observation even though blasts might not be predefined in the structure of a normal WBC. Clinical documents commonly have Subjective and Objective findings, both of which are kinds of Observations. In addition, clinical documents commonly contain Assessments, which are also kinds of Observations. Thus, the establishment of a diagnosis is an Observation. Examples: Recording the results of a Family History Assessment Laboratory test and associated result Physical exam test and associated result Device temperature Soil lead level
1 | _ActClassROI | ActClassROI | Regions of Interest (ROI) within a subject Act. Primarily used for making secondary observations on a subset of a subject observation. The relationship between a ROI and its referenced Act is specified through an ActRelationship of type "subject" (SUBJ), which must always be present.

2 | ROIBND | bounded ROI | A Region of Interest (ROI) specified for a multidimensional observation, such as an Observation Series (OBSSER). The ROI is specified using a set of observation criteria, each delineating the boundary of the region in one of the dimensions in the multidimensional observation. The relationship between a ROI and its referenced Act is specified through an ActRelationship of type subject (SUBJ), which must always be present. Each of the boundary criteria observations is connected with the ROI using ActRelationships of type "has component" (COMP). In each boundary criterion, the Act.code names the dimension and the Observation.value specifies the range of values inside the region. Typically the bounded dimension is continuous, and so the Observation.value will be an interval (IVL) data type. The Observation.value need not be specified if the respective dimension is only named but not constrained. For example, an ROI for the QT interval of a certain beat in ECG Lead II would contain 2 boundary criteria, one naming the interval in time (constrained), and the other naming the interval in ECG Lead II (only named, but not constrained).

2 | ROIOVL | overlay ROI | A Region of Interest (ROI) specified for an image using an overlay shape. Typically used to make reference to specific regions in images, e.g., to specify the location of a radiologic finding in an image or to specify the site of a physical finding by "circling" a region in a schematic picture of a human body. The units of the coordinate values are in pixels. The origin is in the upper left hand corner, with positive X values going to the right and positive Y values going down. The relationship between a ROI and its referenced Act is specified through an ActRelationship of type "subject" (SUBJ), which must always be present.

1 | _SubjectPhysicalPosition | subject physical position | The spatial relationship of a subject whether human, other animal, or plant, to a frame of reference such as gravity or a collection device.

1 | ALRT | detected issue | An observation identifying a potential adverse outcome as a result of an Act or combination of Acts. Examples: Detection of a drug-drug interaction; Identification of a late-submission for an invoice; Requesting discharge for a patient who does not meet hospital-defined discharge criteria. Discussion: This class is commonly used for identifying 'business rule' or 'process' problems that may result in a refusal to carry out a particular request. In some circumstances it may be possible to 'bypass' a problem by modifying the request to acknowledge the issue and/or by providing some form of mitigation. Constraints: the Act or Acts that may cause the the adverse outcome are the target of a subject ActRelationship. The subbtypes of this concept indicate the type of problem being detected (e.g. drug-drug interaction) while the Observation.value is used to repesent a specific problem code (e.g. specific drug-drug interaction id).

1 | BATTERY | battery | Definition: An observation that is composed of a set of observations. These observations typically have a logical or practical grouping for generally accepted clinical or functional purposes, such as observations that are run together because of automation. A battery can define required and optional component observations and, in some cases, will define complex rules that determine whether or not a particular observation is made. BATTERY is a constraint on the Observation class in that it is understood to always be composed of component observations. UsageNotes: The focus in a BATTERY is that it is composed of individual observations. In request (RQO) mood, a battery is a request to perform the component observations. In event (EVN) mood a battery is a reporting of associated set of observation events. In definition mood a battery is the definition of the associated set of observations. Examples: Vital signs, Full blood count, Chemistry panel.

1 | CLNTRL | clinical trial | The set of actions that define an experiment to assess the effectiveness and/or safety of a biopharmaceutical product (food, drug, device, etc.). In definition mood, this set of actions is often embodied in a clinical trial protocol; in event mood, this designates the aggregate act of applying the actions to one or more subjects.

1 | DGIMG | diagnostic image | Class for holding attributes unique to diagnostic images.

1 | GEN | genomic observation | Description:An observation of genomic phenomena.

2 | DETPOL | determinant peptide | Description:A determinant peptide in a polypeptide as described by polypeptide.

2 | EXP | expression level | Description:An expression level of genes/proteins or other expressed genomic entities.

2 | LOC | locus | Description:The position of a gene (or other significant sequence) on the genome.

2 | PHN | phenotype | Description:A genomic phenomenon that is expressed externally in the organism.

2 | POL | polypeptide | Description:A polypeptide resulting from the translation of a gene.

2 | SEQ | bio sequence | Description:A sequence of biomolecule like the DNA, RNA, protein and the like.

2 | SEQVAR | bio sequence variation | Description:A variation in a sequence as described by BioSequence.

1 | INVSTG | investigation | An formalized inquiry into the circumstances surrounding a particular unplanned event or potential event for the purposes of identifying possible causes and contributing factors for the event. This investigation could be conducted at a local institutional level or at the level of a local or national government.

1 | OBSSER | observation series | Container for Correlated Observation Sequences sharing a common frame of reference. All Observations of the same cd must be comparable and relative to the common frame of reference. For example, a 3-channel ECG device records a 12-lead ECG in 4 steps (3 leads at a time). Each of the separate 3-channel recordings would be in their own "OBSCOR". And, all 4 OBSCOR would be contained in one OBSSER because all the times are relative to the same origin (beginning of the recording) and all the ECG signals were from a fixed set of electrodes.

2 | OBSCOR | correlated observation sequences | Container for Observation Sequences (Observations whose values are contained in LIST<>'s) having values correlated with each other. Each contained Observation Sequence LIST<> must be the same length. Values in the LIST<>'s are correlated based on index. E.g. the values in position 2 in all the LIST<>'s are correlated. This is analogous to a table where each column is an Observation Sequence with a LIST<> of values, and each row in the table is a correlation between the columns. For example, a 12-lead ECG would contain 13 sequences: one sequence for time, and a sequence for each of the 12 leads.

1 | POS | position | An observation denoting the physical location of a person or thing based on a reference coordinate system.

2 | POSACC | position accuracy | Description:An observation representing the degree to which the assignment of the spatial coordinates, based on a matching algorithm by a geocoding engine against a reference spatial database, matches true or accepted values.

2 | POSCOORD | position coordinate | Description:An observation representing one of a set of numerical values used to determine the position of a place. The name of the coordinate value is determined by the reference coordinate system.

1 | SPCOBS | specimen observation | An observation on a specimen in a laboratory environment that may affect processing, analysis or result interpretation

1 | VERIF | Verification | An act which describes the process whereby a 'verifying party' validates either the existence of the Role attested to by some Credential or the actual Vetting act and its details.

4.4.2.11535 HL7 v3 Value Set ActClassSupply

URL: https://hl7.org/fhir/R4/v3/ActClassSupply/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActClassSupply) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActClassSupply
Version: | 2014-03-26
Name: | v3.ActClassSupply
Title: | V3 Value SetActClassSupply
Definition: | Supply orders and deliveries are simple Acts that focus on the delivered product. The product is associated with the Supply Act via Participation.typeCode="product". With general Supply Acts, the precise identification of the Material (manufacturer, serial numbers, etc.) is important. Most of the detailed information about the Supply should be represented using the Material class. If delivery needs to be scheduled, tracked, and billed separately, one can associate a Transportation Act with the Supply Act. Pharmacy dispense services are represented as Supply Acts, associated with a SubstanceAdministration Act. The SubstanceAdministration class represents the administration of medication, while dispensing is supply.
OID: | 2.16.840.1.113883.1.11.11535 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.11535.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActClass where concept is-a SPLY

4.4.2.11535.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 1 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActClass

Code | Display | Definition
SPLY | supply | Supply orders and deliveries are simple Acts that focus on the delivered product. The product is associated with the Supply Act via Participation.typeCode="product". With general Supply Acts, the precise identification of the Material (manufacturer, serial numbers, etc.) is important. Most of the detailed information about the Supply should be represented using the Material class. If delivery needs to be scheduled, tracked, and billed separately, one can associate a Transportation Act with the Supply Act. Pharmacy dispense services are represented as Supply Acts, associated with a SubstanceAdministration Act. The SubstanceAdministration class represents the administration of medication, while dispensing is supply.

4.4.2.11580 HL7 v3 Value Set RoleClassManufacturedProduct

URL: https://hl7.org/fhir/R4/v3/RoleClassManufacturedProduct/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-RoleClassManufacturedProduct) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-RoleClassManufacturedProduct
Version: | 2014-03-26
Name: | v3.RoleClassManufacturedProduct
Title: | V3 Value SetRoleClassManufacturedProduct
Definition: | Scoped by the manufacturer
OID: | 2.16.840.1.113883.1.11.11580 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.11580.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-RoleClass where concept is-a MANU

4.4.2.11580.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 2 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-RoleClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-RoleClass

Lvl | Code | Display | Definition
0 | MANU | manufactured product | Scoped by the manufacturer
1 | THER | therapeutic agent | A manufactured material (player) that is used for its therapeutic properties. The manufacturer is the scoper.

4.4.2.11591 HL7 v3 Value Set RoleClassSpecimen

URL: https://hl7.org/fhir/R4/v3/RoleClassSpecimen/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-RoleClassSpecimen) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-RoleClassSpecimen
Version: | 2014-03-26
Name: | v3.RoleClassSpecimen
Title: | V3 Value SetRoleClassSpecimen
Definition: | A role played by a material entity that is a specimen for an act. It is scoped by the source of the specimen.
OID: | 2.16.840.1.113883.1.11.11591 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.11591.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-RoleClass where concept is-a SPEC

4.4.2.11591.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 3 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-RoleClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-RoleClass

Lvl | Code | Display | Definition
0 | SPEC | specimen | A role played by a material entity that is a specimen for an act. It is scoped by the source of the specimen.
1 | ALQT | aliquot | A portion (player) of an original or source specimen (scoper) used for testing or transportation.
1 | ISLT | isolate | A microorganism that has been isolated from other microorganisms or a source matrix.

4.4.2.11623 HL7 v3 Value Set EntityClassDevice

URL: https://hl7.org/fhir/R4/v3/EntityClassDevice/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-EntityClassDevice) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-EntityClassDevice
Version: | 2014-03-26
Name: | v3.EntityClassDevice
Title: | V3 Value SetEntityClassDevice
Definition: | A subtype of ManufacturedMaterial used in an activity, without being substantially changed through that activity. The kind of device is identified by the code attribute inherited from Entity. Usage: This includes durable (reusable) medical equipment as well as disposable equipment.
OID: | 2.16.840.1.113883.1.11.11623 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.11623.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-EntityClass where concept is-a DEV

4.4.2.11623.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 3 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-EntityClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-EntityClass

Lvl | Code | Display | Definition
0 | DEV | device | A subtype of ManufacturedMaterial used in an activity, without being substantially changed through that activity. The kind of device is identified by the code attribute inherited from Entity. Usage: This includes durable (reusable) medical equipment as well as disposable equipment.
1 | CER | certificate representation | A physical artifact that stores information about the granting of authorization.
1 | MODDV | imaging modality | Class to contain unique attributes of diagnostic imaging equipment.

4.4.2.13922 HL7 v3 Value Set EntityClassRoot

URL: https://hl7.org/fhir/R4/v3/EntityClassRoot/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-EntityClassRoot) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-EntityClassRoot
Version: | 2014-03-26
Name: | v3.EntityClassRoot
Title: | V3 Value SetEntityClassRoot
Definition: | Corresponds to the Entity class
OID: | 2.16.840.1.113883.1.11.13922 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.13922.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-EntityClass where concept is-a ENT

4.4.2.13922.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 27 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-EntityClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-EntityClass

Lvl | Code | Display | Definition
0 | ENT | entity | Corresponds to the Entity class
1 | HCE | health chart entity | A health chart included to serve as a document receiving entity in the management of medical records.
1 | LIV | living subject | Anything that essentially has the property of life, independent of current state (a dead human corpse is still essentially a living subject).
2 | NLIV | non-person living subject | A subtype of living subject that includes all living things except the species Homo Sapiens.
3 | ANM | animal | A living subject from the animal kingdom.
3 | MIC | microorganism | All single celled living organisms including protozoa, bacteria, yeast, viruses, etc.
3 | PLNT | plant | A living subject from the order of plants.
2 | PSN | person | A living subject of the species homo sapiens.
1 | MAT | material | Any thing that has extension in space and mass, may be of living or non-living origin.
2 | CHEM | chemical substance | A substance that is fully defined by an organic or inorganic chemical formula, includes mixtures of other chemical substances. Refine using, e.g., IUPAC codes.
2 | FOOD | food | Naturally occurring, processed or manufactured entities that are primarily used as food for humans and animals.
2 | MMAT | manufactured material | Corresponds to the ManufacturedMaterial class
3 | CONT | container | A container of other entities.
4 | HOLD | holder | A type of container that can hold other containers or other holders.
3 | DEV | device | A subtype of ManufacturedMaterial used in an activity, without being substantially changed through that activity. The kind of device is identified by the code attribute inherited from Entity. Usage: This includes durable (reusable) medical equipment as well as disposable equipment.
4 | CER | certificate representation | A physical artifact that stores information about the granting of authorization.
4 | MODDV | imaging modality | Class to contain unique attributes of diagnostic imaging equipment.
1 | ORG | organization | A social or legal structure formed by human beings.
2 | PUB | public institution | An agency of the people of a state often assuming some authority over a certain matter. Includes government, governmental agencies, associations.
2 | STATE | state | A politically organized body of people bonded by territory, culture, or ethnicity, having sovereignty (to a certain extent) granted by other states (enclosing or neighboring states). This includes countries (nations), provinces (e.g., one of the United States of America or a French departement), counties or municipalities. Refine using, e.g., ISO country codes, FIPS-PUB state codes, etc.
3 | NAT | Nation | A politically organized body of people bonded by territory and known as a nation.
1 | PLC | place | A physical place or site with its containing structure. May be natural or man-made. The geographic position of a place may or may not be constant.
2 | CITY | city or town | The territory of a city, town or other municipality.
2 | COUNTRY | country | The territory of a sovereign nation.
2 | COUNTY | county or parish | The territory of a county, parish or other division of a state or province.
2 | PROVINCE | state or province | The territory of a state, province, department or other division of a sovereign country.
1 | RGRP | group | A grouping of resources (personnel, material, or places) to be used for scheduling purposes. May be a pool of like-type resources, a team, or combination of personnel, material and places.

4.4.2.13934 HL7 v3 Value Set EntityClassManufacturedMaterial

URL: https://hl7.org/fhir/R4/v3/EntityClassManufacturedMaterial/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-EntityClassManufacturedMaterial) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-EntityClassManufacturedMaterial
Version: | 2014-03-26
Name: | v3.EntityClassManufacturedMaterial
Title: | V3 Value SetEntityClassManufacturedMaterial
Definition: | Corresponds to the ManufacturedMaterial class
OID: | 2.16.840.1.113883.1.11.13934 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.13934.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-EntityClass where concept is-a MMAT

4.4.2.13934.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 6 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-EntityClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-EntityClass

Lvl | Code | Display | Definition
0 | MMAT | manufactured material | Corresponds to the ManufacturedMaterial class
1 | CONT | container | A container of other entities.
2 | HOLD | holder | A type of container that can hold other containers or other holders.
1 | DEV | device | A subtype of ManufacturedMaterial used in an activity, without being substantially changed through that activity. The kind of device is identified by the code attribute inherited from Entity. Usage: This includes durable (reusable) medical equipment as well as disposable equipment.
2 | CER | certificate representation | A physical artifact that stores information about the granting of authorization.
2 | MODDV | imaging modality | Class to contain unique attributes of diagnostic imaging equipment.

4.4.2.13940 HL7 v3 Value Set RoleClassRoot

URL: https://hl7.org/fhir/R4/v3/RoleClassRoot/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-RoleClassRoot) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-RoleClassRoot
Version: | 2014-03-26
Name: | v3.RoleClassRoot
Title: | V3 Value SetRoleClassRoot
Definition: | Corresponds to the Role class
OID: | 2.16.840.1.113883.1.11.13940 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.13940.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-RoleClass where concept is-a ROL

4.4.2.13940.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 99 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-RoleClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-RoleClass

Lvl | Code | Display | Definition
0 | ROL | role | Corresponds to the Role class
1 | _RoleClassAssociative | RoleClassAssociative | A general association between two entities that is neither partitive nor ontological.
2 | _RoleClassMutualRelationship | RoleClassMutualRelationship | A relationship that is based on mutual behavior of the two Entities as being related. The basis of such relationship may be agreements (e.g., spouses, contract parties) or they may be de facto behavior (e.g. friends) or may be an incidental involvement with each other (e.g. parties over a dispute, siblings, children).
3 | _RoleClassRelationshipFormal | RoleClassRelationshipFormal | A relationship between two entities that is formally recognized, frequently by a contract or similar agreement.

4 | AFFL | affiliate | Player of the Affiliate role has a business/professional relationship with scoper. Player and scoper may be persons or organization. The Affiliate relationship does not imply membership in a group, nor does it exist for resource scheduling purposes. Example: A healthcare provider is affiliated with another provider as a business associate.

4 | AGNT | agent | An entity (player) that acts or is authorized to act on behalf of another entity (scoper).

5 | ASSIGNED | assigned entity | An agent role in which the agent is an Entity acting in the employ of an organization. The focus is on functional role on behalf of the organization, unlike the Employee role where the focus is on the 'Human Resources' relationship between the employee and the organization.

6 | COMPAR | commissioning party | An Entity that is authorized to issue or instantiate permissions, privileges, credentials or other formal/legal authorizations.

6 | SGNOFF | signing authority or officer | The role of a person (player) who is the officer or signature authority for of a scoping entity, usually an organization (scoper).

5 | CON | contact | A person or an organization (player) which provides or receives information regarding another entity (scoper). Examples; patient NOK and emergency contacts; guarantor contact; employer contact.

6 | ECON | emergency contact | An entity to be contacted in the event of an emergency.

6 | NOK | next of kin | An individual designated for notification as the next of kin for a given entity.

5 | GUARD | guardian | Guardian of a ward

4 | CIT | citizen | Citizen of apolitical entity

4 | COVPTY | covered party | A role class played by a person who receives benefit coverage under the terms of a particular insurance policy. The underwriter of that policy is the scoping entity. The covered party receives coverage because of some contractual or other relationship with the holder of that policy. Discussion:This reason for coverage is captured in 'Role.code' and a relationship link with type code of indirect authority should be included using the policy holder role as the source, and the covered party role as the target. Note that a particular policy may cover several individuals one of whom may be, but need not be, the policy holder. Thus the notion of covered party is a role that is distinct from that of the policy holder.

5 | CLAIM | claimant | Description: A role played by a party making a claim for coverage under a policy or program. A claimant must be either a person or organization, or a group of persons or organizations. A claimant is not a named insured or a program eligible. Discussion: With respect to liability insurance such as property and casualty insurance, a claimant must file a claim requesting indemnification for a loss that the claimant considers covered under the policy of a named insured. The claims adjuster for the policy underwriter will review the claim to determine whether the loss meets the benefit coverage criteria under a policy, and base any indemnification or coverage payment on that review. If a third party is liable in whole or part for the loss, the underwriter may pursue third party liability recovery. A claimant may be involved in civil or criminal legal proceedings involving claims against a defendant party that is indemnified by an insurance policy or to protest the finding of a claims adjustor. With respect to life insurance, a beneficiary designated by a named insured becomes a claimant of the proceeds of coverage, as in the case of a life insurance policy. However, a claimant for coverage under life insurance is not necessarily a designated beneficiary. Note: A claimant is not a named insured. However, a named insured may make a claim under a policy, e.g., an insured driver may make a claim for an injury under his or her comprehensive automobile insurance policy. Similarly, a program eligible may make a claim under program, e.g., an unemployed worker may claim benefits under an unemployment insurance program, but parties playing these covered party role classes are not, for purposes of this vocabulary and in an effort to clearly distinguish role classes, considered claimants. In the case of a named insured making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that either a named insured or an individual insured has filed a claim for a loss. In the case of a program eligible, a role type code INJWKR (injured worker) subtypes the class to indicate that the covered party in a workers compensation program is an injured worker, and as such, has filed a "claim" under the program for benefits. Likewise, a covered role type code UNEMP (unemployed worker) subtypes the program eligible class to indicate that the covered party in an unemployment insurance program has filed a claim for unemployment benefits. Example: A claimant under automobile policy that is not the named insured.

5 | NAMED | named insured | Description: A role played by a party to an insurance policy to which the insurer agrees to indemnify for losses, provides benefits for, or renders services. A named insured may be either a person, non-person living subject, or an organization, or a group of persons, non-person living subjects, or organizations. Discussion: The coded concept NAMED should not be used where a more specific child concept in this Specializable value set applies. In some cases, the named insured may not be the policy holder, e.g., where a policy holder purchases life insurance policy in which another party is the named insured and the policy holder is the beneficiary of the policy. Note: The party playing the role of a named insured is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a named insured may make a claim under a policy, e.g., e.g., a party that is the named insured and policy holder under a comprehensive automobile insurance policy may become the claimant for coverage under that policy e.g., if injured in an automobile accident and there is no liable third party. In the case of a named insured making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that a named insured has filed a claim for a loss. Example: The named insured under a comprehensive automobile, disability, or property and casualty policy that is the named insured and may or may not be the policy holder.

6 | DEPEN | dependent | Description: A role played by a person covered under a policy or program based on an association with a subscriber, which is recognized by the policy holder. Note: The party playing the role of a dependent is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a dependent may make a claim under a policy, e.g., a dependent under a health insurance policy may become the claimant for coverage under that policy for wellness examines or if injured and there is no liable third party. In the case of a dependent making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that the dependent has filed a claim for services covered under the health insurance policy. Example: The dependent has an association with the subscriber such as a financial dependency or personal relationship such as that of a spouse, or a natural or adopted child. The policy holder may be required by law to recognize certain associations or may have discretion about the associations. For example, a policy holder may dictate the criteria for the dependent status of adult children who are students, such as requiring full time enrollment, or may recognize domestic partners as dependents. Under certain circumstances, the dependent may be under the indirect authority of a responsible party acting as a surrogate for the subscriber, for example, if the subscriber is differently abled or deceased, a guardian ad Lidem or estate executor may be appointed to assume the subscriberaTMs legal standing in the relationship with the dependent.

6 | INDIV | individual | Description: A role played by a party covered under a policy as the policy holder. An individual may be either a person or an organization. Note: The party playing the role of an individual insured is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a named insured may make a claim under a policy, e.g., a party that is the named insured and policy holder under a comprehensive automobile insurance policy may become the claimant for coverage under that policy if injured in an automobile accident and there is no liable third party. In the case of an individual insured making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that an individual insured has filed a claim for a loss. Example: The individual insured under a comprehensive automobile, disability, or property and casualty policy that is the policy holder.

6 | SUBSCR | subscriber | Description: A role played by a person covered under a policy based on association with a sponsor who is the policy holder, and whose association may provide for the eligibility of dependents for coverage. Discussion: The policy holder

holds the contract with the policy or program underwriter. The subscriber holds a certificate of coverage under the contract. In legal proceedings concerning the policy or program, the terms of the contract takes precedence over the terms of the certificate of coverage if there are any inconsistencies. Note: The party playing the role of a subscriber is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a subscriber may make a claim under a policy, e.g., a subscriber under a health insurance policy may become the claimant for coverage under that policy for wellness examines or if injured and there is no liable third party. In the case of a subscriber making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that the subscriber has filed a claim for services covered under the health insurance policy. Example: An employee or a member of an association.

5 | PROG | program eligible | Description: A role played by a party that meets the eligibility criteria for coverage under a program. A program eligible may be either a person, non-person living subject, or an organization, or a group of persons, non-person living subjects, or organizations. Discussion: A program as typically government administered coverage for parties determined eligible under the terms of the program. Note: The party playing a program eligible is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However a program eligible may make a claim under program, e.g., an unemployed worker may claim benefits under an unemployment insurance program, but parties playing these covered party role classes are not, for purposes of this vocabulary and in an effort to clearly distinguish role classes, considered claimants. In the case of a program eligible, a role type code INJWKR (injured worker) subtypes the class to indicate that the covered party in a workers compensation program is an injured worker, and as such, has filed a "claim" under the program for benefits. Likewise, a covered role type code UNEMP (unemployed worker) subtypes the program eligible class to indicate that the covered party in an unemployment insurance program has filed a claim for unemployment benefits. Example: A party meeting eligibility criteria related to health or financial status, e.g., in the U.S., persons meeting health, demographic, or financial criteria established by state and federal law are eligible for Medicaid.

4 | CRINV | clinical research investigator | A role played by a provider, always a person, who has agency authority from a Clinical Research Sponsor to direct the conduct of a clinical research trial or study on behalf of the sponsor.

4 | CRSPNSR | clinical research sponsor | A role played by an entity, usually an organization, that is the sponsor of a clinical research trial or study. The sponsor commissions the study, bears the expenses, is responsible for satisfying all legal requirements concerning subject safety and privacy, and is generally responsible for collection, storage and analysis of the data generated during the trial. No scoper is necessary, as a clinical research sponsor undertakes the role on its own authority and declaration. Clinical research sponsors are usually educational or other research organizations, government agencies or biopharmaceutical companies.

4 | EMP | employee | A relationship between a person or organization and a person or organization formed for the purpose of exchanging work for compensation. The purpose of the role is to identify the type of relationship the employee has to the employer, rather than the nature of the work actually performed. (Contrast with AssignedEntity.)

5 | MIL | military person | A role played by a member of a military service. Scoper is the military service (e.g. Army, Navy, Air Force, etc.) or, more specifically, the unit (e.g. Company C, 3rd Battalion, 4th Division, etc.)

4 | GUAR | guarantor | A person or organization (player) that serves as a financial guarantor for another person or organization (scoper).

4 | INVSBJ | Investigation Subject | An entity that is the subject of an investigation. This role is scoped by the party responsible for the investigation.

5 | CASEBJ | Case Subject | A person, non-person living subject, or place that is the subject of an investigation related to a notifiable condition (health circumstance that is reportable within the applicable public health jurisdiction)

5 | RESBJ | research subject | Definition:Specifies the administrative functionality within a formal experimental design for which the ResearchSubject role was established. Examples: Screening - role is used for pre-enrollment evaluation portion of the design; enrolled - role is used for subjects admitted to the experimental portion of the design.

4 | LIC | licensed entity | A relationship in which the scoper certifies the player (e. g. a medical care giver, a medical device or a provider organization) to perform certain activities that fall under the jurisdiction of the scoper (e.g., a health authority licensing healthcare providers, a medical quality authority certifying healthcare professionals).

5 | NOT | notary public | notary public

5 | PROV | healthcare provider | An Entity (player) that is authorized to provide health care services by some authorizing agency (scoper).

4 | PAT | patient | A Role of a LivingSubject (player) as an actual or potential recipient of health care services from a healthcare provider organization (scoper). Usage Note: Communication about relationships between patients and specific healthcare practitioners (people) is not done via scoper. Instead this is generally done using the CareProvision act. This allows linkage between patient and a particular healthcare practitioner role and also allows description of the type of care involved in the relationship.

4 | PAYEE | payee | The role of an organization or individual designated to receive payment for a claim against a particular coverage. The scoping entity is the organization that is the submitter of the invoice in question.

4 | PAYOR | invoice payor | The role of an organization that undertakes to accept claims invoices, assess the coverage or payments due for those invoices and pay to the designated payees for those invoices. This role may be either the underwriter or a third-party organization authorized by the underwriter. The scoping entity is the organization that underwrites the claimed coverage.

4 | POLHOLD | policy holder | A role played by a person or organization that holds an insurance policy. The underwriter of that policy is the scoping entity. Discussion:The identifier of the policy is captured in 'Role.id' when the Role is a policy holder. A particular policy may cover several individuals one of whom may be, but need not be, the policy holder. Thus the notion of covered party is a role that is distinct from that of the policy holder.

4 | QUAL | qualified entity | An entity (player) that has been recognized as having certain training/experience or other characteristics that would make said entity an appropriate performer for a certain activity. The scoper is an organization that educates or qualifies entities.

4 | SPNSR | coverage sponsor | A role played by an entity, usually an organization that is the sponsor of an insurance plan or a health program. A sponsor is the party that is ultimately accountable for the coverage by employment contract or by law. A sponsor can be an employer, union, government agency, or association. Fully insured sponsors establish the terms of the plan and contract with health insurance plans to assume the risk and to administer the plan. Self-insured sponsors delegate coverage administration, but not risk, to third-party administrators. Program sponsors designate services to be covered in accordance with statute. Program sponsors may administer the coverage themselves, delegate coverage administration, but not risk to third-party administrators, or contract with health insurance plans to assume the risk and administrator a program. Sponsors qualify individuals who may become a policy holder of the plan; where the sponsor is the policy holder, who may become a subscriber or a dependent to a policy under the plan; or where the sponsor is a government agency, who may become program eligibles under a program. The sponsor role may be further qualified by the SponsorRole.code. Entities playing the sponsor role may also play the role of a Coverage Administrator. Example: An employer, union, government agency, or association.

4 | STD | student | A role played by an individual who is a student of a school, which is the scoping entity.

4 | UNDWRT | underwriter | A role played by a person or an organization. It is the party that accepts fiscal responsibility for insurance plans and the policies created under those plans; administers and accepts fiscal responsibility for a program that provides coverage for services to eligible individuals; and/or has the responsibility to assess the merits of each risk and decide a suitable premium for accepting all or part of the risk. If played by an organization, this role may be further specified by an appropriate RoleCode. Example: A health insurer; Medicaid Program; Lloyd's of London

3 | CAREGIVER | caregiver | A person responsible for the primary care of a patient at home.

3 | PRS | personal relationship | Links two entities with classCode PSN (person) in a personal relationship. The character of the relationship must be defined by a PersonalRelationshipRoleType code. The player and scoper are determined by PersonalRelationshipRoleType code as well.

3 | SELF | self | The "same" roleclass asserts an identity between playing and scoping entities: that they are in fact instances of the same entity and, in the case of discrepancies (e.g different DOB, gender), that one or both are in error. Usage: playing and scoping entities must have same classcode, but need not have identical attributes or values. Example: a provider registry maintains sets of conflicting demographic data for what is reported to be the same individual.

2 | _RoleClassPassive | RoleClassPassive | An association for a playing Entity that is used, known, treated, handled, built, or destroyed, etc. under the auspices of the scoping Entity. The playing Entity is passive in these roles (even though it may be active in other roles), in the sense that the kinds of things done to it in this role happen without an agreement from the playing Entity.

3 | ACCESS | access | A role in which the playing entity (material) provides access to another entity. The principal use case is intravenous (or other bodily) access lines that preexist and need to be referred to for medication routing instructions.

3 | ADJY | adjacency | A physical association whereby two Entities are in some (even lose) spatial relationship with each other such that they touch each other in some way. Examples: the colon is connected (and therefore adjacent) to the jejunum; the colon is adjacent to the liver (even if not actually connected.) UsageConstraints: Adjacency is in principle a symmetrical connection, but scoper and player of the role should, where applicable, be assigned to have scoper be the larger, more central Entity and player the smaller, more distant, appendage.

4 | CONC | connection | An adjacency of two Entities held together by a bond which attaches to each of the two entities. Examples: biceps brachii muscle connected to the radius bone, port 3 on a network switch connected to port 5 on a patch panel. UsageConstraints: See Adjacency for the assignment of scoper (larger, more central) and player (smaller, more distant).

5 | BOND | molecular bond | A connection between two atoms of a molecule. Examples: double bond between first and second C in ethane, peptide bond between two amino-acid, disulfide bridge between two proteins, chelate and ion associations, even the much weaker van-der-Waals bonds can be considered molecular bonds. UsageConstraints: See connection and adjacency for the assignment of player and scoper.

5 | CONY | continuity | A connection between two regional parts. Examples: the connection between ascending aorta and the aortic arc, connection between descending colon and sigmoid. UsageConstraints: See connection and adjacency for the assignment of player and scoper.

3 | ADMM | Administerable Material | A material (player) that can be administered to an Entity (scoper).

3 | BIRTHPL | birthplace | Relates a place (playing Entity) as the location where a living subject (scoping Entity) was born.

3 | DEATHPLC | place of death | Definition: Relates a place (playing Entity) as the location where a living subject (scoping Entity) died.

3 | DST | distributed material | A material (player) distributed by a distributor (scoper) who functions between a manufacturer and a buyer or retailer.

4 | RET | retailed material | Material (player) sold by a retailer (scoper), who also give advice to prospective buyers.

3 | EXLOC | event location | A role played by a place at which the location of an event may be recorded.

4 | SDLOC | service delivery location | A role played by a place at which services may be provided.

5 | DSDLOC | dedicated service delivery location | A role of a place (player) that is intended to house the provision of services. Scoper is the Entity (typically Organization) that provides these services. This is not synonymous with "ownership."

5 | ISDLOC | incidental service delivery location | A role played by a place at which health care services may be provided without prior designation or authorization.

3 | EXPR | exposed entity | A role played by an entity that has been exposed to a person or animal suffering a contagious disease, or with a location from which a toxin has been distributed. The player of the role is normally a person or animal, but it is possible that other entity types could become exposed. The role is scoped by the source of the exposure, and it is quite possible for a person playing the role of exposed party to also become the scoper a role played by another person. That is to say, once a person has become infected, it is possible, perhaps likely, for that person to infect others. Management of exposures and tracking exposed parties is a key function within public health, and within most public health contexts - exposed parties are known as "contacts."

3 | HLD | held entity | Entity that is currently in the possession of a holder (scoper), who holds, or uses it, usually based on some agreement with the owner.

3 | HLTHCHRT | health chart | The role of a material (player) that is the physical health chart belonging to an organization (scoper).

3 | IDENT | identified entity | A role in which the scoping entity designates an identifier for a playing entity.

3 | MANU | manufactured product | Scoped by the manufacturer

4 | THER | therapeutic agent | A manufactured material (player) that is used for its therapeutic properties. The manufacturer is the scoper.

3 | MNT | maintained entity | An entity (player) that is maintained by another entity (scoper). This is typical role held by durable equipment. The scoper assumes responsibility for proper operation, quality, and safety.

3 | OWN | owned entity | An Entity (player) for which someone (scoper) is granted by law the right to call the material (player) his own. This entitles the scoper to make decisions about the disposition of that material.

3 | RGPR | regulated product | A product regulated by some governmentatl orgnization. The role is played by Material and scoped by Organization. Rationale: To support an entity clone used to identify the NDC number for a drug product.

3 | TERR | territory of authority | Relates a place entity (player) as the region over which the scoper (typically an Organization) has certain authority (jurisdiction). For example, the Calgary Regional Health Authority (scoper) has authority over the territory "Region 4 of Alberta" (player) in matters of health.

3 | USED | used entity | Description:An entity (player) that is used by another entity (scoper)

3 | WRTE | warranted product | A role a product plays when a guarantee is given to the purchaser by the seller (scoping entity) stating that the product is reliable and free from known defects and that the seller will repair or replace defective parts within a given time limit and under certain conditions.

1 | _RoleClassOntological | RoleClassOntological | A relationship in which the scoping Entity defines or specifies what the playing Entity is. Thus, the player's "being" (Greek: ontos) is specified.

2 | EQUIV | equivalent entity | Description: Specifies the player Entity (the equivalent Entity) as an Entity that is considered to be equivalent to a reference Entity (scoper). The equivalence is in principle a symmetric relationship, however, it is expected that the scoper is a reference entity which serves as reference entity for multiple different equivalent entities. Examples: An innovator's medicine formulation is the reference for "generics", i.e., formulations manufactured differently but having been proven to be biologically equivalent to the reference medicine. Another example is a reference ingredient that serves as basis for quantity specifications (basis of strength, e.g., metoprolol succinate specified in terms of metoprolol tartrate.)

3 | SAME | same | The "same" role asserts an identity between playing and scoping entities, i.e., that they are in fact two records of the same entity instance, and, in the case of discrepancies (e.g different DOB, gender), that one or both are in error. Usage: playing and scoping entities must have same classCode, but need not have identical attributes or values. Example: a provider registry maintains sets of conflicting demographic data for what is reported to be the same individual.

3 | SUBY | subsumed by | Relates a prevailing record of an Entity (scoper) with another record (player) that it subsumes. Examples: Show a correct new Person object (scoper) that subsumes one or more duplicate Person objects that had accidentally been created for the same physical person. Constraints: Both the player and scoper must have the same classCode.

2 | GEN | has generalization | Relates a specialized material concept (player) to its generalization (scoper).

3 | GRIC | has generic | A special link between pharmaceuticals indicating that the target (scoper) is a generic for the source (player).

2 | INST | instance | An individual piece of material (player) instantiating a class of material (scoper).

1 | _RoleClassPartitive | RoleClassPartitive | An association between two Entities where the playing Entity is considered in some way "part" of the scoping Entity, e.g., as a member, component, ingredient, or content. Being "part" in the broadest sense of the word can mean anything from being an integral structural component to a mere incidental temporary association of a playing Entity with a (generally larger) scoping Entity.

2 | CONT | content | Relates a material as the content (player) to a container (scoper). Unlike ingredients, the content and a container remain separate (not mixed) and the content can be removed from the container. A content is not part of an empty container.

2 | EXPAGTCAR | exposure agent carrier | An exposure agent carrier is an entity that is capable of conveying an exposure agent from one entity to another. The scoper of the role must be the exposure agent (e.g., pathogen).

3 | EXPVECTOR | exposure vector | Description: A vector is a living subject that carries an exposure agent. The vector does not cause the disease itself, but exposes targets to the exposure agent. A mosquito carrying malaria is an example of a vector. The scoper of the role must be the exposure agent (e.g., pathogen).

3 | FOMITE | fomite | Description: A fomite is a non-living entity that is capable of conveying exposure agent from one entity to another. A doorknob contaminated with a Norovirus is an example of a fomite. Anyone touching the doorknob would be exposed to the virus. The scoper of the role must be the exposure agent (e.g., pathogen).

2 | INGR | ingredient | Relates a component (player) to a mixture (scoper). E.g., Glucose and Water are ingredients of D5W, latex may be an ingredient in a tracheal tube.

3 | ACTI | active ingredient | Definition: a therapeutically active ingredient (player) in a mixture (scoper), where the mixture is typically a manufactured pharmaceutical. It is unknown if the quantity of such an ingredient is expressed precisely in terms of the playing ingredient substance, or, if it is specified in terms of a closely related substance (active moiety or reference substance).

4 | ACTIB | active ingredient - basis of strength | Description: Active ingredient, where the ingredient substance (player) is itself the "basis of strength", i.e., where the Role.quantity specifies exactly the quantity of the player substance in the medicine formulation. Examples: Lopressor 50 mg actually contains 50 mg of metoprolol succinate, even though the active moiety is metoprolol, but also: Tenormin 50 mg contain 50 mg of atenolol, as free base, i.e., where the active ingredient atenolol is also the active moiety.

4 | ACTIM | active ingredient - moiety is basis of strength | Description: Active ingredient, where not the ingredient substance (player), but itaTMs active moiety is the "basis of strength", i.e., where the Role.quantity specifies the quantity of the player substance's active moiety in the medicine formulation. Examples: 1 mL of Betopic 5mg/mL eye drops contains 5.6 mg betaxolol hydrochloride equivalent to betaxolol base 5 mg.

4 | ACTIR | active ingredient - reference substance is basis of strength | Description: Active ingredient, where not the ingredient substance (player) but another reference substance with the same active moiety, is the "basis of strength", i.e., where the Role.quantity specifies the quantity of a reference substance, similar but different from the player substance's in the medicine formulation. Examples: Toprol-XL 50 mg contains 47.5 mg of metoprolol succinate equivalent to 50 mg of metoprolol tartrate.

3 | ADJV | adjuvant | A component (player) added to enhance the action of an active ingredient (scoper) (in the manner of a catalyst) but which has no active effect in and of itself. Such ingredients are significant in defining equivalence of products in a way that inactive ingredients are not.

3 | ADTV | additive | An ingredient (player) that is added to a base (scoper), that amounts to a minor part of the overall mixture.

3 | BASE | base | A base ingredient (player) is what comprises the major part of a mixture (scoper). E.g., Water in most i.v. solutions, or Vaseline in salves. Among all ingredients of a material, there should be only one base. A base substance can, in turn, be a mixture.

3 | CNTM | contaminant ingredient | An ingredient whose presence is not intended but may not be reasonably avoided given the circumstances of the mixture's nature or origin.

3 | IACT | inactive ingredient | An ingredient which is not considered therapeutically active, e.g., colors, flavors, stabilizers, or preservatives, fillers, or structural components added to an active ingredient in order to facilitate administration of the active ingredient but without being considered therapeutically active. An inactive ingredient need not be biologically inert, e.g., might be active as an allergen or might have a pleasant taste, but is not an essential constituent delivering the therapeutic effect.

4 | COLR | color additive | A substance (player) influencing the optical aspect of material (scoper).

4 | FLVR | flavor additive | A substance (player) added to a mixture (scoper) to make it taste a certain way. In food the use is obvious, in pharmaceuticals flavors can hide disgusting taste of the active ingredient (important in pediatric treatments).

4 | PRSV | preservative | A substance (player) added to a mixture (scoper) to prevent microorganisms (fungi, bacteria) to spoil the mixture.

4 | STBL | stabilizer | A stabilizer (player) added to a mixture (scoper) in order to prevent the molecular disintegration of the main substance.

3 | MECH | mechanical ingredient | An ingredient (player) of a medication (scoper) that is inseparable from the active ingredients, but has no intended chemical or pharmaceutical effect itself, but which may have some systemic effect on the patient. An example is a collagen matrix used as a base for transplanting skin cells. The collagen matrix can be left permanently in the graft site. Because it is of bovine origin, the patient may exhibit allergies or may have cultural objections to its use.

2 | LOCE | located entity | Relates an entity (player) to a location (scoper) at which it is present in some way. This presence may be limited in time.

3 | STOR | stored entity | Relates an entity (player) (e.g. a device) to a location (scoper) at which it is normally found or stored when not used.

2 | MBR | member | A role played by an entity that is a member of a group. The group provides the scope for this role. Among other uses, groups as used in insurance (groups of covered individuals) and in scheduling where resources may be grouped for scheduling and logistical purposes.
2 | PART | part | Definition: an association between two Entities where the playing Entity (the part) is a component of the whole (scoper) in the sense of an integral structural component, that is distinct from other parts in the same whole, has a distinct function in the whole, and, as an effect, the full integrity of the whole depends (to some degree) on the presence of this part, even though the part may often be separable from the whole. Discussion: Part is defined in opposition to (a) ingredient (not separable), (b) content (not a functional component), and (c) member (not functionally distinct from other members).
3 | ACTM | active moiety | The molecule or ion that is responsible for the intended pharmacological action of the drug substance, excluding those appended or associated parts of the molecule that make the molecule an ester, salt (including a salt with hydrogen or coordination bonds), or other noncovalent derivative (such as a complex, chelate, or clathrate). Examples: heparin-sodium and heparin-potassium have the same active moiety, heparin; the active moiety of morphine-hydrochloride is morphine.
2 | SPEC | specimen | A role played by a material entity that is a specimen for an act. It is scoped by the source of the specimen.
3 | ALQT | aliquot | A portion (player) of an original or source specimen (scoper) used for testing or transportation.
3 | ISLT | isolate | A microorganism that has been isolated from other microorganisms or a source matrix.

4.4.2.13948 HL7 v3 Value Set ActClassClinicalDocument

URL: https://hl7.org/fhir/R4/v3/ActClassClinicalDocument/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActClassClinicalDocument) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActClassClinicalDocument
Version: | 2014-03-26
Name: | v3.ActClassClinicalDocument
Title: | V3 Value SetActClassClinicalDocument
Definition: | A clinical document is a documentation of clinical observations and services, with the following characteristics: (1) Persistence - A clinical document continues to exist in an unaltered state, for a time period defined by local and regulatory requirements; (2) Stewardship - A clinical document is maintained by a person or organization entrusted with its care; (3) Potential for authentication - A clinical document is an assemblage of information that is intended to be legally authenticated; (4) Wholeness - Authentication of a clinical document applies to the whole and does not apply to portions of the document without the full context of the document; (5) Human readability - A clinical document is human readable."
OID: | 2.16.840.1.113883.1.11.13948 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.13948.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActClass where concept is-a DOCCLIN

4.4.2.13948.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 2 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActClass

Lvl | Code | Display | Definition
0 | DOCCLIN | clinical document | A clinical document is a documentation of clinical observations and services, with the following characteristics: Persistence - A clinical document continues to exist in an unaltered state, for a time period defined by local and regulatory requirements; Stewardship - A clinical document is maintained by a person or organization entrusted with its care; Potential for authentication - A clinical document is an assemblage of information that is intended to be legally authenticated; Wholeness - Authentication of a clinical document applies to the whole and does not apply to portions of the document without the full context of the document; Human readability - A clinical document is human readable.
1 | CDALVLONE | CDA Level One clinical document | A clinical document that conforms to Level One of the HL7 Clinical Document Architecture (CDA)

4.4.2.13955 HL7 v3 Value Set ActEncounterCode

URL: https://hl7.org/fhir/R4/v3/ActEncounterCode/vs.html
Vocabulary Work Group | Maturity Level: 2 | Trial Use | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActEncounterCode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActEncounterCode
Version: | 2014-03-26
Name: | v3.ActEncounterCode
Title: | V3 Value SetActEncounterCode
Definition: | Domain provides codes that qualify the ActEncounterClass (ENC)
OID: | 2.16.840.1.113883.1.11.13955 (for OID based terminology systems)

Source Resource | XML / JSON

This value set is used in the following places:

• Resource: Encounter.class (Coding / Extensible)

• Resource: Encounter.classHistory.class (Coding / Extensible)

4.4.2.13955.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActCode where concept is-a _ActEncounterCode

http://terminology.hl7.org/CodeSystem/v3-ActCode

Code | Display

_ActEncounterCode | ActEncounterCode | Domain provides codes that qualify the ActEncounterClass (ENC)

4.4.2.13955.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 11 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActCode

Code | Display | Definition

AMB | ambulatory | A comprehensive term for health care provided in a healthcare facility (e.g. a practitioneraTMs office, clinic setting, or hospital) on a nonresident basis. The term ambulatory usually implies that the patient has come to the location and is not assigned to a bed. Sometimes referred to as an outpatient encounter.

EMER | emergency | A patient encounter that takes place at a dedicated healthcare service delivery location where the patient receives immediate evaluation and treatment, provided until the patient can be discharged or responsibility for the patient's care is transferred elsewhere (for example, the patient could be admitted as an inpatient or transferred to another facility.)

FLD | field | A patient encounter that takes place both outside a dedicated service delivery location and outside a patient's residence. Example locations might include an accident site and at a supermarket.

HH | home health | Healthcare encounter that takes place in the residence of the patient or a designee

IMP | inpatient encounter | A patient encounter where a patient is admitted by a hospital or equivalent facility, assigned to a location where patients generally stay at least overnight and provided with room, board, and continuous nursing service.

ACUTE | inpatient acute | An acute inpatient encounter.

NONAC | inpatient non-acute | Any category of inpatient encounter except 'acute'

OBSENC | observation encounter | An encounter where the patient usually will start in different encounter, such as one in the emergency department (EMER) but then transition to this type of encounter because they require a significant period of treatment and monitoring to determine whether or not their condition warrants an inpatient admission or discharge. In the majority of cases the decision about admission or discharge will occur within a time period determined by local, regional or national regulation, often between 24 and 48 hours.

PRENC | pre-admission | A patient encounter where patient is scheduled or planned to receive service delivery in the future, and the patient is given a pre-admission account number. When the patient comes back for subsequent service, the pre-admission encounter is selected and is encapsulated into the service registration, and a new account number is generated. Usage Note: This is intended to be used in advance of encounter types such as ambulatory, inpatient encounter, virtual, etc.

SS | short stay | An encounter where the patient is admitted to a health care facility for a predetermined length of time, usually less than 24 hours.

VR | virtual | A patient encounter where the patient and the practitioner(s) are not in the same physical location. Examples include telephone conference, email exchange, robotic surgery, and televideo conference.

4.4.2.14006 HL7 v3 Value Set RoleClassAgent

URL: https://hl7.org/fhir/R4/v3/RoleClassAgent/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-RoleClassAgent) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-RoleClassAgent

Version: | 2014-03-26

Name: | v3.RoleClassAgent

Title: | V3 Value SetRoleClassAgent

Definition: | An entity (player) that acts or is authorized to act on behalf of another entity (scoper).

OID: | 2.16.840.1.113883.1.11.14006 (for OID based terminology systems)

Source Resource | XML / JSON

This value set is not currently used

4.4.2.14006.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-RoleClass where concept is-a AGNT

4.4.2.14006.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 8 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-RoleClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-RoleClass

Lvl | Code | Display | Definition
0 | AGNT | agent | An entity (player) that acts or is authorized to act on behalf of another entity (scoper).
1 | ASSIGNED | assigned entity | An agent role in which the agent is an Entity acting in the employ of an organization. The focus is on functional role on behalf of the organization, unlike the Employee role where the focus is on the 'Human Resources' relationship between the employee and the organization.
2 | COMPAR | commissioning party | An Entity that is authorized to issue or instantiate permissions, privileges, credentials or other formal/legal authorizations.
2 | SGNOFF | signing authority or officer | The role of a person (player) who is the officer or signature authority for of a scoping entity, usually an organization (scoper).
1 | CON | contact | A person or an organization (player) which provides or receives information regarding another entity (scoper). Examples; patient NOK and emergency contacts; guarantor contact; employer contact.
2 | ECON | emergency contact | An entity to be contacted in the event of an emergency.
2 | NOK | next of kin | An individual designated for notification as the next of kin for a given entity.
1 | GUARD | guardian | Guardian of a ward

4.4.2.16208 HL7 v3 Value Set ActPharmacySupplyType

URL: https://hl7.org/fhir/R4/v3/ActPharmacySupplyType/vs.html
Vocabulary Work Group | Maturity Level: 1 | Draft | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActPharmacySupplyType) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActPharmacySupplyType
Version: | 2014-03-26
Name: | v3.ActPharmacySupplyType
Title: | V3 Value SetActPharmacySupplyType
Definition: | Identifies types of dispensing events
OID: | 2.16.840.1.113883.1.11.16208 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is used in the following places:

• Resource: MedicationDispense.type (CodeableConcept / Example)

4.4.2.16208.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActCode where concept is-a _ActPharmacySupplyType

http://terminology.hl7.org/CodeSystem/v3-ActCode

Code | Display
_ActPharmacySupplyType | ActPharmacySupplyType | Identifies types of dispensing events

4.4.2.16208.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 22 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActCode

Code | Display | Definition
DF | Daily Fill | A fill providing sufficient supply for one day
EM | Emergency Supply | A supply action where there is no 'valid' order for the supplied medication. E.g. Emergency vacation supply, weekend supply (when prescriber is unavailable to provide a renewal prescription)
SO | Script Owing | An emergency supply where the expectation is that a formal order authorizing the supply will be provided at a later date.
FF | First Fill | The initial fill against an order. (This includes initial fills against refill orders.)
FFC | First Fill - Complete | A first fill where the quantity supplied is equal to one full repetition of the ordered amount. (e.g. If the order was 90 tablets, 3 refills, a complete fill would be for the full 90 tablets).
FFP | First Fill - Part Fill | A first fill where the quantity supplied is less than one full repetition of the ordered amount. (e.g. If the order was 90 tablets, 3 refills, a partial fill might be for only 30 tablets.)
FFSS | first fill, partial strength | A first fill where the strength supplied is less than the ordered strength. (e.g. 10mg for an order of 50mg where a subsequent fill will dispense 40mg tablets).
TF | Trial Fill | A fill where a small portion is provided to allow for determination of the therapy effectiveness and patient tolerance.
FS | Floor stock | A supply action to restock a smaller more local dispensary.
MS | Manufacturer Sample | A supply of a manufacturer sample

RF | Refill | A fill against an order that has already been filled (or partially filled) at least once.
UD | Unit Dose | A supply action that provides sufficient material for a single dose.
RFC | Refill - Complete | A refill where the quantity supplied is equal to one full repetition of the ordered amount. (e.g. If the order was 90 tablets, 3 refills, a complete fill would be for the full 90 tablets.)
RFCS | refill complete partial strength | A refill where the quantity supplied is equal to one full repetition of the ordered amount. (e.g. If the order was 90 tablets, 3 refills, a complete fill would be for the full 90 tablets.) and where the strength supplied is less than the ordered strength (e.g. 10mg for an order of 50mg where a subsequent fill will dispense 40mg tablets).
RFF | Refill (First fill this facility) | The first fill against an order that has already been filled at least once at another facility.
RFFS | refill partial strength (first fill this facility) | The first fill against an order that has already been filled at least once at another facility and where the strength supplied is less than the ordered strength (e.g. 10mg for an order of 50mg where a subsequent fill will dispense 40mg tablets).
RFP | Refill - Part Fill | A refill where the quantity supplied is less than one full repetition of the ordered amount. (e.g. If the order was 90 tablets, 3 refills, a partial fill might be for only 30 tablets.)
RFPS | refill part fill partial strength | A refill where the quantity supplied is less than one full repetition of the ordered amount. (e.g. If the order was 90 tablets, 3 refills, a partial fill might be for only 30 tablets.) and where the strength supplied is less than the ordered strength (e.g. 10mg for an order of 50mg where a subsequent fill will dispense 40mg tablets).
RFS | refill partial strength | A fill against an order that has already been filled (or partially filled) at least once and where the strength supplied is less than the ordered strength (e.g. 10mg for an order of 50mg where a subsequent fill will dispense 40mg tablets).
TB | Trial Balance | A fill where the remainder of a 'complete' fill is provided after a trial fill has been provided.
TBS | trial balance partial strength | A fill where the remainder of a 'complete' fill is provided after a trial fill has been provided and where the strength supplied is less than the ordered strength (e.g. 10mg for an order of 50mg where a subsequent fill will dispense 40mg tablets).
UDE | unit dose equivalent | A supply action that provides sufficient material for a single dose via multiple products. E.g. 2 50mg tablets for a 100mg unit dose.

4.4.2.16226 HL7 v3 Value Set ObservationType

URL: https://hl7.org/fhir/R4/v3/ObservationType/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ObservationType) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ObservationType
Version: | 2014-03-26
Name: | v3.ObservationType
Title: | V3 Value SetObservationType
Definition: | Identifies the kinds of observations that can be performed
OID: | 2.16.840.1.113883.1.11.16226 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.16226.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActCode where concept is-a _ObservationType

• Include all codes defined in http://loinc.org

4.4.2.16226.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 1250 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12

Code | System | Display | Definition
ARTBLD | http://terminology.hl7.org/CodeSystem/v3-ActCode | ActSpecObsArtBldCode | Describes the artificial blood identifier that is associated with the specimen.
DILUTION | http://terminology.hl7.org/CodeSystem/v3-ActCode | ActSpecObsDilutionCode | An observation that reports the dilution of a sample.
AUTO-HIGH | http://terminology.hl7.org/CodeSystem/v3-ActCode | Auto-High Dilution | The dilution of a sample performed by automated equipment. The value is specified by the equipment
AUTO-LOW | http://terminology.hl7.org/CodeSystem/v3-ActCode | Auto-Low Dilution | The dilution of a sample performed by automated equipment. The value is specified by the equipment
PRE | http://terminology.hl7.org/CodeSystem/v3-ActCode | Pre-Dilution | The dilution of the specimen made prior to being loaded onto analytical equipment
RERUN | http://terminology.hl7.org/CodeSystem/v3-ActCode | Rerun Dilution | The value of the dilution of a sample after it had been analyzed at a prior dilution value
EVNFCTS | http://terminology.hl7.org/CodeSystem/v3-ActCode | ActSpecObsEvntfctsCode | Domain provides codes that qualify the ActLabObsEnvfctsCode domain. (Environmental Factors)
INTFR | http://terminology.hl7.org/CodeSystem/v3-ActCode | ActSpecObsInterferenceCode | An observation that relates to factors that may potentially cause interference with the observation

FIBRIN | http://terminology.hl7.org/CodeSystem/v3-ActCode | Fibrin | The Fibrin Index of the specimen. In the case of only differentiating between Absent and Present, recommend using 0 and 1

HEMOLYSIS | http://terminology.hl7.org/CodeSystem/v3-ActCode | Hemolysis | An observation of the hemolysis index of the specimen in g/L

ICTERUS | http://terminology.hl7.org/CodeSystem/v3-ActCode | Icterus | An observation that describes the icterus index of the specimen. It is recommended to use mMol/L of bilirubin

LIPEMIA | http://terminology.hl7.org/CodeSystem/v3-ActCode | Lipemia | An observation used to describe the Lipemia Index of the specimen. It is recommended to use the optical turbidity at 600 nm (in absorbance units).

VOLUME | http://terminology.hl7.org/CodeSystem/v3-ActCode | ActSpecObsVolumeCode | An observation that reports the volume of a sample.

AVAILABLE | http://terminology.hl7.org/CodeSystem/v3-ActCode | Available Volume | The available quantity of specimen. This is the current quantity minus any planned consumption (e.g., tests that are planned)

CONSUMPTION | http://terminology.hl7.org/CodeSystem/v3-ActCode | Consumption Volume | The quantity of specimen that is used each time the equipment uses this substance

CURRENT | http://terminology.hl7.org/CodeSystem/v3-ActCode | Current Volume | The current quantity of the specimen, i.e., initial quantity minus what has been actually used.

INITIAL | http://terminology.hl7.org/CodeSystem/v3-ActCode | Initial Volume | The initial quantity of the specimen in inventory

_ActPatientAnnotationType | http://terminology.hl7.org/CodeSystem/v3-ActCode | ActPatientAnnotationType | Description:Provides a categorization for annotations recorded directly against the patient .

ANNDI | http://terminology.hl7.org/CodeSystem/v3-ActCode | diagnostic image note | Description:A note that is specific to a patient's diagnostic images, either historical, current or planned.

ANNGEN | http://terminology.hl7.org/CodeSystem/v3-ActCode | general note | Description:A general or uncategorized note.

ANNIMM | http://terminology.hl7.org/CodeSystem/v3-ActCode | immunization note | A note that is specific to a patient's immunizations, either historical, current or planned.

ANNLAB | http://terminology.hl7.org/CodeSystem/v3-ActCode | laboratory note | Description:A note that is specific to a patient's laboratory results, either historical, current or planned.

ANNMED | http://terminology.hl7.org/CodeSystem/v3-ActCode | medication note | Description:A note that is specific to a patient's medications, either historical, current or planned.

GENE | http://terminology.hl7.org/CodeSystem/v3-ActCode | gene | Description: A DNA segment that contributes to phenotype/function. In the absence of demonstrated function a gene may be characterized by sequence, transcription or homology

_ImmunizationObservationType | http://terminology.hl7.org/CodeSystem/v3-ActCode | ImmunizationObservationType | Description: Observation codes which describe characteristics of the immunization material.

OBSANTC | http://terminology.hl7.org/CodeSystem/v3-ActCode | antigen count | Description: Indicates the valid antigen count.

OBSANTV | http://terminology.hl7.org/CodeSystem/v3-ActCode | antigen validity | Description: Indicates whether an antigen is valid or invalid.

PAT_ADV_EVNT | http://terminology.hl7.org/CodeSystem/v3-ActCode | patient adverse event | Indicates that the ICSR is describing problems that a patient experienced after receiving a vaccine product.

VAC_PROBLEM | http://terminology.hl7.org/CodeSystem/v3-ActCode | vaccine product problem | Indicates that the ICSR is describing a problem with the actual vaccine product such as physical defects (cloudy, particulate matter) or inability to confer immunity.

21611-9 | http://terminology.hl7.org/CodeSystem/v3-ActCode | age patient qn est | Definition:Estimated age.

21612-7 | http://terminology.hl7.org/CodeSystem/v3-ActCode | age patient qn reported | Definition:Reported age.

29553-5 | http://terminology.hl7.org/CodeSystem/v3-ActCode | age patient qn calc | Definition:Calculated age.

30525-0 | http://terminology.hl7.org/CodeSystem/v3-ActCode | age patient qn definition | Definition:General specification of age with no implied method of determination.

30972-4 | http://terminology.hl7.org/CodeSystem/v3-ActCode | age at onset of adverse event | Definition:Age at onset of associated adverse event; no implied method of determination.

REP_HALF_LIFE | http://terminology.hl7.org/CodeSystem/v3-ActCode | representative half-life | Description:This observation represents an 'average' or 'expected' half-life typical of the product.

SPLCOATING | http://terminology.hl7.org/CodeSystem/v3-ActCode | coating | Definition: A characteristic of an oral solid dosage form of a medicinal product, indicating whether it has one or more coatings such as sugar coating, film coating, or enteric coating. Only coatings to the external surface or the dosage form should be considered (for example, coatings to individual pellets or granules inside a capsule or tablet are excluded from consideration). Constraints: The Observation.value must be a Boolean (BL) with true for the presence or false for the absence of one or more coatings on a solid dosage form.

SPLCOLOR | http://terminology.hl7.org/CodeSystem/v3-ActCode | color | Definition: A characteristic of an oral solid dosage form of a medicinal product, specifying the color or colors that most predominantly define the appearance of the dose form. SPLCOLOR is not an FDA specification for the actual color of solid dosage forms or the names of colors that can appear in labeling. Constraints: The Observation.value must be a single coded value or a list of multiple coded values, specifying one or more distinct colors that approximate of the color(s) of distinct areas of the solid dosage form, such as the different sides of a tablet or one-part capsule, or the different halves of a two-part capsule. Bands on banded capsules, regardless of the color, are not considered when assigning an SPLCOLOR. Imprints on the dosage form, regardless of their color are not considered when assigning an SPLCOLOR. If more than one color exists on a particular side or half, then the most predominant color on that side or half is recorded. If the gelatin capsule shell is colorless and transparent, use the predominant color of the contents that appears through the colorless and transparent capsule shell. Colors can include: Black;Gray;White;Red;Pink;Purple;Green;Yellow;Orange;Brown;Blue;Turquoise.

SPLIMAGE | http://terminology.hl7.org/CodeSystem/v3-ActCode | image | Description: A characteristic representing a single file reference that contains two or more views of the same dosage form of the product; in most cases this should represent front and back views of the dosage form, but occasionally additional views might be needed in order to capture all of the important physical characteristics of the dosage form. Any imprint and/or symbol should be clearly identifiable, and the viewer should not normally need to rotate the image in order to read it. Images that are submitted with SPL should be included in the same directory as the SPL file.

SPLIMPRINT | http://terminology.hl7.org/CodeSystem/v3-ActCode | imprint | Definition: A characteristic of an oral solid dosage form of a medicinal product, specifying the alphanumeric text that appears on the solid dosage form, including text that is embossed, debossed, engraved or printed with ink. The presence of other non-textual distinguishing marks or symbols is recorded by SPLSYMBOL. Examples: Included in SPLIMPRINT are alphanumeric text that appears on the bands of banded capsules and logos and other symbols that can be interpreted as letters or numbers. Constraints: The Observation.value must be of type Character String (ST). Excluded from SPLIMPRINT are internal and external cut-outs in the form of alphanumeric text and the letter 'R' with a

circle around it (when referring to a registered trademark) and the letters 'TM' (when referring to a 'trade mark'). To record text, begin on either side or part of the dosage form. Start at the top left and progress as one would normally read a book. Enter a semicolon to show separation between words or line divisions.

SPLSCORING | http://terminology.hl7.org/CodeSystem/v3-ActCode | scoring | Definition: A characteristic of an oral solid dosage form of a medicinal product, specifying the number of equal pieces that the solid dosage form can be divided into using score line(s). Example: One score line creating two equal pieces is given a value of 2, two parallel score lines creating three equal pieces is given a value of 3. Constraints: Whether three parallel score lines create four equal pieces or two intersecting score lines create two equal pieces using one score line and four equal pieces using both score lines, both have the scoring value of 4. Solid dosage forms that are not scored are given a value of 1. Solid dosage forms that can only be divided into unequal pieces are given a null-value with nullFlavor other (OTH).

SPLSHAPE | http://terminology.hl7.org/CodeSystem/v3-ActCode | shape | Description: A characteristic of an oral solid dosage form of a medicinal product, specifying the two dimensional representation of the solid dose form, in terms of the outside perimeter of a solid dosage form when the dosage form, resting on a flat surface, is viewed from directly above, including slight rounding of corners. SPLSHAPE does not include embossing, scoring, debossing, or internal cut-outs. SPLSHAPE is independent of the orientation of the imprint and logo. Shapes can include: Triangle (3 sided); Square; Round; Semicircle; Pentagon (5 sided); Diamond; Double circle; Bullet; Hexagon (6 sided); Rectangle; Gear; Capsule; Heptagon (7 sided); Trapezoid; Oval; Clover; Octagon (8 sided); Tear; Freeform.

SPLSIZE | http://terminology.hl7.org/CodeSystem/v3-ActCode | size | Definition: A characteristic of an oral solid dosage form of a medicinal product, specifying the longest single dimension of the solid dosage form as a physical quantity in the dimension of length (e.g., 3 mm). The length is should be specified in millimeters and should be rounded to the nearest whole millimeter. Example: SPLSIZE for a rectangular shaped tablet is the length and SPLSIZE for a round shaped tablet is the diameter.

SPLSYMBOL | http://terminology.hl7.org/CodeSystem/v3-ActCode | symbol | Definition: A characteristic of an oral solid dosage form of a medicinal product, to describe whether or not the medicinal product has a mark or symbol appearing on it for easy and definite recognition. Score lines, letters, numbers, and internal and external cut-outs are not considered marks or symbols. See SPLSCORING and SPLIMPRINT for these characteristics. Constraints: The Observation.value must be a Boolean (BL) with <u>true</u> indicating the presence and <u>false</u> for the absence of marks or symbols. Example:

AIRTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | airborne transmission | Communication of an agent from a living subject or environmental source to a living subject through indirect contact via oral or nasal inhalation.

ANANTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | animal to animal transmission | Communication of an agent from one animal to another proximate animal.

ANHUMTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | animal to human transmission | Communication of an agent from an animal to a proximate person.

BDYFLDTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | body fluid contact transmission | Communication of an agent from one living subject to another living subject through direct contact with any body fluid.

BLDTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | blood borne transmission | Communication of an agent to a living subject through direct contact with blood or blood products whether the contact with blood is part of a therapeutic procedure or not.

DERMTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | transdermal transmission | Communication of an agent from a living subject or environmental source to a living subject via agent migration through intact skin.

ENVTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | environmental exposure transmission | Communication of an agent from an environmental surface or source to a living subject by direct contact.

FECTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | fecal-oral transmission | Communication of an agent from a living subject or environmental source to a living subject through oral contact with material contaminated by person or animal fecal material.

FOMTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | fomite transmission | Communication of an agent from an non-living material to a living subject through direct contact.

FOODTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | food-borne transmission | Communication of an agent from a food source to a living subject via oral consumption.

HUMHUMTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | human to human transmission | Communication of an agent from a person to a proximate person.

INDTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | indeterminate disease transmission mode | Communication of an agent to a living subject via an undetermined route.

LACTTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | lactation transmission | Communication of an agent from one living subject to another living subject through direct contact with mammalian milk or colostrum.

NOSTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | nosocomial transmission | Communication of an agent from any entity to a living subject while the living subject is in the patient role in a healthcare facility.

PARTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | parenteral transmission | Communication of an agent from a living subject or environmental source to a living subject where the acquisition of the agent is not via the alimentary canal.

PLACTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | transplacental transmission | Communication of an agent from a living subject to the progeny of that living subject via agent migration across the maternal-fetal placental membranes while in utero.

SEXTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | sexual transmission | Communication of an agent from one living subject to another living subject through direct contact with genital or oral tissues as part of a sexual act.

TRNSFTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | transfusion transmission | Communication of an agent from one living subject to another living subject through direct contact with blood or blood products where the contact with blood is part of a therapeutic procedure.

VECTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | vector-borne transmission | Communication of an agent from a living subject acting as a required intermediary in the agent transmission process to a recipient living subject via direct contact.

WATTRNS | http://terminology.hl7.org/CodeSystem/v3-ActCode | water-borne transmission | Communication of an agent from a contaminated water source to a living subject whether the water is ingested as a food or not. The route of entry of the water may be through any bodily orifice.

_ObservationQualityMeasureAttribute | http://terminology.hl7.org/CodeSystem/v3-ActCode | ObservationQualityMeasureAttribute | Codes used to define various metadata aspects of a health quality measure.

AGGREGATE | http://terminology.hl7.org/CodeSystem/v3-ActCode | aggregate measure observation | Indicates that the observation is carrying out an aggregation calculation, contained in the value element.

CMPMSRMTH | http://terminology.hl7.org/CodeSystem/v3-ActCode | composite measure method | Indicates what method is used in a quality measure to combine the component measure results included in an composite measure.

CMPMSRSCRWGHT | http://terminology.hl7.org/CodeSystem/v3-ActCode | component measure scoring weight | An attribute of a quality measure describing the weight this component measure score is to carry in determining the overall composite measure final score. The value is real value greater than 0 and less than 1.0. Each component measure score will be multiplied by its CMPMSRSCRWGHT and then summed with the other component measures to determine the final overall composite measure score. The sum across all CMPMSRSCRWGHT values within a single composite measure SHALL be 1.0. The value assigned is scoped to the composite measure referencing this component measure only.

COPY | http://terminology.hl7.org/CodeSystem/v3-ActCode | copyright | Identifies the organization(s) who own the intellectual property represented by the eMeasure.

CRS | http://terminology.hl7.org/CodeSystem/v3-ActCode | clinical recommendation statement | Summary of relevant clinical guidelines or other clinical recommendations supporting this eMeasure.

DEF | http://terminology.hl7.org/CodeSystem/v3-ActCode | definition | Description of individual terms, provided as needed.

DISC | http://terminology.hl7.org/CodeSystem/v3-ActCode | disclaimer | Disclaimer information for the eMeasure.

FINALDT | http://terminology.hl7.org/CodeSystem/v3-ActCode | finalized date/time | The timestamp when the eMeasure was last packaged in the Measure Authoring Tool.

GUIDE | http://terminology.hl7.org/CodeSystem/v3-ActCode | guidance | Used to allow measure developers to provide additional guidance for implementers to understand greater specificity than could be provided in the logic for data criteria.

IDUR | http://terminology.hl7.org/CodeSystem/v3-ActCode | improvement notation | Information on whether an increase or decrease in score is the preferred result (e.g., a higher score indicates better quality OR a lower score indicates better quality OR quality is within a range).

ITMCNT | http://terminology.hl7.org/CodeSystem/v3-ActCode | items counted | Describes the items counted by the measure (e.g., patients, encounters, procedures, etc.)

KEY | http://terminology.hl7.org/CodeSystem/v3-ActCode | keyword | A significant word that aids in discoverability.

MEDT | http://terminology.hl7.org/CodeSystem/v3-ActCode | measurement end date | The end date of the measurement period.

MSD | http://terminology.hl7.org/CodeSystem/v3-ActCode | measurement start date | The start date of the measurement period.

MSRADJ | http://terminology.hl7.org/CodeSystem/v3-ActCode | risk adjustment | The method of adjusting for clinical severity and conditions present at the start of care that can influence patient outcomes for making valid comparisons of outcome measures across providers. Indicates whether an eMeasure is subject to the statistical process for reducing, removing, or clarifying the influences of confounding factors to allow more useful comparisons.

MSRAGG | http://terminology.hl7.org/CodeSystem/v3-ActCode | rate aggregation | Describes how to combine information calculated based on logic in each of several populations into one summarized result. It can also be used to describe how to risk adjust the data based on supplemental data elements described in the eMeasure. (e.g., pneumonia hospital measures antibiotic selection in the ICU versus non-ICU and then the roll-up of the two). Open Issue: The description does NOT align well with the definition used in the HQMF specfication; correct the MSGAGG definition, and the possible distinction of MSRAGG as a child of AGGREGATE.

MSRIMPROV | http://terminology.hl7.org/CodeSystem/v3-ActCode | health quality measure improvement notation | Information on whether an increase or decrease in score is the preferred result. This should reflect information on which way is better, an increase or decrease in score.

MSRJUR | http://terminology.hl7.org/CodeSystem/v3-ActCode | jurisdiction | The list of jurisdiction(s) for which the measure applies.

MSRRPTR | http://terminology.hl7.org/CodeSystem/v3-ActCode | reporter type | Type of person or organization that is expected to report the issue.

MSRRPTTIME | http://terminology.hl7.org/CodeSystem/v3-ActCode | timeframe for reporting | The maximum time that may elapse following completion of the measure until the measure report must be sent to the receiver.

MSRSCORE | http://terminology.hl7.org/CodeSystem/v3-ActCode | measure scoring | Indicates how the calculation is performed for the eMeasure (e.g., proportion, continuous variable, ratio)

MSRSET | http://terminology.hl7.org/CodeSystem/v3-ActCode | health quality measure care setting | Location(s) in which care being measured is rendered Usage Note: MSRSET is used rather than RoleCode because the setting applies to what is being measured, as opposed to participating directly in the health quality measure documantion itself).

MSRTOPIC | http://terminology.hl7.org/CodeSystem/v3-ActCode | health quality measure topic type | health quality measure topic type

MSRTP | http://terminology.hl7.org/CodeSystem/v3-ActCode | measurement period | The time period for which the eMeasure applies.

MSRTYPE | http://terminology.hl7.org/CodeSystem/v3-ActCode | measure type | Indicates whether the eMeasure is used to examine a process or an outcome over time (e.g., Structure, Process, Outcome).

RAT | http://terminology.hl7.org/CodeSystem/v3-ActCode | rationale | Succinct statement of the need for the measure. Usually includes statements pertaining to Importance criterion: impact, gap in care and evidence.

REF | http://terminology.hl7.org/CodeSystem/v3-ActCode | reference | Identifies bibliographic citations or references to clinical practice guidelines, sources of evidence, or other relevant materials supporting the intent and rationale of the eMeasure.

SDE | http://terminology.hl7.org/CodeSystem/v3-ActCode | supplemental data elements | Comparison of results across strata can be used to show where disparities exist or where there is a need to expose differences in results. For example, Centers for Medicare & Medicaid Services (CMS) in the U.S. defines four required Supplemental Data Elements (payer, ethnicity, race, and gender), which are variables used to aggregate data into various subgroups. Additional supplemental data elements required for risk adjustment or other purposes of data aggregation can be included in the Supplemental Data Element section.

STRAT | http://terminology.hl7.org/CodeSystem/v3-ActCode | stratification | Describes the strata for which the measure is to be evaluated. There are three examples of reasons for stratification based on existing work. These include: (1) evaluate the measure based on different age groupings within the population described in the measure (e.g., evaluate the whole [age 14-25] and each sub-stratum [14-19] and [20-25]); (2) evaluate the eMeasure based on either a specific condition, a specific discharge location, or both; (3) evaluate the eMeasure based on different locations within a facility (e.g., evaluate the overall rate for all intensive care units and also some strata include additional findings [specific birth weights for neonatal intensive care units]).

TRANF | http://terminology.hl7.org/CodeSystem/v3-ActCode | transmission format | Can be a URL or hyperlinks that link to the transmission formats that are specified for a particular reporting program.

USE | http://terminology.hl7.org/CodeSystem/v3-ActCode | notice of use | Usage notes.

TIME_ABSOLUTE | http://terminology.hl7.org/CodeSystem/v3-ActCode | absolute time sequence | A sequence of values in the "absolute" time domain. This is the same time domain that all HL7 timestamps use. It is time as measured by the Gregorian calendar

TIME_RELATIVE | http://terminology.hl7.org/CodeSystem/v3-ActCode | relative time sequence | A sequence of values in a "relative" time domain. The time is measured relative to the earliest effective time in the Observation Series containing this sequence.

REPRESENTATIVE_BEAT | http://terminology.hl7.org/CodeSystem/v3-ActCode | ECG representative beat waveforms | This Observation Series type contains waveforms of a "representative beat" (a.k.a. "median beat" or "average beat"). The waveform samples are measured in relative time, relative to the beginning of the beat as defined by the Observation Series effective time. The waveforms are not directly acquired from the subject, but rather algorithmically derived from the "rhythm" waveforms.
RHYTHM | http://terminology.hl7.org/CodeSystem/v3-ActCode | ECG rhythm waveforms | This Observation type contains ECG "rhythm" waveforms. The waveform samples are measured in absolute time (a.k.a. "subject time" or "effective time"). These waveforms are usually "raw" with some minimal amount of noise reduction and baseline filtering applied.
_PatientImmunizationRelatedObservationType | http://terminology.hl7.org/CodeSystem/v3-ActCode | PatientImmunizationRelatedObservationType | Description: Reporting codes that are related to an immunization event.
CLSSRM | http://terminology.hl7.org/CodeSystem/v3-ActCode | classroom | Description: The class room associated with the patient during the immunization event.
GRADE | http://terminology.hl7.org/CodeSystem/v3-ActCode | grade | Description: The school grade or level the patient was in when immunized.
SCHL | http://terminology.hl7.org/CodeSystem/v3-ActCode | school | Description: The school the patient attended when immunized.
SCHLDIV | http://terminology.hl7.org/CodeSystem/v3-ActCode | school division | Description: The school division or district associated with the patient during the immunization event.
TEACHER | http://terminology.hl7.org/CodeSystem/v3-ActCode | teacher | Description: The patient's teacher when immunized.
DENEX | http://terminology.hl7.org/CodeSystem/v3-ActCode | denominator exclusions | Criteria which specify subjects who should be removed from the eMeasure population and denominator before determining if numerator criteria are met. Denominator exclusions are used in proportion and ratio measures to help narrow the denominator.
DENEXCEP | http://terminology.hl7.org/CodeSystem/v3-ActCode | denominator exceptions | Criteria which specify the removal of a subject, procedure or unit of measurement from the denominator, only if the numerator criteria are not met. Denominator exceptions allow for adjustment of the calculated score for those providers with higher risk populations. Denominator exceptions are used only in proportion eMeasures. They are not appropriate for ratio or continuous variable eMeasures. Denominator exceptions allow for the exercise of clinical judgment and should be specifically defined where capturing the information in a structured manner fits the clinical workflow. Generic denominator exception reasons used in proportion eMeasures fall into three general categories: Medical reasons Patient (or subject) reasons System reasons
DENOM | http://terminology.hl7.org/CodeSystem/v3-ActCode | denominator | Criteria for specifying the entities to be evaluated by a specific quality measure, based on a shared common set of characteristics (within a specific measurement set to which a given measure belongs). The denominator can be the same as the initial population, or it may be a subset of the initial population to further constrain it for the purpose of the eMeasure. Different measures within an eMeasure set may have different denominators. Continuous Variable eMeasures do not have a denominator, but instead define a measure population.
IPOP | http://terminology.hl7.org/CodeSystem/v3-ActCode | initial population | Criteria for specifying the entities to be evaluated by a specific quality measure, based on a shared common set of characteristics (within a specific measurement set to which a given measure belongs).
IPPOP | http://terminology.hl7.org/CodeSystem/v3-ActCode | initial patient population | Criteria for specifying the patients to be evaluated by a specific quality measure, based on a shared common set of characteristics (within a specific measurement set to which a given measure belongs). Details often include information based upon specific age groups, diagnoses, diagnostic and procedure codes, and enrollment periods.
MSROBS | http://terminology.hl7.org/CodeSystem/v3-ActCode | measure observation | Defines the observation to be performed for each patient or event in the measure population. Measure observations for each case in the population are aggregated to determine the overall measure score for the population. Examples: the median time from arrival in the Emergency Room to departure the median time from decision to admit to a hospital to the actual admission for Emergency Room patients
MSRPOPL | http://terminology.hl7.org/CodeSystem/v3-ActCode | measure population | Criteria for specifying the measure population as a narrative description (e.g., all patients seen in the Emergency Department during the measurement period). This is used only in continuous variable eMeasures.
MSRPOPLEX | http://terminology.hl7.org/CodeSystem/v3-ActCode | measure population exclusions | Criteria for specifying subjects who should be removed from the eMeasure's Initial Population and Measure Population. Measure Population Exclusions are used in Continuous Variable measures to help narrow the Measure Population before determining the value(s) of the continuous variable(s).
NUMER | http://terminology.hl7.org/CodeSystem/v3-ActCode | numerator | Criteria for specifying the processes or outcomes expected for each patient, procedure, or other unit of measurement defined in the denominator for proportion measures, or related to (but not directly derived from) the denominator for ratio measures (e.g., a numerator listing the number of central line blood stream infections and a denominator indicating the days per thousand of central line usage in a specific time period).
NUMEX | http://terminology.hl7.org/CodeSystem/v3-ActCode | numerator exclusions | Criteria for specifying instances that should not be included in the numerator data. (e.g., if the number of central line blood stream infections per 1000 catheter days were to exclude infections with a specific bacterium, that bacterium would be listed as a numerator exclusion). Numerator Exclusions are used only in ratio eMeasures.
PREFSTRENGTH | http://terminology.hl7.org/CodeSystem/v3-ActCode | preference strength | An observation about how important a preference is to the target of the preference.
ADVERSE_REACTION | http://terminology.hl7.org/CodeSystem/v3-ActCode | Adverse Reaction | Indicates that the observation is of an unexpected negative occurrence in the subject suspected to result from the subject's exposure to one or more agents. Observation values would be the symptom resulting from the reaction.
ASSERTION | http://terminology.hl7.org/CodeSystem/v3-ActCode | Assertion | Description:Refines classCode OBS to indicate an observation in which observation.value contains a finding or other nominalized statement, where the encoded information in Observation.value is not altered by Observation.code. For instance, observation.code="ASSERTION" and observation.value="fracture of femur present" is an assertion of a clinical finding of femur fracture.
CASESER | http://terminology.hl7.org/CodeSystem/v3-ActCode | case seriousness criteria | Definition:An observation that provides a characterization of the level of harm to an investigation subject as a result of a reaction or event.
CDIO | http://terminology.hl7.org/CodeSystem/v3-ActCode | case disease imported observation | An observation that states whether the disease was likely acquired outside the jurisdiction of observation, and if so, the nature of the inter-jurisdictional relationship. OpenIssue: This code could be moved to LOINC if it can be done before there are significant implemenations using it.
CRIT | http://terminology.hl7.org/CodeSystem/v3-ActCode | criticality | A clinical judgment as to the worst case result of a future exposure (including substance administration). When the worst case result is assessed to have a life-threatening or organ system threatening potential, it is considered to be of high criticality.

CTMO | http://terminology.hl7.org/CodeSystem/v3-ActCode | case transmission mode observation | An observation that states the mechanism by which disease was acquired by the living subject involved in the public health case. OpenIssue: This code could be moved to LOINC if it can be done before there are significant implemenations using it.
DX | http://terminology.hl7.org/CodeSystem/v3-ActCode | ObservationDiagnosisTypes | Includes all codes defining types of indications such as diagnosis, symptom and other indications such as contrast agents for lab tests.
ADMDX | http://terminology.hl7.org/CodeSystem/v3-ActCode | admitting diagnosis | Admitting diagnosis are the diagnoses documented for administrative purposes as the basis for a hospital admission.
DISDX | http://terminology.hl7.org/CodeSystem/v3-ActCode | discharge diagnosis | Discharge diagnosis are the diagnoses documented for administrative purposes as the time of hospital discharge.
INTDX | http://terminology.hl7.org/CodeSystem/v3-ActCode | intermediate diagnosis | Intermediate diagnoses are those diagnoses documented for administrative purposes during the course of a hospital stay.
NOI | http://terminology.hl7.org/CodeSystem/v3-ActCode | nature of injury | The type of injury that the injury coding specifies.
GISTIER | http://terminology.hl7.org/CodeSystem/v3-ActCode | GIS tier | Description: Accuracy determined as per the GIS tier code system.
HHOBS | http://terminology.hl7.org/CodeSystem/v3-ActCode | household situation observation | Indicates that the observation is of a person’s living situation in a household including the household composition and circumstances.
ISSUE | http://terminology.hl7.org/CodeSystem/v3-ActCode | detected issue | There is a clinical issue for the therapy that makes continuation of the therapy inappropriate. Open Issue: The definition of this code does not correctly represent the concept space of its specializations (children)
NAT | http://terminology.hl7.org/CodeSystem/v3-ActCode | Insufficient authorization | The requesting party has insufficient authorization to invoke the interaction.
SUPPRESSED | http://terminology.hl7.org/CodeSystem/v3-ActCode | record suppressed | Description: One or more records in the query response have been suppressed due to consent or privacy restrictions.
VALIDAT | http://terminology.hl7.org/CodeSystem/v3-ActCode | validation issue | Description:The specified element did not pass business-rule validation.
KEY204 | http://terminology.hl7.org/CodeSystem/v3-ActCode | Unknown key identifier | The ID of the patient, order, etc., was not found. Used for transactions other than additions, e.g. transfer of a non-existent patient.
KEY205 | http://terminology.hl7.org/CodeSystem/v3-ActCode | Duplicate key identifier | The ID of the patient, order, etc., already exists. Used in response to addition transactions (Admit, New Order, etc.).
COMPLY | http://terminology.hl7.org/CodeSystem/v3-ActCode | Compliance Alert | There may be an issue with the patient complying with the intentions of the proposed therapy
DUPTHPY | http://terminology.hl7.org/CodeSystem/v3-ActCode | Duplicate Therapy Alert | The proposed therapy appears to duplicate an existing therapy
DUPTHPCLS | http://terminology.hl7.org/CodeSystem/v3-ActCode | duplicate therapeutic alass alert | Description:The proposed therapy appears to have the same intended therapeutic benefit as an existing therapy, though the specific mechanisms of action vary.
DUPTHPGEN | http://terminology.hl7.org/CodeSystem/v3-ActCode | duplicate generic alert | Description:The proposed therapy appears to have the same intended therapeutic benefit as an existing therapy and uses the same mechanisms of action as the existing therapy.
ABUSE | http://terminology.hl7.org/CodeSystem/v3-ActCode | commonly abused/misused alert | Description:The proposed therapy is frequently misused or abused and therefore should be used with caution and/or monitoring.
FRAUD | http://terminology.hl7.org/CodeSystem/v3-ActCode | potential fraud | Description:The request is suspected to have a fraudulent basis.
PLYDOC | http://terminology.hl7.org/CodeSystem/v3-ActCode | Poly-orderer Alert | A similar or identical therapy was recently ordered by a different practitioner.
PLYPHRM | http://terminology.hl7.org/CodeSystem/v3-ActCode | Poly-supplier Alert | This patient was recently supplied a similar or identical therapy from a different pharmacy or supplier.
DOSE | http://terminology.hl7.org/CodeSystem/v3-ActCode | Dosage problem | Proposed dosage instructions for therapy differ from standard practice.
DOSECOND | http://terminology.hl7.org/CodeSystem/v3-ActCode | dosage-condition alert | Description:Proposed dosage is inappropriate due to patient's medical condition.
DOSEDUR | http://terminology.hl7.org/CodeSystem/v3-ActCode | Dose-Duration Alert | Proposed length of therapy differs from standard practice.
DOSEDURH | http://terminology.hl7.org/CodeSystem/v3-ActCode | Dose-Duration High Alert | Proposed length of therapy is longer than standard practice
DOSEDURHIND | http://terminology.hl7.org/CodeSystem/v3-ActCode | Dose-Duration High for Indication Alert | Proposed length of therapy is longer than standard practice for the identified indication or diagnosis
DOSEDURL | http://terminology.hl7.org/CodeSystem/v3-ActCode | Dose-Duration Low Alert | Proposed length of therapy is shorter than that necessary for therapeutic effect
DOSEDURLIND | http://terminology.hl7.org/CodeSystem/v3-ActCode | Dose-Duration Low for Indication Alert | Proposed length of therapy is shorter than standard practice for the identified indication or diagnosis
DOSEH | http://terminology.hl7.org/CodeSystem/v3-ActCode | High Dose Alert | Proposed dosage exceeds standard practice
DOSEHINDA | http://terminology.hl7.org/CodeSystem/v3-ActCode | High Dose for Age Alert | Proposed dosage exceeds standard practice for the patient's age
DOSEHIND | http://terminology.hl7.org/CodeSystem/v3-ActCode | High Dose for Indication Alert | High Dose for Indication Alert
DOSEHINDSA | http://terminology.hl7.org/CodeSystem/v3-ActCode | High Dose for Height/Surface Area Alert | Proposed dosage exceeds standard practice for the patient's height or body surface area
DOSEHINDW | http://terminology.hl7.org/CodeSystem/v3-ActCode | High Dose for Weight Alert | Proposed dosage exceeds standard practice for the patient's weight
DOSEIVL | http://terminology.hl7.org/CodeSystem/v3-ActCode | Dose-Interval Alert | Proposed dosage interval/timing differs from standard practice
DOSEIVLIND | http://terminology.hl7.org/CodeSystem/v3-ActCode | Dose-Interval for Indication Alert | Proposed dosage interval/timing differs from standard practice for the identified indication or diagnosis
DOSEL | http://terminology.hl7.org/CodeSystem/v3-ActCode | Low Dose Alert | Proposed dosage is below suggested therapeutic levels

DOSELINDA | http://terminology.hl7.org/CodeSystem/v3-ActCode | Low Dose for Age Alert | Proposed dosage is below suggested therapeutic levels for the patient's age
DOSELIND | http://terminology.hl7.org/CodeSystem/v3-ActCode | Low Dose for Indication Alert | Low Dose for Indication Alert
DOSELINDSA | http://terminology.hl7.org/CodeSystem/v3-ActCode | Low Dose for Height/Surface Area Alert | Proposed dosage is below suggested therapeutic levels for the patient's height or body surface area
DOSELINDW | http://terminology.hl7.org/CodeSystem/v3-ActCode | Low Dose for Weight Alert | Proposed dosage is below suggested therapeutic levels for the patient's weight
MDOSE | http://terminology.hl7.org/CodeSystem/v3-ActCode | maximum dosage reached | Description:The maximum quantity of this drug allowed to be administered within a particular time-range (month, year, lifetime) has been reached or exceeded.
OBSA | http://terminology.hl7.org/CodeSystem/v3-ActCode | Observation Alert | Proposed therapy may be inappropriate or contraindicated due to conditions or characteristics of the patient
AGE | http://terminology.hl7.org/CodeSystem/v3-ActCode | Age Alert | Proposed therapy may be inappropriate or contraindicated due to patient age
ADALRT | http://terminology.hl7.org/CodeSystem/v3-ActCode | adult alert | Proposed therapy is outside of the standard practice for an adult patient.
GEALRT | http://terminology.hl7.org/CodeSystem/v3-ActCode | geriatric alert | Proposed therapy is outside of standard practice for a geriatric patient.
PEALRT | http://terminology.hl7.org/CodeSystem/v3-ActCode | pediatric alert | Proposed therapy is outside of the standard practice for a pediatric patient.
COND | http://terminology.hl7.org/CodeSystem/v3-ActCode | Condition Alert | Proposed therapy may be inappropriate or contraindicated due to an existing/recent patient condition or diagnosis
HGHT | http://terminology.hl7.org/CodeSystem/v3-ActCode
LACT | http://terminology.hl7.org/CodeSystem/v3-ActCode | Lactation Alert | Proposed therapy may be inappropriate or contraindicated when breast-feeding
PREG | http://terminology.hl7.org/CodeSystem/v3-ActCode | Pregnancy Alert | Proposed therapy may be inappropriate or contraindicated during pregnancy
WGHT | http://terminology.hl7.org/CodeSystem/v3-ActCode
CREACT | http://terminology.hl7.org/CodeSystem/v3-ActCode | common reaction alert | Description:Proposed therapy may be inappropriate or contraindicated because of a common but non-patient specific reaction to the product. Example:There is no record of a specific sensitivity for the patient, but the presence of the sensitivity is common and therefore caution is warranted.
GEN | http://terminology.hl7.org/CodeSystem/v3-ActCode | Genetic Alert | Proposed therapy may be inappropriate or contraindicated due to patient genetic indicators.
GEND | http://terminology.hl7.org/CodeSystem/v3-ActCode | Gender Alert | Proposed therapy may be inappropriate or contraindicated due to patient gender.
LAB | http://terminology.hl7.org/CodeSystem/v3-ActCode | Lab Alert | Proposed therapy may be inappropriate or contraindicated due to recent lab test results
REACT | http://terminology.hl7.org/CodeSystem/v3-ActCode | Reaction Alert | Proposed therapy may be inappropriate or contraindicated based on the potential for a patient reaction to the proposed product
ALGY | http://terminology.hl7.org/CodeSystem/v3-ActCode | Allergy Alert | Proposed therapy may be inappropriate or contraindicated because of a recorded patient allergy to the proposed product. (Allergies are immune based reactions.)
INT | http://terminology.hl7.org/CodeSystem/v3-ActCode | Intolerance Alert | Proposed therapy may be inappropriate or contraindicated because of a recorded patient intolerance to the proposed product. (Intolerances are non-immune based sensitivities.)
RREACT | http://terminology.hl7.org/CodeSystem/v3-ActCode | Related Reaction Alert | Proposed therapy may be inappropriate or contraindicated because of a potential patient reaction to a cross-sensitivity related product.
RALG | http://terminology.hl7.org/CodeSystem/v3-ActCode | Related Allergy Alert | Proposed therapy may be inappropriate or contraindicated because of a recorded patient allergy to a cross-sensitivity related product. (Allergies are immune based reactions.)
RAR | http://terminology.hl7.org/CodeSystem/v3-ActCode | Related Prior Reaction Alert | Proposed therapy may be inappropriate or contraindicated because of a recorded prior adverse reaction to a cross-sensitivity related product.
RINT | http://terminology.hl7.org/CodeSystem/v3-ActCode | Related Intolerance Alert | Proposed therapy may be inappropriate or contraindicated because of a recorded patient intolerance to a cross-sensitivity related product. (Intolerances are non-immune based sensitivities.)
BUS | http://terminology.hl7.org/CodeSystem/v3-ActCode | business constraint violation | Description:A local business rule relating multiple elements has been violated.
CODE_INVAL | http://terminology.hl7.org/CodeSystem/v3-ActCode | code is not valid | Description:The specified code is not valid against the list of codes allowed for the element.
CODE_DEPREC | http://terminology.hl7.org/CodeSystem/v3-ActCode | code has been deprecated | Description:The specified code has been deprecated and should no longer be used. Select another code from the code system.
FORMAT | http://terminology.hl7.org/CodeSystem/v3-ActCode | invalid format | Description:The element does not follow the formatting or type rules defined for the field.
ILLEGAL | http://terminology.hl7.org/CodeSystem/v3-ActCode | illegal | Description:The request is missing elements or contains elements which cause it to not meet the legal standards for actioning.
LEN_RANGE | http://terminology.hl7.org/CodeSystem/v3-ActCode | length out of range | Description:The length of the data specified falls out of the range defined for the element.
LEN_LONG | http://terminology.hl7.org/CodeSystem/v3-ActCode | length is too long | Description:The length of the data specified is greater than the maximum length defined for the element.
LEN_SHORT | http://terminology.hl7.org/CodeSystem/v3-ActCode | length is too short | Description:The length of the data specified is less than the minimum length defined for the element.
MISSCOND | http://terminology.hl7.org/CodeSystem/v3-ActCode | conditional element missing | Description:The specified element must be specified with a non-null value under certain conditions. In this case, the conditions are true but the element is still missing or null.
MISSMAND | http://terminology.hl7.org/CodeSystem/v3-ActCode | mandatory element missing | Description:The specified element is mandatory and was not included in the instance.
NODUPS | http://terminology.hl7.org/CodeSystem/v3-ActCode | duplicate values are not permitted | Description:More than one element with the same value exists in the set. Duplicates not permission in this set in a set.

NOPERSIST | http://terminology.hl7.org/CodeSystem/v3-ActCode | element will not be persisted | Description: Element in submitted message will not persist in data storage based on detected issue.
REP_RANGE | http://terminology.hl7.org/CodeSystem/v3-ActCode | repetitions out of range | Description:The number of repeating elements falls outside the range of the allowed number of repetitions.
MAXOCCURS | http://terminology.hl7.org/CodeSystem/v3-ActCode | repetitions above maximum | Description:The number of repeating elements is above the maximum number of repetitions allowed.
MINOCCURS | http://terminology.hl7.org/CodeSystem/v3-ActCode | repetitions below minimum | Description:The number of repeating elements is below the minimum number of repetitions allowed.
KEY206 | http://terminology.hl7.org/CodeSystem/v3-ActCode | non-matching identification | Description: Metadata associated with the identification (e.g. name or gender) does not match the identification being verified.
OBSOLETE | http://terminology.hl7.org/CodeSystem/v3-ActCode | obsolete record returned | Description: One or more records in the query response have a status of 'obsolete'.
FOOD | http://terminology.hl7.org/CodeSystem/v3-ActCode | Food Interaction Alert | Proposed therapy may interact with certain foods
TPROD | http://terminology.hl7.org/CodeSystem/v3-ActCode | Therapeutic Product Alert | Proposed therapy may interact with an existing or recent therapeutic product
DRG | http://terminology.hl7.org/CodeSystem/v3-ActCode | Drug Interaction Alert | Proposed therapy may interact with an existing or recent drug therapy
NHP | http://terminology.hl7.org/CodeSystem/v3-ActCode | Natural Health Product Alert | Proposed therapy may interact with existing or recent natural health product therapy
NONRX | http://terminology.hl7.org/CodeSystem/v3-ActCode | Non-Prescription Interaction Alert | Proposed therapy may interact with a non-prescription drug (e.g. alcohol, tobacco, Aspirin)
PREVINEF | http://terminology.hl7.org/CodeSystem/v3-ActCode | previously ineffective | Definition:The same or similar treatment has previously been attempted with the patient without achieving a positive effect.
DACT | http://terminology.hl7.org/CodeSystem/v3-ActCode | drug action detected issue | Description:Proposed therapy may be contraindicated or ineffective based on an existing or recent drug therapy.
TIME | http://terminology.hl7.org/CodeSystem/v3-ActCode | timing detected issue | Description:Proposed therapy may be inappropriate or ineffective based on the proposed start or end time.
ALRTENDLATE | http://terminology.hl7.org/CodeSystem/v3-ActCode | end too late alert | Definition:Proposed therapy may be inappropriate or ineffective because the end of administration is too close to another planned therapy.
ALRTSTRTLATE | http://terminology.hl7.org/CodeSystem/v3-ActCode | start too late alert | Definition:Proposed therapy may be inappropriate or ineffective because the start of administration is too late after the onset of the condition.
ENDLATE | http://terminology.hl7.org/CodeSystem/v3-ActCode | End Too Late Alert | Proposed therapy may be inappropriate or ineffective because the end of administration is too close to another planned therapy
STRTLATE | http://terminology.hl7.org/CodeSystem/v3-ActCode | Start Too Late Alert | Proposed therapy may be inappropriate or ineffective because the start of administration is too late after the onset of the condition
ALLDONE | http://terminology.hl7.org/CodeSystem/v3-ActCode | already performed | Definition:The requested action has already been performed and so this request has no effect
FULFIL | http://terminology.hl7.org/CodeSystem/v3-ActCode | fulfillment alert | Definition:The therapy being performed is in some way out of alignment with the requested therapy.
NOTACTN | http://terminology.hl7.org/CodeSystem/v3-ActCode | no longer actionable | Definition:The status of the request being fulfilled has changed such that it is no longer actionable. This may be because the request has expired, has already been completely fulfilled or has been otherwise stopped or disabled. (Not used for 'suspended' orders.)
NOTEQUIV | http://terminology.hl7.org/CodeSystem/v3-ActCode | not equivalent alert | Definition:The therapy being performed is not sufficiently equivalent to the therapy which was requested.
NOTEQUIVGEN | http://terminology.hl7.org/CodeSystem/v3-ActCode | not generically equivalent alert | Definition:The therapy being performed is not generically equivalent (having the identical biological action) to the therapy which was requested.
NOTEQUIVTHER | http://terminology.hl7.org/CodeSystem/v3-ActCode | not therapeutically equivalent alert | Definition:The therapy being performed is not therapeutically equivalent (having the same overall patient effect) to the therapy which was requested.
TIMING | http://terminology.hl7.org/CodeSystem/v3-ActCode | event timing incorrect alert | Definition:The therapy is being performed at a time which diverges from the time the therapy was requested
INTERVAL | http://terminology.hl7.org/CodeSystem/v3-ActCode | outside requested time | Definition:The therapy action is being performed outside the bounds of the time period requested
MINFREQ | http://terminology.hl7.org/CodeSystem/v3-ActCode | too soon within frequency based on the usage | Definition:The therapy action is being performed too soon after the previous occurrence based on the requested frequency
HELD | http://terminology.hl7.org/CodeSystem/v3-ActCode | held/suspended alert | Definition:There should be no actions taken in fulfillment of a request that has been held or suspended.
TOOLATE | http://terminology.hl7.org/CodeSystem/v3-ActCode | Refill Too Late Alert | The patient is receiving a subsequent fill significantly later than would be expected based on the amount previously supplied and the therapy dosage instructions
TOOSOON | http://terminology.hl7.org/CodeSystem/v3-ActCode | Refill Too Soon Alert | The patient is receiving a subsequent fill significantly earlier than would be expected based on the amount previously supplied and the therapy dosage instructions
HISTORIC | http://terminology.hl7.org/CodeSystem/v3-ActCode | record recorded as historical | Description: While the record was accepted in the repository, there is a more recent version of a record of this type.
PATPREF | http://terminology.hl7.org/CodeSystem/v3-ActCode | violates stated preferences | Definition:The proposed therapy goes against preferences or consent constraints recorded in the patient's record.
PATPREFALT | http://terminology.hl7.org/CodeSystem/v3-ActCode | violates stated preferences, alternate available | Definition:The proposed therapy goes against preferences or consent constraints recorded in the patient's record. An alternate therapy meeting those constraints is available.
KSUBJ | http://terminology.hl7.org/CodeSystem/v3-ActCode | knowledge subject | Categorization of types of observation that capture the main clinical knowledge subject which may be a medication, a laboratory test, a disease.
KSUBT | http://terminology.hl7.org/CodeSystem/v3-ActCode | knowledge subtopic | Categorization of types of observation that capture a knowledge subtopic which might be treatment, etiology, or prognosis.
OINT | http://terminology.hl7.org/CodeSystem/v3-ActCode | intolerance | Hypersensitivity resulting in an adverse reaction upon exposure to an agent.
ALG | http://terminology.hl7.org/CodeSystem/v3-ActCode | Allergy | Hypersensitivity to an agent caused by an immunologic response to an initial exposure

DALG | http://terminology.hl7.org/CodeSystem/v3-ActCode | Drug Allergy | An allergy to a pharmaceutical product.
EALG | http://terminology.hl7.org/CodeSystem/v3-ActCode | Environmental Allergy | An allergy to a substance other than a drug or a food. E.g. Latex, pollen, etc.
FALG | http://terminology.hl7.org/CodeSystem/v3-ActCode | Food Allergy | An allergy to a substance generally consumed for nutritional purposes.
DINT | http://terminology.hl7.org/CodeSystem/v3-ActCode | Drug Intolerance | Hypersensitivity resulting in an adverse reaction upon exposure to a drug.
DNAINT | http://terminology.hl7.org/CodeSystem/v3-ActCode | Drug Non-Allergy Intolerance | Hypersensitivity to an agent caused by a mechanism other than an immunologic response to an initial exposure
EINT | http://terminology.hl7.org/CodeSystem/v3-ActCode | Environmental Intolerance | Hypersensitivity resulting in an adverse reaction upon exposure to environmental conditions.
ENAINT | http://terminology.hl7.org/CodeSystem/v3-ActCode | Environmental Non-Allergy Intolerance | Hypersensitivity to an agent caused by a mechanism other than an immunologic response to an initial exposure
FINT | http://terminology.hl7.org/CodeSystem/v3-ActCode | Food Intolerance | Hypersensitivity resulting in an adverse reaction upon exposure to food.
FNAINT | http://terminology.hl7.org/CodeSystem/v3-ActCode | Food Non-Allergy Intolerance | Hypersensitivity to an agent caused by a mechanism other than an immunologic response to an initial exposure
NAINT | http://terminology.hl7.org/CodeSystem/v3-ActCode | Non-Allergy Intolerance | Hypersensitivity to an agent caused by a mechanism other than an immunologic response to an initial exposure
SEV | http://terminology.hl7.org/CodeSystem/v3-ActCode | Severity Observation | A subjective evaluation of the seriousness or intensity associated with another observation.
FDACOATING | http://terminology.hl7.org/CodeSystem/v3-ActCode | coating | FDA label coating
FDACOLOR | http://terminology.hl7.org/CodeSystem/v3-ActCode | color | FDA label color
FDAIMPRINTCD | http://terminology.hl7.org/CodeSystem/v3-ActCode | imprint code | FDA label imprint code
FDALOGO | http://terminology.hl7.org/CodeSystem/v3-ActCode | logo | FDA label logo
FDASCORING | http://terminology.hl7.org/CodeSystem/v3-ActCode | scoring | FDA label scoring
FDASHAPE | http://terminology.hl7.org/CodeSystem/v3-ActCode | shape | FDA label shape
FDASIZE | http://terminology.hl7.org/CodeSystem/v3-ActCode | size | FDA label size
1-8 | http://loinc.org | Acyclovir [Susceptibility]
10-9 | http://loinc.org | Amdinocillin [Susceptibility] by Serum bactericidal titer
100-8 | http://loinc.org | Cefoperazone [Susceptibility] by Minimum inhibitory concentration (MIC)
1000-9 | http://loinc.org | DBG Ab [Presence] in Serum or Plasma from Blood product unit
10000-8 | http://loinc.org | R wave duration in lead AVR
10001-6 | http://loinc.org | R wave duration in lead I
10002-4 | http://loinc.org | R wave duration in lead II
10003-2 | http://loinc.org | R wave duration in lead III
10004-0 | http://loinc.org | R wave duration in lead V1
10005-7 | http://loinc.org | R wave duration in lead V2
10006-5 | http://loinc.org | R wave duration in lead V3
10007-3 | http://loinc.org | R wave duration in lead V4
10008-1 | http://loinc.org | R wave duration in lead V5
10009-9 | http://loinc.org | R wave duration in lead V6
1001-7 | http://loinc.org | DBG Ab [Presence] in Serum or Plasma from Donor
10010-7 | http://loinc.org | R' wave amplitude in lead AVF
10011-5 | http://loinc.org | R' wave amplitude in lead AVL
10012-3 | http://loinc.org | R' wave amplitude in lead AVR
10013-1 | http://loinc.org | R' wave amplitude in lead I
10014-9 | http://loinc.org | R' wave amplitude in lead II
10015-6 | http://loinc.org | R' wave amplitude in lead III
10016-4 | http://loinc.org | R' wave amplitude in lead V1
10017-2 | http://loinc.org | R' wave amplitude in lead V2
10018-0 | http://loinc.org | R' wave amplitude in lead V3
10019-8 | http://loinc.org | R' wave amplitude in lead V4
1002-5 | http://loinc.org | DBG Ab [Presence] in Serum or Plasma
10020-6 | http://loinc.org | R' wave amplitude in lead V5
10021-4 | http://loinc.org | R' wave amplitude in lead V6
10022-2 | http://loinc.org | R' wave duration in lead AVF
10023-0 | http://loinc.org | R' wave duration in lead AVL
10024-8 | http://loinc.org | R' wave duration in lead AVR
10025-5 | http://loinc.org | R' wave duration in lead I
10026-3 | http://loinc.org | R' wave duration in lead II
10027-1 | http://loinc.org | R' wave duration in lead III
10028-9 | http://loinc.org | R' wave duration in lead V1
10029-7 | http://loinc.org | R' wave duration in lead V2
1003-3 | http://loinc.org | Indirect antiglobulin test.complement specific reagent [Presence] in Serum or Plasma
10030-5 | http://loinc.org | R' wave duration in lead V3
10031-3 | http://loinc.org | R' wave duration in lead V4
10032-1 | http://loinc.org | R' wave duration in lead V5
10033-9 | http://loinc.org | R' wave duration in lead V6
10034-7 | http://loinc.org | S wave amplitude in lead AVF
10035-4 | http://loinc.org | S wave amplitude in lead AVL
10036-2 | http://loinc.org | S wave amplitude in lead AVR
10037-0 | http://loinc.org | S wave amplitude in lead I

10038-8 | http://loinc.org | S wave amplitude in lead II
10039-6 | http://loinc.org | S wave amplitude in lead III
1004-1 | http://loinc.org | Direct antiglobulin test.complement specific reagent [Presence] on Red Blood Cells
10040-4 | http://loinc.org | S wave amplitude in lead V1
10041-2 | http://loinc.org | S wave amplitude in lead V2
10042-0 | http://loinc.org | S wave amplitude in lead V3
10043-8 | http://loinc.org | S wave amplitude in lead V4
10044-6 | http://loinc.org | S wave amplitude in lead V5
10045-3 | http://loinc.org | S wave amplitude in lead V6
10046-1 | http://loinc.org | S wave duration in lead AVF
10047-9 | http://loinc.org | S wave duration in lead AVL
10048-7 | http://loinc.org | S wave duration in lead AVR
10049-5 | http://loinc.org | S wave duration in lead I
1005-8 | http://loinc.org | Indirect antiglobulin test.IgG specific reagent [Presence] in Serum or Plasma
10050-3 | http://loinc.org | S wave duration in lead II
10051-1 | http://loinc.org | S wave duration in lead III
10052-9 | http://loinc.org | S wave duration in lead V1
10053-7 | http://loinc.org | S wave duration in lead V2
10054-5 | http://loinc.org | S wave duration in lead V3
10055-2 | http://loinc.org | S wave duration in lead V4
10056-0 | http://loinc.org | S wave duration in lead V5
10057-8 | http://loinc.org | S wave duration in lead V6
10058-6 | http://loinc.org | S' wave amplitude in lead AVF
10059-4 | http://loinc.org | S' wave amplitude in lead AVL
1006-6 | http://loinc.org | Direct antiglobulin test.IgG specific reagent [Interpretation] on Red Blood Cells
10060-2 | http://loinc.org | S' wave amplitude in lead AVR
10061-0 | http://loinc.org | S' wave amplitude in lead I
10062-8 | http://loinc.org | S' wave amplitude in lead II
10063-6 | http://loinc.org | S' wave amplitude in lead III
10064-4 | http://loinc.org | S' wave amplitude in lead V1
10065-1 | http://loinc.org | S' wave amplitude in lead V2
10066-9 | http://loinc.org | S' wave amplitude in lead V3
10067-7 | http://loinc.org | S' wave amplitude in lead V4
10068-5 | http://loinc.org | S' wave amplitude in lead V5
10069-3 | http://loinc.org | S' wave amplitude in lead V6
1007-4 | http://loinc.org | Direct antiglobulin test.poly specific reagent [Presence] on Red Blood Cells
10070-1 | http://loinc.org | S' wave duration in lead AVF
10071-9 | http://loinc.org | S' wave duration in lead AVL
10072-7 | http://loinc.org | S' wave duration in lead AVR
10073-5 | http://loinc.org | S' wave duration in lead I
10074-3 | http://loinc.org | S' wave duration in lead II
10075-0 | http://loinc.org | S' wave duration in lead III
10076-8 | http://loinc.org | S' wave duration in lead V1
10077-6 | http://loinc.org | S' wave duration in lead V2
10078-4 | http://loinc.org | S' wave duration in lead V3
10079-2 | http://loinc.org | S' wave duration in lead V4
1008-2 | http://loinc.org | Indirect antiglobulin test.poly specific reagent [Presence] in Serum or Plasma
10080-0 | http://loinc.org | S' wave duration in lead V5
10081-8 | http://loinc.org | S' wave duration in lead V6
10082-6 | http://loinc.org | ST initial amplitude 6 ms in lead AVF
10083-4 | http://loinc.org | ST initial amplitude 6 ms in lead AVL
10084-2 | http://loinc.org | ST initial amplitude 6 ms in lead AVR
10085-9 | http://loinc.org | ST initial amplitude 6 ms in lead I
10086-7 | http://loinc.org | ST initial amplitude 6 ms in lead II
10087-5 | http://loinc.org | ST initial amplitude 6 ms in lead III
10088-3 | http://loinc.org | ST initial amplitude 6 ms in lead V1
10089-1 | http://loinc.org | ST initial amplitude 6 ms in lead V2
1009-0 | http://loinc.org | Deprecated Direct antiglobulin test.poly specific reagent [Presence] on Red Blood Cells
10090-9 | http://loinc.org | ST initial amplitude 6 ms in lead V3
10091-7 | http://loinc.org | ST initial amplitude 6 ms in lead V4
10092-5 | http://loinc.org | ST initial amplitude 6 ms in lead V5
10093-3 | http://loinc.org | ST initial amplitude 6 ms in lead V6
10094-1 | http://loinc.org | ST slope in lead AVF
10095-8 | http://loinc.org | ST slope in lead AVL
10096-6 | http://loinc.org | ST slope in lead AVR
10097-4 | http://loinc.org | ST slope in lead I
10098-2 | http://loinc.org | ST slope in lead II
10099-0 | http://loinc.org | ST slope in lead III
101-6 | http://loinc.org | Cefoperazone [Susceptibility] by Disk diffusion (KB)
1010-8 | http://loinc.org | E sup(w) Ab [Presence] in Serum or Plasma from Blood product unit
10100-6 | http://loinc.org | ST slope in lead V1
10101-4 | http://loinc.org | ST slope in lead V2

10102-2 | http://loinc.org | ST slope in lead V3
10103-0 | http://loinc.org | ST slope in lead V4
10104-8 | http://loinc.org | ST slope in lead V5
10105-5 | http://loinc.org | ST slope in lead V6
10106-3 | http://loinc.org | ST wave end displacement in lead AVF
10107-1 | http://loinc.org | ST wave end displacement in lead AVL
10108-9 | http://loinc.org | ST wave end displacement in lead AVR
10109-7 | http://loinc.org | ST wave end displacement in lead I
1011-6 | http://loinc.org | E sup(w) Ab [Presence] in Serum or Plasma from Donor
10110-5 | http://loinc.org | ST wave end displacement in lead II
10111-3 | http://loinc.org | ST wave end displacement in lead III
10112-1 | http://loinc.org | ST wave end displacement in lead V1
10113-9 | http://loinc.org | ST wave end displacement in lead V2
10114-7 | http://loinc.org | ST wave end displacement in lead V3
10115-4 | http://loinc.org | ST wave end displacement in lead V4
10116-2 | http://loinc.org | ST wave end displacement in lead V5
10117-0 | http://loinc.org | ST wave end displacement in lead V6
10118-8 | http://loinc.org | ST wave mid displacement in lead AVF
10119-6 | http://loinc.org | ST wave mid displacement in lead AVL
1012-4 | http://loinc.org | E sup(w) Ab [Presence] in Serum or Plasma
10120-4 | http://loinc.org | ST wave mid displacement in lead AVR
10121-2 | http://loinc.org | ST wave mid displacement in lead I
10122-0 | http://loinc.org | ST wave mid displacement in lead II
10123-8 | http://loinc.org | ST wave mid displacement in lead III
10124-6 | http://loinc.org | ST wave mid displacement in lead V1
10125-3 | http://loinc.org | ST wave mid displacement in lead V2
10126-1 | http://loinc.org | ST wave mid displacement in lead V3
10127-9 | http://loinc.org | ST wave mid displacement in lead V4
10128-7 | http://loinc.org | ST wave mid displacement in lead V5
10129-5 | http://loinc.org | ST wave mid displacement in lead V6
1013-2 | http://loinc.org | E sup(w) Ag [Presence] on Red Blood Cells from Blood product unit
10130-3 | http://loinc.org | T' wave amplitude in lead AVF
10131-1 | http://loinc.org | T' wave amplitude in lead AVL
10132-9 | http://loinc.org | T' wave amplitude in lead AVR
10133-7 | http://loinc.org | T' wave amplitude in lead I
10134-5 | http://loinc.org | T' wave amplitude in lead II
10135-2 | http://loinc.org | T' wave amplitude in lead III
10136-0 | http://loinc.org | T' wave amplitude in lead V1
10137-8 | http://loinc.org | T' wave amplitude in lead V2
10138-6 | http://loinc.org | T' wave amplitude in lead V3
10139-4 | http://loinc.org | T' wave amplitude in lead V4
1014-0 | http://loinc.org | E sup(w) Ag [Presence] on Red Blood Cells from Donor
10140-2 | http://loinc.org | T' wave amplitude in lead V5
10141-0 | http://loinc.org | T' wave amplitude in lead V6
10142-8 | http://loinc.org | T wave amplitude in lead AVF
10143-6 | http://loinc.org | T wave amplitude in lead AVL
10144-4 | http://loinc.org | T wave amplitude in lead AVR
10145-1 | http://loinc.org | T wave amplitude in lead I
10146-9 | http://loinc.org | T wave amplitude in lead II
10147-7 | http://loinc.org | T wave amplitude in lead III
10148-5 | http://loinc.org | T wave amplitude in lead V1
10149-3 | http://loinc.org | T wave amplitude in lead V2
1015-7 | http://loinc.org | E sup(w) Ag [Presence] on Red Blood Cells
10150-1 | http://loinc.org | T wave amplitude in lead V3
10151-9 | http://loinc.org | T wave amplitude in lead V4
10152-7 | http://loinc.org | T wave amplitude in lead V5
10153-5 | http://loinc.org | T wave amplitude in lead V6
10154-3 | http://loinc.org | Chief complaint Narrative - Reported
10155-0 | http://loinc.org | History of allergies, reported
10156-8 | http://loinc.org | History of Childhood diseases Narrative
10157-6 | http://loinc.org | History of family member diseases Narrative
10158-4 | http://loinc.org | History of Functional status Narrative
10159-2 | http://loinc.org | History of Industrial exposure Narrative
1016-5 | http://loinc.org | E Ab [Presence] in Serum or Plasma from Blood product unit
10160-0 | http://loinc.org | History of Medication use Narrative
10161-8 | http://loinc.org | History of Occupational exposure Narrative
10162-6 | http://loinc.org | History of pregnancies Narrative
10163-4 | http://loinc.org | History of pregnancies
10164-2 | http://loinc.org | History of Present illness Narrative
10165-9 | http://loinc.org | Deprecated History of psychiatric symptoms and diseases Narrative
10166-7 | http://loinc.org | History of Social function Narrative
10167-5 | http://loinc.org | History of Surgical procedures Narrative

10168-3 | http://loinc.org | History of Cardiovascular system disorders Narrative
10169-1 | http://loinc.org | History of Ear disorders Narrative
1017-3 | http://loinc.org | E Ab [Presence] in Serum or Plasma from Donor
10170-9 | http://loinc.org | History of Endocrine system disorders Narrative
10171-7 | http://loinc.org | History of Eyes disorders Narrative
10172-5 | http://loinc.org | History of Hematologic system disorders Narrative
10173-3 | http://loinc.org | History of Musculoskeletal system disorders Narrative
10174-1 | http://loinc.org | History of Nose disorders Narrative
10175-8 | http://loinc.org | History of Oral cavity disorders Narrative
10176-6 | http://loinc.org | History of Reproductive system disorders Narrative
10177-4 | http://loinc.org | History of Respiratory system disorders Narrative
10178-2 | http://loinc.org | History of Skin disorders Narrative
10179-0 | http://loinc.org | History of Throat and Neck disorders Narrative
1018-1 | http://loinc.org | E Ab [Presence] in Serum or Plasma
10180-8 | http://loinc.org | History of Throat and Neck disorders
10181-6 | http://loinc.org | History of Urinary tract disorders Narrative
10182-4 | http://loinc.org | History of Travel Narrative
10183-2 | http://loinc.org | Hospital discharge medications Narrative
10184-0 | http://loinc.org | Hospital discharge physical findings Narrative
10185-7 | http://loinc.org | Hospital discharge procedures Narrative
10186-5 | http://loinc.org | Identifying information Narrative Observed
10187-3 | http://loinc.org | Review of systems Narrative - Reported
10188-1 | http://loinc.org | Review of systems overview Narrative - Reported
10189-9 | http://loinc.org | Review of systems overview - Reported
1019-9 | http://loinc.org | E Ag [Presence] on Red Blood Cells from Blood product unit
10190-7 | http://loinc.org | Mental status Narrative
10191-5 | http://loinc.org | Physical findings of Abdomen Narrative
10192-3 | http://loinc.org | Physical findings of Back Narrative
10193-1 | http://loinc.org | Physical findings of Breasts Narrative
10194-9 | http://loinc.org | Physical findings of Neurologic deep tendon reflexes Narrative
10195-6 | http://loinc.org | Physical findings of Ear Narrative
10196-4 | http://loinc.org | Physical findings of Extremities Narrative
10197-2 | http://loinc.org | Physical findings of Eye Narrative
10198-0 | http://loinc.org | Physical findings of Genitourinary tract Narrative
10199-8 | http://loinc.org | Physical findings of Head Narrative
102-4 | http://loinc.org | Cefoperazone [Susceptibility] by Serum bactericidal titer
1020-7 | http://loinc.org | E Ag [Presence] on Red Blood Cells from Donor
10200-4 | http://loinc.org | Physical findings of Heart Narrative
10201-2 | http://loinc.org | Physical findings of Mouth and Throat and Teeth Narrative
10202-0 | http://loinc.org | Physical findings of Nervous system Narrative
10203-8 | http://loinc.org | Physical findings of Nose Narrative
10204-6 | http://loinc.org | Physical findings of Pelvis Narrative
10205-3 | http://loinc.org | Physical findings of Rectum Narrative
10206-1 | http://loinc.org | Physical findings of Skin Narrative
10207-9 | http://loinc.org | Physical findings of Thorax and Lungs Narrative
10208-7 | http://loinc.org | Physical findings of Vessels Narrative
10209-5 | http://loinc.org | Physical findings of Neurologic balance and Coordination Narrative
1021-5 | http://loinc.org | E Ag [Presence] on Red Blood Cells
10210-3 | http://loinc.org | Physical findings of General status Narrative
10211-1 | http://loinc.org | Physical findings of Sense of touch Narrative
10212-9 | http://loinc.org | Physical findings of Strength Narrative
10213-7 | http://loinc.org | Surgical operation note anesthesia Narrative
10214-5 | http://loinc.org | Surgical operation note anesthesia duration
10215-2 | http://loinc.org | Surgical operation note findings Narrative
10216-0 | http://loinc.org | Surgical operation note fluids Narrative
10217-8 | http://loinc.org | Surgical operation note indications [Interpretation] Narrative
10218-6 | http://loinc.org | Surgical operation note postoperative diagnosis Narrative
10219-4 | http://loinc.org | Surgical operation note preoperative diagnosis Narrative
1022-3 | http://loinc.org | Fy sup(a) Ab [Presence] in Serum or Plasma from Blood product unit
10220-2 | http://loinc.org | Surgical operation note prep time duration
10221-0 | http://loinc.org | Surgical operation note specimens taken Narrative
10222-8 | http://loinc.org | Surgical operation note surgical complications [Interpretation] Narrative
10223-6 | http://loinc.org | Surgical operation note surgical procedure Narrative
10224-4 | http://loinc.org | Hemodynamic method special circumstances
10225-1 | http://loinc.org | Cardiac measurement device Institution inventory number
10226-9 | http://loinc.org | Oxygen content in Intravascular space
10227-7 | http://loinc.org | Cardiac measurement device Vendor model number
10228-5 | http://loinc.org | Cardiac measurement device Vendor serial number
10229-3 | http://loinc.org | Hemodynamic method Type of
1023-1 | http://loinc.org | Fy sup(a) Ab [Presence] in Serum or Plasma from Donor
10230-1 | http://loinc.org | Left ventricular Ejection fraction
10231-9 | http://loinc.org | Right ventricular Ejection fraction

10232-7 | http://loinc.org | Oxygen content in Aorta root
10233-5 | http://loinc.org | Oxygen content in Left atrium
10234-3 | http://loinc.org | Oxygen content in Right atrium
10235-0 | http://loinc.org | Oxygen content in High right atrium
10236-8 | http://loinc.org | Oxygen content in Low right atrium
10237-6 | http://loinc.org | Oxygen content in Mid right atrium
10238-4 | http://loinc.org | Oxygen content in Left ventricle
10239-2 | http://loinc.org | Oxygen content in Right ventricular outflow tract
1024-9 | http://loinc.org | Fy sup(a) Ab [Presence] in Serum or Plasma
10240-0 | http://loinc.org | Oxygen content in Right ventricle
10241-8 | http://loinc.org | Oxygen content in Coronary sinus
10242-6 | http://loinc.org | Oxygen content in Ductus arteriosus
10243-4 | http://loinc.org | Oxygen content in Inferior vena cava
10244-2 | http://loinc.org | Oxygen content in Left pulmonary artery
10245-9 | http://loinc.org | Oxygen content in Main pulmonary artery
10246-7 | http://loinc.org | Oxygen content in Right pulmonary artery
10247-5 | http://loinc.org | Oxygen content in Pulmonary wedge
10248-3 | http://loinc.org | Oxygen content in Superior vena cava
10249-1 | http://loinc.org | Heart rate device Institution inventory number
1025-6 | http://loinc.org | Fy sup(a) Ag [Presence] on Red Blood Cells from Blood product unit
10250-9 | http://loinc.org | Heart rate device Vendor model number
10251-7 | http://loinc.org | Heart rate device Vendor serial number
10252-5 | http://loinc.org | Biliary drain site
10253-3 | http://loinc.org | Type of Biliary drain
10254-1 | http://loinc.org | Type of Peritoneal drain
10255-8 | http://loinc.org | Type of Subarachnoid drain
10256-6 | http://loinc.org | Biliary drain Institution inventory number
10257-4 | http://loinc.org | Chest tube Institution inventory number
10258-2 | http://loinc.org | Enteral tube Institution inventory number
10259-0 | http://loinc.org | Gastrostomy tube Institution inventory number
1026-4 | http://loinc.org | Fy sup(a) Ag [Presence] on Red Blood Cells from Donor
10260-8 | http://loinc.org | GI tube Institution inventory number
10261-6 | http://loinc.org | Intravascular tube Institution inventory number
10262-4 | http://loinc.org | IO tube Institution inventory number
10263-2 | http://loinc.org | Nasogastric tube Institution inventory number
10264-0 | http://loinc.org | Oral tube Institution inventory number
10265-7 | http://loinc.org | Peritoneal drain Institution inventory number
10266-5 | http://loinc.org | Peritoneal tube Institution inventory number
10267-3 | http://loinc.org | Rate control device Institution inventory number
10268-1 | http://loinc.org | Stool collection device Institution inventory number
10269-9 | http://loinc.org | Subarachnoid drain Institution inventory number
1027-2 | http://loinc.org | Fy sup(a) Ag [Presence] on Red Blood Cells
10270-7 | http://loinc.org | Synovial drain Institution inventory number
10271-5 | http://loinc.org | Tube or drain Institution inventory number
10272-3 | http://loinc.org | Upper GI tube Institution inventory number
10273-1 | http://loinc.org | Bladder irrigation tube Institution inventory number
10274-9 | http://loinc.org | Urine tube Institution inventory number
10275-6 | http://loinc.org | Wound drain device Institution inventory number
10276-4 | http://loinc.org | Peritoneal drain site
10277-2 | http://loinc.org | Stool collection site
10278-0 | http://loinc.org | Subarachnoid drain site
10279-8 | http://loinc.org | Biliary drain Vendor model number
1028-0 | http://loinc.org | Fy sup(b) Ab [Presence] in Serum or Plasma from Blood product unit
10280-6 | http://loinc.org | Chest tube Vendor model number
10281-4 | http://loinc.org | Enteral tube Vendor model number
10282-2 | http://loinc.org | Gastrostomy tube Vendor model number
10283-0 | http://loinc.org | GI tube Vendor model number
10284-8 | http://loinc.org | Intravascular tube Vendor model number
10285-5 | http://loinc.org | IO tube Vendor model number
10286-3 | http://loinc.org | Nasogastric tube Vendor model number
10287-1 | http://loinc.org | Oral tube Vendor model number
10288-9 | http://loinc.org | Peritoneal drain Vendor model number
10289-7 | http://loinc.org | Peritoneal tube Vendor model number
1029-8 | http://loinc.org | Fy sup(b) Ab [Presence] in Serum or Plasma from Donor
10290-5 | http://loinc.org | Rate control device Vendor model number
10291-3 | http://loinc.org | Stool collection device Vendor model number
10292-1 | http://loinc.org | Subarachnoid drain Vendor model number
10293-9 | http://loinc.org | Synovial drain Vendor model number
10294-7 | http://loinc.org | Tube or drain Vendor model number
10295-4 | http://loinc.org | Upper GI tube Vendor model number
10296-2 | http://loinc.org | Bladder irrigation tube Vendor model number
10297-0 | http://loinc.org | Urine tube Vendor model number

10298-8 | http://loinc.org | Wound drain device Vendor model number
10299-6 | http://loinc.org | Biliary drain Vendor serial number
103-2 | http://loinc.org | Ceforanide [Susceptibility] by Minimum lethal concentration (MLC)
1030-6 | http://loinc.org | Fy sup(b) Ab [Presence] in Serum or Plasma
10300-2 | http://loinc.org | Chest tube Vendor serial number
10301-0 | http://loinc.org | Enteral tube Vendor serial number
10302-8 | http://loinc.org | Gastrostomy tube Vendor serial number
10303-6 | http://loinc.org | GI tube Vendor serial number
10304-4 | http://loinc.org | Intravascular tube Vendor serial number
10305-1 | http://loinc.org | IO tube Vendor serial number
10306-9 | http://loinc.org | Nasogastric tube Vendor serial number
10307-7 | http://loinc.org | Oral tube Vendor serial number
10308-5 | http://loinc.org | Peritoneal drain Vendor serial number
10309-3 | http://loinc.org | Peritoneal tube Vendor serial number
1031-4 | http://loinc.org | Fy sup(b) Ag [Presence] on Red Blood Cells from Blood product unit
10310-1 | http://loinc.org | Rate control device Vendor serial number
10311-9 | http://loinc.org | Stool collection device Vendor serial number
10312-7 | http://loinc.org | Subarachnoid drain Vendor serial number
10313-5 | http://loinc.org | Synovial drain Vendor serial number
10314-3 | http://loinc.org | Tube or drain Vendor serial number
10315-0 | http://loinc.org | Upper GI tube Vendor serial number
10316-8 | http://loinc.org | Bladder irrigation tube Vendor serial number
10317-6 | http://loinc.org | Urine tube Vendor serial number
10318-4 | http://loinc.org | Wound drain device Vendor serial number
10319-2 | http://loinc.org | Enema device Institution inventory number
1032-2 | http://loinc.org | Fy sup(b) Ag [Presence] on Red Blood Cells from Donor
10320-0 | http://loinc.org | Enema device Vendor model number
10321-8 | http://loinc.org | Enema device Vendor serial number
10322-6 | http://loinc.org | Potassium intake 24 hour
10323-4 | http://loinc.org | Wound drain fluid [Appearance] Lower GI tract
10324-2 | http://loinc.org | Breath rate device Institution inventory number
10325-9 | http://loinc.org | Breath rate device Vendor model number
10326-7 | http://loinc.org | Breath rate device Vendor serial number
10327-5 | http://loinc.org | Eosinophils/100 leukocytes in Sputum by Manual count
10328-3 | http://loinc.org | Lymphocytes/100 leukocytes in Cerebral spinal fluid by Manual count
10329-1 | http://loinc.org | Monocytes/100 leukocytes in Cerebral spinal fluid by Manual count
1033-0 | http://loinc.org | Fy sup(b) Ag [Presence] on Red Blood Cells
10330-9 | http://loinc.org | Monocytes/100 leukocytes in Body fluid by Manual count
10331-7 | http://loinc.org | Rh [Type] in Blood
10332-5 | http://loinc.org | Cortisol [Mass/volume] in Serum or Plasma --pre 250 ug corticotropin IM
10333-3 | http://loinc.org | Appearance of Cerebral spinal fluid
10334-1 | http://loinc.org | Cancer Ag 125 [Units/volume] in Serum or Plasma
10335-8 | http://loinc.org | Color of Cerebral spinal fluid
10336-6 | http://loinc.org | Gonadotropin peptide [Mass/volume] in Urine
10337-4 | http://loinc.org | Procollagen type I [Mass/volume] in Serum
10338-2 | http://loinc.org | Barbiturates [Mass/volume] in Serum or Plasma
10339-0 | http://loinc.org | Fluoxetine+Norfluoxetine [Mass/volume] in Serum or Plasma
1034-8 | http://loinc.org | Fetal cell screen [Interpretation] in Blood
10340-8 | http://loinc.org | Molindone [Mass/volume] in Serum or Plasma
10341-6 | http://loinc.org | Norpropoxyphene [Mass/volume] in Serum or Plasma
10342-4 | http://loinc.org | Sulfamethoxazole [Mass/volume] in Serum or Plasma
10343-2 | http://loinc.org | Temazepam [Mass/volume] in Serum or Plasma
10344-0 | http://loinc.org | Tranylcypromine [Mass/volume] in Serum or Plasma
10345-7 | http://loinc.org | Trihexyphenidyl [Mass/volume] in Urine
10346-5 | http://loinc.org | Hemoglobin A [Units/volume] in Blood by Electrophoresis
10347-3 | http://loinc.org | Babesia microti identified in Blood by Light microscopy
10348-1 | http://loinc.org | Bordetella parapertussis Ab [Presence] in Serum
10349-9 | http://loinc.org | Brucella sp Ab [Units/volume] in Serum
1035-5 | http://loinc.org | Fresh frozen plasma given [Volume]
10350-7 | http://loinc.org | Herpes simplex virus IgM Ab [Titer] in Serum by Immunoassay
10351-5 | http://loinc.org | HIV 1 RNA [Units/volume] (viral load) in Serum or Plasma by Probe with amplification
10352-3 | http://loinc.org | Bacteria identified in Genital specimen by Aerobe culture
10353-1 | http://loinc.org | Bacteria identified in Nose by Aerobe culture
10354-9 | http://loinc.org | Bacteria identified in Urethra by Culture
10355-6 | http://loinc.org | Microscopic observation [Identifier] in Bone marrow by Wright Giemsa stain
10356-4 | http://loinc.org | Deprecated Microscopic observation [Identifier] in Stool by Trichrome stain
10357-2 | http://loinc.org | Microscopic observation [Identifier] in Wound by Gram stain
10358-0 | http://loinc.org | Teichoate Ab [Titer] in Serum by Immune diffusion (ID)
10359-8 | http://loinc.org | Asialoganglioside GM1 IgM Ab [Titer] in Serum
1036-3 | http://loinc.org | G Ab [Presence] in Serum or Plasma from Blood product unit
10360-6 | http://loinc.org | DNA single strand IgG Ab [Units/volume] in Serum
10361-4 | http://loinc.org | DNA single strand IgM Ab [Units/volume] in Serum

10362-2 | http://loinc.org | Endomysium IgA Ab [Presence] in Serum
10363-0 | http://loinc.org | Barbiturates [Presence] in Unspecified specimen
10364-8 | http://loinc.org | Cotinine [Mass/mass] in Hair
10365-5 | http://loinc.org | Cotinine [Mass/volume] in Serum or Plasma
10366-3 | http://loinc.org | Cotinine [Mass/volume] in Urine
10367-1 | http://loinc.org | Ethanol [Mass/volume] in Gastric fluid
10368-9 | http://loinc.org | Lead [Mass/volume] in Capillary blood
10369-7 | http://loinc.org | Opiates [Mass/mass] in Hair
1037-1 | http://loinc.org | G Ab [Presence] in Serum or Plasma from Donor
10370-5 | http://loinc.org | Phencyclidine [Mass/mass] in Hair
10371-3 | http://loinc.org | Bite cells [Presence] in Blood by Light microscopy
10372-1 | http://loinc.org | Blister cells [Presence] in Blood by Light microscopy
10373-9 | http://loinc.org | Fragments [Presence] in Blood by Light microscopy
10374-7 | http://loinc.org | Helmet cells [Presence] in Blood by Light microscopy
10375-4 | http://loinc.org | Irregularly contracted cells [Presence] in Blood by Light microscopy
10376-2 | http://loinc.org | Oval macrocytes [Presence] in Blood by Light microscopy
10377-0 | http://loinc.org | Pencil cells [Presence] in Blood by Light microscopy
10378-8 | http://loinc.org | Polychromasia [Presence] in Blood by Light microscopy
10379-6 | http://loinc.org | Erythrocytes.dual population [Presence] in Blood by Light microscopy
1038-9 | http://loinc.org | G Ab [Presence] in Serum or Plasma
10380-4 | http://loinc.org | Stomatocytes [Presence] in Blood by Light microscopy
10381-2 | http://loinc.org | Target cells [Presence] in Blood by Light microscopy
10382-0 | http://loinc.org | A,B variant NOS Ag [Presence] on Red Blood Cells from Blood product unit
10383-8 | http://loinc.org | A,B variant NOS Ag [Presence] on Red Blood Cells from Donor
10384-6 | http://loinc.org | A,B variant NOS Ag [Presence] on Red Blood Cells
10385-3 | http://loinc.org | Albumin concentration given
10386-1 | http://loinc.org | Albumin given [Volume]
10387-9 | http://loinc.org | Autologous erythrocytes given [Volume]
10388-7 | http://loinc.org | Autologous whole blood given [Volume]
10389-5 | http://loinc.org | Blood product.other [Type]
1039-7 | http://loinc.org | G Ag [Presence] on Red Blood Cells from Blood product unit
10390-3 | http://loinc.org | Blood product special preparation [Type]
10391-1 | http://loinc.org | Cytomegalovirus immune globulin given [Volume]
10392-9 | http://loinc.org | Cryoprecipitate given [Volume]
10393-7 | http://loinc.org | Factor IX given [Type]
10394-5 | http://loinc.org | Factor IX given [Volume]
10395-2 | http://loinc.org | Factor VIII given [Type]
10396-0 | http://loinc.org | Factor VIII given [Volume]
10397-8 | http://loinc.org | Hepatitis B immune globulin given [Volume]
10398-6 | http://loinc.org | I Ag [Presence] on Red Blood Cells from Blood product unit
10399-4 | http://loinc.org | I Ag [Presence] on Red Blood Cells from Donor
104-0 | http://loinc.org | Ceforanide [Susceptibility] by Minimum inhibitory concentration (MIC)
1040-5 | http://loinc.org | G Ag [Presence] on Red Blood Cells from Donor
10400-0 | http://loinc.org | I Ag [Presence] on Red Blood Cells
10401-8 | http://loinc.org | Immune serum globulin given [Type]
10402-6 | http://loinc.org | Immune serum globulin given [Volume]
10403-4 | http://loinc.org | Inject immune serum globulin [Volume]
10404-2 | http://loinc.org | Inject Rh immune globulin [Volume]
10405-9 | http://loinc.org | Inject varicella zoster virus immune globulin [Volume]
10406-7 | http://loinc.org | little i Ag [Presence] on Red Blood Cells from Blood product unit
10407-5 | http://loinc.org | little i Ag [Presence] on Red Blood Cells from Donor
10408-3 | http://loinc.org | little i Ag [Presence] on Red Blood Cells
10409-1 | http://loinc.org | Pentaspan given [Volume]
1041-3 | http://loinc.org | G Ag [Presence] on Red Blood Cells
10410-9 | http://loinc.org | Plasma given [Type]
10411-7 | http://loinc.org | Plasma given [Volume]
10412-5 | http://loinc.org | Platelets given [Type]
10413-3 | http://loinc.org | Rh immune globulin given [Volume]
10414-1 | http://loinc.org | Transfuse albumin [Volume]
10415-8 | http://loinc.org | Transfuse blood exchange transfusion [Volume]
10416-6 | http://loinc.org | Transfuse blood product other [Volume]
10417-4 | http://loinc.org | Transfuse cryoprecipitate [Volume]
10418-2 | http://loinc.org | Transfuse factor IX [Volume]
10419-0 | http://loinc.org | Transfuse factor VIII [Volume]
1042-1 | http://loinc.org | H Ab [Presence] in Serum or Plasma from Blood product unit
10420-8 | http://loinc.org | Transfuse immune serum globulin [Volume]
10421-6 | http://loinc.org | Transfuse Pentaspan [Volume]
10422-4 | http://loinc.org | Transfuse plasma [Volume]
10423-2 | http://loinc.org | Transfuse platelets [Volume]
10424-0 | http://loinc.org | Transfuse erythrocytes [Volume]
10425-7 | http://loinc.org | Transfuse Rh immune globulin [Volume]
10426-5 | http://loinc.org | Transfuse whole blood [Volume]

10427-3 | http://loinc.org | Transfuse whole blood autologous [Volume]
10428-1 | http://loinc.org | Varicella zoster virus immune globulin given [Volume]
10429-9 | http://loinc.org | AE 1 Ag [Presence] in Tissue by Immune stain
1043-9 | http://loinc.org | H Ab [Presence] in Serum or Plasma from Donor
10430-7 | http://loinc.org | AE 3 Ag [Presence] in Tissue by Immune stain
10431-5 | http://loinc.org | B-cell Ag [Presence] in Tissue by Immune stain
10432-3 | http://loinc.org | CD30 Ag [Presence] in Tissue by Immune stain
10433-1 | http://loinc.org | BR-2 Ag [Presence] in Tissue by Immune stain
10434-9 | http://loinc.org | Complement C3 Ag [Presence] in Tissue by Immune stain
10435-6 | http://loinc.org | C5B-9 Ag [Presence] in Tissue by Immune stain
10436-4 | http://loinc.org | CD15 Ag [Presence] in Tissue by Immune stain
10437-2 | http://loinc.org | CD16 Ag [Presence] in Tissue by Immune stain
10438-0 | http://loinc.org | CD20 Ag [Presence] in Tissue by Immune stain
10439-8 | http://loinc.org | CD3 Ag [Presence] in Tissue by Immune stain
1044-7 | http://loinc.org | H Ab [Presence] in Serum
10440-6 | http://loinc.org | Deprecated CD30 Ag [Presence] in Tissue by Immune stain
10441-4 | http://loinc.org | CD34 Ag [Presence] in Tissue by Immune stain
10442-2 | http://loinc.org | CD56 Ag [Presence] in Tissue by Immune stain
10443-0 | http://loinc.org | CD43 T-cell monocyte+myeloid cell Ag [Presence] in Tissue by Immune stain
10444-8 | http://loinc.org | CD57 Ag [Presence] in Tissue by Immune stain
10445-5 | http://loinc.org | CD11c Ag [Presence] in Tissue by Immune stain
10446-3 | http://loinc.org | Leukocyte common Ag [Presence] in Tissue by Immune stain
10447-1 | http://loinc.org | M-5 Ag [Presence] in Tissue by Immune stain
10448-9 | http://loinc.org | T-cell Ag [Presence] in Tissue by Immune stain
10449-7 | http://loinc.org | Glucose [Mass/volume] in Serum or Plasma --1 hour post meal
1045-4 | http://loinc.org | H NOS Ab [Presence] in Serum or Plasma from Blood product unit
10450-5 | http://loinc.org | Glucose [Mass/volume] in Serum or Plasma --10 hours fasting
10451-3 | http://loinc.org | Proinsulin [Moles/volume] in Serum or Plasma --12 hours fasting
10452-1 | http://loinc.org | Xylose [Mass/volume] in Serum or Plasma --1 hour post 25 g xylose PO
10453-9 | http://loinc.org | Xylose [Mass/volume] in Serum or Plasma --1 hour post dose xylose PO
10454-7 | http://loinc.org | Xylose [Mass/volume] in Serum or Plasma --2 hours post 25 g xylose PO
10455-4 | http://loinc.org | Xylose [Mass/volume] in Serum or Plasma --30 minutes post 25 g xylose PO
10456-2 | http://loinc.org | Xylose [Mass/volume] in Serum or Plasma --6 hours fasting
10457-0 | http://loinc.org | Actin Ag [Presence] in Tissue by Immune stain
10458-8 | http://loinc.org | Alkaline phosphatase.placental Ag [Presence] in Tissue by Immune stain
10459-6 | http://loinc.org | Alpha-1-Fetoprotein Ag [Presence] in Tissue by Immune stain
1046-2 | http://loinc.org | H NOS Ab [Presence] in Serum or Plasma from Donor
10460-4 | http://loinc.org | Lactalbumin alpha Ag [Presence] in Tissue by Immune stain
10461-2 | http://loinc.org | Alpha-1-Antichymotrypsin Ag [Presence] in Tissue by Immune stain
10462-0 | http://loinc.org | Alpha 1 antitrypsin Ag [Presence] in Tissue by Immune stain
10463-8 | http://loinc.org | Amyloid A component Ag [Presence] in Tissue by Immune stain
10464-6 | http://loinc.org | Amyloid P component Ag [Presence] in Tissue by Immune stain
10465-3 | http://loinc.org | Amyloid.prealbumin Ag [Presence] in Tissue by Immune stain
10466-1 | http://loinc.org | Anion gap 3 in Serum or Plasma
10467-9 | http://loinc.org | Beta-2-Microglobulin amyloid Ag [Presence] in Tissue by Immune stain
10468-7 | http://loinc.org | Calcitonin Ag [Presence] in Tissue by Immune stain
10469-5 | http://loinc.org | Carcinoembryonic Ag [Presence] in Tissue by Immune stain
1047-0 | http://loinc.org | H NOS Ab [Presence] in Serum or Plasma
10470-3 | http://loinc.org | Choriogonadotropin Ag [Presence] in Tissue by Immune stain
10471-1 | http://loinc.org | Chromogranin A Ag [Presence] in Tissue by Immune stain
10472-9 | http://loinc.org | Chromogranin Ag [Presence] in Tissue by Immune stain
10473-7 | http://loinc.org | Chymotrypsin Ag [Presence] in Tissue by Immune stain
10474-5 | http://loinc.org | Collagen type 4 Ag [Presence] in Tissue by Immune stain
10475-2 | http://loinc.org | Corticotropin Ag [Presence] in Tissue by Immune stain
10476-0 | http://loinc.org | Desmin Ag [Presence] in Tissue by Immune stain
10477-8 | http://loinc.org | Enolase.neuron specific Ag [Presence] in Tissue by Immune stain
10478-6 | http://loinc.org | Eosinophil major basic protein Ag [Presence] in Tissue by Immune stain
10479-4 | http://loinc.org | CD227 Ag [Presence] in Tissue by Immune stain
1048-8 | http://loinc.org | H NOS Ag [Presence] on Red Blood Cells from Blood product unit
10480-2 | http://loinc.org | Estrogen+Progesterone receptor Ag [Presence] in Tissue by Immune stain
10481-0 | http://loinc.org | Follitropin.alpha subunit Ag [Presence] in Tissue by Immune stain
10482-8 | http://loinc.org | Follitropin.beta subunit Ag [Presence] in Tissue by Immune stain
10483-6 | http://loinc.org | Gastrin Ag [Presence] in Tissue by Immune stain
10484-4 | http://loinc.org | Glial fibrillary acidic protein Ag [Presence] in Tissue by Immune stain
10485-1 | http://loinc.org | Glucagon Ag [Presence] in Tissue by Immune stain
10486-9 | http://loinc.org | Hemoglobin Ag [Presence] in Tissue by Immune stain
10487-7 | http://loinc.org | HMB-45 Ag [Presence] in Tissue by Immune stain
10488-5 | http://loinc.org | IgA Ag [Presence] in Tissue by Immune stain
10489-3 | http://loinc.org | IgA.heavy chain Ag [Presence] in Tissue by Immune stain
1049-6 | http://loinc.org | H NOS Ag [Presence] on Red Blood Cells from Donor
10490-1 | http://loinc.org | IgE Ag [Presence] in Tissue by Immune stain
10491-9 | http://loinc.org | IgG Ag [Presence] in Tissue by Immune stain

10492-7 | http://loinc.org | IgG.heavy chain Ag [Presence] in Tissue by Immune stain
10493-5 | http://loinc.org | IgM Ag [Presence] in Tissue by Immune stain
10494-3 | http://loinc.org | IgM.heavy chain Ag [Presence] in Tissue by Immune stain
10495-0 | http://loinc.org | Insulin Ag [Presence] in Tissue by Immune stain
10496-8 | http://loinc.org | Kappa light chains Ag [Presence] in Tissue by Immune stain
10497-6 | http://loinc.org | Immunoglobulin light chains.kappa amyloid Ag [Presence] in Tissue by Immune stain
10498-4 | http://loinc.org | Keratin Ag [Presence] in Tissue by Immune stain
10499-2 | http://loinc.org | Lambda light chains Ag [Presence] in Tissue by Immune stain
105-7 | http://loinc.org | Ceforanide [Susceptibility] by Disk diffusion (KB)
1050-4 | http://loinc.org | H NOS Ag [Presence] on Red Blood Cells
10500-7 | http://loinc.org | Immunoglobulin light chains.lambda amyloid Ag [Presence] in Tissue by Immune stain
10501-5 | http://loinc.org | Lutropin [Units/volume] in Serum or Plasma
10502-3 | http://loinc.org | Lutropin Ag [Presence] in Tissue by Immune stain
10503-1 | http://loinc.org | Lysozyme Ag [Presence] in Tissue by Immune stain
10504-9 | http://loinc.org | Myelin basic protein Ag [Presence] in Tissue by Immune stain
10505-6 | http://loinc.org | Myoglobin Ag [Presence] in Tissue by Immune stain
10506-4 | http://loinc.org | Peanut agglutinin Ag [Presence] in Tissue by Immune stain
10507-2 | http://loinc.org | Prolactin Ag [Presence] in Tissue by Immune stain
10508-0 | http://loinc.org | Prostate specific Ag [Presence] in Tissue by Immune stain
10509-8 | http://loinc.org | Prostatic acid phosphatase Ag [Presence] in Tissue by Immune stain
1051-2 | http://loinc.org | Hemolytic disease of newborn screen [Interpretation] in Blood
10510-6 | http://loinc.org | S-100 Ag Ag [Presence] in Tissue by Immune stain
10511-4 | http://loinc.org | Serotonin Ag [Presence] in Tissue by Immune stain
10512-2 | http://loinc.org | Somatostatin Ag [Presence] in Tissue by Immune stain
10513-0 | http://loinc.org | Somatotropin Ag [Presence] in Tissue by Immune stain
10514-8 | http://loinc.org | Synaptophysin Ag [Presence] in Tissue by Immune stain
10515-5 | http://loinc.org | Thyroglobulin Ag [Presence] in Tissue by Immune stain
10516-3 | http://loinc.org | Thyrotropin Ag [Presence] in Tissue by Immune stain
10517-1 | http://loinc.org | Trypsin Ag [Presence] in Tissue by Immune stain
10518-9 | http://loinc.org | Ulex europaeus I lectin Ag [Presence] in Tissue by Immune stain
10519-7 | http://loinc.org | Vimentin Ag [Presence] in Tissue by Immune stain
1052-0 | http://loinc.org | Deprecated I (intermediate) subtype [Presence] in Serum or Plasma from Blood product unit
10520-5 | http://loinc.org | Coagulation factor VI Ag [Presence] in Tissue by Immune stain
10521-3 | http://loinc.org | Coagulation factor VIII Ag [Presence] in Tissue by Immune stain
10522-1 | http://loinc.org | Coagulation factor XIII Ag [Presence] in Tissue by Immune stain
10523-9 | http://loinc.org | Fibrinogen Ag [Presence] in Tissue by Immune stain
10524-7 | http://loinc.org | Microscopic observation [Identifier] in Cervix by Cyto stain
10525-4 | http://loinc.org | Microscopic observation [Identifier] in Unspecified specimen by Cyto stain
10526-2 | http://loinc.org | Microscopic observation [Identifier] in Sputum by Cyto stain
10527-0 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Cyto stain
10528-8 | http://loinc.org | Acetophenazine [Mass/volume] in Serum or Plasma
10529-6 | http://loinc.org | Amoxapine metabolite [Mass/volume] in Serum or Plasma
1053-8 | http://loinc.org | Deprecated I (intermediate) subtype [Presence] in serum or plasma from donor
10530-4 | http://loinc.org | Aprobarbital [Mass/volume] in Serum or Plasma
10531-2 | http://loinc.org | Bretylium [Mass/volume] in Serum or Plasma
10532-0 | http://loinc.org | Deprecated Desethylamiodarone [Mass/volume] in Serum or Plasma
10533-8 | http://loinc.org | Propoxyphene+Acetaminophen [Mass/volume] in Serum or Plasma
10534-6 | http://loinc.org | Diazoxide [Mass/volume] in Serum or Plasma
10535-3 | http://loinc.org | Digoxin [Mass/volume] in Serum or Plasma
10536-1 | http://loinc.org | Dipyridamole [Mass/volume] in Serum or Plasma
10537-9 | http://loinc.org | Deprecated Doxepin+Nordoxepin [Mass/volume] in Serum or Plasma
10538-7 | http://loinc.org | Deprecated Fluoxetine+Norfluoxetine [Mass/volume] in Serum or Plasma
10539-5 | http://loinc.org | Glipizide [Mass/volume] in Serum or Plasma
1054-6 | http://loinc.org | Deprecated I (intermediate) subtype [Presence] in srum
10540-3 | http://loinc.org | Glyburide [Mass/volume] in Serum or Plasma
10541-1 | http://loinc.org | Mepivacaine [Mass/volume] in Serum or Plasma
10542-9 | http://loinc.org | Metharbital [Mass/volume] in Serum or Plasma
10543-7 | http://loinc.org | Methsuximide+Normethsuximide [Mass/volume] in Serum or Plasma
10544-5 | http://loinc.org | Normeperidine [Presence] in Serum or Plasma
10545-2 | http://loinc.org | Normephenytoin [Mass/volume] in Serum or Plasma
10546-0 | http://loinc.org | Normethsuximide [Mass/volume] in Serum or Plasma
10547-8 | http://loinc.org | Primidone+Phenobarbital [Mass/volume] in Serum or Plasma
10548-6 | http://loinc.org | Phenytoin Free/Phenytoin.total in Serum or Plasma
10549-4 | http://loinc.org | Pirmenol [Mass/volume] in Serum or Plasma
1055-3 | http://loinc.org | Deprecated I (intermediate) subtype [Presence] on Red Blood Cells from Blood product unit
10550-2 | http://loinc.org | Deprecated Temazepam [Mass/volume] in Serum or Plasma
10551-0 | http://loinc.org | Triamterene [Mass/volume] in Serum or Plasma
10552-8 | http://loinc.org | Tricyclic antidepressants [Mass/volume] in Serum or Plasma
10553-6 | http://loinc.org | Prostatic acid phosphatase [Enzymatic activity/volume] in Genital fluid
10554-4 | http://loinc.org | Deprecated Prostatic acid phosphatase [Enzymatic activity/volume] in Seminal plasma
10555-1 | http://loinc.org | Acrosin [Entitic Catalytic Activity] in Spermatozoa
10556-9 | http://loinc.org | Deprecated Adenosine triphosphatase [Enzymatic activity/volume] in Seminal plasma

10557-7 | http://loinc.org | Adenosine triphosphate [Moles/volume] in Semen
10558-5 | http://loinc.org | Albumin [Moles/volume] in Semen
10559-3 | http://loinc.org | Deprecated Calcium [Molecules/volume] in Semen
1056-1 | http://loinc.org | Deprecated I (intermediate) subtype [Presence] on Red Blood Cells from donor
10560-1 | http://loinc.org | Carcinoembryonic Ag [Units/volume] in Semen
10561-9 | http://loinc.org | Carnitine [Moles/volume] in Semen
10562-7 | http://loinc.org | Cells [#/volume] in Cervical mucus
10563-5 | http://loinc.org | Cells other than spermatozoa [#/volume] in Semen
10564-3 | http://loinc.org | Cervical mucus [Volume]
10565-0 | http://loinc.org | Choriogonadotropin [Units/volume] in Semen
10566-8 | http://loinc.org | Deprecated Choriogonadotropin [Molecules/volume] in Semen
10567-6 | http://loinc.org | Citrate [Moles/volume] in Semen
10568-4 | http://loinc.org | Clarity of Semen
10569-2 | http://loinc.org | Color of Semen
1057-9 | http://loinc.org | Deprecated I (intermediate) subtype [Presence] on Red Blood Cells
10570-0 | http://loinc.org | Consistency of Cervical mucus
10571-8 | http://loinc.org | Deprecated Consistency of Semen
10572-6 | http://loinc.org | Deprecated Duration^post ejaculation
10573-4 | http://loinc.org | Fern[ing [Type] in Cervical mucus
10574-2 | http://loinc.org | Fructose [Moles/volume] in Semen
10575-9 | http://loinc.org | Deprecated Gamma glutamyl transferase [Enzymatic activity/volume] in Semen
10576-7 | http://loinc.org | Germ cells.immature [#/volume] in Semen
10577-5 | http://loinc.org | Glucosidase [Enzymatic activity/volume] in Seminal plasma
10578-3 | http://loinc.org | Glycerophosphocholine [Moles/volume] in Seminal plasma
10579-1 | http://loinc.org | Leukocytes [#/volume] in Semen
1058-7 | http://loinc.org | I Ab [Presence] in Serum or Plasma from Blood product unit
10580-9 | http://loinc.org | Liquefaction [Time] in Semen
10581-7 | http://loinc.org | Deprecated Number of entities [#] in Spermatozoa
10582-5 | http://loinc.org | pH of Cervical mucus
10583-3 | http://loinc.org | Prostaglandin F1 alpha [Moles/volume] in Semen
10584-1 | http://loinc.org | Deprecated Protein [Mass/volume] in Semen
10585-8 | http://loinc.org | Round cells [#/volume] in Semen
10586-6 | http://loinc.org | Deprecated Volume of Semen
10587-4 | http://loinc.org | Sexual abstinence duration
10588-2 | http://loinc.org | Spermatogonia [#/volume] in Semen
10589-0 | http://loinc.org | Spermatids [#/volume] in Semen by Streptoavidin-biotin method (SAB)
1059-5 | http://loinc.org | I Ab [Presence] in Serum or Plasma from Donor
10590-8 | http://loinc.org | Spermatids [#/volume] in Semen by Sertoli cell barrier method (SCB)
10591-6 | http://loinc.org | Primary Spermatocytes [#/volume] in Semen
10592-4 | http://loinc.org | Secondary Spermatocytes [#/volume] in Semen
10593-2 | http://loinc.org | Spermatozoa Pyriform Head/100 spermatozoa in Semen
10594-0 | http://loinc.org | Spermatozoa Tapering Head/100 spermatozoa in Semen
10595-7 | http://loinc.org | Deprecated Spermatozoa [#/volume] in Semen
10596-5 | http://loinc.org | Deprecated Spermatozoa Ab in cervical mucosa
10597-3 | http://loinc.org | Deprecated Spermatozoa Ab in semen
10598-1 | http://loinc.org | Deprecated Spermatozoa Ab in serum
10599-9 | http://loinc.org | Spermatozoa penetration [Presence] in Seminal fluid and Cervical mucosa by Kremer test
106-5 | http://loinc.org | Ceforanide [Susceptibility] by Serum bactericidal titer
1060-3 | http://loinc.org | I Ab [Presence] in Serum or Plasma
10600-5 | http://loinc.org | Spermatozoa penetration [Presence] in Seminal fluid and Cervical mucosa
10601-3 | http://loinc.org | Spermatozoa penetration [Presence] in Seminal fluid and Cervical mucosa --post coitus
10602-1 | http://loinc.org | Spermatozoa Abnormal Head/100 spermatozoa in Semen
10603-9 | http://loinc.org | Spermatozoa Abnormal Midpiece/100 spermatozoa in Semen
10604-7 | http://loinc.org | Spermatozoa Abnormal Tail/100 spermatozoa in Semen
10605-4 | http://loinc.org | Spermatozoa Agglutinated/100 spermatozoa in Semen
10606-2 | http://loinc.org | Spermatozoa Amorphous Head/100 spermatozoa in Semen
10607-0 | http://loinc.org | Spermatozoa Coiled Tail/100 spermatozoa in Semen
10608-8 | http://loinc.org | Spermatozoa Cytoplasmic Droplet/100 spermatozoa in Semen
10609-6 | http://loinc.org | Spermatozoa Duplicate Head/100 spermatozoa in Semen
1061-1 | http://loinc.org | I NOS Ab [Presence] in Serum or Plasma from Blood product unit
10610-4 | http://loinc.org | Spermatozoa Duplicate Tail/100 spermatozoa in Semen
10611-2 | http://loinc.org | Spermatozoa Immotile/100 spermatozoa in Semen
10612-0 | http://loinc.org | Spermatozoa Large Oval Head/100 spermatozoa in Semen
10613-8 | http://loinc.org | Spermatozoa Viable/100 spermatozoa in Semen
10614-6 | http://loinc.org | Spermatozoa Motile w IgA/100 spermatozoa in Semen by Immunobead
10615-3 | http://loinc.org | Spermatozoa Motile w IgA/100 spermatozoa in Semen by Mixed antiglobulin reaction
10616-1 | http://loinc.org | Spermatozoa Motile w IgG/100 spermatozoa in Semen by Immunobead
10617-9 | http://loinc.org | Spermatozoa Motile w IgG/100 spermatozoa in Semen by Mixed antiglobulin reaction
10618-7 | http://loinc.org | Spermatozoa Motile w IgM/100 spermatozoa in Semen by Immunobead
10619-5 | http://loinc.org | Spermatozoa Motile w IgM/100 spermatozoa in Semen by Mixed antiglobulin reaction
1062-9 | http://loinc.org | I NOS Ab [Presence] in Serum or Plasma from Donor
10620-3 | http://loinc.org | Spermatozoa Nonprogressive/100 spermatozoa in Semen

10621-1 | http://loinc.org | Spermatozoa Normal Head/100 spermatozoa in Semen
10622-9 | http://loinc.org | Spermatozoa Normal/100 spermatozoa in Semen
10623-7 | http://loinc.org | Spermatozoa Pin Head/100 spermatozoa in Semen
10624-5 | http://loinc.org | Spermatozoa Rapid/100 spermatozoa in Semen
10625-2 | http://loinc.org | Spermatozoa Round Head/100 spermatozoa in Semen
10626-0 | http://loinc.org | Spermatozoa Slow/100 spermatozoa in Semen
10627-8 | http://loinc.org | Spermatozoa Small Oval Head/100 spermatozoa in Semen
10628-6 | http://loinc.org | Spermatozoa Tail Swelling/100 spermatozoa in Semen
10629-4 | http://loinc.org | Spermatozoa Vacuolated Head/100 spermatozoa in Semen
1063-7 | http://loinc.org | I NOS Ab [Presence] in Serum or Plasma
10630-2 | http://loinc.org | Spinnbarkeit [Length] in Cervical mucus
10631-0 | http://loinc.org | Testosterone [Moles/volume] in Semen
10632-8 | http://loinc.org | Time until next menstrual period
10633-6 | http://loinc.org | Zinc [Moles/volume] in Semen
10634-4 | http://loinc.org | Complement C1 esterase inhibitor.functional/Complement C1 esterase inhibitor.total in Serum or Plasma
10635-1 | http://loinc.org | Acanthamoeba sp identified in Eye by Organism specific culture
10636-9 | http://loinc.org | Acanthamoeba sp identified in Eye by Wet preparation
10637-7 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Night blue stain
10638-5 | http://loinc.org | Thermophilic Actinomycetes colony count [#/volume] in Unspecified specimen by Organism specific culture
10639-3 | http://loinc.org | Thermophilic Actinomycetes identified in Unspecified specimen by Organism specific culture
1064-5 | http://loinc.org | I NOS Ag [Presence] on Red Blood Cells from Blood product unit
10640-1 | http://loinc.org | Deprecated Adenovirus 40+41 [Identifier] in Stool by Electron microscopy
10641-9 | http://loinc.org | Amoeba identified in Aspirate by Immune stain
10642-7 | http://loinc.org | Amoeba identified in Aspirate by Wet preparation
10643-5 | http://loinc.org | Amoeba identified in Stool by Organism specific culture
10644-3 | http://loinc.org | Arthropod identified in Unspecified specimen
10645-0 | http://loinc.org | Aspergillus fumigatus Ag [Presence] in Tissue by Immunofluorescence
10646-8 | http://loinc.org | Astrovirus [Identifier] in Stool by Electron microscopy
10647-6 | http://loinc.org | Babesia sp identified in Blood by Thick film
10648-4 | http://loinc.org | Babesia sp identified in Blood by Thin film
10649-2 | http://loinc.org | Calicivirus [Identifier] in Stool by Electron microscopy
1065-2 | http://loinc.org | I NOS Ag [Presence] on Red Blood Cells from Donor
10650-0 | http://loinc.org | Candida sp DNA [Presence] in Blood by Probe with amplification
10651-8 | http://loinc.org | Chlamydophila pneumoniae Ag [Presence] in Sputum or Bronchial
10652-6 | http://loinc.org | Chlamydophila pneumoniae DNA [Presence] in Sputum or Bronchial by Probe with amplification
10653-4 | http://loinc.org | Clotrimazole [Susceptibility] by Minimum inhibitory concentration (MIC)
10654-2 | http://loinc.org | Clotrimazole [Susceptibility] by Minimum lethal concentration (MLC)
10655-9 | http://loinc.org | Coccidia identified in Duodenal fluid by Acid fast stain
10656-7 | http://loinc.org | Coccidia identified in Stool by Acid fast stain
10657-5 | http://loinc.org | Deprecated Cryptococcus neoformans Ag [Presence] in Tissue by Immunofluorescence
10658-3 | http://loinc.org | Cyanobacterium identified in Water by Light microscopy
10659-1 | http://loinc.org | Cyclospora sp identified in Stool by Acid fast stain
1066-0 | http://loinc.org | I NOS Ag [Presence] on Red Blood Cells
10660-9 | http://loinc.org | Cytomegalovirus Ag [Presence] in Tissue by Immune stain
10661-7 | http://loinc.org | Dinoflagellate identified in Water by Light microscopy
10662-5 | http://loinc.org | Filaria identified in Blood by Concentration
10663-3 | http://loinc.org | Filaria identified in Blood by Thick film
10664-1 | http://loinc.org | Filaria identified in Blood by Thin film
10665-8 | http://loinc.org | Fungus colony count [#/volume] in Unspecified specimen by Environmental culture
10666-6 | http://loinc.org | Fungus identified in Tissue by Fontana-Masson stain
10667-4 | http://loinc.org | Fungus identified in Unspecified specimen by Animal inoculation
10668-2 | http://loinc.org | Fungus identified in Unspecified specimen by Environmental culture
10669-0 | http://loinc.org | Fungus identified in Unspecified specimen by Sticky tape for environmental fungus
1067-8 | http://loinc.org | Jk sup(a) Ab [Presence] in Serum or Plasma from Blood product unit
10670-8 | http://loinc.org | Giardia lamblia [Presence] in Stool by Organism specific culture
10671-6 | http://loinc.org | Helminth identified in Unspecified specimen
10672-4 | http://loinc.org | Helminth+Arthropod identified in Unspecified specimen
10673-2 | http://loinc.org | Hepatitis B virus core Ag [Presence] in Tissue by Immune stain
10674-0 | http://loinc.org | Hepatitis B virus surface Ag [Presence] in Tissue by Immune stain
10675-7 | http://loinc.org | Hepatitis B virus surface Ag [Presence] in Tissue by Orcein stain
10676-5 | http://loinc.org | Hepatitis C virus RNA [Units/volume] (viral load) in Serum or Plasma by Probe with amplification
10677-3 | http://loinc.org | Herpes simplex virus 1 Ag [Presence] in Tissue by Immune stain
10678-1 | http://loinc.org | Herpes simplex virus 1+2 Ag [Presence] in Tissue by Immune stain
10679-9 | http://loinc.org | Herpes simplex virus 2 Ag [Presence] in Tissue by Immune stain
1068-6 | http://loinc.org | Jk sup(a) Ab [Presence] in Serum or Plasma from Donor
10680-7 | http://loinc.org | Herpes simplex virus identified in Cerebral spinal fluid by Electron microscopy
10681-5 | http://loinc.org | Herpes simplex virus identified in Tissue by Electron microscopy
10682-3 | http://loinc.org | Deprecated HIV 1 RNA [Units/volume] in Serum or Plasma by Probe with amplification
10683-1 | http://loinc.org | Hydatid cyst identified in Aspirate by Immune stain
10684-9 | http://loinc.org | Hydatid cyst identified in Liver by Wet preparation

10685-6 | http://loinc.org | Hydatid cyst identified in Lung tissue by Wet preparation
10686-4 | http://loinc.org | Leishmania sp identified in Tissue by Giemsa stain
10687-2 | http://loinc.org | Leishmania sp identified in Tissue by Organism specific culture
10688-0 | http://loinc.org | Microscopic observation [Identifier] in Hair by KOH preparation
10689-8 | http://loinc.org | Microscopic observation [Identifier] in Nail by KOH preparation
1069-4 | http://loinc.org | Jk sup(a) Ab [Presence] in Serum or Plasma
10690-6 | http://loinc.org | Microsporidia identified in Duodenal fluid by Trichrome stain modified
10691-4 | http://loinc.org | Mushroom.toxic identified by Inspection
10692-2 | http://loinc.org | Mushroom.toxic identified in Food by Light microscopy
10693-0 | http://loinc.org | Mushroom.toxic identified in Vomitus by Light microscopy
10694-8 | http://loinc.org | Naegleria sp identified in Tissue by Organism specific culture
10695-5 | http://loinc.org | Naegleria sp identified in Tissue by Wet preparation
10696-3 | http://loinc.org | Norovirus [Identifier] in Stool by Electron microscopy
10697-1 | http://loinc.org | Nystatin [Susceptibility] by Minimum inhibitory concentration (MIC)
10698-9 | http://loinc.org | Nystatin [Susceptibility] by Minimum lethal concentration (MLC)
10699-7 | http://loinc.org | Onchocerca sp identified in Tissue by Wet preparation
107-3 | http://loinc.org | Cefotaxime [Susceptibility] by Minimum lethal concentration (MLC)
1070-2 | http://loinc.org | Jk sup(a) Ag [Presence] on Red Blood Cells from Blood product unit
10700-3 | http://loinc.org | Orthopoxvirus [Identifier] in Skin by Electron microscopy
10701-1 | http://loinc.org | Ova and parasites identified in Stool by Concentration
10702-9 | http://loinc.org | Ova and parasites identified in Stool by Immune stain
10703-7 | http://loinc.org | Ova and parasites identified in Stool by Kinyoun iron hematoxylin stain
10704-5 | http://loinc.org | Ova and parasites identified in Stool by Light microscopy
10705-2 | http://loinc.org | Human papilloma virus Ag [Presence] in Tissue by Immune stain
10706-0 | http://loinc.org | Picornavirus [Identifier] in Stool by Electron microscopy
10707-8 | http://loinc.org | Toxic Plant identified in Plant specimen by Inspection
10708-6 | http://loinc.org | Toxic Plant identified in Vomitus by Light microscopy
10709-4 | http://loinc.org | Plasmodium falciparum Ag [Units/volume] in Blood by Immunofluorescence
1071-0 | http://loinc.org | Jk sup(a) Ag [Presence] on Red Blood Cells from Donor
10710-2 | http://loinc.org | Plasmodium sp identified in Blood by Thin film
10711-0 | http://loinc.org | Plasmodium vivax Ag [Units/volume] in Blood by Immunofluorescence
10712-8 | http://loinc.org | Pneumocystis sp identified in Lung tissue
10713-6 | http://loinc.org | Prototheca identified in Unspecified specimen by Culture
10714-4 | http://loinc.org | Deprecated Rotavirus [Identifier] in Stool by Electron microscopy
10715-1 | http://loinc.org | Schistosoma sp identified in Urine sediment by Light microscopy
10716-9 | http://loinc.org | Schistosoma sp identified in Unspecified specimen by Organism specific culture
10717-7 | http://loinc.org | Streptococcus pneumoniae Ab [Units/volume] in Serum by Latex agglutination
10718-5 | http://loinc.org | Strongyloides sp Ab [Units/volume] in Serum by Immunoassay
10719-3 | http://loinc.org | Taenia solium adult Ab [Units/volume] in Serum by Immunoblot
1072-8 | http://loinc.org | Jk sup(a) Ag [Presence] on Red Blood Cells
10720-1 | http://loinc.org | Terbinafine [Susceptibility] by Minimum inhibitory concentration (MIC)
10721-9 | http://loinc.org | Terbinafine [Susceptibility] by Minimum lethal concentration (MLC)
10722-7 | http://loinc.org | Torovirus [Identifier] in Stool by Electron microscopy
10723-5 | http://loinc.org | Toxoplasma gondii IgA Ab [Units/volume] in Serum by Immunoassay
10724-3 | http://loinc.org | Toxoplasma gondii IgE Ab [Units/volume] in Serum by Immunoassay
10725-0 | http://loinc.org | Toxoplasma gondii DNA [Units/volume] in Body fluid by Probe with amplification
10726-8 | http://loinc.org | Toxoplasma gondii [Presence] in Tissue by Giemsa stain
10727-6 | http://loinc.org | Toxoplasma gondii identified in Tissue
10728-4 | http://loinc.org | Trichomonas sp identified in Genital specimen by Organism specific culture
10729-2 | http://loinc.org | Trypanosoma sp identified in Blood by Light microscopy
1073-6 | http://loinc.org | Jk sup(b) Ab [Presence] in Serum or Plasma from Blood product unit
10730-0 | http://loinc.org | Trypanosoma sp identified in Blood by Organism specific culture
10731-8 | http://loinc.org | Trypanosoma sp identified in Blood by Thick film
10732-6 | http://loinc.org | Trypanosoma sp identified in Blood by Thin film
10733-4 | http://loinc.org | Trypanosoma sp identified in Blood by Wet preparation
10734-2 | http://loinc.org | Varicella zoster virus identified in Skin by Electron microscopy
10735-9 | http://loinc.org | Viral sequencing [Identifier] in Serum by Sequencing
10736-7 | http://loinc.org | Virus identified in Cerebral spinal fluid by Electron microscopy
10737-5 | http://loinc.org | Virus identified in Stool by Electron microscopy
10738-3 | http://loinc.org | Virus identified in Tissue by Electron microscopy
10739-1 | http://loinc.org | Virus identified in Unspecified specimen by Electron microscopy
1074-4 | http://loinc.org | Jk sup(b) Ab [Presence] in Serum or Plasma from Donor
10740-9 | http://loinc.org | Aluminum.microscopic observation [Identifier] in Bone by Histomorphometry stain
10741-7 | http://loinc.org | Amyloid.microscopic observation [Identifier] in Brain by Thioflavine-S stain
10742-5 | http://loinc.org | Amyloid.microscopic observation [Identifier] in Tissue by Bennhold stain.Putchler modified
10743-3 | http://loinc.org | Amyloid.microscopic observation [Identifier] in Tissue by Highman stain
10744-1 | http://loinc.org | Amyloid.microscopic observation [Identifier] in Tissue by Vassar-culling stain
10745-8 | http://loinc.org | Bile.microscopic observation [Identifier] in Tissue by Fouchet stain
10746-6 | http://loinc.org | Calcium.microscopic observation [Identifier] in Tissue by Von Kossa stain
10747-4 | http://loinc.org | Collagen fibers+Elastic fibers.microscopic observation [Identifier] in Tissue by Lawson-Van Gieson stain
10748-2 | http://loinc.org | Collagen fibers+Elastic fibers.microscopic observation [Identifier] in Tissue by Verhoeff-Van Gieson stain

10749-0 | http://loinc.org | Collagen fibers.microscopic observation [Identifier] in Tissue by Van Gieson stain
1075-1 | http://loinc.org | Jk sup(b) Ab [Presence] in Serum or Plasma
10750-8 | http://loinc.org | Connective tissue.microscopic observation [Identifier] in Tissue by Trichrome stain.Masson
10751-6 | http://loinc.org | Copper.microscopic observation [Identifier] in Tissue by Rhodamine stain
10752-4 | http://loinc.org | Fat.microscopic observation [Identifier] in Milk by Sudan IV stain
10753-2 | http://loinc.org | Fat.microscopic observation [Identifier] in Stool by Sudan IV stain
10754-0 | http://loinc.org | Fat.microscopic observation [Identifier] in Tissue by Sudan IV stain
10755-7 | http://loinc.org | Fungus.microscopic observation [Identifier] in Tissue by Methenamine silver stain.Grocott
10756-5 | http://loinc.org | Glial fibers.microscopic observation [Identifier] in Tissue by Holzer stain
10757-3 | http://loinc.org | Hematologic+Nuclear elements.microscopic observation [Identifier] in Tissue by Giemsa stain.May-Grunwald
10758-1 | http://loinc.org | Iron.microscopic observation [Identifier] in Bone by Histomorphometry stain
10759-9 | http://loinc.org | Iron.microscopic observation [Identifier] in Sputum by Gomori stain
1076-9 | http://loinc.org | Jk sup(b) Ag [Presence] on Red Blood Cells from Blood product unit
10760-7 | http://loinc.org | Iron.microscopic observation [Identifier] in Tissue by Gomori stain
10761-5 | http://loinc.org | Iron.microscopic observation [Identifier] in Tissue by Other stain
10762-3 | http://loinc.org | Microscopic observation [Identifier] in Blood by Hemosiderin stain
10763-1 | http://loinc.org | Microscopic observation [Identifier] in Body fluid by Sudan black stain
10764-9 | http://loinc.org | Microscopic observation [Identifier] in Sputum by Silver stain
10765-6 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Acetate esterase stain
10766-4 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Alcian blue stain
10767-2 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Alcian blue stain.sulfated
10768-0 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Alcian blue stain.with periodic acid-Schiff
10769-8 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Alizarin red S stain
1077-7 | http://loinc.org | Jk sup(b) Ag [Presence] on Red Blood Cells from Donor
10770-6 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Argentaffin stain
10771-4 | http://loinc.org | Deprecated Microscopic observation [Identifier] in Tissue by Rhodamine-auramine fluorochrome stain
10772-2 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Azure-eosin stain
10773-0 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Basic fuchsin stain
10774-8 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Bielschowsky stain
10775-5 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Bleach stain
10776-3 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Bodian stain
10777-1 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Brown and Brenn stain
10778-9 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Butyrate esterase stain
10779-7 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Carmine stain.Best
1078-5 | http://loinc.org | Jk sup(b) Ag [Presence] on Red Blood Cells
10780-5 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Chloracetate esterase stain
10781-3 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Churukian-Schenk stain
10782-1 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Congo red stain
10783-9 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Crystal violet stain
10784-7 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Esterase stain.non-specific
10785-4 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Fite-Faraco stain
10786-2 | http://loinc.org | Deprecated Microscopic observation [Identifier] in Tissue by Giemsa stain.3 micron
10787-0 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Gridley stain
10788-8 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Hansel stain
10789-6 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Hematoxylin-eosin-Harris regressive stain
1079-3 | http://loinc.org | Js sup(a) Ab [Presence] in Serum or Plasma from Blood product unit
10790-4 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Hematoxylin-eosin-Mayers progressive stain
10791-2 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Mallory-Heidenhain stain
10792-0 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Methenamine silver stain.Jones
10793-8 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Methyl green stain
10794-6 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Methyl green-pyronine Y stain
10795-3 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Methyl violet stain
10796-1 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Mucicarmine stain
10797-9 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Neutral red stain
10798-7 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Oil red O stain
10799-5 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Pentachrome stain.Movat
108-1 | http://loinc.org | Cefotaxime [Susceptibility] by Minimum inhibitory concentration (MIC)
1080-1 | http://loinc.org | Js sup(a) Ab [Presence] in Serum or Plasma from Donor
10800-1 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Periodic acid-Schiff stain with diatase digestion
10801-9 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Phosphotungstic acid Hematoxylin (PTAH) Stain
10802-7 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Prussian blue stain
10803-5 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Quinacrine fluorescent stain
10804-3 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Reticulin stain
10805-0 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Safranin stain
10806-8 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Schmorl stain
10807-6 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Sevier-Munger stain
10808-4 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Silver impregnation stain.Dieterle
10809-2 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Silver nitrate stain
1081-9 | http://loinc.org | Js sup(a) Ab [Presence] in Serum or Plasma
10810-0 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Silver stain.Fontana-Masson
10811-8 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Silver stain.Grimelius

10812-6 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Steiner stain
10813-4 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Sudan black stain
10814-2 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Supravital stain
10815-9 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Tetrachrome stain
10816-7 | http://loinc.org | Deprecated Microscopic observation [Identifier] in Tissue by Toluidine blue O stain
10817-5 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Trichrome stain.Gomori-Wheatley
10818-3 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Trichrome stain.Masson modified
10819-1 | http://loinc.org | Microscopic observation [Identifier] in Tissue by Wade stain
1082-7 | http://loinc.org | Js sup(a) Ag [Presence] on Red Blood Cells from Blood product unit
10820-9 | http://loinc.org | Deprecated Microscopic observation [Identifier] in Tissue by Warthin-Starry stain
10821-7 | http://loinc.org | Deprecated Microscopic observation [Identifier] in Tissue by Wright Giemsa stain
10822-5 | http://loinc.org | Mucin.microscopic observation [Identifier] in Tissue by Mucicarmine stain.Mayer
10823-3 | http://loinc.org | Mucopolysaccharides.microscopic observation [Identifier] in Tissue by Colloidal ferric oxide stain.Hale
10824-1 | http://loinc.org | Myelin+Myelin breakdown products.microscopic observation [Identifier] in Tissue by Luxol fast blue/Periodic acid-Schiff stain
10825-8 | http://loinc.org | Myelin+Nerve cells.microscopic observation [Identifier] in Tissue by Luxol fast blue/Cresyl violet stain
10826-6 | http://loinc.org | Nissel.microscopic observation [Identifier] in Tissue by Cresyl echt violet stain
10827-4 | http://loinc.org | Reticulum.microscopic observation [Identifier] in Tissue by Gomori stain
10828-2 | http://loinc.org | Urate crystals.microscopic observation [type] in Tissue by De Galantha stain
10829-0 | http://loinc.org | Silicon [Mass/volume] in Serum or Plasma
1083-5 | http://loinc.org | Js sup(a) Ag [Presence] on Red Blood Cells from Donor
10830-8 | http://loinc.org | Deprecated Surgical operation note complications
10831-6 | http://loinc.org | Surgical operation note complications [Interpretation]
10832-4 | http://loinc.org | Glucose [Mass/volume] in Serum or Plasma --15 minutes post 50 g lactose PO
10833-2 | http://loinc.org | Insulin [Mass/volume] in Serum or Plasma --7 hours post 75 g glucose PO
10834-0 | http://loinc.org | Globulin [Mass/volume] in Serum by calculation
10835-7 | http://loinc.org | Lipoprotein a [Mass/volume] in Serum or Plasma
10836-5 | http://loinc.org | Niacin [Mass/volume] in Blood
10837-3 | http://loinc.org | Organic acids [Presence] in Serum or Plasma
10838-1 | http://loinc.org | Phosphoserine [Mass/volume] in Serum or Plasma
10839-9 | http://loinc.org | Troponin I.cardiac [Mass/volume] in Serum or Plasma
1084-3 | http://loinc.org | Js sup(a) Ag [Presence] on Red Blood Cells
10840-7 | http://loinc.org | Atropine [Mass/volume] in Urine
10841-5 | http://loinc.org | Methazolamide [Mass/volume] in Serum or Plasma
10842-3 | http://loinc.org | Deprecated HLA-DQ1 [Presence]
10843-1 | http://loinc.org | Deprecated HLA-DQ2 [Presence]
10844-9 | http://loinc.org | Deprecated HLA-DQ3 [Presence]
10845-6 | http://loinc.org | Deprecated HLA-DQ4 [Presence]
10846-4 | http://loinc.org | Borrelia burgdorferi DNA [Presence] in Blood by Probe with amplification
10847-2 | http://loinc.org | Borrelia burgdorferi DNA [Presence] in Body fluid by Probe with amplification
10848-0 | http://loinc.org | Chlamydia sp IgG Ab [Titer] in Serum by Immunofluorescence
10849-8 | http://loinc.org | Chlamydia sp IgM Ab [Titer] in Serum by Immunofluorescence
1085-0 | http://loinc.org | Js sup(b) Ab [Presence] in Serum or Plasma from Blood product unit
10850-6 | http://loinc.org | Cyclospora cayetanensis [Presence] in Unspecified specimen
10851-4 | http://loinc.org | Escherichia coli O157:H7 [Presence] in Stool by Organism specific culture
10852-2 | http://loinc.org | Deprecated Fungus identified in Blood by Culture
10853-0 | http://loinc.org | Isospora belli [Presence] in Unspecified specimen by Acid fast stain.Kinyoun modified
10854-8 | http://loinc.org | Deprecated Midrofilaria identified in Blood by Concentration
10855-5 | http://loinc.org | Ova and parasites identified in Duodenal fluid or Gastric fluid by Light microscopy
10856-3 | http://loinc.org | Microscopic observation [Identifier] in Genital mucus by Gram stain
10857-1 | http://loinc.org | Microsporidia identified in Unspecified specimen by Light microscopy
10858-9 | http://loinc.org | Teichoate Ab [Presence] in Serum
10859-7 | http://loinc.org | Trypanosoma sp [Identifier] in Blood by Acridine Orange + Giemsa Stain
1086-8 | http://loinc.org | Js sup(b) Ab [Presence] in Serum or Plasma from Donor
10860-5 | http://loinc.org | Varicella zoster virus [Presence] in Unspecified specimen by Organism specific culture
10861-3 | http://loinc.org | Progesterone receptor [Mass/mass] in Tissue
10862-1 | http://loinc.org | Basement membrane Ab [Titer] in Serum
10863-9 | http://loinc.org | Endomysium IgA Ab [Titer] in Serum
10864-7 | http://loinc.org | Immune complex [Units/volume] in Serum or Plasma by Raji cell assay
10865-4 | http://loinc.org | Intercellular substance Ab [Presence] in Serum
10866-2 | http://loinc.org | Sulfatide IgG Ab [Titer] in Serum
10867-0 | http://loinc.org | Sulfatide IgM Ab [Titer] in Serum
10868-8 | http://loinc.org | Bacitracin [Susceptibility] by Disk diffusion (KB)
10869-6 | http://loinc.org | Xylose [Mass/volume] in Blood --3 hours post 25 g xylose PO
1087-6 | http://loinc.org | Js sup(b) Ab [Presence] in Serum or Plasma
10870-4 | http://loinc.org | Xylose [Mass/volume] in Blood --4 hours post 25 g xylose PO
10871-2 | http://loinc.org | Xylose [Mass/volume] in Blood --5 hours post 25 g xylose PO
10872-0 | http://loinc.org | Xylose [Mass/volume] in Blood --baseline
10873-8 | http://loinc.org | Beta-2-Microglobulin [Mass/time] in 24 hour Urine
10874-6 | http://loinc.org | Bombesin [Mass/volume] in Plasma
10875-3 | http://loinc.org | Carnitine esters [Mass/volume] in Urine
10876-1 | http://loinc.org | Carnitine esters [Mass/volume] in Serum or Plasma

10877-9 | http://loinc.org | Carnitine free (C0) [Mass/volume] in Urine
10878-7 | http://loinc.org | Iodine.protein bound [Mass/volume] in Serum or Plasma
10879-5 | http://loinc.org | Isovalerylglycine [Mass/volume] in Urine
1088-4 | http://loinc.org | Js sup(b) Ag [Presence] on Red Blood Cells from Blood product unit
10880-3 | http://loinc.org | Magnesium [Mass/mass] in Stool
10881-1 | http://loinc.org | Pentacarboxylporphyrins [Mass/volume] in Red Blood Cells
10882-9 | http://loinc.org | Pentacarboxylporphyrins [Mass/volume] in Serum or Plasma
10883-7 | http://loinc.org | Phenolphthalein [Mass/mass] in Stool
10884-5 | http://loinc.org | Phosphate [Mass/mass] in Stool
10885-2 | http://loinc.org | Porphyrins [Mass/time] in 24 hour Urine
10886-0 | http://loinc.org | Prostate Specific Ag Free [Mass/volume] in Serum or Plasma
10887-8 | http://loinc.org | Pyridinoline [Mass/time] in 24 hour Urine
10888-6 | http://loinc.org | Lipase [Enzymatic activity/volume] in Urine
10889-4 | http://loinc.org | Bisacodyl [Mass/mass] in Stool
1089-2 | http://loinc.org | Js sup(b) Ag [Presence] on Red Blood Cells from Donor
10890-2 | http://loinc.org | Deprecated Normethsuximide [Mass/volume] in Serum or Plasma
10891-0 | http://loinc.org | Oxyphenisatin [Mass/mass] in Stool
10892-8 | http://loinc.org | Pentoxifylline [Mass/volume] in Serum or Plasma
10893-6 | http://loinc.org | Trenbolone [Mass/volume] in Urine
10894-4 | http://loinc.org | Aspergillus niger Ab [Presence] in Serum by Immune diffusion (ID)
10895-1 | http://loinc.org | Clostridium difficile toxin B [Presence] in Stool
10896-9 | http://loinc.org | Eastern equine encephalitis virus IgG Ab [Titer] in Serum by Immunofluorescence
10897-7 | http://loinc.org | Eastern equine encephalitis virus IgG Ab [Titer] in Cerebral spinal fluid by Immunofluorescence
10898-5 | http://loinc.org | Eastern equine encephalitis virus IgM Ab [Titer] in Serum by Immunofluorescence

4.4.2.16508 HL7 v3 Value Set ActIncidentCode

URL: https://hl7.org/fhir/R4/v3/ActIncidentCode/vs.html
Vocabulary Work Group | Maturity Level: 2 | Trial Use | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActIncidentCode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActIncidentCode
Version: | 2014-03-26
Name: | v3.ActIncidentCode
Title: | V3 Value SetActIncidentCode
Definition: | Set of codes indicating the type of incident or accident.
OID: | 2.16.840.1.113883.1.11.16508 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is used in the following places:

• Resource: ExplanationOfBenefit.accident.type (CodeableConcept / Extensible)
• Resource: Claim.accident.type (CodeableConcept / Extensible)

4.4.2.16508.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActCode where concept is-a _ActIncidentCode

http://terminology.hl7.org/CodeSystem/v3-ActCode

Code | Display
_ActIncidentCode | ActIncidentCode | Set of codes indicating the type of incident or accident.

4.4.2.16508.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 4 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActCode

Code | Display | Definition
MVA | Motor vehicle accident | Incident or accident as the result of a motor vehicle accident
SCHOOL | School Accident | Incident or accident is the result of a school place accident.
SPT | Sporting Accident | Incident or accident is the result of a sporting accident.
WPA | Workplace accident | Incident or accident is the result of a work place accident

4.4.2.16621 HL7 v3 Value Set ActSubstanceAdminSubstitutionCode

URL: https://hl7.org/fhir/R4/v3/ActSubstanceAdminSubstitutionCode/vs.html
Vocabulary Work Group | Maturity Level: 1 | Draft | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActSubstanceAdminSubstitutionCode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActSubstanceAdminSubstitutionCode
Version: | 2014-03-26
Name: | v3.ActSubstanceAdminSubstitutionCode
Title: | V3 Value SetActSubstanceAdminSubstitutionCode
Definition: | No Description Provided
OID: | 2.16.840.1.113883.1.11.16621 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is used in the following places:

• Resource: MedicationRequest.substitution.allowed[x] (boolean|CodeableConcept / Example)

• Resource: MedicationDispense.substitution.type (CodeableConcept / Example)

4.4.2.16621.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-substanceAdminSubstitution where concept is-a _ActSubstanceAdminSubstitutionCode

http://terminology.hl7.org/CodeSystem/v3-substanceAdminSubstitution

Code | Display
_ActSubstanceAdminSubstitutionCode | ActSubstanceAdminSubstitutionCode | Description: Substitution occurred or is permitted with another product that may potentially have different ingredients, but having the same biological and therapeutic effects.

4.4.2.16621.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 9 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-substanceAdminSubstitution version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-substanceAdminSubstitution

Code | Display | Definition
E | equivalent | Description: Substitution occurred or is permitted with another bioequivalent and therapeutically equivalent product.
EC | equivalent composition | Description: Substitution occurred or is permitted with another product that is a: pharmaceutical alternative containing the same active ingredient but is formulated with different salt, ester pharmaceutical equivalent that has the same active ingredient, strength, dosage form and route of administration Examples: Pharmaceutical alternative: Erythromycin Ethylsuccinate for Erythromycin Stearate Pharmaceutical equivalent: Lisonpril for Zestril
BC | brand composition | Description: Substitution occurred or is permitted between equivalent Brands but not Generics Examples: Zestril for Prinivil Coumadin for Jantoven
G | generic composition | Description: Substitution occurred or is permitted between equivalent Generics but not Brands Examples: Lisnopril (Lupin Corp) for Lisnopril (Wockhardt Corp)
TE | therapeutic alternative | Description: Substitution occurred or is permitted with another product having the same therapeutic objective and safety profile. Examples: ranitidine for Tagamet
TB | therapeutic brand | Description: Substitution occurred or is permitted between therapeutically equivalent Brands but not Generics > Examples: Zantac for Tagamet
TG | therapeutic generic | Description: Substitution occurred or is permitted between therapeutically equivalent Generics but not Brands > Examples: Ranitidine for cimetidine
F | formulary | Description: This substitution was performed or is permitted based on formulary guidelines.
N | none | No substitution occurred or is permitted.

4.4.2.16643 HL7 v3 Value Set SeverityObservation

URL: https://hl7.org/fhir/R4/v3/SeverityObservation/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-SeverityObservation) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-SeverityObservation
Version: | 2014-03-26
Name: | v3.SeverityObservation
Title: | V3 Value SetSeverityObservation
Definition: | Potential values for observations of severity.
OID: | 2.16.840.1.113883.1.11.16643 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is used in the following places:

• ConceptMap: Translation target in v3 map for DetectedIssueSeverity

4.4.2.16643.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ObservationValue where concept is-a _SeverityObservation

http://terminology.hl7.org/CodeSystem/v3-ObservationValue

Code | Display
_SeverityObservation | SeverityObservation | Potential values for observations of severity.

4.4.2.16643.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 3 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ObservationValue version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ObservationValue

Code | Display | Definition
H | High | Indicates the condition may be life-threatening or has the potential to cause permanent injury.
L | Low | Indicates the condition may result in some adverse consequences but is unlikely to substantially affect the situation of the subject.
M | Moderate | Indicates the condition may result in noticable adverse adverse consequences but is unlikely to be life-threatening or cause permanent injury.

4.4.2.16926 HL7 v3 Value Set x_BasicConfidentialityKind

URL: https://hl7.org/fhir/R4/v3/x_BasicConfidentialityKind/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-x_BasicConfidentialityKind) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-xBasicConfidentialityKind
Version: | 2014-03-26
Name: | v3.x_BasicConfidentialityKind
Title: | V3 Value Setx_BasicConfidentialityKind
Definition: | Description: Used to enumerate the typical confidentiality constraints placed upon a clinical document. Usage Note: x_BasicConfidentialityKind is a subset of Confidentiality codes that are used as metadata indicating the receiver responsibility to comply with normally applicable jurisdictional privacy law or disclosure authorization; that the receiver may not disclose this information except as directed by the information custodian, who may be the information subject; or that the receiver may not disclose this information except as directed by the information custodian, who may be the information subject.
OID: | 2.16.840.1.113883.1.11.16926 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.16926.1 Content Logical Definition

This value set includes codes from the following code systems:

http://terminology.hl7.org/CodeSystem/v3-Confidentiality

Code | Display
N | normal | Definition: Privacy metadata indicating that the information is typical, non-stigmatizing health information, which presents typical risk of harm if disclosed without authorization. Examples: In the US, this includes what HIPAA identifies as the minimum necessary protected health information (PHI) given a covered purpose of use (treatment, payment, or operations). Includes typical, non-stigmatizing health information disclosed in an application for health, workers compensation, disability, or life insurance. Map: Partial Map to ISO 13606-4 Sensitivity Level (3) Clinical Care: Default for normal clinical care access (i.e. most clinical staff directly caring for the patient should be able to access nearly all of the EHR). Maps to normal confidentiality for treatment information but not to ancillary care, payment and operations. Usage Note: This metadata indicates that the receiver may be obligated to comply with applicable jurisdictional privacy law or disclosure authorization.
R | restricted | Privacy metadata indicating highly sensitive, potentially stigmatizing information, which presents a high risk to the information subject if disclosed without authorization. May be pre-empted by jurisdictional law, e.g., for public health reporting or emergency treatment. Examples: Includes information that is additionally protected such as sensitive conditions mental health, HIV, substance abuse, domestic violence, child abuse, genetic disease, and reproductive health; or sensitive demographic information such as a patient's standing as an employee or a celebrity. May be used to indicate proprietary or classified information that is not related to an individual, e.g., secret ingredients in a therapeutic substance; or the name of a manufacturer. Map: Partial Map to ISO 13606-4 Sensitivity Level (3) Clinical Care: Default for normal clinical care access (i.e. most clinical staff directly caring for the patient should be able to access nearly all of the EHR). Maps to normal confidentiality for treatment information but not to ancillary care, payment and operations.. Usage Note: This metadata indicates that the receiver may be obligated to comply with applicable, prevailing (default) jurisdictional privacy law or disclosure authorization..
V | very restricted | . Privacy metadata indicating that the information is extremely sensitive and likely stigmatizing health information that presents a very high risk if disclosed without authorization. This information must be kept in the highest confidence. Examples: Includes information about a victim of abuse, patient requested information sensitivity, and taboo subjects relating to health status that must be discussed with the patient by an attending provider before sharing with the patient. May also include information held under â€œlegal lockâ€? or attorney-client privilege Map: This metadata indicates that the receiver may not disclose this information except as directed by the information custodian, who may be the information subject. Usage

Note: This metadata indicates that the receiver may not disclose this information except as directed by the information custodian, who may be the information subject.

4.4.2.16927 HL7 v3 Value Set RoleClassServiceDeliveryLocation

URL: https://hl7.org/fhir/R4/v3/RoleClassServiceDeliveryLocation/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-RoleClassServiceDeliveryLocation) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-RoleClassServiceDeliveryLocation
Version: | 2014-03-26
Name: | v3.RoleClassServiceDeliveryLocation
Title: | V3 Value SetRoleClassServiceDeliveryLocation
Definition: | A role played by a place at which services may be provided.
OID: | 2.16.840.1.113883.1.11.16927 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.16927.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-RoleClass where concept is-a SDLOC

4.4.2.16927.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 3 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-RoleClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-RoleClass

Lvl | Code | Display | Definition
0 | SDLOC | service delivery location | A role played by a place at which services may be provided.
1 | DSDLOC | dedicated service delivery location | A role of a place (player) that is intended to house the provision of services. Scoper is the Entity (typically Organization) that provides these services. This is not synonymous with "ownership."
1 | ISDLOC | incidental service delivery location | A role played by a place at which health care services may be provided without prior designation or authorization.

4.4.2.17660 HL7 v3 Value Set ServiceDeliveryLocationRoleType

URL: https://hl7.org/fhir/R4/v3/ServiceDeliveryLocationRoleType/vs.html
Vocabulary Work Group | Maturity Level: 3 | Trial Use | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ServiceDeliveryLocationRoleType) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ServiceDeliveryLocationRoleType
Version: | 2014-03-26
Name: | v3.ServiceDeliveryLocationRoleType
Title: | V3 Value SetServiceDeliveryLocationRoleType
Definition: | A role of a place that further classifies the setting (e.g., accident site, road side, work site, community location) in which services are delivered.
OID: | 2.16.840.1.113883.1.11.17660 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is used in the following places:

• Resource: ServiceRequest.locationCode (CodeableConcept / Example)

• Profile: ServiceRequest-Genetics: ServiceRequest.locationCode (CodeableConcept / Example)

• Resource: Location.type (CodeableConcept / Extensible)

4.4.2.17660.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-RoleCode where concept is-a _ServiceDeliveryLocationRoleType

http://terminology.hl7.org/CodeSystem/v3-RoleCode

Code | Display
_ServiceDeliveryLocationRoleType | ServiceDeliveryLocationRoleType | A role of a place that further classifies the setting (e.g., accident site, road side, work site, community location) in which services are delivered.

4.4.2.17660.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 123 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-RoleCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-RoleCode

Code | Display | Definition
DX | Diagnostics or therapeutics unit | A practice setting where diagnostic procedures or therapeutic interventions are performed
CVDX | Cardiovascular diagnostics or therapeutics unit | A practice setting where cardiovascular diagnostic procedures or therapeutic interventions are performed (e.g., cardiac catheterization lab, echocardiography suite)
CATH | Cardiac catheterization lab | Cardiac catheterization lab
ECHO | Echocardiography lab | Echocardiography lab
GIDX | Gastroenterology diagnostics or therapeutics lab | A practice setting where GI procedures (such as endoscopies) are performed
ENDOS | Endoscopy lab | (X12N 261QD0000N)
RADDX | Radiology diagnostics or therapeutics unit | A practice setting where radiology services (diagnostic or therapeutic) are provided (X12N 261QR0200N)
RADO | Radiation oncology unit | (X12N 261QX0203N)
RNEU | Neuroradiology unit | Neuroradiology unit
HOSP | Hospital | An acute care institution that provides medical, surgical, or psychiatric care and treatment for the sick or the injured.
CHR | Chronic Care Facility | (1) A hospital including a physical plant and personnel that provides multidisciplinary diagnosis and treatment for diseases that have one or more of the following characteristics: is permanent; leaves residual disability; is caused by nonreversible pathological alteration; requires special training of the patient for rehabilitation; and/or may be expected to require a long period of supervision or care. In addition, patients require the safety, security, and shelter of these specialized inpatient or partial hospitalization settings. (2) A hospital that provides medical and skilled nursing services to patients with long-term illnesses who are not in an acute phase but who require an intensity of services not available in nursing homes
GACH | Hospitals; General Acute Care Hospital | (X12N 282N00000N)
MHSP | Military Hospital | A health care facility operated by the Department of Defense or other military operation.
PSYCHF | Psychatric Care Facility | Healthcare facility that cares for patients with psychiatric illness(s).
RH | Rehabilitation hospital | (X12N 283X00000N)
RHAT | addiction treatment center | Description: A location that plays the role of delivering services which may include life training and/or social support to people with addictions.
RHII | intellectual impairment center | Description: A location that plays the role of delivering services which may include adaptation, rehabilitation and social integration services for people with intellectual and/or pervasive development disorders such as autism or severe behaviour disorder.
RHMAD | parents with adjustment difficulties center | Description: A location that plays the role of delivering services which may social support services for adolescents who are pregnant or have child and are experiencing adaptation issues/difficulties in their current or eventual parenting role.
RHPI | physical impairment center | Description: A location that plays the role of delivering services which may include adaptation, rehabilitation and social integration services for people with physical impairments.
RHPIH | physical impairment - hearing center | Description: A location that plays the role of delivering services for people with hearing impairments.
RHPIMS | physical impairment - motor skills center | Description: A location that plays the role of delivering services for people with motor skill impairments.
RHPIVS | physical impairment - visual skills center | Description: A location that plays the role of delivering services for people with visual skill impairments.
RHYAD | youths with adjustment difficulties center | Description: A location that plays the role of delivering services which may include life training and/or social support services for the adaption, rehabilitation and social integration of youths with adjustment difficulties.
HU | Hospital unit | Hospital unit
BMTU | Bone marrow transplant unit | Bone marrow transplant unit
CCU | Coronary care unit | Coronary care unit
CHEST | Chest unit | A specialty unit in hospital that focuses on chronic respirator patients and pulmonary failure
EPIL | Epilepsy unit | Epilepsy unit
ER | Emergency room | The section of a health care facility for providing rapid treatment to victims of sudden illness or trauma.
ETU | Emergency trauma unit | Emergency trauma unit
HD | Hemodialysis unit | Hemodialysis unit
HLAB | hospital laboratory | Description: A location that plays the role of delivering services which may include tests done based on clinical specimens to get health information about a patient as pertaining to the diagnosis, treatment and prevention of disease. Hospital laboratories may be further divided into specialized units such as Anatomic Pathology, Microbiology, and Biochemistry.
INLAB | inpatient laboratory | Description: A location that plays the role of delivering services which may include tests are done on clinical specimens to get health information about a patient pertaining to the diagnosis, treatment, and prevention of disease for a hospital visit longer than one day.
OUTLAB | outpatient laboratory | Description: A location that plays the role of delivering services which may include tests are done on clinical specimens to get health information about a patient pertaining to the diagnosis, treatment, and prevention of disease for same day visits.
HRAD | radiology unit | Description: A location that plays the role of delivering services which may include the branch of medicine that uses ionizing and non-ionizing radiation to diagnose and treat diseases. The radiology unit may be further divided into

subspecialties such as Imaging, Cardiovascular, Thoracic, and Ultrasound.
HUSCS | specimen collection site | Description: A location that plays the role of delivering services which may include collecting specimens and/or samples from patients for laboratory testing purposes, but does not perform any tests or analysis functions.
ICU | Intensive care unit | Intensive care unit
PEDICU | Pediatric intensive care unit | Pediatric intensive care unit
PEDNICU | Pediatric neonatal intensive care unit | Pediatric neonatal intensive care unit
INPHARM | inpatient pharmacy | Description: A location that plays the role of delivering services which may include providing judicious, safe, efficacious, appropriate and cost effective use of medicines for treatment of patients for visits longer than one day. The distinction between inpatient pharmacies and retail (or outpatient) pharmacies is that they are part of a patient's continuity of care while staying in the hospital.
MBL | medical laboratory | Description: A location that plays the role of delivering services which include biochemistry, hematology, microbiology, immunochemistry, and toxicology.
NCCS | Neurology critical care and stroke unit | Neurology critical care and stroke unit
NS | Neurosurgery unit | Neurosurgery unit
OUTPHARM | outpatient pharmacy | Description: A location that plays the role of delivering services which may include providing judicious, safe, efficacious, appropriate and cost effective use of medicines for treatment of patients for outpatient visits and may also be used for discharge prescriptions.
PEDU | Pediatric unit | Pediatric unit
PHU | Psychiatric hospital unit | (X12N 273R00000N)
RHU | Rehabilitation hospital unit | Rehabilitation hospital unit
SLEEP | Sleep disorders unit | (X12N 261QA1200N)
NCCF | Nursing or custodial care facility | Nursing or custodial care facility
SNF | Skilled nursing facility | (X12N 314000000N)
OF | Outpatient facility | Outpatient facility
ALL | Allergy clinic | Allergy clinic
AMPUT | Amputee clinic | Amputee clinic
BMTC | Bone marrow transplant clinic | Bone marrow transplant clinic
BREAST | Breast clinic | Breast clinic
CANC | Child and adolescent neurology clinic | Child and adolescent neurology clinic
CAPC | Child and adolescent psychiatry clinic | Child and adolescent psychiatry clinic
CARD | Ambulatory Health Care Facilities; Clinic/Center; Rehabilitation: Cardiac Facilities | Ambulatory Health Care Facilities; Clinic/Center; Rehabilitation: Cardiac Facilities
PEDCARD | Pediatric cardiology clinic | Pediatric cardiology clinic
COAG | Coagulation clinic | Coagulation clinic
CRS | Colon and rectal surgery clinic | Colon and rectal surgery clinic
DERM | Dermatology clinic | Dermatology clinic
ENDO | Endocrinology clinic | Endocrinology clinic
PEDE | Pediatric endocrinology clinic | Pediatric endocrinology clinic
ENT | Otorhinolaryngology clinic | Otorhinolaryngology clinic
FMC | Family medicine clinic | Family medicine clinic
GI | Gastroenterology clinic | Gastroenterology clinic
PEDGI | Pediatric gastroenterology clinic | Pediatric gastroenterology clinic
GIM | General internal medicine clinic | General internal medicine clinic
GYN | Gynecology clinic | Gynecology clinic
HEM | Hematology clinic | Hematology clinic
PEDHEM | Pediatric hematology clinic | Pediatric hematology clinic
HTN | Hypertension clinic | Hypertension clinic
IEC | Impairment evaluation center | Focuses on assessing disability
INFD | Infectious disease clinic | Infectious disease clinic
PEDID | Pediatric infectious disease clinic | Pediatric infectious disease clinic
INV | Infertility clinic | Infertility clinic
LYMPH | Lympedema clinic | Lympedema clinic
MGEN | Medical genetics clinic | Medical genetics clinic
NEPH | Nephrology clinic | Nephrology clinic
PEDNEPH | Pediatric nephrology clinic | Pediatric nephrology clinic
NEUR | Neurology clinic | Neurology clinic
OB | Obstetrics clinic | Obstetrics clinic
OMS | Oral and maxillofacial surgery clinic | Oral and maxillofacial surgery clinic
ONCL | Medical oncology clinic | Medical oncology clinic
PEDHO | Pediatric oncology clinic | Pediatric oncology clinic
OPH | Opthalmology clinic | Opthalmology clinic
OPTC | optometry clinic | Description: A location that plays the role of delivering services which may include examination, diagnosis, treatment, management, and prevention of diseases and disorders of the eye as well as prescribing and fitting appropriate corrective lenses (glasses or contact lenses) as needed. Optometry clinics may also provide tests for visual field screening, measuring intra-ocular pressure and ophthalmoscopy, as and when required.
ORTHO | Orthopedics clinic | Orthopedics clinic
HAND | Hand clinic | Hand clinic
PAINCL | Pain clinic | (X12N 261QP3300N)
PC | Primary care clinic | (X12N 261QP2300N)
PEDC | Pediatrics clinic | Pediatrics clinic
PEDRHEUM | Pediatric rheumatology clinic | Pediatric rheumatology clinic
POD | Podiatry clinic | (X12N 261QP1100N)
PREV | Preventive medicine clinic | Preventive medicine clinic

PROCTO | Proctology clinic | Proctology clinic
PROFF | Provider's Office | Location where healthcare service was delivered, identified as the healthcare provider's practice office.
PROS | Prosthodontics clinic | Prosthodontics clinic
PSI | Psychology clinic | Psychology clinic
PSY | Psychiatry clinic | Psychiatry clinic
RHEUM | Rheumatology clinic | Rheumatology clinic
SPMED | Sports medicine clinic | Sports medicine clinic
SU | Surgery clinic | Surgery clinic
PLS | Plastic surgery clinic | Plastic surgery clinic
URO | Urology clinic | Urology clinic
TR | Transplant clinic | Transplant clinic
TRAVEL | Travel and geographic medicine clinic | Travel and geographic medicine clinic
WND | Wound clinic | Wound clinic
RTF | Residential treatment facility | Residential treatment facility
PRC | Pain rehabilitation center | Pain rehabilitation center
SURF | Substance use rehabilitation facility | (X12N 324500000N)
DADDR | Delivery Address | Location address where medical supplies were transported to for use.
MOBL | Mobile Unit | Location (mobile) where healthcare service was delivered.
AMB | Ambulance | Location (mobile) where healthcare service was delivered, identified specifically as an ambulance.
PHARM | Pharmacy | Location where healthcare service was delivered, identified as a pharmacy.
ACC | accident site | Location of an accident where healthcare service was delivered, such as a roadside.
COMM | Community Location | Community location where healthcare is delivered.
CSC | community service center | Description: A location that plays the role of delivering services which may include providing front-line services to the population of a defined geographic area such as: healthcare services and social services, preventive or curative, rehabilitation or reintegration.
PTRES | Patient's Residence | location where healthcare was delivered which is the residence of the Patient.
SCHOOL | school | Location where healthcare service was delivered, identified as a school or educational facility.
UPC | underage protection center | Description: A location that plays the role of delivering services which may include: social emergency services required for a young person as required under any jurisdictional youth laws, child placement, and family mediation in the defined geographical area the SDL is responsible for. It may provide expertise in a judiciary setting on child custody, adoption and biological history research.
WORK | work site | Location where healthcare service was delivered, identified as a work place.

4.4.2.17893 HL7 v3 Value Set ActClassROI

URL: https://hl7.org/fhir/R4/v3/ActClassROI/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActClassROI) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActClassROI
Version: | 2014-03-26
Name: | v3.ActClassROI
Title: | V3 Value SetActClassROI
Definition: | Regions of Interest (ROI) within a subject Act. Primarily used for making secondary observations on a subset of a subject observation. The relationship between a ROI and its referenced Act is specified through an ActRelationship of type "subject" (SUBJ), which must always be present.
OID: | 2.16.840.1.113883.1.11.17893 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.17893.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActClass where concept is-a _ActClassROI

http://terminology.hl7.org/CodeSystem/v3-ActClass

Code | Display
_ActClassROI | ActClassROI | Regions of Interest (ROI) within a subject Act. Primarily used for making secondary observations on a subset of a subject observation. The relationship between a ROI and its referenced Act is specified through an ActRelationship of type "subject" (SUBJ), which must always be present.

4.4.2.17893.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 2 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActClass

Code | Display | Definition
ROIBND | bounded ROI | A Region of Interest (ROI) specified for a multidimensional observation, such as an Observation Series (OBSSER). The ROI is specified using a set of observation criteria, each delineating the boundary of the region in one of the

dimensions in the multidimensional observation. The relationship between a ROI and its referenced Act is specified through an ActRelationship of type subject (SUBJ), which must always be present. Each of the boundary criteria observations is connected with the ROI using ActRelationships of type "has component" (COMP). In each boundary criterion, the Act.code names the dimension and the Observation.value specifies the range of values inside the region. Typically the bounded dimension is continuous, and so the Observation.value will be an interval (IVL) data type. The Observation.value need not be specified if the respective dimension is only named but not constrained. For example, an ROI for the QT interval of a certain beat in ECG Lead II would contain 2 boundary criteria, one naming the interval in time (constrained), and the other naming the interval in ECG Lead II (only named, but not constrained).

ROIOVL | overlay ROI | A Region of Interest (ROI) specified for an image using an overlay shape. Typically used to make reference to specific regions in images, e.g., to specify the location of a radiologic finding in an image or to specify the site of a physical finding by "circling" a region in a schematic picture of a human body. The units of the coordinate values are in pixels. The origin is in the upper left hand corner, with positive X values going to the right and positive Y values going down. The relationship between a ROI and its referenced Act is specified through an ActRelationship of type "subject" (SUBJ), which must always be present.

4.4.2.18938 HL7 v3 Value Set ActClassDocument

URL: https://hl7.org/fhir/R4/v3/ActClassDocument/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActClassDocument) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActClassDocument
Version: | 2014-03-26
Name: | v3.ActClassDocument
Title: | V3 Value SetActClassDocument
Definition: | Specialization of Act to add the characteristics unique to document management services.
OID: | 2.16.840.1.113883.1.11.18938 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.18938.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActClass where concept is-a DOC

4.4.2.18938.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 3 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActClass

Lvl | Code | Display | Definition

0 | DOC | document | The notion of a document comes particularly from the paper world, where it corresponds to the contents recorded on discrete pieces of paper. In the electronic world, a document is a kind of composition that bears resemblance to their paper world counter-parts. Documents typically are meant to be human-readable. HL7's notion of document differs from that described in the W3C XML Recommendation, in which a document refers specifically to the contents that fall between the root element's start-tag and end-tag. Not all XML documents are HL7 documents.

1 | DOCCLIN | clinical document | A clinical document is a documentation of clinical observations and services, with the following characteristics: Persistence - A clinical document continues to exist in an unaltered state, for a time period defined by local and regulatory requirements; Stewardship - A clinical document is maintained by a person or organization entrusted with its care; Potential for authentication - A clinical document is an assemblage of information that is intended to be legally authenticated; Wholeness - Authentication of a clinical document applies to the whole and does not apply to portions of the document without the full context of the document; Human readability - A clinical document is human readable.

2 | CDALVLONE | CDA Level One clinical document | A clinical document that conforms to Level One of the HL7 Clinical Document Architecture (CDA)

4.4.2.18977 HL7 v3 Value Set ActRelationshipConditional

URL: https://hl7.org/fhir/R4/v3/ActRelationshipConditional/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActRelationshipConditional) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActRelationshipConditional
Version: | 2014-03-26
Name: | v3.ActRelationshipConditional
Title: | V3 Value SetActRelationshipConditional
Definition: | Specifies under what circumstances (target Act) the source-Act may, must, must not or has occurred
OID: | 2.16.840.1.113883.1.11.18977 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.18977.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType where concept is-a _ActRelationshipConditional

http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType

Code | Display

_ActRelationshipConditional | ActRelationshipConditional | Specifies under what circumstances (target Act) the source-Act may, must, must not or has occurred

4.4.2.18977.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 17 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActRelationshipType

Code | Display | Definition

CIND | has contra-indication | A contraindication is just a negation of a reason, i.e. it gives a condition under which the action is not to be done. Both, source and target can be any kind of service; target service is in criterion mood. How the strength of a contraindication is expressed (e.g., relative, absolute) is left as an open issue. The priorityNumber attribute could be used.

PRCN | has pre-condition | A requirement to be true before a service is performed. The target can be any service in criterion mood. For multiple pre-conditions a conjunction attribute (AND, OR, XOR) is applicable.

RSON | has reason | Description: The reason or rationale for a service. A reason link is weaker than a trigger, it only suggests that some service may be or might have been a reason for some action, but not that this reason requires/required the action to be taken. Also, as opposed to the trigger, there is no strong timely relation between the reason and the action. As well as providing various types of information about the rationale for a service, the RSON act relationship is routinely used between a SBADM act and an OBS act to describe the indication for use of a medication. Child concepts may be used to describe types of indication. Discussion: In prior releases, the code "SUGG" (suggests) was expressed as "an inversion of the reason link." That code has been retired in favor of the inversion indicator that is an attribute of ActRelationship.

BLOCK | blocks | Definition: The source act is performed to block the effects of the target act. This act relationship should be used when describing near miss type incidents where potential harm could have occurred, but the action described in the source act blocked the potential harmful effects of the incident actually occurring.

DIAG | diagnoses | Description: The source act is intended to help establish the presence of a (an adverse) situation described by the target act. This is not limited to diseases but can apply to any adverse situation or condition of medical or technical nature.

IMM | immunization against | Description: The source act is intented to provide immunity against the effects of the target act (the target act describes an infectious disease)

ACTIMM | active immunization against | Description: The source act is intended to provide active immunity against the effects of the target act (the target act describes an infectious disease)

PASSIMM | passive immunization against | Description: The source act is intended to provide passive immunity against the effects of the target act (the target act describes an infectious disease).

MITGT | mitigates | The source act removes or lessens the occurrence or effect of the target act.

RCVY | recovers | Definition: The source act is performed to recover from the effects of the target act.

PRYLX | prophylaxis of | Description: The source act is intended to reduce the risk of of an adverse situation to emerge as described by the target act. This is not limited to diseases but can apply to any adverse situation or condition of medical or technical nature.

TREAT | treats | Description: The source act is intended to improve a pre-existing adverse situation described by the target act. This is not limited to diseases but can apply to any adverse situation or condition of medical or technical nature.

ADJUNCT | adjunctive treatment | Description: The source act is intended to offer an additional treatment for the management or cure of a pre-existing adverse situation described by the target act. This is not limited to diseases but can apply to any adverse situation or condition of medical or technical nature. It is not a requirement that the non-adjunctive treatment is explicitly specified.

MTREAT | maintenance treatment | Description: The source act is intended to provide long term maintenance improvement or management of a pre-existing adverse situation described by the target act. This is not limited to diseases but can apply to any adverse situation or condition of medical or technical nature.

PALLTREAT | palliates | Description: The source act is intended to provide palliation for the effects of the target act.

SYMP | symptomatic relief | Description: The source act is intented to provide symptomatic relief for the effects of the target act.

TRIG | has trigger | A pre-condition that if true should result in the source Act being executed. The target is in typically in criterion mood. When reported after the fact (i.e. the criterion has been met) it may be in Event mood. A delay between the trigger and the triggered action can be specified. Discussion: This includes the concept of a required act for a service or financial instrument such as an insurance plan or policy. In such cases, the trigger is the occurrence of a specific condition such as coverage limits being exceeded.

4.4.2.19032 HL7 v3 Value Set ParticipationIndirectTarget

URL: https://hl7.org/fhir/R4/v3/ParticipationIndirectTarget/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ParticipationIndirectTarget) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ParticipationIndirectTarget

Version: | 2014-03-26
Name: | v3.ParticipationIndirectTarget
Title: | V3 Value SetParticipationIndirectTarget
Definition: | Target that is not substantially present in the act and which is not directly affected by the act, but which will be a focus of the record or documentation of the act.
OID: | 2.16.840.1.113883.1.11.19032 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.19032.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ParticipationType where concept is-a IND

4.4.2.19032.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 8 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ParticipationType version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ParticipationType

Lvl | Code | Display | Definition
0 | IND | indirect target | Target that is not substantially present in the act and which is not directly affected by the act, but which will be a focus of the record or documentation of the act.
1 | BEN | beneficiary | Target on behalf of whom the service happens, but that is not necessarily present in the service. Can occur together with direct target to indicate that a target is both, as in the case where the patient is the indirect beneficiary of a service rendered to a family member, e.g. counseling or given home care instructions. This concept includes a participant, such as a covered party, who derives benefits from a service act covered by a coverage act. Note that the semantic role of the intended recipient who benefits from the happening denoted by the verb in the clause. Thus, a patient who has no coverage under a policy or program may be a beneficiary of a health service while not being the beneficiary of coverage for that service.
1 | CAGNT | causative agent | Definition: A factor, such as a microorganism, chemical substance, or form of radiation, whose presence, excessive presence, or (in deficiency diseases) relative absence is essential, in whole or in part, for the occurrence of a condition. Constraint: The use of this participation is limited to observations.
1 | COV | coverage target | The target participation for an individual in a health care coverage act in which the target role is either the policy holder of the coverage, or a covered party under the coverage.
1 | GUAR | guarantor party | The target person or organization contractually recognized by the issuer as a participant who has assumed fiscal responsibility for another personaTMs financial obligations by guaranteeing to pay for amounts owed to a particular account Example:The subscriber of the patientaTMs health insurance policy signs a contract with the provider to be fiscally responsible for the patient billing account balance amount owed.
1 | HLD | holder | Participant who posses an instrument such as a financial contract (insurance policy) usually based on some agreement with the author.
1 | RCT | record target | The record target indicates whose medical record holds the documentation of this act. This is especially important when the subject of a service is not the patient himself.
1 | RCV | receiver | The person (or organization) who receives the product of an Act.

4.4.2.19105 HL7 v3 Value Set RoleClassPassive

URL: https://hl7.org/fhir/R4/v3/RoleClassPassive/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-RoleClassPassive) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-RoleClassPassive
Version: | 2014-03-26
Name: | v3.RoleClassPassive
Title: | V3 Value SetRoleClassPassive
Definition: | An association for a playing Entity that is used, known, treated, handled, built, or destroyed, etc. under the auspices of the scoping Entity. The playing Entity is passive in these roles (even though it may be active in other roles), in the sense that the kinds of things done to it in this role happen without an agreement from the playing Entity.
OID: | 2.16.840.1.113883.1.11.19105 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.19105.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-RoleClass where concept is-a _RoleClassPassive

http://terminology.hl7.org/CodeSystem/v3-RoleClass

Code | Display
_RoleClassPassive | RoleClassPassive | An association for a playing Entity that is used, known, treated, handled, built, or destroyed, etc. under the auspices of the scoping Entity. The playing Entity is passive in these roles (even though it may be active

in other roles), in the sense that the kinds of things done to it in this role happen without an agreement from the playing Entity.

4.4.2.19105.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 26 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-RoleClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-RoleClass

Code | Display | Definition

ACCESS | access | A role in which the playing entity (material) provides access to another entity. The principal use case is intravenous (or other bodily) access lines that preexist and need to be referred to for medication routing instructions.

ADJY | adjacency | A physical association whereby two Entities are in some (even lose) spatial relationship with each other such that they touch each other in some way. Examples: the colon is connected (and therefore adjacent) to the jejunum; the colon is adjacent to the liver (even if not actually connected.) UsageConstraints: Adjacency is in principle a symmetrical connection, but scoper and player of the role should, where applicable, be assigned to have scoper be the larger, more central Entity and player the smaller, more distant, appendage.

CONC | connection | An adjacency of two Entities held together by a bond which attaches to each of the two entities. Examples: biceps brachii muscle connected to the radius bone, port 3 on a network switch connected to port 5 on a patch panel. UsageConstraints: See Adjacency for the assignment of scoper (larger, more central) and player (smaller, more distant).

BOND | molecular bond | A connection between two atoms of a molecule. Examples: double bond between first and second C in ethane, peptide bond between two amino-acid, disulfide bridge between two proteins, chelate and ion associations, even the much weaker van-der-Waals bonds can be considered molecular bonds. UsageConstraints: See connection and adjacency for the assignment of player and scoper.

CONY | continuity | A connection between two regional parts. Examples: the connection between ascending aorta and the aortic arc, connection between descending colon and sigmoid. UsageConstraints: See connection and adjacency for the assignment of player and scoper.

ADMM | Administerable Material | A material (player) that can be administered to an Entity (scoper).

BIRTHPL | birthplace | Relates a place (playing Entity) as the location where a living subject (scoping Entity) was born.

DEATHPLC | place of death | Definition: Relates a place (playing Entity) as the location where a living subject (scoping Entity) died.

DST | distributed material | A material (player) distributed by a distributor (scoper) who functions between a manufacturer and a buyer or retailer.

RET | retailed material | Material (player) sold by a retailer (scoper), who also give advice to prospective buyers.

EXLOC | event location | A role played by a place at which the location of an event may be recorded.

SDLOC | service delivery location | A role played by a place at which services may be provided.

DSDLOC | dedicated service delivery location | A role of a place (player) that is intended to house the provision of services. Scoper is the Entity (typically Organization) that provides these services. This is not synonymous with "ownership."

ISDLOC | incidental service delivery location | A role played by a place at which health care services may be provided without prior designation or authorization.

EXPR | exposed entity | A role played by an entity that has been exposed to a person or animal suffering a contagious disease, or with a location from which a toxin has been distributed. The player of the role is normally a person or animal, but it is possible that other entity types could become exposed. The role is scoped by the source of the exposure, and it is quite possible for a person playing the role of exposed party to also become the scoper a role played by another person. That is to say, once a person has become infected, it is possible, perhaps likely, for that person to infect others. Management of exposures and tracking exposed parties is a key function within public health, and within most public health contexts - exposed parties are known as "contacts."

HLD | held entity | Entity that is currently in the possession of a holder (scoper), who holds, or uses it, usually based on some agreement with the owner.

HLTHCHRT | health chart | The role of a material (player) that is the physical health chart belonging to an organization (scoper).

IDENT | identified entity | A role in which the scoping entity designates an identifier for a playing entity.

MANU | manufactured product | Scoped by the manufacturer

THER | therapeutic agent | A manufactured material (player) that is used for its therapeutic properties. The manufacturer is the scoper.

MNT | maintained entity | An entity (player) that is maintained by another entity (scoper). This is typical role held by durable equipment. The scoper assumes responsibility for proper operation, quality, and safety.

OWN | owned entity | An Entity (player) for which someone (scoper) is granted by law the right to call the material (player) his own. This entitles the scoper to make decisions about the disposition of that material.

RGPR | regulated product | A product regulated by some governmentatl orgnization. The role is played by Material and scoped by Organization. Rationale: To support an entity clone used to identify the NDC number for a drug product.

TERR | territory of authority | Relates a place entity (player) as the region over which the scoper (typically an Organization) has certain authority (jurisdiction). For example, the Calgary Regional Health Authority (scoper) has authority over the territory "Region 4 of Alberta" (player) in matters of health.

USED | used entity | Description:An entity (player) that is used by another entity (scoper)

WRTE | warranted product | A role a product plays when a guarantee is given to the purchaser by the seller (scoping entity) stating that the product is reliable and free from known defects and that the seller will repair or replace defective parts within a given time limit and under certain conditions.

4.4.2.19313 HL7 v3 Value Set RoleClassAssociative

URL: https://hl7.org/fhir/R4/v3/RoleClassAssociative/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-RoleClassAssociative) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-RoleClassAssociative

Version: | 2014-03-26
Name: | v3.RoleClassAssociative
Title: | V3 Value SetRoleClassAssociative
Definition: | A general association between two entities that is neither partitive nor ontological.
OID: | 2.16.840.1.113883.1.11.19313 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.19313.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-RoleClass where concept is-a _RoleClassAssociative

http://terminology.hl7.org/CodeSystem/v3-RoleClass

Code | Display
_RoleClassAssociative | RoleClassAssociative | A general association between two entities that is neither partitive nor ontological.

4.4.2.19313.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 65 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-RoleClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-RoleClass

Code | Display | Definition
AFFL | affiliate | Player of the Affiliate role has a business/professional relationship with scoper. Player and scoper may be persons or organization. The Affiliate relationship does not imply membership in a group, nor does it exist for resource scheduling purposes. Example: A healthcare provider is affiliated with another provider as a business associate.
AGNT | agent | An entity (player) that acts or is authorized to act on behalf of another entity (scoper).
ASSIGNED | assigned entity | An agent role in which the agent is an Entity acting in the employ of an organization. The focus is on functional role on behalf of the organization, unlike the Employee role where the focus is on the 'Human Resources' relationship between the employee and the organization.
COMPAR | commissioning party | An Entity that is authorized to issue or instantiate permissions, privileges, credentials or other formal/legal authorizations.
SGNOFF | signing authority or officer | The role of a person (player) who is the officer or signature authority for of a scoping entity, usually an organization (scoper).
CON | contact | A person or an organization (player) which provides or receives information regarding another entity (scoper). Examples; patient NOK and emergency contacts; guarantor contact; employer contact.
ECON | emergency contact | An entity to be contacted in the event of an emergency.
NOK | next of kin | An individual designated for notification as the next of kin for a given entity.
GUARD | guardian | Guardian of a ward
CIT | citizen | Citizen of apolitical entity
COVPTY | covered party | A role class played by a person who receives benefit coverage under the terms of a particular insurance policy. The underwriter of that policy is the scoping entity. The covered party receives coverage because of some contractual or other relationship with the holder of that policy. Discussion:This reason for coverage is captured in 'Role.code' and a relationship link with type code of indirect authority should be included using the policy holder role as the source, and the covered party role as the target. Note that a particular policy may cover several individuals one of whom may be, but need not be, the policy holder. Thus the notion of covered party is a role that is distinct from that of the policy holder.
CLAIM | claimant | Description: A role played by a party making a claim for coverage under a policy or program. A claimant must be either a person or organization, or a group of persons or organizations. A claimant is not a named insured or a program eligible. Discussion: With respect to liability insurance such as property and casualty insurance, a claimant must file a claim requesting indemnification for a loss that the claimant considers covered under the policy of a named insured. The claims adjuster for the policy underwriter will review the claim to determine whether the loss meets the benefit coverage criteria under a policy, and base any indemnification or coverage payment on that review. If a third party is liable in whole or part for the loss, the underwriter may pursue third party liability recovery. A claimant may be involved in civil or criminal legal proceedings involving claims against a defendant party that is indemnified by an insurance policy or to protest the finding of a claims adjustor. With respect to life insurance, a beneficiary designated by a named insured becomes a claimant of the proceeds of coverage, as in the case of a life insurance policy. However, a claimant for coverage under life insurance is not necessarily a designated beneficiary. Note: A claimant is not a named insured. However, a named insured may make a claim under a policy, e.g., an insured driver may make a claim for an injury under his or her comprehensive automobile insurance policy. Similarly, a program eligible may make a claim under program, e.g., an unemployed worker may claim benefits under an unemployment insurance program, but parties playing these covered party role classes are not, for purposes of this vocabulary and in an effort to clearly distinguish role classes, considered claimants. In the case of a named insured making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that either a named insured or an individual insured has filed a claim for a loss. In the case of a program eligible, a role type code INJWKR (injured worker) subtypes the class to indicate that the covered party in a workers compensation program is an injured worker, and as such, has filed a "claim" under the program for benefits. Likewise, a covered role type code UNEMP (unemployed worker) subtypes the program eligible class to indicate that the covered party in an unemployment insurance program has filed a claim for unemployment benefits. Example: A claimant under automobile policy that is not the named insured.
NAMED | named insured | Description: A role played by a party to an insurance policy to which the insurer agrees to indemnify for losses, provides benefits for, or renders services. A named insured may be either a person, non-person living subject, or an organization, or a group of persons, non-person living subjects, or organizations. Discussion: The coded concept NAMED should not be used where a more specific child concept in this Specializable value set applies. In some cases, the named insured may not be the policy holder, e.g., where a policy holder purchases life insurance policy in which another party is the named insured and the policy holder is the beneficiary of the policy. Note: The party playing the role of a named insured is not a claimant in the sense

conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a named insured may make a claim under a policy, e.g., e.g., a party that is the named insured and policy holder under a comprehensive automobile insurance policy may become the claimant for coverage under that policy e.g., if injured in an automobile accident and there is no liable third party. In the case of a named insured making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that a named insured has filed a claim for a loss. Example: The named insured under a comprehensive automobile, disability, or property and casualty policy that is the named insured and may or may not be the policy holder.

DEPEN | dependent | Description: A role played by a person covered under a policy or program based on an association with a subscriber, which is recognized by the policy holder. Note: The party playing the role of a dependent is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a dependent may make a claim under a policy, e.g., a dependent under a health insurance policy may become the claimant for coverage under that policy for wellness examines or if injured and there is no liable third party. In the case of a dependent making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that the dependent has filed a claim for services covered under the health insurance policy. Example: The dependent has an association with the subscriber such as a financial dependency or personal relationship such as that of a spouse, or a natural or adopted child. The policy holder may be required by law to recognize certain associations or may have discretion about the associations. For example, a policy holder may dictate the criteria for the dependent status of adult children who are students, such as requiring full time enrollment, or may recognize domestic partners as dependents. Under certain circumstances, the dependent may be under the indirect authority of a responsible party acting as a surrogate for the subscriber, for example, if the subscriber is differently abled or deceased, a guardian ad Lidem or estate executor may be appointed to assume the subscriberaTMs legal standing in the relationship with the dependent.

INDIV | individual | Description: A role played by a party covered under a policy as the policy holder. An individual may be either a person or an organization. Note: The party playing the role of an individual insured is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a named insured may make a claim under a policy, e.g., a party that is the named insured and policy holder under a comprehensive automobile insurance policy may become the claimant for coverage under that policy if injured in an automobile accident and there is no liable third party. In the case of an individual insured making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that an individual insured has filed a claim for a loss. Example: The individual insured under a comprehensive automobile, disability, or property and casualty policy that is the policy holder.

SUBSCR | subscriber | Description: A role played by a person covered under a policy based on association with a sponsor who is the policy holder, and whose association may provide for the eligibility of dependents for coverage. Discussion: The policy holder holds the contract with the policy or program underwriter. The subscriber holds a certificate of coverage under the contract. In legal proceedings concerning the policy or program, the terms of the contract takes precedence over the terms of the certificate of coverage if there are any inconsistencies. Note: The party playing the role of a subscriber is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a subscriber may make a claim under a policy, e.g., a subscriber under a health insurance policy may become the claimant for coverage under that policy for wellness examines or if injured and there is no liable third party. In the case of a subscriber making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that the subscriber has filed a claim for services covered under the health insurance policy. Example: An employee or a member of an association.

PROG | program eligible | Description: A role played by a party that meets the eligibility criteria for coverage under a program. A program eligible may be either a person, non-person living subject, or an organization, or a group of persons, non-person living subjects, or organizations. Discussion: A program as typically government administered coverage for parties determined eligible under the terms of the program. Note: The party playing a program eligible is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However a program eligible may make a claim under program, e.g., an unemployed worker may claim benefits under an unemployment insurance program, but parties playing these covered party role classes are not, for purposes of this vocabulary and in an effort to clearly distinguish role classes, considered claimants. In the case of a program eligible, a role type code INJWKR (injured worker) subtypes the class to indicate that the covered party in a workers compensation program is an injured worker, and as such, has filed a "claim" under the program for benefits. Likewise, a covered role type code UNEMP (unemployed worker) subtypes the program eligible class to indicate that the covered party in an unemployment insurance program has filed a claim for unemployment benefits. Example: A party meeting eligibility criteria related to health or financial status, e.g., in the U.S., persons meeting health, demographic, or financial criteria established by state and federal law are eligible for Medicaid.

CRINV | clinical research investigator | A role played by a provider, always a person, who has agency authority from a Clinical Research Sponsor to direct the conduct of a clinical research trial or study on behalf of the sponsor.

CRSPNSR | clinical research sponsor | A role played by an entity, usually an organization, that is the sponsor of a clinical research trial or study. The sponsor commissions the study, bears the expenses, is responsible for satisfying all legal requirements concerning subject safety and privacy, and is generally responsible for collection, storage and analysis of the data generated during the trial. No scoper is necessary, as a clinical research sponsor undertakes the role on its own authority and declaration. Clinical research sponsors are usually educational or other research organizations, government agencies or biopharmaceutical companies.

EMP | employee | A relationship between a person or organization and a person or organization formed for the purpose of exchanging work for compensation. The purpose of the role is to identify the type of relationship the employee has to the employer, rather than the nature of the work actually performed. (Contrast with AssignedEntity.)

MIL | military person | A role played by a member of a military service. Scoper is the military service (e.g. Army, Navy, Air Force, etc.) or, more specifically, the unit (e.g. Company C, 3rd Battalion, 4th Division, etc.)

GUAR | guarantor | A person or organization (player) that serves as a financial guarantor for another person or organization (scoper).

INVSBJ | Investigation Subject | An entity that is the subject of an investigation. This role is scoped by the party responsible for the investigation.

CASEBJ | Case Subject | A person, non-person living subject, or place that is the subject of an investigation related to a notifiable condition (health circumstance that is reportable within the applicable public health jurisdiction)

RESBJ | research subject | Definition:Specifies the administrative functionality within a formal experimental design for which the ResearchSubject role was established. Examples: Screening - role is used for pre-enrollment evaluation portion of the design; enrolled - role is used for subjects admitted to the experimental portion of the design.

LIC | licensed entity | A relationship in which the scoper certifies the player (e. g. a medical care giver, a medical device or a provider organization) to perform certain activities that fall under the jurisdiction of the scoper (e.g., a health authority licensing healthcare providers, a medical quality authority certifying healthcare professionals).

NOT | notary public | notary public

PROV | healthcare provider | An Entity (player) that is authorized to provide health care services by some authorizing agency (scoper).

PAT | patient | A Role of a LivingSubject (player) as an actual or potential recipient of health care services from a healthcare provider organization (scoper). Usage Note: Communication about relationships between patients and specific healthcare practitioners (people) is not done via scoper. Instead this is generally done using the CareProvision act. This allows linkage between patient and a particular healthcare practitioner role and also allows description of the type of care involved in the relationship.

PAYEE | payee | The role of an organization or individual designated to receive payment for a claim against a particular coverage. The scoping entity is the organization that is the submitter of the invoice in question.

PAYOR | invoice payor | The role of an organization that undertakes to accept claims invoices, assess the coverage or payments due for those invoices and pay to the designated payees for those invoices. This role may be either the underwriter or a third-party organization authorized by the underwriter. The scoping entity is the organization that underwrites the claimed coverage.

POLHOLD | policy holder | A role played by a person or organization that holds an insurance policy. The underwriter of that policy is the scoping entity. Discussion:The identifier of the policy is captured in 'Role.id' when the Role is a policy holder. A particular policy may cover several individuals one of whom may be, but need not be, the policy holder. Thus the notion of covered party is a role that is distinct from that of the policy holder.

QUAL | qualified entity | An entity (player) that has been recognized as having certain training/experience or other characteristics that would make said entity an appropriate performer for a certain activity. The scoper is an organization that educates or qualifies entities.

SPNSR | coverage sponsor | A role played by an entity, usually an organization that is the sponsor of an insurance plan or a health program. A sponsor is the party that is ultimately accountable for the coverage by employment contract or by law. A sponsor can be an employer, union, government agency, or association. Fully insured sponsors establish the terms of the plan and contract with health insurance plans to assume the risk and to administer the plan. Self-insured sponsors delegate coverage administration, but not risk, to third-party administrators. Program sponsors designate services to be covered in accordance with statute. Program sponsors may administer the coverage themselves, delegate coverage administration, but not risk to third-party administrators, or contract with health insurance plans to assume the risk and administrator a program. Sponsors qualify individuals who may become a policy holder of the plan; where the sponsor is the policy holder, who may become a subscriber or a dependent to a policy under the plan; or where the sponsor is a government agency, who may become program eligibles under a program. The sponsor role may be further qualified by the SponsorRole.code. Entities playing the sponsor role may also play the role of a Coverage Administrator. Example: An employer, union, government agency, or association.

STD | student | A role played by an individual who is a student of a school, which is the scoping entity.

UNDWRT | underwriter | A role played by a person or an organization. It is the party that accepts fiscal responsibility for insurance plans and the policies created under those plans; administers and accepts fiscal responsibility for a program that provides coverage for services to eligible individuals; and/or has the responsibility to assess the merits of each risk and decide a suitable premium for accepting all or part of the risk. If played by an organization, this role may be further specified by an appropriate RoleCode. Example: A health insurer; Medicaid Program; Lloyd's of London

CAREGIVER | caregiver | A person responsible for the primary care of a patient at home.

PRS | personal relationship | Links two entities with classCode PSN (person) in a personal relationship. The character of the relationship must be defined by a PersonalRelationshipRoleType code. The player and scoper are determined by PersonalRelationshipRoleType code as well.

SELF | self | The "same" roleclass asserts an identity between playing and scoping entities: that they are in fact instances of the same entity and, in the case of discrepancies (e.g different DOB, gender), that one or both are in error. Usage: playing and scoping entities must have same classcode, but need not have identical attributes or values. Example: a provider registry maintains sets of conflicting demographic data for what is reported to be the same individual.

ACCESS | access | A role in which the playing entity (material) provides access to another entity. The principal use case is intravenous (or other bodily) access lines that preexist and need to be referred to for medication routing instructions.

ADJY | adjacency | A physical association whereby two Entities are in some (even lose) spatial relationship with each other such that they touch each other in some way. Examples: the colon is connected (and therefore adjacent) to the jejunum; the colon is adjacent to the liver (even if not actually connected.) UsageConstraints: Adjacency is in principle a symmetrical connection, but scoper and player of the role should, where applicable, be assigned to have scoper be the larger, more central Entity and player the smaller, more distant, appendage.

CONC | connection | An adjacency of two Entities held together by a bond which attaches to each of the two entities. Examples: biceps brachii muscle connected to the radius bone, port 3 on a network switch connected to port 5 on a patch panel. UsageConstraints: See Adjacency for the assignment of scoper (larger, more central) and player (smaller, more distant).

BOND | molecular bond | A connection between two atoms of a molecule. Examples: double bond between first and second C in ethane, peptide bond between two amino-acid, disulfide bridge between two proteins, chelate and ion associations, even the much weaker van-der-Waals bonds can be considered molecular bonds. UsageConstraints: See connection and adjacency for the assignment of player and scoper.

CONY | continuity | A connection between two regional parts. Examples: the connection between ascending aorta and the aortic arc, connection between descending colon and sigmoid. UsageConstraints: See connection and adjacency for the assignment of player and scoper.

ADMM | Administerable Material | A material (player) that can be administered to an Entity (scoper).

BIRTHPL | birthplace | Relates a place (playing Entity) as the location where a living subject (scoping Entity) was born.

DEATHPLC | place of death | Definition: Relates a place (playing Entity) as the location where a living subject (scoping Entity) died.

DST | distributed material | A material (player) distributed by a distributor (scoper) who functions between a manufacturer and a buyer or retailer.

RET | retailed material | Material (player) sold by a retailer (scoper), who also give advice to prospective buyers.

EXLOC | event location | A role played by a place at which the location of an event may be recorded.

SDLOC | service delivery location | A role played by a place at which services may be provided.

DSDLOC | dedicated service delivery location | A role of a place (player) that is intended to house the provision of services. Scoper is the Entity (typically Organization) that provides these services. This is not synonymous with "ownership."

ISDLOC | incidental service delivery location | A role played by a place at which health care services may be provided without prior designation or authorization.

EXPR | exposed entity | A role played by an entity that has been exposed to a person or animal suffering a contagious disease, or with a location from which a toxin has been distributed. The player of the role is normally a person or animal, but it is possible that other entity types could become exposed. The role is scoped by the source of the exposure, and it is quite possible for a person playing the role of exposed party to also become the scoper a role played by another person. That is to say, once a person has become infected, it is possible, perhaps likely, for that person to infect others. Management of exposures and tracking exposed

parties is a key function within public health, and within most public health contexts - exposed parties are known as "contacts."
HLD | held entity | Entity that is currently in the possession of a holder (scoper), who holds, or uses it, usually based on some agreement with the owner.
HLTHCHRT | health chart | The role of a material (player) that is the physical health chart belonging to an organization (scoper).
IDENT | identified entity | A role in which the scoping entity designates an identifier for a playing entity.
MANU | manufactured product | Scoped by the manufacturer
THER | therapeutic agent | A manufactured material (player) that is used for its therapeutic properties. The manufacturer is the scoper.
MNT | maintained entity | An entity (player) that is maintained by another entity (scoper). This is typical role held by durable equipment. The scoper assumes responsibility for proper operation, quality, and safety.
OWN | owned entity | An Entity (player) for which someone (scoper) is granted by law the right to call the material (player) his own. This entitles the scoper to make decisions about the disposition of that material.
RGPR | regulated product | A product regulated by some governmentatl orgnization. The role is played by Material and scoped by Organization. Rationale: To support an entity clone used to identify the NDC number for a drug product.
TERR | territory of authority | Relates a place entity (player) as the region over which the scoper (typically an Organization) has certain authority (jurisdiction). For example, the Calgary Regional Health Authority (scoper) has authority over the territory "Region 4 of Alberta" (player) in matters of health.
USED | used entity | Description:An entity (player) that is used by another entity (scoper)
WRTE | warranted product | A role a product plays when a guarantee is given to the purchaser by the seller (scoping entity) stating that the product is reliable and free from known defects and that the seller will repair or replace defective parts within a given time limit and under certain conditions.

4.4.2.19316 HL7 v3 Value Set RoleClassMutualRelationship

URL: https://hl7.org/fhir/R4/v3/RoleClassMutualRelationship/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-RoleClassMutualRelationship) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-RoleClassMutualRelationship
Version: | 2014-03-26
Name: | v3.RoleClassMutualRelationship
Title: | V3 Value SetRoleClassMutualRelationship
Definition: | A relationship that is based on mutual behavior of the two Entities as being related. The basis of such relationship may be agreements (e.g., spouses, contract parties) or they may bede facto behavior (e.g. friends) or may be an incidental involvement with each other (e.g. parties over a dispute, siblings, children).
OID: | 2.16.840.1.113883.1.11.19316 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.19316.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-RoleClass where concept is-a _RoleClassMutualRelationship

http://terminology.hl7.org/CodeSystem/v3-RoleClass

Code | Display
_RoleClassMutualRelationship | RoleClassMutualRelationship | A relationship that is based on mutual behavior of the two Entities as being related. The basis of such relationship may be agreements (e.g., spouses, contract parties) or they may be de facto behavior (e.g. friends) or may be an incidental involvement with each other (e.g. parties over a dispute, siblings, children).

4.4.2.19316.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 39 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-RoleClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-RoleClass

Code | Display | Definition
AFFL | affiliate | Player of the Affiliate role has a business/professional relationship with scoper. Player and scoper may be persons or organization. The Affiliate relationship does not imply membership in a group, nor does it exist for resource scheduling purposes. Example: A healthcare provider is affiliated with another provider as a business associate.
AGNT | agent | An entity (player) that acts or is authorized to act on behalf of another entity (scoper).
ASSIGNED | assigned entity | An agent role in which the agent is an Entity acting in the employ of an organization. The focus is on functional role on behalf of the organization, unlike the Employee role where the focus is on the 'Human Resources' relationship between the employee and the organization.
COMPAR | commissioning party | An Entity that is authorized to issue or instantiate permissions, privileges, credentials or other formal/legal authorizations.
SGNOFF | signing authority or officer | The role of a person (player) who is the officer or signature authority for of a scoping entity, usually an organization (scoper).

CON | contact | A person or an organization (player) which provides or receives information regarding another entity (scoper). Examples; patient NOK and emergency contacts; guarantor contact; employer contact.

ECON | emergency contact | An entity to be contacted in the event of an emergency.

NOK | next of kin | An individual designated for notification as the next of kin for a given entity.

GUARD | guardian | Guardian of a ward

CIT | citizen | Citizen of apolitical entity

COVPTY | covered party | A role class played by a person who receives benefit coverage under the terms of a particular insurance policy. The underwriter of that policy is the scoping entity. The covered party receives coverage because of some contractual or other relationship with the holder of that policy. Discussion:This reason for coverage is captured in 'Role.code' and a relationship link with type code of indirect authority should be included using the policy holder role as the source, and the covered party role as the target. Note that a particular policy may cover several individuals one of whom may be, but need not be, the policy holder. Thus the notion of covered party is a role that is distinct from that of the policy holder.

CLAIM | claimant | Description: A role played by a party making a claim for coverage under a policy or program. A claimant must be either a person or organization, or a group of persons or organizations. A claimant is not a named insured or a program eligible. Discussion: With respect to liability insurance such as property and casualty insurance, a claimant must file a claim requesting indemnification for a loss that the claimant considers covered under the policy of a named insured. The claims adjuster for the policy underwriter will review the claim to determine whether the loss meets the benefit coverage criteria under a policy, and base any indemnification or coverage payment on that review. If a third party is liable in whole or part for the loss, the underwriter may pursue third party liability recovery. A claimant may be involved in civil or criminal legal proceedings involving claims against a defendant party that is indemnified by an insurance policy or to protest the finding of a claims adjustor. With respect to life insurance, a beneficiary designated by a named insured becomes a claimant of the proceeds of coverage, as in the case of a life insurance policy. However, a claimant for coverage under life insurance is not necessarily a designated beneficiary. Note: A claimant is not a named insured. However, a named insured may make a claim under a policy, e.g., an insured driver may make a claim for an injury under his or her comprehensive automobile insurance policy. Similarly, a program eligible may make a claim under program, e.g., an unemployed worker may claim benefits under an unemployment insurance program, but parties playing these covered party role classes are not, for purposes of this vocabulary and in an effort to clearly distinguish role classes, considered claimants. In the case of a named insured making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that either a named insured or an individual insured has filed a claim for a loss. In the case of a program eligible, a role type code INJWKR (injured worker) subtypes the class to indicate that the covered party in a workers compensation program is an injured worker, and as such, has filed a "claim" under the program for benefits. Likewise, a covered role type code UNEMP (unemployed worker) subtypes the program eligible class to indicate that the covered party in an unemployment insurance program has filed a claim for unemployment benefits. Example: A claimant under automobile policy that is not the named insured.

NAMED | named insured | Description: A role played by a party to an insurance policy to which the insurer agrees to indemnify for losses, provides benefits for, or renders services. A named insured may be either a person, non-person living subject, or an organization, or a group of persons, non-person living subjects, or organizations. Discussion: The coded concept NAMED should not be used where a more specific child concept in this Specializable value set applies. In some cases, the named insured may not be the policy holder, e.g., where a policy holder purchases life insurance policy in which another party is the named insured and the policy holder is the beneficiary of the policy. Note: The party playing the role of a named insured is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a named insured may make a claim under a policy, e.g., e.g., a party that is the named insured and policy holder under a comprehensive automobile insurance policy may become the claimant for coverage under that policy e.g., if injured in an automobile accident and there is no liable third party. In the case of a named insured making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that a named insured has filed a claim for a loss. Example: The named insured under a comprehensive automobile, disability, or property and casualty policy that is the named insured and may or may not be the policy holder.

DEPEN | dependent | Description: A role played by a person covered under a policy or program based on an association with a subscriber, which is recognized by the policy holder. Note: The party playing the role of a dependent is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a dependent may make a claim under a policy, e.g., a dependent under a health insurance policy may become the claimant for coverage under that policy for wellness examines or if injured and there is no liable third party. In the case of a dependent making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that the dependent has filed a claim for services covered under the health insurance policy. Example: The dependent has an association with the subscriber such as a financial dependency or personal relationship such as that of a spouse, or a natural or adopted child. The policy holder may be required by law to recognize certain associations or may have discretion about the associations. For example, a policy holder may dictate the criteria for the dependent status of adult children who are students, such as requiring full time enrollment, or may recognize domestic partners as dependents. Under certain circumstances, the dependent may be under the indirect authority of a responsible party acting as a surrogate for the subscriber, for example, if the subscriber is differently abled or deceased, a guardian ad Lidem or estate executor may be appointed to assume the subscriberaTMs legal standing in the relationship with the dependent.

INDIV | individual | Description: A role played by a party covered under a policy as the policy holder. An individual may be either a person or an organization. Note: The party playing the role of an individual insured is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a named insured may make a claim under a policy, e.g., a party that is the named insured and policy holder under a comprehensive automobile insurance policy may become the claimant for coverage under that policy if injured in an automobile accident and there is no liable third party. In the case of an individual insured making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that an individual insured has filed a claim for a loss. Example: The individual insured under a comprehensive automobile, disability, or property and casualty policy that is the policy holder.

SUBSCR | subscriber | Description: A role played by a person covered under a policy based on association with a sponsor who is the policy holder, and whose association may provide for the eligibility of dependents for coverage. Discussion: The policy holder holds the contract with the policy or program underwriter. The subscriber holds a certificate of coverage under the contract. In legal proceedings concerning the policy or program, the terms of the contract takes precedence over the terms of the certificate of coverage if there are any inconsistencies. Note: The party playing the role of a subscriber is not a claimant in the sense conveyed by the RoleClassCoveredParty CLAIM (claimant). However, a subscriber may make a claim under a policy, e.g., a subscriber under a health insurance policy may become the claimant for coverage under that policy for wellness examines or if injured and there is no liable third party. In the case of a subscriber making a claim, a role type code INSCLM (insured claimant) subtypes the class to indicate that the subscriber has filed a claim for services covered under the health insurance policy. Example: An employee or a member of an association.

PROG | program eligible | Description: A role played by a party that meets the eligibility criteria for coverage under a program. A program eligible may be either a person, non-person living subject, or an organization, or a group of persons, non-person living subjects, or organizations. Discussion: A program as typically government administered coverage for parties determined eligible under the terms of the program. Note: The party playing a program eligible is not a claimant in the sense conveyed by the

RoleClassCoveredParty CLAIM (claimant). However a program eligible may make a claim under program, e.g., an unemployed worker may claim benefits under an unemployment insurance program, but parties playing these covered party role classes are not, for purposes of this vocabulary and in an effort to clearly distinguish role classes, considered claimants. In the case of a program eligible, a role type code INJWKR (injured worker) subtypes the class to indicate that the covered party in a workers compensation program is an injured worker, and as such, has filed a "claim" under the program for benefits. Likewise, a covered role type code UNEMP (unemployed worker) subtypes the program eligible class to indicate that the covered party in an unemployment insurance program has filed a claim for unemployment benefits. Example: A party meeting eligibility criteria related to health or financial status, e.g., in the U.S., persons meeting health, demographic, or financial criteria established by state and federal law are eligible for Medicaid.

CRINV | clinical research investigator | A role played by a provider, always a person, who has agency authority from a Clinical Research Sponsor to direct the conduct of a clinical research trial or study on behalf of the sponsor.

CRSPNSR | clinical research sponsor | A role played by an entity, usually an organization, that is the sponsor of a clinical research trial or study. The sponsor commissions the study, bears the expenses, is responsible for satisfying all legal requirements concerning subject safety and privacy, and is generally responsible for collection, storage and analysis of the data generated during the trial. No scoper is necessary, as a clinical research sponsor undertakes the role on its own authority and declaration. Clinical research sponsors are usually educational or other research organizations, government agencies or biopharmaceutical companies.

EMP | employee | A relationship between a person or organization and a person or organization formed for the purpose of exchanging work for compensation. The purpose of the role is to identify the type of relationship the employee has to the employer, rather than the nature of the work actually performed. (Contrast with AssignedEntity.)

MIL | military person | A role played by a member of a military service. Scoper is the military service (e.g. Army, Navy, Air Force, etc.) or, more specifically, the unit (e.g. Company C, 3rd Battalion, 4th Division, etc.)

GUAR | guarantor | A person or organization (player) that serves as a financial guarantor for another person or organization (scoper).

INVSBJ | Investigation Subject | An entity that is the subject of an investigation. This role is scoped by the party responsible for the investigation.

CASEBJ | Case Subject | A person, non-person living subject, or place that is the subject of an investigation related to a notifiable condition (health circumstance that is reportable within the applicable public health jurisdiction)

RESBJ | research subject | Definition:Specifies the administrative functionality within a formal experimental design for which the ResearchSubject role was established. Examples: Screening - role is used for pre-enrollment evaluation portion of the design; enrolled - role is used for subjects admitted to the experimental portion of the design.

LIC | licensed entity | A relationship in which the scoper certifies the player (e. g. a medical care giver, a medical device or a provider organization) to perform certain activities that fall under the jurisdiction of the scoper (e.g., a health authority licensing healthcare providers, a medical quality authority certifying healthcare professionals).

NOT | notary public | notary public

PROV | healthcare provider | An Entity (player) that is authorized to provide health care services by some authorizing agency (scoper).

PAT | patient | A Role of a LivingSubject (player) as an actual or potential recipient of health care services from a healthcare provider organization (scoper). Usage Note: Communication about relationships between patients and specific healthcare practitioners (people) is not done via scoper. Instead this is generally done using the CareProvision act. This allows linkage between patient and a particular healthcare practitioner role and also allows description of the type of care involved in the relationship.

PAYEE | payee | The role of an organization or individual designated to receive payment for a claim against a particular coverage. The scoping entity is the organization that is the submitter of the invoice in question.

PAYOR | invoice payor | The role of an organization that undertakes to accept claims invoices, assess the coverage or payments due for those invoices and pay to the designated payees for those invoices. This role may be either the underwriter or a third-party organization authorized by the underwriter. The scoping entity is the organization that underwrites the claimed coverage.

POLHOLD | policy holder | A role played by a person or organization that holds an insurance policy. The underwriter of that policy is the scoping entity. Discussion:The identifier of the policy is captured in 'Role.id' when the Role is a policy holder. A particular policy may cover several individuals one of whom may be, but need not be, the policy holder. Thus the notion of covered party is a role that is distinct from that of the policy holder.

QUAL | qualified entity | An entity (player) that has been recognized as having certain training/experience or other characteristics that would make said entity an appropriate performer for a certain activity. The scoper is an organization that educates or qualifies entities.

SPNSR | coverage sponsor | A role played by an entity, usually an organization that is the sponsor of an insurance plan or a health program. A sponsor is the party that is ultimately accountable for the coverage by employment contract or by law. A sponsor can be an employer, union, government agency, or association. Fully insured sponsors establish the terms of the plan and contract with health insurance plans to assume the risk and to administer the plan. Self-insured sponsors delegate coverage administration, but not risk, to third-party administrators. Program sponsors designate services to be covered in accordance with statute. Program sponsors may administer the coverage themselves, delegate coverage administration, but not risk to third-party administrators, or contract with health insurance plans to assume the risk and administrator a program. Sponsors qualify individuals who may become a policy holder of the plan; where the sponsor is the policy holder, who may become a subscriber or a dependent to a policy under the plan; or where the sponsor is a government agency, who may become program eligibles under a program. The sponsor role may be further qualified by the SponsorRole.code. Entities playing the sponsor role may also play the role of a Coverage Administrator. Example: An employer, union, government agency, or association.

STD | student | A role played by an individual who is a student of a school, which is the scoping entity.

UNDWRT | underwriter | A role played by a person or an organization. It is the party that accepts fiscal responsibility for insurance plans and the policies created under those plans; administers and accepts fiscal responsibility for a program that provides coverage for services to eligible individuals; and/or has the responsibility to assess the merits of each risk and decide a suitable premium for accepting all or part of the risk. If played by an organization, this role may be further specified by an appropriate RoleCode. Example: A health insurer; Medicaid Program; Lloyd's of London

CAREGIVER | caregiver | A person responsible for the primary care of a patient at home.

PRS | personal relationship | Links two entities with classCode PSN (person) in a personal relationship. The character of the relationship must be defined by a PersonalRelationshipRoleType code. The player and scoper are determined by PersonalRelationshipRoleType code as well.

SELF | self | The "same" roleclass asserts an identity between playing and scoping entities: that they are in fact instances of the same entity and, in the case of discrepancies (e.g different DOB, gender), that one or both are in error. Usage: playing and scoping entities must have same classcode, but need not have identical attributes or values. Example: a provider registry maintains sets of conflicting demographic data for what is reported to be the same individual.

4.4.2.19377 HL7 v3 Value Set SubstanceAdminSubstitutionReason

URL: https://hl7.org/fhir/R4/v3/SubstanceAdminSubstitutionReason/vs.html
Vocabulary Work Group | Maturity Level: 1 | Draft | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-SubstanceAdminSubstitutionReason) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-SubstanceAdminSubstitutionReason
Version: | 2014-03-26
Name: | v3.SubstanceAdminSubstitutionReason
Title: | V3 Value SetSubstanceAdminSubstitutionReason
Definition: | No Description Provided
OID: | 2.16.840.1.113883.1.11.19377 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is used in the following places:

• Resource: MedicationRequest.substitution.reason (CodeableConcept / Example)

• Resource: MedicationDispense.substitution.reason (CodeableConcept / Example)

4.4.2.19377.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActReason where concept is-a _SubstanceAdminSubstitutionReason

http://terminology.hl7.org/CodeSystem/v3-ActReason

Code | Display
_SubstanceAdminSubstitutionReason | SubstanceAdminSubstitutionReason | SubstanceAdminSubstitutionReason

4.4.2.19377.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 4 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActReason version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActReason

Code | Display | Definition
CT | continuing therapy | Indicates that the decision to substitute or to not substitute was driven by a desire to maintain consistency with a pre-existing therapy. I.e. The performer provided the same item/service as had been previously provided rather than providing exactly what was ordered, or rather than substituting with a lower-cost equivalent.
FP | formulary policy | Indicates that the decision to substitute or to not substitute was driven by a policy expressed within the formulary.
OS | out of stock | In the case of 'substitution', indicates that the substitution occurred because the ordered item was not in stock. In the case of 'no substitution', indicates that a cheaper equivalent was not substituted because it was not in stock.
RR | regulatory requirement | Indicates that the decision to substitute or to not substitute was driven by a jurisdictional regulatory requirement mandating or prohibiting substitution.

4.4.2.19398 HL7 v3 Value Set ActInvoiceGroupCode

URL: https://hl7.org/fhir/R4/v3/ActInvoiceGroupCode/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActInvoiceGroupCode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActInvoiceGroupCode
Version: | 2014-03-26
Name: | v3.ActInvoiceGroupCode
Title: | V3 Value SetActInvoiceGroupCode
Definition: | Type of invoice element that is used to assist in describing an Invoice that is either submitted for adjudication or for which is returned on adjudication results. Invoice elements of this type signify a grouping of one or more children (detail) invoice elements. They do not have intrinsic costing associated with them, but merely reflect the sum of all costing for it's immediate children invoice elements.
OID: | 2.16.840.1.113883.1.11.19398 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.19398.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActCode where concept is-a _ActInvoiceGroupCode

http://terminology.hl7.org/CodeSystem/v3-ActCode

Code | Display

_ActInvoiceGroupCode | ActInvoiceGroupCode | Type of invoice element that is used to assist in describing an Invoice that is either submitted for adjudication or for which is returned on adjudication results. Invoice elements of this type signify a grouping of one or more children (detail) invoice elements. They do not have intrinsic costing associated with them, but merely reflect the sum of all costing for it's immediate children invoice elements.

4.4.2.19398.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 17 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActCode

Code | Display | Definition

CPNDDRGING | compound drug invoice group | A grouping of invoice element groups and details including the ones specifying the compound ingredients being invoiced. It may also contain generic detail items such as markup.

CPNDINDING | compound ingredient invoice group | A grouping of invoice element details including the one specifying an ingredient drug being invoiced. It may also contain generic detail items such as tax or markup.

CPNDSUPING | compound supply invoice group | A grouping of invoice element groups and details including the ones specifying the compound supplies being invoiced. It may also contain generic detail items such as markup.

DRUGING | drug invoice group | A grouping of invoice element details including the one specifying the drug being invoiced. It may also contain generic detail items such as markup.

FRAMEING | frame invoice group | A grouping of invoice element details including the ones specifying the frame fee and the frame dispensing cost that are being invoiced.

LENSING | lens invoice group | A grouping of invoice element details including the ones specifying the lens fee and the lens dispensing cost that are being invoiced.

PRDING | product invoice group | A grouping of invoice element details including the one specifying the product (good or supply) being invoiced. It may also contain generic detail items such as tax or discount.

CPINV | clinical product invoice | Clinical product invoice where the Invoice Grouping contains one or more billable item and is supported by clinical product(s). For example, a crutch or a wheelchair.

CSINV | clinical service invoice | Clinical Services Invoice which can be used to describe a single service, multiple services or repeated services. [1] Single Clinical services invoice where the Invoice Grouping contains one billable item and is supported by one clinical service. For example, a single service for an office visit or simple clinical procedure (e.g. knee mobilization). [2] Multiple Clinical services invoice where the Invoice Grouping contains more than one billable item, supported by one or more clinical services. The services can be distinct and over multiple dates, but for the same patient. This type of invoice includes a series of treatments which must be adjudicated together. For example, an adjustment and ultrasound for a chiropractic session where fees are associated for each of the services and adjudicated (invoiced) together. [3] Repeated Clinical services invoice where the Invoice Grouping contains one or more billable item, supported by the same clinical service repeated over a period of time. For example, the same Chiropractic adjustment (service or treatment) delivered on 3 separate occasions over a period of time at the discretion of the provider (e.g. month).

CSPINV | clinical service and product | A clinical Invoice Grouping consisting of one or more services and one or more product. Billing for these service(s) and product(s) are supported by multiple clinical billable events (acts). All items in the Invoice Grouping must be adjudicated together to be acceptable to the Adjudicator. For example , a brace (product) invoiced together with the fitting (service).

FININV | financial invoice | Invoice Grouping without clinical justification. These will not require identification of participants and associations from a clinical context such as patient and provider. Examples are interest charges and mileage.

OHSINV | oral health service | A clinical Invoice Grouping consisting of one or more oral health services. Billing for these service(s) are supported by multiple clinical billable events (acts). All items in the Invoice Grouping must be adjudicated together to be acceptable to the Adjudicator.

PAINV | preferred accommodation invoice | HealthCare facility preferred accommodation invoice.

RXCINV | Rx compound invoice | Pharmacy dispense invoice for a compound.

RXDINV | Rx dispense invoice | Pharmacy dispense invoice not involving a compound

SBFINV | sessional or block fee invoice | Clinical services invoice where the Invoice Group contains one billable item for multiple clinical services in one or more sessions.

VRXINV | vision dispense invoice | Vision dispense invoice for up to 2 lens (left and right), frame and optional discount. Eye exams are invoiced as a clinical service invoice.

4.4.2.19563 HL7 v3 Value Set PersonalRelationshipRoleType

URL: https://hl7.org/fhir/R4/v3/PersonalRelationshipRoleType/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-PersonalRelationshipRoleType) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-PersonalRelationshipRoleType
Version: | 2014-03-26
Name: | v3.PersonalRelationshipRoleType
Title: | V3 Value SetPersonalRelationshipRoleType
Definition: | Types of personal relationships between two living subjects. Example: Parent, sibling, unrelated friend, neighbor
OID: | 2.16.840.1.113883.1.11.19563 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.19563.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-RoleCode where concept is-a _PersonalRelationshipRoleType

http://terminology.hl7.org/CodeSystem/v3-RoleCode

Code | Display
_PersonalRelationshipRoleType | PersonalRelationshipRoleType | PersonalRelationshipRoleType

4.4.2.19563.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 109 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-RoleCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-RoleCode

Code | Display | Definition
FAMMEMB | family member | A relationship between two people characterizing their "familial" relationship
CHILD | child | The player of the role is a child of the scoping entity.
CHLDADOPT | adopted child | The player of the role is a child taken into a family through legal means and raised by the scoping person (parent) as his or her own child.
DAUADOPT | adopted daughter | The player of the role is a female child taken into a family through legal means and raised by the scoping person (parent) as his or her own child.
SONADOPT | adopted son | The player of the role is a male child taken into a family through legal means and raised by the scoping person (parent) as his or her own child.
CHLDFOST | foster child | The player of the role is a child receiving parental care and nurture from the scoping person (parent) but not related to him or her through legal or blood ties.
DAUFOST | foster daughter | The player of the role is a female child receiving parental care and nurture from the scoping person (parent) but not related to him or her through legal or blood ties.
SONFOST | foster son | The player of the role is a male child receiving parental care and nurture from the scoping person (parent) but not related to him or her through legal or blood ties.
DAUC | daughter | Description: The player of the role is a female child (of any type) of scoping entity (parent)
DAU | natural daughter | The player of the role is a female offspring of the scoping entity (parent).
STPDAU | stepdaughter | The player of the role is a daughter of the scoping person's spouse by a previous union.
NCHILD | natural child | The player of the role is an offspring of the scoping entity as determined by birth.
SON | natural son | The player of the role is a male offspring of the scoping entity (parent).
SONC | son | Description: The player of the role is a male child (of any type) of scoping entity (parent)
STPSON | stepson | The player of the role is a son of the scoping person's spouse by a previous union.
STPCHLD | step child | The player of the role is a child of the scoping person's spouse by a previous union.
EXT | extended family member | Description: A family member not having an immediate genetic or legal relationship e.g. Aunt, cousin, great grandparent, grandchild, grandparent, niece, nephew or uncle.
AUNT | aunt | The player of the role is a sister of the scoping person's mother or father.
MAUNT | maternal aunt | Description:The player of the role is a biological sister of the scoping person's biological mother.
PAUNT | paternal aunt | Description:The player of the role is a biological sister of the scoping person's biological father.
COUSN | cousin | The player of the role is a relative of the scoping person descended from a common ancestor, such as a grandparent, by two or more steps in a diverging line.
MCOUSN | maternal cousin | Description:The player of the role is a biological relative of the scoping person descended from a common ancestor on the player's mother's side, such as a grandparent, by two or more steps in a diverging line.
PCOUSN | paternal cousin | Description:The player of the role is a biological relative of the scoping person descended from a common ancestor on the player's father's side, such as a grandparent, by two or more steps in a diverging line.
GGRPRN | great grandparent | The player of the role is a parent of the scoping person's grandparent.
GGRFTH | great grandfather | The player of the role is the father of the scoping person's grandparent.
MGGRFTH | maternal great-grandfather | Description:The player of the role is the biological father of the scoping person's biological mother's parent.
PGGRFTH | paternal great-grandfather | Description:The player of the role is the biological father of the scoping person's biological father's parent.
GGRMTH | great grandmother | The player of the role is the mother of the scoping person's grandparent.
MGGRMTH | maternal great-grandmother | Description:The player of the role is the biological mother of the scoping person's biological mother's parent.
PGGRMTH | paternal great-grandmother | Description:The player of the role is the biological mother of the scoping person's biological father's parent.
MGGRPRN | maternal great-grandparent | Description:The player of the role is a biological parent of the scoping person's biological mother's parent.

PGGRPRN | paternal great-grandparent | Description:The player of the role is a biological parent of the scoping person's biological father's parent.
GRNDCHILD | grandchild | The player of the role is a child of the scoping person's son or daughter.
GRNDDAU | granddaughter | The player of the role is a daughter of the scoping person's son or daughter.
GRNDSON | grandson | The player of the role is a son of the scoping person's son or daughter.
GRPRN | grandparent | The player of the role is a parent of the scoping person's mother or father.
GRFTH | grandfather | The player of the role is the father of the scoping person's mother or father.
MGRFTH | maternal grandfather | Description:The player of the role is the biological father of the scoping person's biological mother.
PGRFTH | paternal grandfather | Description:The player of the role is the biological father of the scoping person's biological father.
GRMTH | grandmother | The player of the role is the mother of the scoping person's mother or father.
MGRMTH | maternal grandmother | Description:The player of the role is the biological mother of the scoping person's biological mother.
PGRMTH | paternal grandmother | Description:The player of the role is the biological mother of the scoping person's biological father.
MGRPRN | maternal grandparent | Description:The player of the role is the biological parent of the scoping person's biological mother.
PGRPRN | paternal grandparent | Description:The player of the role is the biological parent of the scoping person's biological father.
INLAW | inlaw | A relationship between an individual and a member of their spousal partner's immediate family.
CHLDINLAW | child-in-law | The player of the role is the spouse of scoping person's child.
DAUINLAW | daughter in-law | The player of the role is the wife of scoping person's son.
SONINLAW | son in-law | The player of the role is the husband of scoping person's daughter.
PRNINLAW | parent in-law | The player of the role is the parent of scoping person's husband or wife.
FTHINLAW | father-in-law | The player of the role is the father of the scoping person's husband or wife.
MTHINLAW | mother-in-law | The player of the role is the mother of the scoping person's husband or wife.
SIBINLAW | sibling in-law | The player of the role is: (1) a sibling of the scoping person's spouse, or (2) the spouse of the scoping person's sibling, or (3) the spouse of a sibling of the scoping person's spouse.
BROINLAW | brother-in-law | The player of the role is: (1) a brother of the scoping person's spouse, or (2) the husband of the scoping person's sister, or (3) the husband of a sister of the scoping person's spouse.
SISINLAW | sister-in-law | The player of the role is: (1) a sister of the scoping person's spouse, or (2) the wife of the scoping person's brother, or (3) the wife of a brother of the scoping person's spouse.
NIENEPH | niece/nephew | The player of the role is a child of scoping person's brother or sister or of the brother or sister of the scoping person's spouse.
NEPHEW | nephew | The player of the role is a son of the scoping person's brother or sister or of the brother or sister of the scoping person's spouse.
NIECE | niece | The player of the role is a daughter of the scoping person's brother or sister or of the brother or sister of the scoping person's spouse.
UNCLE | uncle | The player of the role is a brother of the scoping person's mother or father.
MUNCLE | maternal uncle | Description:The player of the role is a biological brother of the scoping person's biological mother.
PUNCLE | paternal uncle | Description:The player of the role is a biological brother of the scoping person's biological father.
PRN | parent | The player of the role is one who begets, gives birth to, or nurtures and raises the scoping entity (child).
ADOPTP | adoptive parent | The player of the role (parent) has taken the scoper (child) into their family through legal means and raises them as his or her own child.
ADOPTF | adoptive father | The player of the role (father) is a male who has taken the scoper (child) into their family through legal means and raises them as his own child.
ADOPTM | adoptive mother | The player of the role (father) is a female who has taken the scoper (child) into their family through legal means and raises them as her own child.
FTH | father | The player of the role is a male who begets or raises or nurtures the scoping entity (child).
FTHFOST | foster father | The player of the role (parent) who is a male state-certified caregiver responsible for the scoper (child) who has been placed in the parent's care. The placement of the child is usually arranged through the government or a social-service agency, and temporary. The state, via a jurisdiction recognized child protection agency, stands as in loco parentis to the child, making all legal decisions while the foster parent is responsible for the day-to-day care of the specified child.
NFTH | natural father | The player of the role is a male who begets the scoping entity (child).
NFTHF | natural father of fetus | Indicates the biologic male parent of a fetus.
STPFTH | stepfather | The player of the role is the husband of scoping person's mother and not the scoping person's natural father.
MTH | mother | The player of the role is a female who conceives, gives birth to, or raises and nurtures the scoping entity (child).
GESTM | gestational mother | The player is a female whose womb carries the fetus of the scoper. Generally used when the gestational mother and natural mother are not the same.
MTHFOST | foster mother | The player of the role (parent) who is a female state-certified caregiver responsible for the scoper (child) who has been placed in the parent's care. The placement of the child is usually arranged through the government or a social-service agency, and temporary. The state, via a jurisdiction recognized child protection agency, stands as in loco parentis to the child, making all legal decisions while the foster parent is responsible for the day-to-day care of the specified child.
NMTH | natural mother | The player of the role is a female who conceives or gives birth to the scoping entity (child).
NMTHF | natural mother of fetus | The player is the biologic female parent of the scoping fetus.
STPMTH | stepmother | The player of the role is the wife of scoping person's father and not the scoping person's natural mother.
NPRN | natural parent | natural parent
PRNFOST | foster parent | The player of the role (parent) who is a state-certified caregiver responsible for the scoper (child) who has been placed in the parent's care. The placement of the child is usually arranged through the government or a social-service agency, and temporary. The state, via a jurisdiction recognized child protection agency, stands as in loco parentis to the child, making all legal decisions while the foster parent is responsible for the day-to-day care of the specified child.
STPPRN | step parent | The player of the role is the spouse of the scoping person's parent and not the scoping person's natural parent.
SIB | sibling | The player of the role shares one or both parents in common with the scoping entity.
BRO | brother | The player of the role is a male sharing one or both parents in common with the scoping entity.

HBRO | half-brother | The player of the role is a male related to the scoping entity by sharing only one biological parent.
NBRO | natural brother | The player of the role is a male having the same biological parents as the scoping entity.
TWINBRO | twin brother | The scoper was carried in the same womb as the male player and shares common biological parents.
FTWINBRO | fraternal twin brother | The scoper was carried in the same womb as the male player and shares common biological parents but is the product of a distinct egg/sperm pair.
ITWINBRO | identical twin brother | The male scoper is an offspring of the same egg-sperm pair as the male player.
STPBRO | stepbrother | The player of the role is a son of the scoping person's stepparent.
HSIB | half-sibling | The player of the role is related to the scoping entity by sharing only one biological parent.
HSIS | half-sister | The player of the role is a female related to the scoping entity by sharing only one biological parent.
NSIB | natural sibling | The player of the role has both biological parents in common with the scoping entity.
NSIS | natural sister | The player of the role is a female having the same biological parents as the scoping entity.
TWINSIS | twin sister | The scoper was carried in the same womb as the female player and shares common biological parents.
FTWINSIS | fraternal twin sister | The scoper was carried in the same womb as the female player and shares common biological parents but is the product of a distinct egg/sperm pair.
ITWINSIS | identical twin sister | The female scoper is an offspring of the same egg-sperm pair as the female player.
TWIN | twin | The scoper and player were carried in the same womb and shared common biological parents.
FTWIN | fraternal twin | The scoper and player were carried in the same womb and share common biological parents but are the product of distinct egg/sperm pairs.
ITWIN | identical twin | The scoper and player are offspring of the same egg-sperm pair.
SIS | sister | The player of the role is a female sharing one or both parents in common with the scoping entity.
STPSIS | stepsister | The player of the role is a daughter of the scoping person's stepparent.
STPSIB | step sibling | The player of the role is a child of the scoping person's stepparent.
SIGOTHR | significant other | A person who is important to one's well being; especially a spouse or one in a similar relationship. (The player is the one who is important)
DOMPART | domestic partner | The player of the role cohabits with the scoping person but is not the scoping person's spouse.
FMRSPS | former spouse | Player of the role was previously joined to the scoping person in marriage and this marriage is now dissolved and inactive. Usage Note: This is significant to indicate as some jurisdictions have different legal requirements for former spouse to access the patient's record, from a general friend.
SPS | spouse | The player of the role is a marriage partner of the scoping person.
HUSB | husband | The player of the role is a man joined to a woman (scoping person) in marriage.
WIFE | wife | The player of the role is a woman joined to a man (scoping person) in marriage.
FRND | unrelated friend | The player of the role is a person who is known, liked, and trusted by the scoping person.
NBOR | neighbor | The player of the role lives near or next to the scoping person.
ONESELF | self | The relationship that a person has with his or her self.
ROOM | Roommate | One who shares living quarters with the subject.

4.4.2.19579 HL7 v3 Value Set FamilyMember

URL: https://hl7.org/fhir/R4/v3/FamilyMember/vs.html
Vocabulary Work Group | Maturity Level: 1 | Draft | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-FamilyMember) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-FamilyMember
Version: | 2014-03-26
Name: | v3.FamilyMember
Title: | V3 Value SetFamilyMember
Definition: | A relationship between two people characterizing their "familial" relationship
OID: | 2.16.840.1.113883.1.11.19579 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is used in the following places:

• Resource: FamilyMemberHistory.relationship (CodeableConcept / Example)
• Profile: FamilyMemberHistory-Genetic: FamilyMemberHistory.relationship (CodeableConcept / Example)

4.4.2.19579.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-RoleCode where concept is-a FAMMEMB

4.4.2.19579.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 105 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-RoleCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-RoleCode

Lvl | Code | Display | Definition
0 | FAMMEMB | family member | A relationship between two people characterizing their "familial" relationship
1 | CHILD | child | The player of the role is a child of the scoping entity.
2 | CHLDADOPT | adopted child | The player of the role is a child taken into a family through legal means and raised by the scoping person (parent) as his or her own child.

3 | DAUADOPT | adopted daughter | The player of the role is a female child taken into a family through legal means and raised by the scoping person (parent) as his or her own child.
3 | SONADOPT | adopted son | The player of the role is a male child taken into a family through legal means and raised by the scoping person (parent) as his or her own child.
2 | CHLDFOST | foster child | The player of the role is a child receiving parental care and nurture from the scoping person (parent) but not related to him or her through legal or blood ties.
3 | DAUFOST | foster daughter | The player of the role is a female child receiving parental care and nurture from the scoping person (parent) but not related to him or her through legal or blood ties.
3 | SONFOST | foster son | The player of the role is a male child receiving parental care and nurture from the scoping person (parent) but not related to him or her through legal or blood ties.
2 | DAUC | daughter | Description: The player of the role is a female child (of any type) of scoping entity (parent)
3 | DAU | natural daughter | The player of the role is a female offspring of the scoping entity (parent).
3 | STPDAU | stepdaughter | The player of the role is a daughter of the scoping person's spouse by a previous union.
2 | NCHILD | natural child | The player of the role is an offspring of the scoping entity as determined by birth.
3 | SON | natural son | The player of the role is a male offspring of the scoping entity (parent).
2 | SONC | son | Description: The player of the role is a male child (of any type) of scoping entity (parent)
3 | STPSON | stepson | The player of the role is a son of the scoping person's spouse by a previous union.
2 | STPCHLD | step child | The player of the role is a child of the scoping person's spouse by a previous union.
1 | EXT | extended family member | Description: A family member not having an immediate genetic or legal relationship e.g. Aunt, cousin, great grandparent, grandchild, grandparent, niece, nephew or uncle.
2 | AUNT | aunt | The player of the role is a sister of the scoping person's mother or father.
3 | MAUNT | maternal aunt | Description:The player of the role is a biological sister of the scoping person's biological mother.
3 | PAUNT | paternal aunt | Description:The player of the role is a biological sister of the scoping person's biological father.
2 | COUSN | cousin | The player of the role is a relative of the scoping person descended from a common ancestor, such as a grandparent, by two or more steps in a diverging line.
3 | MCOUSN | maternal cousin | Description:The player of the role is a biological relative of the scoping person descended from a common ancestor on the player's mother's side, such as a grandparent, by two or more steps in a diverging line.
3 | PCOUSN | paternal cousin | Description:The player of the role is a biological relative of the scoping person descended from a common ancestor on the player's father's side, such as a grandparent, by two or more steps in a diverging line.
2 | GGRPRN | great grandparent | The player of the role is a parent of the scoping person's grandparent.
3 | GGRFTH | great grandfather | The player of the role is the father of the scoping person's grandparent.
4 | MGGRFTH | maternal great-grandfather | Description:The player of the role is the biological father of the scoping person's biological mother's parent.
4 | PGGRFTH | paternal great-grandfather | Description:The player of the role is the biological father of the scoping person's biological father's parent.
3 | GGRMTH | great grandmother | The player of the role is the mother of the scoping person's grandparent.
4 | MGGRMTH | maternal great-grandmother | Description:The player of the role is the biological mother of the scoping person's biological mother's parent.
4 | PGGRMTH | paternal great-grandmother | Description:The player of the role is the biological mother of the scoping person's biological father's parent.
3 | MGGRPRN | maternal great-grandparent | Description:The player of the role is a biological parent of the scoping person's biological mother's parent.
3 | PGGRPRN | paternal great-grandparent | Description:The player of the role is a biological parent of the scoping person's biological father's parent.
2 | GRNDCHILD | grandchild | The player of the role is a child of the scoping person's son or daughter.
3 | GRNDDAU | granddaughter | The player of the role is a daughter of the scoping person's son or daughter.
3 | GRNDSON | grandson | The player of the role is a son of the scoping person's son or daughter.
2 | GRPRN | grandparent | The player of the role is a parent of the scoping person's mother or father.
3 | GRFTH | grandfather | The player of the role is the father of the scoping person's mother or father.
4 | MGRFTH | maternal grandfather | Description:The player of the role is the biological father of the scoping person's biological mother.
4 | PGRFTH | paternal grandfather | Description:The player of the role is the biological father of the scoping person's biological father.
3 | GRMTH | grandmother | The player of the role is the mother of the scoping person's mother or father.
4 | MGRMTH | maternal grandmother | Description:The player of the role is the biological mother of the scoping person's biological mother.
4 | PGRMTH | paternal grandmother | Description:The player of the role is the biological mother of the scoping person's biological father.
3 | MGRPRN | maternal grandparent | Description:The player of the role is the biological parent of the scoping person's biological mother.
3 | PGRPRN | paternal grandparent | Description:The player of the role is the biological parent of the scoping person's biological father.
2 | INLAW | inlaw | A relationship between an individual and a member of their spousal partner's immediate family.
3 | CHLDINLAW | child-in-law | The player of the role is the spouse of scoping person's child.
4 | DAUINLAW | daughter in-law | The player of the role is the wife of scoping person's son.
4 | SONINLAW | son in-law | The player of the role is the husband of scoping person's daughter.
3 | PRNINLAW | parent in-law | The player of the role is the parent of scoping person's husband or wife.
4 | FTHINLAW | father-in-law | The player of the role is the father of the scoping person's husband or wife.
4 | MTHINLAW | mother-in-law | The player of the role is the mother of the scoping person's husband or wife.
3 | SIBINLAW | sibling in-law | The player of the role is: (1) a sibling of the scoping person's spouse, or (2) the spouse of the scoping person's sibling, or (3) the spouse of a sibling of the scoping person's spouse.
4 | BROINLAW | brother-in-law | The player of the role is: (1) a brother of the scoping person's spouse, or (2) the husband of the scoping person's sister, or (3) the husband of a sister of the scoping person's spouse.

4 | SISINLAW | sister-in-law | The player of the role is: (1) a sister of the scoping person's spouse, or (2) the wife of the scoping person's brother, or (3) the wife of a brother of the scoping person's spouse.
2 | NIENEPH | niece/nephew | The player of the role is a child of scoping person's brother or sister or of the brother or sister of the scoping person's spouse.
3 | NEPHEW | nephew | The player of the role is a son of the scoping person's brother or sister or of the brother or sister of the scoping person's spouse.
3 | NIECE | niece | The player of the role is a daughter of the scoping person's brother or sister or of the brother or sister of the scoping person's spouse.
2 | UNCLE | uncle | The player of the role is a brother of the scoping person's mother or father.
3 | MUNCLE | maternal uncle | Description:The player of the role is a biological brother of the scoping person's biological mother.
3 | PUNCLE | paternal uncle | Description:The player of the role is a biological brother of the scoping person's biological father.
1 | PRN | parent | The player of the role is one who begets, gives birth to, or nurtures and raises the scoping entity (child).
2 | ADOPTP | adoptive parent | The player of the role (parent) has taken the scoper (child) into their family through legal means and raises them as his or her own child.
3 | ADOPTF | adoptive father | The player of the role (father) is a male who has taken the scoper (child) into their family through legal means and raises them as his own child.
3 | ADOPTM | adoptive mother | The player of the role (father) is a female who has taken the scoper (child) into their family through legal means and raises them as her own child.
2 | FTH | father | The player of the role is a male who begets or raises or nurtures the scoping entity (child).
3 | FTHFOST | foster father | The player of the role (parent) who is a male state-certified caregiver responsible for the scoper (child) who has been placed in the parent's care. The placement of the child is usually arranged through the government or a social-service agency, and temporary. The state, via a jurisdiction recognized child protection agency, stands as in loco parentis to the child, making all legal decisions while the foster parent is responsible for the day-to-day care of the specified child.
3 | NFTH | natural father | The player of the role is a male who begets the scoping entity (child).
4 | NFTHF | natural father of fetus | Indicates the biologic male parent of a fetus.
3 | STPFTH | stepfather | The player of the role is the husband of scoping person's mother and not the scoping person's natural father.
2 | MTH | mother | The player of the role is a female who conceives, gives birth to, or raises and nurtures the scoping entity (child).
3 | GESTM | gestational mother | The player is a female whose womb carries the fetus of the scoper. Generally used when the gestational mother and natural mother are not the same.
3 | MTHFOST | foster mother | The player of the role (parent) who is a female state-certified caregiver responsible for the scoper (child) who has been placed in the parent's care. The placement of the child is usually arranged through the government or a social-service agency, and temporary. The state, via a jurisdiction recognized child protection agency, stands as in loco parentis to the child, making all legal decisions while the foster parent is responsible for the day-to-day care of the specified child.
3 | NMTH | natural mother | The player of the role is a female who conceives or gives birth to the scoping entity (child).
4 | NMTHF | natural mother of fetus | The player is the biologic female parent of the scoping fetus.
3 | STPMTH | stepmother | The player of the role is the wife of scoping person's father and not the scoping person's natural mother.
2 | NPRN | natural parent | natural parent
2 | PRNFOST | foster parent | The player of the role (parent) who is a state-certified caregiver responsible for the scoper (child) who has been placed in the parent's care. The placement of the child is usually arranged through the government or a social-service agency, and temporary. The state, via a jurisdiction recognized child protection agency, stands as in loco parentis to the child, making all legal decisions while the foster parent is responsible for the day-to-day care of the specified child.
2 | STPPRN | step parent | The player of the role is the spouse of the scoping person's parent and not the scoping person's natural parent.
1 | SIB | sibling | The player of the role shares one or both parents in common with the scoping entity.
2 | BRO | brother | The player of the role is a male sharing one or both parents in common with the scoping entity.
3 | HBRO | half-brother | The player of the role is a male related to the scoping entity by sharing only one biological parent.
3 | NBRO | natural brother | The player of the role is a male having the same biological parents as the scoping entity.
4 | TWINBRO | twin brother | The scoper was carried in the same womb as the male player and shares common biological parents.
5 | FTWINBRO | fraternal twin brother | The scoper was carried in the same womb as the male player and shares common biological parents but is the product of a distinct egg/sperm pair.
5 | ITWINBRO | identical twin brother | The male scoper is an offspring of the same egg-sperm pair as the male player.
3 | STPBRO | stepbrother | The player of the role is a son of the scoping person's stepparent.
2 | HSIB | half-sibling | The player of the role is related to the scoping entity by sharing only one biological parent.
3 | HSIS | half-sister | The player of the role is a female related to the scoping entity by sharing only one biological parent.
2 | NSIB | natural sibling | The player of the role has both biological parents in common with the scoping entity.
3 | NSIS | natural sister | The player of the role is a female having the same biological parents as the scoping entity.
4 | TWINSIS | twin sister | The scoper was carried in the same womb as the female player and shares common biological parents.
5 | FTWINSIS | fraternal twin sister | The scoper was carried in the same womb as the female player and shares common biological parents but is the product of a distinct egg/sperm pair.
5 | ITWINSIS | identical twin sister | The female scoper is an offspring of the same egg-sperm pair as the female player.
3 | TWIN | twin | The scoper and player were carried in the same womb and shared common biological parents.
4 | FTWIN | fraternal twin | The scoper and player were carried in the same womb and share common biological parents but are the product of distinct egg/sperm pairs.
4 | ITWIN | identical twin | The scoper and player are offspring of the same egg-sperm pair.
2 | SIS | sister | The player of the role is a female sharing one or both parents in common with the scoping entity.
3 | STPSIS | stepsister | The player of the role is a daughter of the scoping person's stepparent.
2 | STPSIB | step sibling | The player of the role is a child of the scoping person's stepparent.
1 | SIGOTHR | significant other | A person who is important to one's well being; especially a spouse or one in a similar relationship. (The player is the one who is important)
2 | DOMPART | domestic partner | The player of the role cohabits with the scoping person but is not the scoping person's spouse.
2 | FMRSPS | former spouse | Player of the role was previously joined to the scoping person in marriage and this marriage is now dissolved and inactive. Usage Note: This is significant to indicate as some jurisdictions have different legal requirements for former spouse to access the patient's record, from a general friend.

2 | SPS | spouse | The player of the role is a marriage partner of the scoping person.
3 | HUSB | husband | The player of the role is a man joined to a woman (scoping person) in marriage.
3 | WIFE | wife | The player of the role is a woman joined to a man (scoping person) in marriage.

4.4.2.19584 HL7 v3 Value Set ParticipationTargetSubject

URL: https://hl7.org/fhir/R4/v3/ParticipationTargetSubject/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ParticipationTargetSubject) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ParticipationTargetSubject
Version: | 2014-03-26
Name: | v3.ParticipationTargetSubject
Title: | V3 Value SetParticipationTargetSubject
Definition: | The principle target that the service acts on. E.g. the patient in physical examination, a specimen in a lab observation. May also be a patient's family member (teaching) or a device or room (cleaning, disinfecting, housekeeping). Note: not all direct targets are subjects, consumables, and devices used as tools for a service are not subjects. However, a device may be a subject of a maintenance service.
OID: | 2.16.840.1.113883.1.11.19584 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.19584.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ParticipationType where concept is-a SBJ

4.4.2.19584.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 2 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ParticipationType version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ParticipationType

Lvl | Code | Display | Definition
0 | SBJ | subject | The principle target on which the action happens. Examples: The patient in physical examination, a specimen in a lab observation. May also be a patient's family member (teaching) or a device or room (cleaning, disinfecting, housekeeping). UsageNotes: Not all direct targets are subjects. Consumables and devices used as tools for an act are not subjects. However, a device may be a subject of a maintenance action.
1 | SPC | specimen | The subject of non-clinical (e.g. laboratory) observation services is a specimen.

4.4.2.19665 HL7 v3 Value Set ActClassProcedure

URL: https://hl7.org/fhir/R4/v3/ActClassProcedure/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActClassProcedure) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActClassProcedure
Version: | 2014-03-26
Name: | v3.ActClassProcedure
Title: | V3 Value SetActClassProcedure
Definition: | An Act whose immediate and primary outcome (post-condition) is the alteration of the physical condition of the subject. Examples: : Procedures may involve the disruption of some body surface (e.g. an incision in a surgical procedure), but they also include conservative procedures such as reduction of a luxated join, chiropractic treatment, massage, balneotherapy, acupuncture, shiatsu, etc. Outside of clinical medicine, procedures may be such things as alteration of environments (e.g. straightening rivers, draining swamps, building dams) or the repair or change of machinery etc.
OID: | 2.16.840.1.113883.1.11.19665 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.19665.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActClass where concept is-a PROC

4.4.2.19665.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 4 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActClass

Lvl | Code | Display | Definition

0 | PROC | procedure | An Act whose immediate and primary outcome (post-condition) is the alteration of the physical condition of the subject. Examples: : Procedures may involve the disruption of some body surface (e.g. an incision in a surgical procedure), but they also include conservative procedures such as reduction of a luxated join, chiropractic treatment, massage, balneotherapy, acupuncture, shiatsu, etc. Outside of clinical medicine, procedures may be such things as alteration of environments (e.g. straightening rivers, draining swamps, building dams) or the repair or change of machinery etc.

1 | SBADM | substance administration | The act of introducing or otherwise applying a substance to the subject. Discussion: The effect of the substance is typically established on a biochemical basis, however, that is not a requirement. For example, radiotherapy can largely be described in the same way, especially if it is a systemic therapy such as radio-iodine. This class also includes the application of chemical treatments to an area. Examples: Chemotherapy protocol; Drug prescription; Vaccination record

1 | SBEXT | Substance Extraction | Description: The act of removing a substance from the subject.

2 | SPECCOLLECT | Specimen Collection | A procedure for obtaining a specimen from a source entity.

4.4.2.19676 HL7 v3 Value Set ParticipationInformationTranscriber

URL: https://hl7.org/fhir/R4/v3/ParticipationInformationTranscriber/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ParticipationInformationTranscriber) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ParticipationInformationTranscriber

Version: | 2014-03-26

Name: | v3.ParticipationInformationTranscriber

Title: | V3 Value SetParticipationInformationTranscriber

Definition: | An entity entering the data into the originating system. The data entry entity is collected optionally for internal quality control purposes. This includes the transcriptionist for dictated text transcribed into electronic form.

OID: | 2.16.840.1.113883.1.11.19676 (for OID based terminology systems)

Source Resource | XML / JSON

This value set is not currently used

4.4.2.19676.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ParticipationType where concept is-a TRANS

4.4.2.19676.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 2 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ParticipationType version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ParticipationType

Lvl | Code | Display | Definition

0 | TRANS | Transcriber | An entity entering the data into the originating system. The data entry entity is collected optionally for internal quality control purposes. This includes the transcriptionist for dictated text transcribed into electronic form.

1 | ENT | data entry person | A person entering the data into the originating system. The data entry person is collected optionally for internal quality control purposes. This includes the transcriptionist for dictated text.

4.4.2.19707 HL7 v3 Value Set VerificationMethod

URL: https://hl7.org/fhir/R4/v3/VerificationMethod/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-VerificationMethod) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-VerificationMethod

Version: | 2014-03-26

Name: | v3.VerificationMethod

Title: | V3 Value SetVerificationMethod

Definition: | No Description Provided

OID: | 2.16.840.1.113883.1.11.19707 (for OID based terminology systems)

Source Resource | XML / JSON

This value set is not currently used

4.4.2.19707.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ObservationMethod where concept is-a _VerificationMethod

4.4.2.19707.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 4 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ObservationMethod version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ObservationMethod

Lvl | Code | Display | Definition
0 | _VerificationMethod | VerificationMethod | VerificationMethod
1 | VDOC | document verification | Verification by means of document. Example: Fax, letter, attachment to e-mail.
1 | VREG | registry verification | verification by means of a response to an electronic query Example: query message to a Covered Party registry application or Coverage Administrator.
1 | VTOKEN | electronic token verification | Verification by means of electronic token. Example: smartcard, magnetic swipe card, RFID device.
1 | VVOICE | voice-based verification | Verification by means of voice. Example: By speaking with or calling the Coverage Administrator or Covered Party

4.4.2.19812 HL7 v3 Value Set MilitaryRoleType

URL: https://hl7.org/fhir/R4/v3/MilitaryRoleType/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-MilitaryRoleType) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-MilitaryRoleType
Version: | 2014-03-26
Name: | v3.MilitaryRoleType
Title: | V3 Value SetMilitaryRoleType
Definition: | Definition: A person playing the role of program eligible under a program based on military status. Discussion: This CoveredPartyRoleType.code is typically used when the CoveredPartyRole class code is either "program eligible" or "subscriber" and the person's status as a member of the military meets jurisdictional or program criteria
OID: | 2.16.840.1.113883.1.11.19812 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.19812.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-RoleCode where concept is-a MIL

4.4.2.19812.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 4 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-RoleCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-RoleCode

Lvl | Code | Display | Definition
0 | MIL | military | Definition: A person playing the role of program eligible under a program based on military status. Discussion: This CoveredPartyRoleType.code is typically used when the CoveredPartyRole class code is either "program eligible" or "subscriber" and the person's status as a member of the military meets jurisdictional or program criteria
1 | ACTMIL | active duty military | Description: A person playing the role of program eligible under a program based on active military status. Discussion: This CoveredPartyRoleType.code is typically used when the CoveredPartyRole class code is either "program eligible" or "subscriber" and the persons status as active duty military meets jurisdictional or program criteria.
1 | RETMIL | retired military | Description: A person playing the role of program eligible under a program based on retired military status. Discussion: This CoveredPartyRoleType.code is typically used when the CoveredPartyRole class code is either "program eligible" or "subscriber" and the persons status as retired military meets jurisdictional or program criteria.
1 | VET | veteran | Description: A person playing the role of program eligible under a program based on status as a military veteran. Discussion: This CoveredPartyRoleType.code is typically used when the CoveredPartyRole class code is either "program eligible" or "subscriber" and the persons status as a veteran meets jurisdictional or program criteria.

4.4.2.19846 HL7 v3 Value Set ActTaskCode

URL: https://hl7.org/fhir/R4/v3/ActTaskCode/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActTaskCode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActTaskCode
Version: | 2014-03-26
Name: | v3.ActTaskCode
Title: | V3 Value SetActTaskCode
Definition: | Description: A task or action that a user may perform in a clinical information system.
OID: | 2.16.840.1.113883.1.11.19846 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.19846.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActCode where concept is-a _ActTaskCode

4.4.2.19846.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 37 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActCode

Lvl | Code | Display | Definition
0 | _ActTaskCode | ActTaskCode | Description: A task or action that a user may perform in a clinical information system (e.g., medication order entry, laboratory test results review, problem list entry).
1 | OE | order entry task | A clinician creates a request for a service to be performed for a given patient.
2 | LABOE | laboratory test order entry task | A clinician creates a request for a laboratory test to be done for a given patient.
2 | MEDOE | medication order entry task | A clinician creates a request for the administration of one or more medications to a given patient.
1 | PATDOC | patient documentation task | A person enters documentation about a given patient.
2 | ALLERLREV | allergy list review | Description: A person reviews a list of known allergies of a given patient.
2 | CLINNOTEE | clinical note entry task | A clinician enters a clinical note about a given patient
3 | DIAGLISTE | diagnosis list entry task | A clinician enters a diagnosis for a given patient.
3 | DISCHINSTE | discharge instruction entry | A person provides a discharge instruction to a patient.
3 | DISCHSUME | discharge summary entry task | A clinician enters a discharge summary for a given patient.
3 | PATEDUE | patient education entry | A person provides a patient-specific education handout to a patient.
3 | PATREPE | pathology report entry task | A pathologist enters a report for a given patient.
3 | PROBLISTE | problem list entry task | A clinician enters a problem for a given patient.
3 | RADREPE | radiology report entry task | A radiologist enters a report for a given patient.
2 | IMMLREV | immunization list review | Description: A person reviews a list of immunizations due or received for a given patient.
2 | REMLREV | reminder list review | Description: A person reviews a list of health care reminders for a given patient.
3 | WELLREMLREV | wellness reminder list review | Description: A person reviews a list of wellness or preventive care reminders for a given patient.
1 | PATINFO | patient information review task | A person (e.g., clinician, the patient herself) reviews patient information in the electronic medical record.
2 | ALLERLE | allergy list entry | Description: A person enters a known allergy for a given patient.
2 | CDSREV | clinical decision support intervention review | A person reviews a recommendation/assessment provided automatically by a clinical decision support application for a given patient.
2 | CLINNOTEREV | clinical note review task | A person reviews a clinical note of a given patient.
3 | DISCHSUMREV | discharge summary review task | A person reviews a discharge summary of a given patient.
2 | DIAGLISTREV | diagnosis list review task | A person reviews a list of diagnoses of a given patient.
2 | IMMLE | immunization list entry | Description: A person enters an immunization due or received for a given patient.
2 | LABRREV | laboratory results review task | A person reviews a list of laboratory results of a given patient.
2 | MICRORREV | microbiology results review task | A person reviews a list of microbiology results of a given patient.
3 | MICROORGRREV | microbiology organisms results review task | A person reviews organisms of microbiology results of a given patient.
3 | MICROSENSRREV | microbiology sensitivity test results review task | A person reviews the sensitivity test of microbiology results of a given patient.
2 | MLREV | medication list review task | A person reviews a list of medication orders submitted to a given patient
3 | MARWLREV | medication administration record work list review task | A clinician reviews a work list of medications to be administered to a given patient.
2 | OREV | orders review task | A person reviews a list of orders submitted to a given patient.
2 | PATREPREV | pathology report review task | A person reviews a pathology report of a given patient.
2 | PROBLISTREV | problem list review task | A person reviews a list of problems of a given patient.
2 | RADREPREV | radiology report review task | A person reviews a radiology report of a given patient.
2 | REMLE | reminder list entry | Description: A person enters a health care reminder for a given patient.
3 | WELLREMLE | wellness reminder list entry | Description: A person enters a wellness or preventive care reminder for a given patient.
2 | RISKASSESS | risk assessment instrument task | A person reviews a Risk Assessment Instrument report of a given patient.
3 | FALLRISK | falls risk assessment instrument task | A person reviews a Falls Risk Assessment Instrument report of a given patient.

4.4.2.19855 HL7 v3 Value Set ActCoverageTypeCode

URL: https://hl7.org/fhir/R4/v3/ActCoverageTypeCode/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActCoverageTypeCode) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActCoverageTypeCode
Version: | 2014-03-26
Name: | v3.ActCoverageTypeCode
Title: | V3 Value SetActCoverageTypeCode
Definition: | Definition: Set of codes indicating the type of insurance policy or program that pays for the cost of benefits provided to covered parties.
OID: | 2.16.840.1.113883.1.11.19855 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.19855.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActCode where concept is-a _ActCoverageTypeCode

4.4.2.19855.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 52 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActCode

Lvl | Code | Display | Definition
0 | _ActCoverageTypeCode | ActCoverageTypeCode | Definition: Set of codes indicating the type of insurance policy or program that pays for the cost of benefits provided to covered parties.
1 | _ActInsurancePolicyCode | ActInsurancePolicyCode | Set of codes indicating the type of insurance policy or other source of funds to cover healthcare costs.
2 | EHCPOL | extended healthcare | Private insurance policy that provides coverage in addition to other policies (e.g. in addition to a Public Healthcare insurance policy).
2 | HSAPOL | health spending account | Insurance policy that provides for an allotment of funds replenished on a periodic (e.g. annual) basis. The use of the funds under this policy is at the discretion of the covered party.
2 | AUTOPOL | automobile | Insurance policy for injuries sustained in an automobile accident. Will also typically covered non-named parties to the policy, such as pedestrians and passengers.
3 | COL | collision coverage policy | Definition: An automobile insurance policy under which the insurance company will cover the cost of damages to an automobile owned by the named insured that are caused by accident or intentionally by another party.
3 | UNINSMOT | uninsured motorist policy | Definition: An automobile insurance policy under which the insurance company will indemnify a loss for which another motorist is liable if that motorist is unable to pay because he or she is uninsured. Coverage under the policy applies to bodily injury damages only. Injuries to the covered party caused by a hit-and-run driver are also covered.
2 | PUBLICPOL | public healthcare | Insurance policy funded by a public health system such as a provincial or national health plan. Examples include BC MSP (British Columbia Medical Services Plan) OHIP (Ontario Health Insurance Plan), NHS (National Health Service).
3 | DENTPRG | dental program | Definition: A public or government health program that administers and funds coverage for dental care to assist program eligible who meet financial and health status criteria.
3 | DISEASEPRG | public health program | Definition: A public or government health program that administers and funds coverage for health and social services to assist program eligible who meet financial and health status criteria related to a particular disease. Example: Reproductive health, sexually transmitted disease, and end renal disease programs.
4 | CANPRG | women's cancer detection program | Definition: A program that provides low-income, uninsured, and underserved women access to timely, high-quality screening and diagnostic services, to detect breast and cervical cancer at the earliest stages. Example: To improve women's access to screening for breast and cervical cancers, Congress passed the Breast and Cervical Cancer Mortality Prevention Act of 1990, which guided CDC in creating the National Breast and Cervical Cancer Early Detection Program (NBCCEDP), which provides access to critical breast and cervical cancer screening services for underserved women in the United States. An estimated 7 to 10% of U.S. women of screening age are eligible to receive NBCCEDP services. Federal guidelines establish an eligibility baseline to direct services to uninsured and underinsured women at or below 250% of federal poverty level; ages 18 to 64 for cervical screening; ages 40 to 64 for breast screening.
4 | ENDRENAL | end renal program | Definition: A public or government program that administers publicly funded coverage of kidney dialysis and kidney transplant services. Example: In the U.S., the Medicare End-stage Renal Disease program (ESRD), the National Kidney Foundation (NKF) American Kidney Fund (AKF) The Organ Transplant Fund.
4 | HIVAIDS | HIV-AIDS program | Definition: Government administered and funded HIV-AIDS program for beneficiaries meeting financial and health status criteria. Administration, funding levels, eligibility criteria, covered benefits, provider types, and financial participation are typically set by a regulatory process. Payer responsibilities for administering the program may be delegated to contractors. Example: In the U.S., the Ryan White program, which is administered by the Health Resources and Services Administration.
3 | MANDPOL | mandatory health program | mandatory health program
3 | MENTPRG | mental health program | Definition: Government administered and funded mental health program for beneficiaries meeting financial and mental health status criteria. Administration, funding levels, eligibility criteria, covered benefits, provider types, and financial participation are typically set by a regulatory process. Payer responsibilities for administering the program may be delegated to contractors. Example: In the U.S., states receive funding for substance use programs from the Substance Abuse Mental Health Administration (SAMHSA).

3 | SAFNET | safety net clinic program | Definition: Government administered and funded program to support provision of care to underserved populations through safety net clinics. Example: In the U.S., safety net providers such as federally qualified health centers (FQHC) receive funding under PHSA Section 330 grants administered by the Health Resources and Services Administration.

3 | SUBPRG | substance use program | Definition: Government administered and funded substance use program for beneficiaries meeting financial, substance use behavior, and health status criteria. Beneficiaries may be required to enroll as a result of legal proceedings. Administration, funding levels, eligibility criteria, covered benefits, provider types, and financial participation are typically set by a regulatory process. Payer responsibilities for administering the program may be delegated to contractors. Example: In the U.S., states receive funding for substance use programs from the Substance Abuse Mental Health Administration (SAMHSA).

3 | SUBSIDIZ | subsidized health program | Definition: A government health program that provides coverage for health services to persons meeting eligibility criteria such as income, location of residence, access to other coverages, health condition, and age, the cost of which is to some extent subsidized by public funds.

4 | SUBSIDMC | subsidized managed care program | Definition: A government health program that provides coverage through managed care contracts for health services to persons meeting eligibility criteria such as income, location of residence, access to other coverages, health condition, and age, the cost of which is to some extent subsidized by public funds. Discussion: The structure and business processes for underwriting and administering a subsidized managed care program is further specified by the Underwriter and Payer Role.class and Role.code.

4 | SUBSUPP | subsidized supplemental health program | Definition: A government health program that provides coverage for health services to persons meeting eligibility criteria for a supplemental health policy or program such as income, location of residence, access to other coverages, health condition, and age, the cost of which is to some extent subsidized by public funds. Example: Supplemental health coverage program may cover the cost of a health program or policy financial participations, such as the copays and the premiums, and may provide coverage for services in addition to those covered under the supplemented health program or policy. In the U.S., Medicaid programs may pay the premium for a covered party who is also covered under the Medicare program or a private health policy. Discussion: The structure and business processes for underwriting and administering a subsidized supplemental retiree health program is further specified by the Underwriter and Payer Role.class and Role.code.

2 | WCBPOL | worker's compensation | Insurance policy for injuries sustained in the work place or in the course of employment.

1 | _ActInsuranceTypeCode | ActInsuranceTypeCode | Definition: Set of codes indicating the type of insurance policy. Insurance, in law and economics, is a form of risk management primarily used to hedge against the risk of potential financial loss. Insurance is defined as the equitable transfer of the risk of a potential loss, from one entity to another, in exchange for a premium and duty of care. A policy holder is an individual or an organization enters into a contract with an underwriter which stipulates that, in exchange for payment of a sum of money (a premium), one or more covered parties (insureds) is guaranteed compensation for losses resulting from certain perils under specified conditions. The underwriter analyzes the risk of loss, makes a decision as to whether the risk is insurable, and prices the premium accordingly. A policy provides benefits that indemnify or cover the cost of a loss incurred by a covered party, and may include coverage for services required to remediate a loss. An insurance policy contains pertinent facts about the policy holder, the insurance coverage, the covered parties, and the insurer. A policy may include exemptions and provisions specifying the extent to which the indemnification clause cannot be enforced for intentional tortious conduct of a covered party, e.g., whether the covered parties are jointly or severably insured. Discussion: In contrast to programs, an insurance policy has one or more policy holders, who own the policy. The policy holder may be the covered party, a relative of the covered party, a partnership, or a corporation, e.g., an employer. A subscriber of a self-insured health insurance policy is a policy holder. A subscriber of an employer sponsored health insurance policy is holds a certificate of coverage, but is not a policy holder; the policy holder is the employer. See CoveredRoleType.

2 | _ActHealthInsuranceTypeCode | ActHealthInsuranceTypeCode | Definition: Set of codes indicating the type of health insurance policy that covers health services provided to covered parties. A health insurance policy is a written contract for insurance between the insurance company and the policyholder, and contains pertinent facts about the policy owner (the policy holder), the health insurance coverage, the insured subscribers and dependents, and the insurer. Health insurance is typically administered in accordance with a plan, which specifies (1) the type of health services and health conditions that will be covered under what circumstances (e.g., exclusion of a pre-existing condition, service must be deemed medically necessary; service must not be experimental; service must provided in accordance with a protocol; drug must be on a formulary; service must be prior authorized; or be a referral from a primary care provider); (2) the type and affiliation of providers (e.g., only allopathic physicians, only in network, only providers employed by an HMO); (3) financial participations required of covered parties (e.g., co-pays, coinsurance, deductibles, out-of-pocket); and (4) the manner in which services will be paid (e.g., under indemnity or fee-for-service health plans, the covered party typically pays out-of-pocket and then file a claim for reimbursement, while health plans that have contractual relationships with providers, i.e., network providers, typically do not allow the providers to bill the covered party for the cost of the service until after filing a claim with the payer and receiving reimbursement).

3 | DENTAL | dental care policy | Definition: A health insurance policy that that covers benefits for dental services.

3 | DISEASE | disease specific policy | Definition: A health insurance policy that covers benefits for healthcare services provided for named conditions under the policy, e.g., cancer, diabetes, or HIV-AIDS.

3 | DRUGPOL | drug policy | Definition: A health insurance policy that covers benefits for prescription drugs, pharmaceuticals, and supplies.

3 | HIP | health insurance plan policy | Definition: A health insurance policy that covers healthcare benefits by protecting covered parties from medical expenses arising from health conditions, sickness, or accidental injury as well as preventive care. Health insurance policies explicitly exclude coverage for losses insured under a disability policy, workers' compensation program, liability insurance (including automobile insurance); or for medical expenses, coverage for on-site medical clinics or for limited dental or vision benefits when these are provided under a separate policy. Discussion: Health insurance policies are offered by health insurance plans that typically reimburse providers for covered services on a fee-for-service basis, that is, a fee that is the allowable amount that a provider may charge. This is in contrast to managed care plans, which typically prepay providers a per-member/per-month amount or capitation as reimbursement for all covered services rendered. Health insurance plans include indemnity and healthcare services plans.

3 | LTC | long term care policy | Definition: An insurance policy that covers benefits for long-term care services people need when they no longer can care for themselves. This may be due to an accident, disability, prolonged illness or the simple process of aging. Long-term care services assist with activities of daily living including: Help at home with day-to-day activities, such as cooking, cleaning, bathing and dressing Care in the community, such as in an adult day care facility Supervised care provided in an assisted living facility Skilled care provided in a nursing home

3 | MCPOL | managed care policy | Definition: Government mandated program providing coverage, disability income, and vocational rehabilitation for injuries sustained in the work place or in the course of employment. Employers may either self-fund the program, purchase commercial coverage, or pay a premium to a government entity that administers the program. Employees may be required to pay premiums toward the cost of coverage as well. Managed care policies specifically exclude coverage for

losses insured under a disability policy, workers' compensation program, liability insurance (including automobile insurance); or for medical expenses, coverage for on-site medical clinics or for limited dental or vision benefits when these are provided under a separate policy. Discussion: Managed care policies are offered by managed care plans that contract with selected providers or health care organizations to provide comprehensive health care at a discount to covered parties and coordinate the financing and delivery of health care. Managed care uses medical protocols and procedures agreed on by the medical profession to be cost effective, also known as medical practice guidelines. Providers are typically reimbursed for covered services by a capitated amount on a per member per month basis that may reflect difference in the health status and level of services anticipated to be needed by the member.

4 | POS | point of service policy | Definition: A policy for a health plan that has features of both an HMO and a FFS plan. Like an HMO, a POS plan encourages the use its HMO network to maintain discounted fees with participating providers, but recognizes that sometimes covered parties want to choose their own provider. The POS plan allows a covered party to use providers who are not part of the HMO network (non-participating providers). However, there is a greater cost associated with choosing these non-network providers. A covered party will usually pay deductibles and coinsurances that are substantially higher than the payments when he or she uses a plan provider. Use of non-participating providers often requires the covered party to pay the provider directly and then to file a claim for reimbursement, like in an FFS plan.

4 | HMO | health maintenance organization policy | Definition: A policy for a health plan that provides coverage for health care only through contracted or employed physicians and hospitals located in particular geographic or service areas. HMOs emphasize prevention and early detection of illness. Eligibility to enroll in an HMO is determined by where a covered party lives or works.

4 | PPO | preferred provider organization policy | Definition: A network-based, managed care plan that allows a covered party to choose any health care provider. However, if care is received from a "preferred" (participating in-network) provider, there are generally higher benefit coverage and lower deductibles.

3 | MENTPOL | mental health policy | Definition: A health insurance policy that covers benefits for mental health services and prescriptions.

3 | SUBPOL | substance use policy | Definition: A health insurance policy that covers benefits for substance use services.

3 | VISPOL | vision care policy | Definition: Set of codes for a policy that provides coverage for health care expenses arising from vision services. A health insurance policy that covers benefits for vision care services, prescriptions, and products.

2 | DIS | disability insurance policy | Definition: An insurance policy that provides a regular payment to compensate for income lost due to the covered party's inability to work because of illness or injury.

2 | EWB | employee welfare benefit plan policy | Definition: An insurance policy under a benefit plan run by an employer or employee organization for the purpose of providing benefits other than pension-related to employees and their families. Typically provides health-related benefits, benefits for disability, disease or unemployment, or day care and scholarship benefits, among others. An employer sponsored health policy includes coverage of health care expenses arising from sickness or accidental injury, coverage for on-site medical clinics or for dental or vision benefits, which are typically provided under a separate policy. Coverage excludes health care expenses covered by accident or disability, workers' compensation, liability or automobile insurance.

2 | FLEXP | flexible benefit plan policy | Definition: An insurance policy that covers qualified benefits under a Flexible Benefit plan such as group medical insurance, long and short term disability income insurance, group term life insurance for employees only up to $50,000 face amount, specified disease coverage such as a cancer policy, dental and/or vision insurance, hospital indemnity insurance, accidental death and dismemberment insurance, a medical expense reimbursement plan and a dependent care reimbursement plan. Discussion: See UnderwriterRoleTypeCode flexible benefit plan which is defined as a benefit plan that allows employees to choose from several life, health, disability, dental, and other insurance plans according to their individual needs. Also known as cafeteria plans. Authorized under Section 125 of the Revenue Act of 1978.

2 | LIFE | life insurance policy | Definition: A policy under which the insurer agrees to pay a sum of money upon the occurrence of the covered partys death. In return, the policyholder agrees to pay a stipulated amount called a premium at regular intervals. Life insurance indemnifies the beneficiary for the loss of the insurable interest that a beneficiary has in the life of a covered party. For persons related by blood, a substantial interest established through love and affection, and for all other persons, a lawful and substantial economic interest in having the life of the insured continue. An insurable interest is required when purchasing life insurance on another person. Specific exclusions are often written into the contract to limit the liability of the insurer; for example claims resulting from suicide or relating to war, riot and civil commotion. Discussion:A life insurance policy may be used by the covered party as a source of health care coverage in the case of a viatical settlement, which is the sale of a life insurance policy by the policy owner, before the policy matures. Such a sale, at a price discounted from the face amount of the policy but usually in excess of the premiums paid or current cash surrender value, provides the seller an immediate cash settlement. Generally, viatical settlements involve insured individuals with a life expectancy of less than two years. In countries without state-subsidized healthcare and high healthcare costs (e.g. United States), this is a practical way to pay extremely high health insurance premiums that severely ill people face. Some people are also familiar with life settlements, which are similar transactions but involve insureds with longer life expectancies (two to fifteen years).

3 | ANNU | annuity policy | Definition: A policy that, after an initial premium or premiums, pays out a sum at pre-determined intervals. For example, a policy holder may pay $10,000, and in return receive $150 each month until he dies; or $1,000 for each of 14 years or death benefits if he dies before the full term of the annuity has elapsed.

3 | TLIFE | term life insurance policy | Definition: Life insurance under which the benefit is payable only if the insured dies during a specified period. If an insured dies during that period, the beneficiary receives the death payments. If the insured survives, the policy ends and the beneficiary receives nothing.

3 | ULIFE | universal life insurance policy | Definition: Life insurance under which the benefit is payable upon the insuredaTMs death or diagnosis of a terminal illness. If an insured dies during that period, the beneficiary receives the death payments. If the insured survives, the policy ends and the beneficiary receives nothing

2 | PNC | property and casualty insurance policy | Definition: A type of insurance that covers damage to or loss of the policyholderaTMs property by providing payments for damages to property damage or the injury or death of living subjects. The terms "casualty" and "liability" insurance are often used interchangeably. Both cover the policyholder's legal liability for damages caused to other persons and/or their property.

2 | REI | reinsurance policy | Definition: An agreement between two or more insurance companies by which the risk of loss is proportioned. Thus the risk of loss is spread and a disproportionately large loss under a single policy does not fall on one insurance company. Acceptance by an insurer, called a reinsurer, of all or part of the risk of loss of another insurance company. Discussion: Reinsurance is a means by which an insurance company can protect itself against the risk of losses with other insurance companies. Individuals and corporations obtain insurance policies to provide protection for various risks (hurricanes, earthquakes, lawsuits, collisions, sickness and death, etc.). Reinsurers, in turn, provide insurance to insurance companies. For example, an HMO may purchase a reinsurance policy to protect itself from losing too much money from one insured's particularly expensive health care costs. An insurance company issuing an automobile liability policy, with a limit of $100,000 per accident may reinsure its liability in excess of $10,000. A fire insurance company which issues a large policy generally reinsures a portion of the risk with one or several other companies. Also called risk control insurance or stop-loss insurance.

2 | SURPL | surplus line insurance policy | Definition: A risk or part of a risk for which there is no normal insurance market available. Insurance written by unauthorized insurance companies. Surplus lines insurance is insurance placed with unauthorized insurance companies through licensed surplus lines agents or brokers.

2 | UMBRL | umbrella liability insurance policy | Definition: A form of insurance protection that provides additional liability coverage after the limits of your underlying policy are reached. An umbrella liability policy also protects you (the insured) in many situations not covered by the usual liability policies.

1 | _ActProgramTypeCode | ActProgramTypeCode | Definition: A set of codes used to indicate coverage under a program. A program is an organized structure for administering and funding coverage of a benefit package for covered parties meeting eligibility criteria, typically related to employment, health, financial, and demographic status. Programs are typically established or permitted by legislation with provisions for ongoing government oversight. Regulations may mandate the structure of the program, the manner in which it is funded and administered, covered benefits, provider types, eligibility criteria and financial participation. A government agency may be charged with implementing the program in accordance to the regulation. Risk of loss under a program in most cases would not meet what an underwriter would consider an insurable risk, i.e., the risk is not random in nature, not financially measurable, and likely requires subsidization with government funds. Discussion: Programs do not have policy holders or subscribers. Program eligibles are enrolled based on health status, statutory eligibility, financial status, or age. Program eligibles who are covered parties under the program may be referred to as members, beneficiaries, eligibles, or recipients. Programs risk are underwritten by not for profit organizations such as governmental entities, and the beneficiaries typically do not pay for any or some portion of the cost of coverage. See CoveredPartyRoleType.

2 | CHAR | charity program | Definition: A program that covers the cost of services provided directly to a beneficiary who typically has no other source of coverage without charge.

2 | CRIME | crime victim program | Definition: A program that covers the cost of services provided to crime victims for injuries or losses related to the occurrence of a crime.

2 | EAP | employee assistance program | Definition: An employee assistance program is run by an employer or employee organization for the purpose of providing benefits and covering all or part of the cost for employees to receive counseling, referrals, and advice in dealing with stressful issues in their lives. These may include substance abuse, bereavement, marital problems, weight issues, or general wellness issues. The services are usually provided by a third-party, rather than the company itself, and the company receives only summary statistical data from the service provider. Employee's names and services received are kept confidential.

2 | GOVEMP | government employee health program | Definition: A set of codes used to indicate a government program that is an organized structure for administering and funding coverage of a benefit package for covered parties meeting eligibility criteria, typically related to employment, health and financial status. Government programs are established or permitted by legislation with provisions for ongoing government oversight. Regulation mandates the structure of the program, the manner in which it is funded and administered, covered benefits, provider types, eligibility criteria and financial participation. A government agency is charged with implementing the program in accordance to the regulation Example: Federal employee health benefit program in the U.S.

2 | HIRISK | high risk pool program | Definition: A government program that provides health coverage to individuals who are considered medically uninsurable or high risk, and who have been denied health insurance due to a serious health condition. In certain cases, it also applies to those who have been quoted very high premiums a" again, due to a serious health condition. The pool charges premiums for coverage. Because the pool covers high-risk people, it incurs a higher level of claims than premiums can cover. The insurance industry pays into the pool to make up the difference and help it remain viable.

2 | IND | indigenous peoples health program | Definition: Services provided directly and through contracted and operated indigenous peoples health programs. Example: Indian Health Service in the U.S.

2 | MILITARY | military health program | Definition: A government program that provides coverage for health services to military personnel, retirees, and dependents. A covered party who is a subscriber can choose from among Fee-for-Service (FFS) plans, and their Preferred Provider Organizations (PPO), or Plans offering a Point of Service (POS) Product, or Health Maintenance Organizations. Example: In the U.S., TRICARE, CHAMPUS.

2 | RETIRE | retiree health program | Definition: A government mandated program with specific eligibility requirements based on premium contributions made during employment, length of employment, age, and employment status, e.g., being retired, disabled, or a dependent of a covered party under this program. Benefits typically include ambulatory, inpatient, and long-term care, such as hospice care, home health care and respite care.

2 | SOCIAL | social service program | Definition: A social service program funded by a public or governmental entity. Example: Programs providing habilitation, food, lodging, medicine, transportation, equipment, devices, products, education, training, counseling, alteration of living or work space, and other resources to persons meeting eligibility criteria.

2 | VET | veteran health program | Definition: Services provided directly and through contracted and operated veteran health programs.

4.4.2.19897 HL7 v3 Value Set ActConsentType

URL: https://hl7.org/fhir/R4/v3/ActConsentType/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActConsentType) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActConsentType

Version: | 2014-03-26

Name: | v3.ActConsentType

Title: | V3 Value SetActConsentType

Definition: | Definition: The type of consent directive, e.g., to consent or dissent to collect, access, or use in specific ways within an EHRS or for health information exchange; or to disclose health information for purposes such as research.

OID: | 2.16.840.1.113883.1.11.19897 (for OID based terminology systems)

Source Resource | XML / JSON

This value set is not currently used

4.4.2.19897.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActCode where concept is-a _ActConsentType

4.4.2.19897.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 9 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActCode

Lvl | Code | Display | Definition
0 | _ActConsentType | ActConsentType | Definition: The type of consent directive, e.g., to consent or dissent to collect, access, or use in specific ways within an EHRS or for health information exchange; or to disclose health information for purposes such as research.
1 | ICOL | information collection | Definition: Consent to have healthcare information collected in an electronic health record. This entails that the information may be used in analysis, modified, updated.
1 | IDSCL | information disclosure | Definition: Consent to have collected healthcare information disclosed.
1 | INFA | information access | Definition: Consent to access healthcare information.
2 | INFAO | access only | Definition: Consent to access or "read" only, which entails that the information is not to be copied, screen printed, saved, emailed, stored, re-disclosed or altered in any way. This level ensures that data which is masked or to which access is restricted will not be. Example: Opened and then emailed or screen printed for use outside of the consent directive purpose.
2 | INFASO | access and save only | Definition: Consent to access and save only, which entails that access to the saved copy will remain locked.
1 | IRDSCL | information redisclosure | Definition: Information re-disclosed without the patient's consent.
1 | RESEARCH | research information access | Definition: Consent to have healthcare information in an electronic health record accessed for research purposes.
2 | RSDID | de-identified information access | Definition: Consent to have de-identified healthcare information in an electronic health record that is accessed for research purposes, but without consent to re-identify the information under any circumstance.
2 | RSREID | re-identifiable information access | Definition: Consent to have de-identified healthcare information in an electronic health record that is accessed for research purposes re-identified under specific circumstances outlined in the consent. Example:: Where there is a need to inform the subject of potential health issues.

4.4.2.20224 HL7 v3 Value Set ActClassInvestigation

URL: https://hl7.org/fhir/R4/v3/ActClassInvestigation/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActClassInvestigation) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActClassInvestigation
Version: | 2014-03-26
Name: | v3.ActClassInvestigation
Title: | V3 Value SetActClassInvestigation
Definition: | An formalized inquiry into the circumstances surrounding a particular unplanned event or potential event for the purposes of identifying possible causes and contributing factors for the event. This investigation could be conducted at a local institutional level or at the level of a local or national government.
OID: | 2.16.840.1.113883.1.11.20224 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.20224.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActClass where concept is-a INVSTG

4.4.2.20224.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 1 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActClass version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActClass

Code | Display | Definition
INVSTG | investigation | An formalized inquiry into the circumstances surrounding a particular unplanned event or potential event for the purposes of identifying possible causes and contributing factors for the event. This investigation could be conducted at a local institutional level or at the level of a local or national government.

4.4.2.20425 HL7 v3 Value Set ActConsentDirective

URL: https://hl7.org/fhir/R4/v3/ActConsentDirective/vs.html
Vocabulary Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ActConsentDirective) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ActConsentDirective
Version: | 2014-03-26
Name: | v3.ActConsentDirective
Title: | V3 Value SetActConsentDirective
Definition: | ActConsentDirective codes are used to specify the type of Consent Directive to which a Consent Directive Act conforms.
OID: | 2.16.840.1.113883.1.11.20425 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is used in the following places:

• Resource: Contract.term.offer.decision (CodeableConcept / Extensible)

4.4.2.20425.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActCode where concept is-a _ActConsentDirective

4.4.2.20425.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 10 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActCode

Lvl | Code | Display | Definition

0 | _ActConsentDirective | ActConsentDirective | Specifies the type of agreement between one or more grantor and grantee in which rights and obligations related to one or more shared items of interest are allocated. Usage Note: Such agreements may be considered "consent directives" or "contracts" depending on the context, and are considered closely related or synonymous from a legal perspective. Examples: Healthcare Privacy Consent Directive permitting or restricting in whole or part the collection, access, use, and disclosure of health information, and any associated handling caveats. Healthcare Medical Consent Directive to receive medical procedures after being informed of risks and benefits, thereby reducing the grantee's liability. Research Informed Consent for participation in clinical trials and disclosure of health information after being informed of risks and benefits, thereby reducing the grantee's liability. Substitute decision maker delegation in which the grantee assumes responsibility to act on behalf of the grantor. Contracts in which the agreement requires assent/dissent by the grantor of terms offered by a grantee, a consumer opts out of an "award" system for use of a retailer's marketing or credit card vendor's point collection cards in exchange for allowing purchase tracking and profiling. A mobile device or App privacy policy and terms of service to which a user must agree in whole or in part in order to utilize the service. Agreements between a client and an authorization server or between an authorization server and a resource operator and/or resource owner permitting or restricting e.g., collection, access, use, and disclosure of information, and any associated handling caveats.

1 | EMRGONLY | emergency only | This general consent directive specifically limits disclosure of health information for purpose of emergency treatment. Additional parameters may further limit the disclosure to specific users, roles, duration, types of information, and impose uses obligations. Definition: Opt-in to disclosure of health information for emergency only consent directive.

1 | GRANTORCHOICE | grantor choice | A grantor's terms of agreement to which a grantee may assent or dissent, and which may include an opportunity for a grantee to request restrictions or extensions. Comment: A grantor typically is able to stipulate preferred terms of agreement when the grantor has control over the topic of the agreement, which a grantee must accept in full or may be offered an opportunity to extend or restrict certain terms. Usage Note: If the grantor's term of agreement must be accepted in full, then this is considered "basic consent". If a grantee is offered an opportunity to extend or restrict certain terms, then the agreement is considered "granular consent". Examples: Healthcare: A PHR account holder [grantor] may require any PHR user [grantee] to accept the terms of agreement in full, or may permit a PHR user to extend or restrict terms selected by the account holder or requested by the PHR user. Non-healthcare: The owner of a resource server [grantor] may require any authorization server [grantee] to meet authorization requirements stipulated in the grantor's terms of agreement.

1 | IMPLIED | implied consent | A grantor's presumed assent to the grantee's terms of agreement is based on the grantor's behavior, which may result from not expressly assenting to the consent directive offered, or from having no right to assent or dissent offered by the grantee. Comment: Implied or "implicit" consent occurs when the behavior of the grantor is understood by a reasonable person to signal agreement to the grantee's terms. Usage Note: Implied consent with no opportunity to assent or dissent to certain terms is considered "basic consent". Examples: Healthcare: A patient schedules an appointment with a provider, and either does not take the opportunity to expressly assent or dissent to the provider's consent directive, does not have an opportunity to do so, as in the case where emergency care is required, or simply behaves as though the patient [grantor] agrees to the rights granted to the provider [grantee] in an implicit consent directive. An injured and unconscious patient is deemed to have assented to emergency treatment by those permitted to do so under jurisdictional laws, e.g., Good Samaritan laws. Non-healthcare: Upon receiving a driver's license, the driver is deemed to have assented without explicitly consenting to undergoing field sobriety tests. A corporation that does business in a foreign nation is deemed to have deemed to have assented without explicitly consenting to abide by that nation's laws.

1 | IMPLIEDD | implied consent with opportunity to dissent | A grantor's presumed assent to the grantee's terms of agreement, which is based on the grantor's behavior, and includes a right to dissent to certain terms. Comment: A grantor assenting to the grantee's terms of agreement may or may not exercise a right to dissent to grantor selected terms or to grantee's selected terms to which a grantor may dissent. Usage Note: Implied or "implicit" consent with an "opportunity to dissent" occurs when the grantor's behavior is understood by a reasonable person to signal assent to the grantee's terms of agreement whether the grantor requests or the grantee approves further restrictions, is considered "granular consent". Examples: Healthcare Examples: A healthcare provider deems a patient's assent to disclosure of health information to family members and friends, but offers an opportunity or permits the patient to dissent to such disclosures. A health information exchanges deems a patient to have assented to disclosure of health information for treatment purposes, but offers the patient an opportunity to dissents to disclosure to particular provider organizations. Non-healthcare Examples: A bank deems a banking customer's assent to specified collection,

access, use, or disclosure of financial information as a requirement of holding a bank account, but provides the user an opportunity to limit third-party collection, access, use or disclosure of that information for marketing purposes.

1 | NOCONSENT | no consent | No notification or opportunity is provided for a grantor to assent or dissent to a grantee's terms of agreement. Comment: A "No Consent" policy scheme provides no opportunity for accommodation of an individual's preferences, and may not comply with Fair Information Practice Principles [FIPP] by enabling the data subject to object, access collected information, correct errors, or have accounting of disclosures. Usage Note: The grantee's terms of agreement, may be available to the grantor by reviewing the grantee's privacy policies, but there is no notice by which a grantor is apprised of the policy directly or able to acknowledge. Examples: Healthcare: Without notification or an opportunity to assent or dissent, a patient's health information is automatically included in and available (often according to certain rules) through a health information exchange. Note that this differs from implied consent, where the patient is assumed to have consented. Without notification or an opportunity to assent or dissent, a patient's health information is collected, accessed, used, or disclosed for research, public health, security, fraud prevention, court order, or law enforcement. Non-healthcare: Without notification or an opportunity to assent or dissent, a consumer's healthcare or non-healthcare internet searches are aggregated for secondary uses such as behavioral tracking and profiling. Without notification or an opportunity to assent or dissent, a consumer's location and activities in a shopping mall are tracked by RFID tags on purchased items.

1 | NOPP | notice of privacy practices | Acknowledgement of custodian notice of privacy practices. Usage Notes: This type of consent directive acknowledges a custodian's notice of privacy practices including its permitted collection, access, use and disclosure of health information to users and for purposes of use specified.

1 | OPTIN | opt-in | A grantor's assent to the terms of an agreement offered by a grantee without an opportunity for to dissent to any terms. Comment: Acceptance of a grantee's terms pertaining, for example, to permissible activities, purposes of use, handling caveats, expiry date, and revocation policies. Usage Note: Opt-in with no opportunity for a grantor to restrict certain permissions sought by the grantee is considered "basic consent". Examples: Healthcare: A patient [grantor] signs a provider's [grantee's] consent directive form, which lists permissible collection, access, use, or disclosure activities, purposes of use, handling caveats, and revocation policies. Non-healthcare: An employee [grantor] signs an employer's [grantee's] non-disclosure and non-compete agreement.

1 | OPTINR | opt-in with restrictions | A grantor's assent to the grantee's terms of an agreement with an opportunity for to dissent to certain grantor or grantee selected terms. Comment: A grantor dissenting to the grantee's terms of agreement may or may not exercise a right to assent to grantor's pre-approved restrictions or to grantee's selected terms to which a grantor may dissent. Usage Note: Opt-in with restrictions is considered "granular consent" because the grantor has an opportunity to narrow the permissions sought by the grantee. Examples: Healthcare: A patient assent to grantee's consent directive terms for collection, access, use, or disclosure of health information, and dissents to disclosure to certain recipients as allowed by the provider's pre-approved restriction list. Non-Healthcare: A cell phone user assents to the cell phone's privacy practices and terms of use, but dissents from location tracking by turning off the cell phone's tracking capability.

1 | OPTOUT | op-out | A grantor's dissent to the terms of agreement offered by a grantee without an opportunity for to assent to any terms. Comment: Rejection of a grantee's terms of agreement pertaining, for example, to permissible activities, purposes of use, handling caveats, expiry date, and revocation policies. Usage Note: Opt-out with no opportunity for a grantor to permit certain permissions sought by the grantee is considered "basic consent". Examples: Healthcare: A patient [grantor] declines to sign a provider's [grantee's] consent directive form, which lists permissible collection, access, use, or disclosure activities, purposes of use, handling caveats, revocation policies, and consequences of not assenting. Non-healthcare: An employee [grantor] refuses to sign an employer's [grantee's] agreement not to join unions or participate in a strike where state law protects employee's collective bargaining rights. A citizen [grantor] refuses to enroll in mandatory government [grantee] health insurance based on religious beliefs, which is an exemption.

1 | OPTOUTE | opt-out with exceptions | A grantor's dissent to the grantee's terms of agreement except for certain grantor or grantee selected terms. Comment: A rejection of a grantee's terms of agreement while assenting to certain permissions sought by the grantee or requesting approval of additional grantor terms. Usage Note: Opt-out with exceptions is considered a "granular consent" because the grantor has an opportunity to accept certain permissions sought by the grantee or request additional grantor terms, while rejecting other grantee terms. Examples: Healthcare: A patient [grantor] dissents to a health information exchange consent directive with the exception of disclosure based on a limited "time to live" shared secret [e.g., a token or password], which the patient can give to a provider when seeking care. Non-healthcare: A social media user [grantor] dissents from public access to their account, but assents to access to a circle of friends.

4.4.2.20428 HL7 v3 Value Set InformationSensitivityPolicy

URL: https://hl7.org/fhir/R4/v3/InformationSensitivityPolicy/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-InformationSensitivityPolicy) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-InformationSensitivityPolicy

Version: | 2014-03-26

Name: | v3.InformationSensitivityPolicy

Title: | V3 Value SetInformationSensitivityPolicy

Definition: | Sensitivity codes are not useful for interoperability outside of a policy domain because sensitivity policies are typically localized and vary drastically across policy domains even for the same information category because of differing organizational business rules, security policies, and jurisdictional requirements. For example, an "employee" sensitivity code would make little sense for use outside of a policy domain. "Taboo" would rarely be useful outside of a policy domain unless there are jurisdictional requirements requiring that a provider disclose sensitive information to a patient directly. Sensitivity codes may be more appropriate in a legacy system's Master Files in order to notify those who access a patient's orders and observations about the sensitivity policies that apply. Newer systems may have a security engine that uses a sensitivity policy's criteria directly. The specializable Sensitivity Act.code may be useful in some scenarious if used in combination with a sensitivity identifier and/or Act.title.

OID: | 2.16.840.1.113883.1.11.20428 (for OID based terminology systems)

Source Resource | XML / JSON

This value set is not currently used

4.4.2.20428.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActCode where concept is-a _InformationSensitivityPolicy

4.4.2.20428.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 41 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActCode

Lvl | Code | Display | Definition

0 | _InformationSensitivityPolicy | InformationSensitivityPolicy | A mandate, obligation, requirement, rule, or expectation characterizing the value or importance of a resource and may include its vulnerability. (Based on ISO7498-2:1989. Note: The vulnerability of personally identifiable sensitive information may be based on concerns that the unauthorized disclosure may result in social stigmatization or discrimination.) Description: Types of Sensitivity policy that apply to Acts or Roles. A sensitivity policy is adopted by an enterprise or group of enterprises (a 'policy domain') through a formal data use agreement that stipulates the value, importance, and vulnerability of information. A sensitivity code representing a sensitivity policy may be associated with criteria such as categories of information or sets of information identifiers (e.g., a value set of clinical codes or branch in a code system hierarchy). These criteria may in turn be used for the Policy Decision Point in a Security Engine. A sensitivity code may be used to set the confidentiality code used on information about Acts and Roles to trigger the security mechanisms required to control how security principals (i.e., a person, a machine, a software application) may act on the information (e.g., collection, access, use, or disclosure). Sensitivity codes are never assigned to the transport or business envelope containing patient specific information being exchanged outside of a policy domain as this would disclose the information intended to be protected by the policy. When sensitive information is exchanged with others outside of a policy domain, the confidentiality code on the transport or business envelope conveys the receiver's responsibilities and indicates the how the information is to be safeguarded without unauthorized disclosure of the sensitive information. This ensures that sensitive information is treated by receivers as the sender intends, accomplishing interoperability without point to point negotiations. Usage Note: Sensitivity codes are not useful for interoperability outside of a policy domain because sensitivity policies are typically localized and vary drastically across policy domains even for the same information category because of differing organizational business rules, security policies, and jurisdictional requirements. For example, an employee's sensitivity code would make little sense for use outside of a policy domain. 'Taboo' would rarely be useful outside of a policy domain unless there are jurisdictional requirements requiring that a provider disclose sensitive information to a patient directly. Sensitivity codes may be more appropriate in a legacy system's Master Files in order to notify those who access a patient's orders and observations about the sensitivity policies that apply. Newer systems may have a security engine that uses a sensitivity policy's criteria directly. The specializable InformationSensitivityPolicy Act.code may be useful in some scenarios if used in combination with a sensitivity identifier and/or Act.title.

1 | _ActInformationSensitivityPolicy | ActInformationSensitivityPolicy | Types of sensitivity policies that apply to Acts. Act.confidentialityCode is defined in the RIM as "constraints around appropriate disclosure of information about this Act, regardless of mood." Usage Note: ActSensitivity codes are used to bind information to an Act.confidentialityCode according to local sensitivity policy so that those confidentiality codes can then govern its handling across enterprises. Internally to a policy domain, however, local policies guide the access control system on how end users in that policy domain are able to use information tagged with these sensitivity values.

2 | ETH | substance abuse information sensitivity | Policy for handling alcohol or drug-abuse information, which will be afforded heightened confidentiality. Information handling protocols based on organizational policies related to alcohol or drug-abuse information that is deemed sensitive. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

2 | GDIS | genetic disease information sensitivity | Policy for handling genetic disease information, which will be afforded heightened confidentiality. Information handling protocols based on organizational policies related to genetic disease information that is deemed sensitive. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

2 | HIV | HIV/AIDS information sensitivity | Policy for handling HIV or AIDS information, which will be afforded heightened confidentiality. Information handling protocols based on organizational policies related to HIV or AIDS information that is deemed sensitive. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

2 | MST | military sexual trauma information sensitivity | Policy for handling information related to sexual assault or repeated, threatening sexual harassment that occurred while the patient was in the military, which is afforded heightened confidentiality. Access control concerns for military sexual trauma is based on the patient being subject to control by a higher ranking military perpetrator and/or censure by others within the military unit. Due to the relatively unfettered access to healthcare information by higher ranking military personnel and those who have command over the patient, there is a need to sequester this information outside of the typical controls on access to military health records. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law in addition to this more generic code.

2 | SCA | sickle cell anemia information sensitivity | Policy for handling sickle cell disease information, which is afforded heightened confidentiality. Information handling protocols are based on organizational policies related to sickle cell disease information, which is deemed sensitive. Usage Note: If there is a jurisdictional mandate, then the Act valued with this ActCode should be associated with an Act valued with any applicable laws from the ActPrivacyLaw code system.

2 | SDV | sexual assault, abuse, or domestic violence information sensitivity | Policy for handling sexual assault, abuse, or domestic violence information, which will be afforded heightened confidentiality. Information handling protocols based on organizational policies related to sexual assault, abuse, or domestic violence information that is deemed sensitive. SDV code covers violence perpetrated by related and non-related persons. This code should be specific to physical and mental trauma caused by a related person only. The access control concerns are keeping the patient safe from the perpetrator who may have an abusive psychological control over the patient, may be stalking the patient, or may try to manipulate care givers into allowing the perpetrator to make contact with the patient. The definition needs to be clarified. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

2 | SEX | sexuality and reproductive health information sensitivity | Policy for handling sexuality and reproductive health information, which will be afforded heightened confidentiality. Information handling protocols based on organizational policies

related to sexuality and reproductive health information that is deemed sensitive. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.
2 | SPI | specially protected information sensitivity | Policy for handling information deemed specially protected by law or policy including substance abuse, substance use, psychiatric, mental health, behavioral health, and cognitive disorders, which is afforded heightened confidentiality. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law in addition to this more generic code.
3 | BH | behavioral health information sensitivity | Policy for handling information related to behavioral and emotional disturbances affecting social adjustment and physical health, which is afforded heightened confidentiality. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law in addition to this more generic code.
4 | COGN | cognitive disability information sensitivity | Policy for handling information related to cognitive disability disorders and conditions caused by these disorders, which are afforded heightened confidentiality. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law in addition to this more generic code. Examples may include dementia, traumatic brain injury, attention deficit, hearing and visual disability such as dyslexia and other disorders and related conditions which impair learning and self-sufficiency. However, the cognitive disabilities to which this term may apply versus other behavioral health categories varies by jurisdiction and organizational policy in part due to overlap with other behavioral health conditions. Implementers should constrain to those diagnoses applicable in the domain in which this code is used.
4 | DVD | developmental disability information sensitivity | Policy for handling information related to developmental disability disorders and conditions caused by these disorders, which is afforded heightened confidentiality. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law in addition to this more generic code. A diverse group of chronic conditions that are due to mental or physical impairments impacting activities of daily living, self-care, language acuity, learning, mobility, independent living and economic self-sufficiency. Examples may include Down syndrome and Autism spectrum. However, the developmental disabilities to which this term applies versus other behavioral health categories varies by jurisdiction and organizational policy in part due to overlap with other behavioral health conditions. Implementers should constrain to those diagnoses applicable in the domain in which this code is used.
4 | EMOTDIS | emotional disturbance information sensitivity | Policy for handling information related to emotional disturbance disorders and conditions caused by these disorders, which is afforded heightened confidentiality. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law in addition to this more generic code. Typical used to characterize behavioral and mental health issues of adolescents where the disorder may be temporarily diagnosed in order to avoid the potential and unnecessary stigmatizing diagnoses of disorder long term.
3 | MH | mental health information sensitivity | Policy for handling information related to psychological disorders, which is afforded heightened confidentiality. Mental health information may be deemed specifically sensitive and distinct from physical health, substance use disorders, and behavioral disabilities and disorders in some jurisdictions. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law in addition to this more generic code.
3 | PSY | psychiatry disorder information sensitivity | Policy for handling psychiatry psychiatric disorder information, which is afforded heightened confidentiality. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.
3 | PSYTHPN | psychotherapy note information sensitivity | Policy for handling psychotherapy note information, which is afforded heightened confidentiality. Usage Note: In some jurisdiction, disclosure of psychotherapy notes requires patient consent. If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.
3 | SUD | substance use disorder information sensitivity | Policy for handling information related to alcohol or drug use disorders and conditions caused by these disorders, which is afforded heightened confidentiality. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law in addition to this more generic code.
4 | ETHUD | alcohol use disorder information sensitivity | Policy for handling information related to alcohol use disorders and conditions caused by these disorders, which is afforded heightened confidentiality. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law in addition to this more generic code.
4 | OPIOIDUD | opioid use disorder information sensitivity | Policy for handling information related to opioid use disorders and conditions caused by these disorders, which is afforded heightened confidentiality. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law in addition to this more generic code.
2 | STD | sexually transmitted disease information sensitivity | Policy for handling sexually transmitted disease information, which will be afforded heightened confidentiality. Information handling protocols based on organizational policies related to sexually transmitted disease information that is deemed sensitive. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.
2 | TBOO | taboo | Policy for handling information not to be initially disclosed or discussed with patient except by a physician assigned to patient in this case. Information handling protocols based on organizational policies related to sensitive patient information that must be initially discussed with the patient by an attending physician before being disclosed to the patient. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code. Open Issue: This definition conflates a rule and a characteristic, and there may be a similar issue with ts sibling codes.
2 | VIO | violence information sensitivity | Policy for handling information related to harm by violence, which is afforded heightened confidentiality. Harm by violence is perpetrated by an unrelated person. Access control concerns for information about mental or physical harm resulting from violence caused by an unrelated person may include manipulation of care givers or access to records that enable the perpetrator contact or locate the patient, but the perpetrator will likely not have established abusive psychological control over the patient. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law in addition to this more generic code.
2 | SICKLE | sickle cell | Types of sensitivity policies that apply to Acts. Act.confidentialityCode is defined in the RIM as "constraints around appropriate disclosure of information about this Act, regardless of mood." Usage Note: ActSensitivity codes are used to bind information to an Act.confidentialityCode according to local sensitivity policy so that those confidentiality codes can then govern its handling across enterprises. Internally to a policy domain, however, local policies guide the access control system on how end users in that policy domain are able to use information tagged with these sensitivity values.
1 | _EntitySensitivityPolicyType | EntityInformationSensitivityPolicy | Types of sensitivity policies that may apply to a sensitive attribute on an Entity. Usage Note: EntitySensitivity codes are used to convey a policy that is applicable to sensitive information conveyed by an entity attribute. May be used to bind a Role.confidentialityCode associated with an Entity per organizational policy. Role.confidentialityCode is defined in the RIM as "an indication of the appropriate disclosure of information about this Role with respect to the playing Entity."
2 | DEMO | all demographic information sensitivity | Policy for handling all demographic information about an information subject, which will be afforded heightened confidentiality. Policies may govern sensitivity of information related to all demographic about

an information subject, the disclosure of which could impact the privacy, well-being, or safety of that subject. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

2 | DOB | date of birth information sensitivity | Policy for handling information related to an information subject's date of birth, which will be afforded heightened confidentiality.Policies may govern sensitivity of information related to an information subject's date of birth, the disclosure of which could impact the privacy, well-being, or safety of that subject. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

2 | GENDER | gender and sexual orientation information sensitivity | Policy for handling information related to an information subject's gender and sexual orientation, which will be afforded heightened confidentiality. Policies may govern sensitivity of information related to an information subject's gender and sexual orientation, the disclosure of which could impact the privacy, well-being, or safety of that subject. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

2 | LIVARG | living arrangement information sensitivity | Policy for handling information related to an information subject's living arrangement, which will be afforded heightened confidentiality. Policies may govern sensitivity of information related to an information subject's living arrangement, the disclosure of which could impact the privacy, well-being, or safety of that subject. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

2 | MARST | marital status information sensitivity | Policy for handling information related to an information subject's marital status, which will be afforded heightened confidentiality. Policies may govern sensitivity of information related to an information subject's marital status, the disclosure of which could impact the privacy, well-being, or safety of that subject. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

2 | RACE | race information sensitivity | Policy for handling information related to an information subject's race, which will be afforded heightened confidentiality. Policies may govern sensitivity of information related to an information subject's race, the disclosure of which could impact the privacy, well-being, or safety of that subject. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

2 | REL | religion information sensitivity | Policy for handling information related to an information subject's religious affiliation, which will be afforded heightened confidentiality. Policies may govern sensitivity of information related to an information subject's religion, the disclosure of which could impact the privacy, well-being, or safety of that subject. Usage Notes: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

1 | _RoleInformationSensitivityPolicy | RoleInformationSensitivityPolicy | Types of sensitivity policies that apply to Roles. Usage Notes: RoleSensitivity codes are used to bind information to a Role.confidentialityCode per organizational policy. Role.confidentialityCode is defined in the RIM as "an indication of the appropriate disclosure of information about this Role with respect to the playing Entity."

2 | B | business information sensitivity | Policy for handling trade secrets such as financial information or intellectual property, which will be afforded heightened confidentiality. Description: Since the service class can represent knowledge structures that may be considered a trade or business secret, there is sometimes (though rarely) the need to flag those items as of business level confidentiality. Usage Notes: No patient related information may ever be of this confidentiality level. If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

2 | EMPL | employer information sensitivity | Policy for handling information related to an employer which is deemed classified to protect an employee who is the information subject, and which will be afforded heightened confidentiality. Description: Policies may govern sensitivity of information related to an employer, such as law enforcement or national security, the identity of which could impact the privacy, well-being, or safety of an information subject who is an employee. Usage Notes: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

2 | LOCIS | location information sensitivity | Policy for handling information related to the location of the information subject, which will be afforded heightened confidentiality. Description: Policies may govern sensitivity of information related to the location of the information subject, the disclosure of which could impact the privacy, well-being, or safety of that subject. Usage Notes: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

2 | SSP | sensitive service provider information sensitivity | Policy for handling information related to a provider of sensitive services, which will be afforded heightened confidentiality. Description: Policies may govern sensitivity of information related to providers who deliver sensitive healthcare services in order to protect the privacy, well-being, and safety of the provider and of patients receiving sensitive services. Usage Notes: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

1 | ADOL | adolescent information sensitivity | Policy for handling information related to an adolescent, which will be afforded heightened confidentiality per applicable organizational or jurisdictional policy. An enterprise may have a policy that requires that adolescent patient information be provided heightened confidentiality. Information deemed sensitive typically includes health information and patient role information including patient status, demographics, next of kin, and location. Usage Note: For use within an enterprise in which an adolescent is the information subject. If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

1 | CEL | celebrity information sensitivity | Policy for handling information related to a celebrity (people of public interest (VIP), which will be afforded heightened confidentiality. Celebrities are people of public interest (VIP) about whose information an enterprise may have a policy that requires heightened confidentiality. Information deemed sensitive may include health information and patient role information including patient status, demographics, next of kin, and location. Usage Note: For use within an enterprise in which the information subject is deemed a celebrity or very important person. If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

1 | DIA | diagnosis information sensitivity | Policy for handling information related to a diagnosis, health condition or health problem, which will be afforded heightened confidentiality. Diagnostic, health condition or health problem related information may be deemed sensitive by organizational policy, and require heightened confidentiality. Usage Note: For use within an enterprise that provides heightened confidentiality to diagnostic, health condition or health problem related information deemed sensitive. If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

1 | DRGIS | drug information sensitivity | Policy for handling information related to a drug, which will be afforded heightened confidentiality. Drug information may be deemed sensitive by organizational policy, and require heightened confidentiality. Usage Note: For use within an enterprise that provides heightened confidentiality to drug information deemed sensitive. If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.
1 | EMP | employee information sensitivity | Policy for handling information related to an employee, which will be afforded heightened confidentiality. When a patient is an employee, an enterprise may have a policy that requires heightened confidentiality. Information deemed sensitive typically includes health information and patient role information including patient status, demographics, next of kin, and location. Usage Note: Policy for handling information related to an employee, which will be afforded heightened confidentiality. Description: When a patient is an employee, an enterprise may have a policy that requires heightened confidentiality. Information deemed sensitive typically includes health information and patient role information including patient status, demographics, next of kin, and location.
1 | PDS | patient default information sensitivity | Policy for specially protecting information reported by or about a patient, which is deemed sensitive within the enterprise (i.e., by default regardless of whether the patient requested that the information be deemed sensitive for another reason.) For example information reported by the patient about another person, e.g., a family member, may be deemed sensitive by default. Organizational policy may allow the sensitivity tag to be cleared on patient's request. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law in addition to this more generic code. For example, VA deems employee information sensitive by default. Information about a patient who is being stalked or a victim of abuse or violence may be deemed sensitive by default per a provider organization's policies.
1 | PHY | physician requested information sensitivity | Policy for handling information about a patient, which a physician or other licensed healthcare provider deems sensitive. Once tagged by the provider, this may trigger alerts for follow up actions according to organizational policy or jurisdictional law. Usage Note: For use within an enterprise that provides heightened confidentiality to certain types of information designated by a physician as sensitive. If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code. Use cases in which this code could be used are, e.g., in systems that lack the ability to automatically detect sensitive information and must rely on manual tagging; a system that lacks an applicable sensitivity tag, or for ad hoc situations where criticality of the situation requires that the tagging be done immediately by the provider before coding or transcription of consult notes can be completed, e.g., upon detection of a patient with suicidal tendencies or potential for violence.
1 | PRS | patient requested information sensitivity | Policy for specially protecting information reported by or about a patient, which the patient deems sensitive, and the patient requests that collection, access, use, or disclosure of that information be restricted. For example, a minor patient may request that information about reproductive health not be disclosed to the patient's family or to particular providers and payers. Usage Note: If there is a jurisdictional mandate, then use the applicable ActPrivacyLaw code system, and specify the law rather than or in addition to this more generic code.

4.4.2.20444 HL7 v3 Value Set SecurityPolicy

URL: https://hl7.org/fhir/R4/v3/SecurityPolicy/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-SecurityPolicy) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-SecurityPolicy
Version: | 2014-03-26
Name: | v3.SecurityPolicy
Title: | V3 Value SetSecurityPolicy
Definition: | Types of security policies that further specify the ActClassPolicy value set. Examples: encrypt prohibit redisclosure without consent directive
OID: | 2.16.840.1.113883.1.11.20444 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.20444.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActCode where concept is-a SecurityPolicy

4.4.2.20444.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 51 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActCode

Lvl | Code | Display | Definition
0 | SecurityPolicy | security policy | Types of security policies that further specify the ActClassPolicy value set. Examples: obligation to encrypt refrain from redisclosure without consent
1 | AUTHPOL | authorization policy | Authorisation policies are essentially security policies related to access-control and specify what activities a subject is permitted or forbidden to do, to a set of target objects. They are designed to protect target objects so are interpreted by access control agents or the run-time systems at the target system. A positive authorisation policy defines the actions that a subject is permitted to perform on a target. A negative authorisation policy specifies the actions that a subject is forbidden to perform on a target. Positive authorisation policies may also include filters to transform the parameters associated with their actions. (Based on PONDERS)
2 | ACCESSCONSCHEME | access control scheme | An access control policy specific to the type of access control scheme, which is used to enforce one or more authorization policies. Usage Note: Access control schemes are the type of access control policy,

which is comprised of access control policy rules concerning the provision of the access control service. There are two categories of access control policies, rule-based and identity-based, which are identified in CCITT Rec. X.800 aka ISO 7498-2. Rule-based access control policies are intended to apply to all access requests by any initiator on any target in a security domain. Identity-based access control policies are based on rules specific to an individual initiator, a group of initiators, entities acting on behalf of initiators, or originators acting in a specific role. Context can modify rule-based or identity-based access control policies. Context rules may define the entire policy in effect. Real systems will usually employ a combination of these policy types; if a rule-based policy is used, then an identity-based policy is usually in effect also. An access control scheme may be based on access control lists, capabilities, labels, and context or a combination of these. An access control scheme is a component of an access control mechanism or "service") along with the supporting mechanisms required by that scheme to provide access control decision information (ADI) supplied by the scheme to the access decision facility (ADF also known as a PDP). (Based on ISO/IEC 10181-3:1996) Examples: Attribute Based Access Control (ABAC) Discretionary Access Control (DAC) History Based Access Control (HBAC) Identity Based Access Control (IBAC) Mandatory Access Control (MAC) Organization Based Access Control (OrBAC) Relationship Based Access Control (RelBac) Responsibility Based Access Control (RespBAC) Risk Adaptable Access Control (RAdAC) >

1 | DELEPOL | delegation policy | Delegation policies specify which actions subjects are allowed to delegate to others. A delegation policy thus specifies an authorisation to delegate. Subjects must already possess the access rights to be delegated. Delegation policies are aimed at subjects delegating rights to servers or third parties to perform actions on their behalf and are not meant to be the means by which security administrators would assign rights to subjects. A negative delegation policy identifies what delegations are forbidden. A Delegation policy specifies the authorisation policy from which delegated rights are derived, the grantors, which are the entities which can delegate these access rights, and the grantees, which are the entities to which the access rights can be delegated. There are two types of delegation policy, positive and negative. (Based on PONDERS)

1 | ObligationPolicy | obligation policy | Conveys the mandated workflow action that an information custodian, receiver, or user must perform. Usage Notes: Per ISO 22600-2, ObligationPolicy instances 'are event-triggered and define actions to be performed by manager agent'. Per HL7 Composite Security and Privacy Domain Analysis Model: This value set refers to the action required to receive the permission specified in the privacy rule. Per OASIS XACML, an obligation is an operation specified in a policy or policy that is performed in conjunction with the enforcement of an access control decision.

2 | ANONY | anonymize | Custodian system must remove any information that could result in identifying the information subject.

2 | AOD | accounting of disclosure | Custodian system must make available to an information subject upon request an accounting of certain disclosures of the individualâ€™s protected health information over a period of time. Policy may dictate that the accounting include information about the information disclosed, the date of disclosure, the identification of the receiver, the purpose of the disclosure, the time in which the disclosing entity must provide a response and the time period for which accountings of disclosure can be requested.

2 | AUDIT | audit | Custodian system must monitor systems to ensure that all users are authorized to operate on information objects.

2 | AUDTR | audit trail | Custodian system must monitor and maintain retrievable log for each user and operation on information.

2 | CPLYCC | comply with confidentiality code | Custodian security system must retrieve, evaluate, and comply with the information handling directions of the Confidentiality Code associated with an information target.

2 | CPLYCD | comply with consent directive | Custodian security system must retrieve, evaluate, and comply with applicable information subject consent directives.

2 | CPLYJPP | comply with jurisdictional privacy policy | Custodian security system must retrieve, evaluate, and comply with applicable jurisdictional privacy policies associated with the target information.

2 | CPLYOPP | comply with organizational privacy policy | Custodian security system must retrieve, evaluate, and comply with applicable organizational privacy policies associated with the target information.

2 | CPLYOSP | comply with organizational security policy | Custodian security system must retrieve, evaluate, and comply with the organizational security policies associated with the target information.

2 | CPLYPOL | comply with policy | Custodian security system must retrieve, evaluate, and comply with applicable policies associated with the target information.

2 | DECLASSIFYLABEL | declassify security label | Custodian security system must declassify information assigned security labels by instantiating a new version of the classified information so as to break the binding of the classifying security label when assigning a new security label that marks the information as unclassified in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the previous assignment and binding.

2 | DEID | deidentify | Custodian system must strip information of data that would allow the identification of the source of the information or the information subject.

2 | DELAU | delete after use | Custodian system must remove target information from access after use.

2 | DOWNGRDLABEL | downgrade security label | Custodian security system must downgrade information assigned security labels by instantiating a new version of the classified information so as to break the binding of the classifying security label when assigning a new security label that marks the information as classified at a less protected level in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the previous assignment and binding.

2 | DRIVLABEL | derive security label | Custodian security system must assign and bind security labels derived from compilations of information by aggregation or disaggregation in order to classify information compiled in the information systems under its control for collection, access, use and disclosure in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the previous assignment and binding.

2 | ENCRYPT | encrypt | Custodian system must render information unreadable by algorithmically transforming plaintext into ciphertext. Usage Notes: A mathematical transposition of a file or data stream so that it cannot be deciphered at the receiving end without the proper key. Encryption is a security feature that assures that only the parties who are supposed to be participating in a videoconference or data transfer are able to do so. It can include a password, public and private keys, or a complex combination of all. (Per Infoway.)

3 | ENCRYPTR | encrypt at rest | Custodian system must render information unreadable and unusable by algorithmically transforming plaintext into ciphertext when "at rest" or in storage.

3 | ENCRYPTT | encrypt in transit | Custodian system must render information unreadable and unusable by algorithmically transforming plaintext into ciphertext while "in transit" or being transported by any means.

3 | ENCRYPTU | encrypt in use | Custodian system must render information unreadable and unusable by algorithmically transforming plaintext into ciphertext while in use such that operations permitted on the target information are limited by the license granted to the end user.

2 | HUAPRV | human approval | Custodian system must require human review and approval for permission requested.

2 | LABEL | assign security label | Custodian security system must assign and bind security labels in order to classify information created in the information systems under its control for collection, access, use and disclosure in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the assignment and binding. Usage Note: In security systems, security policy label assignments do not change, they may supersede prior assignments, and such reassignments are always tracked for auditing and other purposes.

2 | MASK | mask | Custodian system must render information unreadable and unusable by algorithmically transforming plaintext into ciphertext. User may be provided a key to decrypt per license or "shared secret".

2 | MINEC | minimum necessary | Custodian must limit access and disclosure to the minimum information required to support an authorized user's purpose of use. Usage Note: Limiting the information available for access and disclosure to that an authorized user or receiver "needs to know" in order to perform permitted workflow or purpose of use.

2 | PERSISTLABEL | persist security label | Custodian security system must persist the binding of security labels to classify information received or imported by information systems under its control for collection, access, use and disclosure in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the assignment and binding.

2 | PRIVMARK | privacy mark | Custodian must create and/or maintain human readable security label tags as required by policy. Map: Aligns with ISO 22600-3 Section A.3.4.3 description of privacy mark: "If present, the privacy-mark is not used for access control. The content of the privacy-mark may be defined by the security policy in force (identified by the security-policy-identifier) which may define a list of values to be used. Alternately, the value may be determined by the originator of the security-label."

2 | PSEUD | pseudonymize | Custodian system must strip information of data that would allow the identification of the source of the information or the information subject. Custodian may retain a key to relink data necessary to reidentify the information subject.

2 | REDACT | redact | Custodian system must remove information, which is not authorized to be access, used, or disclosed from records made available to otherwise authorized users.

2 | UPGRDLABEL | upgrade security label | Custodian security system must declassify information assigned security labels by instantiating a new version of the classified information so as to break the binding of the classifying security label when assigning a new security label that marks the information as classified at a more protected level in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the previous assignment and binding.

1 | RefrainPolicy | refrain policy | Conveys prohibited actions which an information custodian, receiver, or user is not permitted to perform unless otherwise authorized or permitted under specified circumstances. Usage Notes: ISO 22600-2 species that a Refrain Policy "defines actions the subjects must refrain from performing". Per HL7 Composite Security and Privacy Domain Analysis Model: May be used to indicate that a specific action is prohibited based on specific access control attributes e.g., purpose of use, information type, user role, etc.

2 | NOAUTH | no disclosure without subject authorization | Prohibition on disclosure without information subject's authorization.

2 | NOCOLLECT | no collection | Prohibition on collection or storage of the information.

2 | NODSCLCD | no disclosure without consent directive | Prohibition on disclosure without organizational approved patient restriction.

2 | NODSCLCDS | no disclosure without information subject's consent directive | Prohibition on disclosure without a consent directive from the information subject.

2 | NOINTEGRATE | no integration | Prohibition on Integration into other records.

2 | NOLIST | no unlisted entity disclosure | Prohibition on disclosure except to entities on specific access list.

2 | NOMOU | no disclosure without MOU | Prohibition on disclosure without an interagency service agreement or memorandum of understanding (MOU).

2 | NOORGPOL | no disclosure without organizational authorization | Prohibition on disclosure without organizational authorization.

2 | NOPAT | no disclosure to patient, family or caregivers without attending provider's authorization | Prohibition on disclosing information to patient, family or caregivers without attending provider's authorization. Usage Note: The information may be labeled with the ActInformationSensitivity TBOO code, triggering application of this RefrainPolicy code as a handling caveat controlling access. Maps to FHIR NOPAT: Typically, this is used on an Alert resource, when the alert records information on patient abuse or non-compliance. FHIR print name is "keep information from patient". Maps to the French realm - code: INVISIBLE_PATIENT. displayName: Document non visible par le patient codingScheme: 1.2.250.1.213.1.1.4.13 French use case: A label for documents that the author chose to hide from the patient until the content can be disclose to the patient in a face to face meeting between a healthcare professional and the patient (in French law some results like cancer diagnosis or AIDS diagnosis must be announced to the patient by a healthcare professional and should not be find out by the patient alone).

2 | NOPERSISTP | no collection beyond purpose of use | Prohibition on collection of the information beyond time necessary to accomplish authorized purpose of use is prohibited.

2 | NORDSCLCD | no redisclosure without consent directive | Prohibition on redisclosure without patient consent directive.

2 | NORDSCLCDS | no redisclosure without information subject's consent directive | Prohibition on redisclosure without a consent directive from the information subject.

2 | NORDSCLW | no disclosure without jurisdictional authorization | Prohibition on disclosure without authorization under jurisdictional law.

2 | NORELINK | no relinking | Prohibition on associating de-identified or pseudonymized information with other information in a manner that could or does result in disclosing information intended to be masked.

2 | NOREUSE | no reuse beyond purpose of use | Prohibition on use of the information beyond the purpose of use initially authorized.

2 | NOVIP | no unauthorized VIP disclosure | Prohibition on disclosure except to principals with access permission to specific VIP information.

2 | ORCON | no disclosure without originator authorization | Prohibition on disclosure except as permitted by the information originator.

4.4.2.20445 HL7 v3 Value Set ObligationPolicy

URL: https://hl7.org/fhir/R4/v3/ObligationPolicy/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ObligationPolicy) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ObligationPolicy
Version: | 2014-03-26
Name: | v3.ObligationPolicy
Title: | V3 Value SetObligationPolicy
Definition: | Conveys the mandated workflow action that an information custodian, receiver, or user must perform. Examples: encrypt Usage Note: Per OASIS XACML, an obligation is an operation specified in a policy or policy that is performed in conjunction with the enforcement of an access control decision.
OID: | 2.16.840.1.113883.1.11.20445 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.20445.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActCode where concept is-a ObligationPolicy

4.4.2.20445.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 29 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActCode

Lvl | Code | Display | Definition
0 | ObligationPolicy | obligation policy | Conveys the mandated workflow action that an information custodian, receiver, or user must perform. Usage Notes: Per ISO 22600-2, ObligationPolicy instances 'are event-triggered and define actions to be performed by manager agent'. Per HL7 Composite Security and Privacy Domain Analysis Model: This value set refers to the action required to receive the permission specified in the privacy rule. Per OASIS XACML, an obligation is an operation specified in a policy or policy that is performed in conjunction with the enforcement of an access control decision.
1 | ANONY | anonymize | Custodian system must remove any information that could result in identifying the information subject.
1 | AOD | accounting of disclosure | Custodian system must make available to an information subject upon request an accounting of certain disclosures of the individualâ€™s protected health information over a period of time. Policy may dictate that the accounting include information about the information disclosed, the date of disclosure, the identification of the receiver, the purpose of the disclosure, the time in which the disclosing entity must provide a response and the time period for which accountings of disclosure can be requested.
1 | AUDIT | audit | Custodian system must monitor systems to ensure that all users are authorized to operate on information objects.
1 | AUDTR | audit trail | Custodian system must monitor and maintain retrievable log for each user and operation on information.
1 | CPLYCC | comply with confidentiality code | Custodian security system must retrieve, evaluate, and comply with the information handling directions of the Confidentiality Code associated with an information target.
1 | CPLYCD | comply with consent directive | Custodian security system must retrieve, evaluate, and comply with applicable information subject consent directives.
1 | CPLYJPP | comply with jurisdictional privacy policy | Custodian security system must retrieve, evaluate, and comply with applicable jurisdictional privacy policies associated with the target information.
1 | CPLYOPP | comply with organizational privacy policy | Custodian security system must retrieve, evaluate, and comply with applicable organizational privacy policies associated with the target information.
1 | CPLYOSP | comply with organizational security policy | Custodian security system must retrieve, evaluate, and comply with the organizational security policies associated with the target information.
1 | CPLYPOL | comply with policy | Custodian security system must retrieve, evaluate, and comply with applicable policies associated with the target information.
1 | DECLASSIFYLABEL | declassify security label | Custodian security system must declassify information assigned security labels by instantiating a new version of the classified information so as to break the binding of the classifying security label when assigning a new security label that marks the information as unclassified in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the previous assignment and binding.
1 | DEID | deidentify | Custodian system must strip information of data that would allow the identification of the source of the information or the information subject.
1 | DELAU | delete after use | Custodian system must remove target information from access after use.
1 | DOWNGRDLABEL | downgrade security label | Custodian security system must downgrade information assigned security labels by instantiating a new version of the classified information so as to break the binding of the classifying security label when assigning a new security label that marks the information as classified at a less protected level in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the previous assignment and binding.
1 | DRIVLABEL | derive security label | Custodian security system must assign and bind security labels derived from compilations of information by aggregation or disaggregation in order to classify information compiled in the information systems under its control for collection, access, use and disclosure in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the previous assignment and binding.
1 | ENCRYPT | encrypt | Custodian system must render information unreadable by algorithmically transforming plaintext into ciphertext. Usage Notes: A mathematical transposition of a file or data stream so that it cannot be deciphered at the receiving end without the proper key. Encryption is a security feature that assures that only the parties who are supposed to be participating in a videoconference or data transfer are able to do so. It can include a password, public and private keys, or a complex combination of all. (Per Infoway.)
2 | ENCRYPTR | encrypt at rest | Custodian system must render information unreadable and unusable by algorithmically transforming plaintext into ciphertext when "at rest" or in storage.

2 | ENCRYPTT | encrypt in transit | Custodian system must render information unreadable and unusable by algorithmically transforming plaintext into ciphertext while "in transit" or being transported by any means.
2 | ENCRYPTU | encrypt in use | Custodian system must render information unreadable and unusable by algorithmically transforming plaintext into ciphertext while in use such that operations permitted on the target information are limited by the license granted to the end user.
1 | HUAPRV | human approval | Custodian system must require human review and approval for permission requested.
1 | LABEL | assign security label | Custodian security system must assign and bind security labels in order to classify information created in the information systems under its control for collection, access, use and disclosure in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the assignment and binding. Usage Note: In security systems, security policy label assignments do not change, they may supersede prior assignments, and such reassignments are always tracked for auditing and other purposes.
1 | MASK | mask | Custodian system must render information unreadable and unusable by algorithmically transforming plaintext into ciphertext. User may be provided a key to decrypt per license or "shared secret".
1 | MINEC | minimum necessary | Custodian must limit access and disclosure to the minimum information required to support an authorized user's purpose of use. Usage Note: Limiting the information available for access and disclosure to that an authorized user or receiver "needs to know" in order to perform permitted workflow or purpose of use.
1 | PERSISTLABEL | persist security label | Custodian security system must persist the binding of security labels to classify information received or imported by information systems under its control for collection, access, use and disclosure in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the assignment and binding.
1 | PRIVMARK | privacy mark | Custodian must create and/or maintain human readable security label tags as required by policy. Map: Aligns with ISO 22600-3 Section A.3.4.3 description of privacy mark: "If present, the privacy-mark is not used for access control. The content of the privacy-mark may be defined by the security policy in force (identified by the security-policy-identifier) which may define a list of values to be used. Alternately, the value may be determined by the originator of the security-label."
1 | PSEUD | pseudonymize | Custodian system must strip information of data that would allow the identification of the source of the information or the information subject. Custodian may retain a key to relink data necessary to reidentify the information subject.
1 | REDACT | redact | Custodian system must remove information, which is not authorized to be access, used, or disclosed from records made available to otherwise authorized users.
1 | UPGRDLABEL | upgrade security label | Custodian security system must declassify information assigned security labels by instantiating a new version of the classified information so as to break the binding of the classifying security label when assigning a new security label that marks the information as classified at a more protected level in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the previous assignment and binding.

4.4.2.20446 HL7 v3 Value Set RefrainPolicy

URL: https://hl7.org/fhir/R4/v3/RefrainPolicy/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-RefrainPolicy) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-RefrainPolicy
Version: | 2014-03-26
Name: | v3.RefrainPolicy
Title: | V3 Value SetRefrainPolicy
Definition: | Conveys prohibited actions which an information custodian, receiver, or user is not permitted to perform unless otherwise authorized or permitted under specified circumstances. Examples: prohibit redisclosure without consent directive
OID: | 2.16.840.1.113883.1.11.20446 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.20446.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActCode where concept is-a RefrainPolicy

4.4.2.20446.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 18 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActCode

Lvl | Code | Display | Definition
0 | RefrainPolicy | refrain policy | Conveys prohibited actions which an information custodian, receiver, or user is not permitted to perform unless otherwise authorized or permitted under specified circumstances. Usage Notes: ISO 22600-2 species that a Refrain Policy "defines actions the subjects must refrain from performing". Per HL7 Composite Security and Privacy Domain Analysis Model: May be used to indicate that a specific action is prohibited based on specific access control attributes e.g., purpose of use, information type, user role, etc.
1 | NOAUTH | no disclosure without subject authorization | Prohibition on disclosure without information subject's authorization.
1 | NOCOLLECT | no collection | Prohibition on collection or storage of the information.
1 | NODSCLCD | no disclosure without consent directive | Prohibition on disclosure without organizational approved patient restriction.

1 | NODSCLCDS | no disclosure without information subject's consent directive | Prohibition on disclosure without a consent directive from the information subject.
1 | NOINTEGRATE | no integration | Prohibition on Integration into other records.
1 | NOLIST | no unlisted entity disclosure | Prohibition on disclosure except to entities on specific access list.
1 | NOMOU | no disclosure without MOU | Prohibition on disclosure without an interagency service agreement or memorandum of understanding (MOU).
1 | NOORGPOL | no disclosure without organizational authorization | Prohibition on disclosure without organizational authorization.
1 | NOPAT | no disclosure to patient, family or caregivers without attending provider's authorization | Prohibition on disclosing information to patient, family or caregivers without attending provider's authorization. Usage Note: The information may be labeled with the ActInformationSensitivity TBOO code, triggering application of this RefrainPolicy code as a handling caveat controlling access. Maps to FHIR NOPAT: Typically, this is used on an Alert resource, when the alert records information on patient abuse or non-compliance. FHIR print name is "keep information from patient". Maps to the French realm - code: INVISIBLE_PATIENT. displayName: Document non visible par le patient codingScheme: 1.2.250.1.213.1.1.4.13 French use case: A label for documents that the author chose to hide from the patient until the content can be disclose to the patient in a face to face meeting between a healthcare professional and the patient (in French law some results like cancer diagnosis or AIDS diagnosis must be announced to the patient by a healthcare professional and should not be find out by the patient alone).
1 | NOPERSISTP | no collection beyond purpose of use | Prohibition on collection of the information beyond time necessary to accomplish authorized purpose of use is prohibited.
1 | NORDSCLCD | no redisclosure without consent directive | Prohibition on redisclosure without patient consent directive.
1 | NORDSCLCDS | no redisclosure without information subject's consent directive | Prohibition on redisclosure without a consent directive from the information subject.
1 | NORDSCLW | no disclosure without jurisdictional authorization | Prohibition on disclosure without authorization under jurisdictional law.
1 | NORELINK | no relinking | Prohibition on associating de-identified or pseudonymized information with other information in a manner that could or does result in disclosing information intended to be masked.
1 | NOREUSE | no reuse beyond purpose of use | Prohibition on use of the information beyond the purpose of use initially authorized.
1 | NOVIP | no unauthorized VIP disclosure | Prohibition on disclosure except to principals with access permission to specific VIP information.
1 | ORCON | no disclosure without originator authorization | Prohibition on disclosure except as permitted by the information originator.

4.4.2.20448 HL7 v3 Value Set PurposeOfUse

URL: https://hl7.org/fhir/R4/v3/PurposeOfUse/vs.html
Vocabulary Work Group | Maturity Level: 3 | Trial Use | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-PurposeOfUse) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-PurposeOfUse
Version: | 2014-03-26
Name: | v3.PurposeOfUse
Title: | V3 Value SetPurposeOfUse
Definition: | Supports communication of purpose of use at a general level.
OID: | 2.16.840.1.113883.1.11.20448 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is used in the following places:

- Resource: Consent.provision.purpose (Coding / Extensible)
- Resource: Contract.term.action.intent (CodeableConcept / Example)
- Resource: Contract.term.action.reasonCode (CodeableConcept / Example)
- Resource: Provenance.reason (CodeableConcept / Extensible)
- Profile: provenance-relevant-history: Provenance.reason (CodeableConcept / Extensible)
- Resource: AuditEvent.purposeOfEvent (CodeableConcept / Extensible)
- Resource: AuditEvent.agent.purposeOfUse (CodeableConcept / Extensible)

4.4.2.20448.1 Content Logical Definition

This value set includes codes from the following code systems:

- Include codes from http://terminology.hl7.org/CodeSystem/v3-ActReason where concept is-a PurposeOfUse

4.4.2.20448.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 60 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActReason version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActReason

Lvl | Code | Display | Definition
0 | PurposeOfUse | purpose of use | Reason for performing one or more operations on information, which may be permitted by source system's security policy in accordance with one or more privacy policies and consent directives. Usage Notes: The rationale

or purpose for an act relating to the management of personal health information, such as collecting personal health information for research or public health purposes.

1 | HMARKT | healthcare marketing | To perform one or more operations on information for marketing services and products related to health care.

1 | HOPERAT | healthcare operations | To perform one or more operations on information used for conducting administrative and contractual activities related to the provision of health care.

2 | CAREMGT | care management | To perform analytics, evaluation and other secondary uses of treatment and healthcare related information to manage the quality, efficacy, patient safety, population health, and cost effectiveness of healthcare delivery. Explicitly excludes the use of information to organize the delivery of health care for care coordination and case management, or to provide healthcare treatment. Usage Note: The concept of care management is narrower than the list of activities related to more general organizational objectives such as provider profiling, education of healthcare and non-healthcare professionals; insurance underwriting, premium rating, reinsurance; organizational legal, medical review, auditing, compliance and fraud and abuse detection; business planning, development, and restructuring; fund-raising; and customer service. Map: Maps to ISO 14265 Classification Term "Health service management and quality assurance" described as "To inform persons or processes responsible for determining the availability, quality, safety, equity and cost-effectiveness of health care services." There is a semantic gap in concepts. This classification term is described as activities, i.e., "to inform persons" or "to inform processes" rather than the rationale for performing actions/operations on information related to the activity.

2 | DONAT | donation | To perform one or more operations on information used for cadaveric organ, eye or tissue donation.

2 | FRAUD | fraud | To perform one or more operations on information used for fraud detection and prevention processes.

2 | GOV | government | To perform one or more operations on information used within government processes.

2 | HACCRED | health accreditation | To perform one or more operations on information for conducting activities related to meeting accreditation criteria.

2 | HCOMPL | health compliance | To perform one or more operations on information used for conducting activities required to meet a mandate.

2 | HDECD | decedent | To perform one or more operations on information used for handling deceased patient matters.

2 | HDIRECT | directory | To perform one or more operation operations on information used to manage a patient directory. Examples: facility enterprise payer health information exchange patient directory

2 | HDM | healthcare delivery management | To perform one or more actions on information used for conducting administrative and contractual activities by or on behalf of organizational entities responsible for delivery of an individual's benefits in a healthcare program, health plan or insurance. Explicitly excludes the use of information to organize the delivery of health care for care coordination and case management, or to provide healthcare treatment. Usage Note: Examples of activities conducted under this purpose of use: provider profiling, risk adjustment, underwriting, fraud and abuse, quality improvement population health and care management. Aligns with HIPAA Operation POU minus coordination of care or other treatment related activities. Similar to the description in SAMHSA Confidentiality of Substance Use Disorder Patient Records Supplemental notice of proposed rulemaking. Map: Maps to ISO 14265 Classification Term "Administration of care for an individual subject of care" described as "To inform persons or processes responsible for enabling the availability of resources or funding or permissions for providing health care services to the subject of care." However, this classification term is described as activities, i.e., "to inform persons" or "to inform processes" rather than the rationale for performing actions/operations on information related to the activity.

2 | HLEGAL | legal | To perform one or more operations on information for conducting activities required by legal proceeding.

2 | HOUTCOMS | health outcome measure | To perform one or more operations on information used for assessing results and comparative effectiveness achieved by health care practices and interventions.

2 | HPRGRP | health program reporting | To perform one or more operations on information used for conducting activities to meet program accounting requirements.

2 | HQUALIMP | health quality improvement | To perform one or more operations on information used for conducting administrative activities to improve health care quality.

2 | HSYSADMIN | health system administration | To perform one or more operations on information to administer the electronic systems used for the delivery of health care.

3 | LABELING | labeling | To perform one or more operations on information to assign, persist, and manage labels to healthcare data to characterize various aspects, such as its security classification, sensitivity, compartment, integrity, and provenance; applicable privacy, consent, security, provenance, and trust policies; and handling caveats such as purpose of use, obligations, and refrain policies. Label management includes classification of target data by constructing and binding of a label set per applicable policies, security policy information file semantics, and classification guides. Label management also includes process and procedures for subsequent revision of a label for, e.g., reclassification, downgrading classification, and declassification. Label revisions may be triggered by, e.g., expiry of classification period; changes in applicable policy, e.g., revocation of a consent directive; or changes in the governing policy domain in which the data is relocated or a copy of the data is sent. If a label is revised, an audit log should be kept and the provenance of the label changes should be tracked.

3 | METAMGT | metadata management | To perform one or more operations on information to assign, persist, and manage metadata to healthcare data to characterize various aspects used for its indexing, discovery, retrieval, and processing by systems, applications, and end users. For example, master index identifier, media type, and location.

2 | MEMADMIN | member administration | To perform one or more operations on information to administer health care coverage to an enrollee under a policy or program.

2 | MILCDM | military command | To perform one or more operations on information for conducting activities required by military processes, procedures, policies, or law.

2 | PATADMIN | patient administration | To perform one or more operations on information used for operational activities conducted to administer the delivery of health care to a patient.

2 | PATSFTY | patient safety | To perform one or more operations on information in processes related to ensuring the safety of health care.

2 | PERFMSR | performance measure | To perform one or more operations on information used for monitoring performance of recommended health care practices and interventions.

2 | RECORDMGT | records management | To perform one or more operations on information used within the health records management process.

2 | SYSDEV | system development | To perform one or more operations on information to design, develop, implement, test, or deploy a healthcare system or application.

3 | HTEST | test health data | To perform one or more operations on information that is simulated or synthetic health data used for testing system capabilities outside of a production or operational system environment. Usage Note: Data marked with a HTEST security label enables an access control system to permit interfacing systems or end users provisioned with a clearance, which includes a HTEST purpose of use attribute, to test, verify, or validate that a system or application will operate in production as

intended based on design specifications.
2 | TRAIN | training | To perform one or more operations on information used in training and education.
1 | HPAYMT | healthcare payment | To perform one or more operations on information for conducting financial or contractual activities related to payment for provision of health care.
2 | CLMATTCH | claim attachment | To perform one or more operations on information for provision of additional clinical evidence in support of a request for coverage or payment for health services.
2 | COVAUTH | coverage authorization | To perform one or more operations on information for conducting prior authorization or predetermination of coverage for services.
2 | COVERAGE | coverage under policy or program | To perform one or more operations on information for conducting activities related to coverage under a program or policy.
3 | ELIGDTRM | eligibility determination | To perform one or more operations on information used for conducting eligibility determination for coverage in a program or policy. May entail review of financial status or disability assessment.
3 | ELIGVER | eligibility verification | To perform one or more operations on information used for conducting eligibility verification of coverage in a program or policy. May entail provider contacting coverage source (e.g., government health program such as workers compensation or health plan) for confirmation of enrollment, eligibility for specific services, and any applicable copays.
3 | ENROLLM | enrollment | To perform one or more operations on information used for enrolling a covered party in a program or policy. May entail recording of covered party's and any dependent's demographic information and benefit choices.
3 | MILDCRG | military discharge | To perform one or more operations on information for the process of releasing military personnel from their service obligations, which may include determining service merit, discharge benefits, and disability assessment.
2 | REMITADV | remittance advice | To perform one or more operations on information about the amount remitted for a health care claim.
1 | HRESCH | healthcare research | To perform one or more operations on information for conducting scientific investigations to obtain health care knowledge. Use of the data iincludes basic and applied research such as biomedical, population origin or ancestry, translational research, and disease, discipline, specialty specific healthcare research and clinical trial research.
2 | BIORCH | biomedical research | To perform one or more operations on information for conducting scientific investigations to obtain health care knowledge. Use of the data must be related to specified biomedical basic or applied research. For example, research on rare plants to determine whether biologic properties may be useful for pharmaceutical development. May be used in combination with clinical trial and other healthcare research purposes of use.
2 | CLINTRCH | clinical trial research | To perform one or more operations on information for conducting scientific investigations in accordance with clinical trial protocols to obtain health care knowledge.
3 | CLINTRCHNPC | clinical trial research without patient care | To perform one or more operations on information for conducting scientific investigations in accordance with clinical trial protocols to obtain health care knowledge without provision of patient care. May be post-coordinated or used with other purposes of use such as disease, discipline, specialty, population origins or ancestry, translational healthcare research. For example, a clinical trial conducted on laboratory specimens collected from a specified patient population.
3 | CLINTRCHPC | clinical trial research with patient care | To perform one or more operations on information for conducting scientific investigations with patient care in accordance with clinical trial protocols to obtain health care knowledge. May be post-coordinated or used with other purposes of use such as disease, discipline, specialty, population origins or ancestry, translational healthcare research. For example, an "off-label" drug used for cancer therapy administer to a specified patient population.
3 | PRECLINTRCH | preclinical trial research | To perform one or more operations on information in preparation for conducting scientific investigation to obtain health care knowledge, such as research on animals or review of patient health records, to determine the feasibility of a clinical trial study; assist with protocol design; or in preparation for institutional review board or ethics committee approval process. May be post-coordinated or used with other purposes of use such as disease, discipline, specialty, population origins or ancestry, translational healthcare research.
2 | DSRCH | disease specific healthcare research | To perform one or more operations on information for conducting scientific investigations to obtain health care knowledge. Use of the data must be related to specified conditions, diagnosis, or disease healthcare research. For example, conducting cancer research by testing reaction of tumor cells to certain biologics. May be used in combination with clinical trial and other healthcare research purposes of use.
2 | POARCH | population origins or ancestry healthcare research | To perform one or more operations on information, including genealogical pedigrees, historical records, surveys, family health data, health records, and genetic information, for conducting scientific investigations to obtain health care knowledge. Use of the data must be related to population origins and/or ancestry healthcare research. For example, gathering genetic specimens from a specific population in order to determine the ancestry and population origins of that group. May be used in combination with clinical trial and other healthcare research purposes of use.
2 | TRANSRCH | translational healthcare research | To perform one or more operations on information for conducting scientific investigations to obtain health care knowledge related to evidence based medicine during the course of providing healthcare treatment. Sometimes referred to as "bench to bedside", which is the iterative feedback loop between healthcare research and clinical trials with input from information collected in the course of routine provision of healthcare. For example, by extending a patient encounter to conduct a survey related to a research topic such as attitudes about use of a wellness device that a patient agreed to use. May be used in combination with clinical trial and other healthcare research purposes of use.
1 | PATRQT | patient requested | To perform one or more operations on information in response to a patient's request.
2 | FAMRQT | family requested | To perform one or more operations on information in response to a request by a family member authorized by the patient.
2 | PWATRNY | power of attorney | To perform one or more operations on information in response to a request by a person appointed as the patient's legal representative.
2 | SUPNWK | support network | To perform one or more operations on information in response to a request by a person authorized by the patient.
1 | PUBHLTH | public health | To perform one or more operations on information for conducting public health activities, such as the reporting of notifiable conditions.
2 | DISASTER | disaster | To perform one or more operations on information used for provision of immediately needed health care to a population of living subjects located in a disaster zone.
2 | THREAT | threat | To perform one or more operations on information used to prevent injury or disease to living subjects who may be the target of violence.
1 | TREAT | treatment | To perform one or more operations on information for provision of health care.
2 | CLINTRL | clinical trial | To perform health care as part of the clinical trial protocol.

2 | COC | coordination of care | To perform one or more actions on information in order to organize the provision and case management of an individualâ€™s healthcare, including: Monitoring a person's goals, needs, and preferences; acting as the communication link between two or more participants concerned with a person's health and wellness; organizing and facilitating care activities and promoting self-management by advocating for, empowering, and educating a person; and ensuring safe, appropriate, non-duplicative, and effective integrated care. Usage Note: Use when describing these functions: 1. Monitoring a personâ€™s goals, needs, and preferences. 2. Acting as the communication link between two or more participants concerned with a person's health and wellness. 3. Organizing and facilitating care activities and promoting self-management by advocating for, empowering, and educating a person. 4. Ensuring safe, appropriate, non-duplicative, and effective integrated care. The goal is to clearly differentiate this type of coordination of care from HIPAA Operations by specifying that these actions on information are undertaken in the provision of healthcare treatment. For similar uses of this concept, see SAMHSA Confidentiality of Substance Use Disorder Patient Records Supplemental notice of proposed rulemaking, which differentiates concepts of care coordination and case management for the provision of treatment as specifically distinct from activities related to health care delivery management and the operations of organizational entities involved in the delivery of healthcare. Map: Maps to ISO 14265 Classification Terms: "Support of care activities within the provider organisation for an individual subject of care" described as "To inform persons or processes enabling others to provide health care services to the subject of care." "Subject of Care Uses" described as "To inform the subject of care in support of his or her own interests."

2 | ETREAT | Emergency Treatment | To perform one or more operations on information for provision of immediately needed health care for an emergent condition.

3 | BTG | break the glass | To perform policy override operations on information for provision of immediately needed health care for an emergent condition affecting potential harm, death or patient safety by end users who are not provisioned for this purpose of use. Includes override of organizational provisioning policies and may include override of subject of care consent directive restricting access. Map: Partially Maps to ISO 14265 Classification Term "Emergency care provision to an individual subject of care" described as "To inform persons needing to provide health care services to the subject of care urgently, possibly needing to over-ride the policies and consents pertaining to Purpose 1 above." Purpose 1 is equivalent to HL7 treatment purpose of use: "Clinical care provision to an individual subject of care" described as "To inform persons or processes responsible for providing health care services to the subject of care." The ISO description conflates both of the proposed specializations of HL7 ETREAT: break the glass and the typically broader access to health information normally available to providers who are provisioned for emergency workflows on a regular basis, e.g., Emergency Room providers. Examples of greater access than is normally accessible by providers based on the need to know are access to sensitive information for which access typically requires a patient's consent. This is not an override of a patient's dissent to disclose sensitive information in cases where the applicable policy waives the need for that consent to access this information. In US, Title 38 Section 7332 and 42 CFR Part 2 both permit emergency access without the need to override a patient's consent directive; rather, this access is a limitation to the patient's right to dissent from disclosure.

3 | ERTREAT | emergency room treatment | To perform one or more operations on information for provision of immediately needed health care for an emergent condition in an emergency room or similar emergent care context by end users provisioned for this purpose, which does not constitute as policy override such as in a "Break the Glass" purpose of use. Map:Partially Maps to ISO 14265 Classification Term "Emergency care provision to an individual subject of care" described as "To inform persons needing to provide health care services to the subject of care urgently, possibly needing to over-ride the policies and consents pertaining to Purpose 1 above." Purpose 1 is equivalent to HL7 treatment purpose of use: "Clinical care provision to an individual subject of care" described as "To inform persons or processes responsible for providing health care services to the subject of care." The ISO description conflates both of the proposed specializations of HL7 ETREAT: break the glass and the typically broader access to health information normally available to providers who are provisioned for emergency workflows on a regular basis, e.g., Emergency Room providers. Examples of greater access than is normally accessible by providers based on the need to know are access to sensitive information for which access typically requires a patient's consent. This is not an override of a patient's dissent to disclose sensitive information in cases where the applicable policy waives the need for that consent to access this information. In US, Title 38 Section 7332 and 42 CFR Part 2 both permit emergency access without the need to override a patient's consent directive; rather, this access is a limitation to the patient's right to dissent from disclosure. There is a semantic gap in concepts. This classification term is described as activities â€œto inform personsâ€? rather than the rationale for performing actions/operations on information related to the activity.

2 | POPHLTH | population health | To perform one or more operations on information for provision of health care to a population of living subjects, e.g., needle exchange program.

4.4.2.20449 HL7 v3 Value Set GeneralPurposeOfUse

URL: https://hl7.org/fhir/R4/v3/GeneralPurposeOfUse/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-GeneralPurposeOfUse) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-GeneralPurposeOfUse
Version: | 2014-03-26
Name: | v3.GeneralPurposeOfUse
Title: | V3 Value SetGeneralPurposeOfUse
Definition: | Supports communication of purpose of use at a general level.
OID: | 2.16.840.1.113883.1.11.20449 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.20449.1 Content Logical Definition

This value set includes codes from the following code systems:

http://terminology.hl7.org/CodeSystem/v3-ActReason

Code | Display
COVERAGE | coverage under policy or program | To perform one or more operations on information for conducting activities related to coverage under a program or policy.

ETREAT | Emergency Treatment | To perform one or more operations on information for provision of immediately needed health care for an emergent condition.
HMARKT | healthcare marketing | To perform one or more operations on information for marketing services and products related to health care.
HOPERAT | healthcare operations | To perform one or more operations on information used for conducting administrative and contractual activities related to the provision of health care.
HPAYMT | healthcare payment | To perform one or more operations on information for conducting financial or contractual activities related to payment for provision of health care.
HRESCH | healthcare research | To perform one or more operations on information for conducting scientific investigations to obtain health care knowledge. Use of the data iincludes basic and applied research such as biomedical, population origin or ancestry, translational research, and disease, discipline, specialty specific healthcare research and clinical trial research.
PATRQT | patient requested | To perform one or more operations on information in response to a patient's request.
PUBHLTH | public health | To perform one or more operations on information for conducting public health activities, such as the reporting of notifiable conditions.
TREAT | treatment | To perform one or more operations on information for provision of health care.

4.4.2.20471 HL7 v3 Value Set SecurityControlObservationValue

URL: https://hl7.org/fhir/R4/v3/SecurityControlObservationValue/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-SecurityControlObservationValue) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-SecurityControlObservationValue
Version: | 2014-03-26
Name: | v3.SecurityControlObservationValue
Title: | V3 Value SetSecurityControlObservationValue
Definition: | Security observation values used to indicate security control metadata. V:SecurityControl is the union of V:SecurityPolicy,V:ObligationPolicy, V:RefrainPolicy, V:PurposeOfUse, and V:GeneralPurpose of Use used to populate the SecurityControlObservationValue attribute in order to convey one or more nonhierarchical security control metadata dictating handling caveats, purpose of use, dissemination controls and other refrain policies, and obligations to which a custodian or receiver is required to comply.
OID: | 2.16.840.1.113883.1.11.20471 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.20471.1 Content Logical Definition

This value set includes codes from the following code systems:

• Import all the codes that are contained in http://terminology.hl7.org/ValueSet/v3-SecurityPolicy
• Import all the codes that are contained in http://terminology.hl7.org/ValueSet/v3-ObligationPolicy
• Import all the codes that are contained in http://terminology.hl7.org/ValueSet/v3-RefrainPolicy
• Import all the codes that are contained in http://terminology.hl7.org/ValueSet/v3-PurposeOfUse
• Import all the codes that are contained in http://terminology.hl7.org/ValueSet/v3-GeneralPurposeOfUse

4.4.2.20471.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 111 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActReason version 2018-08-12,
http://terminology.hl7.org/ValueSet/v3-GeneralPurposeOfUse version 2014-03-26,
http://terminology.hl7.org/ValueSet/v3-PurposeOfUse version 2014-03-26,
http://terminology.hl7.org/ValueSet/v3-SecurityPolicy version 2014-03-26,
http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12,
http://terminology.hl7.org/ValueSet/v3-RefrainPolicy version 2014-03-26,
http://terminology.hl7.org/ValueSet/v3-ObligationPolicy version 2014-03-26

Code | System | Display | Definition
SecurityPolicy | http://terminology.hl7.org/CodeSystem/v3-ActCode | security policy | Types of security policies that further specify the ActClassPolicy value set. Examples: obligation to encrypt refrain from redisclosure without consent
AUTHPOL | http://terminology.hl7.org/CodeSystem/v3-ActCode | authorization policy | Authorisation policies are essentially security policies related to access-control and specify what activities a subject is permitted or forbidden to do, to a set of target objects. They are designed to protect target objects so are interpreted by access control agents or the run-time systems at the target system. A positive authorisation policy defines the actions that a subject is permitted to perform on a target. A negative authorisation policy specifies the actions that a subject is forbidden to perform on a target. Positive authorisation policies may also include filters to transform the parameters associated with their actions. (Based on PONDERS)
ACCESSCONSCHEME | http://terminology.hl7.org/CodeSystem/v3-ActCode | access control scheme | An access control policy specific to the type of access control scheme, which is used to enforce one or more authorization policies. Usage Note: Access control schemes are the type of access control policy, which is comprised of access control policy rules concerning the provision of

the access control service. There are two categories of access control policies, rule-based and identity-based, which are identified in CCITT Rec. X.800 aka ISO 7498-2. Rule-based access control policies are intended to apply to all access requests by any initiator on any target in a security domain. Identity-based access control policies are based on rules specific to an individual initiator, a group of initiators, entities acting on behalf of initiators, or originators acting in a specific role. Context can modify rule-based or identity-based access control policies. Context rules may define the entire policy in effect. Real systems will usually employ a combination of these policy types; if a rule-based policy is used, then an identity-based policy is usually in effect also. An access control scheme may be based on access control lists, capabilities, labels, and context or a combination of these. An access control scheme is a component of an access control mechanism or "service") along with the supporting mechanisms required by that scheme to provide access control decision information (ADI) supplied by the scheme to the access decision facility (ADF also known as a PDP). (Based on ISO/IEC 10181-3:1996) Examples: Attribute Based Access Control (ABAC) Discretionary Access Control (DAC) History Based Access Control (HBAC) Identity Based Access Control (IBAC) Mandatory Access Control (MAC) Organization Based Access Control (OrBAC) Relationship Based Access Control (RelBac) Responsibility Based Access Control (RespBAC) Risk Adaptable Access Control (RAdAC) >

DELEPOL | http://terminology.hl7.org/CodeSystem/v3-ActCode | delegation policy | Delegation policies specify which actions subjects are allowed to delegate to others. A delegation policy thus specifies an authorisation to delegate. Subjects must already possess the access rights to be delegated. Delegation policies are aimed at subjects delegating rights to servers or third parties to perform actions on their behalf and are not meant to be the means by which security administrators would assign rights to subjects. A negative delegation policy identifies what delegations are forbidden. A Delegation policy specifies the authorisation policy from which delegated rights are derived, the grantors, which are the entities which can delegate these access rights, and the grantees, which are the entities to which the access rights can be delegated. There are two types of delegation policy, positive and negative. (Based on PONDERS)

ObligationPolicy | http://terminology.hl7.org/CodeSystem/v3-ActCode | obligation policy | Conveys the mandated workflow action that an information custodian, receiver, or user must perform. Usage Notes: Per ISO 22600-2, ObligationPolicy instances 'are event-triggered and define actions to be performed by manager agent'. Per HL7 Composite Security and Privacy Domain Analysis Model: This value set refers to the action required to receive the permission specified in the privacy rule. Per OASIS XACML, an obligation is an operation specified in a policy or policy that is performed in conjunction with the enforcement of an access control decision.

ANONY | http://terminology.hl7.org/CodeSystem/v3-ActCode | anonymize | Custodian system must remove any information that could result in identifying the information subject.

AOD | http://terminology.hl7.org/CodeSystem/v3-ActCode | accounting of disclosure | Custodian system must make available to an information subject upon request an accounting of certain disclosures of the individualâ€™s protected health information over a period of time. Policy may dictate that the accounting include information about the information disclosed, the date of disclosure, the identification of the receiver, the purpose of the disclosure, the time in which the disclosing entity must provide a response and the time period for which accountings of disclosure can be requested.

AUDIT | http://terminology.hl7.org/CodeSystem/v3-ActCode | audit | Custodian system must monitor systems to ensure that all users are authorized to operate on information objects.

AUDTR | http://terminology.hl7.org/CodeSystem/v3-ActCode | audit trail | Custodian system must monitor and maintain retrievable log for each user and operation on information.

CPLYCC | http://terminology.hl7.org/CodeSystem/v3-ActCode | comply with confidentiality code | Custodian security system must retrieve, evaluate, and comply with the information handling directions of the Confidentiality Code associated with an information target.

CPLYCD | http://terminology.hl7.org/CodeSystem/v3-ActCode | comply with consent directive | Custodian security system must retrieve, evaluate, and comply with applicable information subject consent directives.

CPLYJPP | http://terminology.hl7.org/CodeSystem/v3-ActCode | comply with jurisdictional privacy policy | Custodian security system must retrieve, evaluate, and comply with applicable jurisdictional privacy policies associated with the target information.

CPLYOPP | http://terminology.hl7.org/CodeSystem/v3-ActCode | comply with organizational privacy policy | Custodian security system must retrieve, evaluate, and comply with applicable organizational privacy policies associated with the target information.

CPLYOSP | http://terminology.hl7.org/CodeSystem/v3-ActCode | comply with organizational security policy | Custodian security system must retrieve, evaluate, and comply with the organizational security policies associated with the target information.

CPLYPOL | http://terminology.hl7.org/CodeSystem/v3-ActCode | comply with policy | Custodian security system must retrieve, evaluate, and comply with applicable policies associated with the target information.

DECLASSIFYLABEL | http://terminology.hl7.org/CodeSystem/v3-ActCode | declassify security label | Custodian security system must declassify information assigned security labels by instantiating a new version of the classified information so as to break the binding of the classifying security label when assigning a new security label that marks the information as unclassified in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the previous assignment and binding.

DEID | http://terminology.hl7.org/CodeSystem/v3-ActCode | deidentify | Custodian system must strip information of data that would allow the identification of the source of the information or the information subject.

DELAU | http://terminology.hl7.org/CodeSystem/v3-ActCode | delete after use | Custodian system must remove target information from access after use.

DOWNGRDLABEL | http://terminology.hl7.org/CodeSystem/v3-ActCode | downgrade security label | Custodian security system must downgrade information assigned security labels by instantiating a new version of the classified information so as to break the binding of the classifying security label when assigning a new security label that marks the information as classified at a less protected level in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the previous assignment and binding.

DRIVLABEL | http://terminology.hl7.org/CodeSystem/v3-ActCode | derive security label | Custodian security system must assign and bind security labels derived from compilations of information by aggregation or disaggregation in order to classify information compiled in the information systems under its control for collection, access, use and disclosure in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the previous assignment and binding.

ENCRYPT | http://terminology.hl7.org/CodeSystem/v3-ActCode | encrypt | Custodian system must render information unreadable by algorithmically transforming plaintext into ciphertext. Usage Notes: A mathematical transposition of a file or data stream so that it cannot be deciphered at the receiving end without the proper key. Encryption is a security feature that assures that only the parties who are supposed to be participating in a videoconference or data transfer are able to do so. It can include a password, public and private keys, or a complex combination of all. (Per Infoway.)

ENCRYPTR | http://terminology.hl7.org/CodeSystem/v3-ActCode | encrypt at rest | Custodian system must render information unreadable and unusable by algorithmically transforming plaintext into ciphertext when "at rest" or in storage.

ENCRYPTT | http://terminology.hl7.org/CodeSystem/v3-ActCode | encrypt in transit | Custodian system must render information unreadable and unusable by algorithmically transforming plaintext into ciphertext while "in transit" or being transported by any means.

ENCRYPTU | http://terminology.hl7.org/CodeSystem/v3-ActCode | encrypt in use | Custodian system must render information unreadable and unusable by algorithmically transforming plaintext into ciphertext while in use such that operations permitted on the target information are limited by the license granted to the end user.

HUAPRV | http://terminology.hl7.org/CodeSystem/v3-ActCode | human approval | Custodian system must require human review and approval for permission requested.

LABEL | http://terminology.hl7.org/CodeSystem/v3-ActCode | assign security label | Custodian security system must assign and bind security labels in order to classify information created in the information systems under its control for collection, access, use and disclosure in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the assignment and binding. Usage Note: In security systems, security policy label assignments do not change, they may supersede prior assignments, and such reassignments are always tracked for auditing and other purposes.

MASK | http://terminology.hl7.org/CodeSystem/v3-ActCode | mask | Custodian system must render information unreadable and unusable by algorithmically transforming plaintext into ciphertext. User may be provided a key to decrypt per license or "shared secret".

MINEC | http://terminology.hl7.org/CodeSystem/v3-ActCode | minimum necessary | Custodian must limit access and disclosure to the minimum information required to support an authorized user's purpose of use. Usage Note: Limiting the information available for access and disclosure to that an authorized user or receiver "needs to know" in order to perform permitted workflow or purpose of use.

PERSISTLABEL | http://terminology.hl7.org/CodeSystem/v3-ActCode | persist security label | Custodian security system must persist the binding of security labels to classify information received or imported by information systems under its control for collection, access, use and disclosure in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the assignment and binding.

PRIVMARK | http://terminology.hl7.org/CodeSystem/v3-ActCode | privacy mark | Custodian must create and/or maintain human readable security label tags as required by policy. Map: Aligns with ISO 22600-3 Section A.3.4.3 description of privacy mark: "If present, the privacy-mark is not used for access control. The content of the privacy-mark may be defined by the security policy in force (identified by the security-policy-identifier) which may define a list of values to be used. Alternately, the value may be determined by the originator of the security-label."

PSEUD | http://terminology.hl7.org/CodeSystem/v3-ActCode | pseudonymize | Custodian system must strip information of data that would allow the identification of the source of the information or the information subject. Custodian may retain a key to relink data necessary to reidentify the information subject.

REDACT | http://terminology.hl7.org/CodeSystem/v3-ActCode | redact | Custodian system must remove information, which is not authorized to be access, used, or disclosed from records made available to otherwise authorized users.

UPGRDLABEL | http://terminology.hl7.org/CodeSystem/v3-ActCode | upgrade security label | Custodian security system must declassify information assigned security labels by instantiating a new version of the classified information so as to break the binding of the classifying security label when assigning a new security label that marks the information as classified at a more protected level in accordance with applicable jurisdictional privacy policies associated with the target information. The system must retain an immutable record of the previous assignment and binding.

RefrainPolicy | http://terminology.hl7.org/CodeSystem/v3-ActCode | refrain policy | Conveys prohibited actions which an information custodian, receiver, or user is not permitted to perform unless otherwise authorized or permitted under specified circumstances. Usage Notes: ISO 22600-2 species that a Refrain Policy "defines actions the subjects must refrain from performing". Per HL7 Composite Security and Privacy Domain Analysis Model: May be used to indicate that a specific action is prohibited based on specific access control attributes e.g., purpose of use, information type, user role, etc.

NOAUTH | http://terminology.hl7.org/CodeSystem/v3-ActCode | no disclosure without subject authorization | Prohibition on disclosure without information subject's authorization.

NOCOLLECT | http://terminology.hl7.org/CodeSystem/v3-ActCode | no collection | Prohibition on collection or storage of the information.

NODSCLCD | http://terminology.hl7.org/CodeSystem/v3-ActCode | no disclosure without consent directive | Prohibition on disclosure without organizational approved patient restriction.

NODSCLCDS | http://terminology.hl7.org/CodeSystem/v3-ActCode | no disclosure without information subject's consent directive | Prohibition on disclosure without a consent directive from the information subject.

NOINTEGRATE | http://terminology.hl7.org/CodeSystem/v3-ActCode | no integration | Prohibition on Integration into other records.

NOLIST | http://terminology.hl7.org/CodeSystem/v3-ActCode | no unlisted entity disclosure | Prohibition on disclosure except to entities on specific access list.

NOMOU | http://terminology.hl7.org/CodeSystem/v3-ActCode | no disclosure without MOU | Prohibition on disclosure without an interagency service agreement or memorandum of understanding (MOU).

NOORGPOL | http://terminology.hl7.org/CodeSystem/v3-ActCode | no disclosure without organizational authorization | Prohibition on disclosure without organizational authorization.

NOPAT | http://terminology.hl7.org/CodeSystem/v3-ActCode | no disclosure to patient, family or caregivers without attending provider's authorization | Prohibition on disclosing information to patient, family or caregivers without attending provider's authorization. Usage Note: The information may be labeled with the ActInformationSensitivity TBOO code, triggering application of this RefrainPolicy code as a handling caveat controlling access. Maps to FHIR NOPAT: Typically, this is used on an Alert resource, when the alert records information on patient abuse or non-compliance. FHIR print name is "keep information from patient". Maps to the French realm - code: INVISIBLE_PATIENT. displayName: Document non visible par le patient codingScheme: 1.2.250.1.213.1.1.4.13 French use case: A label for documents that the author chose to hide from the patient until the content can be disclose to the patient in a face to face meeting between a healthcare professional and the patient (in French law some results like cancer diagnosis or AIDS diagnosis must be announced to the patient by a healthcare professional and should not be find out by the patient alone).

NOPERSISTP | http://terminology.hl7.org/CodeSystem/v3-ActCode | no collection beyond purpose of use | Prohibition on collection of the information beyond time necessary to accomplish authorized purpose of use is prohibited.

NORDSCLCD | http://terminology.hl7.org/CodeSystem/v3-ActCode | no redisclosure without consent directive | Prohibition on redisclosure without patient consent directive.

NORDSCLCDS | http://terminology.hl7.org/CodeSystem/v3-ActCode | no redisclosure without information subject's consent directive | Prohibition on redisclosure without a consent directive from the information subject.

NORDSCLW | http://terminology.hl7.org/CodeSystem/v3-ActCode | no disclosure without jurisdictional authorization | Prohibition on disclosure without authorization under jurisdictional law.
NORELINK | http://terminology.hl7.org/CodeSystem/v3-ActCode | no relinking | Prohibition on associating de-identified or pseudonymized information with other information in a manner that could or does result in disclosing information intended to be masked.
NOREUSE | http://terminology.hl7.org/CodeSystem/v3-ActCode | no reuse beyond purpose of use | Prohibition on use of the information beyond the purpose of use initially authorized.
NOVIP | http://terminology.hl7.org/CodeSystem/v3-ActCode | no unauthorized VIP disclosure | Prohibition on disclosure except to principals with access permission to specific VIP information.
ORCON | http://terminology.hl7.org/CodeSystem/v3-ActCode | no disclosure without originator authorization | Prohibition on disclosure except as permitted by the information originator.
PurposeOfUse | http://terminology.hl7.org/CodeSystem/v3-ActReason | purpose of use | Reason for performing one or more operations on information, which may be permitted by source system's security policy in accordance with one or more privacy policies and consent directives. Usage Notes: The rationale or purpose for an act relating to the management of personal health information, such as collecting personal health information for research or public health purposes.
HMARKT | http://terminology.hl7.org/CodeSystem/v3-ActReason | healthcare marketing | To perform one or more operations on information for marketing services and products related to health care.
HOPERAT | http://terminology.hl7.org/CodeSystem/v3-ActReason | healthcare operations | To perform one or more operations on information used for conducting administrative and contractual activities related to the provision of health care.
CAREMGT | http://terminology.hl7.org/CodeSystem/v3-ActReason | care management | To perform analytics, evaluation and other secondary uses of treatment and healthcare related information to manage the quality, efficacy, patient safety, population health, and cost effectiveness of healthcare delivery. Explicitly excludes the use of information to organize the delivery of health care for care coordination and case management, or to provide healthcare treatment. Usage Note: The concept of care management is narrower than the list of activities related to more general organizational objectives such as provider profiling, education of healthcare and non-healthcare professionals; insurance underwriting, premium rating, reinsurance; organizational legal, medical review, auditing, compliance and fraud and abuse detection; business planning, development, and restructuring; fund-raising; and customer service. Map: Maps to ISO 14265 Classification Term "Health service management and quality assurance" described as "To inform persons or processes responsible for determining the availability, quality, safety, equity and cost-effectiveness of health care services." There is a semantic gap in concepts. This classification term is described as activities, i.e., "to inform persons" or "to inform processes" rather than the rationale for performing actions/operations on information related to the activity.
DONAT | http://terminology.hl7.org/CodeSystem/v3-ActReason | donation | To perform one or more operations on information used for cadaveric organ, eye or tissue donation.
FRAUD | http://terminology.hl7.org/CodeSystem/v3-ActReason | fraud | To perform one or more operations on information used for fraud detection and prevention processes.
GOV | http://terminology.hl7.org/CodeSystem/v3-ActReason | government | To perform one or more operations on information used within government processes.
HACCRED | http://terminology.hl7.org/CodeSystem/v3-ActReason | health accreditation | To perform one or more operations on information for conducting activities related to meeting accreditation criteria.
HCOMPL | http://terminology.hl7.org/CodeSystem/v3-ActReason | health compliance | To perform one or more operations on information used for conducting activities required to meet a mandate.
HDECD | http://terminology.hl7.org/CodeSystem/v3-ActReason | decedent | To perform one or more operations on information used for handling deceased patient matters.
HDIRECT | http://terminology.hl7.org/CodeSystem/v3-ActReason | directory | To perform one or more operation operations on information used to manage a patient directory. Examples: facility enterprise payer health information exchange patient directory
HDM | http://terminology.hl7.org/CodeSystem/v3-ActReason | healthcare delivery management | To perform one or more actions on information used for conducting administrative and contractual activities by or on behalf of organizational entities responsible for delivery of an individual's benefits in a healthcare program, health plan or insurance. Explicitly excludes the use of information to organize the delivery of health care for care coordination and case management, or to provide healthcare treatment. Usage Note: Examples of activities conducted under this purpose of use: provider profiling, risk adjustment, underwriting, fraud and abuse, quality improvement population health and care management. Aligns with HIPAA Operation POU minus coordination of care or other treatment related activities. Similar to the description in SAMHSA Confidentiality of Substance Use Disorder Patient Records Supplemental notice of proposed rulemaking. Map: Maps to ISO 14265 Classification Term "Administration of care for an individual subject of care" described as "To inform persons or processes responsible for enabling the availability of resources or funding or permissions for providing health care services to the subject of care." However, this classification term is described as activities, i.e., "to inform persons" or "to inform processes" rather than the rationale for performing actions/operations on information related to the activity.
HLEGAL | http://terminology.hl7.org/CodeSystem/v3-ActReason | legal | To perform one or more operations on information for conducting activities required by legal proceeding.
HOUTCOMS | http://terminology.hl7.org/CodeSystem/v3-ActReason | health outcome measure | To perform one or more operations on information used for assessing results and comparative effectiveness achieved by health care practices and interventions.
HPRGRP | http://terminology.hl7.org/CodeSystem/v3-ActReason | health program reporting | To perform one or more operations on information used for conducting activities to meet program accounting requirements.
HQUALIMP | http://terminology.hl7.org/CodeSystem/v3-ActReason | health quality improvement | To perform one or more operations on information used for conducting administrative activities to improve health care quality.
HSYSADMIN | http://terminology.hl7.org/CodeSystem/v3-ActReason | health system administration | To perform one or more operations on information to administer the electronic systems used for the delivery of health care.
LABELING | http://terminology.hl7.org/CodeSystem/v3-ActReason | labeling | To perform one or more operations on information to assign, persist, and manage labels to healthcare data to characterize various aspects, such as its security classification, sensitivity, compartment, integrity, and provenance; applicable privacy, consent, security, provenance, and trust policies; and handling caveats such as purpose of use, obligations, and refrain policies. Label management includes classification of target data by constructing and binding of a label set per applicable policies, security policy information file semantics, and classification guides. Label management also includes process and procedures for subsequent revision of a label for, e.g., reclassification, downgrading classification, and declassification. Label revisions may be triggered by, e.g., expiry of classification period; changes in applicable policy, e.g., revocation of a consent directive; or changes in the governing policy domain in which the data is relocated or a copy of the data is sent. If a label is revised, an audit log should be kept and the provenance of the label changes should be tracked.

METAMGT | http://terminology.hl7.org/CodeSystem/v3-ActReason | metadata management | To perform one or more operations on information to assign, persist, and manage metadata to healthcare data to characterize various aspects used for its indexing, discovery, retrieval, and processing by systems, applications, and end users. For example, master index identifier, media type, and location.
MEMADMIN | http://terminology.hl7.org/CodeSystem/v3-ActReason | member administration | To perform one or more operations on information to administer health care coverage to an enrollee under a policy or program.
MILCDM | http://terminology.hl7.org/CodeSystem/v3-ActReason | military command | To perform one or more operations on information for conducting activities required by military processes, procedures, policies, or law.
PATADMIN | http://terminology.hl7.org/CodeSystem/v3-ActReason | patient administration | To perform one or more operations on information used for operational activities conducted to administer the delivery of health care to a patient.
PATSFTY | http://terminology.hl7.org/CodeSystem/v3-ActReason | patient safety | To perform one or more operations on information in processes related to ensuring the safety of health care.
PERFMSR | http://terminology.hl7.org/CodeSystem/v3-ActReason | performance measure | To perform one or more operations on information used for monitoring performance of recommended health care practices and interventions.
RECORDMGT | http://terminology.hl7.org/CodeSystem/v3-ActReason | records management | To perform one or more operations on information used within the health records management process.
SYSDEV | http://terminology.hl7.org/CodeSystem/v3-ActReason | system development | To perform one or more operations on information to design, develop, implement, test, or deploy a healthcare system or application.
HTEST | http://terminology.hl7.org/CodeSystem/v3-ActReason | test health data | To perform one or more operations on information that is simulated or synthetic health data used for testing system capabilities outside of a production or operational system environment. Usage Note: Data marked with a HTEST security label enables an access control system to permit interfacing systems or end users provisioned with a clearance, which includes a HTEST purpose of use attribute, to test, verify, or validate that a system or application will operate in production as intended based on design specifications.
TRAIN | http://terminology.hl7.org/CodeSystem/v3-ActReason | training | To perform one or more operations on information used in training and education.
HPAYMT | http://terminology.hl7.org/CodeSystem/v3-ActReason | healthcare payment | To perform one or more operations on information for conducting financial or contractual activities related to payment for provision of health care.
CLMATTCH | http://terminology.hl7.org/CodeSystem/v3-ActReason | claim attachment | To perform one or more operations on information for provision of additional clinical evidence in support of a request for coverage or payment for health services.
COVAUTH | http://terminology.hl7.org/CodeSystem/v3-ActReason | coverage authorization | To perform one or more operations on information for conducting prior authorization or predetermination of coverage for services.
COVERAGE | http://terminology.hl7.org/CodeSystem/v3-ActReason | coverage under policy or program | To perform one or more operations on information for conducting activities related to coverage under a program or policy.
ELIGDTRM | http://terminology.hl7.org/CodeSystem/v3-ActReason | eligibility determination | To perform one or more operations on information used for conducting eligibility determination for coverage in a program or policy. May entail review of financial status or disability assessment.
ELIGVER | http://terminology.hl7.org/CodeSystem/v3-ActReason | eligibility verification | To perform one or more operations on information used for conducting eligibility verification of coverage in a program or policy. May entail provider contacting coverage source (e.g., government health program such as workers compensation or health plan) for confirmation of enrollment, eligibility for specific services, and any applicable copays.
ENROLLM | http://terminology.hl7.org/CodeSystem/v3-ActReason | enrollment | To perform one or more operations on information used for enrolling a covered party in a program or policy. May entail recording of covered party's and any dependent's demographic information and benefit choices.
MILDCRG | http://terminology.hl7.org/CodeSystem/v3-ActReason | military discharge | To perform one or more operations on information for the process of releasing military personnel from their service obligations, which may include determining service merit, discharge benefits, and disability assessment.
REMITADV | http://terminology.hl7.org/CodeSystem/v3-ActReason | remittance advice | To perform one or more operations on information about the amount remitted for a health care claim.
HRESCH | http://terminology.hl7.org/CodeSystem/v3-ActReason | healthcare research | To perform one or more operations on information for conducting scientific investigations to obtain health care knowledge. Use of the data iincludes basic and applied research such as biomedical, population origin or ancestry, translational research, and disease, discipline, specialty specific healthcare research and clinical trial research.
BIORCH | http://terminology.hl7.org/CodeSystem/v3-ActReason | biomedical research | To perform one or more operations on information for conducting scientific investigations to obtain health care knowledge. Use of the data must be related to specified biomedical basic or applied research. For example, research on rare plants to determine whether biologic properties may be useful for pharmaceutical development. May be used in combination with clinical trial and other healthcare research purposes of use.
CLINTRCH | http://terminology.hl7.org/CodeSystem/v3-ActReason | clinical trial research | To perform one or more operations on information for conducting scientific investigations in accordance with clinical trial protocols to obtain health care knowledge.
CLINTRCHNPC | http://terminology.hl7.org/CodeSystem/v3-ActReason | clinical trial research without patient care | To perform one or more operations on information for conducting scientific investigations in accordance with clinical trial protocols to obtain health care knowledge without provision of patient care. May be post-coordinated or used with other purposes of use such as disease, discipline, specialty, population origins or ancestry, translational healthcare research. For example, a clinical trial conducted on laboratory specimens collected from a specified patient population.
CLINTRCHPC | http://terminology.hl7.org/CodeSystem/v3-ActReason | clinical trial research with patient care | To perform one or more operations on information for conducting scientific investigations with patient care in accordance with clinical trial protocols to obtain health care knowledge. May be post-coordinated or used with other purposes of use such as disease, discipline, specialty, population origins or ancestry, translational healthcare research. For example, an "off-label" drug used for cancer therapy administer to a specified patient population.
PRECLINTRCH | http://terminology.hl7.org/CodeSystem/v3-ActReason | preclinical trial research | To perform one or more operations on information in preparation for conducting scientific investigation to obtain health care knowledge, such as research on animals or review of patient health records, to determine the feasibility of a clinical trial study; assist with protocol design; or in preparation for institutional review board or ethics committee approval process. May be post-coordinated or used with other purposes of use such as disease, discipline, specialty, population origins or ancestry, translational healthcare research.
DSRCH | http://terminology.hl7.org/CodeSystem/v3-ActReason | disease specific healthcare research | To perform one or more operations on information for conducting scientific investigations to obtain health care knowledge. Use of the data must be related to specified conditions, diagnosis, or disease healthcare research. For example, conducting cancer research by testing reaction of tumor cells to certain biologics. May be used in combination with clinical trial and other healthcare research purposes of use.

POARCH | http://terminology.hl7.org/CodeSystem/v3-ActReason | population origins or ancestry healthcare research | To perform one or more operations on information, including genealogical pedigrees, historical records, surveys, family health data, health records, and genetic information, for conducting scientific investigations to obtain health care knowledge. Use of the data must be related to population origins and/or ancestry healthcare research. For example, gathering genetic specimens from a specific population in order to determine the ancestry and population origins of that group. May be used in combination with clinical trial and other healthcare research purposes of use.

TRANSRCH | http://terminology.hl7.org/CodeSystem/v3-ActReason | translational healthcare research | To perform one or more operations on information for conducting scientific investigations to obtain health care knowledge related to evidence based medicine during the course of providing healthcare treatment. Sometimes referred to as "bench to bedside", which is the iterative feedback loop between healthcare research and clinical trials with input from information collected in the course of routine provision of healthcare. For example, by extending a patient encounter to conduct a survey related to a research topic such as attitudes about use of a wellness device that a patient agreed to use. May be used in combination with clinical trial and other healthcare research purposes of use.

PATRQT | http://terminology.hl7.org/CodeSystem/v3-ActReason | patient requested | To perform one or more operations on information in response to a patient's request.

FAMRQT | http://terminology.hl7.org/CodeSystem/v3-ActReason | family requested | To perform one or more operations on information in response to a request by a family member authorized by the patient.

PWATRNY | http://terminology.hl7.org/CodeSystem/v3-ActReason | power of attorney | To perform one or more operations on information in response to a request by a person appointed as the patient's legal representative.

SUPNWK | http://terminology.hl7.org/CodeSystem/v3-ActReason | support network | To perform one or more operations on information in response to a request by a person authorized by the patient.

PUBHLTH | http://terminology.hl7.org/CodeSystem/v3-ActReason | public health | To perform one or more operations on information for conducting public health activities, such as the reporting of notifiable conditions.

DISASTER | http://terminology.hl7.org/CodeSystem/v3-ActReason | disaster | To perform one or more operations on information used for provision of immediately needed health care to a population of living subjects located in a disaster zone.

THREAT | http://terminology.hl7.org/CodeSystem/v3-ActReason | threat | To perform one or more operations on information used to prevent injury or disease to living subjects who may be the target of violence.

TREAT | http://terminology.hl7.org/CodeSystem/v3-ActReason | treatment | To perform one or more operations on information for provision of health care.

CLINTRL | http://terminology.hl7.org/CodeSystem/v3-ActReason | clinical trial | To perform health care as part of the clinical trial protocol.

COC | http://terminology.hl7.org/CodeSystem/v3-ActReason | coordination of care | To perform one or more actions on information in order to organize the provision and case management of an individualâ€™s healthcare, including: Monitoring a person's goals, needs, and preferences; acting as the communication link between two or more participants concerned with a person's health and wellness; organizing and facilitating care activities and promoting self-management by advocating for, empowering, and educating a person; and ensuring safe, appropriate, non-duplicative, and effective integrated care. Usage Note: Use when describing these functions: 1. Monitoring a personâ€™s goals, needs, and preferences. 2. Acting as the communication link between two or more participants concerned with a person's health and wellness. 3. Organizing and facilitating care activities and promoting self-management by advocating for, empowering, and educating a person. 4. Ensuring safe, appropriate, non-duplicative, and effective integrated care. The goal is to clearly differentiate this type of coordination of care from HIPAA Operations by specifying that these actions on information are undertaken in the provision of healthcare treatment. For similar uses of this concept, see SAMHSA Confidentiality of Substance Use Disorder Patient Records Supplemental notice of proposed rulemaking, which differentiates concepts of care coordination and case management for the provision of treatment as specifically distinct from activities related to health care delivery management and the operations of organizational entities involved in the delivery of healthcare. Map: Maps to ISO 14265 Classification Terms: "Support of care activities within the provider organisation for an individual subject of care" described as "To inform persons or processes enabling others to provide health care services to the subject of care." "Subject of Care Uses" described as "To inform the subject of care in support of his or her own interests."

ETREAT | http://terminology.hl7.org/CodeSystem/v3-ActReason | Emergency Treatment | To perform one or more operations on information for provision of immediately needed health care for an emergent condition.

BTG | http://terminology.hl7.org/CodeSystem/v3-ActReason | break the glass | To perform policy override operations on information for provision of immediately needed health care for an emergent condition affecting potential harm, death or patient safety by end users who are not provisioned for this purpose of use. Includes override of organizational provisioning policies and may include override of subject of care consent directive restricting access. Map: Partially Maps to ISO 14265 Classification Term "Emergency care provision to an individual subject of care" described as "To inform persons needing to provide health care services to the subject of care urgently, possibly needing to over-ride the policies and consents pertaining to Purpose 1 above." Purpose 1 is equivalent to HL7 treatment purpose of use: "Clinical care provision to an individual subject of care" described as "To inform persons or processes responsible for providing health care services to the subject of care." The ISO description conflates both of the proposed specializations of HL7 ETREAT: break the glass and the typically broader access to health information normally available to providers who are provisioned for emergency workflows on a regular basis, e.g., Emergency Room providers. Examples of greater access than is normally accessible by providers based on the need to know are access to sensitive information for which access typically requires a patient's consent. This is not an override of a patient's dissent to disclose sensitive information in cases where the applicable policy waives the need for that consent to access this information. In US, Title 38 Section 7332 and 42 CFR Part 2 both permit emergency access without the need to override a patient's consent directive; rather, this access is a limitation to the patient's right to dissent from disclosure.

ERTREAT | http://terminology.hl7.org/CodeSystem/v3-ActReason | emergency room treatment | To perform one or more operations on information for provision of immediately needed health care for an emergent condition in an emergency room or similar emergent care context by end users provisioned for this purpose, which does not constitute as policy override such as in a "Break the Glass" purpose of use. Map:Partially Maps to ISO 14265 Classification Term "Emergency care provision to an individual subject of care" described as "To inform persons needing to provide health care services to the subject of care urgently, possibly needing to over-ride the policies and consents pertaining to Purpose 1 above." Purpose 1 is equivalent to HL7 treatment purpose of use: "Clinical care provision to an individual subject of care" described as "To inform persons or processes responsible for providing health care services to the subject of care." The ISO description conflates both of the proposed specializations of HL7 ETREAT: break the glass and the typically broader access to health information normally available to providers who are provisioned for emergency workflows on a regular basis, e.g., Emergency Room providers. Examples of greater access than is normally accessible by providers based on the need to know are access to sensitive information for which access typically requires a patient's consent. This is not an override of a patient's dissent to disclose sensitive information in cases where the applicable policy waives the need for that consent to access this information. In US, Title 38 Section 7332 and 42 CFR Part 2 both permit emergency access without the need to override a patient's consent directive; rather, this access is a limitation to the patient's

right to dissent from disclosure. There is a semantic gap in concepts. This classification term is described as activities â€œto inform personsâ€? rather than the rationale for performing actions/operations on information related to the activity.
POPHLTH | http://terminology.hl7.org/CodeSystem/v3-ActReason | population health | To perform one or more operations on information for provision of health care to a population of living subjects, e.g., needle exchange program.

4.4.2.20478 HL7 v3 Value Set Compartment

URL: https://hl7.org/fhir/R4/v3/Compartment/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-Compartment) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-Compartment
Version: | 2014-03-26
Name: | v3.Compartment
Title: | V3 Value SetCompartment
Definition: | A named tag set for metadata used to populate a security category label field that "segments" an IT resource per policy by indicating that access and use is restricted to members of a defined community or project. (HL7 Healthcare Privacy and Security Classification System) Usage Note: This is the healthcare analog to the US Intelligence Community's concept of a Special Access Program. Compartment codes may be used in as a field value in an initiator's clearance to indicate permission to access and use an IT Resource with a security label having the same compartment value in security category label field. Map: Aligns with ISO 2382-8 definition of Compartment - "A division of data into isolated blocks with separate security controls for the purpose of reducing risk."
OID: | 2.16.840.1.113883.1.11.20478 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.20478.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ActCode where concept is-a COMPT

4.4.2.20478.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 9 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ActCode version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ActCode

Lvl | Code | Display | Definition
0 | COMPT | compartment | This is the healthcare analog to the US Intelligence Community's concept of a Special Access Program. Compartment codes may be used in as a field value in an initiator's clearance to indicate permission to access and use an IT Resource with a security label having the same compartment value in security category label field. Map: Aligns with ISO 2382-8 definition of Compartment - "A division of data into isolated blocks with separate security controls for the purpose of reducing risk."
1 | ACOCOMPT | accountable care organization compartment | A group of health care entities, which may include health care providers, care givers, hospitals, facilities, health plans, and other health care constituents who coordinate care for reimbursement based on quality metrics for improving outcomes and lowering costs, and may be authorized to access the consumer's health information because of membership in that group. Security Compartment Labels assigned to a consumer's information use in accountable care workflows should be met or exceeded by the Security Compartment attribute claimed by a participant in a an accountable care workflow who is requesting access to that information
1 | CTCOMPT | care team compartment | Care coordination across participants in a care plan requires sharing of a healthcare consumer's information specific to that workflow. A care team member should only have access to that information while participating in that workflow or for other authorized uses. Security Compartment Labels assigned to a consumer's information use in care coordination workflows should be met or exceeded by the Security Compartment attribute claimed by a participant in a care team member workflow who is requesting access to that information
1 | FMCOMPT | financial management compartment | Financial management department members who have access to healthcare consumer information as part of a patient account, billing and claims workflows. Security Compartment Labels assigned to consumer information used in these workflows should be met or exceeded by the Security Compartment attribute claimed by a participant in a financial management workflow who is requesting access to that information.
1 | HRCOMPT | human resource compartment | A security category label field value, which indicates that access and use of an IT resource is restricted to members of human resources department or workflow.
1 | LRCOMPT | legitimate relationship compartment | Providers and care givers who have an established relationship per criteria determined by policy are considered to have an established care provision relations with a healthcare consumer, and may be authorized to access the consumer's health information because of that relationship. Providers and care givers should only have access to that information while participating in legitimate relationship workflows or for other authorized uses. Security Compartment Labels assigned to a consumer's information use in legitimate relationship workflows should be met or exceeded by the Security Compartment attribute claimed by a participant in a legitimate relationship workflow who is requesting access to that information.
1 | PACOMPT | patient administration compartment | Patient administration members who have access to healthcare consumer information as part of a patient administration workflows. Security Compartment Labels assigned to consumer information used in these workflows should be met or exceeded by the Security Compartment attribute claimed by a participant in a patient administration workflow who is requesting access to that information.
1 | RESCOMPT | research project compartment | A security category label field value, which indicates that access and use of an IT resource is restricted to members of a research project.

1 | RMGTCOMPT | records management compartment | A security category label field value, which indicates that access and use of an IT resource is restricted to members of records management department or workflow.

4.4.2.20481 HL7 v3 Value Set SecurityIntegrityObservationValue

URL: https://hl7.org/fhir/R4/v3/SecurityIntegrityObservationValue/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-SecurityIntegrityObservationValue) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-SecurityIntegrityObservationValue
Version: | 2014-03-26
Name: | v3.SecurityIntegrityObservationValue
Title: | V3 Value SetSecurityIntegrityObservationValue
Definition: | No Description Provided
OID: | 2.16.840.1.113883.1.11.20481 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.20481.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ObservationValue where concept is-a _SECINTOBV

http://terminology.hl7.org/CodeSystem/v3-ObservationValue

Code | Display
_SECINTOBV | security integrity | Abstract security observation values used to indicate security integrity metadata. Examples: Codes conveying integrity status, integrity confidence, and provenance.

4.4.2.20481.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 33 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ObservationValue version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ObservationValue

Code | Display | Definition
ABSTRED | abstracted | Security metadata observation values used to indicate the use of a more abstract version of the content, e.g., replacing exact value of an age or date field with a range, or remove the left digits of a credit card number or SSN.
AGGRED | aggregated | Security metadata observation values used to indicate the use of an algorithmic combination of actual values with the result of an aggregate function, e.g., average, sum, or count in order to limit disclosure of an IT resource (data, information object, service, or system capability) to the minimum necessary.
ANONYED | anonymized | Security metadata observation value conveying the alteration integrity of an IT resource (data, information object, service, or system capability) by used to indicate the mechanism by which software systems can strip portions of the resource that could allow the identification of the source of the information or the information subject. No key to relink the data is retained.
MAPPED | mapped | Security metadata observation value used to indicate that the IT resource semantic content has been transformed from one encoding to another. Usage Note: "MAP" code does not indicate the semantic fidelity of the transformed content. To indicate semantic fidelity for maps of HL7 to other code systems, this security alteration integrity observation may be further specified using an Act valued with Value Set: MapRelationship (2.16.840.1.113883.1.11.11052). Semantic fidelity of the mapped IT Resource may also be indicated using a SecurityIntegrityConfidenceObservation.
MASKED | masked | Security metadata observation value conveying the alteration integrity of an IT resource (data, information object, service, or system capability) by indicating the mechanism by which software systems can make data unintelligible (that is, as unreadable and unusable by algorithmically transforming plaintext into ciphertext) such that it can only be accessed or used by authorized users. An authorized user may be provided a key to decrypt per license or "shared secret". Usage Note: "MASKED" may be used, per applicable policy, as a flag to indicate to a user or receiver that some portion of an IT resource has been further encrypted, and may be accessed only by an authorized user or receiver to which a decryption key is provided.
PSEUDED | pseudonymized | Security metadata observation value conveying the alteration integrity of an IT resource (data, information object, service, or system capability), by indicating the mechanism by which software systems can strip portions of the resource that could allow the identification of the source of the information or the information subject. Custodian may retain a key to relink data necessary to reidentify the information subject. Rationale: Personal data which has been processed to make it impossible to know whose data it is. Used particularly for secondary use of health data. In some cases, it may be possible for authorized individuals to restore the identity of the individual, e.g.,for public health case management. Based on ISO/TS 25237:2008 Health informaticsâ€”Pseudonymization
REDACTED | redacted | Security metadata observation value used to indicate the mechanism by which software systems can filter an IT resource (data, information object, service, or system capability) to remove any portion of the resource that is not authorized to be access, used, or disclosed. Usage Note: "REDACTED" may be used, per applicable policy, as a flag to indicate to a user or receiver that some portion of an IT resource has filtered and not included in the content accessed or received.
SUBSETTED | subsetted | Metadata observation used to indicate that some information has been removed from the source object when the view this object contains was constructed because of configuration options when the view was created. The content may

not be suitable for use as the basis of a record update Usage Note: This is not suitable to be used when information is removed for security reasons - see the code REDACTED for this use.

SYNTAC | syntactic transform | Security metadata observation value used to indicate that the IT resource syntax has been transformed from one syntactical representation to another. Usage Note: "SYNTAC" code does not indicate the syntactical correctness of the syntactically transformed IT resource.

TRSLT | translated | Security metadata observation value used to indicate that the IT resource has been translated from one human language to another. Usage Note: "TRSLT" does not indicate the fidelity of the translation or the languages translated. The fidelity of the IT Resource translation may be indicated using a SecurityIntegrityConfidenceObservation. To indicate languages, use the Value Set:HumanLanguage (2.16.840.1.113883.1.11.11526)

VERSIONED | versioned | Security metadata observation value conveying the alteration integrity of an IT resource (data, information object, service, or system capability) which indicates that the resource only retains versions of an IT resource for access and use per applicable policy Usage Note: When this code is used, expectation is that the system has removed historical versions of the data that falls outside the time period deemed to be the effective time of the applicable version.

CRYTOHASH | cryptographic hash function | Security metadata observation value used to indicate the mechanism by which software systems can establish that data was not modified in transit. Rationale: This definition is intended to align with the ISO 22600-2 3.3.19 definition of cryptographic checkvalue: Information which is derived by performing a cryptographic transformation (see cryptography) on the data unit. The derivation of the checkvalue may be performed in one or more steps and is a result of a mathematical function of the key and a data unit. It is usually used to check the integrity of a data unit. Examples: SHA-1 SHA-2 (Secure Hash Algorithm)

DIGSIG | digital signature | Security metadata observation value used to indicate the mechanism by which software systems use digital signature to establish that data has not been modified. Rationale: This definition is intended to align with the ISO 22600-2 3.3.26 definition of digital signature: Data appended to, or a cryptographic transformation (see cryptography) of, a data unit that allows a recipient of the data unit to prove the source and integrity of the data unit and protect against forgery e.g., by the recipient.

HRELIABLE | highly reliable | Security metadata observation value used to indicate that the veracity or trustworthiness of an IT resource (data, information object, service, or system capability) for a specified purpose of use is perceived to be or deemed by policy to be very high.

RELIABLE | reliable | Security metadata observation value used to indicate that the veracity or trustworthiness of an IT resource (data, information object, service, or system capability) for a specified purpose of use is perceived to be or deemed by policy to be adequate.

UNCERTREL | uncertain reliability | Security metadata observation value used to indicate that the veracity or trustworthiness of an IT resource (data, information object, service, or system capability) for a specified purpose of use is perceived to be or deemed by policy to be uncertain.

UNRELIABLE | unreliable | Security metadata observation value used to indicate that the veracity or trustworthiness of an IT resource (data, information object, service, or system capability) for a specified purpose of use is perceived to be or deemed by policy to be inadequate.

CLINAST | clinician asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a clinician.

DEVAST | device asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a device.

HCPAST | healthcare professional asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a healthcare professional.

PACQAST | patient acquaintance asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a patient acquaintance.

PATAST | patient asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a patient.

PAYAST | payer asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a payer.

PROAST | professional asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a professional.

SDMAST | substitute decision maker asserted | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was asserted by a substitute decision maker.

CLINRPT | clinician reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a clinician.

DEVRPT | device reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a device.

HCPRPT | healthcare professional reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a healthcare professional.

PACQRPT | patient acquaintance reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a patient acquaintance.

PATRPT | patient reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a patient.

PAYRPT | payer reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a payer.

PRORPT | professional reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a professional.

SDMRPT | substitute decision maker reported | Security provenance metadata observation value used to indicate that an IT resource (data, information object, service, or system capability) was reported by a substitute decision maker.

4.4.2.20545 HL7 v3 Value Set ProvenanceEventCurrentState-AS

URL: https://hl7.org/fhir/R4/v3/ProvenanceEventCurrentState-AS/vs.html

Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ProvenanceEventCurrentState-AS) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ProvenanceEventCurrentState-AS
Version: | 2014-08-07
Name: | v3.ProvenanceEventCurrentState-AS
Title: | V3 Value SetProvenanceEventCurrentState-AS
Definition: | Specifies the state change of a target Act, using ActStatus codes, from its previous state as a predecessor Act. For example, if the target Act is the result of a predecessor Act being "obsoleted" and replaced with the target Act, the source ProvenanceEventCurrentState Act code would be "obsoleted".
OID: | 2.16.840.1.113883.1.11.20545 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.20545.1 Content Logical Definition

This value set includes codes from the following code systems:

http://terminology.hl7.org/CodeSystem/v3-ActStatus

Code | Display
aborted | aborted | The Act has been terminated prior to the originally intended completion.
cancelled | cancelled | The Act has been abandoned before activation.
completed | completed | An Act that has terminated normally after all of its constituents have been performed.
new | new | An Act that is in the preparatory stages and may not yet be acted upon
nullified | nullified | This Act instance was created in error and has been 'removed' and is treated as though it never existed. A record is retained for audit purposes only.
obsolete | obsolete | This Act instance has been replaced by a new instance.

4.4.2.20546 HL7 v3 Value Set ProvenanceEventCurrentState-DC

URL: https://hl7.org/fhir/R4/v3/ProvenanceEventCurrentState-DC/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ProvenanceEventCurrentState-DC) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ProvenanceEventCurrentState-DC
Version: | 2014-08-07
Name: | v3.ProvenanceEventCurrentState-DC
Title: | V3 Value SetProvenanceEventCurrentState-DC
Definition: | Specifies the state change of a target Act using DocuymentCompletion codes, from its previous state as a predecessor Act. For example, if the target Act is the result of a predecessor Act being "obsoleted" and replaced with the target Act, the source ProvenanceEventCurrentState Act code would be "obsoleted".
OID: | 2.16.840.1.113883.1.11.20546 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.20546.1 Content Logical Definition

This value set includes codes from the following code systems:

http://terminology.hl7.org/CodeSystem/v3-DocumentCompletion

Code | Display
AU | authenticated | A completion status in which a document has been signed manually or electronically by one or more individuals who attest to its accuracy. No explicit determination is made that the assigned individual has performed the authentication. While the standard allows multiple instances of authentication, it would be typical to have a single instance of authentication, usually by the assigned individual.
DI | dictated | A completion status in which information has been orally recorded but not yet transcribed.
DO | documented | A completion status in which document content, other than dictation, has been received but has not been translated into the final electronic format. Examples include paper documents, whether hand-written or typewritten, and intermediate electronic forms, such as voice to text.
LA | legally authenticated | A completion status in which a document has been signed manually or electronically by the individual who is legally responsible for that document. This is the most mature state in the workflow progression.
UC | unsigned completed document | A completion status where the document is complete and there is no expectation that the document will be signed.

4.4.2.20547 HL7 v3 Value Set ProvenanceEventCurrentState

URL: https://hl7.org/fhir/R4/v3/ProvenanceEventCurrentState/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-ProvenanceEventCurrentState) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-ProvenanceEventCurrentState
Version: | 2014-08-07
Name: | v3.ProvenanceEventCurrentState
Title: | V3 Value SetProvenanceEventCurrentState
Definition: | Specifies the state change of a target Act, such as a document or an entry, from its previous state as a predecessor Act. For example, if the target Act is the result of a predecessor Act being "obsoleted" and replaced with the target Act, the source ProvenanceEventCurrentState Act code would be "obsoleted".
OID: | 2.16.840.1.113883.1.11.20547 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.20547.1 Content Logical Definition

This value set includes codes from the following code systems:

• Import all the codes that are contained in http://terminology.hl7.org/ValueSet/v3-ProvenanceEventCurrentState-AS

• Import all the codes that are contained in http://terminology.hl7.org/ValueSet/v3-ProvenanceEventCurrentState-DC

4.4.2.20547.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 11 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-DocumentCompletion version 2018-08-12, http://terminology.hl7.org/ValueSet/v3-ProvenanceEventCurrentState-DC version 2014-08-07, http://terminology.hl7.org/ValueSet/v3-ProvenanceEventCurrentState-AS version 2014-08-07, http://terminology.hl7.org/CodeSystem/v3-ActStatus version 2018-08-12

Code | System | Display | Definition
aborted | http://terminology.hl7.org/CodeSystem/v3-ActStatus | aborted | The Act has been terminated prior to the originally intended completion.
cancelled | http://terminology.hl7.org/CodeSystem/v3-ActStatus | cancelled | The Act has been abandoned before activation.
completed | http://terminology.hl7.org/CodeSystem/v3-ActStatus | completed | An Act that has terminated normally after all of its constituents have been performed.
new | http://terminology.hl7.org/CodeSystem/v3-ActStatus | new | An Act that is in the preparatory stages and may not yet be acted upon
nullified | http://terminology.hl7.org/CodeSystem/v3-ActStatus | nullified | This Act instance was created in error and has been 'removed' and is treated as though it never existed. A record is retained for audit purposes only.
obsolete | http://terminology.hl7.org/CodeSystem/v3-ActStatus | obsolete | This Act instance has been replaced by a new instance.
AU | http://terminology.hl7.org/CodeSystem/v3-DocumentCompletion | authenticated | A completion status in which a document has been signed manually or electronically by one or more individuals who attest to its accuracy. No explicit determination is made that the assigned individual has performed the authentication. While the standard allows multiple instances of authentication, it would be typical to have a single instance of authentication, usually by the assigned individual.
DI | http://terminology.hl7.org/CodeSystem/v3-DocumentCompletion | dictated | A completion status in which information has been orally recorded but not yet transcribed.
DO | http://terminology.hl7.org/CodeSystem/v3-DocumentCompletion | documented | A completion status in which document content, other than dictation, has been received but has not been translated into the final electronic format. Examples include paper documents, whether hand-written or typewritten, and intermediate electronic forms, such as voice to text.
LA | http://terminology.hl7.org/CodeSystem/v3-DocumentCompletion | legally authenticated | A completion status in which a document has been signed manually or electronically by the individual who is legally responsible for that document. This is the most mature state in the workflow progression.
UC | http://terminology.hl7.org/CodeSystem/v3-DocumentCompletion | unsigned completed document | A completion status where the document is complete and there is no expectation that the document will be signed.

4.4.2.20561 HL7 v3 Value Set WorkScheduleODH

URL: https://hl7.org/fhir/R4/v3/WorkScheduleODH/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-WorkScheduleODH) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-WorkScheduleODH
Version: | 2018-08-12
Name: | v3.WorkScheduleODH
Title: | V3 Value SetWorkScheduleODH
Definition: | Describes an individual's typical arrangement of working hours for an occupation.
OID: | 2.16.840.1.113883.1.11.20561 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.20561.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ObservationValue where concept is-a _WorkSchedule

http://terminology.hl7.org/CodeSystem/v3-ObservationValue

Code | Display
_WorkSchedule | _WorkSchedule | Concepts that describe an individual's typical arrangement of working hours for an occupation.

4.4.2.20561.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 9 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ObservationValue version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ObservationValue

Code | Display | Definition
DS | daytime shift | A person who is scheduled for work during daytime hours (for example between 6am and 6pm) on a regular basis.
EMS | early morning shift | Consistent Early morning schedule of 13 hours or less per shift (between 2 am and 2 pm)
ES | evening shift | A person who is scheduled for work during evening hours (for example between 2pm and midnight) on a regular basis.
NS | night shift | Scheduled for work during nighttime hours (for example between 9pm and 8am) on a regular basis.
RSWN | rotating shift with nights | Scheduled for work times that change periodically between days, and/or evenings, and includes some night shifts.
RSWON | rotating shift without nights | Scheduled for work days/times that change periodically between days, but does not include night or evening work.
SS | split shift | Shift consisting of two distinct work periods each day that are separated by a break of a few hours (for example 2 to 4 hours)
VLS | very long shift | Shifts of 17 or more hours.
VS | variable shift | Irregular, unpredictable hours scheduled on a short notice (for example, less than 2 day notice): inconsistent schedule, on-call, as needed, as available.

4.4.2.20562 HL7 v3 Value Set employmentStatusODH

URL: https://hl7.org/fhir/R4/v3/employmentStatusODH/vs.html
Vocabulary Work Group | Maturity Level: N/A | External | Use Context: Any

This value set (http://terminology.hl7.org/ValueSet/v3-employmentStatusODH) is defined as part of HL7 v3.

Summary

Defining URL: | http://terminology.hl7.org/ValueSet/v3-employmentStatusODH
Version: | 2018-08-12
Name: | v3.employmentStatusODH
Title: | V3 Value SetemploymentStatusODH
Definition: | Concepts representing whether a person does or does not currently have a job or is not currently in the labor pool seeking employment.
OID: | 2.16.840.1.113883.1.11.20562 (for OID based terminology systems)
Source Resource | XML / JSON

This value set is not currently used

4.4.2.20562.1 Content Logical Definition

This value set includes codes from the following code systems:

• Include codes from http://terminology.hl7.org/CodeSystem/v3-ObservationValue where concept is-a _EmploymentStatus

http://terminology.hl7.org/CodeSystem/v3-ObservationValue

Code | Display
_EmploymentStatus | _EmploymentStatus | Concepts representing whether a person does or does not currently have a job or is not currently in the labor pool seeking employment.

4.4.2.20562.2 Expansion

This expansion generated 01 Nov 2019

This value set contains 3 concepts

Expansion based on http://terminology.hl7.org/CodeSystem/v3-ObservationValue version 2018-08-12

All codes from system http://terminology.hl7.org/CodeSystem/v3-ObservationValue

Code | Display | Definition

Employed | Employed | Individuals who, during the last week: a) did any work for at least 1 hour as paid or unpaid employees of a business or government organization; worked in their own businesses, professions, or on their own farms; or b) were not working, but who have a job or business from which the individual was temporarily absent because of vacation, illness, bad weather, childcare problems, maternity or paternity leave, labor-management dispute, job training, or other family or personal reasons, regardless of whether or not they were paid for the time off or were seeking other jobs.

NotInLaborForce | Not In Labor Force | Persons not classified as employed or unemployed, meaning those who have no job and are not looking for one.

Unemployed | Unemployed | Persons who currently have no employment, but are available for work and have made specific efforts to find employment.

4.5 Mappings between Value Sets Defined in FHIR

URL: https://hl7.org/fhir/R4/terminologies-conceptmaps.html

4.5 Mappings between Value Sets Defined in FHIR

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative

This table contains a list of all the mappings between value sets that are defined as part of the FHIR specification.

Name | Source | Target

IndicatorToRequestPriority | http://cds-hooks.hl7.org/ValueSet/indicator | Request priority

IndicatorToRequestPriority | http://cds-hooks.hl7.org/ValueSet/indicator | Request priority

FHIR-v3-Address-Use | AddressUse | v3 Code System AddressUse

FHIR-v3-Address-Use | AddressUse | v3 Code System AddressUse

Specimen mapping from v2 table 0487 to SNOMED CT | v2 Specimen Type | http://snomed.info/sct?fhir_vs

Specimen mapping from v2 table 0487 to SNOMED CT | v2 Specimen Type | http://snomed.info/sct?fhir_vs

SNOMED CT to ICD-10-CM mappings for fracture of ulna | http://snomed.info/sct?fhir_vs | http://hl7.org/fhir/sid/icd-10-us

SNOMED CT to ICD-10-CM mappings for fracture of ulna | http://snomed.info/sct?fhir_vs | http://hl7.org/fhir/sid/icd-10-us

v3.AddressType | AddressType | v3 Code System AddressUse

v2.AddressUse | AddressUse | v2 Adress Type

v3.AddressUse | AddressUse | v3 Code System AddressUse

v2.AdministrativeGender | AdministrativeGender | v2 SEX

v3.AdministrativeGender | AdministrativeGender | v3 Code System AdministrativeGender

v3.CompositionStatus | CompositionStatus | v3 Code System ActStatus

v2.ContactPointSystem | ContactPointSystem | v2 Telecommunication Equipment Type

v2.ContactPointUse | ContactPointUse | v2 Telecommunication Use Code

v3.ContactPointUse | ContactPointUse | v3 Code System AddressUse

v3.DataAbsentReason | DataAbsentReason | v3 Code System NullFlavor

v3.DetectedIssueSeverity | DetectedIssueSeverity | V3 Value SetSeverityObservation

v3.DocumentReferenceStatus | DocumentReferenceStatus | v3 Code System ActStatus

v2.NameUse | NameUse | v2 Name Type

v3.NameUse | NameUse | v3 Code System EntityNameUseR2

FHIR-exanple-2 | http://example.org/fhir/example1 | http://example.org/fhir/example2

FHIR-exanple-2 | http://example.org/fhir/example1 | http://example.org/fhir/example2

AccountStatusCanonicalMap | AccountStatus | Canonical Status Codes for FHIR Resources

AllergyIntoleranceClinicalStatusCodesCanonicalMap | AllergyIntolerance Clinical Status Codes | Canonical Status Codes for FHIR Resources

AllergyIntoleranceVerificationStatusCodesCanonicalMap | AllergyIntolerance Verification Status Codes | Canonical Status Codes for FHIR Resources

AppointmentStatusCanonicalMap | AppointmentStatus | Canonical Status Codes for FHIR Resources

CarePlanActivityStatusCanonicalMap | CarePlanActivityStatus | Canonical Status Codes for FHIR Resources

CareTeamStatusCanonicalMap | CareTeamStatus | Canonical Status Codes for FHIR Resources

ChargeItemStatusCanonicalMap | ChargeItemStatus | Canonical Status Codes for FHIR Resources

ClinicalImpressionStatusCanonicalMap | Clinical Impression Status | Canonical Status Codes for FHIR Resources

CompositionStatusCanonicalMap | CompositionStatus | Canonical Status Codes for FHIR Resources

ConditionVerificationStatusCanonicalMap | ConditionVerificationStatus | Canonical Status Codes for FHIR Resources

ConsentStateCanonicalMap | ConsentState | Canonical Status Codes for FHIR Resources

ContractResourcePublicationStatusCodesCanonicalMap | Contract Resource Publication Status codes | Canonical Status Codes for FHIR Resources

ContractResourceStatusCodesCanonicalMap | Contract Resource Status Codes | Canonical Status Codes for FHIR Resources

DeviceUseStatementStatusCanonicalMap | DeviceUseStatementStatus | Canonical Status Codes for FHIR Resources

FHIRDeviceStatusCanonicalMap | FHIRDeviceStatus | Canonical Status Codes for FHIR Resources

DiagnosticReportStatusCanonicalMap | DiagnosticReportStatus | Canonical Status Codes for FHIR Resources

DocumentReferenceStatusCanonicalMap | DocumentReferenceStatus | Canonical Status Codes for FHIR Resources

EncounterLocationStatusCanonicalMap | EncounterLocationStatus | Canonical Status Codes for FHIR Resources

EncounterStatusCanonicalMap | EncounterStatus | Canonical Status Codes for FHIR Resources

EndpointStatusCanonicalMap | EndpointStatus | Canonical Status Codes for FHIR Resources

EpisodeOfCareStatusCanonicalMap | EpisodeOfCareStatus | Canonical Status Codes for FHIR Resources

EventStatusCanonicalMap | EventStatus | Canonical Status Codes for FHIR Resources

ExplanationOfBenefitStatusCanonicalMap | ExplanationOfBenefitStatus | Canonical Status Codes for FHIR Resources

FlagStatusCanonicalMap | FlagStatus | Canonical Status Codes for FHIR Resources

FinancialResourceStatusCodesCanonicalMap | Financial Resource Status Codes | Canonical Status Codes for FHIR Resources

GoalAchievementStatusCanonicalMap | Goal achievement status | Canonical Status Codes for FHIR Resources
GoalLifecycleStatusCanonicalMap | GoalLifecycleStatus | Canonical Status Codes for FHIR Resources
GuidanceResponseStatusCanonicalMap | GuidanceResponseStatus | Canonical Status Codes for FHIR Resources
FamilyHistoryStatusCanonicalMap | FamilyHistoryStatus | Canonical Status Codes for FHIR Resources
ImagingStudyStatusCanonicalMap | ImagingStudyStatus | Canonical Status Codes for FHIR Resources
ImmunizationEvaluationStatusCodesCanonicalMap | Immunization Evaluation Status Codes | Canonical Status Codes for FHIR Resources
ImmunizationStatusCodesCanonicalMap | Immunization Status Codes | Canonical Status Codes for FHIR Resources
InvoiceStatusCanonicalMap | InvoiceStatus | Canonical Status Codes for FHIR Resources
ListStatusCanonicalMap | ListStatus | Canonical Status Codes for FHIR Resources
LocationStatusCanonicalMap | LocationStatus | Canonical Status Codes for FHIR Resources
MeasureReportStatusCanonicalMap | MeasureReportStatus | Canonical Status Codes for FHIR Resources
MedicationAdministration Status CodesCanonicalMap | Medication administration status codes | Canonical Status Codes for FHIR Resources
Medication Status CodesCanonicalMap | Medication status codes | Canonical Status Codes for FHIR Resources
Medication Status CodesCanonicalMap | Medication status codes | Canonical Status Codes for FHIR Resources
MedicationDispense Status CodesCanonicalMap | Medication dispense status codes | Canonical Status Codes for FHIR Resources
MedicationKnowledge Status CodesCanonicalMap | Medication knowledge status codes | Canonical Status Codes for FHIR Resources
medicationrequest StatusCanonicalMap | Medicationrequest status | Canonical Status Codes for FHIR Resources
DeviceMetricOperationalStatusCanonicalMap | DeviceMetricOperationalStatus | Canonical Status Codes for FHIR Resources
ObservationStatusCanonicalMap | ObservationStatus | Canonical Status Codes for FHIR Resources
ParticipationStatusCanonicalMap | ParticipationStatus | Canonical Status Codes for FHIR Resources
BiologicallyDerivedProductStatusCanonicalMap | BiologicallyDerivedProductStatus | Canonical Status Codes for FHIR Resources
PublicationStatusCanonicalMap | PublicationStatus | Canonical Status Codes for FHIR Resources
QuestionnaireResponseStatusCanonicalMap | QuestionnaireResponseStatus | Canonical Status Codes for FHIR Resources
TestReportStatusCanonicalMap | TestReportStatus | Canonical Status Codes for FHIR Resources
RequestStatusCanonicalMap | RequestStatus | Canonical Status Codes for FHIR Resources
ResearchStudyStatusCanonicalMap | ResearchStudyStatus | Canonical Status Codes for FHIR Resources
ResearchSubjectStatusCanonicalMap | ResearchSubjectStatus | Canonical Status Codes for FHIR Resources
SlotStatusCanonicalMap | SlotStatus | Canonical Status Codes for FHIR Resources
SpecimenStatusCanonicalMap | SpecimenStatus | Canonical Status Codes for FHIR Resources
SubscriptionStatusCanonicalMap | SubscriptionStatus | Canonical Status Codes for FHIR Resources
FHIRSubstanceStatusCanonicalMap | FHIRSubstanceStatus | Canonical Status Codes for FHIR Resources
SupplyDeliveryStatusCanonicalMap | SupplyDeliveryStatus | Canonical Status Codes for FHIR Resources
SupplyRequestStatusCanonicalMap | SupplyRequestStatus | Canonical Status Codes for FHIR Resources
TaskStatusCanonicalMap | TaskStatus | Canonical Status Codes for FHIR Resources
statusCanonicalMap | Status | Canonical Status Codes for FHIR Resources
validation-statusCanonicalMap | Validation-status | Canonical Status Codes for FHIR Resources

4.6 Known Identifier Systems

URL: https://hl7.org/fhir/R4/identifier-registry.html

4.6 Known Identifier Systems

FHIR Infrastructure Work Group | Maturity Level: 3 | Standards Status: Trial Use

The following names (URIs) may be used in the system element of the Identifier data type. If a URI is defined here, it SHALL be used in preference to any other identifying mechanism. If an identifier system is not listed here, the correct URI may be determined by working through the following list, in order:

• the HL7 OID Registry

• the documentation associated with the identifier

• consulting the owner of the identifier

• asking on the HL7 vocabulary mailing list

See also the list of known coding systems that can be used in the system element of the Coding data type. Additional identifier systems may be registered on the HL7 FHIR registry at http://hl7.org/fhir/registry .

Implementation Note: This list of URIs is incomplete and subject to change. Some values may be dropped and others will likely be added in the coming months as HL7 institutes formal processes around URIs in vocabulary.

The URI column indicates the correct value to use in the Identifier.system. The OID is provided for compatibility with HL7 v2 /CDA based systems. Note that any URL in http://example.org is reserved for testing and documentation purposes.

identifier | URI | OID (for non-FHIR systems) | Type | Comment
URIs (W3C): when the identifier is a URI | urn:ietf:rfc:3986 | 2.16.840.1.113883.4.873 | | As defined by RFC 3986 (with many schemes defined in many RFCs). For OIDs and UUIDs, use the URN form (urn:oid: (note: lowercase) and urn:uuid:
DICOM Unique Id | urn:dicom:uid | | | An OID issued under DICOM OID rules. DICOM OIDs are represented as plain OIDs, with a prefix of "urn:oid:"
United States Social Security Number | http://hl7.org/fhir/sid/us-ssn | 2.16.840.1.113883.4.1 | SB (US) | The SSN is represented in resources with dashes removed

United States Medicare Number | http://hl7.org/fhir/sid/us-medicare | 2.16.840.1.113883.4.572 | SB (US) | Medicare Numbers (HIC or HICN) are represented without any spaces or dashes
Medicare Beneficiary Identifier (United States) | http://hl7.org/fhir/sid/us-mbi | 2.16.840.1.113883.4.927 | SB (US) | Medicare Beneficiary Identifiers are represented without any spaces or dashes
United States National Provider Identifier | http://hl7.org/fhir/sid/us-npi | 2.16.840.1.113883.4.6 | PRN (US)
Global Trade Item Number (GTIN) | https://www.gs1.org/gtin | 1.3.160 | | Note: GTINs may be used in both Codes and Identifiers
Alaska Driver's License | urn:oid:2.16.840.1.113883.4.3.2 | 2.16.840.1.113883.4.3.2 | DL (US)
Alabama Driver's License | urn:oid:2.16.840.1.113883.4.3.1 | 2.16.840.1.113883.4.3.1 | DL (US)
Arkansas Driver's License | urn:oid:2.16.840.1.113883.4.3.5 | 2.16.840.1.113883.4.3.5 | DL (US)
Arizona Driver's License | urn:oid:2.16.840.1.113883.4.3.4 | 2.16.840.1.113883.4.3.4 | DL (US)
California Driver's License | urn:oid:2.16.840.1.113883.4.3.6 | 2.16.840.1.113883.4.3.6 | DL (US)
Colorado Driver's License | urn:oid:2.16.840.1.113883.4.3.8 | 2.16.840.1.113883.4.3.8 | DL (US)
Connecticut Driver's License | urn:oid:2.16.840.1.113883.4.3.9 | 2.16.840.1.113883.4.3.9 | DL (US)
DC Driver's License | urn:oid:2.16.840.1.113883.4.3.11 | 2.16.840.1.113883.4.3.11 | DL (US)
Delaware Driver's License | urn:oid:2.16.840.1.113883.4.3.10 | 2.16.840.1.113883.4.3.10 | DL (US)
Florida Driver's License | urn:oid:2.16.840.1.113883.4.3.12 | 2.16.840.1.113883.4.3.12 | DL (US)
Georgia Driver's License | urn:oid:2.16.840.1.113883.4.3.13 | 2.16.840.1.113883.4.3.13 | DL (US)
Hawaii Driver's License | urn:oid:2.16.840.1.113883.4.3.15 | 2.16.840.1.113883.4.3.15 | DL (US)
Indiana Driver's License | urn:oid:2.16.840.1.113883.4.3.18 | 2.16.840.1.113883.4.3.18 | DL (US)
Iowa Driver's License | urn:oid:2.16.840.1.113883.4.3.19 | 2.16.840.1.113883.4.3.19 | DL (US)
Idaho Driver's License | urn:oid:2.16.840.1.113883.4.3.16 | 2.16.840.1.113883.4.3.16 | DL (US)
Illinois Driver's License | urn:oid:2.16.840.1.113883.4.3.17 | 2.16.840.1.113883.4.3.17 | DL (US)
Kansas Driver's License | urn:oid:2.16.840.1.113883.4.3.20 | 2.16.840.1.113883.4.3.20 | DL (US)
Kentucky Driver's License | urn:oid:2.16.840.1.113883.4.3.21 | 2.16.840.1.113883.4.3.21 | DL (US)
Louisiana Driver's License | urn:oid:2.16.840.1.113883.4.3.22 | 2.16.840.1.113883.4.3.22 | DL (US)
Massachusetts Driver's License | urn:oid:2.16.840.1.113883.4.3.25 | 2.16.840.1.113883.4.3.25 | DL (US)
Maryland Driver's License | urn:oid:2.16.840.1.113883.4.3.24 | 2.16.840.1.113883.4.3.24 | DL (US)
Maine Driver's License | urn:oid:2.16.840.1.113883.4.3.23 | 2.16.840.1.113883.4.3.23 | DL (US)
Michigan Driver's License | urn:oid:2.16.840.1.113883.4.3.26 | 2.16.840.1.113883.4.3.26 | DL (US)
Minnesota Driver's License | urn:oid:2.16.840.1.113883.4.3.27 | 2.16.840.1.113883.4.3.27 | DL (US)
Missouri Driver's License | urn:oid:2.16.840.1.113883.4.3.29 | 2.16.840.1.113883.4.3.29 | DL (US)
Mississippi Driver's License | urn:oid:2.16.840.1.113883.4.3.28 | 2.16.840.1.113883.4.3.28 | DL (US)
Montana Driver's License | urn:oid:2.16.840.1.113883.4.3.30 | 2.16.840.1.113883.4.3.30 | DL (US)
New York Driver's License | urn:oid:2.16.840.1.113883.4.3.36 | 2.16.840.1.113883.4.3.36 | DL (US)
North Carolina Driver's License | urn:oid:2.16.840.1.113883.4.3.37 | 2.16.840.1.113883.4.3.37 | DL (US)
North Dakota Driver's License | urn:oid:2.16.840.1.113883.4.3.38 | 2.16.840.1.113883.4.3.38 | DL (US)
Nebraska Driver's License | urn:oid:2.16.840.1.113883.4.3.31 | 2.16.840.1.113883.4.3.31 | DL (US)
New Hampshire Driver's License | urn:oid:2.16.840.1.113883.4.3.33 | 2.16.840.1.113883.4.3.33 | DL (US)
New Jersey Driver's License | urn:oid:2.16.840.1.113883.4.3.34 | 2.16.840.1.113883.4.3.34 | DL (US)
New Mexico Driver's License | urn:oid:2.16.840.1.113883.4.3.35 | 2.16.840.1.113883.4.3.35 | DL (US)
Nevada Driver's License | urn:oid:2.16.840.1.113883.4.3.32 | 2.16.840.1.113883.4.3.32 | DL (US)
Ohio Driver's License | urn:oid:2.16.840.1.113883.4.3.39 | 2.16.840.1.113883.4.3.39 | DL (US)
Oklahoma Driver's License | urn:oid:2.16.840.1.113883.4.3.40 | 2.16.840.1.113883.4.3.40 | DL (US)
Oregon Driver's License | urn:oid:2.16.840.1.113883.4.3.41 | 2.16.840.1.113883.4.3.41 | DL (US)
Pennsylvania Driver's License | urn:oid:2.16.840.1.113883.4.3.42 | 2.16.840.1.113883.4.3.42 | DL (US)
Rhode Island Driver's License | urn:oid:2.16.840.1.113883.4.3.44 | 2.16.840.1.113883.4.3.44 | DL (US)
South Carolina Driver's License | urn:oid:2.16.840.1.113883.4.3.45 | 2.16.840.1.113883.4.3.45 | DL (US)
South Dakota Driver's License | urn:oid:2.16.840.1.113883.4.3.46 | 2.16.840.1.113883.4.3.46 | DL (US)
Tennessee Driver's License | urn:oid:2.16.840.1.113883.4.3.47 | 2.16.840.1.113883.4.3.47 | DL (US)
Texas Driver's License | urn:oid:2.16.840.1.113883.4.3.48 | 2.16.840.1.113883.4.3.48 | DL (US)
Utah Driver's License | urn:oid:2.16.840.1.113883.4.3.49 | 2.16.840.1.113883.4.3.49 | DL (US)
Virginia Driver's License | urn:oid:2.16.840.1.113883.4.3.51 | 2.16.840.1.113883.4.3.51 | DL (US)
Vermont Driver's License | urn:oid:2.16.840.1.113883.4.3.50 | 2.16.840.1.113883.4.3.50 | DL (US)
Washington Driver's License | urn:oid:2.16.840.1.113883.4.3.53 | 2.16.840.1.113883.4.3.53 | DL (US)
Wisconsin Driver's License | urn:oid:2.16.840.1.113883.4.3.55 | 2.16.840.1.113883.4.3.55 | DL (US)
West Virginia Driver's License | urn:oid:2.16.840.1.113883.4.3.54 | 2.16.840.1.113883.4.3.54 | DL (US)
Wyoming Driver's License | urn:oid:2.16.840.1.113883.4.3.56 | 2.16.840.1.113883.4.3.56 | DL (US)

4.7 Terminology Service

URL: https://hl7.org/fhir/R4/terminology-service.html

4.7 Terminology Service

FHIR Infrastructure Work Group | Maturity Level: 4 | Standards Status: Trial Use

This specification includes support for the provision of a terminology service - that is, a service that lets healthcare applications make use of codes and value sets without having to become experts in the fine details of code system, value set and concept map resources, and the underlying code systems and terminological principles. A server that supports all the functionality described here can be described as a "FHIR Terminology Service", and SHALL conform to the Terminology Service Capability Statement. Note that servers can declare that they provide terminology services in their capability statement:

```
<CapabilityStatement xmlns="http://hl7.org/fhir">
  <!-- snip -->
  <instantiates value="http://hl7.org/fhir/CapabilityStatement/terminology-server"/>
  <!-- snip -->
</CapabilityStatement>
```

4.7.1 Security

Generally, SSL SHOULD be used for all production health care data exchange. Even though terminology servers do not generally handle patient information directly, observers may still be able to infer information about patients by observing the codes and concepts used in terminology service operations, so encryption is still recommended.

A terminology server may choose not to authenticate the clients/users in any fashion, but might need to do so in order to limit or account for usage, or enforce agreement to licensing terms. For a value set maintenance server that allows terminologies to be edited, some form of authorization and/or authentication would be appropriate. This specification does not require any particular approach to security.

4.7.2 Basic Concepts

A FHIR terminology service is simply a set of functions built on the definitions provided by a collection of CodeSystem, ValueSet and ConceptMap resources, with additional inherently known terminologies providing support.

The terminology service builds on the basic principles for using terminologies in FHIR. Implementers should be familiar with:

• Using codes in FHIR

• The CodeSystem resource

• The ValueSet resource

• The ConceptMap resource

In addition, implementers should be familiar with the operations framework. Further useful information about terminologies may be found in:

• Underlying Principles: HL7 v3 Core Principles

• SNOMED CT technical documentation . Note: "namespace" is used differently here from the way it is used by IHTSDO (see discussion here)

4.7.2.1 External Code Systems

In order to be used with a value set, code systems and their content must be defined somewhere. They can be defined explicitly using the code system resource, or they can be defined elsewhere, and then used in a value set by referencing the correct system url. The FHIR specification defines a set of namespaces for commonly encountered code systems, and defines how some work with FHIR (e.g. SNOMED CT, LOINC, RxNorm). These code systems are often large and have many internally defined properties that are part of their formal definitions. The CodeSystem resource is not an appropriate way to distribute the contents of these code systems; the standard FHIR code system resource simply represents the properties of the code system. Instead, these terminologies provide their own distribution formats, and it is assumed that the content of these code systems are externally known to the terminology server.

Most useful terminology servers will make one or more of these external code systems available for use within the value sets that they manage. The list of additional terminologies that a terminology server supports beyond those defined in its value sets is published to clients by referencing code system resources in the server's Capability Statement.

```
{
 "resourceType" : "CapabilityStatement",
 "extension" : [
   {
     "url" : "http://hl7.org/fhir/StructureDefinition/capabilitystatement-supported-system",
     "valueUri" : "http://loinc.org"
   }]
}
```

This extension is added to the root Capability Statement.

4.7.2.2 Implementation Note

When a terminology server exposes an external code system, it makes a set of services available internally that serve the operational interfaces below. The internal server depends on the following logical information for a terminology:

• its URL (namespace, and how versioning works)

• what codes are valid

• what properties can be used to select codes

• what implicit value sets exist

The FHIR specification itself defines these things for common terminologies (including SNOMED CT, LOINC, RxNorm), and provides the CodeSystem infrastructure for supporting typical relatively simple small code systems.

Implementation Note: Implementers interested in working with existing published terminologies for which the CodeSystem infrastructure is not suitable should discuss their needs with HL7 to get the list above extended.

Note: A terminology service may choose to expose additional external code system specific related functionality such as summation, or structured search, but these services are outside the scope of the FHIR terminology service.

4.7.2.3 Operations across all value sets

For some of the operations below, it can be useful to perform them across all value sets known to the system. For example, $expand using a text filter, and searching all value sets at once. A special value set is defined that means "all value sets known to the server":

`http://hl7.org/fhir/ValueSet/@all`

Technically, this value set automatically imports all the existing value sets on the server. Note that this URL has no fixed meaning - its interpretation is server specific (e.g. whether it includes all versions of all value sets). This URL can only be used as a parameter to the operations described on this page.

4.7.2.4 Terminology Maintenance

The terminology service uses the code systems and value set resources defined on the system - both the implicit ones associated with the external code systems and those explicitly available at the /CodeSystem and /ValueSet endpoints - to serve the operational interface defined below. As code systems and value sets are created, updated or deleted, the outcomes of the operational services change. A terminology server should validate incoming resources and ensure integrity of the terminology services. Typically, servers provide a test and production environment, but there is no explicit notion of this in the interface itself.

4.7.3 Value Set Expansion

A value set describes a set of rules for what codes or concepts are considered to be in the value set. These rules might be simple (e.g. a direct list of codes from a specified version of a code system), or they might be quite complex (e.g. all codes with a particular property from an unspecified version of a code system).

A FHIR-enabled application can simply ask a terminology server to figure out all the details and return a list of the current codes in the value set. This is known as "expanding" the valueset. As a summary, the client passes the server the following information:

• the value set (either by its URL on the RESTful interface, by its logical identifier (ValueSet.url), or directly as a parameter to the call)

• (Optionally) a text filter to use to restrict the codes that are returned (e.g. user input text). It is left to server discretion to choose how to apply the text filter

• (Optionally) a date at which the expansion should be evaluated (usually, this is the current date/time, but there are circumstances where that is not appropriate)

• (Optionally) which page to retrieve - asking the server to break the expansion into a set of chunks

• (Optionally) other parameters that supply additional information about how to perform the expansion

The server returns a value set that contains the current list of codes that meet the filter criteria (or an OperationOutcome with an error if the expansion fails). Note that some value sets expand to many thousands of codes, or even an infinite number, and for these, the server SHOULD return an error code too-costly. In these cases, the client can try again with a more specific text filter to reduce the number of codes returned - this may result in a valid expansion.

For further information, consult the definition of the operation.

The $expand operation has support for paging - for a client to retrieve a big expansion in a set of partial views, in order to present the most optimal user experience. The client specifies both an offset and a count - how many codes per page, and where in the sequence to start. The return expansion specifies the number of concepts in the expansion, and the offset at which this partial view starts. Note that all expansions SHOULD include the total code count, but the offset element SHALL only exist when paging is being used. Expansions that are hierarchical trees of concepts are not subject to paging and the server simply returns the entire expansion.

Some example uses for the expansion operation:

• get a list of codes to display in a User interface (e.g. a drop-down interface)

• a variation on this is to offer the user a text box to type in. As the user types, call the expand operation to provide the user with a list of matching codes/concepts (like a browser search)

• fetch a list of codes to use when generating software programming instructions

• get a list of codes so that software can check whether a code is valid or not in a specific context

Examples

Expanding a value set that is already registered on the server as "23", with a text filter of "abdo":

```
GET [base]/ValueSet/23/$expand?filter=abdo
```

Expanding a value set that is specified by the client (using JSON):

```
POST [base]/ValueSet/$expand
[other headers]

{
  "resourceType" : "Parameters",
  "parameter" : [
     {
     "name" : "valueSet",
     "resource" : {
       "resourceType" : "ValueSet",
     [value set details]
     }
   }
  ]
}
```

The server responds with a value set (this example in XML):

```
HTTP/1.1 200 OK
[other headers]

<ValueSet xmlns="http://hl7.org/fhir">
  <!-- the server SHOULD populate the id with a newly created UUID
    so clients can easily track a particular expansion  -->
  <id value="43770626-f685-4ba8-8d66-fb63e674c467"/>
  <!-- no need for meta, though it is allowed for security labels, profiles -->

  <!-- other value set details -->
  <expansion>
    <!-- when expanded -->
    <timestamp value="20141203T08:50:00+11:00"/>
  <contains>
    <!-- expansion contents -->
  </contains>
  </expansion>
</ValueSet>
```

4.7.4 Concept Lookup / Decomposition

A system can ask a terminology server to return a set of information about a particular system/code combination using the lookup operation. The server returns information for both display and processing purposes. The client passes the server the following information:

• the code value (either a code, or a Coding data type)

• (Optionally) the id or the url of the code system in which the code is being checked

• (Optionally) a date at which the code information should be returned (usually, this is the current date/time, but there are circumstances where that is not appropriate)

• (Optionally) a set of properties to return about the code

The server returns some or all of the following information:

• a human description of the system

• a recommended display for the code

• properties of the code (e.g. status)

• other designations for the code (a value, optionally with language and/or a use code)

• relationships between this code and other codes (parent/child properties, etc.)

• Component properties of the specified code (e.g. to support reasoning) (e.g. decomposition)

The recommended display for the code is a text representation of the code that the terminology server recommends as the default choice to show to the user, though a client may choose out of the other designations if it has reason to.

If the client does not ask for any particular properties to be returned, it is at the discretion of the server to decide which properties to return (though note that the "version" property is always returned if the code system has a version).

Examples

Looking up a code in a code system:

```
GET [base]/CodeSystem/loinc/$lookup?code=1963-8
```

Note that the logical id "loinc" is not a reliable identifier across systems; each server assigns logical ids to code system resources however it sees fit. A more reliable query is this:

```
GET [base]/CodeSystem/$lookup?system=http://loinc.org&code=1963-8&property=code&property=display&property=designations
```

Lookup the code system using a Coding (this example in XML):

```
POST [base]/CodeSystem/$lookup
[other headers]
```

```
<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="coding"/>
    <valueCoding>
      <system value="http://loinc.org"/>
      <code value="1963-8"/>
    </valueCoding>
  </parameter>
</Parameters>
```

The server responds with a set of information (JSON this time):

```
HTTP/1.1 200 OK
[other headers]

{
  "resourceType" : "Parameters",
  "parameter" : [
    {
    "name" : "name",
    "valueString" : "LOINC"
  },
  {
    "name" : "version",
    "valueString" : "2.56"
  },
  {
    "name" : "display",
    "valueString" : "Bicarbonate [Moles/volume] in Serum"
  },
  {
    "name" : "abstract",
    "valueString" : "false"
  }
  ]
}
```

4.7.4.1 Standard Properties

The following properties are defined for all code systems:

Name | Usage
system | The name of the code system
version | The version of the code system used for the look up operation
display | The recommended display for the code, if one is known
definition | The definition for the code
designation | Other designations for the code
lang.X | Designations in language X (where X is an IETF Language code, see BCP-47
parent | Parent codes for this code (for code systems with a defined hierarchy)
child | child codes of this code (for code systems with a defined hierarchy)

In addition, any property codes defined by the code system (CodeSysem.property.code) can be used (and see the definitions for SNOMED CT, LOINC, and RxNorm).

4.7.5 Value Set Validation

One way to determine whether a code is in a value set is to expand the value set (as described above), and then look at the returned codes to see if the code is in the expansion. However, this is not an efficient way to test whether a code is valid, and for some value sets (e.g. with infinite number of members), it cannot work. Instead, a FHIR terminology server provides a "validate-code" operation. The client passes the server the following information:

• the value set (either by its URL on the RESTful interface, by its logical identifier (ValueSet.url), or directly as a parameter to the call)

• the code value (either a code + system, a Coding data type, or a CodeableConcept)

• (Optionally) a date at which the expansion should be evaluated (usually, and by default, this is the current date/time, but there are circumstances where that is not appropriate)

The server returns a true/false indicating whether the code/concept is valid, and a list of errors and warnings associated with it. The server should also return an appropriate display for the concept for use in a UI context.

Note that if the server is passed a CodeableConcept, the server is able to check whether any of the codes are valid against the value set, and also check whether multiple codings are allowed and/or the codings provided are consistent with each other.

Every code system has an implicit value set that is "all the concepts defined in the code system" (CodeSystem.valueSet). For some code systems, these value set URIs are defined in advance (e.g. for LOINC, it is http://loinc.org/vs). However, for some code systems, they are not known. Clients can refer to these implicit value sets by providing the URI for the code system itself.

Examples

Simple validation of a code/system against a known value set:

```
GET [base]/ValueSet/23/$validate-code?system=http://loinc.org&code=1963-8&display=test
```

Validate a CodeableConcept against a client specified value set (this example in JSON):

```
POST [base]/ValueSet/$validate-code
[other headers]

{
  "ResourceType" : "Parameters",
  "parameter" : [
    {
    "name" : "coding",
    "valueCodeableConcept" : {
      "coding" : {
        "system" : "http://loinc.org",
          "code" : "1963-8",
      "display" : "test"
      }
    }
  },
  {
    "name" : "valueSet",
    "resource": {
      "resourceType" : "ValueSet",
    [etc.]
    }
  }
  ]
}
```

The server responds with validation information (JSON this time):

```
HTTP/1.1 200 OK
[other headers]

{
  "resourceType" : "Parameters",
  "parameter" : [
    {
    "name" : "result",
    "valueBoolean" : false
  },
  {
    "name" : "message",
    "valueString" : "The display \"test\" is incorrect"
  },
  {
    "name" : "display",
    "valueString" : "Bicarbonate [Moles/volume] in Serum"
  }
  ]
}
```

4.7.6 Subsumption testing

To test the subsumption relationship between code/Coding A and code/Coding B, perform a $subsumes operation. Subsumption testing is based on the CodeSystem definition of subsumption. The client passes the server the following information:

• the system that identifies the code system in which subsumption testing is to be performed (either by invoking the operation on the code system directly, or referring to it by its canonical URL)

• Concepts A and B - either as codes, or Codings

• (Optionally) the version of the code system to use (mostly, this should not matter)

If the client passes Codings, it is allowed to use code system values that are different from the code system in which subsumption testing is to be performed. In this case, the server SHALL return an error unless the relationships between the various code systems is well defined.

If the concepts can be compared, then the server returns an outcome code:

equivalent	Concepts A and B are equivalent
subsumes	Concept A subsumes Concept B
subsumed-by	Concept A is subsumed by Concept B
not-subsumed	Concepts A and B are not related by any subsumption relationship

Examples

Test whether a SNOMED CT Concept 'Disorder of liver' (235856003) subsumes 'Viral hepatitis' (3738000):

```
GET [base]/CodeSystem/$subsumes?system=http://snomed.info/sct&codeA=235856003&codeB=3738000
```

Or using Codings:

```
POST [base]/CodeSystem/$subsumes
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <!-- Subsumption testing - use SNOMED CT rules -->
  <parameter>
    <name value="system"/>
    <valueUri value="http://snomed.info/sct"/>
  </parameter>
```

```
  <!-- Australian distribution -->
  <parameter>
    <name value="version"/>
    <valueString value="http://snomed.info/sct/32506021000036107/version/20160430"/>
  </parameter>
  <parameter>
    <name value="codingA"/>
    <valueCoding>
      <system value="http://snomed.info/sct"/>
      <code value="235856003"/>
    </valueCoding>
  </parameter>
  <parameter>
    <name value="codingB"/>
    <valueCoding>
      <system value="http://snomed.info/sct"/>
      <code value="3738000"/>
    </valueCoding>
  </parameter>
</Parameters>
```

Server response:

```
HTTP/1.1 200 OK
[other headers]

{
  "resourceType" : "Parameters",
  "parameter" : [
    {
    "name" : "outcome",
    "valueCode" : "subsumes"
  },
  ]
}
```

4.7.7 Batch Validation

It is also possible to validate a set of concepts against their relevant value sets by using the $validate-code operation in a Batch interaction.

Example

A request to validate 2 concepts from a CDA document, with OIDs for value set identifiers:

```
POST [base]
[other headers]

{
  "ResourceType": "Bundle",
  "type": "batch",
  "entry": [{
    "request": {
      "method": "Get",
      "url": "ValueSet/$validate-code?system=http://loinc.org&code=2324-4&uri=urn:oid:1.2.3.4.6"
    }
  },
  {
    "request": {
      "method": "GET",
      "url": "ValueSet/$validate-code?system=http://snomed.info/sct&codes=22298006&uri=urn:oid:1.2.3.4.7"
    }
  }]
}
```

The server responds with a series of validation outcomes (JSON this time):

```
HTTP/1.1 200 OK
[other headers]

{
  "ResourceType": "Bundle",
  "type": "batch-response",
  "entry": [{
    "resource": {
      "resourceType": "Parameters",
      "parameter": [{
        "name": "result",
        "valueBoolean": false
      },
      {
        "name": "message",
        "valueString": "'2324-4' is not a valid LOINC code"
      }]
    }
  },
  {
    "resource": {
      "resourceType": "Parameters",
      "parameter": [{
        "name": "result",
        "valueBoolean": false
```

```
      },
      {
        "name": "message",
        "valueString": "The concept is not in the specified value set (\"Organisms\")"
      },
      {
        "name": "display",
        "valueString": "Myocardial infarction"
      }]
    }]
  }
```

4.7.8 Translations

A client can ask a server to translate a concept from one value set to another. Typically, this is used to translate between code systems (e.g. from LOINC to SNOMED CT, or from a HL7 V3 code to a HL7 V2 code). The client calls the translate operation and passes the following parameters:

• a code + system, Coding, or CodeableConcept

• a Concept Map to use for the translation

• the value set for the context of the source

• the value set for the destination

The client passes either a concept map, or the value sets for the source and destination context. If there is no concept map, then the server may determine the appropriate map to use from context provided in the value sets. If there is no particular context, the appropriate value sets would be the value sets for the entire coding system at question (e.g. from http://snomed.info/sct to http://loinc.org/vs). The server performs the translation as it is able based on the concept maps that it knows about. If no single mapping can be determined, then the server returns an error. Some servers may require a concept map to use for the translation.

Example

Translate from FHIR Composition status to HL7 v3 Act Status (based on this defined concept map):

```
GET [base]/ConceptMap/$translate?system=http://hl7.org/fhir/composition-status
  &code=preliminary&source=http://hl7.org/fhir/ValueSet/composition-status
  &target=http://terminology.hl7.org/ValueSet/v3-ActStatus
```

The server responds with validation information:

```
HTTP/1.1 200 OK
[other headers]

{
  "resourceType" : "Parameters",
  "parameter" : [
    {
    "name" : "result",
    "valueBoolean" : true
    },
    {
      "name" : "outcome",
      "valueCoding" : {
        "system" : "http://terminology.hl7.org/CodeSystem/v3-ActStatus",
        "code" : "active",
      }
    }
  ]
}
```

4.7.9 Batch Translation

It is also possible to translate a set of concepts against their relevant value sets by using the $translate operation in a Batch interaction.

Example

A request to translate 2 concepts from a CDA document, with OIDs for value set identifiers:

```
POST [base]
[other headers]

{
  "ResourceType": "Bundle",
  "type": "batch",
  "entry": [{
    "resource": {
      "ResourceType": "Parameters",
      "parameter": [{
        "name": "concept",
        "valueCodeableConcept": {
          "system": "http://loinc.org",
          "code": "2324-4"
        }
      },
      {
        "name": "target",
```

```
        "valueUri": "urn:oid:1.2.3.4.6"
      }]
    },
    "request": {
      "method": "POST",
      "url": "ConceptMap/$translate"
    }
  },
  {
    "resource": {
      "ResourceType": "Parameters",
      "parameter": [{
        "name": "concept",
        "valueCodeableConcept": {
          "system": "http://snomed.info/sct",
          "code": "22298006"
        }
      },
      {
        "name": "target",
        "valueUri": "urn:oid:1.2.3.4.7"
      }]
    },
    {
      "request": {
        "method": "POST",
        "url": "ConceptMap/$translate"
      }
    }]
  }
```

The server responds with a series of translation outcomes:

```
HTTP/1.1 200 OK
[other headers]

{
  "ResourceType": "Bundle",
  "type": "batch-response",
  "entry": [{
    "resource": {
      "resourceType": "Parameters",
      "parameter": [{
        "name": "result",
        "valueBoolean": false
      },
      {
        "name": "message",
        "valueString": "'2324-4' is not a valid LOINC code"
      }]
    }
  },
  {
    "resource": {
      "resourceType": "Parameters",
      "parameter": [{
        "name": "result",
        "valueBoolean": true
      },
      {
        "name": "outcome",
        "valueCodeableConcept": {
          "coding": {
            "system": "http://example.com/codesystems/example",
            "code": "xxxx"
          }
        }
      }]
    }
  }]
}
```

4.7.10 Maintaining a Closure Table

The 5 operations Expand, Lookup, Validate, Subsumes, and Translate account for most operational requirements associated with terminology use. However, there is one difficult but important use case that they do not address, which is integrating terminologically based logic into application searches.

A typical example of this is a user that wants to find any observations for male patients over the age of 50 who attended a particular clinic within a particular 2-week period, with a diagnosis of gout, and who had an elevated serum creatinine.

In this case, both "diagnosis of gout" and "serum creatinine" involve value set and/or subsumption queries (e.g. against SNOMED CT and LOINC respectively). This search has to be executed by some logical processing engine that knows how to find patient related data in a given persistence store. Often, this is some kind of SQL query, though many other technological choices are available. However, this is done, the challenge with an operation like this is to integrate the terminological knowledge into a search execution that also covers

other relationships expressed in the search criteria.

One approach to this problem would be to use the expand operation above, so that the system executing the search could generate expansions, and then search for these expansions. This has a couple of problems:

• the list of subsumed codes could be very long, and the search operation becomes correspondingly inefficient

• the expansion of the subsumption might not be closed, and so the search operation cannot be correct

An alternative approach is to generate a transitive closure table which lists all the possible transitive subsumption relationships, and allows for rapid execution of these kind of queries . However, this has other problems:

• the subsumption table can be very large (>500000 records for SNOMED CT), even though very few of the codes are used

• subsumption tables are generally built up front, and do not deal with new codes as they are encountered very well

• they still do not offer a solution for non-closed expansions

This is the main reason why most systems do not support post-coordination or other forms of coded expressions.

In FHIR, this problem is solved by building a closure table on the fly as new codes are seen. This technique leaves the FHIR terminology server responsible for the terminological reasoning and the client responsible for the closure table maintenance. To the client, it doesn't matter whether the concept is post-coordinated or not. Here's a description of how the process works:

• the client defines a name associated with a particular context in which it wishes to maintain a subsumption based closure table

• the client registers this name with the FHIR Terminology server using the $closure operation (described below), with only one parameter, the name of the context

• any time the client system encounters a new Coding that is not entered in the closure table, it calls the $closure operation with the context name, and the Coding value it has encountered

• the server returns a ConceptMap resource with a list of new entries (code : system -> code : system) that the client should add to its closure table the server can indicate that entries should be removed from the table by providing a (code : system -> code : system) with equivalence "unmatched" (though it's not known why that would be needed)

• the client copies these entries into its closure table

• to facilitate the initialization process, a client can call $closure with multiple Coding values

The $closure operation takes 2 parameters:

• closure table context name

• concepts to enter into the table (0 or more - 0 codings is a request to (re-)initialize the table)

The operation returns a concept map which has a list of mappings that represent new entries to make in the closure table. The subsumption testing performed when building a closure table is the same as for the $subsumes operation, and is based on the CodeSystem definition of subsumption.

The closure table can be resynchronized by passing an additional "version" parameter, which is a value taken from the version in one of the delta responses. This is a request to replay all the mapping changes since that delta was sent.

4.7.10.1 Initializing a Closure Table

Before it can be used, a closure table has to be initialized. To initialize a closure table, POST the following to [base]/ConceptMap/$closure:

```
{
  "resourceType" : "Parameters",
   "parameter" : [{
     "name" : "name",
     "valueString" : "[name]"
  }]
}
```

A successful response is a 200 OK from the server, with an associated ConceptMap:

```
{
    "resourceType": "ConceptMap",
    "id": "[name]",
    "version": "0",
    "name": "Closure Table [name] Creation",
    "status": "active",
    "experimental": true,
    "date": "2015-12-20T23:10:55Z"
}
```

If there is an error (usually involving the closure name) the server returns a HTTP status 400 with an operation outcome:

```
{
  "resourceType": "OperationOutcome",
  "text": {
    "status": "generated",
    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p>invalid closure name \"invalid-id!\":</p></div>"
  },
  "issue": [
    {
      "severity": "error",
      "details": {
        "text" : "invalid closure name \"invalid-id!\""
      }
    }
  ]
}
```

What closure names are valid is at the discretion of the server.

4.7.10.2 Adding to a Closure Table

When the consumer (client) encounters a new code, it POSTs the following to [base]/ConceptMap/$closure:

```
{
  "resourceType" : "Parameters",
  "parameter" : [{
    "name" : "name",
    "valueString" : "[name]"
  }, {
    "name" : "concept",
    "valueCoding" : {
       "system" : "http://snomed.info/sct",
       "code" : "22298006",
       "display" : "Myocardial infarction"
    }
  }]
}
```

Note that this example only includes one concept, but more than one is allowed:

```
{
  "resourceType" : "Parameters",
  "parameter" : [{
    "name" : "name",
    "valueString" : "[name]"
  }, {
    "name" : "concept",
    "valueCoding" : {
       "system" : "http://snomed.info/sct",
       "code" : "22298006",
       "display" : "Myocardial infarction"
    }
  }, {
    "name" : "concept",
    "valueCoding" : {
       "system" : "http://snomed.info/sct",
       "code" : "128599005",
       "display" : "Structural disorder of heart"
    }
  }]
}
```

The response varies depending on the conditions on the server. Possible responses: If the closure table has not been initialized: Return a 404 Not Found with

```
{
  "resourceType": "OperationOutcome",
  "text": {
    "status": "generated",
    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p>invalid closure name \"[name]\":</p></div>"
  },
  "issue": [
    {
      "severity": "error",
      "details": {
        "text" : "invalid closure name \"[name]\""
      }
    }
  ]
}
```

If the closure table needs to be reinitialized: Return a 422 Unprocessable Entity with

```
{
  "resourceType": "OperationOutcome",
  "text": {
    "status": "generated",
    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><p>closure \"[name\" must be reinitialized</p></div>"
   },
   "issue": [{
       "severity": "error",
       "details": {
         "text" : "closure \"[name]\" must be reinitialized"
```

```
      }
    }
  ]
}
```

The server should only send this when its underlying terminology conditions have been changed (e.g. a new version of SNOMED CT has been loaded). When a client gets this, it's only choice is to initialize the closure table, and process all the codes in the closure table again (the assumption here is that the system has some external source of 'all the codes' so it can rebuild the table again). If the concept(s) submitted are processed ok, but there's no new concepts, or no new entries in the table, return a 200 OK with :

```
{
    "resourceType": "ConceptMap",
    "id": "[name]",
    "version": "[version]",
    "name": "Updates for Closure Table [name]",
    "status": "active",
    "experimental": true,
    "date": "2015-12-20T23:12:55Z"
}
```

If there's new entries in the closure table, the server returns a 200 OK with:

```
{
  "resourceType": "ConceptMap",
  "id": "b87db127-9996-4d0c-bda9-a278d7a24a69",
  "version": "[version]",
  "name": "Updates for Closure Table [name]",
  "status": "active",
  "experimental": true,
  "date": "2015-12-20T23:16:24Z",
  "group": [{
    "source": "http://snomed.info/sct",
    "target": "http://snomed.info/sct",
    "element" : {
      "code": "22298006",
      "target": [{
        "code": "128599005",
        "equivalence": "subsumes"
      }]
    }
  }]
}
```

Notes

• The server can return multiple elements, each with 1 or more targets

• servers may return the relationship represented in either direction

• it's important to understand the relationship the right way around. From the spec: The equivalence is read from target to source (e.g. the target is 'wider' than the source). So in this case, 128599005 (Structural disorder of heart) subsumes 22298006 (Myocardial infarction)

• In the $closure operation, the response never explicitly states that a code is subsumed by itself. Clients should assume that this is implicit

• The version is important. Each new invocation of the $closure operation returns a new version of the concept map. The server must keep track of the versions is has issued for replay (see below)

• As well as entering codes that are actually used, the client also enters search terms into the closure table

• The combination of the system and code is the key to the closure table; if the server encounters two different codes that have the same meaning (e.g. syntactical variation), it should create an "equals" relationship between them

4.7.10.3 Re-running Closure operation

Given the way that the closure operation functions, it's possible for a client to lose a response from the server before it is committed to safe storage (or the client might not have particularly safe storage). For this reason, when a client is starting up, it should check that there have been no missing operations. It can do this by passing the last version (from the Concept Map response) it is sure it processed in the request:

```
{
  "resourceType" : "Parameters",
   "parameter" : [{
     "name" : "name",
     "valueString" : "[name]"
  }, {
     "name" : "version",
     "valueString" : "3"
  }]
 }
```

That's a request to return all the additions to the closure table since version 3. The server returns its latest version in the concept map, along with anything added to the closure table since version 3 (not including version 3)

Notes:

• The client can pass a concept or version, but not both

• These examples use a serially incrementing sequential integer, but this is not required, and clients should not assume that there is any meaning or order in the version. Just recall the last version and treat it as a magic fixed value only meaningful to the server. There is, however, one special value: '0'. Passing a last version of 0 should be understood as resyncing the entire closure table

4.7.10.4 Making use of the Closure Table

The client uses the result of the closure operation to maintain a closure table. Simplistically, it might look like this:

Scope | Source | Target
patient-problems | http://snomed.info/sct|22298006 | http://snomed.info/sct|128599005
patient-problems | http://snomed.info/sct|24595009 | http://snomed.info/sct|90560007
obs-code | http://loinc.org|14682-9 | http://loinc.org|LP41281-4

The client can then use a table like this as part of its general search conditions. Using the example from above: "Find any observations for male patients over the age of 50 who attended a particular clinic within a particular 2-week period, with a diagnosis of gout, and who had an elevated serum creatinine." This query could be done, for instance, with an SQL query like this:

```
Select * from Observations, Patients, Encounters, Conditions, Observations as Obs2 where
  Observations.patient = Patients.Key and Patients.Age > 50 and
  Observations.encounter = Encounters.Key and Encounter.clinic = [key]
    and encounter.date >= [date] and encounter.date <= [date] and
  Conditions.patient = Patients.Key and Conditions.code
    in (select Source From ClosureTable
      where Scope = "patient-problems" and Target = "http://snomed.info/sct|90560007") and
  Obs2.patient = Patients.Key and Obs2.value > 0.19 and Obs2.code
    in (select Source From ClosureTable
      where Scope = "obs-code" and Target = "http://loinc.org|LP41281-4")
```

Note that in real clinical systems, tables are usually far more structured than this example implies, and the query is correspondingly more complex. The closure table would usually be normalised - this example is kept simple to demonstrate the concept.

4.8 Resource CodeSystem - Content

URL: https://hl7.org/fhir/R4/codesystem.html
Vocabulary Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Anonymous | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Conformance Package for further details.

The CodeSystem resource is used to declare the existence of and describe a code system or code system supplement and its key properties, and optionally define a part or all of its content.

4.8.1 Scope and Usage

The FHIR terminology specification is based on two key concepts, originally defined in HL7 v3 Core Principles :

• code system - declares the existence of and describes a code system or code system supplement and its key properties, and optionally defines a part or all of its content. Also known as Ontology, Terminology, or Enumeration

• value set - specifies a set of codes drawn from one or more code systems, intended for use in a particular context. Value sets link between CodeSystem definitions and their use in coded elements

Code systems define which codes (symbols and/or expressions) exist, and how they are understood. Value sets select a set of codes from one or more code systems to specify which codes can be used in a particular context.

The CodeSystem resource is used to declare the existence of a code system, and its key properties:

• Identifying URL and version

• Description, Copyright, publication date, and other metadata

• Some key properties of the code system itself - e.g. whether it exhibits concept permanence, whether it defines a compositional grammar, and whether the codes that it defines are case sensitive

• What filters can be used in value sets that use the code system in a ValueSet.compose element

• What concept properties are defined by the code system

In addition, the CodeSystem resource may list some or all of the concepts in the code system, along with their basic properties (code, display, definition), designations, and additional properties. Code System resources may also be used to define supplements, which extend an existing code system with additional designations and properties.

The CodeSystem resource is not intended to support the process of maintaining code systems. Instead, the focus is on publishing the properties and optionally the content of a code system for use throughout the FHIR eco-system, such as to support value set expansion and validation. Note that the important existing (large)

code systems (SNOMED CT, LOINC, RxNorm, ICD family, etc.) all have their own maintenance systems and distribution formats, and CodeSystem is generally not an efficient way to distribute their content, though it is used as one way of declaring the filters and properties associated with those code systems.

4.8.2 Boundaries and Relationships

• Code systems are used in ValueSet resources

• The Coding data type refers to CodeSystem resources by their canonical URL

• The CodeSystem resource design is based, in part, on the code system functionality described in the HL7 CTS 2 specification. A CTS2 server can be used to maintain code systems which are then published using the CodeSystem resource

• The CodeSystem resource documents the inherent structure and capabilities of code system, whereas the TerminologyCapabilities resource documents what a server hosting the code system is capable of

The CodeSystem resource declares the existence of a code system and its key properties including its preferred identifier. The NamingSystem resource identifies the existence of a code or identifier system, and its possible and preferred identifiers. The key difference between the resources is who creates and manages them - CodeSystem resources are managed by the owner or publisher of the code system, who can properly define the code system features and content. NamingSystem resources, on the other hand, are frequently defined by 3rd parties that encounter the code system in use, and need to describe the use, but do not have the authority to define the features and content. Additionally, there may be multiple authoritative NamingSystem resources for a code system, but ideally there would be only one authoritative CodeSystem resource (identified by its canonical URL) that is provided by the code system publisher, with multiple copies distributed on additional FHIR servers or elsewhere and used where needed.

Both Code System supplements and Concept Maps may be used to define relationships between concepts in different systems. ConceptMaps are assertions of the relationships between different concepts that are associated with particular contexts of use, while CodeSystem supplements are used to define inherent properties and semantics of the concepts in the code system

4.8.3 Background and Context

When using code systems and value sets, proper differentiation between a code system and a value set is important. This is one very common area where significant clinical safety risks occur in practice. Implementers should be familiar with the content in Using Codes in Resources.

4.8.3.1 CodeSystem Identification

A code system has three identifiers. The first two can be used to reference the code system in the FHIR context:

• CodeSystem.id: The logical id on the system that holds the CodeSystem resource instance - this typically is expected to change as the resource moves from server to server. The location URI is constructed by appending the logical id to the server base address where the instance is found and the resource type. This URI should be a resolvable URL by which the resource instance may be retrieved, usually from a FHIR server, and it may be a relative reference typically to the server base URL.

• CodeSystem.url: The canonical URL that never changes for this code system - it is the same in every copy. The element is named url rather than uri for legacy reasons and to strongly encourage providing a resolvable URL as the identifier whenever possible. This canonical URL is used to refer to all instances of this particular code system across all servers and systems. Ideally, this URI should be a URL which resolves to the location of the master version of the code system, though this is not always possible.

For example, the code systems published as part of the FHIR specification all have a location ("literal") URI which is the URL where they may be accessed in the FHIR specification itself. However, while a new version of the FHIR specification is being prepared, code systems that are published in the drafts will not be found in the current FHIR specification version.

Because it is common practice to copy (cache) code systems locally, most references to code systems can use either the canonical URL or the location ("literal") URL.

The third code system identifier is used typically for external references to the code system outside of FHIR:

• CodeSystem.identifier: A system/value pair that is used to identify the code system in other contexts (such as an OID in an HL7 v3 specification)

For further information regarding resource identification, see Resource Identity.

This means that each code system has 2 different URIs that can be used to reference it - its canonical URL (the url element), and its local location from which it may be retrieved (which includes the id element). Because it is common practice to copy (cache) code systems locally, most references to code systems use the canonical URL.

Alternatively, the identifier and version elements may be used to reference this code system in a design, a profile, a CDA template or HL7 v3 message (in the CD data type codeSystem and codeSystemVersion properties). These different contexts may make additional restrictions on the possible values of these

elements. The identifier is generally not needed when using code systems in a FHIR context, where the canonical URL is always the focus.

This resource is referenced by Coding, itself, ConceptMap, TerminologyCapabilities and ValueSet

4.8.4 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
CodeSystem | IN | | DomainResource | Declares the existence of and describes a code system or code system supplement + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: Within a code system definition, all the codes SHALL be unique Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this code system, represented as a URI (globally unique) (Coding.system)
identifier | Σ | 0..* | Identifier | Additional identifier for the code system (business identifier)
version | Σ | 0..1 | string | Business version of the code system (Coding.version)
name | ΣI | 0..1 | string | Name for this code system (computer friendly)
title | Σ | 0..1 | string | Name for this code system (human friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the code system
useContext | ΣTU | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for code system (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this code system is defined
copyright | | 0..1 | markdown | Use and/or publishing restrictions
caseSensitive | Σ | 0..1 | boolean | If code comparison is case sensitive
valueSet | Σ | 0..1 | canonical(ValueSet) | Canonical reference to the value set with entire code system
hierarchyMeaning | Σ | 0..1 | code | grouped-by | is-a | part-of | classified-with CodeSystemHierarchyMeaning (Required)
compositional | Σ | 0..1 | boolean | If code system defines a compositional grammar
versionNeeded | Σ | 0..1 | boolean | If definitions are not stable
content | Σ | 1..1 | code | not-present | example | fragment | complete | supplement CodeSystemContentMode (Required)
supplements | Σ | 0..1 | canonical(CodeSystem) | Canonical URL of Code System this adds designations and properties to
count | Σ | 0..1 | unsignedInt | Total concepts in the code system
filter | Σ | 0..* | BackboneElement | Filter that can be used in a value set
code | Σ | 1..1 | code | Code that identifies the filter
description | Σ | 0..1 | string | How or why the filter is used
operator | Σ | 1..* | code | = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists FilterOperator (Required)
value | Σ | 1..1 | string | What to use for the value
property | Σ | 0..* | BackboneElement | Additional information supplied about each concept
code | Σ | 1..1 | code | Identifies the property on the concepts, and when referred to in operations
uri | Σ | 0..1 | uri | Formal identifier for the property
description | Σ | 0..1 | string | Why the property is defined, and/or what it conveys
type | Σ | 1..1 | code | code | Coding | string | integer | boolean | dateTime | decimal PropertyType (Required)
concept | | 0..* | BackboneElement | Concepts in the code system
code | | 1..1 | code | Code that identifies concept
display | | 0..1 | string | Text to display to the user
definition | | 0..1 | string | Formal definition
designation | | 0..* | BackboneElement | Additional representations for the concept
language | | 0..1 | code | Human language of the designation Common Languages (Preferred but limited to AllLanguages)
use | | 0..1 | Coding | Details how this designation would be used Designation Use (Extensible)
value | | 1..1 | string | The text value for this designation
property | | 0..* | BackboneElement | Property value for the concept
code | | 1..1 | code | Reference to CodeSystem.property.code
value[x] | | 1..1 | | Value of the property for this concept
valueCode | | | code
valueCoding | | | Coding
valueString | | | string
valueInteger | | | integer
valueBoolean | | | boolean
valueDateTime | | | dateTime

valueDecimal | | | decimal
concept | | 0..* | see concept | Child Concepts (is-a/contains/categorizes)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CodeSystem xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this code system, represented as a URI (globally unique)
    (Coding.system) -->
 <identifier><!-- 0..* Identifier Additional identifier for the code system (business identifier) --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the code system (Coding.version) -->
 <name value="[string]"/><!--  0..1 Name for this code system (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this code system (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the code system -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for code system (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this code system is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <caseSensitive value="[boolean]"/><!-- 0..1 If code comparison is case sensitive -->
 <valueSet><!-- 0..1 canonical(ValueSet) Canonical reference to the value set with entire code system --></valueSet>
 <hierarchyMeaning value="[code]"/><!-- 0..1 grouped-by | is-a | part-of | classified-with -->
 <compositional value="[boolean]"/><!-- 0..1 If code system defines a compositional grammar -->
 <versionNeeded value="[boolean]"/><!-- 0..1 If definitions are not stable -->
 <content value="[code]"/><!-- 1..1 not-present | example | fragment | complete | supplement -->
 <supplements><!-- 0..1 canonical(CodeSystem) Canonical URL of Code System this adds designations and properties to
    --></supplements>
 <count value="[unsignedInt]"/><!-- 0..1 Total concepts in the code system -->
 <filter>  <!-- 0..* Filter that can be used in a value set -->
  <code value="[code]"/><!-- 1..1 Code that identifies the filter -->
  <description value="[string]"/><!-- 0..1 How or why the filter is used -->
  <operator value="[code]"/><!-- 1..* = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists
    -->
  <value value="[string]"/><!-- 1..1 What to use for the value -->
 </filter>
 <property>  <!-- 0..* Additional information supplied about each concept -->
  <code value="[code]"/><!-- 1..1 Identifies the property on the concepts, and when referred to in operations -->
  <uri value="[uri]"/><!-- 0..1 Formal identifier for the property -->
  <description value="[string]"/><!-- 0..1 Why the property is defined, and/or what it conveys -->
  <type value="[code]"/><!-- 1..1 code | Coding | string | integer | boolean | dateTime | decimal -->
 </property>
 <concept>  <!-- 0..* Concepts in the code system -->
  <code value="[code]"/><!-- 1..1 Code that identifies concept -->
  <display value="[string]"/><!-- 0..1 Text to display to the user -->
  <definition value="[string]"/><!-- 0..1 Formal definition -->
  <designation>  <!-- 0..* Additional representations for the concept -->
   <language value="[code]"/><!-- 0..1 Human language of the designation -->
   <use><!-- 0..1 Coding Details how this designation would be used --></use>
   <value value="[string]"/><!-- 1..1 The text value for this designation -->
  </designation>
  <property>  <!-- 0..* Property value for the concept -->
   <code value="[code]"/><!-- 1..1 Reference to CodeSystem.property.code -->
   <value[x]><!-- 1..1 code|Coding|string|integer|boolean|dateTime|decimal Value of the property for this concept
    --></value[x]>
  </property>
  <concept><!-- 0..* Content as for CodeSystem.concept Child Concepts (is-a/contains/categorizes) --></concept>
 </concept>
</CodeSystem>
```

JSON Template

```
{
  "resourceType" : "CodeSystem",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this code system, represented as a URI (globally unique) (Coding.system)
  "identifier" : [{ Identifier }], // Additional identifier for the code system (business identifier)
  "version" : "<string>", // Business version of the code system (Coding.version)
  "name" : "<string>", // C? Name for this code system (computer friendly)
  "title" : "<string>", // Name for this code system (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the code system
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for code system (if applicable)
  "purpose" : "<markdown>", // Why this code system is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "caseSensitive" : <boolean>, // If code comparison is case sensitive
  "valueSet" : { canonical(ValueSet) }, // Canonical reference to the value set with entire code system
```

```
  "hierarchyMeaning" : "<code>", // grouped-by | is-a | part-of | classified-with
  "compositional" : <boolean>, // If code system defines a compositional grammar
  "versionNeeded" : <boolean>, // If definitions are not stable
  "content" : "<code>", // R!  not-present | example | fragment | complete | supplement
  "supplements" : { canonical(CodeSystem) }, // Canonical URL of Code System this adds designations and properties to
  "count" : "<unsignedInt>", // Total concepts in the code system
  "filter" : [{ // Filter that can be used in a value set
    "code" : "<code>", // R!  Code that identifies the filter
    "description" : "<string>", // How or why the filter is used
    "operator" : ["<code>"], // R!  = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists
    "value" : "<string>" // R!  What to use for the value
  }],
  "property" : [{ // Additional information supplied about each concept
    "code" : "<code>", // R!  Identifies the property on the concepts, and when referred to in operations
    "uri" : "<uri>", // Formal identifier for the property
    "description" : "<string>", // Why the property is defined, and/or what it conveys
    "type" : "<code>" // R!  code | Coding | string | integer | boolean | dateTime | decimal
  }],
  "concept" : [{ // Concepts in the code system
    "code" : "<code>", // R!  Code that identifies concept
    "display" : "<string>", // Text to display to the user
    "definition" : "<string>", // Formal definition
    "designation" : [{ // Additional representations for the concept
      "language" : "<code>", // Human language of the designation
      "use" : { Coding }, // Details how this designation would be used
      "value" : "<string>" // R!  The text value for this designation
    }],
    "property" : [{ // Property value for the concept
      "code" : "<code>", // R!  Reference to CodeSystem.property.code
      // value[x]: Value of the property for this concept. One of these 7:
      "valueCode" : "<code>"
      "valueCoding" : { Coding }
      "valueString" : "<string>"
      "valueInteger" : <integer>
      "valueBoolean" : <boolean>
      "valueDateTime" : "<dateTime>"
      "valueDecimal" : <decimal>
    }],
    "concept" : [{ Content as for CodeSystem.concept }] // Child Concepts (is-a/contains/categorizes)
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CodeSystem;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CodeSystem.url [ uri ]; # 0..1 Canonical identifier for this code system, represented as a URI (globally unique)
    (Coding.system)
  fhir:CodeSystem.identifier [ Identifier ], ... ; # 0..* Additional identifier for the code system (business
    identifier)
  fhir:CodeSystem.version [ string ]; # 0..1 Business version of the code system (Coding.version)
  fhir:CodeSystem.name [ string ]; # 0..1 Name for this code system (computer friendly)
  fhir:CodeSystem.title [ string ]; # 0..1 Name for this code system (human friendly)
  fhir:CodeSystem.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:CodeSystem.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:CodeSystem.date [ dateTime ]; # 0..1 Date last changed
  fhir:CodeSystem.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:CodeSystem.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:CodeSystem.description [ markdown ]; # 0..1 Natural language description of the code system
  fhir:CodeSystem.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:CodeSystem.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for code system (if applicable)
  fhir:CodeSystem.purpose [ markdown ]; # 0..1 Why this code system is defined
  fhir:CodeSystem.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:CodeSystem.caseSensitive [ boolean ]; # 0..1 If code comparison is case sensitive
  fhir:CodeSystem.valueSet [ canonical(ValueSet) ]; # 0..1 Canonical reference to the value set with entire code system
  fhir:CodeSystem.hierarchyMeaning [ code ]; # 0..1 grouped-by | is-a | part-of | classified-with
  fhir:CodeSystem.compositional [ boolean ]; # 0..1 If code system defines a compositional grammar
  fhir:CodeSystem.versionNeeded [ boolean ]; # 0..1 If definitions are not stable
  fhir:CodeSystem.content [ code ]; # 1..1 not-present | example | fragment | complete | supplement
  fhir:CodeSystem.supplements [ canonical(CodeSystem) ]; # 0..1 Canonical URL of Code System this adds designations and
    properties to
  fhir:CodeSystem.count [ unsignedInt ]; # 0..1 Total concepts in the code system
  fhir:CodeSystem.filter [ # 0..* Filter that can be used in a value set
    fhir:CodeSystem.filter.code [ code ]; # 1..1 Code that identifies the filter
    fhir:CodeSystem.filter.description [ string ]; # 0..1 How or why the filter is used
    fhir:CodeSystem.filter.operator [ code ], ... ; # 1..* = | is-a | descendent-of | is-not-a | regex | in | not-in |
    generalizes | exists
    fhir:CodeSystem.filter.value [ string ]; # 1..1 What to use for the value
  ], ...;
  fhir:CodeSystem.property [ # 0..* Additional information supplied about each concept
    fhir:CodeSystem.property.code [ code ]; # 1..1 Identifies the property on the concepts, and when referred to in
    operations
    fhir:CodeSystem.property.uri [ uri ]; # 0..1 Formal identifier for the property
    fhir:CodeSystem.property.description [ string ]; # 0..1 Why the property is defined, and/or what it conveys
```

```
    fhir:CodeSystem.property.type [ code ]; # 1..1 code | Coding | string | integer | boolean | dateTime | decimal
  ], ...;
  fhir:CodeSystem.concept [ # 0..* Concepts in the code system
    fhir:CodeSystem.concept.code [ code ]; # 1..1 Code that identifies concept
    fhir:CodeSystem.concept.display [ string ]; # 0..1 Text to display to the user
    fhir:CodeSystem.concept.definition [ string ]; # 0..1 Formal definition
    fhir:CodeSystem.concept.designation [ # 0..* Additional representations for the concept
      fhir:CodeSystem.concept.designation.language [ code ]; # 0..1 Human language of the designation
      fhir:CodeSystem.concept.designation.use [ Coding ]; # 0..1 Details how this designation would be used
      fhir:CodeSystem.concept.designation.value [ string ]; # 1..1 The text value for this designation
    ], ...;
    fhir:CodeSystem.concept.property [ # 0..* Property value for the concept
      fhir:CodeSystem.concept.property.code [ code ]; # 1..1 Reference to CodeSystem.property.code
      # CodeSystem.concept.property.value[x] : 1..1 Value of the property for this concept. One of these 7
        fhir:CodeSystem.concept.property.valueCode [ code ]
        fhir:CodeSystem.concept.property.valueCoding [ Coding ]
        fhir:CodeSystem.concept.property.valueString [ string ]
        fhir:CodeSystem.concept.property.valueInteger [ integer ]
        fhir:CodeSystem.concept.property.valueBoolean [ boolean ]
        fhir:CodeSystem.concept.property.valueDateTime [ dateTime ]
        fhir:CodeSystem.concept.property.valueDecimal [ decimal ]
    ], ...;
    fhir:CodeSystem.concept.concept [ See CodeSystem.concept ], ... ; # 0..* Child Concepts (is-a/contains/categorizes)
  ], ...;
]
```

Changes since R3

CodeSystem

CodeSystem.identifier | Max Cardinality changed from 1 to *

CodeSystem.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

CodeSystem.experimental | No longer marked as Modifier

CodeSystem.valueSet | Type changed from uri to canonical(ValueSet)

CodeSystem.hierarchyMeaning | Change value set from http://hl7.org/fhir/ValueSet/codesystem-hierarchy-meaning to http://hl7.org/fhir/ValueSet/codesystem-hierarchy-meaning|4.0.1

CodeSystem.content | Change value set from http://hl7.org/fhir/ValueSet/codesystem-content-mode to http://hl7.org/fhir/ValueSet/codesystem-content-mode|4.0.1

CodeSystem.supplements | Added Element

CodeSystem.filter.operator | Change value set from http://hl7.org/fhir/ValueSet/filter-operator to http://hl7.org/fhir/ValueSet/filter-operator|4.0.1

CodeSystem.property.type | Change value set from http://hl7.org/fhir/ValueSet/concept-property-type to http://hl7.org/fhir/ValueSet/concept-property-type|4.0.1

CodeSystem.concept.designation.language | Change binding strength from extensible to preferred

CodeSystem.concept.property.value[x] | Add Type decimal

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

CodeSystem | IN | | DomainResource | Declares the existence of and describes a code system or code system supplement + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: Within a code system definition, all the codes SHALL be unique Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | Σ | 0..1 | uri | Canonical identifier for this code system, represented as a URI (globally unique) (Coding.system)

identifier | Σ | 0..* | Identifier | Additional identifier for the code system (business identifier)

version | Σ | 0..1 | string | Business version of the code system (Coding.version)

name | ΣI | 0..1 | string | Name for this code system (computer friendly)

title | Σ | 0..1 | string | Name for this code system (human friendly)

status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | Σ | 0..1 | boolean | For testing purposes, not real usage

date | Σ | 0..1 | dateTime | Date last changed

publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)

contact | Σ | 0..* | ContactDetail | Contact details for the publisher

description | | 0..1 | markdown | Natural language description of the code system

useContext | ΣTU | 0..* | UsageContext | The context that the content is intended to support

jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for code system (if applicable) Jurisdiction (Extensible)

purpose | | 0..1 | markdown | Why this code system is defined

copyright | | 0..1 | markdown | Use and/or publishing restrictions

caseSensitive | Σ | 0..1 | boolean | If code comparison is case sensitive

valueSet | Σ | 0..1 | canonical(ValueSet) | Canonical reference to the value set with entire code system

hierarchyMeaning | Σ | 0..1 | code | grouped-by | is-a | part-of | classified-with CodeSystemHierarchyMeaning (Required)

compositional | Σ | 0..1 | boolean | If code system defines a compositional grammar

versionNeeded | Σ | 0..1 | boolean | If definitions are not stable

content | Σ | 1..1 | code | not-present | example | fragment | complete | supplement CodeSystemContentMode (Required)

supplements | Σ | 0..1 | canonical(CodeSystem) | Canonical URL of Code System this adds designations and properties to

count | Σ | 0..1 | unsignedInt | Total concepts in the code system
filter | Σ | 0..* | BackboneElement | Filter that can be used in a value set
code | Σ | 1..1 | code | Code that identifies the filter
description | Σ | 0..1 | string | How or why the filter is used
operator | Σ | 1..* | code | = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists FilterOperator (Required)
value | Σ | 1..1 | string | What to use for the value
property | Σ | 0..* | BackboneElement | Additional information supplied about each concept
code | Σ | 1..1 | code | Identifies the property on the concepts, and when referred to in operations
uri | Σ | 0..1 | uri | Formal identifier for the property
description | Σ | 0..1 | string | Why the property is defined, and/or what it conveys
type | Σ | 1..1 | code | code | Coding | string | integer | boolean | dateTime | decimal PropertyType (Required)
concept | | 0..* | BackboneElement | Concepts in the code system
code | | 1..1 | code | Code that identifies concept
display | | 0..1 | string | Text to display to the user
definition | | 0..1 | string | Formal definition
designation | | 0..* | BackboneElement | Additional representations for the concept
language | | 0..1 | code | Human language of the designation Common Languages (Preferred but limited to AllLanguages)
use | | 0..1 | Coding | Details how this designation would be used Designation Use (Extensible)
value | | 1..1 | string | The text value for this designation
property | | 0..* | BackboneElement | Property value for the concept
code | | 1..1 | code | Reference to CodeSystem.property.code
value[x] | | 1..1 | | Value of the property for this concept
valueCode | | | code
valueCoding | | | Coding
valueString | | | string
valueInteger | | | integer
valueBoolean | | | boolean
valueDateTime | | | dateTime
valueDecimal | | | decimal
concept | | 0..* | see concept | Child Concepts (is-a/contains/categorizes)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CodeSystem xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this code system, represented as a URI (globally unique)
    (Coding.system) -->
 <identifier><!-- 0..* Identifier Additional identifier for the code system (business identifier) --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the code system (Coding.version) -->
 <name value="[string]"/><!--  0..1 Name for this code system (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this code system (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the code system -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for code system (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this code system is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <caseSensitive value="[boolean]"/><!-- 0..1 If code comparison is case sensitive -->
 <valueSet><!-- 0..1 canonical(ValueSet) Canonical reference to the value set with entire code system --></valueSet>
 <hierarchyMeaning value="[code]"/><!-- 0..1 grouped-by | is-a | part-of | classified-with -->
 <compositional value="[boolean]"/><!-- 0..1 If code system defines a compositional grammar -->
 <versionNeeded value="[boolean]"/><!-- 0..1 If definitions are not stable -->
 <content value="[code]"/><!-- 1..1 not-present | example | fragment | complete | supplement -->
 <supplements><!-- 0..1 canonical(CodeSystem) Canonical URL of Code System this adds designations and properties to
    --></supplements>
 <count value="[unsignedInt]"/><!-- 0..1 Total concepts in the code system -->
 <filter>  <!-- 0..* Filter that can be used in a value set -->
  <code value="[code]"/><!-- 1..1 Code that identifies the filter -->
  <description value="[string]"/><!-- 0..1 How or why the filter is used -->
  <operator value="[code]"/><!-- 1..* = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists
    -->
  <value value="[string]"/><!-- 1..1 What to use for the value -->
 </filter>
 <property>  <!-- 0..* Additional information supplied about each concept -->
  <code value="[code]"/><!-- 1..1 Identifies the property on the concepts, and when referred to in operations -->
  <uri value="[uri]"/><!-- 0..1 Formal identifier for the property -->
  <description value="[string]"/><!-- 0..1 Why the property is defined, and/or what it conveys -->
  <type value="[code]"/><!-- 1..1 code | Coding | string | integer | boolean | dateTime | decimal -->
 </property>
 <concept>  <!-- 0..* Concepts in the code system -->
  <code value="[code]"/><!-- 1..1 Code that identifies concept -->
  <display value="[string]"/><!-- 0..1 Text to display to the user -->
  <definition value="[string]"/><!-- 0..1 Formal definition -->
  <designation>  <!-- 0..* Additional representations for the concept -->
   <language value="[code]"/><!-- 0..1 Human language of the designation -->
```

```
   <use><!-- 0..1 Coding Details how this designation would be used --></use>
   <value value="[string]"/><!-- 1..1 The text value for this designation -->
  </designation>
  <property>  <!-- 0..* Property value for the concept -->
   <code value="[code]"/><!-- 1..1 Reference to CodeSystem.property.code -->
   <value[x]><!-- 1..1 code|Coding|string|integer|boolean|dateTime|decimal Value of the property for this concept
    --></value[x]>
  </property>
  <concept><!-- 0..* Content as for CodeSystem.concept Child Concepts (is-a/contains/categorizes) --></concept>
 </concept>
</CodeSystem>
```

JSON Template

```
{
  "resourceType" : "CodeSystem",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this code system, represented as a URI (globally unique) (Coding.system)
  "identifier" : [{ Identifier }], // Additional identifier for the code system (business identifier)
  "version" : "<string>", // Business version of the code system (Coding.version)
  "name" : "<string>", // C? Name for this code system (computer friendly)
  "title" : "<string>", // Name for this code system (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the code system
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for code system (if applicable)
  "purpose" : "<markdown>", // Why this code system is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "caseSensitive" : <boolean>, // If code comparison is case sensitive
  "valueSet" : { canonical(ValueSet) }, // Canonical reference to the value set with entire code system
  "hierarchyMeaning" : "<code>", // grouped-by | is-a | part-of | classified-with
  "compositional" : <boolean>, // If code system defines a compositional grammar
  "versionNeeded" : <boolean>, // If definitions are not stable
  "content" : "<code>", // R!  not-present | example | fragment | complete | supplement
  "supplements" : { canonical(CodeSystem) }, // Canonical URL of Code System this adds designations and properties to
  "count" : "<unsignedInt>", // Total concepts in the code system
  "filter" : [{ // Filter that can be used in a value set
    "code" : "<code>", // R!  Code that identifies the filter
    "description" : "<string>", // How or why the filter is used
    "operator" : ["<code>"], // R!  = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists
    "value" : "<string>" // R!  What to use for the value
  }],
  "property" : [{ // Additional information supplied about each concept
    "code" : "<code>", // R!  Identifies the property on the concepts, and when referred to in operations
    "uri" : "<uri>", // Formal identifier for the property
    "description" : "<string>", // Why the property is defined, and/or what it conveys
    "type" : "<code>" // R!  code | Coding | string | integer | boolean | dateTime | decimal
  }],
  "concept" : [{ // Concepts in the code system
    "code" : "<code>", // R!  Code that identifies concept
    "display" : "<string>", // Text to display to the user
    "definition" : "<string>", // Formal definition
    "designation" : [{ // Additional representations for the concept
      "language" : "<code>", // Human language of the designation
      "use" : { Coding }, // Details how this designation would be used
      "value" : "<string>" // R!  The text value for this designation
    }],
    "property" : [{ // Property value for the concept
      "code" : "<code>", // R!  Reference to CodeSystem.property.code
      // value[x]: Value of the property for this concept. One of these 7:
      "valueCode" : "<code>"
      "valueCoding" : { Coding }
      "valueString" : "<string>"
      "valueInteger" : <integer>
      "valueBoolean" : <boolean>
      "valueDateTime" : "<dateTime>"
      "valueDecimal" : <decimal>
    }],
    "concept" : [{ Content as for CodeSystem.concept }] // Child Concepts (is-a/contains/categorizes)
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CodeSystem;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CodeSystem.url [ uri ]; # 0..1 Canonical identifier for this code system, represented as a URI (globally unique)
    (Coding.system)
  fhir:CodeSystem.identifier [ Identifier ], ... ; # 0..* Additional identifier for the code system (business
    identifier)
```

```
  fhir:CodeSystem.version [ string ]; # 0..1 Business version of the code system (Coding.version)
  fhir:CodeSystem.name [ string ]; # 0..1 Name for this code system (computer friendly)
  fhir:CodeSystem.title [ string ]; # 0..1 Name for this code system (human friendly)
  fhir:CodeSystem.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:CodeSystem.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:CodeSystem.date [ dateTime ]; # 0..1 Date last changed
  fhir:CodeSystem.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:CodeSystem.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:CodeSystem.description [ markdown ]; # 0..1 Natural language description of the code system
  fhir:CodeSystem.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:CodeSystem.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for code system (if applicable)
  fhir:CodeSystem.purpose [ markdown ]; # 0..1 Why this code system is defined
  fhir:CodeSystem.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:CodeSystem.caseSensitive [ boolean ]; # 0..1 If code comparison is case sensitive
  fhir:CodeSystem.valueSet [ canonical(ValueSet) ]; # 0..1 Canonical reference to the value set with entire code system
  fhir:CodeSystem.hierarchyMeaning [ code ]; # 0..1 grouped-by | is-a | part-of | classified-with
  fhir:CodeSystem.compositional [ boolean ]; # 0..1 If code system defines a compositional grammar
  fhir:CodeSystem.versionNeeded [ boolean ]; # 0..1 If definitions are not stable
  fhir:CodeSystem.content [ code ]; # 1..1 not-present | example | fragment | complete | supplement
  fhir:CodeSystem.supplements [ canonical(CodeSystem) ]; # 0..1 Canonical URL of Code System this adds designations and
    properties to
  fhir:CodeSystem.count [ unsignedInt ]; # 0..1 Total concepts in the code system
  fhir:CodeSystem.filter [ # 0..* Filter that can be used in a value set
    fhir:CodeSystem.filter.code [ code ]; # 1..1 Code that identifies the filter
    fhir:CodeSystem.filter.description [ string ]; # 0..1 How or why the filter is used
    fhir:CodeSystem.filter.operator [ code ], ... ; # 1..* = | is-a | descendent-of | is-not-a | regex | in | not-in |
    generalizes | exists
    fhir:CodeSystem.filter.value [ string ]; # 1..1 What to use for the value
  ], ...;
  fhir:CodeSystem.property [ # 0..* Additional information supplied about each concept
    fhir:CodeSystem.property.code [ code ]; # 1..1 Identifies the property on the concepts, and when referred to in
    operations
    fhir:CodeSystem.property.uri [ uri ]; # 0..1 Formal identifier for the property
    fhir:CodeSystem.property.description [ string ]; # 0..1 Why the property is defined, and/or what it conveys
    fhir:CodeSystem.property.type [ code ]; # 1..1 code | Coding | string | integer | boolean | dateTime | decimal
  ], ...;
  fhir:CodeSystem.concept [ # 0..* Concepts in the code system
    fhir:CodeSystem.concept.code [ code ]; # 1..1 Code that identifies concept
    fhir:CodeSystem.concept.display [ string ]; # 0..1 Text to display to the user
    fhir:CodeSystem.concept.definition [ string ]; # 0..1 Formal definition
    fhir:CodeSystem.concept.designation [ # 0..* Additional representations for the concept
      fhir:CodeSystem.concept.designation.language [ code ]; # 0..1 Human language of the designation
      fhir:CodeSystem.concept.designation.use [ Coding ]; # 0..1 Details how this designation would be used
      fhir:CodeSystem.concept.designation.value [ string ]; # 1..1 The text value for this designation
    ], ...;
    fhir:CodeSystem.concept.property [ # 0..* Property value for the concept
      fhir:CodeSystem.concept.property.code [ code ]; # 1..1 Reference to CodeSystem.property.code
      # CodeSystem.concept.property.value[x] : 1..1 Value of the property for this concept. One of these 7
        fhir:CodeSystem.concept.property.valueCode [ code ]
        fhir:CodeSystem.concept.property.valueCoding [ Coding ]
        fhir:CodeSystem.concept.property.valueString [ string ]
        fhir:CodeSystem.concept.property.valueInteger [ integer ]
        fhir:CodeSystem.concept.property.valueBoolean [ boolean ]
        fhir:CodeSystem.concept.property.valueDateTime [ dateTime ]
        fhir:CodeSystem.concept.property.valueDecimal [ decimal ]
    ], ...;
    fhir:CodeSystem.concept.concept [ See CodeSystem.concept ], ... ; # 0..* Child Concepts (is-a/contains/categorizes)
  ], ...;
]
```

Changes since Release 3

CodeSystem
CodeSystem.identifier | Max Cardinality changed from 1 to *
CodeSystem.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
CodeSystem.experimental | No longer marked as Modifier
CodeSystem.valueSet | Type changed from uri to canonical(ValueSet)
CodeSystem.hierarchyMeaning | Change value set from http://hl7.org/fhir/ValueSet/codesystem-hierarchy-meaning to http://hl7.org/fhir/ValueSet/codesystem-hierarchy-meaning|4.0.1
CodeSystem.content | Change value set from http://hl7.org/fhir/ValueSet/codesystem-content-mode to http://hl7.org/fhir/ValueSet/codesystem-content-mode|4.0.1
CodeSystem.supplements | Added Element
CodeSystem.filter.operator | Change value set from http://hl7.org/fhir/ValueSet/filter-operator to http://hl7.org/fhir/ValueSet/filter-operator|4.0.1
CodeSystem.property.type | Change value set from http://hl7.org/fhir/ValueSet/concept-property-type to http://hl7.org/fhir/ValueSet/concept-property-type|4.0.1
CodeSystem.concept.designation.language | Change binding strength from extensible to preferred
CodeSystem.concept.property.value[x] | Add Type decimal

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

4.8.4.1 Terminology Bindings

Path | Definition | Type | Reference
CodeSystem.status | The lifecycle status of an artifact. | Required | PublicationStatus
CodeSystem.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
CodeSystem.hierarchyMeaning | The meaning of the hierarchy of concepts in a code system. | Required | CodeSystemHierarchyMeaning
CodeSystem.content | The extent of the content of the code system (the concepts and codes it defines) are represented in a code system resource. | Required | CodeSystemContentMode
CodeSystem.filter.operator | The kind of operation to perform as a part of a property based filter. | Required | FilterOperator
CodeSystem.property.type | The type of a property value. | Required | PropertyType
CodeSystem.concept.designation.language | A human language. | Preferred, but limited to AllLanguages | CommonLanguages
CodeSystem.concept.designation.use | Details of how a designation would be used. | Extensible | DesignationUse

4.8.4.2 Constraints

id | Level | Location | Description | Expression
csd-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
csd-1 | Rule | (base) | Within a code system definition, all the codes SHALL be unique | concept.code.combine($this.descendants().concept.code).isDistinct()

4.8.5 Versioning Code Systems

Most code systems evolve over time, due to corrections, clarifications, and changes to approach or underlying knowledge or reality. If these changes lead to the meanings of existing codes changing significantly, then the interpretation of the code system becomes version dependent. This significantly complicates implementation based on the code system, to the point where it is not clear that safety can be assured, so changing the meaning of an existing code SHOULD be avoided whenever possible. It is preferable to assign a new identifier to a code system when any concepts in it have a significant change in meaning (for example, the German diagnostic classification code system ICD10GM2009 has a different system to ICD10GM2008), but this also can have substantial impact on implementation, so is often not practical - for instance, SNOMED CT has a complex version release framework, which may lead to variations in meaning of concepts, but there is only one identifier for SNOMED CT.

For this reason, a code system MAY provide a version identifier which can be specified in CodeSystem.version. The version specific identifier SHOULD be provided whenever there are potentially significant changes in meaning across multiple releases of a code system. There is no particular format requirement for the version identifier, though HL7 recommends a date-based approach.

When the CodeSystem.versionNeeded is 'true', then the version identifier SHALL be used in Coding instances that refer to the code system.

Where the terminology does not clearly define what string should be used to identify code system versions, the recommendation is to use as the version string the date (expressed in FHIR date format) on which the version of the code system that is being used was officially published.

4.8.6 Multi-part Code Systems

The simple case for a code system is that the entire code system - all the concepts and their codes, designations and properties are distributed in a single CodeSystem resource. However, for a variety of reasons, code systems may be distributed by the code system authority in a set of fragments, and other authorities may issue additional designations and properties in supplements.

4.8.7 Code System Fragments

If the CodeSystem.content value is fragment, then the resource describes part of a code system.

Code Systems may be distributed in fragments for the following reasons:

• Different IP distribution rules for different parts of the code system

• Special purpose modules for specific purposes

• Distribution of proposed content for evaluation

The following rules apply to code system fragments:

• All fragments SHALL have the same CodeSystem.url

• Fragments can only be published by the code system authority, or according to a process defined by the authority, if they have defined one

• Fragments cannot contain any codes, concepts or properties that would not be found in a complete representation of the code system, if one exists

• Publishing a code system in multiple fragments can create confusion for terminology servers and terminology service consumers. Code System authorities that publish fragments should be careful and communicate their intent clearly

4.8.8 Code System Supplements

If the CodeSystem.content value is supplement, then the resource describes a code system supplement. The following rules apply to code system supplements:

• The CodeSystem.supplements element SHALL have a value, which is the URL of the code system being supplemented

• The CodeSystem.url for a supplement SHALL never appear in a Coding.system

• The CodeSystem.url for a supplement must be under the control of the authority creating or publishing the supplement (e.g. not in the same space as the code system being supplemented, unless the supplement is being issued by the same authority as the original code system

• A codesystem supplement cannot define any new CodeSystem.concept.code. i.e.: all CodeSystem.concept.code in the supplement must be a code from the "supplemented" code system

If a supplement needs to define new concepts/codes to use as property values, it can be paired with a new (possibly contained) Code System and use the Coding type for the property values.

The impact of Code System supplements on value set expansion - and therefore value set validation - is subject to ongoing experimentation and implementation testing, and further clarification and additional rules might be proposed in future versions of this specification.

4.8.9 Display, Definition and Designations

Concepts have both a display and a definition. The display is a short text that represents the meaning of the concept to human users, while the definition is a more formal statement of the meaning of the concept, which is often longer. All concepts SHOULD have a display and a definition, though it is not mandatory because there are many cases where no such properties are provided, in spite of their utility and importance for clear and safe communication.

In addition to the display and definition, a concept can have one or more designation elements. The display is equivalent to a special designation with an implied designation.use meaning "primary code/designation" and a language equal to the Resource Language. The designations can provide additional displays for other language, as well as designations for other purposes. When using concepts, applications use the display and designation unless the language or usage in context provides a reason to use one of the designations.

4.8.10 Properties

Each code system can define one or more concept properties. Each concept defined by the code system may have one or more values for each concept property defined by the code system. Typical uses for properties include:

• Tracking administrative status (inactive, deprecation date)

• Providing additional statements about the meaning of the concept

• Defining structured relationships with other concepts in the code system

• Assigning scoring values to the concepts

Properties are identified by their master URI (CodeSystem.property.uri), and then, by their code (CodeSystem.property.code), which is used both internally within the code system resource (CodeSystem.concept.property.code) and also externally, in the following places:

• ConceptMap: ConceptMap.element.target.dependsOn.property and ConceptMap.element.target.product.property

• ValueSet: ValueSet.compose.include.filter.property can refer to any defined code system property

• $lookup operation: In Parameters.parameter.name when returning information about a code

• $translate operation: In Parameters.parameter.part.name for dependencies and products

• $find-matches operation: In Parameters.parameter.name when providing codes, and in Parameters.parameter.name when asking for codes

Properties are defined using the following elements:

Name	Details	Description
code	code	Used to identify the property, in the places shown in the list immediately above this table
uri	optional uri	Reference to the formal meaning of the property. One possible source of meaning is the Concept Properties code system. This part of the definition is optional, but is recommended to provide an additional level of definitional consistency
description	optional string	A description of the property- why it is defined, and how its value might be used
type	code	Coding	string	integer	boolean	dateTime	The type of the property value. Properties of type "code" contain a code defined by the code system (e.g. a reference to another defined concept)

Note that properties provide a common view of concept relationships that is common across all code systems. Some code systems define properties with more sophistication, such as groups of properties, or subsumption

relationships between properties (e.g. SNOMED CT). Servers providing support for these code systems will need to know full details about the underlying relationships in order to provide the correct information about concepts and their properties, but this information does not surface in the resources or operations defined by the FHIR specification.

4.8.11 Concept Status

Many Code Systems have a 'status' associated with the concept. This may categorize the concept as:

- Experimental - provided for trial, but may be removed in the future
- Active - in normal use
- Deprecated - planned to be removed from use
- Retired - still present for historical reasons, but no longer allowed to be used

There is wide variation in the life cycles supported by the different code systems, the words they use to describe the various status values they use, and some code systems have additional status values. HL7 uses Active and Retired. In addition to these status codes, concepts may also be labeled as "Abstract' (not to be used in some circumstances), and have dates associated with their retirement or deprecation. All this information is represented as properties of the concepts. In order to assist with consistency between code systems, the following basic property URIs are defined:

http://hl7.org/fhir/concept-properties#status : code | A property that indicates the status of the concept. If the property is identified by this URL, then it SHALL use at least these status values (where appropriate): active - the concept is for normal use experimental - provided for trial, but may be removed in the future deprecated - planned to be removed from use retired - still present for historical reasons, but no longer allowed to be used The HL7 defined code systems (whether defined by the FHIR project, the V3 framework, or the V2 standard) all use this property to define their status. HL7 uses "active" (and considers this the default status), and deprecated and retired (= inactive)

http://hl7.org/fhir/concept-properties#retirementDate : date | Date Concept was retired

http://hl7.org/fhir/concept-properties#deprecationDate : date | Date Concept was deprecated

http://hl7.org/fhir/concept-properties#parent : code | An immediate parent of the concept in the hierarchy

http://hl7.org/fhir/concept-properties#child : code | An immediate child of the concept in the hierarchy

http://hl7.org/fhir/concept-properties#notSelectable : boolean | This concept is a grouping concept and not intended to be used in the normal use of the code system (though may be used for filters etc.). This is also known as 'Abstract'

The parent and child properties are used when performing concept lookup (using the $lookup operation), and when using properties to indicate parent/child relationships (see below).

4.8.12 Concept Hierarchy

Code systems may be presented hierarchically, using nested concept elements, where the hierarchy has a defined meaning specified in hierarchyMeaning. The code system hierarchy is a single tree, where concepts have only one parent.

Some code systems define concepts that have multiple parents. If a code system has concepts like this (typically, concepts that are subsumed by more than one other concept), the code system SHOULD NOT be represented using the hierarchy structure in the Code System resource, and the relationships between concepts should be defined using properties. If the hierarchy is used, implementers SHALL use a property to indicate additional parents that are not represented in the structural (nested concepts) hierarchy.

Operations based on the codeSystem resource SHALL have the same result whether or not the relationships are represented explicitly as properties or implicitly using the CodeSystem resource hierarchy.

4.8.13 Subsumption Testing

The words 'subsume', 'subsumes', 'subsumed' and 'subsumption' are defined in relation to the type of hierarchy (i.e. the value of CodeSystem.hierarchyMeaning) identified for the code system that is being represented by the CodeSystem resource. Concept A is considered to be subsumed by Concept B if it comes under Concept B in the hierarchy, or if a parent/child relationship is declared in the properties, and the hierarchyMeaning is "is-a").

Where a CodeSystem does not declare its hierarchy meaning directly, then the code system documentation must be consulted manually to determine how subsumption is determined. If there is no definition, none of the subsumption based features can be used with the code system.

Subsumption based logic arises explicitly or implicitly in the following places in the FHIR specification:

- CodeSystem $subsumes operation
- CodeSystem $lookup operation
- ConceptMap $closure operation
- Search by subsumption
- ValueSet $expand operation
- ValueSet $validate-code operation

4.8.14 Filters

The following filters are defined for all code systems:

Property Name | Operation | Value | Definition | Notes
[property] | = | [string] | Includes all codes that have a property value equal to the specified string, where [property] is the code for any defined property
[property] | in | [string,string...] | Includes all codes that have a property value equal to one of the specified strings, where [property] is the code for any defined property | The values cannot include ",", since it is being used as a delimiter

This draft section about the relationships between rich terminologies and structured content is still undergoing review.

4.8.15 Implicit Code Systems

Some other parts of the FHIR infrastructure define set of concepts that may also be treated as code systems. This is most useful when mapping between systems using Concept Maps, but might also be useful for other code system related functionality (e.g. subsetting use Value Sets). This table summarizes how to treat these items as a terminology:

StructureDefinition | The StructureDefinition.url (canonical URL) is the system. Each .snapshot.element.id in the snapshot is a code in the code system
Questionnaire | The Questionnaire.url (canonical URL) is the system. Each .item.linkId in the snapshot is a code in the code system. Items with no linkId cannot be addressed
Medication | Medication resources are a bit different, since they don't have a canonical URL, and there are not multiple items in a resource. So to refer to a medication resource, the system is [base]/Medication, where base is the server address. The Logical Id of the resource is the code

The use of these implicit code systems is not yet well tested, so this section remains informative.

For medications, the relationship between the code system and the medication resource is complex. Many medication related code systems (e.g. RxNorm, DM+D, AMT, etc.) represent more complex information than just code, display name and code system. These often contain information about the content of the medication A similar principle applies to observation definitions, which overlap significantly with LOINC and other related code systems, which contain a set of observation definitions, that contain methods, units, etc. Other similar arrangements exist for Location and many other kinds of resource.

In FHIR, these are handled by splitting the concept into two distinct parts - the Terminology, (Code System & ValueSet resources) is used to manage the codes, display names and relationships. A separate "detail" resource (e.g., Medication resource instances for RxNorm code details, ObservationDefinition instances for LOINC code details, or Location, etc.) is used to convey detailed information (dose form & strength, allowed data type or permitted values, address & hours of operation, etc.). One "detail" resource instance is created for each code.

This division accomplishes several things:

• It allows generic systems that support terminology management to perform standard terminology operations on code systems dealing with complex structures - code lookup, validation, subsumption testing, mapping and translation.

• It allows information to be exchanged about individual medications, data elements and locations. Codes can't be retrieved individually in FHIR - it is necessary to retrieve the entire resource. By packaging the detailed information in separate resources, independent retrieval and update is possible.

• It supports use-cases for sharing medication, location, observation type and similar information in circumstances where the code may be unknown, unavailable or occasionally non-existent (e.g., custom compounds, non-registered locations). Having a distinct resource supports these capabilities, which would not be possible using CodeSystem/ValueSet.

Note that this division in FHIR does not imply that a similar division is required in the internal representation used by systems exposing a FHIR interface. Similarly, some systems may choose to only expose or maintain one aspect of such information types (i.e. only the discrete resource instances or only the value set).

The linkage between the "detail" resource and the Terminology resources is accomplished via the code element (or equivalent) on the detail resource. As well, the "name" or "title" on the detail resource generally corresponds with the display name on the matching code. Most detail resources will also have an "identifier" element. This can be set to the same value and namespace as the code, but if the only identifier a resource has is its defining code, it may be better to omit the identifier entirely. .

4.8.16 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
code TU | token | A code defined in the code system | CodeSystem.concept.code
content-mode TU | token | not-present | example | fragment | complete | supplement | CodeSystem.content
context TU | token | A use context assigned to the code system | (CodeSystem.useContext.value as CodeableConcept)
context-quantity TU | quantity | A quantity- or range-valued use context assigned to the code system | (CodeSystem.useContext.value as Quantity) | (CodeSystem.useContext.value as Range)
context-type TU | token | A type of use context assigned to the code system | CodeSystem.useContext.code

context-type-quantity TU | composite | A use context type and quantity- or range-based value assigned to the code system | On CodeSystem.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value TU | composite | A use context type and value assigned to the code system | On CodeSystem.useContext: context-type: code context: value.as(CodeableConcept)
date TU | date | The code system publication date | CodeSystem.date
description TU | string | The description of the code system | CodeSystem.description
identifier TU | token | External identifier for the code system | CodeSystem.identifier
jurisdiction TU | token | Intended jurisdiction for the code system | CodeSystem.jurisdiction
language TU | token | A language in which a designation is provided | CodeSystem.concept.designation.language
name TU | string | Computationally friendly name of the code system | CodeSystem.name
publisher TU | string | Name of the publisher of the code system | CodeSystem.publisher
status TU | token | The current status of the code system | CodeSystem.status
supplements TU | reference | Find code system supplements for the referenced code system | CodeSystem.supplements (CodeSystem)
system TU | uri | The system for any codes defined by this code system (same as 'url') | CodeSystem.url
title TU | string | The human-friendly name of the code system | CodeSystem.title
url TU | uri | The uri that identifies the code system | CodeSystem.url
version TU | token | The business version of the code system | CodeSystem.version

4.8.17 Resource CodeSystem - Examples

URL: https://hl7.org/fhir/R4/codesystem-examples.html

4.8.17 Resource CodeSystem - Examples

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Simple Code System | example | XML | JSON | Turtle
NHIN Purpose of Use | nhin-purposeofuse | XML | JSON | Turtle
Canonical Definition for SNOMED CT | snomedct | XML | JSON | Turtle
Code system summary | summary | XML | JSON | Turtle
Example Code System Supplement | example-supplement | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

4.8.18 Resource CodeSystem - Detailed Descriptions

URL: https://hl7.org/fhir/R4/codesystem-definitions.html

4.8.18 Resource CodeSystem - Detailed Descriptions

Vocabulary Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the CodeSystem resource.

CodeSystem
Element Id | CodeSystem
Definition | The CodeSystem resource is used to declare the existence of and describe a code system or code system supplement and its key properties, and optionally define a part or all of its content.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element csd-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}') csd-1 Rule Within a code system definition, all the codes SHALL be unique concept.code.combine($this.descendants().concept.code).isDistinct()
Defined on this element
csd-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
csd-1 | Rule | Within a code system definition, all the codes SHALL be unique | concept.code.combine($this.descendants().concept.code).isDistinct()
CodeSystem.url
Element Id | CodeSystem.url
Definition | An absolute URI that is used to identify this code system when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this code system is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the code system is stored on different servers. This is used in Coding.system.
Cardinality | 0..1
Type | uri
Requirements | Allows the code system to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the

meta.source element to indicate where the current master source of the resource can be found.
CodeSystem.identifier
Element Id | CodeSystem.identifier
Definition | A formal identifier that is used to identify this code system when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this code system outside of FHIR, where it is not possible to use the logical URI. Note that HL7 defines at least three identifiers for many of its code systems - the FHIR canonical URL, the OID and the V2 Table 0396 mnemonic code.
CodeSystem.version
Element Id | CodeSystem.version
Definition | The identifier that is used to identify this version of the code system when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the code system author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. This is used in Coding.version.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different code system instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the code system with the format [url]|[version].
CodeSystem.name
Element Id | CodeSystem.name
Definition | A natural language name identifying the code system. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
CodeSystem.title
Element Id | CodeSystem.title
Definition | A short, descriptive, user-friendly title for the code system.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
CodeSystem.status
Element Id | CodeSystem.status
Definition | The date (and optionally time) when the code system resource was created or revised.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of code systems that are appropriate for use versus not.
CodeSystem.experimental
Element Id | CodeSystem.experimental
Definition | A Boolean value to indicate that this code system is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level code system.
Summary | true
Comments | Allows filtering of code systems that are appropriate for use versus not.
CodeSystem.date
Element Id | CodeSystem.date
Definition | The date (and optionally time) when the code system was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the code system changes.
Cardinality | 0..1

Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the code system. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
CodeSystem.publisher
Element Id | CodeSystem.publisher
Definition | The name of the organization or individual that published the code system.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the code system. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the code system is the organization or individual primarily responsible for the maintenance and upkeep of the code system. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the code system. This item SHOULD be populated unless the information is available from context.
CodeSystem.contact
Element Id | CodeSystem.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
CodeSystem.description
Element Id | CodeSystem.description
Definition | A free text natural language description of the code system from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Alternate Names | Scope; Purpose
Comments | This description can be used to capture details such as why the code system was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the code system as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the code system is presumed to be the predominant language in the place the code system was created).
CodeSystem.useContext
Element Id | CodeSystem.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate code system instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
CodeSystem.jurisdiction
Element Id | CodeSystem.jurisdiction
Definition | A legal or geographic region in which the code system is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the code system to be used in jurisdictions other than those for which it was originally designed or intended.
CodeSystem.purpose
Element Id | CodeSystem.purpose
Definition | Explanation of why this code system is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the code system. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this code system.
CodeSystem.copyright
Element Id | CodeSystem.copyright
Definition | A copyright statement relating to the code system and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the code system.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the code system and/or its content.
Alternate Names | License; Restrictions
Comments | ... Sometimes, the copyright differs between the code system and the codes that are included. The copyright statement should clearly differentiate between these when required.
CodeSystem.caseSensitive
Element Id | CodeSystem.caseSensitive

Definition | If code comparison is case sensitive when codes within this system are compared to each other.
Cardinality | 0..1
Type | boolean
Summary | true
Comments | If this value is missing, then it is not specified whether a code system is case sensitive or not. When the rule is not known, Postel's law should be followed: produce codes with the correct case, and accept codes in any case. This element is primarily provided to support validation software.
CodeSystem.valueSet
Element Id | CodeSystem.valueSet
Definition | Canonical reference to the value set that contains the entire code system.
Cardinality | 0..1
Type | canonical(ValueSet)
Summary | true
Comments | The definition of the value set SHALL include all codes from this code system and only codes from this code system, and it SHALL be immutable.
CodeSystem.hierarchyMeaning
Element Id | CodeSystem.hierarchyMeaning
Definition | The meaning of the hierarchy of concepts as represented in this resource.
Cardinality | 0..1
Terminology Binding | CodeSystemHierarchyMeaning (Required)
Type | code
Summary | true
Comments | Note that other representations might have a different hierarchy or none at all, and represent the information using properties.
CodeSystem.compositional
Element Id | CodeSystem.compositional
Definition | The code system defines a compositional (post-coordination) grammar.
Cardinality | 0..1
Type | boolean
Alternate Names | grammar
Summary | true
Comments | Note that the code system resource does not define what the compositional grammar is, only whether or not there is one.
CodeSystem.versionNeeded
Element Id | CodeSystem.versionNeeded
Definition | This flag is used to signify that the code system does not commit to concept permanence across versions. If true, a version must be specified when referencing this code system.
Cardinality | 0..1
Type | boolean
Summary | true
Comments | Best practice is that code systems do not redefine concepts, or that if concepts are redefined, a new code system definition is created. But this is not always possible, so some code systems may be defined as 'versionNeeded'. Most code systems occasionally refine the displays defined for concepts between versions. Contexts in which the concept display values are validated may require that the version be specified for some code systems irrespective of the value of this property.
CodeSystem.content
Element Id | CodeSystem.content
Definition | The extent of the content of the code system (the concepts and codes it defines) are represented in this resource instance.
Cardinality | 1..1
Terminology Binding | CodeSystemContentMode (Required)
Type | code
Summary | true
CodeSystem.supplements
Element Id | CodeSystem.supplements
Definition | The canonical URL of the code system that this code system supplement is adding designations and properties to.
Cardinality | 0..1
Type | canonical(CodeSystem)
Summary | true
Comments | The most common use of a code system supplement is to add additional language support.
CodeSystem.count
Element Id | CodeSystem.count
Definition | The total number of concepts defined by the code system. Where the code system has a compositional grammar, the basis of this count is defined by the system steward.
Cardinality | 0..1
Type | unsignedInt
Summary | true
Comments | The count of concepts defined in this resource cannot be more than this value but may be less for several reasons - see the content element.
CodeSystem.filter
Element Id | CodeSystem.filter
Definition | A filter that can be used in a value set compose statement when selecting concepts using a filter.
Cardinality | 0..*
Summary | true

Comments | Note that filters defined in code systems usually require custom code on the part of any terminology engine that will make them available for use in value set filters. For this reason, they are generally only seen in high value published terminologies.
CodeSystem.filter.code
Element Id | CodeSystem.filter.code
Definition | The code that identifies this filter when it is used as a filter in ValueSet.compose.include.filter.
Cardinality | 1..1
Type | code
Summary | true
CodeSystem.filter.description
Element Id | CodeSystem.filter.description
Definition | A description of how or why the filter is used.
Cardinality | 0..1
Type | string
Summary | true
CodeSystem.filter.operator
Element Id | CodeSystem.filter.operator
Definition | A list of operators that can be used with the filter.
Cardinality | 1..*
Terminology Binding | FilterOperator (Required)
Type | code
Summary | true
CodeSystem.filter.value
Element Id | CodeSystem.filter.value
Definition | A description of what the value for the filter should be.
Cardinality | 1..1
Type | string
Summary | true
CodeSystem.property
Element Id | CodeSystem.property
Definition | A property defines an additional slot through which additional information can be provided about a concept.
Cardinality | 0..*
Summary | true
CodeSystem.property.code
Element Id | CodeSystem.property.code
Definition | A code that is used to identify the property. The code is used internally (in CodeSystem.concept.property.code) and also externally, such as in property filters.
Cardinality | 1..1
Type | code
Summary | true
CodeSystem.property.uri
Element Id | CodeSystem.property.uri
Definition | Reference to the formal meaning of the property. One possible source of meaning is the Concept Properties code system.
Cardinality | 0..1
Type | uri
Summary | true
CodeSystem.property.description
Element Id | CodeSystem.property.description
Definition | A description of the property- why it is defined, and how its value might be used.
Cardinality | 0..1
Type | string
Summary | true
CodeSystem.property.type
Element Id | CodeSystem.property.type
Definition | The type of the property value. Properties of type "code" contain a code defined by the code system (e.g. a reference to another defined concept).
Cardinality | 1..1
Terminology Binding | PropertyType (Required)
Type | code
Summary | true
CodeSystem.concept
Element Id | CodeSystem.concept
Definition | Concepts that are in the code system. The concept definitions are inherently hierarchical, but the definitions must be consulted to determine what the meanings of the hierarchical relationships are.
Cardinality | 0..*
Comments | If this is empty, it means that the code system resource does not represent the content of the code system.
CodeSystem.concept.code
Element Id | CodeSystem.concept.code
Definition | A code - a text symbol - that uniquely identifies the concept within the code system.
Cardinality | 1..1
Type | code
CodeSystem.concept.display

Element Id | CodeSystem.concept.display
Definition | A human readable string that is the recommended default way to present this concept to a user.
Cardinality | 0..1
Type | string
CodeSystem.concept.definition
Element Id | CodeSystem.concept.definition
Definition | The formal definition of the concept. The code system resource does not make formal definitions required, because of the prevalence of legacy systems. However, they are highly recommended, as without them there is no formal meaning associated with the concept.
Cardinality | 0..1
Type | string
CodeSystem.concept.designation
Element Id | CodeSystem.concept.designation
Definition | Additional representations for the concept - other languages, aliases, specialized purposes, used for particular purposes, etc.
Cardinality | 0..*
Requirements | Many concept definition systems support multiple representations, in multiple languages, and for multiple purposes.
Comments | Concepts have both a display and an array of designation. The display is equivalent to a special designation with an implied designation.use of "primary code" and a language equal to the Resource Language.
CodeSystem.concept.designation.language
Element Id | CodeSystem.concept.designation.language
Definition | The language this designation is defined for.
Cardinality | 0..1
Terminology Binding | Common Languages (Preferred but limited to All Languages)
Type | code
Comments | In the absence of a language, the resource language applies.
CodeSystem.concept.designation.use
Element Id | CodeSystem.concept.designation.use
Definition | A code that details how this designation would be used.
Cardinality | 0..1
Terminology Binding | Designation Use (Extensible)
Type | Coding
Comments | If no use is provided, the designation can be assumed to be suitable for general display to a human user.
CodeSystem.concept.designation.value
Element Id | CodeSystem.concept.designation.value
Definition | The text value for this designation.
Cardinality | 1..1
Type | string
CodeSystem.concept.property
Element Id | CodeSystem.concept.property
Definition | A property value for this concept.
Cardinality | 0..*
CodeSystem.concept.property.code
Element Id | CodeSystem.concept.property.code
Definition | A code that is a reference to CodeSystem.property.code.
Cardinality | 1..1
Type | code
CodeSystem.concept.property.value[x]
Element Id | CodeSystem.concept.property.value[x]
Definition | The value of this property.
Cardinality | 1..1
Type | code|Coding|string|integer|boolean|dateTime|decimal
[x] Note | See Choice of Data Types for further information about how to use [x]
CodeSystem.concept.concept
Element Id | CodeSystem.concept.concept
Definition | Defines children of a concept to produce a hierarchy of concepts. The nature of the relationships is variable (is-a/contains/categorizes) - see hierarchyMeaning.
Cardinality | 0..*
Type | See CodeSystem.concept

4.8.19 Resource CodeSystem - Mappings

URL: https://hl7.org/fhir/R4/codesystem-mappings.html

4.8.19 Resource CodeSystem - Mappings

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the codesystem resource (see Mappings to Other Standards for further information & status).

4.8.19.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

CodeSystem
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

4.8.19.2 Workflow Pattern (http://hl7.org/fhir/workflow)

CodeSystem
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright

4.8.19.3 RIM Mapping (http://hl7.org/v3)

CodeSystem | N/A

4.8.20 Resource CodeSystem - Extensions & Profiles

URL: https://hl7.org/fhir/R4/codesystem-profiles.html

4.8.20 Resource CodeSystem - Extensions & Profiles

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the CodeSystem resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

4.8.20.1 Profiles

Profile | Description | Context
ShareableCodeSystem | Enforces the minimum information set for the value set metadata required by HL7 and other organizations that share and publish value sets | Shareable CodeSystem

4.8.20.2 Extensions

codesystem-alternate | CodeSystem.concept | alternate | for CodeSystem HL7 Extensions
codesystem-author | CodeSystem | author | for CodeSystem HL7 Extensions
codesystem-concept-comments | CodeSystem.concept | concept-comments | for CodeSystem HL7 Extensions
codesystem-conceptOrder | CodeSystem.concept | conceptOrder | for CodeSystem HL7 Extensions
codesystem-effectiveDate | CodeSystem | effectiveDate | for CodeSystem HL7 Extensions
codesystem-expirationDate | CodeSystem | expirationDate | for CodeSystem HL7 Extensions
codesystem-history | CodeSystem | history | for CodeSystem HL7 Extensions
codesystem-keyWord | CodeSystem | keyWord | for CodeSystem HL7 Extensions
codesystem-label | CodeSystem.concept | label | for CodeSystem HL7 Extensions
codesystem-map | CodeSystem | map | for CodeSystem HL7 Extensions
codesystem-otherName | CodeSystem | otherName | for CodeSystem HL7 Extensions
codesystem-replacedby | CodeSystem.concept | replacedby | for CodeSystem HL7 Extensions
codesystem-sourceReference | CodeSystem | sourceReference | for CodeSystem HL7 Extensions
codesystem-trusted-expansion | CodeSystem | trusted-expansion | for CodeSystem HL7 Extensions
codesystem-usage | CodeSystem | usage | for CodeSystem HL7 Extensions
codesystem-warning | CodeSystem | warning | for CodeSystem HL7 Extensions
codesystem-workflowStatus | CodeSystem | workflowStatus | for CodeSystem HL7 Extensions
ordinalValue | CodeSystem.concept | Ordinal Value | for General Extensions for use by FHIR Implementers

replaces | CodeSystem | replaces | for General Extensions for use by FHIR Implementers
resource-approvalDate | CodeSystem | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | CodeSystem | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | CodeSystem | lastReviewDate | for Resource HL7 Extensions
valueset-special-status | CodeSystem | special-status | for ValueSet HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

valueset-supplement | supplement | for ValueSet HL7 Extensions
valueset-system | system | for ValueSet HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

4.8.20.3 Search Extensions

No Search Extensions defined for this resource

4.8.21 Resource CodeSystem - Operations

URL: https://hl7.org/fhir/R4/codesystem-operations.html

4.8.21 Resource CodeSystem - Operations

Vocabulary Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This resource has 4 operations operations associated with it:

$lookup | Concept Look Up & Decomposition | Normative
$validate-code | Code System based Validation | Normative
$subsumes | Subsumption Testing | Normative
$find-matches | Finding codes based on supplied properties | Trial Use

For more information about operations, including how they are invoked, see Operations.

4.8.21.1 Operation $lookup on CodeSystem

URL: https://hl7.org/fhir/R4/codesystem-operation-lookup.html

4.8.21.1 Operation $lookup on CodeSystem

Vocabulary Work Group | Maturity Level: N | Normative (from v4.0.1) | Compartments: N/A

| This page has been approved as part of an ANSI standard. See the Conformance Package for further details.

Given a code/system, or a Coding, get additional details about the concept, including definition, status, designations, and properties. One of the products of this operation is a full decomposition of a code from a structured terminology.

When invoking this operation, a client SHALL provide both a system and a code, either using the system+code parameters, or in the coding parameter. Other parameters are optional.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/CodeSystem-lookup
```

Formal Definition (as a OperationDefinition).

URL: [base]/CodeSystem/$lookup

This is an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
code | 0..1 | code | | | The code that is to be located. If a code is provided, a system must be provided
system | 0..1 | uri | | | The system for the code that is to be located
version | 0..1 | string | | | The version of the system, if one was provided in the source data
coding | 0..1 | Coding | | | A coding to look up
date | 0..1 | dateTime | | | The date for which the information should be returned. Normally, this is the current conditions (which is the default value) but under some circumstances, systems need to acccess this information as it would have been in the past. A typical example of this would be where code selection is constrained to the set of codes that were available when the patient was treated, not when the record is being edited. Note that which date is appropriate is a matter for implementation policy.
displayLanguage | 0..1 | code | | | The requested language for display (see $expand.displayLanguage)

property | 0..* | code | | | A property that the client wishes to be returned in the output. If no properties are specified, the server chooses what to return. The following properties are defined for all code systems: url, name, version (code system info) and code information: display, definition, designation, parent and child, and for designations, lang.X where X is a designation language code. Some of the properties are returned explicit in named parameters (when the names match), and the rest (except for lang.X) in the property parameter group

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

name | 1..1 | string | | | A display name for the code system

version | 0..1 | string | | | The version that these details are based on

display | 1..1 | string | | | The preferred display for this concept

designation | 0..* | | | | Additional representations for this concept

designation.language | 0..1 | code | | | The language this designation is defined for

designation.use | 0..1 | Coding | | | A code that details how this designation would be used

designation.value | 1..1 | string | | | The text value for this designation

property | 0..* | | | | One or more properties that contain additional information about the code, including status. For complex terminologies (e.g. SNOMED CT, LOINC, medications), these properties serve to decompose the code

property.code | 1..1 | code | | | Identifies the property returned

property.value | 0..1 | code | Coding | string | integer | boolean | dateTime | decimal | | | The value of the property returned

property.description | 0..1 | string | | | Human Readable representation of the property value (e.g. display for a code)

property.subproperty | 0..* | | | | Nested Properties (mainly used for SNOMED CT decomposition, for relationship Groups)

property.subproperty.code | 1..1 | code | | | Identifies the sub-property returned

property.subproperty.value | 1..1 | code | Coding | string | integer | boolean | dateTime | decimal | | | The value of the sub-property returned

property.subproperty.description | 0..1 | string | | | Human Readable representation of the property value (e.g. display for a code)

Note that the $lookup operation is more than just a code system search - the server finds the concept, and gathers the return information from the underlying code system definitions.

4.8.21.1.1 Examples

Request: Looking up a code/system

```
GET [base]/CodeSystem/$lookup?system=http://loinc.org&code=1963-8
```

Request: Lookup using a Coding

```
POST [base]/CodeSystem/$lookup
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="coding"/>
  <valueCoding>
    <system value="http://loinc.org"/>
    <code value="1963-8"/>
  </valueCoding>
  </parameter>
</Parameters>
```

Response: When the request can be processed ok

```
HTTP/1.1 200 OK
[other headers]

{
  "resourceType" : "Parameters",
  "parameter" : [
    {
    "name" : "name",
    "valueString" : "LOINC"
  },
  {
    "name" : "version",
    "valueString" : "2.48"
  },
  {
    "name" : "display",
    "valueString" : "Bicarbonate [Moles/volume] in Serum"
  },
  {
    "name" : "abstract",
    "valueString" : "false"
  },
  {
      "name" : "designation",
    "part" : [
    {
      "name" : "value",
      "valueString" : "Bicarbonate [Moles/volume] in Serum"
    }
    ]
  }
  ]
}
```

Response: An error - e.g. the requested code is not known

```
HTTP/1.1 400 Bad Request
[other headers]

{
  "resourceType": "OperationOutcome",
  "id": "exception",
  "text": {
    "status": "additional",
    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\">Code "ABC-23" not found</div>"
  },
  "issue": [
    {
      "severity": "error",
      "code": "not-found",
      "details": {
        "text": "Code "ABC-23" not found"
      }
    }
  ]
}
```

For more information about operations, including how they are invoked, see Operations.

4.8.21.2 Operation $validate-code on CodeSystem

URL: https://hl7.org/fhir/R4/codesystem-operation-validate-code.html

4.8.21.2 Operation $validate-code on CodeSystem

Vocabulary Work Group | Maturity Level: N | Normative (from v4.0.1) | Compartments: N/A

| This page has been approved as part of an ANSI standard. See the Conformance Package for further details.

Validate that a coded value is in the code system. If the operation is not called at the instance level, one of the parameters "url" or "codeSystem" must be provided. The operation returns a result (true / false), an error message, and the recommended display for the code.

When invoking this operation, a client SHALL provide one (and only one) of the parameters (code+system, coding, or codeableConcept). Other parameters (including version and display) are optional.

The official URL for this operation definition is

`http://hl7.org/fhir/OperationDefinition/CodeSystem-validate-code`

Formal Definition (as a OperationDefinition).

URL: [base]/CodeSystem/$validate-code

URL: [base]/CodeSystem/[id]/$validate-code

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

url | 0..1 | uri | | | CodeSystem URL. The server must know the code system (e.g. it is defined explicitly in the server'scode systems, or it is known implicitly by the server

codeSystem | 0..1 | CodeSystem | | | The codeSystem is provided directly as part of the request. Servers may choose not to accept code systems in this fashion. This parameter is used when the client wants the server to check against a code system that is not stored on the server

code | 0..1 | code | | | The code that is to be validated

version | 0..1 | string | | | The version of the code system, if one was provided in the source data

display | 0..1 | string | | | The display associated with the code, if provided. If a display is provided a code must be provided. If no display is provided, the server cannot validate the display value, but may choose to return a recommended display name in an extension in the outcome. Whether displays are case sensitive is code system dependent

coding | 0..1 | Coding | | | A coding to validate. The system must match the specified code system

codeableConcept | 0..1 | CodeableConcept | | | A full codeableConcept to validate. The server returns true if one of the coding values is in the code system, and may also validate that the codings are not in conflict with each other if more than one is present

date | 0..1 | dateTime | | | The date for which the validation should be checked. Normally, this is the current conditions (which is the default values) but under some circumstances, systems need to validate that a correct code was used at some point in the past. A typical example of this would be where code selection is constrained to the set of codes that were available when the patient was treated, not when the record is being edited. Note that which date is appropriate is a matter for implementation policy.

abstract | 0..1 | boolean | | | If this parameter has a value of true, the client is stating that the validation is being performed in a context where a concept designated as 'abstract' is appropriate/allowed to be used, and the server should regard abstract codes as valid. If this parameter is false, abstract codes are not considered to be valid. Note that. 'abstract' is a property defined by many HL7 code systems that indicates that the concept is a logical grouping concept that is not intended to be used asa 'concrete' concept to in an actual patient/care/process record. This language is borrowed from Object Orienated theory where 'asbtract' objects are never instantiated. However in the general record and terminology eco-system, there are many contexts where it is appropraite to use these codes e.g. as decision making criterion, or when editing value sets themselves. This parameter allows a client to indicate to the server that it is working in such a context.

displayLanguage | 0..1 | code | | | Specifies the language to be used for description when validating the display property

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

result | 1..1 | boolean | | | True if the concept details supplied are valid

message | 0..1 | string | | | Error details, if result = false. If this is provided when result = true, the message carries hints and warnings
display | 0..1 | string | | | A valid display for the concept if the system wishes to display this to a user

4.8.21.2.1 Examples

Request: Simple validation of a code against a known code system

```
GET [base]/CodeSystem/loinc/$validate-code?code=1963-8&display=test
```

Request: Validate a CodeableConcept against a client specified code system

```
POST [base]/CodeSystem/$validate-code
[other headers]

{
  "ResourceType" : "Parameters",
  "parameter" : [
    {
    "name" : "codeableConcept",
    "valueCodeableConcept" : {
      "coding" : {
        "system" : "http://example.org",
          "code" : "code",
      "display" : "test"
      }
    }
  },
  {
    "name" : "codeSystem",
    "resource": {
      "resourceType" : "codeSystem",
    [etc.]
    }
  }
  ]
}
```

Response: When the request can be processed ok

```
HTTP/1.1 200 OK
[other headers]

{
  "resourceType" : "Parameters",
  "parameter" : [
    {
    "name" : "result",
    "valueBoolean" : "false"
  },
  {
    "name" : "message",
    "valueString" : "The display \"test\" is incorrect"
  },
  {
    "name" : "display",
    "valueString" : "Bicarbonate [Moles/volume] in Serum"
  }
  ]
}
```

Response: When the validation cannot be performed. An error like this not returned if the code is not valid, but when the server is unable to determine whether the code is valid

```
HTTP/1.1 400 Bad Request
[other headers]

{
  "resourceType": "OperationOutcome",
  "id": "exception",
  "text": {
    "status": "additional",
    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\">No system parameter provided in request</div>"
  },
  "issue": [
    {
      "severity": "error",
      "code": "required",
      "details": {
        "text": "No system parameter provided in request"
      }
    }
  ]
}
```

For more information about operations, including how they are invoked, see Operations.

4.8.21.3 Operation $subsumes on CodeSystem

URL: https://hl7.org/fhir/R4/codesystem-operation-subsumes.html

4.8.21.3 Operation $subsumes on CodeSystem

Vocabulary Work Group | Maturity Level: N | Normative (from v4.0.1) | Compartments: N/A

| This page has been approved as part of an ANSI standard. See the Conformance Package for further details.

Test the subsumption relationship between code/Coding A and code/Coding B given the semantics of subsumption in the underlying code system (see hierarchyMeaning).

When invoking this operation, a client SHALL provide both a and codes, either as code or Coding parameters. The system parameter is required unless the operation is invoked on an instance of a code system resource. Other parameters are optional.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/CodeSystem-subsumes
```

Formal Definition (as a OperationDefinition).

URL: [base]/CodeSystem/$subsumes

URL: [base]/CodeSystem/[id]/$subsumes

This is an idempotent operation

In Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
codeA	0..1	code			The "A" code that is to be tested. If a code is provided, a system must be provided
codeB	0..1	code			The "B" code that is to be tested. If a code is provided, a system must be provided
system	0..1	uri			The code system in which subsumption testing is to be performed. This must be provided unless the operation is invoked on a code system instance
version	0..1	string			The version of the code system, if one was provided in the source data
codingA	0..1	Coding			The "A" Coding that is to be tested. The code system does not have to match the specified subsumption code system, but the relationships between the code systems must be well established
codingB	0..1	Coding			The "B" Coding that is to be tested. The code system does not have to match the specified subsumption code system, but the relationships between the code systems must be well established

Out Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
outcome	1..1	code	ConceptSubsumptionOutcome (Required)		The subsumption relationship between code/Coding "A" and code/Coding "B". There are 4 possible codes to be returned (equivalent, subsumes, subsumed-by, and not-subsumed) as defined in the concept-subsumption-outcome value set. If the server is unable to determine the relationship between the codes/Codings, then it returns an error response with an OperationOutcome.

4.8.21.3.1 Examples

Request: Using simple parameters: is 'Viral hepatitis' (3738000) a 'Disorder of liver' (235856003)?

```
GET [base]/CodeSystem/$subsumes?system=http://snomed.info/sct&codeA=3738000&codeB=235856003
```

Request: Using Codings

```
POST [base]/CodeSystem/$subsumes
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <!-- Subsumption testing - use SNOMED CT rules -->
  <parameter>
    <name value="system"/>
    <valueUri value="http://snomed.info/sct"/>
  </parameter>
  <!-- Australian distribution -->
  <parameter>
    <name value="version"/>
    <valuString value="http://snomed.info/sct/32506021000036107/version/20160430"/>
  </parameter>
  <parameter>
    <name value="codingA"/>
    <valueCoding>
      <system value="http://snomed.info/sct"/>
      <code value="3738000"/>
    </valueCoding>
  </parameter>
  <parameter>
    <name value="codingB"/>
    <valueCoding>
      <system value="http://snomed.info/sct"/>
      <code value="235856003"/>
    </valueCoding>
  </parameter>
</Parameters>
```

Response: When the request can be processed ok

```
HTTP/1.1 200 OK
[other headers]

{
  "resourceType" : "Parameters",
  "parameter" : [
    {
```

```
    "name" : "outcome",
    "valueCode" : "subsumed-by"
  }
}
```

Response: When the subsumption cannot be tested

```
HTTP/1.1 500 Error
[other headers]

{
  "resourceType": "OperationOutcome",
  "id": "exception",
  "text": {
    "status": "additional",
    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\">Version "XXX" for http://acme.com/fhir/CodeSystem/some-id is no
    longer available</div>"
  },
  "issue": [
    {
      "severity": "error",
      "code": "not-supported",
      "details": {
        "text": "Version "XXX" for http://acme.com/fhir/CodeSystem/some-id is no longer available"
      }
    }
  ]
}
```

For more information about operations, including how they are invoked, see Operations.

4.8.21.4 Operation $find-matches on CodeSystem

URL: https://hl7.org/fhir/R4/codesystem-operation-find-matches.html

4.8.21.4 Operation $find-matches on CodeSystem

Vocabulary Work Group | Maturity Level: 0 | Trial Use | Compartments: N/A

Given a set of properties (and text), return one or more possible matching codes

This operation takes a set of properties, and examines the code system looking for codes in the code system that match a set of known properties.

When looking for matches, there are 3 possible types of match:

• a complete match - a code that represents all the provided properties correctly

• a partial match - a code that represents some of the provided properties correctly, and not others

• a possible match - a code that may represent the provided properties closely, but may capture less or more precise information for some of the properties

The $find-matches operation can be called in one of 2 modes:

• By a human, looking for the best match for a set of properties. In this mode, the server returns a list of complete, possible or partial matches (possibly with comments), so that the user can choose (or not) the most appropriate code

• By a machine (typically in a system interface performing a transformation). In this mode, the server returns only a list of complete and partial matches, but no possible matches. The machine can choose a code from the list (or not) based on what properties are not coded

These modes are differentiated by the 'exact' parameter, so the client can indicate whether it only wants exact matches (including partial matches) or whether potential matches based on text matching are desired

The find-matches operation is still preliminary. The interface can be expected to change as more experience is gained from implementations.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/CodeSystem-find-matches
```

Formal Definition (as a OperationDefinition).

URL: [base]/CodeSystem/$find-matches

URL: [base]/CodeSystem/[id]/$find-matches

This is an idempotent operation

In Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
system	0..1	uri			The system in which composition is to be performed. This must be provided unless the operation is invoked on a code system instance
version	0..1	string			The version of the system for the inferencing to be performed
property	0..*				One or more properties that contain information to be composed into the code
property.code	1..1	code			Identifies the property provided
property.value	0..1	code | Coding | string | integer | boolean | dateTime			The value of the property provided
property.subproperty	0..*				Nested Properties (mainly used for SNOMED CT composition, for relationship Groups)

property.subproperty.code | 1..1 | code | | | Identifies the sub-property provided
property.subproperty.value | 1..1 | code | Coding | string | integer | boolean | dateTime | | | The value of the sub-property provided
exact | 1..1 | boolean | | | Whether the operation is being used by a human ('false'), or a machine ('true'). If the operation is being used by a human, the terminology server can return a list of possible matches, with commentary. For a machine, the server returns complete or partial matches, not possible matches. The default value is 'false'
compositional | 0..1 | boolean | | | Post-coordinated expressions are allowed to be returned in the matching codes (mainly for SNOMED CT). Default = false
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
match | 0..* | | | | Concepts returned by the server as a result of the inferencing operation
match.code | 1..1 | Coding | | | A code that matches the properties provided
match.unmatched | 0..* | | | | One or more properties that contain properties that could not be matched into the code
match.unmatched.code | 1..1 | code | | | Identifies the property provided
match.unmatched.value | 1..1 | code | Coding | string | integer | boolean | dateTime | | | The value of the property provided
match.unmatched.property | 0..* | | | | Nested Properties (mainly used for SNOMED CT composition, for relationship Groups)
match.unmatched.property.code | 1..1 | code | | | Identifies the sub-property provided
match.unmatched.property.value | 1..1 | code | Coding | string | integer | boolean | dateTime | | | The value of the sub-property provided
match.comment | 0..1 | string | | | Information about the quality of the match, if operation is for a human

4.8.21.4.1 Examples

Request:

```
<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="system"/>
    <valueUri value="http://snomed.info/sct"/>
  </parameter>
  <parameter>
    <name value="exact"/>
    <valueBoolean value="true"/>
  </parameter>
  <parameter>
    <name value="property"/>
    <part>
      <name value="code"/>
      <valueCode value="focus"/>
    </part>
    <part>
      <name value="value"/>
      <!-- Measurement finding -->
      <valueCode value="118245000"/>
    </part>
  </parameter>
  <parameter>
    <name value="property"/>
    <part>
      <name value="code"/>
      <!-- Interprets -->
      <valueCode value="363714003"/>
    </part>
    <part>
      <name value="value"/>
      <!-- Bilirubin -->
      <valueCode value="302787001"/>
    </part>
  </parameter>
</Parameters>
```

Response: SNOMED Composition - provide 2 properties, and ask for a single code for the whole

```
HTTP/1.1 200 OK
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
    <parameter>
        <name value="match" />
        <part>
            <name value="code" />
            <valueCoding>
                <system value="http://snomed.info/sct" />
                <code value="365786009" />
                <display value="Bilirubin level - finding" />
            </valueCoding>
        </part>
    </parameter>
</Parameters>
```

For more information about operations, including how they are invoked, see Operations.

4.8.22 Resource CodeSystem - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/codesystem-version-maps.html

No validation errors - all conversions are clean

4.8.22 Resource CodeSystem - R3/R4 Conversions

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for CodeSystem.

Functional status for this map: 4 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors). (see documentation)

4.8.22.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/CodeSystem3to4" = "R3 to R4 Conversions for CodeSystem"

uses "http://hl7.org/fhir/3.0/StructureDefinition/CodeSystem" alias CodeSystemR3 as source
uses "http://hl7.org/fhir/StructureDefinition/CodeSystem" alias CodeSystem as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group CodeSystem(source src : CodeSystemR3, target tgt : CodeSystem) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.copyright -> tgt.copyright;
  src.caseSensitive -> tgt.caseSensitive;
  src.valueSet -> tgt.valueSet;
  src.hierarchyMeaning -> tgt.hierarchyMeaning;
  src.compositional -> tgt.compositional;
  src.versionNeeded -> tgt.versionNeeded;
  src.content -> tgt.content;
  src.count -> tgt.count;
  src.filter as vs0 -> tgt.filter as vt0 then filter(vs0, vt0);
  src.property as vs0 -> tgt.property as vt0 then property(vs0, vt0);
  src.concept as vs0 -> tgt.concept as vt0 then concept(vs0, vt0);
}

group filter(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.description -> tgt.description;
  src.operator -> tgt.operator;
  src.value -> tgt.value;
}

group property(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.uri -> tgt.uri;
  src.description -> tgt.description;
  src.type -> tgt.type;
}

group concept(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.definition -> tgt.definition;
  src.designation as vs0 -> tgt.designation as vt0 then designation(vs0, vt0);
  src.property as vs0 -> tgt.property as vt0 then propertyC(vs0, vt0);
  src.concept as vs0 -> tgt.concept as vt0 then concept(vs0, vt0);
}

group designation(source src, target tgt) extends BackboneElement {
  src.language -> tgt.language;
  src.use -> tgt.use;
  src.value -> tgt.value;
}

group propertyC(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.value : code as vs0 -> tgt.value = create('code') as vt0 then code(vs0, vt0) "valueCode";
  src.value : Coding as vs0 -> tgt.value = create('Coding') as vt0 then Coding(vs0, vt0) "valueCoding";
  src.value : string as vs0 -> tgt.value = create('string') as vt0 then string(vs0, vt0) "valueString";
  src.value : integer as vs0 -> tgt.value = create('integer') as vt0 then integer(vs0, vt0) "valueInteger";
  src.value : boolean as vs0 -> tgt.value = create('boolean') as vt0 then boolean(vs0, vt0) "valueBoolean";
  src.value : dateTime as vs0 -> tgt.value = create('dateTime') as vt0 then dateTime(vs0, vt0) "valueDateTime";
}
```

4.8.22.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/CodeSystem4to3" = "R4 to R3 Conversion for CodeSystem"
```

```
uses "http://hl7.org/fhir/StructureDefinition/CodeSystem" alias CodeSystem as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/CodeSystem" alias CodeSystemR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group CodeSystem(source src : CodeSystem, target tgt : CodeSystemR3) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.copyright -> tgt.copyright;
  src.caseSensitive -> tgt.caseSensitive;
  src.valueSet -> tgt.valueSet;
  src.hierarchyMeaning -> tgt.hierarchyMeaning;
  src.compositional -> tgt.compositional;
  src.versionNeeded -> tgt.versionNeeded;
  src.content -> tgt.content;
  src.count -> tgt.count;
  src.filter as vs0 -> tgt.filter as vt0 then filter(vs0, vt0);
  src.property as vs0 -> tgt.property as vt0 then property(vs0, vt0);
  src.concept as vs0 -> tgt.concept as vt0 then concept(vs0, vt0);
}

group filter(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.description -> tgt.description;
  src.operator -> tgt.operator;
  src.value -> tgt.value;
}

group property(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.uri -> tgt.uri;
  src.description -> tgt.description;
  src.type -> tgt.type;
}

group concept(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.definition -> tgt.definition;
  src.designation as vs0 -> tgt.designation as vt0 then designation(vs0, vt0);
  src.property as vs0 -> tgt.property as vt0 then propertyC(vs0, vt0);
  src.concept as vs0 -> tgt.concept as vt0 then concept(vs0, vt0);
}

group designation(source src, target tgt) extends BackboneElement {
  src.language -> tgt.language;
  src.use -> tgt.use;
  src.value -> tgt.value;
}

group propertyC(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.value : code as vs0 -> tgt.value = create('code') as vt0 then code(vs0, vt0) "valueCode";
  src.value : Coding as vs0 -> tgt.value = create('Coding') as vt0 then Coding(vs0, vt0) "valueCoding";
  src.value : string as vs0 -> tgt.value = create('string') as vt0 then string(vs0, vt0) "valueString";
  src.value : integer as vs0 -> tgt.value = create('integer') as vt0 then integer(vs0, vt0) "valueInteger";
  src.value : boolean as vs0 -> tgt.value = create('boolean') as vt0 then boolean(vs0, vt0) "valueBoolean";
  src.value : dateTime as vs0 -> tgt.value = create('dateTime') as vt0 then dateTime(vs0, vt0) "valueDateTime";
}
```

4.8.22.3 R4 Validation Errors

4.8.23 CodeSystem HL7 Extensions

URL: https://hl7.org/fhir/R4/codesystem-extensions.html

Defines common extensions used with or related to the CodeSystem resource

4.8.23 CodeSystem HL7 Extensions

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative

4.8.23.1 Content

Extensions:

codesystem-concept-comments | concept-comments : A comment that explains how this code is used in this context (where the value set is expected to be used).
codesystem-otherName | otherName : Human readable names for the codesystem.
codesystem-sourceReference | sourceReference : This text is intended to act as a citation to work done elsewhere that is not part of the current stewarding process where the referenced source is in some way a basis of the current value set definition.
codesystem-keyWord | keyWord : Word or words used in an information retrieval system to indicate the content of the value set.
codesystem-usage | usage : Consumers of the value set and the implementations, projects or standards that the author has utilized the value set in.
codesystem-workflowStatus | workflowStatus : Workflow Status is used to represent details of the value set development process while the value set has an Activity Status of Preliminary. The development of a value set often follows a formal workflow process from initiation to completion, and this element carries the state variable for this state machine. The assumption is that when first created a value set would have a workflow state of Draft. Additional workflow states may be used.
codesystem-effectiveDate | effectiveDate : This is the first date-time when the value set version becomes active, so this value is present on Inactive value set versions as well. The start Date_time is expected to be as of 0001 UTC of the Effective Date.
codesystem-expirationDate | expirationDate : The date when the value set version is no longer expected to be used to create new content. This is the first date-time when the value set version becomes Inactive, so this value MUST present on all Inactive value set versions. The start Date_time is expected to be as of 0001 UTC of the Expiration Date.
codesystem-author | author : User or Org actually involved in creating the value set content.
codesystem-trusted-expansion | trusted-expansion : Indicates an authoritative source for performing value set expansions.
codesystem-history | history : Information on changes made to the Value Set Definition over time, and also has a contained audit trail of all such changes.
codesystem-label | label : The label to list in front of a code when presenting a list of possible values in a questionnaire-like fashion.
codesystem-conceptOrder | conceptOrder : Identifies the relative order in which concepts within the value set should be presented to a user.
codesystem-map | map : A reference to a concept map that is relevant for the interpretation of this value set.
codesystem-replacedby | replacedby : A code that replaces this - use this code instead.
codesystem-warning | warning : An extra warning about the correct use of the value set.
codesystem-alternate | alternate : An additional code that may be used to represent the concept.

4.8.23.2 Search Parameters

Search parameters defined by this package. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths | Source
author | string | Optional Extensions Element | f:CodeSystem/f:extension[@url='http://hl7.org/fhir/StructureDefinition/codesystem-author'] | /f:#author | XML / JSON
effective | date | Optional Extensions Element | f:CodeSystem/f:extension[@url='http://hl7.org/fhir/StructureDefinition/codesystem-effectiveDate'] | /f:#effectiveDate | XML / JSON
end | date | Optional Extensions Element | f:CodeSystem/f:extension[@url='http://hl7.org/fhir/StructureDefinition/codesystem-expirationDate'] | /f:#expirationDate | XML / JSON
keyword | string | Optional Extensions Element | f:CodeSystem/f:extension[@url='http://hl7.org/fhir/StructureDefinition/codesystem-keyWord'] | /f:#keyWord | XML / JSON
workflow | token | Optional Extensions Element | f:CodeSystem/f:extension[@url='http://hl7.org/fhir/StructureDefinition/codesystem-workflowStatus'] | /f:#workflowStatus | XML / JSON

4.8.23.3 Codesystem-extensions-CodeSystem-keyword

URL: https://hl7.org/fhir/R4/codesystem-extensions-CodeSystem-keyword.html

4.8.23.3 Codesystem-extensions-CodeSystem-keyword

Search Parameter keyword (type: string) defined as part of the codesystem-extensions Profile.

Vocabulary | Maturity Level: 0 | Informative

Optional Extensions Element

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/codesystem-extensions-CodeSystem-keyword

• Paths : f:CodeSystem/f:extension[@url='http://hl7.org/fhir/StructureDefinition/codesystem-keyWord'] | /f:#keyWord

• Targets :

4.8.23.4 Codesystem-extensions-CodeSystem-workflow

URL: https://hl7.org/fhir/R4/codesystem-extensions-CodeSystem-workflow.html

4.8.23.4 Codesystem-extensions-CodeSystem-workflow

Search Parameter workflow (type: token) defined as part of the codesystem-extensions Profile.

Vocabulary | Maturity Level: 0 | Informative

Optional Extensions Element

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/codesystem-extensions-CodeSystem-workflow

• Paths : f:CodeSystem/f:extension[@url='http://hl7.org/fhir/StructureDefinition/codesystem-workflowStatus'] | /f:#workflowStatus

• Targets :

4.8.23.5 Codesystem-extensions-CodeSystem-effective

URL: https://hl7.org/fhir/R4/codesystem-extensions-CodeSystem-effective.html

4.8.23.5 Codesystem-extensions-CodeSystem-effective

Search Parameter effective (type: date) defined as part of the codesystem-extensions Profile.

Vocabulary | Maturity Level: 0 | Informative

Optional Extensions Element

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/codesystem-extensions-CodeSystem-effective

• Paths : f:CodeSystem/f:extension[@url='http://hl7.org/fhir/StructureDefinition/codesystem-effectiveDate'] | /f:#effectiveDate

• Targets :

4.8.23.6 Codesystem-extensions-CodeSystem-end

URL: https://hl7.org/fhir/R4/codesystem-extensions-CodeSystem-end.html

4.8.23.6 Codesystem-extensions-CodeSystem-end

Search Parameter end (type: date) defined as part of the codesystem-extensions Profile.

Vocabulary | Maturity Level: 0 | Informative

Optional Extensions Element

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/codesystem-extensions-CodeSystem-end

• Paths : f:CodeSystem/f:extension[@url='http://hl7.org/fhir/StructureDefinition/codesystem-expirationDate'] | /f:#expirationDate

• Targets :

4.8.23.7 Codesystem-extensions-CodeSystem-author

URL: https://hl7.org/fhir/R4/codesystem-extensions-CodeSystem-author.html

4.8.23.7 Codesystem-extensions-CodeSystem-author

Search Parameter author (type: string) defined as part of the codesystem-extensions Profile.

Vocabulary | Maturity Level: 0 | Informative

Optional Extensions Element

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/codesystem-extensions-CodeSystem-author

• Paths : f:CodeSystem/f:extension[@url='http://hl7.org/fhir/StructureDefinition/codesystem-author'] | /f:#author

• Targets :

4.8.24 Shareable CodeSystem

URL: https://hl7.org/fhir/R4/codesystem-shareable.html

Enforces the minimum information set for the value set metadata required by HL7 and other organizations that share and publish value sets

4.8.24 Shareable CodeSystem

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative

4.8.24.1 Content

Profiles:

ShareableCodeSystem | Enforces the minimum information set for the value set metadata required by HL7 and other organizations that share and publish value sets

4.8.24.2 StructureDefinition: ShareableCodeSystem

URL: https://hl7.org/fhir/R4/shareablecodesystem.html

4.8.24.2 StructureDefinition: ShareableCodeSystem

Vocabulary Work Group | Maturity Level: 2 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/shareablecodesystem`

Enforces the minimum information set for the value set metadata required by HL7 and other organizations that share and publish value sets

This profile was published on [no date] as a draft by HL7.

4.8.24.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from CodeSystem.

Summary

Mandatory: 9 elements

This structure is derived from CodeSystem.

Name	Flags	Card.	Type	Description & Constraints
CodeSystem		1..1		
url		1..1	uri	
version		1..1	string	
name		1..1	string	
status		1..1	code	
experimental		1..1	boolean	
publisher		1..1	string	
description		1..1	markdown	
caseSensitive		0..1	boolean	
concept		1..*	BackboneElement	
code		1..1	code	
display		0..1	string	
definition		0..1	string	
concept		0..*		Unknown reference to #CodeSystem.concept

Documentation for this format

Name	Flags	Card.	Type	Description & Constraints
CodeSystem	I	1..1		Declares the existence of and describes a code system or code system supplement
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension	I	0..*	Extension	Additional content defined by implementations
modifierExtension	?!I	0..*	Extension	Extensions that cannot be ignored
url	ΣI	1..1	uri	Canonical identifier for this code system, represented as a URI (globally unique) (Coding.system)
identifier	ΣI	0..*	Identifier	Additional identifier for the code system (business identifier)
version	ΣI	1..1	string	Business version of the code system (Coding.version)
name	ΣI	1..1	string	Name for this code system (computer friendly)
title	ΣI	0..1	string	Name for this code system (human friendly)
status	?!ΣI	1..1	code	draft | active | retired | unknown Binding: PublicationStatus (required)
experimental	ΣI	1..1	boolean	For testing purposes, not real usage
date	ΣI	0..1	dateTime	Date last changed
publisher	ΣI	1..1	string	Name of the publisher (organization or individual)
contact	ΣI	0..*	ContactDetail	Contact details for the publisher
description	I	1..1	markdown	Natural language description of the code system
useContext	ΣI	0..*	UsageContext	The context that the content is intended to support
jurisdiction	ΣI	0..*	CodeableConcept	Intended jurisdiction for code system (if applicable) Binding: Jurisdiction (extensible)
purpose	I	0..1	markdown	Why this code system is defined
copyright	I	0..1	markdown	Use and/or publishing restrictions
caseSensitive	ΣI	0..1	boolean	If code comparison is case sensitive

valueSet | ΣI | 0..1 | canonical(ValueSet) | Canonical reference to the value set with entire code system
hierarchyMeaning | ΣI | 0..1 | code | grouped-by | is-a | part-of | classified-with Binding: CodeSystemHierarchyMeaning (required)
compositional | ΣI | 0..1 | boolean | If code system defines a compositional grammar
versionNeeded | ΣI | 0..1 | boolean | If definitions are not stable
content | ΣI | 1..1 | code | not-present | example | fragment | complete | supplement Binding: CodeSystemContentMode (required)
supplements | ΣI | 0..1 | canonical(CodeSystem) | Canonical URL of Code System this adds designations and properties to
count | ΣI | 0..1 | unsignedInt | Total concepts in the code system
filter | ΣI | 0..* | BackboneElement | Filter that can be used in a value set
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | code | Code that identifies the filter
description | ΣI | 0..1 | string | How or why the filter is used
operator | ΣI | 1..* | code | = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists Binding: FilterOperator (required)
value | ΣI | 1..1 | string | What to use for the value
property | ΣI | 0..* | BackboneElement | Additional information supplied about each concept
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | code | Identifies the property on the concepts, and when referred to in operations
uri | ΣI | 0..1 | uri | Formal identifier for the property
description | ΣI | 0..1 | string | Why the property is defined, and/or what it conveys
type | ΣI | 1..1 | code | code | Coding | string | integer | boolean | dateTime | decimal Binding: PropertyType (required)
concept | I | 1..* | BackboneElement | Concepts in the code system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 1..1 | code | Code that identifies concept
display | I | 0..1 | string | Text to display to the user
definition | I | 0..1 | string | Formal definition
designation | I | 0..* | BackboneElement | Additional representations for the concept
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
language | I | 0..1 | code | Human language of the designation Binding: Common Languages (preferred) Max Binding: All Languages
use | I | 0..1 | Coding | Details how this designation would be used Binding: Designation Use (extensible)
value | I | 1..1 | string | The text value for this designation
property | I | 0..* | BackboneElement | Property value for the concept
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 1..1 | code | Reference to CodeSystem.property.code
value[x] | I | 1..1 | | Value of the property for this concept
valueCode | | | code
valueCoding | | | Coding
valueString | | | string
valueInteger | | | integer
valueBoolean | | | boolean
valueDateTime | | | dateTime
valueDecimal | | | decimal
concept | I | 0..* | | Unknown reference to #CodeSystem.concept Child Concepts (is-a/contains/categorizes)
Documentation for this format

```
<!-- Shareable CodeSystem -->

<CodeSystem xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  1..1
     Canonical identifier for this code system, represented as a URI (globally unique) (Coding.system) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the code system (business identifier) --></identifier>
 <version value="[string]"/><!--  1..1 Business version of the code system (Coding.version) -->
 <name value="[string]"/><!--  1..1 Name for this code system (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this code system (human friendly) -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  1..1 For testing purposes, not real usage -->
```

```
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  1..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  1..1 Natural language description of the code system -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for code system (if applicable)   --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this code system is defined -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <caseSensitive value="[boolean]"/><!--  0..1 If code comparison is case sensitive -->
 <valueSet value="[canonical]"/><!--  0..1 Canonical reference to the value set with entire code system -->
 <hierarchyMeaning value="[code]"/><!--  0..1 grouped-by | is-a | part-of | classified-with   -->
 <compositional value="[boolean]"/><!--  0..1 If code system defines a compositional grammar -->
 <versionNeeded value="[boolean]"/><!--  0..1 If definitions are not stable -->
 <content value="[code]"/><!--  1..1 not-present | example | fragment | complete | supplement   -->
 <supplements value="[canonical]"/><!--  0..1 Canonical URL of Code System this adds designations and properties to -->
 <count value="[unsignedInt]"/><!--  0..1 Total concepts in the code system -->
 <filter>  0..* BackboneElement  <!--  0..* Filter that can be used in a value set -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code value="[code]"/><!--  1..1 Code that identifies the filter -->
  <description value="[string]"/><!--  0..1 How or why the filter is used -->
  <operator value="[code]"/><!--  1..* = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists
       -->
  <value value="[string]"/><!--  1..1 What to use for the value -->
 </filter>
 <property>  0..* BackboneElement  <!--  0..* Additional information supplied about each concept -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code value="[code]"/><!--  1..1
      Identifies the property on the concepts, and when referred to in operations -->
  <uri value="[uri]"/><!--  0..1 Formal identifier for the property -->
  <description value="[string]"/><!--  0..1 Why the property is defined, and/or what it conveys -->
  <type value="[code]"/><!--  1..1 code | Coding | string | integer | boolean | dateTime | decimal   -->
 </property>
 <concept>  1..* BackboneElement  <!--  1..* Concepts in the code system -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code value="[code]"/><!--  1..1 Code that identifies concept -->
  <display value="[string]"/><!--  0..1 Text to display to the user -->
  <definition value="[string]"/><!--  0..1 Formal definition -->
  <designation>  0..* BackboneElement  <!--  0..* Additional representations for the concept -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <language value="[code]"/><!--  0..1 Human language of the designation   -->
   <use><!--  0..1 Coding Details how this designation would be used   --></use>
   <value value="[string]"/><!--  1..1 The text value for this designation -->
  </designation>
  <property>  0..* BackboneElement  <!--  0..* Property value for the concept -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <code value="[code]"/><!--  1..1 Reference to CodeSystem.property.code -->
   <value[x]><!--  1..1 code|Coding|string|integer|
     boolean|dateTime|decimal
       Value of the property for this concept --></value[x]>
  </property>
  <concept><!-- See #CodeSystem.concept  Child Concepts (is-a/contains/categorizes) --></concept>
 </concept>
</CodeSystem>
```

```
{ // Shareable CodeSystem
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C? R! Canonical identifier for this code system, represented as a URI (globally unique)
    (Coding.system)
    "identifier" : [{ Identifier }], // C?Additional identifier for the code system (business identifier)
    "version" : "<string>", // C? R! Business version of the code system (Coding.version)
    "name" : "<string>", // C? R! Name for this code system (computer friendly)
    "title" : "<string>", // C?Name for this code system (human friendly)
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C? R! For testing purposes, not real usage
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C? R! Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C? R! Natural language description of the code system
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for code system (if applicable)
    "purpose" : "<markdown>", // C?Why this code system is defined
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "caseSensitive" : <boolean>, // C?If code comparison is case sensitive
    "valueSet" : "<canonical>", // C?Canonical reference to the value set with entire code system
    "hierarchyMeaning" : "<code>", // C?grouped-by | is-a | part-of | classified-with
    "compositional" : <boolean>, // C?If code system defines a compositional grammar
    "versionNeeded" : <boolean>, // C?If definitions are not stable
    "content" : "<code>", // C? R! not-present | example | fragment | complete | supplement
```

```
  "supplements" : "<canonical>", // C?Canonical URL of Code System this adds designations and properties to
  "count" : "<unsignedInt>", // C?Total concepts in the code system
  "filter" : [{ BackboneElement }], // C?Filter that can be used in a value set
  "property" : [{ BackboneElement }], // C?Additional information supplied about each concept
  "concept" : [{ BackboneElement }] // C? R! Concepts in the code system
}
```

This structure is derived from CodeSystem.

Summary

Mandatory: 9 elements

Differential View

This structure is derived from CodeSystem.

Name | Flags | Card. | Type | Description & Constraints
CodeSystem | | 1..1
url | | 1..1 | uri
version | | 1..1 | string
name | | 1..1 | string
status | | 1..1 | code
experimental | | 1..1 | boolean
publisher | | 1..1 | string
description | | 1..1 | markdown
caseSensitive | | 0..1 | boolean
concept | | 1..* | BackboneElement
code | | 1..1 | code
display | | 0..1 | string
definition | | 0..1 | string
concept | | 0..* | | Unknown reference to #CodeSystem.concept
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
CodeSystem | I | 1..1 | | Declares the existence of and describes a code system or code system supplement
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 1..1 | uri | Canonical identifier for this code system, represented as a URI (globally unique) (Coding.system)
identifier | ΣI | 0..* | Identifier | Additional identifier for the code system (business identifier)
version | ΣI | 1..1 | string | Business version of the code system (Coding.version)
name | ΣI | 1..1 | string | Name for this code system (computer friendly)
title | ΣI | 0..1 | string | Name for this code system (human friendly)
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 1..1 | boolean | For testing purposes, not real usage
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 1..1 | string | Name of the publisher (organization or individual)
contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | I | 1..1 | markdown | Natural language description of the code system
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for code system (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this code system is defined
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
caseSensitive | ΣI | 0..1 | boolean | If code comparison is case sensitive
valueSet | ΣI | 0..1 | canonical(ValueSet) | Canonical reference to the value set with entire code system
hierarchyMeaning | ΣI | 0..1 | code | grouped-by | is-a | part-of | classified-with Binding: CodeSystemHierarchyMeaning (required)
compositional | ΣI | 0..1 | boolean | If code system defines a compositional grammar
versionNeeded | ΣI | 0..1 | boolean | If definitions are not stable
content | ΣI | 1..1 | code | not-present | example | fragment | complete | supplement Binding: CodeSystemContentMode (required)
supplements | ΣI | 0..1 | canonical(CodeSystem) | Canonical URL of Code System this adds designations and properties to
count | ΣI | 0..1 | unsignedInt | Total concepts in the code system
filter | ΣI | 0..* | BackboneElement | Filter that can be used in a value set
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | code | Code that identifies the filter
description | ΣI | 0..1 | string | How or why the filter is used
operator | ΣI | 1..* | code | = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists Binding: FilterOperator (required)
value | ΣI | 1..1 | string | What to use for the value

property | ΣI | 0..* | BackboneElement | Additional information supplied about each concept
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | ΣI | 1..1 | code | Identifies the property on the concepts, and when referred to in operations
uri | ΣI | 0..1 | uri | Formal identifier for the property
description | ΣI | 0..1 | string | Why the property is defined, and/or what it conveys
type | ΣI | 1..1 | code | code | Coding | string | integer | boolean | dateTime | decimal Binding: PropertyType (required)
concept | I | 1..* | BackboneElement | Concepts in the code system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 1..1 | code | Code that identifies concept
display | I | 0..1 | string | Text to display to the user
definition | I | 0..1 | string | Formal definition
designation | I | 0..* | BackboneElement | Additional representations for the concept
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
language | I | 0..1 | code | Human language of the designation Binding: Common Languages (preferred) Max Binding: All Languages
use | I | 0..1 | Coding | Details how this designation would be used Binding: Designation Use (extensible)
value | I | 1..1 | string | The text value for this designation
property | I | 0..* | BackboneElement | Property value for the concept
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 1..1 | code | Reference to CodeSystem.property.code
value[x] | I | 1..1 | | Value of the property for this concept
valueCode | | | code
valueCoding | | | Coding
valueString | | | string
valueInteger | | | integer
valueBoolean | | | boolean
valueDateTime | | | dateTime
valueDecimal | | | decimal
concept | I | 0..* | | Unknown reference to #CodeSystem.concept Child Concepts (is-a/contains/categorizes)
Documentation for this format

XML Template

```
<!-- Shareable CodeSystem -->

<CodeSystem xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  1..1
     Canonical identifier for this code system, represented as a URI (globally unique) (Coding.system) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the code system (business identifier) --></identifier>
 <version value="[string]"/><!--  1..1 Business version of the code system (Coding.version) -->
 <name value="[string]"/><!--  1..1 Name for this code system (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this code system (human friendly) -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  1..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  1..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  1..1 Natural language description of the code system -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for code system (if applicable)   --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this code system is defined -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <caseSensitive value="[boolean]"/><!--  0..1 If code comparison is case sensitive -->
 <valueSet value="[canonical]"/><!--  0..1 Canonical reference to the value set with entire code system -->
 <hierarchyMeaning value="[code]"/><!--  0..1 grouped-by | is-a | part-of | classified-with   -->
 <compositional value="[boolean]"/><!--  0..1 If code system defines a compositional grammar -->
 <versionNeeded value="[boolean]"/><!--  0..1 If definitions are not stable -->
 <content value="[code]"/><!--  1..1 not-present | example | fragment | complete | supplement   -->
 <supplements value="[canonical]"/><!--  0..1 Canonical URL of Code System this adds designations and properties to -->
 <count value="[unsignedInt]"/><!--  0..1 Total concepts in the code system -->
 <filter>  0..* BackboneElement  <!--  0..* Filter that can be used in a value set -->
```

```
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code value="[code]"/><!--  1..1 Code that identifies the filter -->
  <description value="[string]"/><!--  0..1 How or why the filter is used -->
  <operator value="[code]"/><!--  1..* = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists
       -->
  <value value="[string]"/><!--  1..1 What to use for the value -->
 </filter>
 <property>  0..* BackboneElement  <!--  0..* Additional information supplied about each concept -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code value="[code]"/><!--  1..1
      Identifies the property on the concepts, and when referred to in operations -->
  <uri value="[uri]"/><!--  0..1 Formal identifier for the property -->
  <description value="[string]"/><!--  0..1 Why the property is defined, and/or what it conveys -->
  <type value="[code]"/><!--  1..1 code | Coding | string | integer | boolean | dateTime | decimal   -->
 </property>
 <concept>  1..* BackboneElement  <!--  1..* Concepts in the code system -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code value="[code]"/><!--  1..1 Code that identifies concept -->
  <display value="[string]"/><!--  0..1 Text to display to the user -->
  <definition value="[string]"/><!--  0..1 Formal definition -->
  <designation>  0..* BackboneElement  <!--  0..* Additional representations for the concept -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <language value="[code]"/><!--  0..1 Human language of the designation   -->
   <use><!--  0..1 Coding Details how this designation would be used   --></use>
   <value value="[string]"/><!--  1..1 The text value for this designation -->
  </designation>
  <property>  0..* BackboneElement  <!--  0..* Property value for the concept -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <code value="[code]"/><!--  1..1 Reference to CodeSystem.property.code -->
   <value[x]><!--  1..1 code|Coding|string|integer|
     boolean|dateTime|decimal
       Value of the property for this concept --></value[x]>
  </property>
  <concept><!-- See #CodeSystem.concept  Child Concepts (is-a/contains/categorizes) --></concept>
 </concept>
</CodeSystem>
```

JSON Template

```
{ // Shareable CodeSystem
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C? R! Canonical identifier for this code system, represented as a URI (globally unique)
    (Coding.system)
    "identifier" : [{ Identifier }], // C?Additional identifier for the code system (business identifier)
    "version" : "<string>", // C? R! Business version of the code system (Coding.version)
    "name" : "<string>", // C? R! Name for this code system (computer friendly)
    "title" : "<string>", // C?Name for this code system (human friendly)
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C? R! For testing purposes, not real usage
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C? R! Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C? R! Natural language description of the code system
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for code system (if applicable)
    "purpose" : "<markdown>", // C?Why this code system is defined
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "caseSensitive" : <boolean>, // C?If code comparison is case sensitive
    "valueSet" : "<canonical>", // C?Canonical reference to the value set with entire code system
    "hierarchyMeaning" : "<code>", // C?grouped-by | is-a | part-of | classified-with
    "compositional" : <boolean>, // C?If code system defines a compositional grammar
    "versionNeeded" : <boolean>, // C?If definitions are not stable
    "content" : "<code>", // C? R! not-present | example | fragment | complete | supplement
    "supplements" : "<canonical>", // C?Canonical URL of Code System this adds designations and properties to
    "count" : "<unsignedInt>", // C?Total concepts in the code system
    "filter" : [{ BackboneElement }], // C?Filter that can be used in a value set
    "property" : [{ BackboneElement }], // C?Additional information supplied about each concept
    "concept" : [{ BackboneElement }] // C? R! Concepts in the code system
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

4.8.24.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
CodeSystem.language	Common Languages	preferred	Common Languages
CodeSystem.status	PublicationStatus	required	PublicationStatus
CodeSystem.jurisdiction	Jurisdiction	extensible	Jurisdiction
CodeSystem.hierarchyMeaning	CodeSystemHierarchyMeaning	required	CodeSystemHierarchyMeaning

CodeSystem.content | CodeSystemContentMode | required | CodeSystemContentMode
CodeSystem.filter.operator | FilterOperator | required | FilterOperator
CodeSystem.property.type | PropertyType | required | PropertyType
CodeSystem.concept.designation.language | Common Languages | preferred | Common Languages
CodeSystem.concept.designation.use | Designation Use | extensible | Designation Use

4.8.25 JSON Schema for CodeSystem

URL: https://hl7.org/fhir/R4/codesystem.schema.json.html

4.8.25 JSON Schema for CodeSystem

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for CodeSystem, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/CodeSystem",
  "$ref": "#/definitions/CodeSystem",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "CodeSystem": {
      "description": "The CodeSystem resource is used to declare the existence of and describe a code system or code
    system supplement and its key properties, and optionally define a part or all of its content.",
      "properties": {
        "resourceType": {
          "description": "This is a CodeSystem resource",
          "const": "CodeSystem"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
```

containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this code system when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this code system is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the code system is stored on different servers. This is used in [Coding](datatypes.html#Coding).system.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this code system when it is represented in other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the code system when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the code system author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. This is used in [Coding](datatypes.html#Coding).version.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the code system. This name should be usable as an identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the code system.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The date (and optionally time) when the code system resource was created or revised.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this code system is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date (and optionally time) when the code system was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the code system changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {

```
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the code system.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the code system from a consumer\u0027s
perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate code system instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the code system is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this code system is needed and why it has been designed as it has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the code system and/or its contents. Copyright statements
are generally legal restrictions on the use and publishing of the code system.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "caseSensitive": {
      "description": "If code comparison is case sensitive when codes within this system are compared to each
other.",
      "$ref": "#/definitions/boolean"
    },
    "_caseSensitive": {
      "description": "Extensions for caseSensitive",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueSet": {
      "description": "Canonical reference to the value set that contains the entire code system.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "hierarchyMeaning": {
      "description": "The meaning of the hierarchy of concepts as represented in this resource.",
      "enum": [
        "grouped-by",
        "is-a",
        "part-of",
        "classified-with"
      ]
    },
    "_hierarchyMeaning": {
      "description": "Extensions for hierarchyMeaning",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "compositional": {
      "description": "The code system defines a compositional (post-coordination) grammar.",
      "$ref": "#/definitions/boolean"
```

```
    },
    "_compositional": {
      "description": "Extensions for compositional",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "versionNeeded": {
      "description": "This flag is used to signify that the code system does not commit to concept permanence across
 versions. If true, a version must be specified when referencing this code system.",
      "$ref": "#/definitions/boolean"
    },
    "_versionNeeded": {
      "description": "Extensions for versionNeeded",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "content": {
      "description": "The extent of the content of the code system (the concepts and codes it defines) are
represented in this resource instance.",
      "enum": [
        "not-present",
        "example",
        "fragment",
        "complete",
        "supplement"
      ]
    },
    "_content": {
      "description": "Extensions for content",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "supplements": {
      "description": "The canonical URL of the code system that this code system supplement is adding designations
and properties to.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "count": {
      "description": "The total number of concepts defined by the code system. Where the code system has a
compositional grammar, the basis of this count is defined by the system steward.",
      "$ref": "#/definitions/unsignedInt"
    },
    "_count": {
      "description": "Extensions for count",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "filter": {
      "description": "A filter that can be used in a value set compose statement when selecting concepts using a
filter.",
      "items": {
        "$ref": "#/definitions/CodeSystem_Filter"
      },
      "type": "array"
    },
    "property": {
      "description": "A property defines an additional slot through which additional information can be provided
about a concept.",
      "items": {
        "$ref": "#/definitions/CodeSystem_Property"
      },
      "type": "array"
    },
    "concept": {
      "description": "Concepts that are in the code system. The concept definitions are inherently hierarchical, but
 the definitions must be consulted to determine what the meanings of the hierarchical relationships are.",
      "items": {
        "$ref": "#/definitions/CodeSystem_Concept"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"CodeSystem_Filter": {
  "description": "The CodeSystem resource is used to declare the existence of and describe a code system or code
system supplement and its key properties, and optionally define a part or all of its content.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
```

```
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "The code that identifies this filter when it is used as a filter in
[[[ValueSet]]].compose.include.filter.",
      "$ref": "#/definitions/code"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "A description of how or why the filter is used.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "operator": {
      "description": "A list of operators that can be used with the filter.",
      "items": {
        "$ref": "#/definitions/code"
      },
      "type": "array"
    },
    "_operator": {
      "description": "Extensions for operator",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "value": {
      "description": "A description of what the value for the filter should be.",
      "$ref": "#/definitions/string"
    },
    "_value": {
      "description": "Extensions for value",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"CodeSystem_Property": {
  "description": "The CodeSystem resource is used to declare the existence of and describe a code system or code
system supplement and its key properties, and optionally define a part or all of its content.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
```

```
      },
      "type": "array"
    },
    "code": {
      "description": "A code that is used to identify the property. The code is used internally (in
CodeSystem.concept.property.code) and also externally, such as in property filters.",
      "$ref": "#/definitions/code"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "uri": {
      "description": "Reference to the formal meaning of the property. One possible source of meaning is the
[Concept Properties](codesystem-concept-properties.html) code system.",
      "$ref": "#/definitions/uri"
    },
    "_uri": {
      "description": "Extensions for uri",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "A description of the property- why it is defined, and how its value might be used.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "The type of the property value. Properties of type \"code\" contain a code defined by the code
 system (e.g. a reference to another defined concept).",
      "enum": [
        "code",
        "Coding",
        "string",
        "integer",
        "boolean",
        "dateTime",
        "decimal"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"CodeSystem_Concept": {
  "description": "The CodeSystem resource is used to declare the existence of and describe a code system or code
system supplement and its key properties, and optionally define a part or all of its content.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "A code - a text symbol - that uniquely identifies the concept within the code system.",
      "$ref": "#/definitions/code"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
```

```
    },
    "display": {
      "description": "A human readable string that is the recommended default way to present this concept to a
user.",
      "$ref": "#/definitions/string"
    },
    "_display": {
      "description": "Extensions for display",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "definition": {
      "description": "The formal definition of the concept. The code system resource does not make formal
definitions required, because of the prevalence of legacy systems. However, they are highly recommended, as without
them there is no formal meaning associated with the concept.",
      "$ref": "#/definitions/string"
    },
    "_definition": {
      "description": "Extensions for definition",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "designation": {
      "description": "Additional representations for the concept - other languages, aliases, specialized purposes,
used for particular purposes, etc.",
      "items": {
        "$ref": "#/definitions/CodeSystem_Designation"
      },
      "type": "array"
    },
    "property": {
      "description": "A property value for this concept.",
      "items": {
        "$ref": "#/definitions/CodeSystem_Property1"
      },
      "type": "array"
    },
    "concept": {
      "description": "Defines children of a concept to produce a hierarchy of concepts. The nature of the
relationships is variable (is-a/contains/categorizes) - see hierarchyMeaning.",
      "items": {
        "$ref": "#/definitions/CodeSystem_Concept"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"CodeSystem_Designation": {
  "description": "The CodeSystem resource is used to declare the existence of and describe a code system or code
system supplement and its key properties, and optionally define a part or all of its content.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "language": {
      "description": "The language this designation is defined for.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "use": {
      "description": "A code that details how this designation would be used.",
```

```
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "value": {
      "description": "The text value for this designation.",
      "$ref": "#/definitions/string"
    },
    "_value": {
      "description": "Extensions for value",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"CodeSystem_Property1": {
  "description": "The CodeSystem resource is used to declare the existence of and describe a code system or code
system supplement and its key properties, and optionally define a part or all of its content.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "A code that is a reference to CodeSystem.property.code.",
      "$ref": "#/definitions/code"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueCode": {
      "description": "The value of this property.",
      "pattern": "^[^\\s]+(\\s[^\\s]+)*$",
      "type": "string"
    },
    "_valueCode": {
      "description": "Extensions for valueCode",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueCoding": {
      "description": "The value of this property.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "valueString": {
      "description": "The value of this property.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_valueString": {
      "description": "Extensions for valueString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueInteger": {
      "description": "The value of this property.",
      "pattern": "^-?([0]|([1-9][0-9]*))$",
      "type": "number"
    },
    "_valueInteger": {
      "description": "Extensions for valueInteger",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueBoolean": {
      "description": "The value of this property.",
      "pattern": "^true|false$",
      "type": "boolean"
```

```
        },
        "_valueBoolean": {
          "description": "Extensions for valueBoolean",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "valueDateTime": {
          "description": "The value of this property.",
          "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
    1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00))?)?)?$",
          "type": "string"
        },
        "_valueDateTime": {
          "description": "Extensions for valueDateTime",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "valueDecimal": {
          "description": "The value of this property.",
          "pattern": "^-?(0|[1-9][0-9]*)(\\.[0-9]+)?([eE][+-]?[0-9]+)?$",
          "type": "number"
        },
        "_valueDecimal": {
          "description": "Extensions for valueDecimal",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```


4.8.26 Resource CodeSystem - Maturity Dependency View

URL: https://hl7.org/fhir/R4/codesystem-dependencies.html

4.8.26 Resource CodeSystem - Maturity Dependency View

Vocabulary Work Group | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| Some of the content on this page (marked clearly) has been approved as part of an ANSI standard. See the Conformance Package for further details.

Dependency Graph for CodeSystem FMM level 5

Name | Card. | Type | Dependency Analysis
CodeSystem | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext TU | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
copyright | 0..1 | markdown
caseSensitive | 0..1 | boolean
valueSet | 0..1 | canonical(ValueSet)
hierarchyMeaning | 0..1 | code
compositional | 0..1 | boolean
versionNeeded | 0..1 | boolean
content | 1..1 | code
supplements | 0..1 | canonical(CodeSystem)
count | 0..1 | unsignedInt
filter | 0..* | BackboneElement
code | 1..1 | code
description | 0..1 | string
operator | 1..* | code
value | 1..1 | string
property | 0..* | BackboneElement
code | 1..1 | code
uri | 0..1 | uri
description | 0..1 | string
type | 1..1 | code
concept | 0..* | BackboneElement
code | 1..1 | code

display | 0..1 | string
definition | 0..1 | string
designation | 0..* | BackboneElement
language | 0..1 | code
use | 0..1 | Coding
value | 1..1 | string
property | 0..* | BackboneElement
code | 1..1 | code
value[x] | 1..1
valueCode | | code
valueCoding | | Coding
valueString | | string
valueInteger | | integer
valueBoolean | | boolean
valueDateTime | | dateTime
valueDecimal | | decimal
concept | 0..* | see concept

4.9 Resource ValueSet - Content

URL: https://hl7.org/fhir/R4/valueset.html
Vocabulary Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Anonymous | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Conformance Package for further details.

A ValueSet resource instance specifies a set of codes drawn from one or more code systems, intended for use in a particular context. Value sets link between CodeSystem definitions and their use in coded elements.

4.9.1 Scope and Usage

The FHIR terminology specification is based on two key concepts, originally defined in HL7 v3 Core Principles :

• CodeSystem - declares the existence of and describes a code system or code system supplement and its key properties, and optionally defines a part or all of its content. Also known as Ontology, Terminology, or Enumeration

• ValueSet - specifies a set of codes drawn from one or more code systems, intended for use in a particular context. Value sets link between CodeSystem definitions and their use in coded elements

Value sets have 2 aspects:

• .compose: A definition of which codes are intended to be in the value set ("intension")

• .expansion: The list of codes that are actually in the value set under a given set of conditions ("extension") - see Value Set Expansion

The ValueSet resource can carry either the .compose or the .expansion, both of them, or neither of them (if only the metadata is being represented). There is an "$expand" operation which can be used to ask a server to generate an expansion given the composition rules, in a particular context, and a "$validate-code" operation which can be used to ask a server to check whether a given code or concept is in the value set in a particular context.

4.9.2 Boundaries and Relationships

Value Sets are used by many resources:

• Value sets use CodeSystem resources by referring to them via their canonical reference.

• Value sets are used in StructureDefinition, OperationDefinition, Questionnaire, and other resources to specify the allowable contents for coded elements, or business rules for data processing

• ConceptMap resources describe mappings between value sets

• DataRequirement data types to specify data processing conditions

For a full list of uses, see below.

The Characteristics of the ValueSet resource are derived from Formal Value Set Definitions:

• The ValueSet resource design is based on the functionality described in the OMG CTS 2 specification, along with metadata in the HL7 Value Set Definition specification. Value set resources can be converted to CTS2 value set or code system instances.

• The value set resource is aligned with the Value Set Definition (VSD) project. Not all of the elements defined by the VSD are part of the base resource - some are defined as part of the ValueSet Extensions. In the ValueSet resource, the compose element is the VSD "Content Logical definition".

4.9.3 Background and Context

When using value sets, proper differentiation between a code system and a value set is important. This is one very common area where significant clinical safety risks occur in practice. Implementers should be familiar

with the content in Using Codes in Resources.

4.9.3.1 ValueSet Identification

A value set has 3 identifiers:

• ValueSet.id: the logical id on the system that holds the value set - this changes as it moves from server to server (this id, with the server address prepended, is called the 'literal identity' of the resource)

• ValueSet.url: the canonical URL that never changes for this value set - it is the same in every copy. The element is named url rather than uri for legacy reasons and to strongly encourage providing a resolvable URL as the identifier whenever possible. Ideally, it should be a literal URL that is the location of the master version of the value set, though this is not always possible

• ValueSet.identifier: A system/value pair that is used to identify the value set in other contexts (such as an OID in an HL7 v3 specification)

In addition, any expansion for the value set also has ValueSet.expansion.identifier which uniquely identifies each expansion. For further information regarding resource identification, see Resource Identity.

This means that each value set has 2 different URLs that can be used to reference it - its canonical URL (the url element), and its local location from which it may be retrieved (which includes the id element). Because it is common practice to copy (cache) value sets locally, most references to value sets use the canonical URL.

For example, the value sets published as part of FHIR all have a location ("literal") URI which is the URL where they may be accessed in the FHIR specification itself. Note, though, that while a new version of the FHIR Specification is being prepared, value sets that are published in the drafts will not be found in the current published FHIR specification at their canonical URL.

Alternatively, the identifier and version elements may be used to reference this value set in a design, a profile, a CDA template or HL7 v3 message (in the CD data type valueSet and valueSetVersion properties). These different contexts may make additional restrictions on the possible values of these elements. The identifier is generally not needed when using value sets in a FHIR context, where the canonical URL is always the focus.

4.9.3.2 Intensional vs Extensional

A value set may be described as intensional or extensional. The terms intensional and extensional come from the fields of mathematical logic and set theory.

An intensional value set is typically algorithmically defined. That is, the code group is defined as a rule e.g. all codes with the word diabetes in their description). The key benefit of intensional code groups is that they can be dynamically updated. Dynamic updating helps healthcare organizations keep current when new drugs (and their associated codes) become available or codes for diseases and other clinical concepts change. An intensional value set designed to contain all of the drugs in the beta blocker category can automatically receive a new beta blocker’s code as soon as it hits the market.

Extensional value sets, meanwhile, are enumerated lists of codes where each code is listed individually. This gives the author and user of the value set more control over the which codes are in the value set, but there is a greater maintenance burden to ensure that the value set is kept up to date.

This resource is referenced by DataRequirement, ElementDefinition, CodeSystem, ConceptMap, ObservationDefinition, OperationDefinition, Questionnaire, ResearchElementDefinition and itself

4.9.4 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints

ValueSet | IN | | DomainResource | A set of codes drawn from one or more code systems + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | Σ | 0..1 | uri | Canonical identifier for this value set, represented as a URI (globally unique)

identifier | Σ | 0..* | Identifier | Additional identifier for the value set (business identifier)

version | Σ | 0..1 | string | Business version of the value set

name | ΣI | 0..1 | string | Name for this value set (computer friendly)

title | Σ | 0..1 | string | Name for this value set (human friendly)

status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | Σ | 0..1 | boolean | For testing purposes, not real usage

date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the value set
useContext | ΣTU | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for value set (if applicable) Jurisdiction (Extensible)
immutable | Σ | 0..1 | boolean | Indicates whether or not any change to the content logical definition may occur
purpose | | 0..1 | markdown | Why this value set is defined
copyright | | 0..1 | markdown | Use and/or publishing restrictions
compose | | 0..1 | BackboneElement | Content logical definition of the value set (CLD)
lockedDate | Σ | 0..1 | date | Fixed date for references with no specified version (transitive)
inactive | Σ | 0..1 | boolean | Whether inactive codes are in the value set
include | ΣI | 1..* | BackboneElement | Include one or more codes from a code system or other value set(s) + Rule: A value set include/exclude SHALL have a value set or a system + Rule: A value set with concepts or filters SHALL include a system + Rule: Cannot have both concept and filter
system | ΣI | 0..1 | uri | The system the codes come from
version | Σ | 0..1 | string | Specific version of the code system referred to
concept | I | 0..* | BackboneElement | A concept defined in the system
code | | 1..1 | code | Code or expression from system
display | | 0..1 | string | Text to display for this code for this value set in this valueset
designation | | 0..* | BackboneElement | Additional representations for this concept
language | | 0..1 | code | Human language of the designation Common Languages (Preferred but limited to AllLanguages)
use | | 0..1 | Coding | Types of uses of designations Designation Use (Extensible)
value | | 1..1 | string | The text value for this designation
filter | ΣI | 0..* | BackboneElement | Select codes/concepts by their properties (including relationships)
property | Σ | 1..1 | code | A property/filter defined by the code system
op | Σ | 1..1 | code | = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists FilterOperator (Required)
value | Σ | 1..1 | string | Code from the system, or regex criteria, or boolean value for exists
valueSet | ΣI | 0..* | canonical(ValueSet) | Select the contents included in this value set
exclude | I | 0..* | see include | Explicitly exclude codes from a code system or other value sets
expansion | | 0..1 | BackboneElement | Used when the value set is "expanded"
identifier | | 0..1 | uri | Identifies the value set expansion (business identifier)
timestamp | | 1..1 | dateTime | Time ValueSet expansion happened
total | | 0..1 | integer | Total number of codes in the expansion
offset | | 0..1 | integer | Offset at which this resource starts
parameter | | 0..* | BackboneElement | Parameter that controlled the expansion process
name | | 1..1 | string | Name as assigned by the client or server
value[x] | | 0..1 | | Value of the named parameter
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueDecimal | | | decimal
valueUri | | | uri
valueCode | | | code
valueDateTime | | | dateTime
contains | I | 0..* | BackboneElement | Codes in the value set + Rule: SHALL have a code or a display + Rule: Must have a code if not abstract + Rule: Must have a system if a code is present
system | | 0..1 | uri | System value for the code
abstract | | 0..1 | boolean | If user cannot select this entry
inactive | | 0..1 | boolean | If concept is inactive in the code system
version | | 0..1 | string | Version in which this code/display is defined
code | I | 0..1 | code | Code - if blank, this is not a selectable code
display | I | 0..1 | string | User display for the concept
designation | | 0..* | see designation | Additional representations for this item
contains | | 0..* | see contains | Codes contained under this entry
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ValueSet xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this value set, represented as a URI (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the value set (business identifier) --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the value set -->
 <name value="[string]"/><!--  0..1 Name for this value set (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this value set (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the value set -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for value set (if applicable) --></jurisdiction>
 <immutable value="[boolean]"/><!-- 0..1 Indicates whether or not any change to the content logical definition may occur
```

```
    -->
 <purpose value="[markdown]"/><!-- 0..1 Why this value set is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <compose>  <!-- 0..1 Content logical definition of the value set (CLD) -->
  <lockedDate value="[date]"/><!-- 0..1 Fixed date for references with no specified version (transitive) -->
  <inactive value="[boolean]"/><!-- 0..1 Whether inactive codes are in the value set -->
  <include>  <!-- 1..* Include one or more codes from a code system or other value set(s) -->
   <system value="[uri]"/><!--  0..1 The system the codes come from -->
   <version value="[string]"/><!-- 0..1 Specific version of the code system referred to -->
   <concept>  <!--  0..* A concept defined in the system -->
    <code value="[code]"/><!-- 1..1 Code or expression from system -->
    <display value="[string]"/><!-- 0..1 Text to display for this code for this value set in this valueset -->
    <designation>  <!-- 0..* Additional representations for this concept -->
     <language value="[code]"/><!-- 0..1 Human language of the designation -->
     <use><!-- 0..1 Coding Types of uses of designations --></use>
     <value value="[string]"/><!-- 1..1 The text value for this designation -->
    </designation>
   </concept>
   <filter>  <!--  0..* Select codes/concepts by their properties (including relationships) -->
    <property value="[code]"/><!-- 1..1 A property/filter defined by the code system -->
    <op value="[code]"/><!-- 1..1 = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists -->
    <value value="[string]"/><!-- 1..1 Code from the system, or regex criteria, or boolean value for exists -->
   </filter>
   <valueSet><!--  0..* canonical(ValueSet) Select the contents included in this value set --></valueSet>
  </include>
  <exclude><!--  0..* Content as for ValueSet.compose.include Explicitly exclude codes from a code system or other value
     sets --></exclude>
 </compose>
 <expansion>  <!-- 0..1 Used when the value set is "expanded" -->
  <identifier value="[uri]"/><!-- 0..1 Identifies the value set expansion (business identifier) -->
  <timestamp value="[dateTime]"/><!-- 1..1 Time ValueSet expansion happened -->
  <total value="[integer]"/><!-- 0..1 Total number of codes in the expansion -->
  <offset value="[integer]"/><!-- 0..1 Offset at which this resource starts -->
  <parameter>  <!-- 0..* Parameter that controlled the expansion process -->
   <name value="[string]"/><!-- 1..1 Name as assigned by the client or server -->
   <value[x]><!-- 0..1 string|boolean|integer|decimal|uri|code|dateTime Value of the named parameter --></value[x]>
  </parameter>
  <contains>  <!-- 0..* Codes in the value set -->
   <system value="[uri]"/><!-- 0..1 System value for the code -->
   <abstract value="[boolean]"/><!-- 0..1 If user cannot select this entry -->
   <inactive value="[boolean]"/><!-- 0..1 If concept is inactive in the code system -->
   <version value="[string]"/><!-- 0..1 Version in which this code/display is defined -->
   <code value="[code]"/><!--  0..1 Code - if blank, this is not a selectable code -->
   <display value="[string]"/><!--  0..1 User display for the concept -->
   <designation><!-- 0..* Content as for ValueSet.compose.include.concept.designation Additional representations for
    this item --></designation>
   <contains><!-- 0..* Content as for ValueSet.expansion.contains Codes contained under this entry --></contains>
  </contains>
 </expansion>
</ValueSet>
```

JSON Template

```
{
  "resourceType" : "ValueSet",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this value set, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the value set (business identifier)
  "version" : "<string>", // Business version of the value set
  "name" : "<string>", // C? Name for this value set (computer friendly)
  "title" : "<string>", // Name for this value set (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the value set
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for value set (if applicable)
  "immutable" : <boolean>, // Indicates whether or not any change to the content logical definition may occur
  "purpose" : "<markdown>", // Why this value set is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "compose" : { // Content logical definition of the value set (CLD)
    "lockedDate" : "<date>", // Fixed date for references with no specified version (transitive)
    "inactive" : <boolean>, // Whether inactive codes are in the value set
    "include" : [{ // R!  Include one or more codes from a code system or other value set(s)
      "system" : "<uri>", // C? The system the codes come from
      "version" : "<string>", // Specific version of the code system referred to
      "concept" : [{ // C? A concept defined in the system
        "code" : "<code>", // R!  Code or expression from system
        "display" : "<string>", // Text to display for this code for this value set in this valueset
        "designation" : [{ // Additional representations for this concept
          "language" : "<code>", // Human language of the designation
          "use" : { Coding }, // Types of uses of designations
          "value" : "<string>" // R!  The text value for this designation
        }]
      }],
      "filter" : [{ // C? Select codes/concepts by their properties (including relationships)
        "property" : "<code>", // R!  A property/filter defined by the code system
```

```
      "op" : "<code>", // R!  = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists
      "value" : "<string>" // R!  Code from the system, or regex criteria, or boolean value for exists
    }],
    "valueSet" : [{ canonical(ValueSet) }] // C? Select the contents included in this value set
  }],
  "exclude" : [{ Content as for ValueSet.compose.include }] // C? Explicitly exclude codes from a code system or other
   value sets
 },
 "expansion" : { // Used when the value set is "expanded"
  "identifier" : "<uri>", // Identifies the value set expansion (business identifier)
  "timestamp" : "<dateTime>", // R!  Time ValueSet expansion happened
  "total" : <integer>, // Total number of codes in the expansion
  "offset" : <integer>, // Offset at which this resource starts
  "parameter" : [{ // Parameter that controlled the expansion process
    "name" : "<string>", // R!  Name as assigned by the client or server
    // value[x]: Value of the named parameter. One of these 7:
    "valueString" : "<string>"
    "valueBoolean" : <boolean>
    "valueInteger" : <integer>
    "valueDecimal" : <decimal>
    "valueUri" : "<uri>"
    "valueCode" : "<code>"
    "valueDateTime" : "<dateTime>"
  }],
  "contains" : [{ // Codes in the value set
    "system" : "<uri>", // System value for the code
    "abstract" : <boolean>, // If user cannot select this entry
    "inactive" : <boolean>, // If concept is inactive in the code system
    "version" : "<string>", // Version in which this code/display is defined
    "code" : "<code>", // C? Code - if blank, this is not a selectable code
    "display" : "<string>", // C? User display for the concept
    "designation" : [{ Content as for ValueSet.compose.include.concept.designation }], // Additional representations
  for this item
    "contains" : [{ Content as for ValueSet.expansion.contains }] // Codes contained under this entry
  }]
 }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ValueSet;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ValueSet.url [ uri ]; # 0..1 Canonical identifier for this value set, represented as a URI (globally unique)
  fhir:ValueSet.identifier [ Identifier ], ... ; # 0..* Additional identifier for the value set (business identifier)
  fhir:ValueSet.version [ string ]; # 0..1 Business version of the value set
  fhir:ValueSet.name [ string ]; # 0..1 Name for this value set (computer friendly)
  fhir:ValueSet.title [ string ]; # 0..1 Name for this value set (human friendly)
  fhir:ValueSet.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ValueSet.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ValueSet.date [ dateTime ]; # 0..1 Date last changed
  fhir:ValueSet.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ValueSet.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ValueSet.description [ markdown ]; # 0..1 Natural language description of the value set
  fhir:ValueSet.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:ValueSet.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for value set (if applicable)
  fhir:ValueSet.immutable [ boolean ]; # 0..1 Indicates whether or not any change to the content logical definition may
    occur
  fhir:ValueSet.purpose [ markdown ]; # 0..1 Why this value set is defined
  fhir:ValueSet.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ValueSet.compose [ # 0..1 Content logical definition of the value set (CLD)
    fhir:ValueSet.compose.lockedDate [ date ]; # 0..1 Fixed date for references with no specified version (transitive)
    fhir:ValueSet.compose.inactive [ boolean ]; # 0..1 Whether inactive codes are in the value set
    fhir:ValueSet.compose.include [ # 1..* Include one or more codes from a code system or other value set(s)
      fhir:ValueSet.compose.include.system [ uri ]; # 0..1 The system the codes come from
      fhir:ValueSet.compose.include.version [ string ]; # 0..1 Specific version of the code system referred to
      fhir:ValueSet.compose.include.concept [ # 0..* A concept defined in the system
        fhir:ValueSet.compose.include.concept.code [ code ]; # 1..1 Code or expression from system
        fhir:ValueSet.compose.include.concept.display [ string ]; # 0..1 Text to display for this code for this value
    set in this valueset
        fhir:ValueSet.compose.include.concept.designation [ # 0..* Additional representations for this concept
          fhir:ValueSet.compose.include.concept.designation.language [ code ]; # 0..1 Human language of the designation
          fhir:ValueSet.compose.include.concept.designation.use [ Coding ]; # 0..1 Types of uses of designations
          fhir:ValueSet.compose.include.concept.designation.value [ string ]; # 1..1 The text value for this designation
        ], ...;
      ], ...;
      fhir:ValueSet.compose.include.filter [ # 0..* Select codes/concepts by their properties (including relationships)
        fhir:ValueSet.compose.include.filter.property [ code ]; # 1..1 A property/filter defined by the code system
        fhir:ValueSet.compose.include.filter.op [ code ]; # 1..1 = | is-a | descendent-of | is-not-a | regex | in | not-
    in | generalizes | exists
        fhir:ValueSet.compose.include.filter.value [ string ]; # 1..1 Code from the system, or regex criteria, or
    boolean value for exists
      ], ...;
      fhir:ValueSet.compose.include.valueSet [ canonical(ValueSet) ], ... ; # 0..* Select the contents included in this
    value set
```

```
    ], ...;
    fhir:ValueSet.compose.exclude [ See ValueSet.compose.include ], ... ; # 0..* Explicitly exclude codes from a code
    system or other value sets
  ];
  fhir:ValueSet.expansion [ # 0..1 Used when the value set is "expanded"
    fhir:ValueSet.expansion.identifier [ uri ]; # 0..1 Identifies the value set expansion (business identifier)
    fhir:ValueSet.expansion.timestamp [ dateTime ]; # 1..1 Time ValueSet expansion happened
    fhir:ValueSet.expansion.total [ integer ]; # 0..1 Total number of codes in the expansion
    fhir:ValueSet.expansion.offset [ integer ]; # 0..1 Offset at which this resource starts
    fhir:ValueSet.expansion.parameter [ # 0..* Parameter that controlled the expansion process
      fhir:ValueSet.expansion.parameter.name [ string ]; # 1..1 Name as assigned by the client or server
      # ValueSet.expansion.parameter.value[x] : 0..1 Value of the named parameter. One of these 7
        fhir:ValueSet.expansion.parameter.valueString [ string ]
        fhir:ValueSet.expansion.parameter.valueBoolean [ boolean ]
        fhir:ValueSet.expansion.parameter.valueInteger [ integer ]
        fhir:ValueSet.expansion.parameter.valueDecimal [ decimal ]
        fhir:ValueSet.expansion.parameter.valueUri [ uri ]
        fhir:ValueSet.expansion.parameter.valueCode [ code ]
        fhir:ValueSet.expansion.parameter.valueDateTime [ dateTime ]
    ], ...;
    fhir:ValueSet.expansion.contains [ # 0..* Codes in the value set
      fhir:ValueSet.expansion.contains.system [ uri ]; # 0..1 System value for the code
      fhir:ValueSet.expansion.contains.abstract [ boolean ]; # 0..1 If user cannot select this entry
      fhir:ValueSet.expansion.contains.inactive [ boolean ]; # 0..1 If concept is inactive in the code system
      fhir:ValueSet.expansion.contains.version [ string ]; # 0..1 Version in which this code/display is defined
      fhir:ValueSet.expansion.contains.code [ code ]; # 0..1 Code - if blank, this is not a selectable code
      fhir:ValueSet.expansion.contains.display [ string ]; # 0..1 User display for the concept
      fhir:ValueSet.expansion.contains.designation [ See ValueSet.compose.include.concept.designation ], ... ; # 0..*
    Additional representations for this item
      fhir:ValueSet.expansion.contains.contains [ See ValueSet.expansion.contains ], ... ; # 0..* Codes contained under
    this entry
    ], ...;
  ];
]
```

Changes since R3

ValueSet

ValueSet.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

ValueSet.experimental | No longer marked as Modifier

ValueSet.compose.include.concept.designation.language | Change binding strength from extensible to preferred

ValueSet.compose.include.filter | No longer marked as Modifier

ValueSet.compose.include.filter.op | Change value set from http://hl7.org/fhir/ValueSet/filter-operator to http://hl7.org/fhir/ValueSet/filter-operator|4.0.1

ValueSet.compose.include.filter.value | Type changed from code to string

ValueSet.compose.include.valueSet | Type changed from uri to canonical(ValueSet)

ValueSet.expansion.identifier | Min Cardinality changed from 1 to 0

ValueSet.expansion.parameter.value[x] | Add Type dateTime

ValueSet.expansion.contains.abstract | Default Value "false" removed

ValueSet.expansion.contains.inactive | Default Value "false" removed

ValueSet.extensible | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 8 tests that all execute ok. All tests pass round-trip testing and 8 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

ValueSet | IN | | DomainResource | A set of codes drawn from one or more code systems + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | Σ | 0..1 | uri | Canonical identifier for this value set, represented as a URI (globally unique)

identifier | Σ | 0..* | Identifier | Additional identifier for the value set (business identifier)

version | Σ | 0..1 | string | Business version of the value set

name | ΣI | 0..1 | string | Name for this value set (computer friendly)

title | Σ | 0..1 | string | Name for this value set (human friendly)

status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | Σ | 0..1 | boolean | For testing purposes, not real usage

date | Σ | 0..1 | dateTime | Date last changed

publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)

contact | Σ | 0..* | ContactDetail | Contact details for the publisher

description | | 0..1 | markdown | Natural language description of the value set

useContext | ΣTU | 0..* | UsageContext | The context that the content is intended to support

jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for value set (if applicable) Jurisdiction (Extensible)

immutable | Σ | 0..1 | boolean | Indicates whether or not any change to the content logical definition may occur

purpose | | 0..1 | markdown | Why this value set is defined

copyright | | 0..1 | markdown | Use and/or publishing restrictions

compose | | 0..1 | BackboneElement | Content logical definition of the value set (CLD)

lockedDate | Σ | 0..1 | date | Fixed date for references with no specified version (transitive)
inactive | Σ | 0..1 | boolean | Whether inactive codes are in the value set
include | ΣI | 1..* | BackboneElement | Include one or more codes from a code system or other value set(s) + Rule: A value set include/exclude SHALL have a value set or a system + Rule: A value set with concepts or filters SHALL include a system + Rule: Cannot have both concept and filter
system | ΣI | 0..1 | uri | The system the codes come from
version | Σ | 0..1 | string | Specific version of the code system referred to
concept | I | 0..* | BackboneElement | A concept defined in the system
code | | 1..1 | code | Code or expression from system
display | | 0..1 | string | Text to display for this code for this value set in this valueset
designation | | 0..* | BackboneElement | Additional representations for this concept
language | | 0..1 | code | Human language of the designation Common Languages (Preferred but limited to AllLanguages)
use | | 0..1 | Coding | Types of uses of designations Designation Use (Extensible)
value | | 1..1 | string | The text value for this designation
filter | ΣI | 0..* | BackboneElement | Select codes/concepts by their properties (including relationships)
property | Σ | 1..1 | code | A property/filter defined by the code system
op | Σ | 1..1 | code | = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists FilterOperator (Required)
value | Σ | 1..1 | string | Code from the system, or regex criteria, or boolean value for exists
valueSet | ΣI | 0..* | canonical(ValueSet) | Select the contents included in this value set
exclude | I | 0..* | see include | Explicitly exclude codes from a code system or other value sets
expansion | | 0..1 | BackboneElement | Used when the value set is "expanded"
identifier | | 0..1 | uri | Identifies the value set expansion (business identifier)
timestamp | | 1..1 | dateTime | Time ValueSet expansion happened
total | | 0..1 | integer | Total number of codes in the expansion
offset | | 0..1 | integer | Offset at which this resource starts
parameter | | 0..* | BackboneElement | Parameter that controlled the expansion process
name | | 1..1 | string | Name as assigned by the client or server
value[x] | | 0..1 | | Value of the named parameter
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueDecimal | | | decimal
valueUri | | | uri
valueCode | | | code
valueDateTime | | | dateTime
contains | I | 0..* | BackboneElement | Codes in the value set + Rule: SHALL have a code or a display + Rule: Must have a code if not abstract + Rule: Must have a system if a code is present
system | | 0..1 | uri | System value for the code
abstract | | 0..1 | boolean | If user cannot select this entry
inactive | | 0..1 | boolean | If concept is inactive in the code system
version | | 0..1 | string | Version in which this code/display is defined
code | I | 0..1 | code | Code - if blank, this is not a selectable code
display | I | 0..1 | string | User display for the concept
designation | | 0..* | see designation | Additional representations for this item
contains | | 0..* | see contains | Codes contained under this entry
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ValueSet xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this value set, represented as a URI (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the value set (business identifier) --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the value set -->
 <name value="[string]"/><!--  0..1 Name for this value set (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this value set (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the value set -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for value set (if applicable) --></jurisdiction>
 <immutable value="[boolean]"/><!-- 0..1 Indicates whether or not any change to the content logical definition may occur
     -->
 <purpose value="[markdown]"/><!-- 0..1 Why this value set is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <compose>  <!-- 0..1 Content logical definition of the value set (CLD) -->
  <lockedDate value="[date]"/><!-- 0..1 Fixed date for references with no specified version (transitive) -->
  <inactive value="[boolean]"/><!-- 0..1 Whether inactive codes are in the value set -->
  <include>  <!-- 1..* Include one or more codes from a code system or other value set(s) -->
   <system value="[uri]"/><!--  0..1 The system the codes come from -->
   <version value="[string]"/><!-- 0..1 Specific version of the code system referred to -->
   <concept>  <!--  0..* A concept defined in the system -->
    <code value="[code]"/><!-- 1..1 Code or expression from system -->
    <display value="[string]"/><!-- 0..1 Text to display for this code for this value set in this valueset -->
```

```
    <designation>  <!-- 0..* Additional representations for this concept -->
     <language value="[code]"/><!-- 0..1 Human language of the designation -->
     <use><!-- 0..1 Coding Types of uses of designations --></use>
     <value value="[string]"/><!-- 1..1 The text value for this designation -->
    </designation>
   </concept>
   <filter>  <!--  0..* Select codes/concepts by their properties (including relationships) -->
    <property value="[code]"/><!-- 1..1 A property/filter defined by the code system -->
    <op value="[code]"/><!-- 1..1 = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists -->
    <value value="[string]"/><!-- 1..1 Code from the system, or regex criteria, or boolean value for exists -->
   </filter>
   <valueSet><!--  0..* canonical(ValueSet) Select the contents included in this value set --></valueSet>
  </include>
  <exclude><!--  0..* Content as for ValueSet.compose.include Explicitly exclude codes from a code system or other value
     sets --></exclude>
 </compose>
 <expansion>  <!-- 0..1 Used when the value set is "expanded" -->
  <identifier value="[uri]"/><!-- 0..1 Identifies the value set expansion (business identifier) -->
  <timestamp value="[dateTime]"/><!-- 1..1 Time ValueSet expansion happened -->
  <total value="[integer]"/><!-- 0..1 Total number of codes in the expansion -->
  <offset value="[integer]"/><!-- 0..1 Offset at which this resource starts -->
  <parameter>  <!-- 0..* Parameter that controlled the expansion process -->
   <name value="[string]"/><!-- 1..1 Name as assigned by the client or server -->
   <value[x]><!-- 0..1 string|boolean|integer|decimal|uri|code|dateTime Value of the named parameter --></value[x]>
  </parameter>
  <contains>  <!-- 0..* Codes in the value set -->
   <system value="[uri]"/><!-- 0..1 System value for the code -->
   <abstract value="[boolean]"/><!-- 0..1 If user cannot select this entry -->
   <inactive value="[boolean]"/><!-- 0..1 If concept is inactive in the code system -->
   <version value="[string]"/><!-- 0..1 Version in which this code/display is defined -->
   <code value="[code]"/><!--  0..1 Code - if blank, this is not a selectable code -->
   <display value="[string]"/><!--  0..1 User display for the concept -->
   <designation><!-- 0..* Content as for ValueSet.compose.include.concept.designation Additional representations for
    this item --></designation>
   <contains><!-- 0..* Content as for ValueSet.expansion.contains Codes contained under this entry --></contains>
  </contains>
 </expansion>
</ValueSet>
```

JSON Template

```
{
  "resourceType" : "ValueSet",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this value set, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the value set (business identifier)
  "version" : "<string>", // Business version of the value set
  "name" : "<string>", // C? Name for this value set (computer friendly)
  "title" : "<string>", // Name for this value set (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the value set
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for value set (if applicable)
  "immutable" : <boolean>, // Indicates whether or not any change to the content logical definition may occur
  "purpose" : "<markdown>", // Why this value set is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "compose" : { // Content logical definition of the value set (CLD)
    "lockedDate" : "<date>", // Fixed date for references with no specified version (transitive)
    "inactive" : <boolean>, // Whether inactive codes are in the value set
    "include" : [{ // R!  Include one or more codes from a code system or other value set(s)
      "system" : "<uri>", // C? The system the codes come from
      "version" : "<string>", // Specific version of the code system referred to
      "concept" : [{ // C? A concept defined in the system
        "code" : "<code>", // R!  Code or expression from system
        "display" : "<string>", // Text to display for this code for this value set in this valueset
        "designation" : [{ // Additional representations for this concept
          "language" : "<code>", // Human language of the designation
          "use" : { Coding }, // Types of uses of designations
          "value" : "<string>" // R!  The text value for this designation
        }]
      }],
      "filter" : [{ // C? Select codes/concepts by their properties (including relationships)
        "property" : "<code>", // R!  A property/filter defined by the code system
        "op" : "<code>", // R!  = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists
        "value" : "<string>" // R!  Code from the system, or regex criteria, or boolean value for exists
      }],
      "valueSet" : [{ canonical(ValueSet) }] // C? Select the contents included in this value set
    }],
    "exclude" : [{ Content as for ValueSet.compose.include }] // C? Explicitly exclude codes from a code system or other
     value sets
  },
  "expansion" : { // Used when the value set is "expanded"
    "identifier" : "<uri>", // Identifies the value set expansion (business identifier)
    "timestamp" : "<dateTime>", // R!  Time ValueSet expansion happened
    "total" : <integer>, // Total number of codes in the expansion
```

```
    "offset" : <integer>, // Offset at which this resource starts
    "parameter" : [{ // Parameter that controlled the expansion process
      "name" : "<string>", // R!  Name as assigned by the client or server
      // value[x]: Value of the named parameter. One of these 7:
      "valueString" : "<string>"
      "valueBoolean" : <boolean>
      "valueInteger" : <integer>
      "valueDecimal" : <decimal>
      "valueUri" : "<uri>"
      "valueCode" : "<code>"
      "valueDateTime" : "<dateTime>"
    }],
    "contains" : [{ // Codes in the value set
      "system" : "<uri>", // System value for the code
      "abstract" : <boolean>, // If user cannot select this entry
      "inactive" : <boolean>, // If concept is inactive in the code system
      "version" : "<string>", // Version in which this code/display is defined
      "code" : "<code>", // C? Code - if blank, this is not a selectable code
      "display" : "<string>", // C? User display for the concept
      "designation" : [{ Content as for ValueSet.compose.include.concept.designation }], // Additional representations
    for this item
      "contains" : [{ Content as for ValueSet.expansion.contains }] // Codes contained under this entry
    }]
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ValueSet;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ValueSet.url [ uri ]; # 0..1 Canonical identifier for this value set, represented as a URI (globally unique)
  fhir:ValueSet.identifier [ Identifier ], ... ; # 0..* Additional identifier for the value set (business identifier)
  fhir:ValueSet.version [ string ]; # 0..1 Business version of the value set
  fhir:ValueSet.name [ string ]; # 0..1 Name for this value set (computer friendly)
  fhir:ValueSet.title [ string ]; # 0..1 Name for this value set (human friendly)
  fhir:ValueSet.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ValueSet.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ValueSet.date [ dateTime ]; # 0..1 Date last changed
  fhir:ValueSet.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ValueSet.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ValueSet.description [ markdown ]; # 0..1 Natural language description of the value set
  fhir:ValueSet.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:ValueSet.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for value set (if applicable)
  fhir:ValueSet.immutable [ boolean ]; # 0..1 Indicates whether or not any change to the content logical definition may
    occur
  fhir:ValueSet.purpose [ markdown ]; # 0..1 Why this value set is defined
  fhir:ValueSet.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ValueSet.compose [ # 0..1 Content logical definition of the value set (CLD)
    fhir:ValueSet.compose.lockedDate [ date ]; # 0..1 Fixed date for references with no specified version (transitive)
    fhir:ValueSet.compose.inactive [ boolean ]; # 0..1 Whether inactive codes are in the value set
    fhir:ValueSet.compose.include [ # 1..* Include one or more codes from a code system or other value set(s)
      fhir:ValueSet.compose.include.system [ uri ]; # 0..1 The system the codes come from
      fhir:ValueSet.compose.include.version [ string ]; # 0..1 Specific version of the code system referred to
      fhir:ValueSet.compose.include.concept [ # 0..* A concept defined in the system
        fhir:ValueSet.compose.include.concept.code [ code ]; # 1..1 Code or expression from system
        fhir:ValueSet.compose.include.concept.display [ string ]; # 0..1 Text to display for this code for this value
    set in this valueset
        fhir:ValueSet.compose.include.concept.designation [ # 0..* Additional representations for this concept
          fhir:ValueSet.compose.include.concept.designation.language [ code ]; # 0..1 Human language of the designation
          fhir:ValueSet.compose.include.concept.designation.use [ Coding ]; # 0..1 Types of uses of designations
          fhir:ValueSet.compose.include.concept.designation.value [ string ]; # 1..1 The text value for this designation
        ], ...;
      ], ...;
      fhir:ValueSet.compose.include.filter [ # 0..* Select codes/concepts by their properties (including relationships)
        fhir:ValueSet.compose.include.filter.property [ code ]; # 1..1 A property/filter defined by the code system
        fhir:ValueSet.compose.include.filter.op [ code ]; # 1..1 = | is-a | descendent-of | is-not-a | regex | in | not-
    in | generalizes | exists
        fhir:ValueSet.compose.include.filter.value [ string ]; # 1..1 Code from the system, or regex criteria, or
    boolean value for exists
      ], ...;
      fhir:ValueSet.compose.include.valueSet [ canonical(ValueSet) ], ... ; # 0..* Select the contents included in this
    value set
    ], ...;
    fhir:ValueSet.compose.exclude [ See ValueSet.compose.include ], ... ; # 0..* Explicitly exclude codes from a code
    system or other value sets
  ];
  fhir:ValueSet.expansion [ # 0..1 Used when the value set is "expanded"
    fhir:ValueSet.expansion.identifier [ uri ]; # 0..1 Identifies the value set expansion (business identifier)
    fhir:ValueSet.expansion.timestamp [ dateTime ]; # 1..1 Time ValueSet expansion happened
    fhir:ValueSet.expansion.total [ integer ]; # 0..1 Total number of codes in the expansion
    fhir:ValueSet.expansion.offset [ integer ]; # 0..1 Offset at which this resource starts
    fhir:ValueSet.expansion.parameter [ # 0..* Parameter that controlled the expansion process
      fhir:ValueSet.expansion.parameter.name [ string ]; # 1..1 Name as assigned by the client or server
      # ValueSet.expansion.parameter.value[x] : 0..1 Value of the named parameter. One of these 7
```

```
        fhir:ValueSet.expansion.parameter.valueString [ string ]
        fhir:ValueSet.expansion.parameter.valueBoolean [ boolean ]
        fhir:ValueSet.expansion.parameter.valueInteger [ integer ]
        fhir:ValueSet.expansion.parameter.valueDecimal [ decimal ]
        fhir:ValueSet.expansion.parameter.valueUri [ uri ]
        fhir:ValueSet.expansion.parameter.valueCode [ code ]
        fhir:ValueSet.expansion.parameter.valueDateTime [ dateTime ]
    ], ...;
    fhir:ValueSet.expansion.contains [ # 0..* Codes in the value set
      fhir:ValueSet.expansion.contains.system [ uri ]; # 0..1 System value for the code
      fhir:ValueSet.expansion.contains.abstract [ boolean ]; # 0..1 If user cannot select this entry
      fhir:ValueSet.expansion.contains.inactive [ boolean ]; # 0..1 If concept is inactive in the code system
      fhir:ValueSet.expansion.contains.version [ string ]; # 0..1 Version in which this code/display is defined
      fhir:ValueSet.expansion.contains.code [ code ]; # 0..1 Code - if blank, this is not a selectable code
      fhir:ValueSet.expansion.contains.display [ string ]; # 0..1 User display for the concept
      fhir:ValueSet.expansion.contains.designation [ See ValueSet.compose.include.concept.designation ], ... ; # 0..*
    Additional representations for this item
      fhir:ValueSet.expansion.contains.contains [ See ValueSet.expansion.contains ], ... ; # 0..* Codes contained under
    this entry
    ], ...;
  ];
]
```

Changes since Release 3

ValueSet
ValueSet.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
ValueSet.experimental | No longer marked as Modifier
ValueSet.compose.include.concept.designation.language | Change binding strength from extensible to preferred
ValueSet.compose.include.filter | No longer marked as Modifier
ValueSet.compose.include.filter.op | Change value set from http://hl7.org/fhir/ValueSet/filter-operator to http://hl7.org/fhir/ValueSet/filter-operator|4.0.1
ValueSet.compose.include.filter.value | Type changed from code to string
ValueSet.compose.include.valueSet | Type changed from uri to canonical(ValueSet)
ValueSet.expansion.identifier | Min Cardinality changed from 1 to 0
ValueSet.expansion.parameter.value[x] | Add Type dateTime
ValueSet.expansion.contains.abstract | Default Value "false" removed
ValueSet.expansion.contains.inactive | Default Value "false" removed
ValueSet.extensible | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 8 tests that all execute ok. All tests pass round-trip testing and 8 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

4.9.4.1 Terminology Bindings

Path | Definition | Type | Reference
ValueSet.status | The lifecycle status of an artifact. | Required | PublicationStatus
ValueSet.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
ValueSet.compose.include.concept.designation.language | A human language. | Preferred, but limited to AllLanguages | CommonLanguages
ValueSet.compose.include.concept.designation.use | Details of how a designation would be used. | Extensible | DesignationUse
ValueSet.compose.include.filter.op | The kind of operation to perform as a part of a property based filter. | Required | FilterOperator

4.9.4.2 Constraints

id | Level | Location | Description | Expression
vsd-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
vsd-1 | Rule | ValueSet.compose.include | A value set include/exclude SHALL have a value set or a system | valueSet.exists() or system.exists()
vsd-2 | Rule | ValueSet.compose.include | A value set with concepts or filters SHALL include a system | (concept.exists() or filter.exists()) implies system.exists()
vsd-3 | Rule | ValueSet.compose.include | Cannot have both concept and filter | concept.empty() or filter.empty()
vsd-6 | Rule | ValueSet.expansion.contains | SHALL have a code or a display | code.exists() or display.exists()
vsd-9 | Rule | ValueSet.expansion.contains | Must have a code if not abstract | code.exists() or abstract = true
vsd-10 | Rule | ValueSet.expansion.contains | Must have a system if a code is present | code.empty() or system.exists()

4.9.5 Composition Rules

A value set can be a simple list of included codes, or it can be some kind of general selection criteria using the facilities provided by the code system. For these value sets:

• Multiple include statements are cumulative - e.g. the value set contains the union of all the includes

• Within an include, all the criterion apply -e.g. the value set contains the intersection of the criterion

• Within an include, a single system with selection criteria may be listed, and/or one or more value sets may be listed

• valueSet(s) only: Codes are 'selected' for inclusion if they are in all the referenced value sets

• If a System only is specified, the following rules apply: no concept or filter: All codes in the system are included concept: Only the enumerated codes are selected filter: Any codes meeting the filter criteria are selected

• valueSet and System: Codes are 'selected' for inclusion if they are selected by the code system selection (after checking for concept and filter) and if they are in all the referenced value sets

• If the system reference is not version specific and filters are present, then the contents of the value set are open and change over time as the underlying code systems are updated

Implementation Note: Use of this capability is subject to future clarification and conformance requirements based on implementation experience.

• Using the property filters is only possible where the code system in use defines the relevant properties. Note that in some cases the underlying code system defines the logical concepts but not the syntax for exercising them. In such cases, the literal definitions may be provided by a third party

• In addition to include rules, codes may be excluded. Rules for interpretation of exclude statements match those for includes, but codes in the exclude statements are never in the value set

• Value sets may include abstract codes - that is, codes designated by the underlying code system as not for use as a selectable concept in a particular context. These abstract codes are typically used as a grouping/searching mechanism, and can be included either by enumerating them, or by using a filter.

• Any compose.exclude SHALL be processed such that excluded codes are not found in the expansion

4.9.6 Display and Designations

Concepts used in ValueSets can have a display, which is a short text that represents the meaning of the concept to human users in the context of the value set (which often has narrower meaning and therefore is amenable to shorter displays. If a display is not provided, the value set uses the display from the code system (which is the preferred approach, because overriding the display can lead to very unsafe outcomes).

When a value set enumerates codes, it is sometimes useful to define an alternative display for the code that is to be used wherever the value set is expanded and used in a UI. This facility is provided to cover the following circumstances:

• The system that defines the code or expression doesn't provide a display for this code (or any codes).

• The system that defines the code or expression defines multiple choices for display.

• The system provides a very long display name that is unnecessary or inappropriate in the context of this value set (e.g. a display name of "Glucose [Mass/volume] in Serum or Plasma --10 PM specimen" for LOINC code 48991-4, when the value set only includes Glucose mass/vol in serum/plasma codes). As the display names get longer, this becomes more important.

Note that care must be taken in order to avoid "changing the meaning" of the concept by implying that it means something other than the explicit definition of the concept in the underlying code system (e.g., in the case above, using a display of "Glucose Concentration at 10pm"). For this reason, some contexts of use do not allow a display to be associated with a specific code in a value set.

Any display name for a concept provided in the value set is for display to a human user. The display in the Coding that results from a user selecting a concept from the expansion must be taken from the underlying code system definition, even if a value set is referenced explicitly in the Coding (e.g. by an extension). The correct display for a code can be determined by a $lookup operation. Any alternative display specified in the value set would go in CodeableConcept.text, perhaps appended to the UI label for the matching data element.

As an example, the LOINC code 55423-8 has a display value of "Number of steps in unspecified time Pedometer". A value set for a pick list in a patient generated data form might choose a simpler name:

```
{
  "resourceType" : "ValueSet",
  "compose" : {
    "include" : [{
      "system" : "http://loinc.org",
      "concept" : [{
        "code" : "55423-8",
        "display" : "Step Count"
      }]
    }]
  }
}
```

The expansion generated by a terminology server will have this:

```
{
  "resourceType" : "ValueSet",
  "expansion" : {
```

```
    "contains" : [{
      "system" : "http://loinc.org",
      "code" : "55423-8",
      "display" : "Step Count"
    }]
  }
}
```

The expansion display is taken from the value set, and this is what is displayed in the pick list. Once the user picks the code, it will appear in the Observation.code like this:

```
{
  "resourceType" : "Observation",
  "code" : {
    "coding" : [{
      "system" : "http://loinc.org",
      "code" : "55423-8",
      "display" : "Number of steps in unspecified time Pedometer"
    }],
    "text" : "Step Count"
  }
}
```

Note that the correct value for the display is not in the expansion above. The client can either omit the display, look it up using $lookup, or the server might pre-populate it in the expansion:

```
{
  "resourceType" : "ValueSet",
  "expansion" : {
    "contains" : [{
      "system" : "http://loinc.org",
      "code" : "55423-8",
      "display" : "Step Count",
      "designation" : [{
        "use" : {
          "system" : "http://snomed.info/sct",
          "code" : "900000000000003001"
        },
        "value" : "Number of steps in unspecified time Pedometer"
      }]
    }]
  }
}
```

Irrespective of this, the display in the expansion always goes in CodeableConcept.text.

In addition to the display, a concept can have one or more designation elements. The display is equivalent to a special designation with an implied designation.use of "primary code" and a language equal to the Resource Language. The designations can provide additional displays for other languages, as well as designations for other purposes. When using concepts, applications use the display unless the language or usage in context provides a reason to use one of the designations.

4.9.7 Value Sets with multiple code systems

Value sets may select codes from multiple code systems - either by including codes from different systems, or importing other value sets that include them. A typical use for crossing code systems is when including a set of codes, and adding a few additional codes to cover cases not catered to by the included codes (e.g. Data missing or workflow error codes).

Best Practice Note: Mixing definitional systems offers the potential for confusing, overlapping, and inconsistent definitions. Creating value sets that cross code systems should be done with care to avoid creating definitional confusion.

Value sets should only include well differentiated concepts, but many value sets and code systems do not have well differentiated concepts because of various real world constraints.

4.9.7.1 Code systems Note

Each Code System defines which filters can be used in ValueSet.compose.include.filter. All code systems have base filters and any additional filters defined in (CodeSystem.filter).

This specification also defines filters for various published code systems:

- LOINC Filters
- Using CT Filters
- RxNorm Filters
- UCUM Filters
- Using Metathesaurus Filters
- CPT Filters
- NDF-RT Filters

4.9.8 Value Set Expansion

A value set can be "expanded", where the definition of the value set is used to create a simple collection of codes suitable for use for data entry or validation. There is a defined operation $expand to ask a server to perform this expansion. Expansions are most useful when a value set includes all the codes in a code system, or a set of codes by filter.

A resource that represents a value set expansion includes the same identification details as the definition of the value set, and MAY include the definition of the value set (.compose). In addition, it has an .expansion element which contains the list of codes that constitute the value set expansion. Each contained code can include nested contained codes - see below for further discussion.

When a request for an expansion is received (e.g., for the $expand operation), the following process should be followed:

• If the value set already has an expansion (e.g., a stored expansion): Check that the parameters associated with the $expand operation are consistent with the parameters recorded in the expansion If the parameters are inconsistent, then either generate a new expansion, or, if there is no definition (.compose), return an error Additional parameters may be applied to an existing expansion if the server has validated that the result will be the same as if the expansion was generated directly from the value set definition (e.g. text filter, designations included).

• Otherwise: For each compose.include: If there is a system, identify the correct version of the code system, and then: If there are no codes or filters, add every code in the code system to the result set. If codes are listed, check that they are valid, and check their active status, and if ok, add them to the result set (the parameters to the $expand operation may be used to control whether active codes are included). If any filters are present, process them in order (as explained above), and add the intersection of their results to the result set. For each valueSet, find the referenced value set by ValueSet.url, expand that (e.g., using the $expand operation: GET [base]/ValueSet/$expand$url=[compose.include.valueSet]), and add it to the result set. This means that expansion across imports is a recursive process. Add the intersection of the result set from the system and the result sets from the value sets to the expansion

• For each compose.exclude, follow the same process as for compose.include, but remove any codes from the result set, instead of adding them.

The final "result set" is then represented in expansion. Note that the expansion structure is inherently ordered; this specification does not fix the meaning of use of the order. The conceptual approach described should not be understood to prohibit any implementation approach in these regards. In addition, note that the method described above is a conceptual approach; individual servers may choose to follow alternative approaches that are more efficient, as long as the outcome is the same.

The impact of Code System supplements on value set expansion - and therefore value set validation - is subject to ongoing experimentation and implementation testing, and further clarification and additional rules might be proposed in future versions of this specification.

4.9.8.1 Hierarchical Expansions

The expansion MAY be hierarchical - that is, it may have contains element that contain their own sub-elements, to any level of depth. This specification does not fix the meaning of the hierarchy: there is no implication about the logical relationship between the nested contain elements, and the structure cannot be used for logical inferencing. The structure exists to provide navigational assistance for helping human users to locate codes in the expansion. There is an example of use of a hierarchical expansion.

Note that the CodeSystem resource and ValueSet.compose offer no direct support for defining hierarchies and groups; applications may want to take advantage of the expand-rules and expand-group extensions as an implementation feature. An expansion may include entries in the expansion that only serve an arbitrary grouping purpose, to make it easier for a human to use the list. These entries have no system or code, and must be marked as abstract.

4.9.8.2 Uniqueness of Concepts in Expansions

Value set definitions may lead to more than one instruction to include a given concept in the value set across the includes and imports. No matter how many times the definitions include a concept, it is only present in the value set once, and will only appear once in a flat expansion of the value set. Note, however, that a concept may appear more than once in a nested hierarchy when the expansion is prepared for UI use (irrespective of how many times it is included in the definitions). Note that uniqueness is based on system/version/code; it is possible to include the same concept from different versions of a code system in the same expansion, though this is generally confusing for users and should be avoided.

The codes in the expansion should be treated as case sensitive - implementers should use the correct case. Implementers can consult the definition of the underlying code systems to determine whether the code system that defines the code is case sensitive or not.

4.9.8.3 Expansion Identifiers

It is important that expansions be identified properly. Any value set definition may produce an infinite number of expansions, depending on the operation parameters. Any expansions produced must be clearly identified so that there is no confusion. The following rules apply:

• The canonical URL for the expansion is the same as the value set it was expanded from

• Each expansion SHALL have a unique identifier in ValueSet.expansion.identifier

• The result of an $expand operation may use the same identifier in ValueSet.expansion.identifier as a previous expansion, but if it does, the canonical representation of the value set expansion SHALL be identical (e.g. a cached response)

4.9.8.4 Expansion Parameters

The expansion contains a set of parameters in ValueSet.expansion.parameter that record what controlled the expansion process. These parameters may be used by users of expanded value sets to check whether the expansion is suitable for a particular purpose, or to pick the correct expansion. The server decides which parameters to include here, but at a minimum, the list SHOULD include all of the parameters that affect the $expand operation. If the expansion will be persisted all of these parameters SHALL be included. If the codeSystem on the server has a specified version then this version SHALL be provided as a parameter in the expansion (note that not all code systems have a version).

The following parameters are predefined by the $expand operation, and are suitable for use in the expansion parameters:

filter | A text filter that is applied to restrict the codes that are returned (this is useful in a UI context). The interpretation of this is delegated to the server in order to allow to determine the most optimal search approach for the context. The server can document the way this parameter works in TerminologyCapabilities..expansion.textFilter. Typical usage of this parameter includes functionality like: using left matching e.g. "acut ast" allowing for wild cards such as %, &, ? searching on definition as well as display(s) allowing for search conditions (and / or / exclusions) Text Search engines such as Lucene or Solr, long with their considerable functionality, might also be used. The optional text search might also be code system specific, and servers might have different implementations for different code systems

date | The date for which the expansion should be generated. if a date is provided, it means that the server should use the value set / code system definitions as they were on the given date, or return an error if this is not possible. Normally, the date is the current conditions (which is the default value) but under some circumstances, systems need to generate an expansion as it would have been in the past. A typical example of this would be where code selection is constrained to the set of codes that were available when the patient was treated, not when the record is being edited. Note that which date is appropriate is a matter for implementation policy.

offset | Paging support - where to start if a subset is desired (default = 0). Offset is number of records (not number of pages)

count | Paging support - how many codes should be provided in a partial page view. Paging only applies to flat expansions - servers ignore paging if the expansion is not flat. If count = 0, the client is asking how large the expansion is. Servers SHOULD honor this request for hierarchical expansions as well, and simply return the overall count

includeDesignations | Controls whether concept designations are to be included or excluded in value set expansions

designation | A token that specifies a system+code that is either a use or a language. Designations that match by language or use are included in the expansion. If no designation is specified, it is at the server discretion which designations to return

includeDefinition | Controls whether the value set definition is included or excluded in value set expansions

activeOnly | Controls whether inactive concepts are included or excluded in value set expansions. Note that if the value set explicitly specifies that inactive codes are included, this parameter can still remove them from a specific expansion, but this parameter cannot include them if the value set excludes them

excludeNested | Controls whether or not the value set expansion nests codes or not (i.e. ValueSet.expansion.contains.contains)

excludeNotForUI | Controls whether or not the value set expansion is assembled for a user interface use or not. Value sets intended for User Interface might include 'abstract' codes or have nested contains with items with no code or abstract = true, with the sole purpose of helping a user navigate through the list efficiently, where as a value set not generated for UI use might be flat, and only contain the selectable codes in the value set. The exact implications of 'for UI' depend on the code system, and what properties it exposes for a terminology server to use. In the FHIR Specification itself, the value set expansions are generated with excludeNotForUI = false, and the expansions used when generated schema / code etc, or performing validation, are all excludeNotForUI = true.

excludePostCoordinated | Controls whether or not the value set expansion includes post coordinated codes

displayLanguage | Specifies the language to be used for description in the expansions i.e. the language to be used for ValueSet.expansion.contains.display

exclude-system | Code system, or a particular version of a code system to be excluded from the value set expansion. The format is the same as a canonical URL: [system]|[version] - e.g. http://loinc.org|2.56

system-version | Specifies a version to use for a system, if the value set does not specify which one to use. The format is the same as a canonical URL: [system]|[version] - e.g. http://loinc.org|2.56

check-system-version | Edge Case: Specifies a version to use for a system. If a value set specifies a different version, an error is returned instead of the expansion. The format is the same as a canonical URL: [system]|[version] - e.g. http://loinc.org|2.56

force-system-version | Edge Case: Specifies a version to use for a system. This parameter overrides any specified version in the value set (and any it depends on). The format is the same as a canonical URL: [system]|[version] - e.g. http://loinc.org|2.56. Note that this has obvious safety issues, in that it may result in a value set expansion giving a different list of codes that is both wrong and unsafe, and implementers should only use this capability reluctantly. It primarily exists to deal with situations where specifications have fallen into decay as time passes. If the value is override, the version used SHALL explicitly be represented in the expansion parameters

The count and offset parameters are important. If the expansion is a page out of the whole expansion, the offset and count parameters SHALL be populated. Clients can reliably use the count/offset parameters to determine whether the whole expansion is returned.

Other parameters that servers may be required to use:

[canonical]#CodeSystem.content | [content] : The content value for the code system for the canonical URL. Applications generating expansions SHALL use this parameter if the CodeSystem.content value is "fragment" or "

[canonical1]#supplement | [canonical2] : Indicates that the specified supplement (canonical2) contributed to the content of the expansion for the code system canonical1 (by influencing selection, or providing designations). Applications generating expansions SHALL use this parameter to record that a supplement was used during the expansion process

Beyond this, servers MAY define their own parameters, though Terminology server authors are requested to bring additional parameters to HL7 (via 'Propose a change' link below) in the interests of interoperability.

Servers can also create and store Provenance statements about the expansion, or AuditEvent records of the expansion process if further transparency is required. These resources can contain considerable detail about the various inputs to the process, and any significant decisions by the expansion engine. Further details around this (and profiles on the relevant resources) may be provided in future versions of this specification or related implementation guides.

Request for Feedback: The existing set of parameters are intended to cover the vast majority of use cases, but there are some cases where the parameters do not provide enough control to a client, particularly with regard to combinations of parameters, and the interplay between code system versions and other parameters. Implementers may need to define their own value sets to meet these requirements.

Ongoing feedback is welcome at [link to be provided]

4.9.8.5 Storing Expansions

Whether to store expanded value sets, or simply to store their definitions and expand on the fly is a matter for system deployment. Some servers, including public value sets servers, only store expansions. However, any system that stores an expansion must be concerned with how to determine whether the expansion is still current, and this requires deep knowledge of how the expansion was created. A system with a dedicated terminology server that returns expansions on demand avoids this problem, but leaves open the question of how to audit the specific expansion that was used for a particular case. One solution to this is to use a dedicated terminology server, and have the clients ask for expansions on demand based on the value set definitions, and for the server to store (and reuse as appropriate) the returned expansion (when it reuses the expansion, ValueSet.expansion.identifier will be the same). If expansions are shared, users need to be aware of how expansion identifiers (which may be server specific) work.

4.9.8.6 Searching for codes in value sets

The search parameters defined on ValueSet include the code parameter. This is intended to allow a consumer to find all the value sets that include a particular code. However, fully evaluating this search parameter is extremely onerous for a server. Further, whether a code is in a value set depends on the context in which expansions are performed (see the $expand operation). For this reason, the degree to which the search parameter is supported can be declared in the Terminology Capability statement.

4.9.9 Combinations of .compose and .expansion

Use cases for the different combinations:

• .compose only: Provides the value set definition (CLD). This is the expected format from the value set publisher.

• .expansion only: Provides the value set expansion, which might or might not be persisted (see 4.8.7 Value Set Expansion above). The "expansion only" format may be returned in the $expand operation output (at the terminology server's discretion).

• Both .compose and .expansion: The terminology server often may include the definition as well as the expansion in the $expand operation output. The resource instance including the expansion might or might not be persisted.

• Neither .compose nor .expansion: This is a valid format for ValueSet resource instances. A primary use case is the return from a summary search (i.e. _summary=true). This may be especially useful to reduce the performance overhead when returning a summary of extensionally defined value sets (which may include a large number of concepts in the value set definition).

4.9.10 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
code TU | token | This special parameter searches for codes in the value set. See additional notes on the ValueSet resource | ValueSet.expansion.contains.code | ValueSet.compose.include.concept.code
context TU | token | A use context assigned to the value set | (ValueSet.useContext.value as CodeableConcept)
context-quantity TU | quantity | A quantity- or range-valued use context assigned to the value set | (ValueSet.useContext.value as Quantity) | (ValueSet.useContext.value as Range)
context-type TU | token | A type of use context assigned to the value set | ValueSet.useContext.code
context-type-quantity TU | composite | A use context type and quantity- or range-based value assigned to the value set | On ValueSet.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value TU | composite | A use context type and value assigned to the value set | On ValueSet.useContext: context-type: code context: value.as(CodeableConcept)
date TU | date | The value set publication date | ValueSet.date
description TU | string | The description of the value set | ValueSet.description
expansion TU | uri | Identifies the value set expansion (business identifier) | ValueSet.expansion.identifier
identifier TU | token | External identifier for the value set | ValueSet.identifier

jurisdiction TU | token | Intended jurisdiction for the value set | ValueSet.jurisdiction
name TU | string | Computationally friendly name of the value set | ValueSet.name
publisher TU | string | Name of the publisher of the value set | ValueSet.publisher
reference TU | uri | A code system included or excluded in the value set or an imported value set | ValueSet.compose.include.system
status TU | token | The current status of the value set | ValueSet.status
title TU | string | The human-friendly name of the value set | ValueSet.title
url TU | uri | The uri that identifies the value set | ValueSet.url
version TU | token | The business version of the value set | ValueSet.version

4.9.11 Resource ValueSet - Examples

URL: https://hl7.org/fhir/R4/valueset-examples.html

4.9.11 Resource ValueSet - Examples

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

As well as the specific example below, there are many value sets published as part of defining other resources. See:

- FHIR Valuesets
- HL7 v2 Tables
- HL7 v3 Code systems and Value sets

Example Name | id | Format
ValueSet Example using enumeration of codes (extensional) | example-extensional | XML | JSON | Turtle
ValueSet Example using code filter (intensional) | example-intensional | XML | JSON | Turtle
ValueSet Example using code filter (intensional) defined in a code system resource | example-filter | XML | JSON | Turtle
ValueSet Example using a prepared expansion | example-expansion | XML | JSON | Turtle
ValueSet Example showing how to specify a hierarchy for UI using extensions | example-hierarchical | XML | JSON | Turtle
Common UCUM codes | ucum-common | XML | JSON | Turtle
All CPT codes | cpt-all | XML | JSON | Turtle
Yes/No/Don't know, for capturing a common answer set in questionnaires | yesnodontknow | XML | JSON | Turtle
NHIN Purpose of Use | nhin-purposeofuse | XML | JSON | Turtle
Valueset including inactive concepts | inactive | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

4.9.12 Resource ValueSet - Detailed Descriptions

URL: https://hl7.org/fhir/R4/valueset-definitions.html

4.9.12 Resource ValueSet - Detailed Descriptions

Vocabulary Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the ValueSet resource.

ValueSet
Element Id | ValueSet
Definition | A ValueSet resource instance specifies a set of codes drawn from one or more code systems, intended for use in a particular context. Value sets link between CodeSystem definitions and their use in coded elements.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element vsd-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
vsd-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
ValueSet.url
Element Id | ValueSet.url
Definition | An absolute URI that is used to identify this value set when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this value set is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the value set is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the value set to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the

meta.source element to indicate where the current master source of the resource can be found.
ValueSet.identifier
Element Id | ValueSet.identifier
Definition | A formal identifier that is used to identify this value set when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this value set outside of FHIR, where it is not possible to use the logical URI.
ValueSet.version
Element Id | ValueSet.version
Definition | The identifier that is used to identify this version of the value set when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the value set author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different value set instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the value set with the format [url]|[version].
ValueSet.name
Element Id | ValueSet.name
Definition | A natural language name identifying the value set. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.A name should be provided unless the value set is a contained resource (e.g. an anonymous value set in a profile). Most registries will require a name.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
ValueSet.title
Element Id | ValueSet.title
Definition | A short, descriptive, user-friendly title for the value set.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
ValueSet.status
Element Id | ValueSet.status
Definition | The status of this value set. Enables tracking the life-cycle of the content. The status of the value set applies to the value set definition (ValueSet.compose) and the associated ValueSet metadata. Expansions do not have a state.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of value sets that are appropriate for use versus not.See also the valueset-workflowStatus extension for additional status information related to the editorial process.
ValueSet.experimental
Element Id | ValueSet.experimental
Definition | A Boolean value to indicate that this value set is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level value set.
Summary | true
Comments | Allows filtering of value sets that are appropriate for use versus not.
ValueSet.date
Element Id | ValueSet.date
Definition | The date (and optionally time) when the value set was created or revised (e.g. the 'content logical definition').
Cardinality | 0..1

Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the value set. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
ValueSet.publisher
Element Id | ValueSet.publisher
Definition | The name of the organization or individual that published the value set.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the value set. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the value set is the organization or individual primarily responsible for the maintenance and upkeep of the value set. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the value set. This item SHOULD be populated unless the information is available from context.
ValueSet.contact
Element Id | ValueSet.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
ValueSet.description
Element Id | ValueSet.description
Definition | A free text natural language description of the value set from a consumer's perspective. The textual description specifies the span of meanings for concepts to be included within the Value Set Expansion, and also may specify the intended use and limitations of the Value Set.
Cardinality | 0..1
Type | markdown
Alternate Names | Scope; Purpose
Comments | This description can be used to capture details such as why the value set was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the value set as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the value set is presumed to be the predominant language in the place the value set was created).The description is not intended to describe the semantics of the Value Set - there are no intrinsic semantics separate from the codes contained in its expansion. The description should capture its intended use, which is needed for ensuring integrity for its use in models across future changes. A description should be provided unless the value set is a contained resource (e.g. an anonymous value set in a profile). Most registries will require a description.
ValueSet.useContext
Element Id | ValueSet.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate value set instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
ValueSet.jurisdiction
Element Id | ValueSet.jurisdiction
Definition | A legal or geographic region in which the value set is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the value set to be used in jurisdictions other than those for which it was originally designed or intended.
ValueSet.immutable
Element Id | ValueSet.immutable
Definition | If this is set to 'true', then no new versions of the content logical definition can be created. Note: Other metadata might still change.
Cardinality | 0..1
Type | boolean
Summary | true
Comments | Normally immutability is set to 'false', which is the default assumption if it is not populated. Note that the implication is that if this is set to 'true', there may be only one ValueSet version for this definition. Immutability tends to be set to 'true' in one of two cases: - Where the value set, by the nature of its usage, cannot change. For example "All specializations of ACT in ActClassCode" - Where there's no safe way to express the "Purpose" such that someone else could safely make changes to the value set definition. Source workflow control must guarantee that the same URI always yields the same definition.
ValueSet.purpose
Element Id | ValueSet.purpose
Definition | Explanation of why this value set is needed and why it has been designed as it has.

Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the value set. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this value set.
ValueSet.copyright
Element Id | ValueSet.copyright
Definition | A copyright statement relating to the value set and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the value set.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the value set and/or its content.
Alternate Names | License; Restrictions
Comments | nullFrequently, the copyright differs between the value set and the codes that are included. The copyright statement should clearly differentiate between these when required.
ValueSet.compose
Element Id | ValueSet.compose
Definition | A set of criteria that define the contents of the value set by including or excluding codes selected from the specified code system(s) that the value set draws from. This is also known as the Content Logical Definition (CLD).
Cardinality | 0..1
Alternate Names | Content Logical Definition; CLD
ValueSet.compose.lockedDate
Element Id | ValueSet.compose.lockedDate
Definition | The Locked Date is the effective date that is used to determine the version of all referenced Code Systems and Value Set Definitions included in the compose that are not already tied to a specific version.
Cardinality | 0..1
Type | date
Summary | true
Comments | With a defined lockedDate the value set is considered "Locked". Otherwise, the value set may have different expansions as underlying code systems and/or value sets evolve. The interpretation of lockedDate is often dependent on the context - e.g. a SNOMED CT derived value set with a lockedDate will have a different expansion in USA than in UK. If a value set specifies a version for include and exclude statements, and also specifies a locked date, the specified versions need to be available that date, or the value set will not be usable.
ValueSet.compose.inactive
Element Id | ValueSet.compose.inactive
Definition | Whether inactive codes - codes that are not approved for current use - are in the value set. If inactive = true, inactive codes are to be included in the expansion, if inactive = false, the inactive codes will not be included in the expansion. If absent, the behavior is determined by the implementation, or by the applicable $expand parameters (but generally, inactive codes would be expected to be included).
Cardinality | 0..1
Type | boolean
Summary | true
Comments | Note that in the FHIR terminology framework, "deprecated" does not mean inactive, but in some code systems, e.g. LOINC, "deprecated" does mean inactive. Code systems should define what codes are considered to be inactive. If this is not clearly defined (including in the FHIR code system resource), then all codes are assumed to be active. The Value Set Definition specification defines an ActiveOnly element, which is the reverse of this element e.g. (ValueSet.compose.inactive=FALSE) is the same as (VSD.ActiveOnly=TRUE).
ValueSet.compose.include
Element Id | ValueSet.compose.include
Definition | Include one or more codes from a code system or other value set(s).
Cardinality | 1..*
Summary | true
Comments | All the conditions in an include must be true. If a system is listed, all the codes from the system are listed. If one or more filters are listed, all of the filters must apply. If one or more value sets are listed, the codes must be in all the value sets. E.g. each include is 'include all the codes that meet all these conditions'.
Invariants | Defined on this element vsd-1 Rule A value set include/exclude SHALL have a value set or a system valueSet.exists() or system.exists() vsd-2 Rule A value set with concepts or filters SHALL include a system (concept.exists() or filter.exists()) implies system.exists() vsd-3 Rule Cannot have both concept and filter concept.empty() or filter.empty()
Defined on this element
vsd-1 | Rule | A value set include/exclude SHALL have a value set or a system | valueSet.exists() or system.exists()
vsd-2 | Rule | A value set with concepts or filters SHALL include a system | (concept.exists() or filter.exists()) implies system.exists()
vsd-3 | Rule | Cannot have both concept and filter | concept.empty() or filter.empty()
ValueSet.compose.include.system
Element Id | ValueSet.compose.include.system
Definition | An absolute URI which is the code system from which the selected codes come from.
Cardinality | 0..1
Type | uri
Summary | true
Comments | If there are no codes or filters, the entire code system is included. Note that the set of codes that are included may contain abstract codes. See ''Coding.system'' for further documentation about the correct value for the system element.
Invariants | Affect this element vsd-1 Rule A value set include/exclude SHALL have a value set or a system valueSet.exists() or system.exists() vsd-2 Rule A value set with concepts or filters SHALL include a system (concept.exists() or filter.exists()) implies system.exists()
Affect this element

vsd-1 | Rule | A value set include/exclude SHALL have a value set or a system | valueSet.exists() or system.exists()
vsd-2 | Rule | A value set with concepts or filters SHALL include a system | (concept.exists() or filter.exists()) implies system.exists()
ValueSet.compose.include.version
Element Id | ValueSet.compose.include.version
Definition | The version of the code system that the codes are selected from, or the special version '*' for all versions.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | This is used when selecting the descendants of a concept - they may change between versions. If no version is specified, then the exact contents of the value set might not be known until a context of use binds it to a particular version. The special value '*' means all versions; It is at server discretion regarding expansions and which versions must be supported.
ValueSet.compose.include.concept
Element Id | ValueSet.compose.include.concept
Definition | Specifies a concept to be included or excluded.
Cardinality | 0..*
Comments | The list of concepts is considered ordered, though the order might not have any particular significance. Typically, the order of an expansion follows that defined in the compose element.
Invariants | Affect this element vsd-3 Rule Cannot have both concept and filter concept.empty() or filter.empty()
Affect this element
vsd-3 | Rule | Cannot have both concept and filter | concept.empty() or filter.empty()
ValueSet.compose.include.concept.code
Element Id | ValueSet.compose.include.concept.code
Definition | Specifies a code for the concept to be included or excluded.
Cardinality | 1..1
Type | code
Comments | Expressions are allowed if defined by the underlying code system.
ValueSet.compose.include.concept.display
Element Id | ValueSet.compose.include.concept.display
Definition | The text to display to the user for this concept in the context of this valueset. If no display is provided, then applications using the value set use the display specified for the code by the system.
Cardinality | 0..1
Type | string
Comments | The value set resource allows for an alternative display to be specified for when this concept is used in this particular value set. See notes in the value set narrative about the correct use of this element.
ValueSet.compose.include.concept.designation
Element Id | ValueSet.compose.include.concept.designation
Definition | Additional representations for this concept when used in this value set - other languages, aliases, specialized purposes, used for particular purposes, etc.
Cardinality | 0..*
Comments | Concepts have both a display and an array of designation. The display is equivalent to a special designation with an implied designation.use of "primary code" and a language equal to the Resource Language.
ValueSet.compose.include.concept.designation.language
Element Id | ValueSet.compose.include.concept.designation.language
Definition | The language this designation is defined for.
Cardinality | 0..1
Terminology Binding | Common Languages (Preferred but limited to All Languages)
Type | code
Comments | In the absence of a language, the resource language applies.
ValueSet.compose.include.concept.designation.use
Element Id | ValueSet.compose.include.concept.designation.use
Definition | A code that represents types of uses of designations.
Cardinality | 0..1
Terminology Binding | Designation Use (Extensible)
Type | Coding
Comments | If no use is provided, the designation can be assumed to be suitable for general display to a human user.
ValueSet.compose.include.concept.designation.value
Element Id | ValueSet.compose.include.concept.designation.value
Definition | The text value for this designation.
Cardinality | 1..1
Type | string
ValueSet.compose.include.filter
Element Id | ValueSet.compose.include.filter
Definition | Select concepts by specify a matching criterion based on the properties (including relationships) defined by the system, or on filters defined by the system. If multiple filters are specified, they SHALL all be true.
Cardinality | 0..*
Summary | true
Comments | Selecting codes by specifying filters based on properties is only possible where the underlying code system defines appropriate properties. Note that in some cases, the underlying code system defines the logical concepts but not the literal codes for the concepts. In such cases, the literal definitions may be provided by a third party.
Invariants | Affect this element vsd-3 Rule Cannot have both concept and filter concept.empty() or filter.empty()
Affect this element

vsd-3 | Rule | Cannot have both concept and filter | concept.empty() or filter.empty()
ValueSet.compose.include.filter.property
Element Id | ValueSet.compose.include.filter.property
Definition | A code that identifies a property or a filter defined in the code system.
Cardinality | 1..1
Type | code
Summary | true
ValueSet.compose.include.filter.op
Element Id | ValueSet.compose.include.filter.op
Definition | The kind of operation to perform as a part of the filter criteria.
Cardinality | 1..1
Terminology Binding | FilterOperator (Required)
Type | code
Summary | true
Comments | In case filter.property represents a property of the system, the operation applies to the selected property. In case filter.property represents a filter of the system, the operation SHALL match one of the CodeSystem.filter.operator values.
ValueSet.compose.include.filter.value
Element Id | ValueSet.compose.include.filter.value
Definition | The match value may be either a code defined by the system, or a string value, which is a regex match on the literal string of the property value (if the filter represents a property defined in CodeSystem) or of the system filter value (if the filter represents a filter defined in CodeSystem) when the operation is 'regex', or one of the values (true and false), when the operation is 'exists'.
Cardinality | 1..1
Type | string
Summary | true
Comments | Use regex matching with care - full regex matching on every SNOMED CT term is prohibitive, for example.
ValueSet.compose.include.valueSet
Element Id | ValueSet.compose.include.valueSet
Definition | Selects the concepts found in this value set (based on its value set definition). This is an absolute URI that is a reference to ValueSet.url. If multiple value sets are specified this includes the union of the contents of all of the referenced value sets.
Cardinality | 0..*
Type | canonical(ValueSet)
Summary | true
Comments | The value set URI is either a logical reference to a defined value set such as a SNOMED CT reference set, or a direct reference to a value set definition using ValueSet.url. The reference might not refer to an actual FHIR ValueSet resource; in this case, whatever is referred to is an implicit definition of a value set that needs to be clear about how versions are resolved.
Invariants | Affect this element vsd-1 Rule A value set include/exclude SHALL have a value set or a system valueSet.exists() or system.exists()
Affect this element
vsd-1 | Rule | A value set include/exclude SHALL have a value set or a system | valueSet.exists() or system.exists()
ValueSet.compose.exclude
Element Id | ValueSet.compose.exclude
Definition | Exclude one or more codes from the value set based on code system filters and/or other value sets.
Cardinality | 0..*
Type | See ValueSet.compose.include
Comments | Usually this is used to selectively exclude codes that were included by subsumption in the inclusions. Any display names specified for the codes are ignored.
Invariants | Affect this element vsd-2 Rule A value set with concepts or filters SHALL include a system (concept.exists() or filter.exists()) implies system.exists()
Affect this element
vsd-2 | Rule | A value set with concepts or filters SHALL include a system | (concept.exists() or filter.exists()) implies system.exists()
ValueSet.expansion
Element Id | ValueSet.expansion
Definition | A value set can also be "expanded", where the value set is turned into a simple collection of enumerated codes. This element holds the expansion, if it has been performed.
Cardinality | 0..1
Comments | Expansion is performed to produce a collection of codes that are ready to use for data entry or validation. Value set expansions are always considered to be stateless - they are a record of the set of codes in the value set at a point in time under a given set of conditions, and are not subject to ongoing maintenance. Expansion.parameter is a simplified list of parameters - a subset of the features of the Parameters resource.
ValueSet.expansion.identifier
Element Id | ValueSet.expansion.identifier
Definition | An identifier that uniquely identifies this expansion of the valueset, based on a unique combination of the provided parameters, the system default parameters, and the underlying system code system versions etc. Systems may re-use the same identifier as long as those factors remain the same, and the expansion is the same, but are not required to do so. This is a business identifier.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | uri
Comments | Typically, this uri is a UUID (e.g. urn:uuid:8230ff20-c97a-4167-a59d-dc2cb9df16dd).
ValueSet.expansion.timestamp
Element Id | ValueSet.expansion.timestamp

Definition | The time at which the expansion was produced by the expanding system.
Cardinality | 1..1
Type | dateTime
Comments | This SHOULD be a fully populated instant, but in some circumstances, value sets are expanded by hand, and the expansion is published without that precision.
ValueSet.expansion.total
Element Id | ValueSet.expansion.total
Definition | The total number of concepts in the expansion. If the number of concept nodes in this resource is less than the stated number, then the server can return more using the offset parameter.
Cardinality | 0..1
Type | integer
Comments | Paging only applies to flat expansions.
ValueSet.expansion.offset
Element Id | ValueSet.expansion.offset
Definition | If paging is being used, the offset at which this resource starts. I.e. this resource is a partial view into the expansion. If paging is not being used, this element SHALL NOT be present.
Cardinality | 0..1
Type | integer
Meaning if Missing | Paging is not being used
Comments | Paging only applies to flat expansions. If a filter is applied, the count is the number of concepts that matched the filter, not the number of concepts in an unfiltered view of the expansion.
ValueSet.expansion.parameter
Element Id | ValueSet.expansion.parameter
Definition | A parameter that controlled the expansion process. These parameters may be used by users of expanded value sets to check whether the expansion is suitable for a particular purpose, or to pick the correct expansion.
Cardinality | 0..*
Comments | The server decides which parameters to include here, but at a minimum, the list SHOULD include all of the parameters that affect the $expand operation. If the expansion will be persisted all of these parameters SHALL be included. If the codeSystem on the server has a specified version then this version SHALL be provided as a parameter in the expansion (note that not all code systems have a version).
ValueSet.expansion.parameter.name
Element Id | ValueSet.expansion.parameter.name
Definition | Name of the input parameter to the $expand operation; may be a server-assigned name for additional default or other server-supplied parameters used to control the expansion process.
Cardinality | 1..1
Type | string
Comments | The names are assigned at the discretion of the server.
ValueSet.expansion.parameter.value[x]
Element Id | ValueSet.expansion.parameter.value[x]
Definition | The value of the parameter.
Cardinality | 0..1
Type | string|boolean|integer|decimal|uri|code|dateTime
[x] Note | See Choice of Data Types for further information about how to use [x]
ValueSet.expansion.contains
Element Id | ValueSet.expansion.contains
Definition | The codes that are contained in the value set expansion.
Cardinality | 0..*
Invariants | Defined on this element vsd-6 Rule SHALL have a code or a display code.exists() or display.exists() vsd-9 Rule Must have a code if not abstract code.exists() or abstract = true vsd-10 Rule Must have a system if a code is present code.empty() or system.exists()
Defined on this element
vsd-6 | Rule | SHALL have a code or a display | code.exists() or display.exists()
vsd-9 | Rule | Must have a code if not abstract | code.exists() or abstract = true
vsd-10 | Rule | Must have a system if a code is present | code.empty() or system.exists()
ValueSet.expansion.contains.system
Element Id | ValueSet.expansion.contains.system
Definition | An absolute URI which is the code system in which the code for this item in the expansion is defined.
Cardinality | 0..1
Type | uri
ValueSet.expansion.contains.abstract
Element Id | ValueSet.expansion.contains.abstract
Definition | If true, this entry is included in the expansion for navigational purposes, and the user cannot select the code directly as a proper value.
Cardinality | 0..1
Type | boolean
Meaning if Missing | If this is not populated, systems can assume that the concept it not abstract. This SHALL always be populated for abstract concepts
Comments | This should not be understood to exclude its use for searching (e.g. by subsumption testing). The client should know whether it is appropriate for the user to select an abstract code or not.
ValueSet.expansion.contains.inactive
Element Id | ValueSet.expansion.contains.inactive
Definition | If the concept is inactive in the code system that defines it. Inactive codes are those that are no longer to be used, but are maintained by the code system for understanding legacy data. It might not be known or specified whether an concept is inactive (and it may depend on the context of use).

Cardinality | 0..1
Type | boolean
Comments | This should only have a value if the concept is inactive.
ValueSet.expansion.contains.version
Element Id | ValueSet.expansion.contains.version
Definition | The version of the code system from this code was taken. Note that a well-maintained code system does not need the version reported, because the meaning of codes is consistent across versions. However this cannot consistently be assured, and when the meaning is not guaranteed to be consistent, the version SHOULD be exchanged.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Comments | The exact value of the version string is specified by the system from which the code is derived.
ValueSet.expansion.contains.code
Element Id | ValueSet.expansion.contains.code
Definition | The code for this item in the expansion hierarchy. If this code is missing the entry in the hierarchy is a place holder (abstract) and does not represent a valid code in the value set.
Cardinality | 0..1
Type | code
Invariants | Affect this element vsd-6 Rule SHALL have a code or a display code.exists() or display.exists()
Affect this element
vsd-6 | Rule | SHALL have a code or a display | code.exists() or display.exists()
ValueSet.expansion.contains.display
Element Id | ValueSet.expansion.contains.display
Definition | The recommended display for this item in the expansion.
Cardinality | 0..1
Type | string
Invariants | Affect this element vsd-6 Rule SHALL have a code or a display code.exists() or display.exists()
Affect this element
vsd-6 | Rule | SHALL have a code or a display | code.exists() or display.exists()
ValueSet.expansion.contains.designation
Element Id | ValueSet.expansion.contains.designation
Definition | Additional representations for this item - other languages, aliases, specialized purposes, used for particular purposes, etc. These are relevant when the conditions of the expansion do not fix to a single correct representation.
Cardinality | 0..*
Type | See ValueSet.compose.include.concept.designation
Comments | The designations provided must be based on the value set and code system definitions.
ValueSet.expansion.contains.contains
Element Id | ValueSet.expansion.contains.contains
Definition | Other codes and entries contained under this entry in the hierarchy.
Cardinality | 0..*
Type | See ValueSet.expansion.contains
Comments | If the expansion uses this element, there is no implication about the logical relationship between them, and the structure cannot be used for logical inferencing. The structure exists to provide navigational assistance for helping human users to locate codes in the expansion.

4.9.13 Resource ValueSet - Mappings

URL: https://hl7.org/fhir/R4/valueset-mappings.html

4.9.13 Resource ValueSet - Mappings

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the valueset resource (see Mappings to Other Standards for further information & status).

4.9.13.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ValueSet
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]
lockedDate | FiveWs.init

4.9.13.2 Workflow Pattern (http://hl7.org/fhir/workflow)

ValueSet
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright

4.9.13.3 RIM Mapping (http://hl7.org/v3)

ValueSet | N/A

4.9.14 Resource ValueSet - Extensions & Profiles

URL: https://hl7.org/fhir/R4/valueset-profiles.html

4.9.14 Resource ValueSet - Extensions & Profiles

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the ValueSet resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

4.9.14.1 Profiles

Profile | Description | Context
ShareableValueSet | Enforces the minimum information set for the value set metadata required by HL7 and other organizations that share and publish value sets | Shareable ValueSet

4.9.14.2 Extensions

ordinalValue | ValueSet.compose.include.concept | Ordinal Value | for General Extensions for use by FHIR Implementers
replaces | ValueSet | replaces | for General Extensions for use by FHIR Implementers
resource-approvalDate | ValueSet | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | ValueSet | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | ValueSet | lastReviewDate | for Resource HL7 Extensions
valueset-activityStatusDate | ValueSet | activityStatusDate | for ValueSet HL7 Extensions
valueset-author | ValueSet | author | for ValueSet HL7 Extensions
valueset-authoritativeSource | ValueSet | authoritativeSource | for ValueSet HL7 Extensions
valueset-caseSensitive | ValueSet.expansion.contains | caseSensitive | for ValueSet HL7 Extensions
valueset-concept-comments | ValueSet.compose.include.concept | concept-comments | for ValueSet HL7 Extensions
valueset-concept-definition | ValueSet.compose.include.concept, ValueSet.expansion.contains | concept-definition | for ValueSet HL7 Extensions
valueset-conceptOrder | ValueSet.compose.include.concept, ValueSet.expansion.contains | conceptOrder | for ValueSet HL7 Extensions
valueset-deprecated | ValueSet.compose.include.concept | deprecated | for ValueSet HL7 Extensions
valueset-effectiveDate | ValueSet | effectiveDate | for ValueSet HL7 Extensions
valueset-expand-group | ValueSet.compose.include | expand-group | for ValueSet HL7 Extensions
valueset-expand-rules | ValueSet.compose.include | expand-rules | for ValueSet HL7 Extensions
valueset-expansionSource | ValueSet.expansion | expansionSource | for ValueSet HL7 Extensions
valueset-expirationDate | ValueSet | expirationDate | for ValueSet HL7 Extensions
valueset-expression | ValueSet | expression | for ValueSet HL7 Extensions
valueset-extensible | ValueSet | extensible | for ValueSet HL7 Extensions
valueset-keyWord | ValueSet | keyWord | for ValueSet HL7 Extensions
valueset-label | ValueSet.compose.include.concept, ValueSet.expansion.contains | label | for ValueSet HL7 Extensions
valueset-map | ValueSet | map | for ValueSet HL7 Extensions
valueset-otherName | ValueSet | otherName | for ValueSet HL7 Extensions
valueset-parameterSource | ValueSet.expansion.parameter | parameterSource | for ValueSet HL7 Extensions
valueset-rules-text | ValueSet | rules-text | for ValueSet HL7 Extensions
valueset-sourceReference | ValueSet | sourceReference | for ValueSet HL7 Extensions
valueset-special-status | ValueSet | special-status | for ValueSet HL7 Extensions
valueset-steward | ValueSet | steward | for ValueSet HL7 Extensions
valueset-supplement | ValueSet | supplement | for ValueSet HL7 Extensions
valueset-system | ValueSet.compose.include.system | system | for ValueSet HL7 Extensions

valueset-systemName | ValueSet.compose.include | systemName | for ValueSet HL7 Extensions
valueset-systemRef | ValueSet.compose.include | systemRef | for ValueSet HL7 Extensions
valueset-toocostly | ValueSet.expansion | toocostly | for ValueSet HL7 Extensions
valueset-trusted-expansion | ValueSet | trusted-expansion | for ValueSet HL7 Extensions
valueset-unclosed | ValueSet.expansion | unclosed | for ValueSet HL7 Extensions
valueset-usage | ValueSet | usage | for ValueSet HL7 Extensions
valueset-warning | ValueSet | warning | for ValueSet HL7 Extensions
valueset-workflowStatus | ValueSet | workflowStatus | for ValueSet HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

11179-permitted-value-valueset | permitted-value-valueset | for ISO 11179 Element Definition Profile
elementdefinition-allowedUnits | allowedUnits | for Common extensions for ElementDefinition data type
elementdefinition-inheritedExtensibleValueSet | inheritedExtensibleValueSet | for Common extensions for ElementDefinition data type
elementdefinition-maxValueSet | maxValueSet | for Common extensions for ElementDefinition data type
elementdefinition-minValueSet | minValueSet | for Common extensions for ElementDefinition data type
questionnaire-unitValueSet | unitValueSet | for Core extensions for Questionnaire
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

4.9.14.3 Search Extensions

No Search Extensions defined for this resource

4.9.15 Resource ValueSet - Operations

URL: https://hl7.org/fhir/R4/valueset-operations.html

4.9.15 Resource ValueSet - Operations

Vocabulary Work Group | Maturity Level: N | Normative (from v4.0.0) | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This resource has 2 operations operations associated with it:

$expand | Value Set Expansion | Normative
$validate-code | Value Set based Validation | Normative

For more information about operations, including how they are invoked, see Operations.

4.9.15.1 Operation $expand on ValueSet

URL: https://hl7.org/fhir/R4/valueset-operation-expand.html

4.9.15.1 Operation $expand on ValueSet

Vocabulary Work Group | Maturity Level: N | Normative (from v4.0.1) | Compartments: N/A

| This page has been approved as part of an ANSI standard. See the Conformance Package for further details.

The definition of a value set is used to create a simple collection of codes suitable for use for data entry or validation.

If the operation is not called at the instance level, one of the in parameters url, context or valueSet must be provided. An expanded value set will be returned, or an OperationOutcome with an error message.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/ValueSet-expand
```

Formal Definition (as a OperationDefinition).

URL: [base]/ValueSet/$expand

URL: [base]/ValueSet/[id]/$expand

This is an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
url | 0..1 | uri | | | A canonical reference to a value set. The server must know the value set (e.g. it is defined explicitly in the server's value sets, or it is defined implicitly by some code system known to the server
valueSet | 0..1 | ValueSet | | | The value set is provided directly as part of the request. Servers may choose not to accept value sets in this fashion

valueSetVersion | 0..1 | string | | | The identifier that is used to identify a specific version of the value set to be used when generating the expansion. This is an arbitrary value managed by the value set author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available.

context | 0..1 | uri | | | The context of the value set, so that the server can resolve this to a value set to expand. The recommended format for this URI is [Structure Definition URL]#[name or path into structure definition] e.g. http://hl7.org/fhir/StructureDefinition/observation-hspc-height-hspcheight#Observation.interpretation. Other forms may be used but are not defined. This form is only usable if the terminology server also has access to the conformance registry that the server is using, but can be used to delegate the mapping from an application context to a binding at run-time

contextDirection | 0..1 | code | | | If a context is provided, a context direction may also be provided. Valid values are: 'incoming': the codes a client can use for PUT/POST operations, and 'outgoing', the codes a client might receive from the server. The purpose is to inform the server whether to use the value set associated with the context for reading or writing purposes (note: for most elements, this is the same value set, but there are a few elements where the reading and writing value sets are different)

filter | 0..1 | string | | | A text filter that is applied to restrict the codes that are returned (this is useful in a UI context). The interpretation of this is delegated to the server in order to allow to determine the most optimal search approach for the context. The server can document the way this parameter works in TerminologyCapabilities..expansion.textFilter. Typical usage of this parameter includes functionality like: using left matching e.g. "acut ast" allowing for wild cards such as %, &, ? searching on definition as well as display(s) allowing for search conditions (and / or / exclusions) Text Search engines such as Lucene or Solr, long with their considerable functionality, might also be used. The optional text search might also be code system specific, and servers might have different implementations for different code systems

date | 0..1 | dateTime | | | The date for which the expansion should be generated. if a date is provided, it means that the server should use the value set / code system definitions as they were on the given date, or return an error if this is not possible. Normally, the date is the current conditions (which is the default value) but under some circumstances, systems need to generate an expansion as it would have been in the past. A typical example of this would be where code selection is constrained to the set of codes that were available when the patient was treated, not when the record is being edited. Note that which date is appropriate is a matter for implementation policy.

offset | 0..1 | integer | | | Paging support - where to start if a subset is desired (default = 0). Offset is number of records (not number of pages)

count | 0..1 | integer | | | Paging support - how many codes should be provided in a partial page view. Paging only applies to flat expansions - servers ignore paging if the expansion is not flat. If count = 0, the client is asking how large the expansion is. Servers SHOULD honor this request for hierarchical expansions as well, and simply return the overall count

includeDesignations | 0..1 | boolean | | | Controls whether concept designations are to be included or excluded in value set expansions

designation | 0..* | string | | | A token that specifies a system+code that is either a use or a language. Designations that match by language or use are included in the expansion. If no designation is specified, it is at the server discretion which designations to return

includeDefinition | 0..1 | boolean | | | Controls whether the value set definition is included or excluded in value set expansions

activeOnly | 0..1 | boolean | | | Controls whether inactive concepts are included or excluded in value set expansions. Note that if the value set explicitly specifies that inactive codes are included, this parameter can still remove them from a specific expansion, but this parameter cannot include them if the value set excludes them

excludeNested | 0..1 | boolean | | | Controls whether or not the value set expansion nests codes or not (i.e. ValueSet.expansion.contains.contains)

excludeNotForUI | 0..1 | boolean | | | Controls whether or not the value set expansion is assembled for a user interface use or not. Value sets intended for User Interface might include 'abstract' codes or have nested contains with items with no code or abstract = true, with the sole purpose of helping a user navigate through the list efficiently, where as a value set not generated for UI use might be flat, and only contain the selectable codes in the value set. The exact implications of 'for UI' depend on the code system, and what properties it exposes for a terminology server to use. In the FHIR Specification itself, the value set expansions are generated with excludeNotForUI = false, and the expansions used when generated schema / code etc, or performing validation, are all excludeNotForUI = true.

excludePostCoordinated | 0..1 | boolean | | | Controls whether or not the value set expansion includes post coordinated codes

displayLanguage | 0..1 | code | | | Specifies the language to be used for description in the expansions i.e. the language to be used for ValueSet.expansion.contains.display

exclude-system | 0..* | canonical | | | Code system, or a particular version of a code system to be excluded from the value set expansion. The format is the same as a canonical URL: [system]|[version] - e.g. http://loinc.org|2.56

system-version | 0..* | canonical | | | Specifies a version to use for a system, if the value set does not specify which one to use. The format is the same as a canonical URL: [system]|[version] - e.g. http://loinc.org|2.56

check-system-version | 0..* | canonical | | | Edge Case: Specifies a version to use for a system. If a value set specifies a different version, an error is returned instead of the expansion. The format is the same as a canonical URL: [system]|[version] - e.g. http://loinc.org|2.56

force-system-version | 0..* | canonical | | | Edge Case: Specifies a version to use for a system. This parameter overrides any specified version in the value set (and any it depends on). The format is the same as a canonical URL: [system]|[version] - e.g. http://loinc.org|2.56. Note that this has obvious safety issues, in that it may result in a value set expansion giving a different list of codes that is both wrong and unsafe, and implementers should only use this capability reluctantly. It primarily exists to deal with situations where specifications have fallen into decay as time passes. If the value is override, the version used SHALL explicitly be represented in the expansion parameters

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | ValueSet | | | The result of the expansion. Servers generating expansions SHOULD ensure that all the parameters that affect the contents of the expansion are recorded in the ValueSet.expansion.parameter list Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The value set expansion returned by this query should be treated as a transient result that will change over time (whether it does or not depends on how the value set is specified), so applications should repeat the operation each time the value set is used.

If the expansion is too large (at the discretion of the server), the server MAY return an error (OperationOutcome with code too-costly). Clients can work through large flat expansions in a set of pages (partial views of the full expansion) instead of just getting the full expansion in a single exchange by using

offset and count parameters, or use the count parameter to request a subset of the expansion for limited purposes. Servers are not obliged to support paging, but if they do, SHALL support both the offset and count parameters. Hierarchical expansions are not subject to paging and servers simply return the entire expansion.

Different servers may return different results from expanding a value set for the following reasons:

• The underlying code systems are different (e.g. different versions, possibly with different defined behavior)

• The server optimizes filter includes differently, such as sorting by code frequency

• Servers introduce arbitrary groups to assist a user to navigate the lists based either on extensions in the definition, or additional knowledge available to the server

When a server cannot correctly expand a value set because it does not fully understand the code systems (e.g. it has the wrong version, or incomplete definitions) then it SHALL return an error. If the value set itself is unbounded due to the inclusion of post-coordinated value sets (e.g. SNOMED CT, UCUM), then the extension http://hl7.org/fhir/StructureDefinition/valueset-unclosed can be used to indicate that the expansion is incomplete

4.9.15.1.1 Examples

Request: Expanding a value set that is already registered on the server as "23", with a text filter of "abdo"

```
GET [base]/ValueSet/23/$expand?filter=abdo
```

Request: Expanding a value set by it's canonical URL (version 1.1), with a text filter of "abdo"

```
GET [base]/ValueSet/$expand?url=http://acme.com/fhir/ValueSet/23&amp;valueSetVerson=1.1&amp;filter=abdo
```

Request: Expanding a value set that is specified by the client (using JSON)

```
POST [base]/ValueSet/23/$expand
[other headers]

{
  "resourceType" : "Parameters",
  "parameter" : [
     {
     "name" : "valueSet",
     "resource" : {
       "resourceType" : "ValueSet",
     [value set details]
     }
   }
  ]
}
```

Request: Expanding a value set for a particular element, for what a client is allowed to PUT/POST. The server determines the appropriate value set to use

```
GET [base]/ValueSet/$expand?context=DiagnosticReport.category&amp;contextDirection=incoming
```

Request: Like the last example, but profile specific. Note the need to encode the #. When no contextDirection is specified, server discretion applies

```
GET [base]/ValueSet/$expand?context=http://fhir.org/guides/argonaut-clinicalnotes/StructureDefinition/argo-
    diagnosticreport%23DiagnosticReport.category
```

Request: Expanding a value set for a particular date (for an old record), for a german user

```
GET [base]/ValueSet/23/$expand?date=2014-02-23&amp;displayLanguage=de
```

Request: Expanding a value set, in the 3rd page

```
GET [base]/ValueSet/23/$expand?offset=200&amp;count=100
```

Request: Expanding a value set. Asking not to get any SNOMED CT codes, and requiring that any LOINC codes come from 2.62 (if any other version of LOINC is in use, it's an error)

```
GET [base]/ValueSet/23/$expand?exclude-system=http://snomed.info/sct&amp;check-system-version=http://loinc.org|2.62
```

Response: General pattern of response. Note: This specification also provides a file with 100s of expansions for code generators in both JSON and XML

```
HTTP/1.1 200 OK
[other headers]

<ValueSet xmlns="http://hl7.org/fhir">
  <!-- the server SHOULD populate the id with a newly created UUID
    so clients can easily track a particular expansion  -->
  <id value="43770626-f685-4ba8-8d66-fb63e674c467"/>
  <!-- no need for meta, though it is allowed for security labels, profiles -->

  <!-- other value set details -->
  <expansion>
    <!-- when expanded -->
    <timestamp value="20141203T08:50:00+11:00"/>
    <contains>
      <!-- expansion contents -->
    </contains>
  </expansion>
</ValueSet>
```

For more information about operations, including how they are invoked, see Operations.

4.9.15.2 Operation $validate-code on ValueSet

URL: https://hl7.org/fhir/R4/valueset-operation-validate-code.html

4.9.15.2 Operation $validate-code on ValueSet

Vocabulary Work Group | Maturity Level: N | Normative (from v4.0.1) | Compartments: N/A

| This page has been approved as part of an ANSI standard. See the Conformance Package for further details.

Validate that a coded value is in the set of codes allowed by a value set.

If the operation is not called at the instance level, one of the in parameters url, context or valueSet must be provided. One (and only one) of the in parameters code, coding, or codeableConcept must be provided. The operation returns a result (true / false), an error message, and the recommended display for the code.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/ValueSet-validate-code
```

Formal Definition (as a OperationDefinition).

URL: [base]/ValueSet/$validate-code

URL: [base]/ValueSet/[id]/$validate-code

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

url | 0..1 | uri | | | Value set Canonical URL. The server must know the value set (e.g. it is defined explicitly in the server's value sets, or it is defined implicitly by some code system known to the server

context | 0..1 | uri | | | The context of the value set, so that the server can resolve this to a value set to validate against. The recommended format for this URI is [Structure Definition URL]#[name or path into structure definition] e.g. http://hl7.org/fhir/StructureDefinition/observation-hspc-height-hspcheight#Observation.interpretation. Other forms may be used but are not defined. This form is only usable if the terminology server also has access to the conformance registry that the server is using, but can be used to delegate the mapping from an application context to a binding at run-time

valueSet | 0..1 | ValueSet | | | The value set is provided directly as part of the request. Servers may choose not to accept value sets in this fashion. This parameter is used when the client wants the server to expand a value set that is not stored on the server

valueSetVersion | 0..1 | string | | | The identifier that is used to identify a specific version of the value set to be used when validating the code. This is an arbitrary value managed by the value set author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available.

code | 0..1 | code | | | The code that is to be validated. If a code is provided, a system or a context must be provided (if a context is provided, then the server SHALL ensure that the code is not ambiguous without a system)

system | 0..1 | uri | | | The system for the code that is to be validated

systemVersion | 0..1 | string | | | The version of the system, if one was provided in the source data

display | 0..1 | string | | | The display associated with the code, if provided. If a display is provided a code must be provided. If no display is provided, the server cannot validate the display value, but may choose to return a recommended display name using the display parameter in the outcome. Whether displays are case sensitive is code system dependent

coding | 0..1 | Coding | | | A coding to validate

codeableConcept | 0..1 | CodeableConcept | | | A full codeableConcept to validate. The server returns true if one of the coding values is in the value set, and may also validate that the codings are not in conflict with each other if more than one is present

date | 0..1 | dateTime | | | The date for which the validation should be checked. Normally, this is the current conditions (which is the default values) but under some circumstances, systems need to validate that a correct code was used at some point in the past. A typical example of this would be where code selection is constrained to the set of codes that were available when the patient was treated, not when the record is being edited. Note that which date is appropriate is a matter for implementation policy.

abstract | 0..1 | boolean | | | If this parameter has a value of true, the client is stating that the validation is being performed in a context where a concept designated as 'abstract' is appropriate/allowed to be used, and the server should regard abstract codes as valid. If this parameter is false, abstract codes are not considered to be valid. Note that. 'abstract' is a property defined by many HL7 code systems that indicates that the concept is a logical grouping concept that is not intended to be used asa 'concrete' concept to in an actual patient/care/process record. This language is borrowed from Object Orienated theory where 'asbtract' objects are never instantiated. However in the general record and terminology eco-system, there are many contexts where it is appropraite to use these codes e.g. as decision making criterion, or when editing value sets themselves. This parameter allows a client to indicate to the server that it is working in such a context.

displayLanguage | 0..1 | code | | | Specifies the language to be used for description when validating the display property

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

result | 1..1 | boolean | | | True if the concept details supplied are valid

message | 0..1 | string | | | Error details, if result = false. If this is provided when result = true, the message carries hints and warnings

display | 0..1 | string | | | A valid display for the concept if the system wishes to display this to a user

Note: the correct behaviour of validation with regard to language for Coding.display items is currently undefined, and further development and testing may lead to specific requirements or recommendations in subsequent releases

4.9.15.2.1 Examples

Request: Simple validation of a code/system against a known value set

```
GET [base]/ValueSet/23/$validate-code?system=http://loinc.org&code=1963-8&display=test
```

Request: Validate a CodeableConcept against a client specified value set

```
POST [base]/ValueSet/$validate-code
[other headers]

{
  "ResourceType" : "Parameters",
  "parameter" : [
    {
    "name" : "codeableConcept",
    "valueCodeableConcept" : {
      "coding" : {
        "system" : "http://loinc.org",
          "code" : "1963-8",
      "display" : "test"
      }
    }
  },
  {
    "name" : "valueSet",
    "resource": {
      "resourceType" : "ValueSet",
    [etc]
    }
  }
  ]
}
```

Response:

```
HTTP/1.1 200 OK
[other headers]

{
  "resourceType" : "Parameters",
  "parameter" : [
    {
    "name" : "result",
    "valueBoolean" : "false"
  },
  {
    "name" : "message",
    "valueString" : "The display \"test\" is incorrect"
  },
  {
    "name" : "display",
    "valueString" : "Bicarbonate [Moles/volume] in Serum"
  }
  ]
}
```

For more information about operations, including how they are invoked, see Operations.

4.9.16 Resource ValueSet - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/valueset-version-maps.html

No validation errors - all conversions are clean

4.9.16 Resource ValueSet - R3/R4 Conversions

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for ValueSet.

Functional status for this map: 8 tests that all execute ok. All tests pass round-trip testing and 8 r3 resources are invalid (0 errors). (see documentation)

4.9.16.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/ValueSet3to4" = "R3 to R4 Conversions for ValueSet"

uses "http://hl7.org/fhir/3.0/StructureDefinition/ValueSet" alias ValueSetR3 as source
uses "http://hl7.org/fhir/StructureDefinition/ValueSet" alias ValueSet as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group ValueSet(source src : ValueSetR3, target tgt : ValueSet) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
```

```
  src.immutable -> tgt.immutable;
  src.purpose -> tgt.purpose;
  src.copyright -> tgt.copyright;
  src.extensible as vs ->  tgt.extension as ext,  ext.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    ValueSet.extensible',  ext.value = vs;
  src.compose as vs0 -> tgt.compose as vt0 then compose(vs0, vt0);
  src.expansion as vs0 -> tgt.expansion as vt0 then expansion(vs0, vt0);
}

group compose(source src, target tgt) extends BackboneElement {
  src.lockedDate -> tgt.lockedDate;
  src.inactive -> tgt.inactive;
  src.include as vs0 -> tgt.include as vt0 then include(vs0, vt0);
  src.exclude as vs0 -> tgt.exclude as vt0 then include(vs0, vt0);
}

group include(source src, target tgt) extends BackboneElement {
  src.system -> tgt.system;
  src.version -> tgt.version;
  src.concept as vs0 -> tgt.concept as vt0 then concept(vs0, vt0);
  src.filter as vs0 -> tgt.filter as vt0 then filter(vs0, vt0);
  src.valueSet -> tgt.valueSet;
}

group concept(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.designation as vs0 -> tgt.designation as vt0 then designation(vs0, vt0);
}

group designation(source src, target tgt) extends BackboneElement {
  src.language -> tgt.language;
  src.use -> tgt.use;
  src.value -> tgt.value;
}

group filter(source src, target tgt) extends BackboneElement {
  src.property -> tgt.property;
  src.op -> tgt.op;
  src.value -> tgt.value;
}

group expansion(source src, target tgt) extends BackboneElement {
  src.identifier -> tgt.identifier;
  src.timestamp -> tgt.timestamp;
  src.total -> tgt.total;
  src.offset -> tgt.offset;
  src.parameter as vs0 -> tgt.parameter as vt0 then parameter(vs0, vt0);
  src.contains as vs0 -> tgt.contains as vt0 then contains(vs0, vt0);
}

group parameter(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.value : string as vs0 -> tgt.value = create('string') as vt0 then string(vs0, vt0) "valueString";
  src.value : boolean as vs0 -> tgt.value = create('boolean') as vt0 then boolean(vs0, vt0) "valueBoolean";
  src.value : integer as vs0 -> tgt.value = create('integer') as vt0 then integer(vs0, vt0) "valueInteger";
  src.value : decimal as vs0 -> tgt.value = create('decimal') as vt0 then decimal(vs0, vt0) "valueDecimal";
  src.value : url as vs0 -> tgt.value = create('url') as vt0 then url(vs0, vt0) "valueUrl";
  src.value : code as vs0 -> tgt.value = create('code') as vt0 then code(vs0, vt0) "valueCode";
  src.value : dateTime as vs0 -> tgt.value = create('dateTime') as vt0 then dateTime(vs0, vt0) "valueDateTime";
}

group contains(source src, target tgt) extends BackboneElement {
  src.system -> tgt.system;
  src.abstract -> tgt.abstract;
  src.inactive -> tgt.inactive;
  src.version -> tgt.version;
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.designation as vs0 -> tgt.designation as vt0 then designation(vs0, vt0);
  src.contains as vs0 -> tgt.contains as vt0 then contains(vs0, vt0);
}
```

4.9.16.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/ValueSet4to3" = "R4 to R3 Conversion for ValueSet"

uses "http://hl7.org/fhir/StructureDefinition/ValueSet" alias ValueSet as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/ValueSet" alias ValueSetR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group ValueSet(source src : ValueSet, target tgt : ValueSetR3) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
```

```
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.immutable -> tgt.immutable;
  src.purpose -> tgt.purpose;
  src.copyright -> tgt.copyright;
  src.extension as ext where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-ValueSet.extensible' then {
    ext.value : boolean as vs0 -> tgt.extensible = vs0 "extensible2";
  } "extensible";
  src.compose as vs0 -> tgt.compose as vt0 then compose(vs0, vt0);
  src.expansion as vs0 -> tgt.expansion as vt0 then expansion(vs0, vt0);
}

group compose(source src, target tgt) extends BackboneElement {
  src.lockedDate -> tgt.lockedDate;
  src.inactive -> tgt.inactive;
  src.include as vs0 -> tgt.include as vt0 then include(vs0, vt0);
  src.exclude as vs0 -> tgt.exclude as vt0 then include(vs0, vt0);
}

group include(source src, target tgt) extends BackboneElement {
  src.system -> tgt.system;
  src.version -> tgt.version;
  src.concept as vs0 -> tgt.concept as vt0 then concept(vs0, vt0);
  src.filter as vs0 -> tgt.filter as vt0 then filter(vs0, vt0);
  src.valueSet -> tgt.valueSet;
}

group concept(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.designation as vs0 -> tgt.designation as vt0 then designation(vs0, vt0);
}

group designation(source src, target tgt) extends BackboneElement {
  src.language -> tgt.language;
  src.use -> tgt.use;
  src.value -> tgt.value;
}

group filter(source src, target tgt) extends BackboneElement {
  src.property -> tgt.property;
  src.op -> tgt.op;
  src.value -> tgt.value;
}

group expansion(source src, target tgt) extends BackboneElement {
  src.identifier -> tgt.identifier;
  src.timestamp -> tgt.timestamp;
  src.total -> tgt.total;
  src.offset -> tgt.offset;
  src.parameter as vs0 -> tgt.parameter as vt0 then parameter(vs0, vt0);
  src.contains as vs0 -> tgt.contains as vt0 then contains(vs0, vt0);
}

group parameter(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.value : string as vs0 -> tgt.value = create('string') as vt0 then string(vs0, vt0) "valueString";
  src.value : boolean as vs0 -> tgt.value = create('boolean') as vt0 then boolean(vs0, vt0) "valueBoolean";
  src.value : integer as vs0 -> tgt.value = create('integer') as vt0 then integer(vs0, vt0) "valueInteger";
  src.value : decimal as vs0 -> tgt.value = create('decimal') as vt0 then decimal(vs0, vt0) "valueDecimal";
  src.value : url as vs0 -> tgt.value = create('url') as vt0 then url(vs0, vt0) "valueUrl";
  src.value : code as vs0 -> tgt.value = create('code') as vt0 then code(vs0, vt0) "valueCode";
  src.value : dateTime as vs0 -> tgt.value = create('dateTime') as vt0 then dateTime(vs0, vt0) "valueDateTime";
}

group contains(source src, target tgt) extends BackboneElement {
  src.system -> tgt.system;
  src.abstract -> tgt.abstract;
  src.inactive -> tgt.inactive;
  src.version -> tgt.version;
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.designation as vs0 -> tgt.designation as vt0 then designation(vs0, vt0);
  src.contains as vs0 -> tgt.contains as vt0 then contains(vs0, vt0);
}
```

4.9.16.3 R4 Validation Errors

4.9.17 ValueSet HL7 Extensions

URL: https://hl7.org/fhir/R4/valueset-extensions.html

Defines common extensions used with or related to the ValueSet resource

4.9.17 ValueSet HL7 Extensions

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative

4.9.17.1 Content

Extensions:

valueset-concept-definition | concept-definition : A definition that describes the meaning of this code when used as part of this value set.

valueset-concept-comments | concept-comments : A comment that explains how this code is used in this context (where the value set is expected to be used).

valueset-otherName | otherName : Human readable names for the valueset.

valueset-sourceReference | sourceReference : This text is intended to act as a citation to work done elsewhere that is not part of the current stewarding process where the referenced source is in some way a basis of the current value set definition.

valueset-keyWord | keyWord : Word or words used in an information retrieval system to indicate the content of the value set.

valueset-usage | usage : Consumers of the value set and the implementations, projects or standards that the author has utilized the value set in.

valueset-workflowStatus | workflowStatus : Workflow Status is used to represent details of the value set development process while in a single Activity Status.

valueset-activityStatusDate | activityStatusDate : The date when the associated Value Set Definition Version activity status is in effect.

valueset-effectiveDate | effectiveDate : This is the first date-time when the value set version becomes active, so this value is present on Inactive value set versions as well. The start Date_time is expected to be as of 0001 UTC of the Effective Date.

valueset-expirationDate | expirationDate : The date when the value set version is no longer expected to be used to create new content. This is the first date-time when the value set version becomes Inactive, so this value SHALL present on all Inactive value set versions. The start Date_time is expected to be as of 0001 UTC of the Expiration Date.

valueset-author | author : The entity or set of entities that create and may modify the Value Set Definition content. The name of a group or an individual, along with contact details.

valueset-steward | steward : The entity that is responsible for the content of the Value Set Definition. This is a textual description of the organizational entity responsible for the content and maintenance.

valueset-trusted-expansion | trusted-expansion : Indicates an authoritative source for performing value set expansions.

valueset-label | label : The label to list in front of a code when presenting a list of possible values in a questionnaire-like fashion.

valueset-conceptOrder | conceptOrder : Identifies the relative order in which concepts within the value set should be presented to a user.

valueset-systemName | systemName : The human-readable name for the code system.

valueset-systemRef | systemRef : The formal URI for the code system. I.e. ValueSet.codeSystem.system (or its equivalent).

valueset-caseSensitive | caseSensitive : If this a case sensitive code.

valueset-reference | reference : A logical reference (e.g. a reference to ValueSet.url) that identifies the value set/version that identifies the set of possible coded values this coding was chosen from or constrained by.

valueset-expansionSource | expansionSource : The logical URL of the ValueSet definition that was used to generate this expansion.

valueset-unclosed | unclosed : Marks that the expansion is incomplete, and values other than those listed may be valid. This may be used because post-coordinated codes are allowed, and no practical expansion can be produced.

valueset-toocostly | toocostly : Marks that the expansion is incomplete, because the full value set is too large to represent, and the client asked for an incomplete fragment.

valueset-map | map : A reference to a concept map that is relevant for the interpretation of this value set.

valueset-warning | warning : An extra warning about the correct use of the value set.

valueset-system | system : Allows a direct reference to the code system for FHIR query.

valueset-supplement | supplement : This extension declares that a value set depends on a particular supplement and should not be used in its absence.

valueset-parameterSource | parameterSource : Declares what the source of this parameter is.

valueset-expand-rules | expand-rules : Defines how concepts are processed into the expansion when it's for UI presentation.

valueset-expand-group | expand-group : This extension declares a group of concepts that is generated into the ValueSet.expansion.contains hierarchy when the expansion is generated for a UI. THere is no inherent assigned meaning to the hierarchy; it is used to help the user navigate the concepts. Each group has a display and/or a code, and a list of members, which are either concepts in the value set, or other groups (by code).

valueset-deprecated | deprecated : if ture, indicates that the concept is deprecated from the value set - that is, it should not be used, and is planned to be withdrawn.

valueset-special-status | special-status : A special note for implementers about the status of the resource.

valueset-extensible | extensible : Whether this is intended to be used with an extensible binding or not.

valueset-authoritativeSource | authoritativeSource : A reference to the authoritative accessible, persisted source of truth of the entire Value Set Definition, including textual information and available versions.

valueset-expression | expression : An expression that provides an alternative definition of the content of the value set. There are two different ways to use this expression extension: If both an expression and a compose element is present, the compose is understood the make the same statement as the expression. If there is no compose, the expression is the only definition of the value set, and the value set can only be processed by a server that understands the expression syntax, it that is computable.

valueset-rules-text | rules-text : An expression that provides an alternative definition of the content of the value set in some form that is not computable - e.g instructions that could only be followed by a human.

4.9.17.2 Search Parameters

Search parameters defined by this package. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths | Source

author | string | Optional Extensions Element | f:ValueSet/f:extension[@url='http://hl7.org/fhir/StructureDefinition/valueset-author'] | /f:#author | XML / JSON

effective | date | Optional Extensions Element | f:ValueSet/f:extension[@url='http://hl7.org/fhir/StructureDefinition/valueset-effectiveDate'] | /f:#effectiveDate | XML / JSON
end | date | Optional Extensions Element | f:ValueSet/f:extension[@url='http://hl7.org/fhir/StructureDefinition/valueset-expirationDate'] | /f:#expirationDate | XML / JSON
keyword | string | Optional Extensions Element | f:ValueSet/f:extension[@url='http://hl7.org/fhir/StructureDefinition/valueset-keyWord'] | /f:#keyWord | XML / JSON
workflow | token | Optional Extensions Element | f:ValueSet/f:extension[@url='http://hl7.org/fhir/StructureDefinition/valueset-workflowStatus'] | /f:#workflowStatus | XML / JSON

4.9.17.3 Valueset-extensions-ValueSet-keyword

URL: https://hl7.org/fhir/R4/valueset-extensions-ValueSet-keyword.html

4.9.17.3 Valueset-extensions-ValueSet-keyword

Search Parameter keyword (type: string) defined as part of the valueset-extensions Profile.

Vocabulary | Maturity Level: 0 | Informative

Optional Extensions Element

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/valueset-extensions-ValueSet-keyword

• Paths : f:ValueSet/f:extension[@url='http://hl7.org/fhir/StructureDefinition/valueset-keyWord'] | /f:#keyWord

• Targets :

4.9.17.4 Valueset-extensions-ValueSet-workflow

URL: https://hl7.org/fhir/R4/valueset-extensions-ValueSet-workflow.html

4.9.17.4 Valueset-extensions-ValueSet-workflow

Search Parameter workflow (type: token) defined as part of the valueset-extensions Profile.

Vocabulary | Maturity Level: 0 | Informative

Optional Extensions Element

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/valueset-extensions-ValueSet-workflow

• Paths : f:ValueSet/f:extension[@url='http://hl7.org/fhir/StructureDefinition/valueset-workflowStatus'] | /f:#workflowStatus

• Targets :

4.9.17.5 Valueset-extensions-ValueSet-effective

URL: https://hl7.org/fhir/R4/valueset-extensions-ValueSet-effective.html

4.9.17.5 Valueset-extensions-ValueSet-effective

Search Parameter effective (type: date) defined as part of the valueset-extensions Profile.

Vocabulary | Maturity Level: 0 | Informative

Optional Extensions Element

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/valueset-extensions-ValueSet-effective

• Paths : f:ValueSet/f:extension[@url='http://hl7.org/fhir/StructureDefinition/valueset-effectiveDate'] | /f:#effectiveDate

• Targets :

4.9.17.6 Valueset-extensions-ValueSet-end

URL: https://hl7.org/fhir/R4/valueset-extensions-ValueSet-end.html

4.9.17.6 Valueset-extensions-ValueSet-end

Search Parameter end (type: date) defined as part of the valueset-extensions Profile.

Vocabulary | Maturity Level: 0 | Informative

Optional Extensions Element

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/valueset-extensions-ValueSet-end

• Paths : f:ValueSet/f:extension[@url='http://hl7.org/fhir/StructureDefinition/valueset-expirationDate'] | /f:#expirationDate

• Targets :

4.9.17.7 Valueset-extensions-ValueSet-author

URL: https://hl7.org/fhir/R4/valueset-extensions-ValueSet-author.html

4.9.17.7 Valueset-extensions-ValueSet-author

Search Parameter author (type: string) defined as part of the valueset-extensions Profile.

Vocabulary | Maturity Level: 0 | Informative

Optional Extensions Element

Other Details:

• Full URL : http://hl7.org/fhir/SearchParameter/valueset-extensions-ValueSet-author

• Paths : f:ValueSet/f:extension[@url='http://hl7.org/fhir/StructureDefinition/valueset-author'] | /f:#author

• Targets :

4.9.18 Shareable ValueSet

URL: https://hl7.org/fhir/R4/valueset-shareable.html

Enforces the minimum information set for the value set metadata required by HL7 and other organizations that share and publish value sets

4.9.18 Shareable ValueSet

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative

4.9.18.1 Content

Profiles:
ShareableValueSet | Enforces the minimum information set for the value set metadata required by HL7 and other organizations that share and publish value sets

4.9.18.2 StructureDefinition: ShareableValueSet

URL: https://hl7.org/fhir/R4/shareablevalueset.html

4.9.18.2 StructureDefinition: ShareableValueSet

Vocabulary Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/shareablevalueset`

Enforces the minimum information set for the value set metadata required by HL7 and other organizations that share and publish value sets

This profile was published on [no date] as a draft by HL7.

4.9.18.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary

• Differential Table

• Snapshot Table

• XML Template

• JSON Template

• All

This structure is derived from ValueSet.

Summary

Mandatory: 7 elements

This structure is derived from ValueSet.

Name | Flags | Card. | Type | Description & Constraints
ValueSet | | 1..1
url | | 1..1 | uri
version | | 1..1 | string
name | | 1..1 | string
status | | 1..1 | code
experimental | | 1..1 | boolean
publisher | | 1..1 | string
description | | 1..1 | markdown
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
ValueSet | I | 1..1 | | A set of codes drawn from one or more code systems
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 1..1 | uri | Canonical identifier for this value set, represented as a URI (globally unique)
identifier | ΣI | 0..* | Identifier | Additional identifier for the value set (business identifier)
version | ΣI | 1..1 | string | Business version of the value set
name | ΣI | 1..1 | string | Name for this value set (computer friendly)
title | ΣI | 0..1 | string | Name for this value set (human friendly)
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 1..1 | boolean | For testing purposes, not real usage
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 1..1 | string | Name of the publisher (organization or individual)
contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | I | 1..1 | markdown | Natural language description of the value set
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for value set (if applicable) Binding: Jurisdiction (extensible)
immutable | ΣI | 0..1 | boolean | Indicates whether or not any change to the content logical definition may occur
purpose | I | 0..1 | markdown | Why this value set is defined
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
compose | I | 0..1 | BackboneElement | Content logical definition of the value set (CLD)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
lockedDate | ΣI | 0..1 | date | Fixed date for references with no specified version (transitive)
inactive | ΣI | 0..1 | boolean | Whether inactive codes are in the value set
include | ΣI | 1..* | BackboneElement | Include one or more codes from a code system or other value set(s) vsd-1: A value set include/exclude SHALL have a value set or a system vsd-2: A value set with concepts or filters SHALL include a system vsd-3: Cannot have both concept and filter
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
system | ΣI | 0..1 | uri | The system the codes come from
version | ΣI | 0..1 | string | Specific version of the code system referred to
concept | I | 0..* | BackboneElement | A concept defined in the system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 1..1 | code | Code or expression from system
display | I | 0..1 | string | Text to display for this code for this value set in this valueset
designation | I | 0..* | BackboneElement | Additional representations for this concept
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
language | I | 0..1 | code | Human language of the designation Binding: Common Languages (preferred) Max Binding: All Languages
use | I | 0..1 | Coding | Types of uses of designations Binding: Designation Use (extensible)
value | I | 1..1 | string | The text value for this designation
filter | ΣI | 0..* | BackboneElement | Select codes/concepts by their properties (including relationships)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
property | ΣI | 1..1 | code | A property/filter defined by the code system
op | ΣI | 1..1 | code | = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists Binding: FilterOperator (required)
value | ΣI | 1..1 | string | Code from the system, or regex criteria, or boolean value for exists
valueSet | ΣI | 0..* | canonical(ValueSet) | Select the contents included in this value set
exclude | I | 0..* | | Unknown reference to #ValueSet.compose.include Explicitly exclude codes from a code system or other value sets
expansion | I | 0..1 | BackboneElement | Used when the value set is "expanded"
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
identifier | I | 0..1 | uri | Identifies the value set expansion (business identifier)
timestamp | I | 1..1 | dateTime | Time ValueSet expansion happened
total | I | 0..1 | integer | Total number of codes in the expansion
offset | I | 0..1 | integer | Offset at which this resource starts

parameter | I | 0..* | BackboneElement | Parameter that controlled the expansion process
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
name | I | 1..1 | string | Name as assigned by the client or server
value[x] | I | 0..1 | | Value of the named parameter
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueDecimal | | | decimal
valueUri | | | uri
valueCode | | | code
valueDateTime | | | dateTime
contains | I | 0..* | BackboneElement | Codes in the value set vsd-6: SHALL have a code or a display vsd-9: Must have a code if not abstract vsd-10: Must have a system if a code is present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
system | I | 0..1 | uri | System value for the code
abstract | I | 0..1 | boolean | If user cannot select this entry
inactive | I | 0..1 | boolean | If concept is inactive in the code system
version | I | 0..1 | string | Version in which this code/display is defined
code | I | 0..1 | code | Code - if blank, this is not a selectable code
display | I | 0..1 | string | User display for the concept
designation | I | 0..* | | Unknown reference to #ValueSet.compose.include.concept.designation Additional representations for this item
contains | I | 0..* | | Unknown reference to #ValueSet.expansion.contains Codes contained under this entry
Documentation for this format

```
<!-- Shareable ValueSet -->

<ValueSet xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  1..1 Canonical identifier for this value set, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the value set (business identifier) --></identifier>
 <version value="[string]"/><!--  1..1 Business version of the value set -->
 <name value="[string]"/><!--  1..1 Name for this value set (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this value set (human friendly) -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  1..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  1..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  1..1 Natural language description of the value set -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for value set (if applicable)   --></jurisdiction>
 <immutable value="[boolean]"/><!--  0..1 Indicates whether or not any change to the content logical definition may
    occur -->
 <purpose value="[markdown]"/><!--  0..1 Why this value set is defined -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <compose>  0..1 BackboneElement  <!--  0..1 Content logical definition of the value set (CLD) -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <lockedDate value="[date]"/><!--  0..1 Fixed date for references with no specified version (transitive) -->
  <inactive value="[boolean]"/><!--  0..1 Whether inactive codes are in the value set -->
  <include>  1..* BackboneElement  <!--  1..* Include one or more codes from a code system or other value set(s) -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <system value="[uri]"/><!--  0..1 The system the codes come from -->
   <version value="[string]"/><!--  0..1 Specific version of the code system referred to -->
   <concept>  0..* BackboneElement  <!--  0..* A concept defined in the system -->
    <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
    <code value="[code]"/><!--  1..1 Code or expression from system -->
    <display value="[string]"/><!--  0..1
        Text to display for this code for this value set in this valueset -->
    <designation>  0..* BackboneElement  <!--  0..* Additional representations for this concept -->
     <modifierExtension><!--  0..*
       Extension  Extensions that cannot be ignored even if unrecognized  --></modifierExtension>
     <language value="[code]"/><!--  0..1 Human language of the designation   -->
     <use><!--  0..1 Coding Types of uses of designations   --></use>
     <value value="[string]"/><!--  1..1 The text value for this designation -->
```

```
    </designation>
   </concept>
   <filter>  0..* BackboneElement  <!--  0..* Select codes/concepts by their properties (including relationships) -->
    <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
    <property value="[code]"/><!--  1..1 A property/filter defined by the code system -->
    <op value="[code]"/><!--  1..1 = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists
    -->
    <value value="[string]"/><!--  1..1
        Code from the system, or regex criteria, or boolean value for exists -->
   </filter>
   <valueSet value="[canonical]"/><!--  0..* Select the contents included in this value set -->
  </include>
  <exclude><!-- See #ValueSet.compose.include  Explicitly exclude codes from a code system or other value sets
    --></exclude>
 </compose>
 <expansion>  0..1 BackboneElement  <!--  0..1 Used when the value set is "expanded" -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <identifier value="[uri]"/><!--  0..1 Identifies the value set expansion (business identifier) -->
  <timestamp value="[dateTime]"/><!--  1..1 Time ValueSet expansion happened -->
  <total value="[integer]"/><!--  0..1 Total number of codes in the expansion -->
  <offset value="[integer]"/><!--  0..1 Offset at which this resource starts -->
  <parameter>  0..* BackboneElement  <!--  0..* Parameter that controlled the expansion process -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <name value="[string]"/><!--  1..1 Name as assigned by the client or server -->
   <value[x]><!--  0..1 string|boolean|integer|
     decimal|uri|code|dateTime Value of the named parameter --></value[x]>
  </parameter>
  <contains>  0..* BackboneElement  <!--  0..* Codes in the value set -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <system value="[uri]"/><!--  0..1 System value for the code -->
   <abstract value="[boolean]"/><!--  0..1 If user cannot select this entry -->
   <inactive value="[boolean]"/><!--  0..1 If concept is inactive in the code system -->
   <version value="[string]"/><!--  0..1 Version in which this code/display is defined -->
   <code value="[code]"/><!--  0..1 Code - if blank, this is not a selectable code -->
   <display value="[string]"/><!--  0..1 User display for the concept -->
   <designation><!-- See #ValueSet.compose.include.concept.designation  Additional representations for this item
    --></designation>
   <contains><!-- See #ValueSet.expansion.contains  Codes contained under this entry --></contains>
  </contains>
 </expansion>
</ValueSet>
```

```
{ // Shareable ValueSet
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C? R! Canonical identifier for this value set, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the value set (business identifier)
    "version" : "<string>", // C? R! Business version of the value set
    "name" : "<string>", // C? R! Name for this value set (computer friendly)
    "title" : "<string>", // C?Name for this value set (human friendly)
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C? R! For testing purposes, not real usage
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C? R! Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C? R! Natural language description of the value set
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for value set (if applicable)
    "immutable" : <boolean>, // C?Indicates whether or not any change to the content logical definition may occur
    "purpose" : "<markdown>", // C?Why this value set is defined
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "compose" : { BackboneElement }, // C?Content logical definition of the value set (CLD)
    "expansion" : { BackboneElement } // C?Used when the value set is "expanded"
  }
```

This structure is derived from ValueSet.

Summary

Mandatory: 7 elements

Differential View

This structure is derived from ValueSet.

Name	Flags	Card.	Type	Description & Constraints
ValueSet		1..1		
url		1..1	uri	
version		1..1	string	
name		1..1	string	
status		1..1	code	

experimental | | 1..1 | boolean
publisher | | 1..1 | string
description | | 1..1 | markdown
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
ValueSet | I | 1..1 | | A set of codes drawn from one or more code systems
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 1..1 | uri | Canonical identifier for this value set, represented as a URI (globally unique)
identifier | ΣI | 0..* | Identifier | Additional identifier for the value set (business identifier)
version | ΣI | 1..1 | string | Business version of the value set
name | ΣI | 1..1 | string | Name for this value set (computer friendly)
title | ΣI | 0..1 | string | Name for this value set (human friendly)
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 1..1 | boolean | For testing purposes, not real usage
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 1..1 | string | Name of the publisher (organization or individual)
contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | I | 1..1 | markdown | Natural language description of the value set
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for value set (if applicable) Binding: Jurisdiction (extensible)
immutable | ΣI | 0..1 | boolean | Indicates whether or not any change to the content logical definition may occur
purpose | I | 0..1 | markdown | Why this value set is defined
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
compose | I | 0..1 | BackboneElement | Content logical definition of the value set (CLD)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
lockedDate | ΣI | 0..1 | date | Fixed date for references with no specified version (transitive)
inactive | ΣI | 0..1 | boolean | Whether inactive codes are in the value set
include | ΣI | 1..* | BackboneElement | Include one or more codes from a code system or other value set(s) vsd-1: A value set include/exclude SHALL have a value set or a system vsd-2: A value set with concepts or filters SHALL include a system vsd-3: Cannot have both concept and filter
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
system | ΣI | 0..1 | uri | The system the codes come from
version | ΣI | 0..1 | string | Specific version of the code system referred to
concept | I | 0..* | BackboneElement | A concept defined in the system
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 1..1 | code | Code or expression from system
display | I | 0..1 | string | Text to display for this code for this value set in this valueset
designation | I | 0..* | BackboneElement | Additional representations for this concept
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
language | I | 0..1 | code | Human language of the designation Binding: Common Languages (preferred) Max Binding: All Languages
use | I | 0..1 | Coding | Types of uses of designations Binding: Designation Use (extensible)
value | I | 1..1 | string | The text value for this designation
filter | ΣI | 0..* | BackboneElement | Select codes/concepts by their properties (including relationships)
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
property | ΣI | 1..1 | code | A property/filter defined by the code system
op | ΣI | 1..1 | code | = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists Binding: FilterOperator (required)
value | ΣI | 1..1 | string | Code from the system, or regex criteria, or boolean value for exists
valueSet | ΣI | 0..* | canonical(ValueSet) | Select the contents included in this value set
exclude | I | 0..* | | Unknown reference to #ValueSet.compose.include Explicitly exclude codes from a code system or other value sets
expansion | I | 0..1 | BackboneElement | Used when the value set is "expanded"
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations

modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
identifier | I | 0..1 | uri | Identifies the value set expansion (business identifier)
timestamp | I | 1..1 | dateTime | Time ValueSet expansion happened
total | I | 0..1 | integer | Total number of codes in the expansion
offset | I | 0..1 | integer | Offset at which this resource starts
parameter | I | 0..* | BackboneElement | Parameter that controlled the expansion process
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
name | I | 1..1 | string | Name as assigned by the client or server
value[x] | I | 0..1 | | Value of the named parameter
valueString | | | string
valueBoolean | | | boolean
valueInteger | | | integer
valueDecimal | | | decimal
valueUri | | | uri
valueCode | | | code
valueDateTime | | | dateTime
contains | I | 0..* | BackboneElement | Codes in the value set vsd-6: SHALL have a code or a display vsd-9: Must have a code if not abstract vsd-10: Must have a system if a code is present
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
system | I | 0..1 | uri | System value for the code
abstract | I | 0..1 | boolean | If user cannot select this entry
inactive | I | 0..1 | boolean | If concept is inactive in the code system
version | I | 0..1 | string | Version in which this code/display is defined
code | I | 0..1 | code | Code - if blank, this is not a selectable code
display | I | 0..1 | string | User display for the concept
designation | I | 0..* | | Unknown reference to #ValueSet.compose.include.concept.designation Additional representations for this item
contains | I | 0..* | | Unknown reference to #ValueSet.expansion.contains Codes contained under this entry
Documentation for this format

XML Template

```
<!-- Shareable ValueSet -->

<ValueSet xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  1..1 Canonical identifier for this value set, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the value set (business identifier) --></identifier>
 <version value="[string]"/><!--  1..1 Business version of the value set -->
 <name value="[string]"/><!--  1..1 Name for this value set (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this value set (human friendly) -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  1..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  1..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  1..1 Natural language description of the value set -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for value set (if applicable)   --></jurisdiction>
 <immutable value="[boolean]"/><!--  0..1 Indicates whether or not any change to the content logical definition may
    occur -->
 <purpose value="[markdown]"/><!--  0..1 Why this value set is defined -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <compose>  0..1 BackboneElement  <!--  0..1 Content logical definition of the value set (CLD) -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <lockedDate value="[date]"/><!--  0..1 Fixed date for references with no specified version (transitive) -->
  <inactive value="[boolean]"/><!--  0..1 Whether inactive codes are in the value set -->
  <include>  1..* BackboneElement  <!--  1..* Include one or more codes from a code system or other value set(s) -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <system value="[uri]"/><!--  0..1 The system the codes come from -->
   <version value="[string]"/><!--  0..1 Specific version of the code system referred to -->
   <concept>  0..* BackboneElement  <!--  0..* A concept defined in the system -->
    <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
    <code value="[code]"/><!--  1..1 Code or expression from system -->
    <display value="[string]"/><!--  0..1
```

```
        Text to display for this code for this value set in this valueset -->
    <designation>  0..* BackboneElement  <!--  0..* Additional representations for this concept -->
     <modifierExtension><!--  0..*
       Extension  Extensions that cannot be ignored even if unrecognized  --></modifierExtension>
     <language value="[code]"/><!--  0..1 Human language of the designation   -->
     <use><!--  0..1 Coding Types of uses of designations   --></use>
     <value value="[string]"/><!--  1..1 The text value for this designation -->
    </designation>
   </concept>
   <filter>  0..* BackboneElement  <!--  0..* Select codes/concepts by their properties (including relationships) -->
    <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
    <property value="[code]"/><!--  1..1 A property/filter defined by the code system -->
    <op value="[code]"/><!--  1..1 = | is-a | descendent-of | is-not-a | regex | in | not-in | generalizes | exists
    -->
    <value value="[string]"/><!--  1..1
        Code from the system, or regex criteria, or boolean value for exists -->
   </filter>
   <valueSet value="[canonical]"/><!--  0..* Select the contents included in this value set -->
  </include>
  <exclude><!-- See #ValueSet.compose.include  Explicitly exclude codes from a code system or other value sets
    --></exclude>
 </compose>
 <expansion>  0..1 BackboneElement  <!--  0..1 Used when the value set is "expanded" -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <identifier value="[uri]"/><!--  0..1 Identifies the value set expansion (business identifier) -->
  <timestamp value="[dateTime]"/><!--  1..1 Time ValueSet expansion happened -->
  <total value="[integer]"/><!--  0..1 Total number of codes in the expansion -->
  <offset value="[integer]"/><!--  0..1 Offset at which this resource starts -->
  <parameter>  0..* BackboneElement  <!--  0..* Parameter that controlled the expansion process -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <name value="[string]"/><!--  1..1 Name as assigned by the client or server -->
   <value[x]><!--  0..1 string|boolean|integer|
     decimal|uri|code|dateTime Value of the named parameter --></value[x]>
  </parameter>
  <contains>  0..* BackboneElement  <!--  0..* Codes in the value set -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <system value="[uri]"/><!--  0..1 System value for the code -->
   <abstract value="[boolean]"/><!--  0..1 If user cannot select this entry -->
   <inactive value="[boolean]"/><!--  0..1 If concept is inactive in the code system -->
   <version value="[string]"/><!--  0..1 Version in which this code/display is defined -->
   <code value="[code]"/><!--  0..1 Code - if blank, this is not a selectable code -->
   <display value="[string]"/><!--  0..1 User display for the concept -->
   <designation><!-- See #ValueSet.compose.include.concept.designation  Additional representations for this item
    --></designation>
   <contains><!-- See #ValueSet.expansion.contains  Codes contained under this entry --></contains>
  </contains>
 </expansion>
</ValueSet>
```

JSON Template

```
{ // Shareable ValueSet
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C? R! Canonical identifier for this value set, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the value set (business identifier)
    "version" : "<string>", // C? R! Business version of the value set
    "name" : "<string>", // C? R! Name for this value set (computer friendly)
    "title" : "<string>", // C?Name for this value set (human friendly)
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C? R! For testing purposes, not real usage
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C? R! Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C? R! Natural language description of the value set
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for value set (if applicable)
    "immutable" : <boolean>, // C?Indicates whether or not any change to the content logical definition may occur
    "purpose" : "<markdown>", // C?Why this value set is defined
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "compose" : { BackboneElement }, // C?Content logical definition of the value set (CLD)
    "expansion" : { BackboneElement } // C?Used when the value set is "expanded"
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

4.9.18.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
ValueSet.language	Common Languages	preferred	Common Languages
ValueSet.status	PublicationStatus	required	PublicationStatus

ValueSet.jurisdiction | Jurisdiction | extensible | Jurisdiction
ValueSet.compose.include.concept.designation.language | Common Languages | preferred | Common Languages
ValueSet.compose.include.concept.designation.use | Designation Use | extensible | Designation Use
ValueSet.compose.include.filter.op | FilterOperator | required | FilterOperator

4.9.18.2.3 Constraints

Id | Path | Details | Requirements
vsd-1 | ValueSet.compose.include | A value set include/exclude SHALL have a value set or a system Expression : valueSet.exists() or system.exists()
vsd-2 | ValueSet.compose.include | A value set with concepts or filters SHALL include a system Expression : (concept.exists() or filter.exists()) implies system.exists()
vsd-3 | ValueSet.compose.include | Cannot have both concept and filter Expression : concept.empty() or filter.empty()
vsd-6 | ValueSet.expansion.contains | SHALL have a code or a display Expression : code.exists() or display.exists()
vsd-9 | ValueSet.expansion.contains | Must have a code if not abstract Expression : code.exists() or abstract = true
vsd-10 | ValueSet.expansion.contains | Must have a system if a code is present Expression : code.empty() or system.exists()

4.9.19 JSON Schema for ValueSet

URL: https://hl7.org/fhir/R4/valueset.schema.json.html

4.9.19 JSON Schema for ValueSet

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for ValueSet, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ValueSet",
  "$ref": "#/definitions/ValueSet",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ValueSet": {
      "description": "A ValueSet resource instance specifies a set of codes drawn from one or more code systems,
    intended for use in a particular context. Value sets link between [[[CodeSystem]]] definitions and their use in
    [coded elements](terminologies.html).",
      "properties": {
        "resourceType": {
          "description": "This is a ValueSet resource",
          "const": "ValueSet"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
```

```
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this value set when it is referenced in a
specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
and SHOULD be a literal address at which at which an authoritative instance of this value set is (or will be)
published. This URL can be the target of a canonical reference. It SHALL remain the same when the value set is
stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this value set when it is represented in other
formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the value set when it is referenced in
 a specification, model, design or instance. This is an arbitrary value managed by the value set author and is not
expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not
available. There is also no expectation that versions can be placed in a lexicographical sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the value set. This name should be usable as an identifier
 for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the value set.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this value set. Enables tracking the life-cycle of the content. The status of
the value set applies to the value set definition (ValueSet.compose) and the associated ValueSet metadata.
Expansions do not have a state.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this value set is authored for testing purposes (or
```

```
education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date (and optionally time) when the value set was created or revised (e.g. the
\u0027content logical definition\u0027).",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the value set.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the value set from a consumer\u0027s perspective.
The textual description specifies the span of meanings for concepts to be included within the Value Set Expansion,
and also may specify the intended use and limitations of the Value Set.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate value set instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the value set is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "immutable": {
      "description": "If this is set to \u0027true\u0027, then no new versions of the content logical definition can
 be created.  Note: Other metadata might still change.",
      "$ref": "#/definitions/boolean"
    },
    "_immutable": {
      "description": "Extensions for immutable",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "purpose": {
      "description": "Explanation of why this value set is needed and why it has been designed as it has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the value set and/or its contents. Copyright statements are
generally legal restrictions on the use and publishing of the value set.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "compose": {
      "description": "A set of criteria that define the contents of the value set by including or excluding codes
selected from the specified code system(s) that the value set draws from. This is also known as the Content Logical
Definition (CLD).",
      "$ref": "#/definitions/ValueSet_Compose"
    },
```

```
    "expansion": {
      "description": "A value set can also be \"expanded\", where the value set is turned into a simple collection
of enumerated codes. This element holds the expansion, if it has been performed.",
      "$ref": "#/definitions/ValueSet_Expansion"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"ValueSet_Compose": {
  "description": "A ValueSet resource instance specifies a set of codes drawn from one or more code systems,
intended for use in a particular context. Value sets link between [[[CodeSystem]]] definitions and their use in
[coded elements](terminologies.html).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "lockedDate": {
      "description": "The Locked Date is  the effective date that is used to determine the version of all referenced
 Code Systems and Value Set Definitions included in the compose that are not already tied to a specific version.",
      "$ref": "#/definitions/date"
    },
    "_lockedDate": {
      "description": "Extensions for lockedDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "inactive": {
      "description": "Whether inactive codes - codes that are not approved for current use - are in the value set.
If inactive \u003d true, inactive codes are to be included in the expansion, if inactive \u003d false, the inactive
codes will not be included in the expansion. If absent, the behavior is determined by the implementation, or by the
applicable $expand parameters (but generally, inactive codes would be expected to be included).",
      "$ref": "#/definitions/boolean"
    },
    "_inactive": {
      "description": "Extensions for inactive",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "include": {
      "description": "Include one or more codes from a code system or other value set(s).",
      "items": {
        "$ref": "#/definitions/ValueSet_Include"
      },
      "type": "array"
    },
    "exclude": {
      "description": "Exclude one or more codes from the value set based on code system filters and/or other value
sets.",
      "items": {
        "$ref": "#/definitions/ValueSet_Include"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "include"
  ]
},
"ValueSet_Include": {
  "description": "A ValueSet resource instance specifies a set of codes drawn from one or more code systems,
intended for use in a particular context. Value sets link between [[[CodeSystem]]] definitions and their use in
```

```
[coded elements](terminologies.html).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "system": {
      "description": "An absolute URI which is the code system from which the selected codes come from.",
      "$ref": "#/definitions/uri"
    },
    "_system": {
      "description": "Extensions for system",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "The version of the code system that the codes are selected from, or the special version
\u0027*\u0027 for all versions.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "concept": {
      "description": "Specifies a concept to be included or excluded.",
      "items": {
        "$ref": "#/definitions/ValueSet_Concept"
      },
      "type": "array"
    },
    "filter": {
      "description": "Select concepts by specify a matching criterion based on the properties (including
relationships) defined by the system, or on filters defined by the system. If multiple filters are specified, they
SHALL all be true.",
      "items": {
        "$ref": "#/definitions/ValueSet_Filter"
      },
      "type": "array"
    },
    "valueSet": {
      "description": "Selects the concepts found in this value set (based on its value set definition). This is an
absolute URI that is a reference to ValueSet.url.  If multiple value sets are specified this includes the union of
the contents of all of the referenced value sets.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"ValueSet_Concept": {
  "description": "A ValueSet resource instance specifies a set of codes drawn from one or more code systems,
intended for use in a particular context. Value sets link between [[[CodeSystem]]] definitions and their use in
[coded elements](terminologies.html).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
```

```
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "Specifies a code for the concept to be included or excluded.",
      "$ref": "#/definitions/code"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "display": {
      "description": "The text to display to the user for this concept in the context of this valueset. If no
display is provided, then applications using the value set use the display specified for the code by the system.",
      "$ref": "#/definitions/string"
    },
    "_display": {
      "description": "Extensions for display",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "designation": {
      "description": "Additional representations for this concept when used in this value set - other languages,
aliases, specialized purposes, used for particular purposes, etc.",
      "items": {
        "$ref": "#/definitions/ValueSet_Designation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"ValueSet_Designation": {
  "description": "A ValueSet resource instance specifies a set of codes drawn from one or more code systems,
intended for use in a particular context. Value sets link between [[[CodeSystem]]] definitions and their use in
[coded elements](terminologies.html).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"

      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "language": {
      "description": "The language this designation is defined for.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
```

```
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "use": {
      "description": "A code that represents types of uses of designations.",
      "$ref": "Coding.schema.json#/definitions/Coding"
    },
    "value": {
      "description": "The text value for this designation.",
      "$ref": "#/definitions/string"
    },
    "_value": {
      "description": "Extensions for value",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"ValueSet_Filter": {
  "description": "A ValueSet resource instance specifies a set of codes drawn from one or more code systems,
intended for use in a particular context. Value sets link between [[[CodeSystem]]] definitions and their use in
[coded elements](terminologies.html).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "property": {
      "description": "A code that identifies a property or a filter defined in the code system.",
      "$ref": "#/definitions/code"
    },
    "_property": {
      "description": "Extensions for property",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "op": {
      "description": "The kind of operation to perform as a part of the filter criteria.",
      "enum": [
        "\u003d",
        "is-a",
        "descendent-of",
        "is-not-a",
        "regex",
        "in",
        "not-in",
        "generalizes",
        "exists"
      ]
    },
    "_op": {
      "description": "Extensions for op",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "value": {
      "description": "The match value may be either a code defined by the system, or a string value, which is a
regex match on the literal string of the property value  (if the filter represents a property defined in CodeSystem)
 or of the system filter value (if the filter represents a filter defined in CodeSystem) when the operation is
\u0027regex\u0027, or one of the values (true and false), when the operation is \u0027exists\u0027.",
      "$ref": "#/definitions/string"
    },
    "_value": {
      "description": "Extensions for value",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
```

```
  "additionalProperties": false
},
"ValueSet_Expansion": {
  "description": "A ValueSet resource instance specifies a set of codes drawn from one or more code systems, intended for use in a particular context. Value sets link between [[[CodeSystem]]] definitions and their use in [coded elements](terminologies.html).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "An identifier that uniquely identifies this expansion of the valueset, based on a unique combination of the provided parameters, the system default parameters, and the underlying system code system versions etc. Systems may re-use the same identifier as long as those factors remain the same, and the expansion is the same, but are not required to do so. This is a business identifier.",
      "$ref": "#/definitions/uri"
    },
    "_identifier": {
      "description": "Extensions for identifier",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "timestamp": {
      "description": "The time at which the expansion was produced by the expanding system.",
      "$ref": "#/definitions/dateTime"
    },
    "_timestamp": {
      "description": "Extensions for timestamp",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "total": {
      "description": "The total number of concepts in the expansion. If the number of concept nodes in this resource is less than the stated number, then the server can return more using the offset parameter.",
      "$ref": "#/definitions/integer"
    },
    "_total": {
      "description": "Extensions for total",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "offset": {
      "description": "If paging is being used, the offset at which this resource starts.  I.e. this resource is a partial view into the expansion. If paging is not being used, this element SHALL NOT be present.",
      "$ref": "#/definitions/integer"
    },
    "_offset": {
      "description": "Extensions for offset",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "parameter": {
      "description": "A parameter that controlled the expansion process. These parameters may be used by users of expanded value sets to check whether the expansion is suitable for a particular purpose, or to pick the correct expansion.",
      "items": {
        "$ref": "#/definitions/ValueSet_Parameter"
      },
      "type": "array"
    },
    "contains": {
      "description": "The codes that are contained in the value set expansion.",
      "items": {
        "$ref": "#/definitions/ValueSet_Contains"
      },
      "type": "array"
    }
```

```
  },
  "additionalProperties": false
},
"ValueSet_Parameter": {
  "description": "A ValueSet resource instance specifies a set of codes drawn from one or more code systems,
intended for use in a particular context. Value sets link between [[[CodeSystem]]] definitions and their use in
[coded elements](terminologies.html).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "Name of the input parameter to the $expand operation; may be a server-assigned name for
additional default or other server-supplied parameters used to control the expansion process.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueString": {
      "description": "The value of the parameter.",
      "pattern": "^[ \\r\\n\\t\\S]+$",
      "type": "string"
    },
    "_valueString": {
      "description": "Extensions for valueString",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueBoolean": {
      "description": "The value of the parameter.",
      "pattern": "^true|false$",
      "type": "boolean"
    },
    "_valueBoolean": {
      "description": "Extensions for valueBoolean",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueInteger": {
      "description": "The value of the parameter.",
      "pattern": "^-?([0]|([1-9][0-9]*))$",
      "type": "number"
    },
    "_valueInteger": {
      "description": "Extensions for valueInteger",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDecimal": {
      "description": "The value of the parameter.",
      "pattern": "^-?(0|[1-9][0-9]*)(\\.[0-9]+)?([eE][+-]?[0-9]+)?$",
      "type": "number"
    },
    "_valueDecimal": {
      "description": "Extensions for valueDecimal",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueUri": {
      "description": "The value of the parameter.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_valueUri": {
      "description": "Extensions for valueUri",
```

```
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueCode": {
      "description": "The value of the parameter.",
      "pattern": "^[^\\s]+(\\s[^\\s]+)*$",
      "type": "string"
    },
    "_valueCode": {
      "description": "Extensions for valueCode",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "valueDateTime": {
      "description": "The value of the parameter.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_valueDateTime": {
      "description": "Extensions for valueDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"ValueSet_Contains": {
  "description": "A ValueSet resource instance specifies a set of codes drawn from one or more code systems,
intended for use in a particular context. Value sets link between [[[CodeSystem]]] definitions and their use in
[coded elements](terminologies.html).",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "system": {
      "description": "An absolute URI which is the code system in which the code for this item in the expansion is
defined.",
      "$ref": "#/definitions/uri"
    },
    "_system": {
      "description": "Extensions for system",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "abstract": {
      "description": "If true, this entry is included in the expansion for navigational purposes, and the user
cannot select the code directly as a proper value.",
      "$ref": "#/definitions/boolean"
    },
    "_abstract": {
      "description": "Extensions for abstract",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "inactive": {
      "description": "If the concept is inactive in the code system that defines it. Inactive codes are those that
are no longer to be used, but are maintained by the code system for understanding legacy data. It might not be known
 or specified whether an concept is inactive (and it may depend on the context of use).",
      "$ref": "#/definitions/boolean"
    },
    "_inactive": {
      "description": "Extensions for inactive",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "The version of the code system from this code was taken. Note that a well-maintained code
system does not need the version reported, because the meaning of codes is consistent across versions. However this
```

```
    cannot consistently be assured, and when the meaning is not guaranteed to be consistent, the version SHOULD be
    exchanged.",
          "$ref": "#/definitions/string"
        },
        "_version": {
          "description": "Extensions for version",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "code": {
          "description": "The code for this item in the expansion hierarchy. If this code is missing the entry in the
    hierarchy is a place holder (abstract) and does not represent a valid code in the value set.",
          "$ref": "#/definitions/code"
        },
        "_code": {
          "description": "Extensions for code",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "display": {
          "description": "The recommended display for this item in the expansion.",
          "$ref": "#/definitions/string"
        },
        "_display": {
          "description": "Extensions for display",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "designation": {
          "description": "Additional representations for this item - other languages, aliases, specialized purposes,
    used for particular purposes, etc. These are relevant when the conditions of the expansion do not fix to a single
    correct representation.",
          "items": {
            "$ref": "#/definitions/ValueSet_Designation"
          },
          "type": "array"
        },
        "contains": {
          "description": "Other codes and entries contained under this entry in the hierarchy.",
          "items": {
            "$ref": "#/definitions/ValueSet_Contains"
          },
          "type": "array"
        }
      },
      "additionalProperties": false
    }
  }
}
```


4.9.20 Resource ValueSet - Maturity Dependency View

URL: https://hl7.org/fhir/R4/valueset-dependencies.html

4.9.20 Resource ValueSet - Maturity Dependency View

Vocabulary Work Group | Maturity Level: N | Normative (from v4.0.0) | Compartments: Not linked to any defined compartments

| This page has been approved as part of an ANSI standard. See the Conformance Package for further details.

Dependency Graph for ValueSet FMM level 5

Name | Card. | Type | Dependency Analysis
ValueSet | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext TU | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
immutable | 0..1 | boolean
purpose | 0..1 | markdown
copyright | 0..1 | markdown
compose | 0..1 | BackboneElement
lockedDate | 0..1 | date
inactive | 0..1 | boolean
include | 1..* | BackboneElement
system | 0..1 | uri
version | 0..1 | string

concept | 0..* | BackboneElement
code | 1..1 | code
display | 0..1 | string
designation | 0..* | BackboneElement
language | 0..1 | code
use | 0..1 | Coding
value | 1..1 | string
filter | 0..* | BackboneElement
property | 1..1 | code
op | 1..1 | code
value | 1..1 | string
valueSet | 0..* | canonical(ValueSet)
exclude | 0..* | see include
expansion | 0..1 | BackboneElement
identifier | 0..1 | uri
timestamp | 1..1 | dateTime
total | 0..1 | integer
offset | 0..1 | integer
parameter | 0..* | BackboneElement
name | 1..1 | string
value[x] | 0..1
valueString | | string
valueBoolean | | boolean
valueInteger | | integer
valueDecimal | | decimal
valueUri | | uri
valueCode | | code
valueDateTime | | dateTime
contains | 0..* | BackboneElement
system | 0..1 | uri
abstract | 0..1 | boolean
inactive | 0..1 | boolean
version | 0..1 | string
code | 0..1 | code
display | 0..1 | string
designation | 0..* | see designation
contains | 0..* | see contains

4.10 Resource ConceptMap - Content

URL: https://hl7.org/fhir/R4/conceptmap.html
Vocabulary Work Group | Maturity Level: 3 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

A statement of relationships from one set of concepts to one or more other concepts - either concepts in code systems, or data element/data element concepts, or classes in class models.

4.10.1 Scope and Usage

A concept map defines a mapping from a set of concepts defined in a code system (commonly referred to as the "system") to one or more concepts defined in other code systems. In the mapping context, a system can be a typical code system based on a recognized standard or local terminology (in any of its forms), or in some cases it may be an "implicit" code system that is not based on a recognized terminology but still represents a set of "concepts" that can be usefully mapped. Mappings are one way - from the source to the target system. In many cases, the reverse mappings are valid, but this cannot be assumed to be the case.

Mappings between code system concepts are only intended to be defined in the context of a particular business usage. The business use case is normally defined by the specification of the source and target value sets. The mappings may be useful in other contexts, but this must be determined based on the context of use and meaning; it cannot be taken for granted automatically. An example where the usage context is important for choosing the correct mapping is mapping from a clinical terminology (e.g. SNOMED CT) to a classification (e.g. ICD-10) for either data analysis or billing. Mappings in the data analysis context would be targeted for an appropriate classification (often at a higher level), whereas in the billing context there may be specific requirements to be met (e.g. leaf level codes only) that could result in multiple mappings for a single source concept and then require additional information beyond the source concept itself in order to select the correct final mapping.

Note that all code systems (explicit or implicit) represented in FHIR have URI identifiers for value sets (also either explicit or implicit) that include the entire code system, and these "all codes" value sets can be used for mappings that are valid in all use contexts that are appropriate for the code system.

Each mapping for a concept from the source to target system includes an equivalence property that specifies how similar the mapping is (or, in some cases, that there is no valid mapping). There is one element for each

concept or field in the source value set or system that needs to be mapped. Each source concept may have multiple targets:

• because there are multiple possible mappings (e.g., ambiguous)

• to specify a correct map, and specify other mappings as invalid

• when there are multiple mappings depending on the values of other elements (dependsOn)

There SHOULD be at least one target for each element, but some incomplete concept maps might not have a target for each concept.

A key concept for the ConceptMap resource is the $translate operation. This operation is a formal definition of an API by which a terminology server can allow clients to ask for a translation to be done based on the content in the ConceptMap resource. As such it also provides useful perspective on the operational use of ConceptMap resources in any context.

4.10.2 Boundaries and Relationships

While ConceptMap resources are not referred to directly from any other resource, they may be included and used in ImplementationGuide resources, and provide background knowledge that is useful in many contexts, including operations defined in this specification.

In addition to ConceptMap, there is also the StructureMap resource. The ConceptMap resource defines relationships between concepts in their own right, along with grading of their equivalencies, while the StructureMap defines an executable transform for instances that conform to a known structure.

Both Code System supplements and Concept Maps may be used to define relationships between concepts in different systems. ConceptMaps are assertions of the relationships between different concepts that are associated with particular contexts of use, while CodeSystem supplements are used to define inherent properties and semantics of the concepts in the code system

4.10.3 Background and Context

Further discussion of the issues involved in mapping between concept definition systems can be found in the HL7 v3 Core Principles document and the functionality described in the OMG CTS 2 specification.

This resource is referenced by itself

4.10.4 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
ConceptMap | ITU | | DomainResource | A map from one set of concepts to one or more other concepts + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this concept map, represented as a URI (globally unique)
identifier | Σ | 0..1 | Identifier | Additional identifier for the concept map
version | Σ | 0..1 | string | Business version of the concept map
name | ΣI | 0..1 | string | Name for this concept map (computer friendly)
title | Σ | 0..1 | string | Name for this concept map (human friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the concept map
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for concept map (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this concept map is defined
copyright | | 0..1 | markdown | Use and/or publishing restrictions
source[x] | Σ | 0..1 | | The source value set that contains the concepts that are being mapped
sourceUri | | | uri
sourceCanonical | | | canonical(ValueSet)
target[x] | Σ | 0..1 | | The target value set which provides context for the mappings
targetUri | | | uri
targetCanonical | | | canonical(ValueSet)

group | | 0..* | BackboneElement | Same source and target systems
source | | 0..1 | uri | Source system where concepts to be mapped are defined
sourceVersion | | 0..1 | string | Specific version of the code system
target | | 0..1 | uri | Target system that the concepts are to be mapped to
targetVersion | | 0..1 | string | Specific version of the code system
element | | 1..* | BackboneElement | Mappings for a concept from the source set
code | | 0..1 | code | Identifies element being mapped
display | | 0..1 | string | Display for the code
target | I | 0..* | BackboneElement | Concept in target system for element + Rule: If the map is narrower or inexact, there SHALL be some comments
code | | 0..1 | code | Code that identifies the target element
display | | 0..1 | string | Display for the code
equivalence | ?! | 1..1 | code | relatedto | equivalent | equal | wider | subsumes | narrower | specializes | inexact | unmatched | disjoint ConceptMapEquivalence (Required)
comment | I | 0..1 | string | Description of status/issues in mapping
dependsOn | | 0..* | BackboneElement | Other elements required for this mapping (from context)
property | | 1..1 | uri | Reference to property mapping depends on
system | | 0..1 | canonical(CodeSystem) | Code System (if necessary)
value | | 1..1 | string | Value of the referenced element
display | | 0..1 | string | Display for the code (if value is a code)
product | | 0..* | see dependsOn | Other concepts that this mapping also produces
unmapped | I | 0..1 | BackboneElement | What to do when there is no mapping for the source concept + Rule: If the mode is 'fixed', a code must be provided + Rule: If the mode is 'other-map', a url must be provided
mode | | 1..1 | code | provided | fixed | other-map ConceptMapGroupUnmappedMode (Required)
code | | 0..1 | code | Fixed code when mode = fixed
display | | 0..1 | string | Display for the code
url | | 0..1 | canonical(ConceptMap) | canonical reference to an additional ConceptMap to use for mapping if the source concept is unmapped
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ConceptMap xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this concept map, represented as a URI (globally unique) -->
 <identifier><!-- 0..1 Identifier Additional identifier for the concept map --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the concept map -->
 <name value="[string]"/><!--  0..1 Name for this concept map (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this concept map (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the concept map -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for concept map (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this concept map is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <source[x]><!-- 0..1 uri|canonical(ValueSet) The source value set that contains the concepts that are being mapped
    --></source[x]>
 <target[x]><!-- 0..1 uri|canonical(ValueSet) The target value set which provides context for the mappings
    --></target[x]>
 <group>  <!-- 0..* Same source and target systems -->
  <source value="[uri]"/><!-- 0..1 Source system where concepts to be mapped are defined -->
  <sourceVersion value="[string]"/><!-- 0..1 Specific version of the  code system -->
  <target value="[uri]"/><!-- 0..1 Target system that the concepts are to be mapped to -->
  <targetVersion value="[string]"/><!-- 0..1 Specific version of the  code system -->
  <element>  <!-- 1..* Mappings for a concept from the source set -->
   <code value="[code]"/><!-- 0..1 Identifies element being mapped -->
   <display value="[string]"/><!-- 0..1 Display for the code -->
   <target>  <!-- 0..* Concept in target system for element -->
    <code value="[code]"/><!-- 0..1 Code that identifies the target element -->
    <display value="[string]"/><!-- 0..1 Display for the code -->
    <equivalence value="[code]"/><!-- 1..1 relatedto | equivalent | equal | wider | subsumes | narrower | specializes |
    inexact | unmatched | disjoint -->
    <comment value="[string]"/><!--  0..1 Description of status/issues in mapping -->
    <dependsOn>  <!-- 0..* Other elements required for this mapping (from context) -->
     <property value="[uri]"/><!-- 1..1 Reference to property mapping depends on -->
     <system><!-- 0..1 canonical(CodeSystem) Code System (if necessary) --></system>
     <value value="[string]"/><!-- 1..1 Value of the referenced element -->
     <display value="[string]"/><!-- 0..1 Display for the code (if value is a code) -->
    </dependsOn>
    <product><!-- 0..* Content as for ConceptMap.group.element.target.dependsOn Other concepts that this mapping also
    produces --></product>
   </target>
  </element>
  <unmapped>  <!-- 0..1 What to do when there is no mapping for the source concept -->
   <mode value="[code]"/><!-- 1..1 provided | fixed | other-map -->
   <code value="[code]"/><!-- 0..1 Fixed code when mode = fixed -->
   <display value="[string]"/><!-- 0..1 Display for the code -->
```

```
   <url><!-- 0..1 canonical(ConceptMap) canonical reference to an additional ConceptMap to use for mapping if the source
     concept is unmapped --></url>
  </unmapped>
 </group>
</ConceptMap>
```

JSON Template

```
{
  "resourceType" : "ConceptMap",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this concept map, represented as a URI (globally unique)
  "identifier" : { Identifier }, // Additional identifier for the concept map
  "version" : "<string>", // Business version of the concept map
  "name" : "<string>", // C? Name for this concept map (computer friendly)
  "title" : "<string>", // Name for this concept map (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the concept map
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for concept map (if applicable)
  "purpose" : "<markdown>", // Why this concept map is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  // source[x]: The source value set that contains the concepts that are being mapped. One of these 2:
  "sourceUri" : "<uri>",
  "sourceCanonical" : { canonical(ValueSet) },
  // target[x]: The target value set which provides context for the mappings. One of these 2:
  "targetUri" : "<uri>",
  "targetCanonical" : { canonical(ValueSet) },
  "group" : [{ // Same source and target systems
    "source" : "<uri>", // Source system where concepts to be mapped are defined
    "sourceVersion" : "<string>", // Specific version of the  code system
    "target" : "<uri>", // Target system that the concepts are to be mapped to
    "targetVersion" : "<string>", // Specific version of the  code system
    "element" : [{ // R!  Mappings for a concept from the source set
      "code" : "<code>", // Identifies element being mapped
      "display" : "<string>", // Display for the code
      "target" : [{ // Concept in target system for element
        "code" : "<code>", // Code that identifies the target element
        "display" : "<string>", // Display for the code
        "equivalence" : "<code>", // R!  relatedto | equivalent | equal | wider | subsumes | narrower | specializes |
    inexact | unmatched | disjoint
        "comment" : "<string>", // C? Description of status/issues in mapping
        "dependsOn" : [{ // Other elements required for this mapping (from context)
          "property" : "<uri>", // R!  Reference to property mapping depends on
          "system" : { canonical(CodeSystem) }, // Code System (if necessary)
          "value" : "<string>", // R!  Value of the referenced element
          "display" : "<string>" // Display for the code (if value is a code)
        }],
        "product" : [{ Content as for ConceptMap.group.element.target.dependsOn }] // Other concepts that this mapping
    also produces
      }]
    }],
    "unmapped" : { // What to do when there is no mapping for the source concept
      "mode" : "<code>", // R!  provided | fixed | other-map
      "code" : "<code>", // Fixed code when mode = fixed
      "display" : "<string>", // Display for the code
      "url" : { canonical(ConceptMap) } // canonical reference to an additional ConceptMap to use for mapping if the
    source concept is unmapped
    }
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ConceptMap;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ConceptMap.url [ uri ]; # 0..1 Canonical identifier for this concept map, represented as a URI (globally unique)
  fhir:ConceptMap.identifier [ Identifier ]; # 0..1 Additional identifier for the concept map
  fhir:ConceptMap.version [ string ]; # 0..1 Business version of the concept map
  fhir:ConceptMap.name [ string ]; # 0..1 Name for this concept map (computer friendly)
  fhir:ConceptMap.title [ string ]; # 0..1 Name for this concept map (human friendly)
  fhir:ConceptMap.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ConceptMap.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ConceptMap.date [ dateTime ]; # 0..1 Date last changed
  fhir:ConceptMap.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ConceptMap.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ConceptMap.description [ markdown ]; # 0..1 Natural language description of the concept map
  fhir:ConceptMap.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:ConceptMap.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for concept map (if applicable)
  fhir:ConceptMap.purpose [ markdown ]; # 0..1 Why this concept map is defined
```

```
  fhir:ConceptMap.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  # ConceptMap.source[x] : 0..1 The source value set that contains the concepts that are being mapped. One of these 2
    fhir:ConceptMap.sourceUri [ uri ]
    fhir:ConceptMap.sourceCanonical [ canonical(ValueSet) ]
  # ConceptMap.target[x] : 0..1 The target value set which provides context for the mappings. One of these 2
    fhir:ConceptMap.targetUri [ uri ]
    fhir:ConceptMap.targetCanonical [ canonical(ValueSet) ]
  fhir:ConceptMap.group [ # 0..* Same source and target systems
    fhir:ConceptMap.group.source [ uri ]; # 0..1 Source system where concepts to be mapped are defined
    fhir:ConceptMap.group.sourceVersion [ string ]; # 0..1 Specific version of the  code system
    fhir:ConceptMap.group.target [ uri ]; # 0..1 Target system that the concepts are to be mapped to
    fhir:ConceptMap.group.targetVersion [ string ]; # 0..1 Specific version of the  code system
    fhir:ConceptMap.group.element [ # 1..* Mappings for a concept from the source set
      fhir:ConceptMap.group.element.code [ code ]; # 0..1 Identifies element being mapped
      fhir:ConceptMap.group.element.display [ string ]; # 0..1 Display for the code
      fhir:ConceptMap.group.element.target [ # 0..* Concept in target system for element
        fhir:ConceptMap.group.element.target.code [ code ]; # 0..1 Code that identifies the target element
        fhir:ConceptMap.group.element.target.display [ string ]; # 0..1 Display for the code
        fhir:ConceptMap.group.element.target.equivalence [ code ]; # 1..1 relatedto | equivalent | equal | wider |
    subsumes | narrower | specializes | inexact | unmatched | disjoint
        fhir:ConceptMap.group.element.target.comment [ string ]; # 0..1 Description of status/issues in mapping
        fhir:ConceptMap.group.element.target.dependsOn [ # 0..* Other elements required for this mapping (from context)
          fhir:ConceptMap.group.element.target.dependsOn.property [ uri ]; # 1..1 Reference to property mapping depends
    on
          fhir:ConceptMap.group.element.target.dependsOn.system [ canonical(CodeSystem) ]; # 0..1 Code System (if
    necessary)
          fhir:ConceptMap.group.element.target.dependsOn.value [ string ]; # 1..1 Value of the referenced element
          fhir:ConceptMap.group.element.target.dependsOn.display [ string ]; # 0..1 Display for the code (if value is a
    code)
        ], ...;
        fhir:ConceptMap.group.element.target.product [ See ConceptMap.group.element.target.dependsOn ], ... ; # 0..*
    Other concepts that this mapping also produces
      ], ...;
    ], ...;
    fhir:ConceptMap.group.unmapped [ # 0..1 What to do when there is no mapping for the source concept
      fhir:ConceptMap.group.unmapped.mode [ code ]; # 1..1 provided | fixed | other-map
      fhir:ConceptMap.group.unmapped.code [ code ]; # 0..1 Fixed code when mode = fixed
      fhir:ConceptMap.group.unmapped.display [ string ]; # 0..1 Display for the code
      fhir:ConceptMap.group.unmapped.url [ canonical(ConceptMap) ]; # 0..1 canonical reference to an additional
    ConceptMap to use for mapping if the source concept is unmapped
    ];
  ], ...;
]
```

Changes since R3

ConceptMap

ConceptMap.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

ConceptMap.experimental | No longer marked as Modifier

ConceptMap.source[x] | Add Type canonical(ValueSet) Remove Type Reference(ValueSet)

ConceptMap.target[x] | Add Type canonical(ValueSet) Remove Type Reference(ValueSet)

ConceptMap.group.element.target.equivalence | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/concept-map-equivalence to http://hl7.org/fhir/ValueSet/concept-map-equivalence|4.0.1 Default Value "equivalent" removed

ConceptMap.group.element.target.dependsOn.system | Type changed from uri to canonical(CodeSystem)

ConceptMap.group.element.target.dependsOn.value | Renamed from code to value

ConceptMap.group.unmapped.mode | Change value set from http://hl7.org/fhir/ValueSet/conceptmap-unmapped-mode to http://hl7.org/fhir/ValueSet/conceptmap-unmapped-mode|4.0.1

ConceptMap.group.unmapped.url | Type changed from uri to canonical(ConceptMap)

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 4 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

ConceptMap | ITU | | DomainResource | A map from one set of concepts to one or more other concepts + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | Σ | 0..1 | uri | Canonical identifier for this concept map, represented as a URI (globally unique)

identifier | Σ | 0..1 | Identifier | Additional identifier for the concept map

version | Σ | 0..1 | string | Business version of the concept map

name | ΣI | 0..1 | string | Name for this concept map (computer friendly)

title | Σ | 0..1 | string | Name for this concept map (human friendly)

status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | Σ | 0..1 | boolean | For testing purposes, not real usage

date | Σ | 0..1 | dateTime | Date last changed

publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)

contact | Σ | 0..* | ContactDetail | Contact details for the publisher

description | | 0..1 | markdown | Natural language description of the concept map

useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for concept map (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this concept map is defined
copyright | | 0..1 | markdown | Use and/or publishing restrictions
source[x] | Σ | 0..1 | | The source value set that contains the concepts that are being mapped
sourceUri | | | uri
sourceCanonical | | | canonical(ValueSet)
target[x] | Σ | 0..1 | | The target value set which provides context for the mappings
targetUri | | | uri
targetCanonical | | | canonical(ValueSet)
group | | 0..* | BackboneElement | Same source and target systems
source | | 0..1 | uri | Source system where concepts to be mapped are defined
sourceVersion | | 0..1 | string | Specific version of the code system
target | | 0..1 | uri | Target system that the concepts are to be mapped to
targetVersion | | 0..1 | string | Specific version of the code system
element | | 1..* | BackboneElement | Mappings for a concept from the source set
code | | 0..1 | code | Identifies element being mapped
display | | 0..1 | string | Display for the code
target | I | 0..* | BackboneElement | Concept in target system for element + Rule: If the map is narrower or inexact, there SHALL be some comments
code | | 0..1 | code | Code that identifies the target element
display | | 0..1 | string | Display for the code
equivalence | ?! | 1..1 | code | relatedto | equivalent | equal | wider | subsumes | narrower | specializes | inexact | unmatched | disjoint ConceptMapEquivalence (Required)
comment | I | 0..1 | string | Description of status/issues in mapping
dependsOn | | 0..* | BackboneElement | Other elements required for this mapping (from context)
property | | 1..1 | uri | Reference to property mapping depends on
system | | 0..1 | canonical(CodeSystem) | Code System (if necessary)
value | | 1..1 | string | Value of the referenced element
display | | 0..1 | string | Display for the code (if value is a code)
product | | 0..* | see dependsOn | Other concepts that this mapping also produces
unmapped | I | 0..1 | BackboneElement | What to do when there is no mapping for the source concept + Rule: If the mode is 'fixed', a code must be provided + Rule: If the mode is 'other-map', a url must be provided
mode | | 1..1 | code | provided | fixed | other-map ConceptMapGroupUnmappedMode (Required)
code | | 0..1 | code | Fixed code when mode = fixed
display | | 0..1 | string | Display for the code
url | | 0..1 | canonical(ConceptMap) | canonical reference to an additional ConceptMap to use for mapping if the source concept is unmapped
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ConceptMap xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this concept map, represented as a URI (globally unique) -->
 <identifier><!-- 0..1 Identifier Additional identifier for the concept map --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the concept map -->
 <name value="[string]"/><!--  0..1 Name for this concept map (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this concept map (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the concept map -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for concept map (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this concept map is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <source[x]><!-- 0..1 uri|canonical(ValueSet) The source value set that contains the concepts that are being mapped
    --></source[x]>
 <target[x]><!-- 0..1 uri|canonical(ValueSet) The target value set which provides context for the mappings
    --></target[x]>
 <group>  <!-- 0..* Same source and target systems -->
  <source value="[uri]"/><!-- 0..1 Source system where concepts to be mapped are defined -->
  <sourceVersion value="[string]"/><!-- 0..1 Specific version of the  code system -->
  <target value="[uri]"/><!-- 0..1 Target system that the concepts are to be mapped to -->
  <targetVersion value="[string]"/><!-- 0..1 Specific version of the  code system -->
  <element>  <!-- 1..* Mappings for a concept from the source set -->
   <code value="[code]"/><!-- 0..1 Identifies element being mapped -->
   <display value="[string]"/><!-- 0..1 Display for the code -->
   <target>  <!-- 0..* Concept in target system for element -->
    <code value="[code]"/><!-- 0..1 Code that identifies the target element -->
    <display value="[string]"/><!-- 0..1 Display for the code -->
    <equivalence value="[code]"/><!-- 1..1 relatedto | equivalent | equal | wider | subsumes | narrower | specializes |
    inexact | unmatched | disjoint -->
    <comment value="[string]"/><!--  0..1 Description of status/issues in mapping -->
    <dependsOn>  <!-- 0..* Other elements required for this mapping (from context) -->
     <property value="[uri]"/><!-- 1..1 Reference to property mapping depends on -->
```

```
     <system><!-- 0..1 canonical(CodeSystem) Code System (if necessary) --></system>
     <value value="[string]"/><!-- 1..1 Value of the referenced element -->
     <display value="[string]"/><!-- 0..1 Display for the code (if value is a code) -->
    </dependsOn>
    <product><!-- 0..* Content as for ConceptMap.group.element.target.dependsOn Other concepts that this mapping also
    produces --></product>
   </target>
  </element>
  <unmapped>  <!-- 0..1 What to do when there is no mapping for the source concept -->
   <mode value="[code]"/><!-- 1..1 provided | fixed | other-map -->
   <code value="[code]"/><!-- 0..1 Fixed code when mode = fixed -->
   <display value="[string]"/><!-- 0..1 Display for the code -->
   <url><!-- 0..1 canonical(ConceptMap) canonical reference to an additional ConceptMap to use for mapping if the source
     concept is unmapped --></url>
  </unmapped>
 </group>
</ConceptMap>
```

JSON Template

```
{
  "resourceType" : "ConceptMap",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this concept map, represented as a URI (globally unique)
  "identifier" : { Identifier }, // Additional identifier for the concept map
  "version" : "<string>", // Business version of the concept map
  "name" : "<string>", // C? Name for this concept map (computer friendly)
  "title" : "<string>", // Name for this concept map (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the concept map
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for concept map (if applicable)
  "purpose" : "<markdown>", // Why this concept map is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  // source[x]: The source value set that contains the concepts that are being mapped. One of these 2:
  "sourceUri" : "<uri>",
  "sourceCanonical" : { canonical(ValueSet) },
  // target[x]: The target value set which provides context for the mappings. One of these 2:
  "targetUri" : "<uri>",
  "targetCanonical" : { canonical(ValueSet) },
  "group" : [{ // Same source and target systems
    "source" : "<uri>", // Source system where concepts to be mapped are defined
    "sourceVersion" : "<string>", // Specific version of the  code system
    "target" : "<uri>", // Target system that the concepts are to be mapped to
    "targetVersion" : "<string>", // Specific version of the  code system
    "element" : [{ // R!  Mappings for a concept from the source set
      "code" : "<code>", // Identifies element being mapped
      "display" : "<string>", // Display for the code
      "target" : [{ // Concept in target system for element
        "code" : "<code>", // Code that identifies the target element
        "display" : "<string>", // Display for the code
        "equivalence" : "<code>", // R!  relatedto | equivalent | equal | wider | subsumes | narrower | specializes |
    inexact | unmatched | disjoint
        "comment" : "<string>", // C? Description of status/issues in mapping
        "dependsOn" : [{ // Other elements required for this mapping (from context)
          "property" : "<uri>", // R!  Reference to property mapping depends on
          "system" : { canonical(CodeSystem) }, // Code System (if necessary)
          "value" : "<string>", // R!  Value of the referenced element
          "display" : "<string>" // Display for the code (if value is a code)
        }],
        "product" : [{ Content as for ConceptMap.group.element.target.dependsOn }] // Other concepts that this mapping
    also produces
      }]
    }],
    "unmapped" : { // What to do when there is no mapping for the source concept
      "mode" : "<code>", // R!  provided | fixed | other-map
      "code" : "<code>", // Fixed code when mode = fixed
      "display" : "<string>", // Display for the code
      "url" : { canonical(ConceptMap) } // canonical reference to an additional ConceptMap to use for mapping if the
    source concept is unmapped
    }
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ConceptMap;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ConceptMap.url [ uri ]; # 0..1 Canonical identifier for this concept map, represented as a URI (globally unique)
  fhir:ConceptMap.identifier [ Identifier ]; # 0..1 Additional identifier for the concept map
```

```
  fhir:ConceptMap.version [ string ]; # 0..1 Business version of the concept map
  fhir:ConceptMap.name [ string ]; # 0..1 Name for this concept map (computer friendly)
  fhir:ConceptMap.title [ string ]; # 0..1 Name for this concept map (human friendly)
  fhir:ConceptMap.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ConceptMap.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ConceptMap.date [ dateTime ]; # 0..1 Date last changed
  fhir:ConceptMap.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ConceptMap.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ConceptMap.description [ markdown ]; # 0..1 Natural language description of the concept map
  fhir:ConceptMap.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:ConceptMap.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for concept map (if applicable)
  fhir:ConceptMap.purpose [ markdown ]; # 0..1 Why this concept map is defined
  fhir:ConceptMap.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  # ConceptMap.source[x] : 0..1 The source value set that contains the concepts that are being mapped. One of these 2
    fhir:ConceptMap.sourceUri [ uri ]
    fhir:ConceptMap.sourceCanonical [ canonical(ValueSet) ]
  # ConceptMap.target[x] : 0..1 The target value set which provides context for the mappings. One of these 2
    fhir:ConceptMap.targetUri [ uri ]
    fhir:ConceptMap.targetCanonical [ canonical(ValueSet) ]
  fhir:ConceptMap.group [ # 0..* Same source and target systems
    fhir:ConceptMap.group.source [ uri ]; # 0..1 Source system where concepts to be mapped are defined
    fhir:ConceptMap.group.sourceVersion [ string ]; # 0..1 Specific version of the  code system
    fhir:ConceptMap.group.target [ uri ]; # 0..1 Target system that the concepts are to be mapped to
    fhir:ConceptMap.group.targetVersion [ string ]; # 0..1 Specific version of the  code system
    fhir:ConceptMap.group.element [ # 1..* Mappings for a concept from the source set
      fhir:ConceptMap.group.element.code [ code ]; # 0..1 Identifies element being mapped
      fhir:ConceptMap.group.element.display [ string ]; # 0..1 Display for the code
      fhir:ConceptMap.group.element.target [ # 0..* Concept in target system for element
        fhir:ConceptMap.group.element.target.code [ code ]; # 0..1 Code that identifies the target element
        fhir:ConceptMap.group.element.target.display [ string ]; # 0..1 Display for the code
        fhir:ConceptMap.group.element.target.equivalence [ code ]; # 1..1 relatedto | equivalent | equal | wider |
    subsumes | narrower | specializes | inexact | unmatched | disjoint
        fhir:ConceptMap.group.element.target.comment [ string ]; # 0..1 Description of status/issues in mapping
        fhir:ConceptMap.group.element.target.dependsOn [ # 0..* Other elements required for this mapping (from context)
          fhir:ConceptMap.group.element.target.dependsOn.property [ uri ]; # 1..1 Reference to property mapping depends
    on
          fhir:ConceptMap.group.element.target.dependsOn.system [ canonical(CodeSystem) ]; # 0..1 Code System (if
    necessary)
          fhir:ConceptMap.group.element.target.dependsOn.value [ string ]; # 1..1 Value of the referenced element
          fhir:ConceptMap.group.element.target.dependsOn.display [ string ]; # 0..1 Display for the code (if value is a
    code)
        ], ...;
        fhir:ConceptMap.group.element.target.product [ See ConceptMap.group.element.target.dependsOn ], ... ; # 0..*
    Other concepts that this mapping also produces
      ], ...;
    ], ...;
    fhir:ConceptMap.group.unmapped [ # 0..1 What to do when there is no mapping for the source concept
      fhir:ConceptMap.group.unmapped.mode [ code ]; # 1..1 provided | fixed | other-map
      fhir:ConceptMap.group.unmapped.code [ code ]; # 0..1 Fixed code when mode = fixed
      fhir:ConceptMap.group.unmapped.display [ string ]; # 0..1 Display for the code
      fhir:ConceptMap.group.unmapped.url [ canonical(ConceptMap) ]; # 0..1 canonical reference to an additional
    ConceptMap to use for mapping if the source concept is unmapped
    ];
  ], ...;
]
```

Changes since Release 3

ConceptMap

ConceptMap.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

ConceptMap.experimental | No longer marked as Modifier

ConceptMap.source[x] | Add Type canonical(ValueSet) Remove Type Reference(ValueSet)

ConceptMap.target[x] | Add Type canonical(ValueSet) Remove Type Reference(ValueSet)

ConceptMap.group.element.target.equivalence | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/concept-map-equivalence to http://hl7.org/fhir/ValueSet/concept-map-equivalence|4.0.1 Default Value "equivalent" removed

ConceptMap.group.element.target.dependsOn.system | Type changed from uri to canonical(CodeSystem)

ConceptMap.group.element.target.dependsOn.value | Renamed from code to value

ConceptMap.group.unmapped.mode | Change value set from http://hl7.org/fhir/ValueSet/conceptmap-unmapped-mode to http://hl7.org/fhir/ValueSet/conceptmap-unmapped-mode|4.0.1

ConceptMap.group.unmapped.url | Type changed from uri to canonical(ConceptMap)

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 4 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

4.10.4.1 Terminology Bindings

Path | Definition | Type | Reference

ConceptMap.status | The lifecycle status of an artifact. | Required | PublicationStatus

ConceptMap.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
ConceptMap.group.element.target.equivalence | The degree of equivalence between concepts. | Required | ConceptMapEquivalence
ConceptMap.group.unmapped.mode | Defines which action to take if there is no match in the group. | Required | ConceptMapGroupUnmappedMode

4.10.4.2 Constraints

id | Level | Location | Description | Expression
cmd-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
cmd-1 | Rule | ConceptMap.group.element.target | If the map is narrower or inexact, there SHALL be some comments | comment.exists() or equivalence.empty() or ((equivalence != 'narrower') and (equivalence != 'inexact'))
cmd-2 | Rule | ConceptMap.group.unmapped | If the mode is 'fixed', a code must be provided | (mode = 'fixed') implies code.exists()
cmd-3 | Rule | ConceptMap.group.unmapped | If the mode is 'other-map', a url must be provided | (mode = 'other-map') implies url.exists()

4.10.5 Notes

• The value of the system, version and code elements are the same as used by the Coding data type

• When a mapping equivalence is characterized as "narrower", some explanation of the scope difference SHALL be provided in the comments

• The concept map is a statement of mapping in a single direction. The existence of a matching mapping in the reverse direction cannot be assumed to exist automatically, but only through human review.

• For simplicity, it is better if is only one element for each source concept. if there is more than one, then just as one element can have multiple targets, multiple occurrences of element with the same element.code simply aggregate their target elements as if there was only one element

4.10.6 Grouping Mappings

The concept mappings in element are arranged into groups that share common source/target systems. These groups have no semantic significance; they exist to make the representation more concise. Concept maps may contain more than one group with the same source and target - this would be a less concise representation but may be useful in order to maintain a fixed order for the concepts that are mapped.

Concepts that are labeled as 'unmatched' are considered to be unmatched in the target value set, irrespective of whether they are contained in a group with a stated target system or not. Groups that contain no target system may only contained 'unmatched' concepts. There is no difference in the meaning of an unmatched target whether or not there is a stated target system.

4.10.7 Mapping Dependencies

In the simple case, a mapping is made from one code to another. E.g. from "home address" in one code system to "address - home" in another. But in other cases, the mapping is not a simple one to one mapping. A typical example might be mapping between a radiology imaging request that has 2 elements: modality, and body site, and another radiology imaging request that pre-coordinates both of these into a single code modality+body site. In this case, it is not possible to map from one code to another. Instead, the mapping might contain entries like this:

Source Concept | DependsOn.property | DependsOn Concept | Target Concept
X-Ray | Bodysite | Chest | Chest-XRay

When attempting to translate the source to the target, an application must also provide a value for the element identified in dependsOn.property so that the correct mapping can be performed. To support the reverse operation, product is the converse, and works in reverse.

To see a real example of a mapping with dependencies, check the Specimen Type Mapping.

4.10.8 Handling Unmapped Codes

If there is no explicit mapping for a code - that is, the engine processing the ConceptMap finds a group with a matching source system, but not matching element for the code it is translating, processing proceeds to the unmapped element that specifies what should happen.

The unmapped element can specify one of the following different actions to take if there is no mapping for a concept:

provided | Use the code source as provided in the $translate request. This is especially useful for translations between versions of the same code system, where only a few codes have changed
fixed | Use the code (and display) explicitly provided in the group.unmapped. This is useful when there's a fall back general code - especially for classifications
other-map | Use the map identified by the canonical URL in url. This is useful when a published concept map needs to be varied for a few specific cases on an institutional bass - the institution can create a simple concept that expresses their special cases, and then refer to the general purpose mappings

Note that this element is not used when there is an element with a matching code value, and an equivalence value of unmatched (or disjoint).

4.10.9 Implicit Code Systems

The ConceptMap resource is intended to map between concepts defined in a code system. It can also be useful to use the ConceptMap resource to define relationships between concepts defined in other kinds of resources. Here are some common kind of conceptual maps:

• Between two elements in different structure definitions (e.g. between CDA and FHIR v2)

• Between a code system and a Medication resource

Though these resources are not explicitly defining code systems, they do define 'concept's that can still usefully be treated as code systems for the sake of subsetting (e.g. ValueSet) and defining relationships (e.g. ConceptMap). See Implicit Code Systems for further information.

Note that this is different from StructureMap because that is intended to define an executional transform between structures, not a conceptual model.

4.10.10 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
context | token | A use context assigned to the concept map | (ConceptMap.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the concept map | (ConceptMap.useContext.value as Quantity) | (ConceptMap.useContext.value as Range)
context-type | token | A type of use context assigned to the concept map | ConceptMap.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the concept map | On ConceptMap.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the concept map | On ConceptMap.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The concept map publication date | ConceptMap.date
dependson | uri | Reference to property mapping depends on | ConceptMap.group.element.target.dependsOn.property
description | string | The description of the concept map | ConceptMap.description
identifier | token | External identifier for the concept map | ConceptMap.identifier
jurisdiction | token | Intended jurisdiction for the concept map | ConceptMap.jurisdiction
name | string | Computationally friendly name of the concept map | ConceptMap.name
other | reference | canonical reference to an additional ConceptMap to use for mapping if the source concept is unmapped | ConceptMap.group.unmapped.url (ConceptMap)
product | uri | Reference to property mapping depends on | ConceptMap.group.element.target.product.property
publisher | string | Name of the publisher of the concept map | ConceptMap.publisher
source | reference | The source value set that contains the concepts that are being mapped | (ConceptMap.source as canonical) (ValueSet)
source-code | token | Identifies element being mapped | ConceptMap.group.element.code
source-system | uri | Source system where concepts to be mapped are defined | ConceptMap.group.source
source-uri | reference | The source value set that contains the concepts that are being mapped | (ConceptMap.source as uri) (ValueSet)
status | token | The current status of the concept map | ConceptMap.status
target | reference | The target value set which provides context for the mappings | (ConceptMap.target as canonical) (ValueSet)
target-code | token | Code that identifies the target element | ConceptMap.group.element.target.code
target-system | uri | Target system that the concepts are to be mapped to | ConceptMap.group.target
target-uri | reference | The target value set which provides context for the mappings | (ConceptMap.target as uri) (ValueSet)
title | string | The human-friendly name of the concept map | ConceptMap.title
url | uri | The uri that identifies the concept map | ConceptMap.url
version | token | The business version of the concept map | ConceptMap.version

4.10.11 Resource ConceptMap - Examples

URL: https://hl7.org/fhir/R4/conceptmap-examples.html

4.10.11 Resource ConceptMap - Examples

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
General ConceptMap Example | 101 | XML | JSON | Turtle
SNOMED CT / ICD-10 mapping example | 103 | XML | JSON | Turtle
Specimen Type v2 -> SNOMED CT Mapping | 102 | XML | JSON | Turtle
Additional ConceptMap Example | example2 | XML | JSON | Turtle
CDS Hooks Indicator mapping to Request Priority | cdshooks-indicator | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

4.10.12 Resource ConceptMap - Detailed Descriptions

URL: https://hl7.org/fhir/R4/conceptmap-definitions.html

4.10.12 Resource ConceptMap - Detailed Descriptions

Vocabulary Work Group | Maturity Level: 3 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the ConceptMap resource.

ConceptMap
Element Id | ConceptMap
Definition | A statement of relationships from one set of concepts to one or more other concepts - either concepts in code systems, or data element/data element concepts, or classes in class models.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element cmd-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
cmd-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
ConceptMap.url
Element Id | ConceptMap.url
Definition | An absolute URI that is used to identify this concept map when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this concept map is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the concept map is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the concept map to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
ConceptMap.identifier
Element Id | ConceptMap.identifier
Definition | A formal identifier that is used to identify this concept map when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..1
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this concept map outside of FHIR, where it is not possible to use the logical URI.
ConceptMap.version
Element Id | ConceptMap.version
Definition | The identifier that is used to identify this version of the concept map when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the concept map author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different concept map instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the concept map with the format [url]|[version].
ConceptMap.name
Element Id | ConceptMap.name
Definition | A natural language name identifying the concept map. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

ConceptMap.title
Element Id | ConceptMap.title
Definition | A short, descriptive, user-friendly title for the concept map.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
ConceptMap.status
Element Id | ConceptMap.status
Definition | The status of this concept map. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of concept maps that are appropriate for use versus not.
ConceptMap.experimental
Element Id | ConceptMap.experimental
Definition | A Boolean value to indicate that this concept map is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level concept map.
Summary | true
Comments | Allows filtering of concept maps that are appropriate for use versus not.
ConceptMap.date
Element Id | ConceptMap.date
Definition | The date (and optionally time) when the concept map was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the concept map changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the concept map. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
ConceptMap.publisher
Element Id | ConceptMap.publisher
Definition | The name of the organization or individual that published the concept map.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the concept map. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the concept map is the organization or individual primarily responsible for the maintenance and upkeep of the concept map. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the concept map. This item SHOULD be populated unless the information is available from context.
ConceptMap.contact
Element Id | ConceptMap.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
ConceptMap.description
Element Id | ConceptMap.description
Definition | A free text natural language description of the concept map from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Comments | The description is not intended to describe the semantics of the concept map. The description should capture its intended use, which is needed for ensuring integrity for its use in models across future changes.
ConceptMap.useContext
Element Id | ConceptMap.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate concept map instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true

Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
ConceptMap.jurisdiction
Element Id | ConceptMap.jurisdiction
Definition | A legal or geographic region in which the concept map is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the concept map to be used in jurisdictions other than those for which it was originally designed or intended.
ConceptMap.purpose
Element Id | ConceptMap.purpose
Definition | Explanation of why this concept map is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the concept map. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this concept map.
ConceptMap.copyright
Element Id | ConceptMap.copyright
Definition | A copyright statement relating to the concept map and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the concept map.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the concept map and/or its content.
Alternate Names | License; Restrictions
Comments | Frequently the copyright differs between the concept map and codes that are included. The copyright statement should clearly differentiate between these when required.
ConceptMap.source[x]
Element Id | ConceptMap.source[x]
Definition | Identifier for the source value set that contains the concepts that are being mapped and provides context for the mappings.
Cardinality | 0..1
Type | uri|canonical(ValueSet)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Should be a version specific reference. URIs SHOULD be absolute. If there is no source or target value set, there is no specified context for the map (not recommended). The source value set may select codes from either an explicit (standard or local) or implicit code system.
ConceptMap.target[x]
Element Id | ConceptMap.target[x]
Definition | The target value set provides context for the mappings. Note that the mapping is made between concepts, not between value sets, but the value set provides important context about how the concept mapping choices are made.
Cardinality | 0..1
Type | uri|canonical(ValueSet)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Should be a version specific reference. URIs SHOULD be absolute. If there is no source or target value set, the is no specified context for the map.
ConceptMap.group
Element Id | ConceptMap.group
Definition | A group of mappings that all have the same source and target system.
Cardinality | 0..*
ConceptMap.group.source
Element Id | ConceptMap.group.source
Definition | An absolute URI that identifies the source system where the concepts to be mapped are defined.
Cardinality | 0..1
Type | uri
Comments | This is not needed if the source value set is specified and it contains concepts from only a single system.
ConceptMap.group.sourceVersion
Element Id | ConceptMap.group.sourceVersion
Definition | The specific version of the code system, as determined by the code system authority.
Cardinality | 0..1
Type | string
Comments | The specification of a particular code system version may be required for code systems which lack concept permanence.
ConceptMap.group.target
Element Id | ConceptMap.group.target
Definition | An absolute URI that identifies the target system that the concepts will be mapped to.
Cardinality | 0..1
Type | uri
Comments | This is not needed if the target value set is specified and it contains concepts from only a single system. The group target may also be omitted if all of the target element equivalence values are 'unmatched'.

ConceptMap.group.targetVersion
Element Id | ConceptMap.group.targetVersion
Definition | The specific version of the code system, as determined by the code system authority.
Cardinality | 0..1
Type | string
Comments | The specification of a particular code system version may be required for code systems which lack concept permanence.
ConceptMap.group.element
Element Id | ConceptMap.group.element
Definition | Mappings for an individual concept in the source to one or more concepts in the target.
Cardinality | 1..*
Alternate Names | code; concept; item
Comments | Generally, the ideal is that there would only be one mapping for each concept in the source value set, but a given concept may be mapped multiple times with different comments or dependencies.
ConceptMap.group.element.code
Element Id | ConceptMap.group.element.code
Definition | Identity (code or path) or the element/item being mapped.
Cardinality | 0..1
Type | code
ConceptMap.group.element.display
Element Id | ConceptMap.group.element.display
Definition | The display for the code. The display is only provided to help editors when editing the concept map.
Cardinality | 0..1
Type | string
Comments | The display is ignored when processing the map.
ConceptMap.group.element.target
Element Id | ConceptMap.group.element.target
Definition | A concept from the target value set that this concept maps to.
Cardinality | 0..*
Comments | Ideally there would only be one map, with equal or equivalent mapping. But multiple maps are allowed for several narrower options, or to assert that other concepts are unmatched.
Invariants | Defined on this element cmd-1 Rule If the map is narrower or inexact, there SHALL be some comments comment.exists() or equivalence.empty() or ((equivalence != 'narrower') and (equivalence != 'inexact'))
Defined on this element
cmd-1 | Rule | If the map is narrower or inexact, there SHALL be some comments | comment.exists() or equivalence.empty() or ((equivalence != 'narrower') and (equivalence != 'inexact'))
ConceptMap.group.element.target.code
Element Id | ConceptMap.group.element.target.code
Definition | Identity (code or path) or the element/item that the map refers to.
Cardinality | 0..1
Type | code
ConceptMap.group.element.target.display
Element Id | ConceptMap.group.element.target.display
Definition | The display for the code. The display is only provided to help editors when editing the concept map.
Cardinality | 0..1
Type | string
Comments | The display is ignored when processing the map.
ConceptMap.group.element.target.equivalence
Element Id | ConceptMap.group.element.target.equivalence
Definition | The equivalence between the source and target concepts (counting for the dependencies and products). The equivalence is read from target to source (e.g. the target is 'wider' than the source).
Cardinality | 1..1
Terminology Binding | ConceptMapEquivalence (Required)
Type | code
Is Modifier | true (Reason: Some of the values mean that there is no mapping from the source to the target, particularly 'disjoint', and the mapping cannot be interpreted without knowing this value as it could mean the elements are equivalent, totally mismatched or anything in between)
Comments | This element is labeled as a modifier because it may indicate that a target does not apply.
ConceptMap.group.element.target.comment
Element Id | ConceptMap.group.element.target.comment
Definition | A description of status/issues in mapping that conveys additional information not represented in the structured data.
Cardinality | 0..1
Type | string
Invariants | Affect this element cmd-1 Rule If the map is narrower or inexact, there SHALL be some comments comment.exists() or equivalence.empty() or ((equivalence != 'narrower') and (equivalence != 'inexact'))
Affect this element
cmd-1 | Rule | If the map is narrower or inexact, there SHALL be some comments | comment.exists() or equivalence.empty() or ((equivalence != 'narrower') and (equivalence != 'inexact'))
ConceptMap.group.element.target.dependsOn
Element Id | ConceptMap.group.element.target.dependsOn
Definition | A set of additional dependencies for this mapping to hold. This mapping is only applicable if the specified element can be resolved, and it has the specified value.
Cardinality | 0..*

ConceptMap.group.element.target.dependsOn.property
Element Id | ConceptMap.group.element.target.dependsOn.property
Definition | A reference to an element that holds a coded value that corresponds to a code system property. The idea is that the information model carries an element somewhere that is labeled to correspond with a code system property.
Cardinality | 1..1
Type | uri
ConceptMap.group.element.target.dependsOn.system
Element Id | ConceptMap.group.element.target.dependsOn.system
Definition | An absolute URI that identifies the code system of the dependency code (if the source/dependency is a value set that crosses code systems).
Cardinality | 0..1
Type | canonical(CodeSystem)
Meaning if Missing | The code is in the source (dependsOn) or target (product) system
ConceptMap.group.element.target.dependsOn.value
Element Id | ConceptMap.group.element.target.dependsOn.value
Definition | Identity (code or path) or the element/item/ValueSet/text that the map depends on / refers to.
Cardinality | 1..1
Type | string
ConceptMap.group.element.target.dependsOn.display
Element Id | ConceptMap.group.element.target.dependsOn.display
Definition | The display for the code. The display is only provided to help editors when editing the concept map.
Cardinality | 0..1
Type | string
Comments | The display is ignored when processing the map.
ConceptMap.group.element.target.product
Element Id | ConceptMap.group.element.target.product
Definition | A set of additional outcomes from this mapping to other elements. To properly execute this mapping, the specified element must be mapped to some data element or source that is in context. The mapping may still be useful without a place for the additional data elements, but the equivalence cannot be relied on.
Cardinality | 0..*
Type | See ConceptMap.group.element.target.dependsOn
ConceptMap.group.unmapped
Element Id | ConceptMap.group.unmapped
Definition | What to do when there is no mapping for the source concept. "Unmapped" does not include codes that are unmatched, and the unmapped element is ignored in a code is specified to have equivalence = unmatched.
Cardinality | 0..1
Comments | This only applies if the source code has a system value that matches the system defined for the group.
Invariants | Defined on this element cmd-2 Rule If the mode is 'fixed', a code must be provided (mode = 'fixed') implies code.exists() cmd-3 Rule If the mode is 'other-map', a url must be provided (mode = 'other-map') implies url.exists()
Defined on this element
cmd-2 | Rule | If the mode is 'fixed', a code must be provided | (mode = 'fixed') implies code.exists()
cmd-3 | Rule | If the mode is 'other-map', a url must be provided | (mode = 'other-map') implies url.exists()
ConceptMap.group.unmapped.mode
Element Id | ConceptMap.group.unmapped.mode
Definition | Defines which action to take if there is no match for the source concept in the target system designated for the group. One of 3 actions are possible: use the unmapped code (this is useful when doing a mapping between versions, and only a few codes have changed), use a fixed code (a default code), or alternatively, a reference to a different concept map can be provided (by canonical URL).
Cardinality | 1..1
Terminology Binding | ConceptMapGroupUnmappedMode (Required)
Type | code
ConceptMap.group.unmapped.code
Element Id | ConceptMap.group.unmapped.code
Definition | The fixed code to use when the mode = 'fixed' - all unmapped codes are mapped to a single fixed code.
Cardinality | 0..1
Type | code
ConceptMap.group.unmapped.display
Element Id | ConceptMap.group.unmapped.display
Definition | The display for the code. The display is only provided to help editors when editing the concept map.
Cardinality | 0..1
Type | string
Comments | The display is ignored when processing the map.
ConceptMap.group.unmapped.url
Element Id | ConceptMap.group.unmapped.url
Definition | The canonical reference to an additional ConceptMap resource instance to use for mapping if this ConceptMap resource contains no matching mapping for the source concept.
Cardinality | 0..1
Type | canonical(ConceptMap)

4.10.13 Resource ConceptMap - Mappings

URL: https://hl7.org/fhir/R4/conceptmap-mappings.html

4.10.13 Resource ConceptMap - Mappings

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the conceptmap resource (see Mappings to Other Standards for further information & status).

4.10.13.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ConceptMap
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

4.10.13.2 Workflow Pattern (http://hl7.org/fhir/workflow)

ConceptMap
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright

4.10.13.3 RIM Mapping (http://hl7.org/v3)

ConceptMap | N/A

4.10.14 Resource ConceptMap - Extensions & Profiles

URL: https://hl7.org/fhir/R4/conceptmap-profiles.html

4.10.14 Resource ConceptMap - Extensions & Profiles

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the ConceptMap resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

4.10.14.1 Profiles

No Profiles defined for this resource

4.10.14.2 Extensions

concept-bidirectional | ConceptMap | bidirectional | for ConceptMap HL7 Extensions
replaces | ConceptMap | replaces | for General Extensions for use by FHIR Implementers
resource-approvalDate | ConceptMap | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | ConceptMap | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | ConceptMap | lastReviewDate | for Resource HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

11179-permitted-value-conceptmap | permitted-value-conceptmap | for ISO 11179 Element Definition Profile
codesystem-map | map | for CodeSystem HL7 Extensions
valueset-map | map | for ValueSet HL7 Extensions
Extensions that refer to Any resource

cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

4.10.14.3 Search Extensions

No Search Extensions defined for this resource

4.10.15 Resource ConceptMap - Operations

URL: https://hl7.org/fhir/R4/conceptmap-operations.html

4.10.15 Resource ConceptMap - Operations

Vocabulary Work Group | Maturity Level: 3 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This resource has 2 operations operations associated with it:

$translate | Concept Translation
$closure | Closure Table Maintenance

For more information about operations, including how they are invoked, see Operations.

4.10.15.1 Operation $translate on ConceptMap

URL: https://hl7.org/fhir/R4/conceptmap-operation-translate.html

4.10.15.1 Operation $translate on ConceptMap

Vocabulary Work Group | Maturity Level: 3 | Trial Use | Compartments: N/A

Translate a code from one value set to another, based on the existing value set and concept maps resources, and/or other additional knowledge available to the server.

One (and only one) of the in parameters (code, coding, codeableConcept) must be provided, to identify the code that is to be translated.

The operation returns a set of parameters including a 'result' for whether there is an acceptable match, and a list of possible matches. Note that the list of matches may include notes of codes for which mapping is specifically excluded, so implementers have to check the match.equivalence for each match.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/ConceptMap-translate
```

Formal Definition (as a OperationDefinition).

URL: [base]/ConceptMap/$translate

URL: [base]/ConceptMap/[id]/$translate

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation
url | 0..1 | uri | | | A canonical URL for a concept map. The server must know the concept map (e.g. it is defined explicitly in the server's concept maps, or it is defined implicitly by some code system known to the server.
conceptMap | 0..1 | ConceptMap | | | The concept map is provided directly as part of the request. Servers may choose not to accept concept maps in this fashion.
conceptMapVersion | 0..1 | string | | | The identifier that is used to identify a specific version of the concept map to be used for the translation. This is an arbitrary value managed by the concept map author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available.
code | 0..1 | code | | | The code that is to be translated. If a code is provided, a system must be provided
system | 0..1 | uri | | | The system for the code that is to be translated
version | 0..1 | string | | | The version of the system, if one was provided in the source data
source | 0..1 | uri | | | Identifies the value set used when the concept (system/code pair) was chosen. May be a logical id, or an absolute or relative location. The source value set is an optional parameter because in some cases, the client cannot know what the source value set is. However, without a source value set, the server may be unable to safely identify an applicable concept map, and would return an error. For this reason, a source value set SHOULD always be provided. Note that servers may be able to identify an appropriate concept map without a source value set if there is a full mapping for the entire code system in the concept map, or by manual intervention
coding | 0..1 | Coding | | | A coding to translate
codeableConcept | 0..1 | CodeableConcept | | | A full codeableConcept to validate. The server can translate any of the coding values (e.g. existing translations) as it chooses
target | 0..1 | uri | | | Identifies the value set in which a translation is sought. May be a logical id, or an absolute or relative location. If there's no target specified, the server should return all known translations, along with their source

targetsystem | 0..1 | uri | | | identifies a target code system in which a mapping is sought. This parameter is an alternative to the target parameter - only one is required. Searching for any translation to a target code system irrespective of the context (e.g. target valueset) may lead to unsafe results, and it is at the discretion of the server to decide when to support this operation
dependency | 0..* | | | | Another element that may help produce the correct mapping
dependency.element | 0..1 | uri | | | The element for this dependency
dependency.concept | 0..1 | CodeableConcept | | | The value for this dependency
reverse | 0..1 | boolean | | | if this is true, then the operation should return all the codes that might be mapped to this code. This parameter reverses the meaning of the source and target parameters
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
result | 1..1 | boolean | | | True if the concept could be translated successfully. The value can only be true if at least one returned match has an equivalence which is not unmatched or disjoint
message | 0..1 | string | | | Error details, for display to a human. If this is provided when result = true, the message carries hints and warnings (e.g. a note that the matches could be improved by providing additional detail)
match | 0..* | | | | A concept in the target value set with an equivalence. Note that there may be multiple matches of equal or differing equivalence, and the matches may include equivalence values that mean that there is no match
match.equivalence | 0..1 | code | | | A code indicating the equivalence of the translation, using values from ConceptMapEquivalence
match.concept | 0..1 | Coding | | | The translation outcome. Note that this would never have userSelected = true, since the process of translations implies that the user is not selecting the code (and only the client could know differently)
match.product | 0..* | | | | Another element that is the product of this mapping
match.product.element | 0..1 | uri | | | The element for this product
match.product.concept | 0..1 | Coding | | | The value for this product
match.source | 0..1 | uri | | | The canonical reference to the concept map from which this mapping comes from

4.10.15.1.1 Examples

Request:

```
GET [base]/ConceptMap/$translate?system=http://hl7.org/fhir/composition-status
  &code=preliminary&source=http://hl7.org/fhir/ValueSet/composition-status
  &target=http://hl7.org/fhir/ValueSet/v3-ActStatus
```

Response:

```
HTTP/1.1 200 OK
[other headers]

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueBoolean": true
    },
    {
      "name": "match",
      "part": [
        {
          "name": "equivalence",
          "valueCode": "equivalent"
        },
        {
          "name": "concept",
          "valueCoding": {
            "system": "http://hl7.org/fhir/v3/ActStatus",
            "code": "active",
            "userSelected": false
          }
        }
      ]
    }
  ]
}
```

For more information about operations, including how they are invoked, see Operations.

4.10.15.2 Operation $closure on ConceptMap

URL: https://hl7.org/fhir/R4/conceptmap-operation-closure.html

4.10.15.2 Operation $closure on ConceptMap

Vocabulary Work Group | Maturity Level: 3 | Trial Use | Compartments: N/A

This operation provides support for ongoing maintenance of a client-side transitive closure table based on server-side terminological logic. For details of how this is used, see Maintaining a Closure Table.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/ConceptMap-closure
```

Formal Definition (as a OperationDefinition).

URL: [base]/$closure

This is not an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
name | 1..1 | string | | | The name that defines the particular context for the subsumption based closure table
concept | 0..* | Coding | | | Concepts to add to the closure table
version | 0..1 | string | | | A request to resynchronise - request to send all new entries since the nominated version was sent by the server
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
return | 1..1 | ConceptMap | | | A list of new entries (code / system --> code/system) that the client should add to its closure table. The only kind of entry mapping equivalences that can be returned are equal, specializes, subsumes and unmatched Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

4.10.15.2.1 Examples

Request:

```
POST [base]/$closure

{
  "resourceType" : "Parameters",
  "parameter" : [
    {
      "name" : "name",
      "valueString" : "patient-problems"
    },
    {
      "name" : "concept",
      "valueCoding" : {
        "system" : "http://snomed.info/sct",
        "code" : "22298006",
      }
    }
  ]
}
```

Response:

```
HTTP/1.1 200 OK
[other headers]

{
  "resourceType": "ConceptMap",
  "identifier": "49088976-d54d-4d19-b868-3d4c18cebabb",
  "version": "8",
  "status": "active",
  "experimental": true,
  "date": "2012-06-13",
  "group": [
    {
      "source": "http://snomed.info/sct",
      "target": "http://snomed.info/sct",
      "element" : {
        "code": "22298006",
        "target": [
          {
            "code": "128599005",
            "equivalence": "subsumes"
          }
        ]
      }
    },
  ]
}
```

For more information about operations, including how they are invoked, see Operations.

4.10.16 Resource ConceptMap - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/conceptmap-version-maps.html

No validation errors - all conversions are clean

4.10.16 Resource ConceptMap - R3/R4 Conversions

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for ConceptMap.

Functional status for this map: 4 tests that all execute ok. All tests pass round-trip testing and 4 r3 resources are invalid (0 errors). (see documentation)

4.10.16.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/ConceptMap3to4" = "R3 to R4 Conversions for ConceptMap"

uses "http://hl7.org/fhir/3.0/StructureDefinition/ConceptMap" alias ConceptMapR3 as source
```

```
uses "http://hl7.org/fhir/StructureDefinition/ConceptMap" alias ConceptMap as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group ConceptMap(source src : ConceptMapR3, target tgt : ConceptMap) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.copyright -> tgt.copyright;
  src.source : uri as vs0 -> tgt.source = create('uri') as vt0 then uri(vs0, vt0);
  src.source : Reference as vs0 -> tgt.source = create('canonical') as vt0 then Reference2Canonical(vs0, vt0);
  src.target : uri as vs0 -> tgt.target = create('uri') as vt0 then uri(vs0, vt0) "targetU";
  src.target : Reference as vs0 -> tgt.target = create('canonical') as vt0 then Reference2Canonical(vs0, vt0);
  src.group as vs0 -> tgt.group as vt0 then group(vs0, vt0);
}

group group(source src, target tgt) extends BackboneElement {
  src.source -> tgt.source;
  src.sourceVersion -> tgt.sourceVersion;
  src.target -> tgt.target;
  src.targetVersion -> tgt.targetVersion;
  src.element as vs0 -> tgt.element as vt0 then element(vs0, vt0);
  src.unmapped as vs0 -> tgt.unmapped as vt0 then unmapped(vs0, vt0);
}

group element(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.target as vs0 -> tgt.target as vt0 then target(vs0, vt0);
}

group target(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.equivalence -> tgt.equivalence;
  src.comment -> tgt.comment;
  src.dependsOn as vs0 -> tgt.dependsOn as vt0 then qualifier(vs0, vt0);
  src.product as vs0 -> tgt.product as vt0 then qualifier(vs0, vt0);
}

group qualifier(source src, target tgt) extends BackboneElement {
  src.property -> tgt.property;
  src.system -> tgt.system;
  src.code -> tgt.value;
  src.display -> tgt.display;
}

group unmapped(source src, target tgt) extends BackboneElement {
  src.mode -> tgt.mode;
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.url -> tgt.url;
}
```

4.10.16.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/ConceptMap4to3" = "R4 to R3 Conversion for ConceptMap"

uses "http://hl7.org/fhir/StructureDefinition/ConceptMap" alias ConceptMap as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/ConceptMap" alias ConceptMapR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group ConceptMap(source src : ConceptMap, target tgt : ConceptMapR3) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
```

```
  src.purpose -> tgt.purpose;
  src.copyright -> tgt.copyright;
  src.source : uri as vs0 -> tgt.source = create('uri') as vt0 then uri(vs0, vt0);
  src.source : canonical as vs0 -> tgt.source = create('Reference') as vt0 then canonical2Reference(vs0, vt0);
  src.target : uri as vs0 -> tgt.target = create('uri') as vt0 then uri(vs0, vt0) "targetU";
  src.target : canonical as vs0 -> tgt.target = create('Reference') as vt0 then canonical2Reference(vs0, vt0);
  src.group as vs0 -> tgt.group as vt0 then group(vs0, vt0);
}

group group(source src, target tgt) extends BackboneElement {
  src.source -> tgt.source;
  src.sourceVersion -> tgt.sourceVersion;
  src.target -> tgt.target;
  src.targetVersion -> tgt.targetVersion;
  src.element as vs0 -> tgt.element as vt0 then element(vs0, vt0);
  src.unmapped as vs0 -> tgt.unmapped as vt0 then unmapped(vs0, vt0);
}

group element(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.target as vs0 -> tgt.target as vt0 then target(vs0, vt0);
}

group target(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.equivalence -> tgt.equivalence;
  src.comment -> tgt.comment;
  src.dependsOn as vs0 -> tgt.dependsOn as vt0 then qualifier(vs0, vt0);
  src.product as vs0 -> tgt.product as vt0 then qualifier(vs0, vt0);
}

group qualifier(source src, target tgt) extends BackboneElement {
  src.property -> tgt.property;
  src.system -> tgt.system;
  src.value -> tgt.code "code";
  src.display -> tgt.display;
}

group unmapped(source src, target tgt) extends BackboneElement {
  src.mode -> tgt.mode;
  src.code -> tgt.code;
  src.display -> tgt.display;
  src.url -> tgt.url;
}
```

4.10.16.3 R4 Validation Errors

4.10.17 ConceptMap HL7 Extensions

URL: https://hl7.org/fhir/R4/conceptmap-extensions.html

Defines common extensions used with or related to the ConceptMap resource

4.10.17 ConceptMap HL7 Extensions

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative

4.10.17.1 Content

Extensions:

concept-bidirectional | bidirectional : Set to true if the concept map can be safely intepreted in reversse.

4.10.18 JSON Schema for ConceptMap

URL: https://hl7.org/fhir/R4/conceptmap.schema.json.html

4.10.18 JSON Schema for ConceptMap

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for ConceptMap, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ConceptMap",
  "$ref": "#/definitions/ConceptMap",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ConceptMap": {
      "description": "A statement of relationships from one set of concepts to one or more other concepts - either
    concepts in code systems, or data element/data element concepts, or classes in class models.",
      "properties": {
```

```
    "resourceType": {
      "description": "This is a ConceptMap resource",
      "const": "ConceptMap"
    },
    "id": {
      "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this concept map when it is referenced in a
specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
and SHOULD be a literal address at which at which an authoritative instance of this concept map is (or will be)
published. This URL can be the target of a canonical reference. It SHALL remain the same when the concept map is
stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this concept map when it is represented in other
formats, or referenced in a specification, model, design or an instance.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "version": {
```

```
      "description": "The identifier that is used to identify this version of the concept map when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the concept map author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the concept map. This name should be usable as an identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the concept map.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this concept map. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this concept map is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the concept map was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the concept map changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the concept map.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the concept map from a consumer\u0027s perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate concept map instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
```

```
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the concept map is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this concept map is needed and why it has been designed as it has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the concept map and/or its contents. Copyright statements
are generally legal restrictions on the use and publishing of the concept map.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "sourceUri": {
      "description": "Identifier for the source value set that contains the concepts that are being mapped and
provides context for the mappings.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_sourceUri": {
      "description": "Extensions for sourceUri",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "sourceCanonical": {
      "description": "Identifier for the source value set that contains the concepts that are being mapped and
provides context for the mappings.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_sourceCanonical": {
      "description": "Extensions for sourceCanonical",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "targetUri": {
      "description": "The target value set provides context for the mappings. Note that the mapping is made between
concepts, not between value sets, but the value set provides important context about how the concept mapping choices
 are made.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_targetUri": {
      "description": "Extensions for targetUri",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "targetCanonical": {
      "description": "The target value set provides context for the mappings. Note that the mapping is made between
concepts, not between value sets, but the value set provides important context about how the concept mapping choices
 are made.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_targetCanonical": {
      "description": "Extensions for targetCanonical",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "group": {
      "description": "A group of mappings that all have the same source and target system.",
      "items": {
        "$ref": "#/definitions/ConceptMap_Group"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"ConceptMap_Group": {
  "description": "A statement of relationships from one set of concepts to one or more other concepts - either
concepts in code systems, or data element/data element concepts, or classes in class models.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
```

```
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "source": {
      "description": "An absolute URI that identifies the source system where the concepts to be mapped are
defined.",
      "$ref": "#/definitions/uri"
    },
    "_source": {
      "description": "Extensions for source",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "sourceVersion": {
      "description": "The specific version of the code system, as determined by the code system authority.",
      "$ref": "#/definitions/string"
    },
    "_sourceVersion": {
      "description": "Extensions for sourceVersion",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "target": {
      "description": "An absolute URI that identifies the target system that the concepts will be mapped to.",
      "$ref": "#/definitions/uri"
    },
    "_target": {
      "description": "Extensions for target",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "targetVersion": {
      "description": "The specific version of the code system, as determined by the code system authority.",
      "$ref": "#/definitions/string"
    },
    "_targetVersion": {
      "description": "Extensions for targetVersion",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "element": {
      "description": "Mappings for an individual concept in the source to one or more concepts in the target.",
      "items": {
        "$ref": "#/definitions/ConceptMap_Element"
      },
      "type": "array"
    },
    "unmapped": {
      "description": "What to do when there is no mapping for the source concept. \"Unmapped\" does not include
codes that are unmatched, and the unmapped element is ignored in a code is specified to have equivalence \u003d
unmatched.",
      "$ref": "#/definitions/ConceptMap_Unmapped"
    }
  },
  "additionalProperties": false,
  "required": [
    "element"
  ]
},
"ConceptMap_Element": {
  "description": "A statement of relationships from one set of concepts to one or more other concepts - either
concepts in code systems, or data element/data element concepts, or classes in class models.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
```

```
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "Identity (code or path) or the element/item being mapped.",
      "$ref": "#/definitions/code"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "display": {
      "description": "The display for the code. The display is only provided to help editors when editing the concept map.",
      "$ref": "#/definitions/string"
    },
    "_display": {
      "description": "Extensions for display",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "target": {
      "description": "A concept from the target value set that this concept maps to.",
      "items": {
        "$ref": "#/definitions/ConceptMap_Target"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"ConceptMap_Target": {
  "description": "A statement of relationships from one set of concepts to one or more other concepts - either concepts in code systems, or data element/data element concepts, or classes in class models.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "Identity (code or path) or the element/item that the map refers to.",
      "$ref": "#/definitions/code"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
```

```
    },
    "display": {
      "description": "The display for the code. The display is only provided to help editors when editing the
concept map.",
      "$ref": "#/definitions/string"
    },
    "_display": {
      "description": "Extensions for display",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "equivalence": {
      "description": "The equivalence between the source and target concepts (counting for the dependencies and
products). The equivalence is read from target to source (e.g. the target is \u0027wider\u0027 than the source).",
      "enum": [
        "relatedto",
        "equivalent",
        "equal",
        "wider",
        "subsumes",
        "narrower",
        "specializes",
        "inexact",
        "unmatched",
        "disjoint"
      ]
    },
    "_equivalence": {
      "description": "Extensions for equivalence",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "comment": {
      "description": "A description of status/issues in mapping that conveys additional information not represented
in  the structured data.",
      "$ref": "#/definitions/string"
    },
    "_comment": {
      "description": "Extensions for comment",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "dependsOn": {
      "description": "A set of additional dependencies for this mapping to hold. This mapping is only applicable if
the specified element can be resolved, and it has the specified value.",
      "items": {
        "$ref": "#/definitions/ConceptMap_DependsOn"
      },
      "type": "array"
    },
    "product": {
      "description": "A set of additional outcomes from this mapping to other elements. To properly execute this
mapping, the specified element must be mapped to some data element or source that is in context. The mapping may
still be useful without a place for the additional data elements, but the equivalence cannot be relied on.",
      "items": {
        "$ref": "#/definitions/ConceptMap_DependsOn"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"ConceptMap_DependsOn": {
  "description": "A statement of relationships from one set of concepts to one or more other concepts - either
concepts in code systems, or data element/data element concepts, or classes in class models.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
```

```
      },
      "type": "array"
    },
    "property": {
      "description": "A reference to an element that holds a coded value that corresponds to a code system property.
 The idea is that the information model carries an element somewhere that is labeled to correspond with a code
system property.",
      "$ref": "#/definitions/uri"
    },
    "_property": {
      "description": "Extensions for property",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "system": {
      "description": "An absolute URI that identifies the code system of the dependency code (if the
source/dependency is a value set that crosses code systems).",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "value": {
      "description": "Identity (code or path) or the element/item/ValueSet/text that the map depends on / refers
to.",
      "$ref": "#/definitions/string"
    },
    "_value": {
      "description": "Extensions for value",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "display": {
      "description": "The display for the code. The display is only provided to help editors when editing the
concept map.",
      "$ref": "#/definitions/string"
    },
    "_display": {
      "description": "Extensions for display",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"ConceptMap_Unmapped": {
  "description": "A statement of relationships from one set of concepts to one or more other concepts - either
concepts in code systems, or data element/data element concepts, or classes in class models.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "mode": {
      "description": "Defines which action to take if there is no match for the source concept in the target system
designated for the group. One of 3 actions are possible: use the unmapped code (this is useful when doing a mapping
between versions, and only a few codes have changed), use a fixed code (a default code), or alternatively, a
reference to a different concept map can be provided (by canonical URL).",
      "enum": [
        "provided",
        "fixed",
        "other-map"
      ]
    },
    "_mode": {
      "description": "Extensions for mode",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "The fixed code to use when the mode \u003d \u0027fixed\u0027  - all unmapped codes are mapped
```

```
    to a single fixed code.",
          "$ref": "#/definitions/code"
        },
        "_code": {
          "description": "Extensions for code",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "display": {
          "description": "The display for the code. The display is only provided to help editors when editing the
    concept map.",
          "$ref": "#/definitions/string"
        },
        "_display": {
          "description": "Extensions for display",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "url": {
          "description": "The canonical reference to an additional ConceptMap resource instance to use for mapping if
    this ConceptMap resource contains no matching mapping for the source concept.",
          "$ref": "canonical.schema.json#/definitions/canonical"
        }
      },
      "additionalProperties": false
    }
  }
}
```

4.10.19 Resource ConceptMap - Maturity Dependency View

URL: https://hl7.org/fhir/R4/conceptmap-dependencies.html

4.10.19 Resource ConceptMap - Maturity Dependency View

Vocabulary Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for ConceptMap FMM level 3

Name | Card. | Type | Dependency Analysis
ConceptMap | | DomainResource
url | 0..1 | uri
identifier | 0..1 | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
copyright | 0..1 | markdown
source[x] | 0..1
sourceUri | | uri
sourceCanonical | | canonical
target[x] | 0..1
targetUri | | uri
targetCanonical | | canonical
group | 0..* | BackboneElement
source | 0..1 | uri
sourceVersion | 0..1 | string
target | 0..1 | uri
targetVersion | 0..1 | string
element | 1..* | BackboneElement
code | 0..1 | code
display | 0..1 | string
target | 0..* | BackboneElement
code | 0..1 | code
display | 0..1 | string
equivalence | 1..1 | code
comment | 0..1 | string
dependsOn | 0..* | BackboneElement
property | 1..1 | uri
system | 0..1 | canonical(CodeSystem)
value | 1..1 | string
display | 0..1 | string
product | 0..* | see dependsOn

unmapped | 0..1 | BackboneElement
mode | 1..1 | code
code | 0..1 | code
display | 0..1 | string
url | 0..1 | canonical(ConceptMap)

4.11 Resource NamingSystem - Content

URL: https://hl7.org/fhir/R4/namingsystem.html
FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

A curated namespace that issues unique symbols within that namespace for the identification of concepts, people, devices, etc. Represents a "System" used within the Identifier and Coding data types.

4.11.1 Scope and Usage

Defines a specific code system or identifier system, so that it can be noted in a registry for other systems to find and understand an identifier.

4.11.2 Boundaries and Relationships

The CodeSystem resource defines the content of a code system, and also its preferred identifier. The NamingSystem resource identifies the existence of a code or identifier system, and its possible and preferred identifiers. The key difference between the resources is who creates and manages them - CodeSystem resources are managed by the owner of the code system resource, who can properly define the features and content of the code system. NamingSystem resources, on the other hand, are frequently defined by 3rd parties that encounter the code system in use, and need to describe the use, but do not have the authority to define the features and content. Additionally, there may be multiple authoritative NamingSystem resources for a code system, but there should only be one CodeSystem resource.

4.11.3 Background and Context

For discussion of policy for creating well maintained OIDs and URIs for resources, see The HL7 Confluence site .

4.11.4 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
NamingSystem | ITU | | DomainResource | System of unique identification + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: Root systems cannot have uuid identifiers + Rule: Can't have more than one preferred identifier for a type Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
name | ΣI | 1..1 | string | Name for this naming system (computer friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
kind | Σ | 1..1 | code | codesystem | identifier | root NamingSystemType (Required)
date | Σ | 1..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
responsible | | 0..1 | string | Who maintains system namespace?
type | | 0..1 | CodeableConcept | e.g. driver, provider, patient, bank etc. IdentifierType (Extensible)
description | | 0..1 | markdown | Natural language description of the naming system
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for naming system (if applicable) Jurisdiction (Extensible)
usage | | 0..1 | string | How/where is it used
uniqueId | Σ | 1..* | BackboneElement | Unique identifiers used for system
type | Σ | 1..1 | code | oid | uuid | uri | other NamingSystemIdentifierType (Required)
value | Σ | 1..1 | string | The unique identifier
preferred | | 0..1 | boolean | Is this the id that should be used for this type
comment | | 0..1 | string | Notes about identifier usage
period | | 0..1 | Period | When is identifier valid?
Documentation for this format

UML Diagram (Legend)

XML Template

```
<NamingSystem xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <name value="[string]"/><!--  1..1 Name for this naming system (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <kind value="[code]"/><!-- 1..1 codesystem | identifier | root -->
 <date value="[dateTime]"/><!-- 1..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <responsible value="[string]"/><!-- 0..1 Who maintains system namespace? -->
 <type><!-- 0..1 CodeableConcept e.g. driver,  provider,  patient, bank etc. --></type>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the naming system -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for naming system (if applicable) --></jurisdiction>
 <usage value="[string]"/><!-- 0..1 How/where is it used -->
 <uniqueId>  <!-- 1..* Unique identifiers used for system -->
  <type value="[code]"/><!-- 1..1 oid | uuid | uri | other -->
  <value value="[string]"/><!-- 1..1 The unique identifier -->
  <preferred value="[boolean]"/><!-- 0..1 Is this the id that should be used for this type -->
  <comment value="[string]"/><!-- 0..1 Notes about identifier usage -->
  <period><!-- 0..1 Period When is identifier valid? --></period>
 </uniqueId>
</NamingSystem>
```

JSON Template

```
{
  "resourceType" : "NamingSystem",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "name" : "<string>", // C? R!  Name for this naming system (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "kind" : "<code>", // R!  codesystem | identifier | root
  "date" : "<dateTime>", // R!  Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "responsible" : "<string>", // Who maintains system namespace?
  "type" : { CodeableConcept }, // e.g. driver,  provider,  patient, bank etc.
  "description" : "<markdown>", // Natural language description of the naming system
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for naming system (if applicable)
  "usage" : "<string>", // How/where is it used
  "uniqueId" : [{ // R!  Unique identifiers used for system
    "type" : "<code>", // R!  oid | uuid | uri | other
    "value" : "<string>", // R!  The unique identifier
    "preferred" : <boolean>, // Is this the id that should be used for this type
    "comment" : "<string>", // Notes about identifier usage
    "period" : { Period } // When is identifier valid?
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:NamingSystem;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:NamingSystem.name [ string ]; # 1..1 Name for this naming system (computer friendly)
  fhir:NamingSystem.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:NamingSystem.kind [ code ]; # 1..1 codesystem | identifier | root
  fhir:NamingSystem.date [ dateTime ]; # 1..1 Date last changed
  fhir:NamingSystem.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:NamingSystem.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:NamingSystem.responsible [ string ]; # 0..1 Who maintains system namespace?
  fhir:NamingSystem.type [ CodeableConcept ]; # 0..1 e.g. driver,  provider,  patient, bank etc.
  fhir:NamingSystem.description [ markdown ]; # 0..1 Natural language description of the naming system
  fhir:NamingSystem.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:NamingSystem.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for naming system (if
    applicable)
  fhir:NamingSystem.usage [ string ]; # 0..1 How/where is it used
  fhir:NamingSystem.uniqueId [ # 1..* Unique identifiers used for system
    fhir:NamingSystem.uniqueId.type [ code ]; # 1..1 oid | uuid | uri | other
    fhir:NamingSystem.uniqueId.value [ string ]; # 1..1 The unique identifier
    fhir:NamingSystem.uniqueId.preferred [ boolean ]; # 0..1 Is this the id that should be used for this type
    fhir:NamingSystem.uniqueId.comment [ string ]; # 0..1 Notes about identifier usage
    fhir:NamingSystem.uniqueId.period [ Period ]; # 0..1 When is identifier valid?
  ], ...;
]
```

Changes since R3

NamingSystem

NamingSystem.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

NamingSystem.kind | Change value set from http://hl7.org/fhir/ValueSet/namingsystem-type to http://hl7.org/fhir/ValueSet/namingsystem-type|4.0.1
NamingSystem.uniqueId.type | Change value set from http://hl7.org/fhir/ValueSet/namingsystem-identifier-type to http://hl7.org/fhir/ValueSet/namingsystem-identifier-type|4.0.1
NamingSystem.replacedBy | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. 1 fail round-trip testing and 3 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
NamingSystem | ITU | | DomainResource | System of unique identification + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: Root systems cannot have uuid identifiers + Rule: Can't have more than one preferred identifier for a type Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
name | ΣI | 1..1 | string | Name for this naming system (computer friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
kind | Σ | 1..1 | code | codesystem | identifier | root NamingSystemType (Required)
date | Σ | 1..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
responsible | | 0..1 | string | Who maintains system namespace?
type | | 0..1 | CodeableConcept | e.g. driver, provider, patient, bank etc. IdentifierType (Extensible)
description | | 0..1 | markdown | Natural language description of the naming system
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for naming system (if applicable) Jurisdiction (Extensible)
usage | | 0..1 | string | How/where is it used
uniqueId | Σ | 1..* | BackboneElement | Unique identifiers used for system
type | Σ | 1..1 | code | oid | uuid | uri | other NamingSystemIdentifierType (Required)
value | Σ | 1..1 | string | The unique identifier
preferred | | 0..1 | boolean | Is this the id that should be used for this type
comment | | 0..1 | string | Notes about identifier usage
period | | 0..1 | Period | When is identifier valid?
Documentation for this format

UML Diagram (Legend)

XML Template

```
<NamingSystem xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <name value="[string]"/><!--  1..1 Name for this naming system (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <kind value="[code]"/><!-- 1..1 codesystem | identifier | root -->
 <date value="[dateTime]"/><!-- 1..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <responsible value="[string]"/><!-- 0..1 Who maintains system namespace? -->
 <type><!-- 0..1 CodeableConcept e.g. driver,  provider,  patient, bank etc. --></type>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the naming system -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for naming system (if applicable) --></jurisdiction>
 <usage value="[string]"/><!-- 0..1 How/where is it used -->
 <uniqueId>  <!-- 1..* Unique identifiers used for system -->
  <type value="[code]"/><!-- 1..1 oid | uuid | uri | other -->
  <value value="[string]"/><!-- 1..1 The unique identifier -->
  <preferred value="[boolean]"/><!-- 0..1 Is this the id that should be used for this type -->
  <comment value="[string]"/><!-- 0..1 Notes about identifier usage -->
  <period><!-- 0..1 Period When is identifier valid? --></period>
 </uniqueId>
</NamingSystem>
```

JSON Template

```
{
  "resourceType" : "NamingSystem",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "name" : "<string>", // C? R!  Name for this naming system (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "kind" : "<code>", // R!  codesystem | identifier | root
  "date" : "<dateTime>", // R!  Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "responsible" : "<string>", // Who maintains system namespace?
  "type" : { CodeableConcept }, // e.g. driver,  provider,  patient, bank etc.
  "description" : "<markdown>", // Natural language description of the naming system
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for naming system (if applicable)
  "usage" : "<string>", // How/where is it used
```

```
  "uniqueId" : [{ // R!  Unique identifiers used for system
    "type" : "<code>", // R!  oid | uuid | uri | other
    "value" : "<string>", // R!  The unique identifier
    "preferred" : <boolean>, // Is this the id that should be used for this type
    "comment" : "<string>", // Notes about identifier usage
    "period" : { Period } // When is identifier valid?
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:NamingSystem;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:NamingSystem.name [ string ]; # 1..1 Name for this naming system (computer friendly)
  fhir:NamingSystem.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:NamingSystem.kind [ code ]; # 1..1 codesystem | identifier | root
  fhir:NamingSystem.date [ dateTime ]; # 1..1 Date last changed
  fhir:NamingSystem.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:NamingSystem.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:NamingSystem.responsible [ string ]; # 0..1 Who maintains system namespace?
  fhir:NamingSystem.type [ CodeableConcept ]; # 0..1 e.g. driver,  provider,  patient, bank etc.
  fhir:NamingSystem.description [ markdown ]; # 0..1 Natural language description of the naming system
  fhir:NamingSystem.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:NamingSystem.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for naming system (if
    applicable)
  fhir:NamingSystem.usage [ string ]; # 0..1 How/where is it used
  fhir:NamingSystem.uniqueId [ # 1..* Unique identifiers used for system
    fhir:NamingSystem.uniqueId.type [ code ]; # 1..1 oid | uuid | uri | other
    fhir:NamingSystem.uniqueId.value [ string ]; # 1..1 The unique identifier
    fhir:NamingSystem.uniqueId.preferred [ boolean ]; # 0..1 Is this the id that should be used for this type
    fhir:NamingSystem.uniqueId.comment [ string ]; # 0..1 Notes about identifier usage
    fhir:NamingSystem.uniqueId.period [ Period ]; # 0..1 When is identifier valid?
  ], ...;
]
```

Changes since Release 3

NamingSystem

NamingSystem.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

NamingSystem.kind | Change value set from http://hl7.org/fhir/ValueSet/namingsystem-type to http://hl7.org/fhir/ValueSet/namingsystem-type|4.0.1

NamingSystem.uniqueId.type | Change value set from http://hl7.org/fhir/ValueSet/namingsystem-identifier-type to http://hl7.org/fhir/ValueSet/namingsystem-identifier-type|4.0.1

NamingSystem.replacedBy | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. 1 fail round-trip testing and 3 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

4.11.4.1 Terminology Bindings

Path | Definition | Type | Reference

NamingSystem.status | The lifecycle status of an artifact. | Required | PublicationStatus

NamingSystem.kind | Identifies the purpose of the naming system. | Required | NamingSystemType

NamingSystem.type | A coded type for an identifier that can be used to determine which identifier to use for a specific purpose. | Extensible | Identifier Type Codes

NamingSystem.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet

NamingSystem.uniqueId.type | Identifies the style of unique identifier used to identify a namespace. | Required | NamingSystemIdentifierType

4.11.4.2 Constraints

id | Level | Location | Description | Expression

nsd-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

nsd-1 | Rule | (base) | Root systems cannot have uuid identifiers | kind != 'root' or uniqueId.all(type != 'uuid')

nsd-2 | Rule | (base) | Can't have more than one preferred identifier for a type | uniqueId.where(preferred = true).select(type).isDistinct()

4.11.4.3 Dealing with duplicate entries

In some cases, the same code or identifier system might accidentally be created more than once in a registry, for example because someone failed to check for an existing entry before adding a new one or knows the same concept with a different name. If this occurs, one of the system entries should be deleted and the

remaining entry should have its uniqueId updated to include any identifiers present on the original entry (and possibly have its descriptive information modified to include additional information from the duplicate entry).

4.11.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
contact | string | Name of an individual to contact | NamingSystem.contact.name
context | token | A use context assigned to the naming system | (NamingSystem.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the naming system | (NamingSystem.useContext.value as Quantity) | (NamingSystem.useContext.value as Range)
context-type | token | A type of use context assigned to the naming system | NamingSystem.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the naming system | On NamingSystem.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the naming system | On NamingSystem.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The naming system publication date | NamingSystem.date
description | string | The description of the naming system | NamingSystem.description
id-type | token | oid | uuid | uri | other | NamingSystem.uniqueId.type
jurisdiction | token | Intended jurisdiction for the naming system | NamingSystem.jurisdiction
kind | token | codesystem | identifier | root | NamingSystem.kind
name | string | Computationally friendly name of the naming system | NamingSystem.name
period | date | When is identifier valid? | NamingSystem.uniqueId.period
publisher | string | Name of the publisher of the naming system | NamingSystem.publisher
responsible | string | Who maintains system namespace? | NamingSystem.responsible
status | token | The current status of the naming system | NamingSystem.status
telecom | token | Contact details for individual or organization | NamingSystem.contact.telecom
type | token | e.g. driver, provider, patient, bank etc. | NamingSystem.type
value | string | The unique identifier | NamingSystem.uniqueId.value

4.11.6 Resource NamingSystem - Examples

URL: https://hl7.org/fhir/R4/namingsystem-examples.html

4.11.6 Resource NamingSystem - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of how to use Naming System for a code system | example | XML | JSON | Turtle
Example of naming system for an identifier | example-id | XML | JSON | Turtle
Common US Driver's license identification | registry | XML | JSON | Turtle
Defined Code systems | terminologies | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

4.11.7 Resource NamingSystem - Detailed Descriptions

URL: https://hl7.org/fhir/R4/namingsystem-definitions.html

4.11.7 Resource NamingSystem - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the NamingSystem resource.

NamingSystem
Element Id | NamingSystem
Definition | A curated namespace that issues unique symbols within that namespace for the identification of concepts, people, devices, etc. Represents a "System" used within the Identifier and Coding data types.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element nsd-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}') nsd-1 Rule Root systems cannot have uuid identifiers kind != 'root' or uniqueId.all(type != 'uuid') nsd-2 Rule Can't have more than one preferred identifier for a type uniqueId.where(preferred = true).select(type).isDistinct()
Defined on this element
nsd-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
nsd-1 | Rule | Root systems cannot have uuid identifiers | kind != 'root' or uniqueId.all(type != 'uuid')
nsd-2 | Rule | Can't have more than one preferred identifier for a type | uniqueId.where(preferred = true).select(type).isDistinct()
NamingSystem.name

Element Id | NamingSystem.name
Definition | A natural language name identifying the naming system. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 1..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.The"symbolic name" for an OID would be captured as an extension.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
NamingSystem.status
Element Id | NamingSystem.status
Definition | The status of this naming system. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of naming systems that are appropriate for use versus not.
NamingSystem.kind
Element Id | NamingSystem.kind
Definition | Indicates the purpose for the naming system - what kinds of things does it make unique?
Cardinality | 1..1
Terminology Binding | NamingSystemType (Required)
Type | code
Summary | true
NamingSystem.date
Element Id | NamingSystem.date
Definition | The date (and optionally time) when the naming system was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the naming system changes.
Cardinality | 1..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the naming system. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
NamingSystem.publisher
Element Id | NamingSystem.publisher
Definition | The name of the organization or individual that published the naming system.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the naming system. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the naming system is the organization or individual primarily responsible for the maintenance and upkeep of the naming system. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the naming system. This item SHOULD be populated unless the information is available from context.
NamingSystem.contact
Element Id | NamingSystem.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
NamingSystem.responsible
Element Id | NamingSystem.responsible
Definition | The name of the organization that is responsible for issuing identifiers or codes for this namespace and ensuring their non-collision.
Cardinality | 0..1
Type | string
Comments | This is the primary organization. Responsibility for some aspects of a namespace may be delegated.
NamingSystem.type
Element Id | NamingSystem.type
Definition | Categorizes a naming system for easier search by grouping related naming systems.
Cardinality | 0..1
Terminology Binding | IdentifierType (Extensible)

Type | CodeableConcept
Comments | This will most commonly be used for identifier namespaces, but categories could potentially be useful for code systems and authorities as well.
NamingSystem.description
Element Id | NamingSystem.description
Definition | A free text natural language description of the naming system from a consumer's perspective. Details about what the namespace identifies including scope, granularity, version labeling, etc.
Cardinality | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the naming system was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the naming system as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the naming system is presumed to be the predominant language in the place the naming system was created).
NamingSystem.useContext
Element Id | NamingSystem.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate naming system instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
NamingSystem.jurisdiction
Element Id | NamingSystem.jurisdiction
Definition | A legal or geographic region in which the naming system is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the naming system to be used in jurisdictions other than those for which it was originally designed or intended.
NamingSystem.usage
Element Id | NamingSystem.usage
Definition | Provides guidance on the use of the namespace, including the handling of formatting characters, use of upper vs. lower case, etc.
Cardinality | 0..1
Type | string
NamingSystem.uniqueId
Element Id | NamingSystem.uniqueId
Definition | Indicates how the system may be identified when referenced in electronic exchange.
Cardinality | 1..*
Summary | true
Comments | Multiple identifiers may exist, either due to duplicate registration, regional rules, needs of different communication technologies, etc.
NamingSystem.uniqueId.type
Element Id | NamingSystem.uniqueId.type
Definition | Identifies the unique identifier scheme used for this particular identifier.
Cardinality | 1..1
Terminology Binding | NamingSystemIdentifierType (Required)
Type | code
Summary | true
Comments | Different identifier types may be used in different types of communications (OIDs for v3, URIs for FHIR, etc.). Other includes RUIDs from v3, standard v2 code name strings, etc.
NamingSystem.uniqueId.value
Element Id | NamingSystem.uniqueId.value
Definition | The string that should be sent over the wire to identify the code system or identifier system.
Cardinality | 1..1
Type | string
Summary | true
Comments | If the value is a URI intended for use as FHIR system identifier, the URI should not contain "" or "?" or "," since this makes escaping very difficult.
NamingSystem.uniqueId.preferred
Element Id | NamingSystem.uniqueId.preferred
Definition | Indicates whether this identifier is the "preferred" identifier of this type.
Cardinality | 0..1
Type | boolean
Meaning if Missing | If there are multiple ids, and one is labeled "preferred", then the assumption is that the others are not preferred. In the absence of any id marked as preferred, no inference can be drawn
NamingSystem.uniqueId.comment
Element Id | NamingSystem.uniqueId.comment
Definition | Notes about the past or intended usage of this identifier.

Cardinality | 0..1
Type | string
Comments | e.g. "must be used in Germany" or "was initially published in error with this value".
NamingSystem.uniqueId.period
Element Id | NamingSystem.uniqueId.period
Definition | Identifies the period of time over which this identifier is considered appropriate to refer to the naming system. Outside of this window, the identifier might be non-deterministic.
Cardinality | 0..1
Type | Period
Comments | Within a registry, a given identifier should only be "active" for a single namespace at a time. (Ideally, an identifier should only ever be associated with a single namespace across all time).

4.11.8 Resource NamingSystem - Mappings

URL: https://hl7.org/fhir/R4/namingsystem-mappings.html

4.11.8 Resource NamingSystem - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

HL7 v2 Mapping

RIM Mapping

Mappings for the namingsystem resource (see Mappings to Other Standards for further information & status).

4.11.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

NamingSystem
status | FiveWs.status
kind | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
responsible | FiveWs.witness
value | FiveWs.identifier

4.11.8.2 Workflow Pattern (http://hl7.org/fhir/workflow)

NamingSystem
status | Definition.status
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction

4.11.8.3 HL7 v2 Mapping (http://hl7.org/v2)

NamingSystem | N/A

4.11.8.4 RIM Mapping (http://hl7.org/v3)

NamingSystem | N/A

4.11.9 Resource NamingSystem - Extensions & Profiles

URL: https://hl7.org/fhir/R4/namingsystem-profiles.html

4.11.9 Resource NamingSystem - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the NamingSystem resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

4.11.9.1 Profiles

No Profiles defined for this resource

4.11.9.2 Extensions

replaces | NamingSystem | replaces | for General Extensions for use by FHIR Implementers

resource-approvalDate | NamingSystem | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | NamingSystem | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | NamingSystem | lastReviewDate | for Resource HL7 Extensions
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

4.11.9.3 Search Extensions

No Search Extensions defined for this resource

4.11.10 Resource NamingSystem - Operations

URL: https://hl7.org/fhir/R4/namingsystem-operations.html

4.11.10 Resource NamingSystem - Operations

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This resource has 1 operation operations associated with it:

$preferred-id | Fetch Preferred it

For more information about operations, including how they are invoked, see Operations.

4.11.10.1 Operation $preferred-id on NamingSystem

URL: https://hl7.org/fhir/R4/namingsystem-operation-preferred-id.html

4.11.10.1 Operation $preferred-id on NamingSystem

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: N/A

This operation returns the preferred identifiers for identifiers, and terminologies. The operation takes 2 parameters:

• a system identifier - either a URI, an OID, or a v2 table 0396 (other) code

• a code for what kind of identifier is desired (URI, OID, v2 table 0396 identifier)

and returns either the requested identifier, or an HTTP errors response with an OperationOutcome because either the provided identifier was not recognized, or the requested identiifer type is not known.

The principle use of this operation is when converting between v2, CDA and FHIR Identifier/CX/II and CodeableConcepts/C(N/W)E/CD but the operation may also find use when converting metadata such as profiles.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/NamingSystem-preferred-id
```

Formal Definition (as a OperationDefinition).

URL: [base]/NamingSystem/$preferred-id

This is an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
id | 1..1 | string | | | The server parses the provided id to see what type it is (mary a URI, an OID as a URI, a plain OID, or a v2 table 0396 code). If the server can't tell what type of identifier it is, it can try it as multiple types. It is an error if more than one system matches the provided identifier
type | 1..1 | code | NamingSystemIdentifierType (Required)
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
result | 1..1 | string | | | OIDs are return as plain OIDs (not the URI form).

Servers handle this request by finding the provided identifier in their known naming systems, and returning the requested identifier type (NamingSystem.uniqueId.type). If there are multiple possible identifiers of the specified type (e.g. multiple OIDs) the server returns an error.

If the server wishes, it can also look through all code systems and value sets it knows about when attempting to find the requested identifier

4.11.10.1.1 Examples

Request: Using GET

```
GET [base]/NamingSystem/$preferred-id?id=http://ns.electronichealth.net.au/id/hi/ihi/1.0&type=oid
```

Request: Using POST

```
POST [base]/NamingSystem/$preferred-id
[other headers]

{
  "resourceType" : "Parameters",
  "parameter" : [ {
     "name" : "id",
     "valueString" : "http://ns.electronichealth.net.au/id/hi/ihi/1.0"
   }, {
     "name" : "type",
     "valueCode" : "oid"
   }
  ]
}
```

Response:

```
HTTP/1.1 200 OK
[other headers]

{
  "resourceType" : "Parameters",
  "parameter" : [ {
     "name" : "result",
     "valueString" : "1.2.36.1.2001.1003.0"
   }
  ]
}
```

For more information about operations, including how they are invoked, see Operations.

4.11.11 Resource NamingSystem - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/namingsystem-version-maps.html

No validation errors - all conversions are clean

4.11.11 Resource NamingSystem - R3/R4 Conversions

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for NamingSystem.

Functional status for this map: 3 tests that all execute ok. 1 fail round-trip testing and 3 r3 resources are invalid (0 errors). (see documentation)

4.11.11.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/NamingSystem3to4" = "R3 to R4 Conversions for NamingSystem"

uses "http://hl7.org/fhir/3.0/StructureDefinition/NamingSystem" alias NamingSystemR3 as source
uses "http://hl7.org/fhir/StructureDefinition/NamingSystem" alias NamingSystem as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group NamingSystem(source src : NamingSystemR3, target tgt : NamingSystem) extends DomainResource <<type+>> {
  src.name -> tgt.name;
  src.status -> tgt.status;
  src.kind -> tgt.kind;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.responsible -> tgt.responsible;
  src.type -> tgt.type;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.usage -> tgt.usage;
  src.uniqueId as vs0 -> tgt.uniqueId as vt0 then {
    vs0.type -> vt0.type;
    vs0.value -> vt0.value;
    vs0.preferred -> vt0.preferred;
    vs0.comment -> vt0.comment;
    vs0.period -> vt0.period;
  };
}
```

4.11.11.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/NamingSystem4to3" = "R4 to R3 Conversion for NamingSystem"

uses "http://hl7.org/fhir/StructureDefinition/NamingSystem" alias NamingSystem as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/NamingSystem" alias NamingSystemR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group NamingSystem(source src : NamingSystem, target tgt : NamingSystemR3) extends DomainResource <<type+>> {
  src.name -> tgt.name;
  src.status -> tgt.status;
  src.kind -> tgt.kind;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.responsible -> tgt.responsible;
  src.type -> tgt.type;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.usage -> tgt.usage;
  src.uniqueId as vs0 -> tgt.uniqueId as vt0 then {
    vs0.type -> vt0.type;
    vs0.value -> vt0.value;
    vs0.preferred -> vt0.preferred;
    vs0.comment -> vt0.comment;
    vs0.period -> vt0.period;
  };
}
```

4.11.11.3 R4 Validation Errors

4.11.12 JSON Schema for NamingSystem

URL: https://hl7.org/fhir/R4/namingsystem.schema.json.html

4.11.12 JSON Schema for NamingSystem

FHIR Infrastructure Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for NamingSystem, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/NamingSystem",
  "$ref": "#/definitions/NamingSystem",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "NamingSystem": {
      "description": "A curated namespace that issues unique symbols within that namespace for the identification of
    concepts, people, devices, etc.  Represents a \"System\" used within the Identifier and Coding data types.",
      "properties": {
        "resourceType": {
          "description": "This is a NamingSystem resource",
          "const": "NamingSystem"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
```

```
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "A natural language name identifying the naming system. This name should be usable as an
identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this naming system. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "kind": {
      "description": "Indicates the purpose for the naming system - what kinds of things does it make unique?",
      "enum": [
        "codesystem",
        "identifier",
        "root"
      ]
    },
    "_kind": {
      "description": "Extensions for kind",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the naming system was published. The date must change
when the business version changes and it must change if the status code changes. In addition, it should change when
the substantive content of the naming system changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the naming system.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
```

```
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "responsible": {
      "description": "The name of the organization that is responsible for issuing identifiers or codes for this
namespace and ensuring their non-collision.",
      "$ref": "#/definitions/string"
    },
    "_responsible": {
      "description": "Extensions for responsible",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "Categorizes a naming system for easier search by grouping related naming systems.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "description": {
      "description": "A free text natural language description of the naming system from a consumer\u0027s
perspective. Details about what the namespace identifies including scope, granularity, version labeling, etc.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate naming system
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the naming system is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "usage": {
      "description": "Provides guidance on the use of the namespace, including the handling of formatting
characters, use of upper vs. lower case, etc.",
      "$ref": "#/definitions/string"
    },
    "_usage": {
      "description": "Extensions for usage",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "uniqueId": {
      "description": "Indicates how the system may be identified when referenced in electronic exchange.",
      "items": {
        "$ref": "#/definitions/NamingSystem_UniqueId"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "uniqueId",
    "resourceType"
  ]
},
"NamingSystem_UniqueId": {
  "description": "A curated namespace that issues unique symbols within that namespace for the identification of
concepts, people, devices, etc.  Represents a \"System\" used within the Identifier and Coding data types.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
```

```
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "type": {
          "description": "Identifies the unique identifier scheme used for this particular identifier.",
          "enum": [
            "oid",
            "uuid",
            "uri",
            "other"
          ]
        },
        "_type": {
          "description": "Extensions for type",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "value": {
          "description": "The string that should be sent over the wire to identify the code system or identifier
    system.",
          "$ref": "#/definitions/string"
        },
        "_value": {
          "description": "Extensions for value",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "preferred": {
          "description": "Indicates whether this identifier is the \"preferred\" identifier of this type.",
          "$ref": "#/definitions/boolean"
        },
        "_preferred": {
          "description": "Extensions for preferred",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "comment": {
          "description": "Notes about the past or intended usage of this identifier.",
          "$ref": "#/definitions/string"
        },
        "_comment": {
          "description": "Extensions for comment",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "period": {
          "description": "Identifies the period of time over which this identifier is considered appropriate to refer to
     the naming system.  Outside of this window, the identifier might be non-deterministic.",
          "$ref": "Period.schema.json#/definitions/Period"
        }
      },
      "additionalProperties": false
    }
  }
}
```


4.11.13 Resource NamingSystem - Maturity Dependency View

URL: https://hl7.org/fhir/R4/namingsystem-dependencies.html

4.11.13 Resource NamingSystem - Maturity Dependency View

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for NamingSystem FMM level 1

Name | Card. | Type | Dependency Analysis
NamingSystem | | DomainResource
name | 1..1 | string
status | 1..1 | code
kind | 1..1 | code
date | 1..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
responsible | 0..1 | string
type | 0..1 | CodeableConcept
description | 0..1 | markdown
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
usage | 0..1 | string
uniqueId | 1..* | BackboneElement

type | 1..1 | code
value | 1..1 | string
preferred | 0..1 | boolean
comment | 0..1 | string
period | 0..1 | Period

4.12 Resource TerminologyCapabilities - Content

URL: https://hl7.org/fhir/R4/terminologycapabilities.html
Vocabulary Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

A TerminologyCapabilities resource documents a set of capabilities (behaviors) of a FHIR Terminology Server that may be used as a statement of actual server functionality or a statement of required or desired server implementation.

4.12.1 Scope and Usage

The TerminologyCapabilities resource provides for a terminology server to describe how the underlying terminology service works.

A client can get the TerminologyCapabilities statement for a Terminology Service by using the capabilities API:

```
GET [base]/metadata?mode=terminology
```

This will return the TerminologyCapabilities resource that applies to the service itself.

4.12.2 Boundaries and Relationships

• The TerminologyCapabilities resource does not describe the API - see CapabilityStatement for this.

• The CodeSystem resource documents the inherent structure and capabilities of code system, whereas the TerminologyCapabilities resource documents what a server hosting the code system is capable of

4.12.3 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints
TerminologyCapabilities | ITU | | DomainResource | A statement of system capabilities + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: A Capability Statement SHALL have at least one of description, software, or implementation element. + Rule: If kind = instance, implementation must be present and software may be present + Rule: If kind = capability, implementation must be absent, software must be present + Rule: If kind = requirements, implementation and software must be absent Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this terminology capabilities, represented as a URI (globally unique)
version | Σ | 0..1 | string | Business version of the terminology capabilities
name | ΣI | 0..1 | string | Name for this terminology capabilities (computer friendly)
title | Σ | 0..1 | string | Name for this terminology capabilities (human friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 1..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the terminology capabilities
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for terminology capabilities (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this terminology capabilities is defined
copyright | Σ | 0..1 | markdown | Use and/or publishing restrictions
kind | ΣI | 1..1 | code | instance | capability | requirements CapabilityStatementKind (Required)
software | ΣI | 0..1 | BackboneElement | Software that is covered by this terminology capability statement
name | Σ | 1..1 | string | A name the software is known by
version | Σ | 0..1 | string | Version covered by this statement
implementation | ΣI | 0..1 | BackboneElement | If this describes a specific instance
description | Σ | 1..1 | string | Describes this specific instance
url | Σ | 0..1 | url | Base URL for the implementation

lockedDate | Σ | 0..1 | boolean | Whether lockedDate is supported
codeSystem | I | 0..* | BackboneElement | A code system supported by the server + Rule: If there is more than one version, a version code must be defined
uri | | 0..1 | canonical(CodeSystem) | URI for the Code System
version | | 0..* | BackboneElement | Version of Code System supported
code | ΣI | 0..1 | string | Version identifier for this version
isDefault | Σ | 0..1 | boolean | If this is the default version for this code system
compositional | | 0..1 | boolean | If compositional grammar is supported
language | | 0..* | code | Language Displays supported
filter | | 0..* | BackboneElement | Filter Properties supported
code | | 1..1 | code | Code of the property supported
op | | 1..* | code | Operations supported for the property
property | | 0..* | code | Properties supported for $lookup
subsumption | | 0..1 | boolean | Whether subsumption is supported
expansion | | 0..1 | BackboneElement | Information about the [ValueSet/$expand](valueset-operation-expand.html) operation
hierarchical | | 0..1 | boolean | Whether the server can return nested value sets
paging | | 0..1 | boolean | Whether the server supports paging on expansion
incomplete | | 0..1 | boolean | Allow request for incomplete expansions?
parameter | | 0..* | BackboneElement | Supported expansion parameter
name | | 1..1 | code | Expansion Parameter name
documentation | | 0..1 | string | Description of support for parameter
textFilter | | 0..1 | markdown | Documentation about text searching works
codeSearch | | 0..1 | code | explicit | all CodeSearchSupport (Required)
validateCode | | 0..1 | BackboneElement | Information about the [ValueSet/$validate-code](valueset-operation-validate-code.html) operation
translations | | 1..1 | boolean | Whether translations are validated
translation | | 0..1 | BackboneElement | Information about the [ConceptMap/$translate](conceptmap-operation-translate.html) operation
needsMap | | 1..1 | boolean | Whether the client must identify the map
closure | | 0..1 | BackboneElement | Information about the [ConceptMap/$closure](conceptmap-operation-closure.html) operation
translation | | 0..1 | boolean | If cross-system closure is supported
Documentation for this format

UML Diagram (Legend)

XML Template

```
<TerminologyCapabilities xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this terminology capabilities, represented as a URI (globally
    unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the terminology capabilities -->
 <name value="[string]"/><!--  0..1 Name for this terminology capabilities (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this terminology capabilities (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 1..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the terminology capabilities -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for terminology capabilities (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this terminology capabilities is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <kind value="[code]"/><!--  1..1 instance | capability | requirements -->
 <software>  <!--  0..1 Software that is covered by this terminology capability statement -->
  <name value="[string]"/><!-- 1..1 A name the software is known by -->
  <version value="[string]"/><!-- 0..1 Version covered by this statement -->
 </software>
 <implementation>  <!--  0..1 If this describes a specific instance -->
  <description value="[string]"/><!-- 1..1 Describes this specific instance -->
  <url value="[url]"/><!-- 0..1 Base URL for the implementation -->
 </implementation>
 <lockedDate value="[boolean]"/><!-- 0..1 Whether lockedDate is supported -->
 <codeSystem>  <!-- 0..* A code system supported by the server -->
  <uri><!-- 0..1 canonical(CodeSystem) URI for the Code System --></uri>
  <version>  <!-- 0..* Version of Code System supported -->
   <code value="[string]"/><!--  0..1 Version identifier for this version -->
   <isDefault value="[boolean]"/><!-- 0..1 If this is the default version for this code system -->
   <compositional value="[boolean]"/><!-- 0..1 If compositional grammar is supported -->
   <language value="[code]"/><!-- 0..* Language Displays supported -->
   <filter>  <!-- 0..* Filter Properties supported -->
    <code value="[code]"/><!-- 1..1 Code of the property supported -->
    <op value="[code]"/><!-- 1..* Operations supported for the property -->
   </filter>
   <property value="[code]"/><!-- 0..* Properties supported for $lookup -->
  </version>
  <subsumption value="[boolean]"/><!-- 0..1 Whether subsumption is supported -->
 </codeSystem>
 <expansion>  <!-- 0..1 Information about the [ValueSet/$expand](valueset-operation-expand.html) operation -->
  <hierarchical value="[boolean]"/><!-- 0..1 Whether the server can return nested value sets -->
```

```
  <paging value="[boolean]"/><!-- 0..1 Whether the server supports paging on expansion -->
  <incomplete value="[boolean]"/><!-- 0..1 Allow request for incomplete expansions? -->
  <parameter>  <!-- 0..* Supported expansion parameter -->
   <name value="[code]"/><!-- 1..1 Expansion Parameter name -->
   <documentation value="[string]"/><!-- 0..1 Description of support for parameter -->
  </parameter>
  <textFilter value="[markdown]"/><!-- 0..1 Documentation about text searching works -->
 </expansion>
 <codeSearch value="[code]"/><!-- 0..1 explicit | all -->
 <validateCode>  <!-- 0..1 Information about the [ValueSet/$validate-code](valueset-operation-validate-code.html)
    operation -->
  <translations value="[boolean]"/><!-- 1..1 Whether translations are validated -->
 </validateCode>
 <translation>  <!-- 0..1 Information about the [ConceptMap/$translate](conceptmap-operation-translate.html) operation
    -->
  <needsMap value="[boolean]"/><!-- 1..1 Whether the client must identify the map -->
 </translation>
 <closure>  <!-- 0..1 Information about the [ConceptMap/$closure](conceptmap-operation-closure.html) operation -->
  <translation value="[boolean]"/><!-- 0..1 If cross-system closure is supported -->
 </closure>
</TerminologyCapabilities>
```

JSON Template

```
{
  "resourceType" : "TerminologyCapabilities",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this terminology capabilities, represented as a URI (globally unique)
  "version" : "<string>", // Business version of the terminology capabilities
  "name" : "<string>", // C? Name for this terminology capabilities (computer friendly)
  "title" : "<string>", // Name for this terminology capabilities (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // R!  Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the terminology capabilities
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for terminology capabilities (if applicable)
  "purpose" : "<markdown>", // Why this terminology capabilities is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "kind" : "<code>", // C? R!  instance | capability | requirements
  "software" : { // C? Software that is covered by this terminology capability statement
    "name" : "<string>", // R!  A name the software is known by
    "version" : "<string>" // Version covered by this statement
  },
  "implementation" : { // C? If this describes a specific instance
    "description" : "<string>", // R!  Describes this specific instance
    "url" : "<url>" // Base URL for the implementation
  },
  "lockedDate" : <boolean>, // Whether lockedDate is supported
  "codeSystem" : [{ // A code system supported by the server
    "uri" : { canonical(CodeSystem) }, // URI for the Code System
    "version" : [{ // Version of Code System supported
      "code" : "<string>", // C? Version identifier for this version
      "isDefault" : <boolean>, // If this is the default version for this code system
      "compositional" : <boolean>, // If compositional grammar is supported
      "language" : ["<code>"], // Language Displays supported
      "filter" : [{ // Filter Properties supported
        "code" : "<code>", // R!  Code of the property supported
        "op" : ["<code>"] // R!  Operations supported for the property
      }],
      "property" : ["<code>"] // Properties supported for $lookup
    }],
    "subsumption" : <boolean> // Whether subsumption is supported
  }],
  "expansion" : { // Information about the [ValueSet/$expand](valueset-operation-expand.html) operation
    "hierarchical" : <boolean>, // Whether the server can return nested value sets
    "paging" : <boolean>, // Whether the server supports paging on expansion
    "incomplete" : <boolean>, // Allow request for incomplete expansions?
    "parameter" : [{ // Supported expansion parameter
      "name" : "<code>", // R!  Expansion Parameter name
      "documentation" : "<string>" // Description of support for parameter
    }],
    "textFilter" : "<markdown>" // Documentation about text searching works
  },
  "codeSearch" : "<code>", // explicit | all
  "validateCode" : { // Information about the [ValueSet/$validate-code](valueset-operation-validate-code.html) operation
    "translations" : <boolean> // R!  Whether translations are validated
  },
  "translation" : { // Information about the [ConceptMap/$translate](conceptmap-operation-translate.html) operation
    "needsMap" : <boolean> // R!  Whether the client must identify the map
  },
  "closure" : { // Information about the [ConceptMap/$closure](conceptmap-operation-closure.html) operation
    "translation" : <boolean> // If cross-system closure is supported
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:TerminologyCapabilities;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:TerminologyCapabilities.url [ uri ]; # 0..1 Canonical identifier for this terminology capabilities, represented
    as a URI (globally unique)
  fhir:TerminologyCapabilities.version [ string ]; # 0..1 Business version of the terminology capabilities
  fhir:TerminologyCapabilities.name [ string ]; # 0..1 Name for this terminology capabilities (computer friendly)
  fhir:TerminologyCapabilities.title [ string ]; # 0..1 Name for this terminology capabilities (human friendly)
  fhir:TerminologyCapabilities.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:TerminologyCapabilities.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:TerminologyCapabilities.date [ dateTime ]; # 1..1 Date last changed
  fhir:TerminologyCapabilities.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:TerminologyCapabilities.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:TerminologyCapabilities.description [ markdown ]; # 0..1 Natural language description of the terminology
    capabilities
  fhir:TerminologyCapabilities.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to
    support
  fhir:TerminologyCapabilities.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for terminology
    capabilities (if applicable)
  fhir:TerminologyCapabilities.purpose [ markdown ]; # 0..1 Why this terminology capabilities is defined
  fhir:TerminologyCapabilities.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:TerminologyCapabilities.kind [ code ]; # 1..1 instance | capability | requirements
  fhir:TerminologyCapabilities.software [ # 0..1 Software that is covered by this terminology capability statement
    fhir:TerminologyCapabilities.software.name [ string ]; # 1..1 A name the software is known by
    fhir:TerminologyCapabilities.software.version [ string ]; # 0..1 Version covered by this statement
  ];
  fhir:TerminologyCapabilities.implementation [ # 0..1 If this describes a specific instance
    fhir:TerminologyCapabilities.implementation.description [ string ]; # 1..1 Describes this specific instance
    fhir:TerminologyCapabilities.implementation.url [ url ]; # 0..1 Base URL for the implementation
  ];
  fhir:TerminologyCapabilities.lockedDate [ boolean ]; # 0..1 Whether lockedDate is supported
  fhir:TerminologyCapabilities.codeSystem [ # 0..* A code system supported by the server
    fhir:TerminologyCapabilities.codeSystem.uri [ canonical(CodeSystem) ]; # 0..1 URI for the Code System
    fhir:TerminologyCapabilities.codeSystem.version [ # 0..* Version of Code System supported
      fhir:TerminologyCapabilities.codeSystem.version.code [ string ]; # 0..1 Version identifier for this version
      fhir:TerminologyCapabilities.codeSystem.version.isDefault [ boolean ]; # 0..1 If this is the default version for
    this code system
      fhir:TerminologyCapabilities.codeSystem.version.compositional [ boolean ]; # 0..1 If compositional grammar is
    supported
      fhir:TerminologyCapabilities.codeSystem.version.language [ code ], ... ; # 0..* Language Displays supported
      fhir:TerminologyCapabilities.codeSystem.version.filter [ # 0..* Filter Properties supported
        fhir:TerminologyCapabilities.codeSystem.version.filter.code [ code ]; # 1..1 Code of the property supported
        fhir:TerminologyCapabilities.codeSystem.version.filter.op [ code ], ... ; # 1..* Operations supported for the
    property
      ], ...;
      fhir:TerminologyCapabilities.codeSystem.version.property [ code ], ... ; # 0..* Properties supported for $lookup
    ], ...;
    fhir:TerminologyCapabilities.codeSystem.subsumption [ boolean ]; # 0..1 Whether subsumption is supported
  ], ...;
  fhir:TerminologyCapabilities.expansion [ # 0..1 Information about the [ValueSet/$expand](valueset-operation-
    expand.html) operation
    fhir:TerminologyCapabilities.expansion.hierarchical [ boolean ]; # 0..1 Whether the server can return nested value
    sets
    fhir:TerminologyCapabilities.expansion.paging [ boolean ]; # 0..1 Whether the server supports paging on expansion
    fhir:TerminologyCapabilities.expansion.incomplete [ boolean ]; # 0..1 Allow request for incomplete expansions?
    fhir:TerminologyCapabilities.expansion.parameter [ # 0..* Supported expansion parameter
      fhir:TerminologyCapabilities.expansion.parameter.name [ code ]; # 1..1 Expansion Parameter name
      fhir:TerminologyCapabilities.expansion.parameter.documentation [ string ]; # 0..1 Description of support for
    parameter
    ], ...;
    fhir:TerminologyCapabilities.expansion.textFilter [ markdown ]; # 0..1 Documentation about text searching works
  ];
  fhir:TerminologyCapabilities.codeSearch [ code ]; # 0..1 explicit | all
  fhir:TerminologyCapabilities.validateCode [ # 0..1 Information about the [ValueSet/$validate-code](valueset-operation-
    validate-code.html) operation
    fhir:TerminologyCapabilities.validateCode.translations [ boolean ]; # 1..1 Whether translations are validated
  ];
  fhir:TerminologyCapabilities.translation [ # 0..1 Information about the [ConceptMap/$translate](conceptmap-operation-
    translate.html) operation
    fhir:TerminologyCapabilities.translation.needsMap [ boolean ]; # 1..1 Whether the client must identify the map
  ];
  fhir:TerminologyCapabilities.closure [ # 0..1 Information about the [ConceptMap/$closure](conceptmap-operation-
    closure.html) operation
    fhir:TerminologyCapabilities.closure.translation [ boolean ]; # 0..1 If cross-system closure is supported
  ];
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = Not Mapped)

Structure

Name | Flags | Card. | Type | Description & Constraints
TerminologyCapabilities | ITU | | DomainResource | A statement of system capabilities + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: A Capability Statement SHALL have at least one of description, software, or implementation element. + Rule: If kind = instance, implementation must be present and software may be present + Rule: If kind = capability, implementation must be absent, software must be present + Rule: If kind = requirements, implementation and software must be absent Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this terminology capabilities, represented as a URI (globally unique)
version | Σ | 0..1 | string | Business version of the terminology capabilities
name | ΣI | 0..1 | string | Name for this terminology capabilities (computer friendly)
title | Σ | 0..1 | string | Name for this terminology capabilities (human friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
date | Σ | 1..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the terminology capabilities
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for terminology capabilities (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this terminology capabilities is defined
copyright | Σ | 0..1 | markdown | Use and/or publishing restrictions
kind | ΣI | 1..1 | code | instance | capability | requirements CapabilityStatementKind (Required)
software | ΣI | 0..1 | BackboneElement | Software that is covered by this terminology capability statement
name | Σ | 1..1 | string | A name the software is known by
version | Σ | 0..1 | string | Version covered by this statement
implementation | ΣI | 0..1 | BackboneElement | If this describes a specific instance
description | Σ | 1..1 | string | Describes this specific instance
url | Σ | 0..1 | url | Base URL for the implementation
lockedDate | Σ | 0..1 | boolean | Whether lockedDate is supported
codeSystem | I | 0..* | BackboneElement | A code system supported by the server + Rule: If there is more than one version, a version code must be defined
uri | | 0..1 | canonical(CodeSystem) | URI for the Code System
version | | 0..* | BackboneElement | Version of Code System supported
code | ΣI | 0..1 | string | Version identifier for this version
isDefault | Σ | 0..1 | boolean | If this is the default version for this code system
compositional | | 0..1 | boolean | If compositional grammar is supported
language | | 0..* | code | Language Displays supported
filter | | 0..* | BackboneElement | Filter Properties supported
code | | 1..1 | code | Code of the property supported
op | | 1..* | code | Operations supported for the property
property | | 0..* | code | Properties supported for $lookup
subsumption | | 0..1 | boolean | Whether subsumption is supported
expansion | | 0..1 | BackboneElement | Information about the [ValueSet/$expand](valueset-operation-expand.html) operation
hierarchical | | 0..1 | boolean | Whether the server can return nested value sets
paging | | 0..1 | boolean | Whether the server supports paging on expansion
incomplete | | 0..1 | boolean | Allow request for incomplete expansions?
parameter | | 0..* | BackboneElement | Supported expansion parameter
name | | 1..1 | code | Expansion Parameter name
documentation | | 0..1 | string | Description of support for parameter
textFilter | | 0..1 | markdown | Documentation about text searching works
codeSearch | | 0..1 | code | explicit | all CodeSearchSupport (Required)
validateCode | | 0..1 | BackboneElement | Information about the [ValueSet/$validate-code](valueset-operation-validate-code.html) operation
translations | | 1..1 | boolean | Whether translations are validated
translation | | 0..1 | BackboneElement | Information about the [ConceptMap/$translate](conceptmap-operation-translate.html) operation
needsMap | | 1..1 | boolean | Whether the client must identify the map
closure | | 0..1 | BackboneElement | Information about the [ConceptMap/$closure](conceptmap-operation-closure.html) operation
translation | | 0..1 | boolean | If cross-system closure is supported
Documentation for this format

UML Diagram (Legend)

XML Template

```
<TerminologyCapabilities xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this terminology capabilities, represented as a URI (globally
    unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the terminology capabilities -->
 <name value="[string]"/><!--  0..1 Name for this terminology capabilities (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this terminology capabilities (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
```

```
 <date value="[dateTime]"/><!-- 1..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the terminology capabilities -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for terminology capabilities (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this terminology capabilities is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <kind value="[code]"/><!--  1..1 instance | capability | requirements -->
 <software>  <!--  0..1 Software that is covered by this terminology capability statement -->
  <name value="[string]"/><!-- 1..1 A name the software is known by -->
  <version value="[string]"/><!-- 0..1 Version covered by this statement -->
 </software>
 <implementation>  <!--  0..1 If this describes a specific instance -->
  <description value="[string]"/><!-- 1..1 Describes this specific instance -->
  <url value="[url]"/><!-- 0..1 Base URL for the implementation -->
 </implementation>
 <lockedDate value="[boolean]"/><!-- 0..1 Whether lockedDate is supported -->
 <codeSystem>  <!-- 0..* A code system supported by the server -->
  <uri><!-- 0..1 canonical(CodeSystem) URI for the Code System --></uri>
  <version>  <!-- 0..* Version of Code System supported -->
   <code value="[string]"/><!--  0..1 Version identifier for this version -->
   <isDefault value="[boolean]"/><!-- 0..1 If this is the default version for this code system -->
   <compositional value="[boolean]"/><!-- 0..1 If compositional grammar is supported -->
   <language value="[code]"/><!-- 0..* Language Displays supported -->
   <filter>  <!-- 0..* Filter Properties supported -->
    <code value="[code]"/><!-- 1..1 Code of the property supported -->
    <op value="[code]"/><!-- 1..* Operations supported for the property -->
   </filter>
   <property value="[code]"/><!-- 0..* Properties supported for $lookup -->
  </version>
  <subsumption value="[boolean]"/><!-- 0..1 Whether subsumption is supported -->
 </codeSystem>
 <expansion>  <!-- 0..1 Information about the [ValueSet/$expand](valueset-operation-expand.html) operation -->
  <hierarchical value="[boolean]"/><!-- 0..1 Whether the server can return nested value sets -->
  <paging value="[boolean]"/><!-- 0..1 Whether the server supports paging on expansion -->
  <incomplete value="[boolean]"/><!-- 0..1 Allow request for incomplete expansions? -->
  <parameter>  <!-- 0..* Supported expansion parameter -->
   <name value="[code]"/><!-- 1..1 Expansion Parameter name -->
   <documentation value="[string]"/><!-- 0..1 Description of support for parameter -->
  </parameter>
  <textFilter value="[markdown]"/><!-- 0..1 Documentation about text searching works -->
 </expansion>
 <codeSearch value="[code]"/><!-- 0..1 explicit | all -->
 <validateCode>  <!-- 0..1 Information about the [ValueSet/$validate-code](valueset-operation-validate-code.html)
    operation -->
  <translations value="[boolean]"/><!-- 1..1 Whether translations are validated -->
 </validateCode>
 <translation>  <!-- 0..1 Information about the [ConceptMap/$translate](conceptmap-operation-translate.html) operation
    -->
  <needsMap value="[boolean]"/><!-- 1..1 Whether the client must identify the map -->
 </translation>
 <closure>  <!-- 0..1 Information about the [ConceptMap/$closure](conceptmap-operation-closure.html) operation -->
  <translation value="[boolean]"/><!-- 0..1 If cross-system closure is supported -->
 </closure>
</TerminologyCapabilities>
```

JSON Template

```
{
  "resourceType" : "TerminologyCapabilities",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this terminology capabilities, represented as a URI (globally unique)
  "version" : "<string>", // Business version of the terminology capabilities
  "name" : "<string>", // C? Name for this terminology capabilities (computer friendly)
  "title" : "<string>", // Name for this terminology capabilities (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // R!  Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the terminology capabilities
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for terminology capabilities (if applicable)
  "purpose" : "<markdown>", // Why this terminology capabilities is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "kind" : "<code>", // C? R!  instance | capability | requirements
  "software" : { // C? Software that is covered by this terminology capability statement
    "name" : "<string>", // R!  A name the software is known by
    "version" : "<string>" // Version covered by this statement
  },
  "implementation" : { // C? If this describes a specific instance
    "description" : "<string>", // R!  Describes this specific instance
    "url" : "<url>" // Base URL for the implementation
  },
  "lockedDate" : <boolean>, // Whether lockedDate is supported
  "codeSystem" : [{ // A code system supported by the server
    "uri" : { canonical(CodeSystem) }, // URI for the Code System
```

```
    "version" : [{ // Version of Code System supported
      "code" : "<string>", // C? Version identifier for this version
      "isDefault" : <boolean>, // If this is the default version for this code system
      "compositional" : <boolean>, // If compositional grammar is supported
      "language" : ["<code>"], // Language Displays supported
      "filter" : [{ // Filter Properties supported
        "code" : "<code>", // R!  Code of the property supported
        "op" : ["<code>"] // R!  Operations supported for the property
      }],
      "property" : ["<code>"] // Properties supported for $lookup
    }],
    "subsumption" : <boolean> // Whether subsumption is supported
  }],
  "expansion" : { // Information about the [ValueSet/$expand](valueset-operation-expand.html) operation
    "hierarchical" : <boolean>, // Whether the server can return nested value sets
    "paging" : <boolean>, // Whether the server supports paging on expansion
    "incomplete" : <boolean>, // Allow request for incomplete expansions?
    "parameter" : [{ // Supported expansion parameter
      "name" : "<code>", // R!  Expansion Parameter name
      "documentation" : "<string>" // Description of support for parameter
    }],
    "textFilter" : "<markdown>" // Documentation about text searching works
  },
  "codeSearch" : "<code>", // explicit | all
  "validateCode" : { // Information about the [ValueSet/$validate-code](valueset-operation-validate-code.html) operation
    "translations" : <boolean> // R!  Whether translations are validated
  },
  "translation" : { // Information about the [ConceptMap/$translate](conceptmap-operation-translate.html) operation
    "needsMap" : <boolean> // R!  Whether the client must identify the map
  },
  "closure" : { // Information about the [ConceptMap/$closure](conceptmap-operation-closure.html) operation
    "translation" : <boolean> // If cross-system closure is supported
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:TerminologyCapabilities;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:TerminologyCapabilities.url [ uri ]; # 0..1 Canonical identifier for this terminology capabilities, represented
    as a URI (globally unique)
  fhir:TerminologyCapabilities.version [ string ]; # 0..1 Business version of the terminology capabilities
  fhir:TerminologyCapabilities.name [ string ]; # 0..1 Name for this terminology capabilities (computer friendly)
  fhir:TerminologyCapabilities.title [ string ]; # 0..1 Name for this terminology capabilities (human friendly)
  fhir:TerminologyCapabilities.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:TerminologyCapabilities.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:TerminologyCapabilities.date [ dateTime ]; # 1..1 Date last changed
  fhir:TerminologyCapabilities.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:TerminologyCapabilities.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:TerminologyCapabilities.description [ markdown ]; # 0..1 Natural language description of the terminology
    capabilities
  fhir:TerminologyCapabilities.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to
    support
  fhir:TerminologyCapabilities.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for terminology
    capabilities (if applicable)
  fhir:TerminologyCapabilities.purpose [ markdown ]; # 0..1 Why this terminology capabilities is defined
  fhir:TerminologyCapabilities.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:TerminologyCapabilities.kind [ code ]; # 1..1 instance | capability | requirements
  fhir:TerminologyCapabilities.software [ # 0..1 Software that is covered by this terminology capability statement
    fhir:TerminologyCapabilities.software.name [ string ]; # 1..1 A name the software is known by
    fhir:TerminologyCapabilities.software.version [ string ]; # 0..1 Version covered by this statement
  ];
  fhir:TerminologyCapabilities.implementation [ # 0..1 If this describes a specific instance
    fhir:TerminologyCapabilities.implementation.description [ string ]; # 1..1 Describes this specific instance
    fhir:TerminologyCapabilities.implementation.url [ url ]; # 0..1 Base URL for the implementation
  ];
  fhir:TerminologyCapabilities.lockedDate [ boolean ]; # 0..1 Whether lockedDate is supported
  fhir:TerminologyCapabilities.codeSystem [ # 0..* A code system supported by the server
    fhir:TerminologyCapabilities.codeSystem.uri [ canonical(CodeSystem) ]; # 0..1 URI for the Code System
    fhir:TerminologyCapabilities.codeSystem.version [ # 0..* Version of Code System supported
      fhir:TerminologyCapabilities.codeSystem.version.code [ string ]; # 0..1 Version identifier for this version
      fhir:TerminologyCapabilities.codeSystem.version.isDefault [ boolean ]; # 0..1 If this is the default version for
    this code system
      fhir:TerminologyCapabilities.codeSystem.version.compositional [ boolean ]; # 0..1 If compositional grammar is
    supported
      fhir:TerminologyCapabilities.codeSystem.version.language [ code ], ... ; # 0..* Language Displays supported
      fhir:TerminologyCapabilities.codeSystem.version.filter [ # 0..* Filter Properties supported
        fhir:TerminologyCapabilities.codeSystem.version.filter.code [ code ]; # 1..1 Code of the property supported
        fhir:TerminologyCapabilities.codeSystem.version.filter.op [ code ], ... ; # 1..* Operations supported for the
    property
      ], ...;
      fhir:TerminologyCapabilities.codeSystem.version.property [ code ], ... ; # 0..* Properties supported for $lookup
    ], ...;
    fhir:TerminologyCapabilities.codeSystem.subsumption [ boolean ]; # 0..1 Whether subsumption is supported
```

```
  ], ...;
  fhir:TerminologyCapabilities.expansion [ # 0..1 Information about the [ValueSet/$expand](valueset-operation-
    expand.html) operation
    fhir:TerminologyCapabilities.expansion.hierarchical [ boolean ]; # 0..1 Whether the server can return nested value
    sets
    fhir:TerminologyCapabilities.expansion.paging [ boolean ]; # 0..1 Whether the server supports paging on expansion
    fhir:TerminologyCapabilities.expansion.incomplete [ boolean ]; # 0..1 Allow request for incomplete expansions?
    fhir:TerminologyCapabilities.expansion.parameter [ # 0..* Supported expansion parameter
      fhir:TerminologyCapabilities.expansion.parameter.name [ code ]; # 1..1 Expansion Parameter name
      fhir:TerminologyCapabilities.expansion.parameter.documentation [ string ]; # 0..1 Description of support for
    parameter
    ], ...;
    fhir:TerminologyCapabilities.expansion.textFilter [ markdown ]; # 0..1 Documentation about text searching works
  ];
  fhir:TerminologyCapabilities.codeSearch [ code ]; # 0..1 explicit | all
  fhir:TerminologyCapabilities.validateCode [ # 0..1 Information about the [ValueSet/$validate-code](valueset-operation-
    validate-code.html) operation
    fhir:TerminologyCapabilities.validateCode.translations [ boolean ]; # 1..1 Whether translations are validated
  ];
  fhir:TerminologyCapabilities.translation [ # 0..1 Information about the [ConceptMap/$translate](conceptmap-operation-
    translate.html) operation
    fhir:TerminologyCapabilities.translation.needsMap [ boolean ]; # 1..1 Whether the client must identify the map
  ];
  fhir:TerminologyCapabilities.closure [ # 0..1 Information about the [ConceptMap/$closure](conceptmap-operation-
    closure.html) operation
    fhir:TerminologyCapabilities.closure.translation [ boolean ]; # 0..1 If cross-system closure is supported
  ];
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = Not Mapped)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

4.12.3.1 Terminology Bindings

Path | Definition | Type | Reference

TerminologyCapabilities.status | The lifecycle status of an artifact. | Required | PublicationStatus

TerminologyCapabilities.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet

TerminologyCapabilities.kind | How a capability statement is intended to be used. | Required | CapabilityStatementKind

TerminologyCapabilities.codeSearch | The degree to which the server supports the code search parameter on ValueSet, if it is supported. | Required | CodeSearchSupport

4.12.3.2 Constraints

id | Level | Location | Description | Expression

tcp-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

tcp-1 | Rule | TerminologyCapabilities.codeSystem | If there is more than one version, a version code must be defined | version.count() > 1 implies version.all(code.exists())

tcp-2 | Rule | (base) | A Capability Statement SHALL have at least one of description, software, or implementation element. | (description.count() + software.count() + implementation.count()) > 0

tcp-3 | Rule | (base) | If kind = instance, implementation must be present and software may be present | (kind != 'instance') or implementation.exists()

tcp-4 | Rule | (base) | If kind = capability, implementation must be absent, software must be present | (kind != 'capability') or (implementation.exists().not() and software.exists())

tcp-5 | Rule | (base) | If kind = requirements, implementation and software must be absent | (kind!='requirements') or (implementation.exists().not() and software.exists().not())

4.12.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

context | token | A use context assigned to the terminology capabilities | (TerminologyCapabilities.useContext.value as CodeableConcept)

context-quantity | quantity | A quantity- or range-valued use context assigned to the terminology capabilities | (TerminologyCapabilities.useContext.value as Quantity) | (TerminologyCapabilities.useContext.value as Range)

context-type | token | A type of use context assigned to the terminology capabilities | TerminologyCapabilities.useContext.code

context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the terminology capabilities | On TerminologyCapabilities.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)

context-type-value | composite | A use context type and value assigned to the terminology capabilities | On TerminologyCapabilities.useContext: context-type: code context: value.as(CodeableConcept)

date | date | The terminology capabilities publication date | TerminologyCapabilities.date

description | string | The description of the terminology capabilities | TerminologyCapabilities.description

jurisdiction | token | Intended jurisdiction for the terminology capabilities | TerminologyCapabilities.jurisdiction

name | string | Computationally friendly name of the terminology capabilities | TerminologyCapabilities.name
publisher | string | Name of the publisher of the terminology capabilities | TerminologyCapabilities.publisher
status | token | The current status of the terminology capabilities | TerminologyCapabilities.status
title | string | The human-friendly name of the terminology capabilities | TerminologyCapabilities.title
url | uri | The uri that identifies the terminology capabilities | TerminologyCapabilities.url
version | token | The business version of the terminology capabilities | TerminologyCapabilities.version

4.12.5 Resource TerminologyCapabilities - Examples

URL: https://hl7.org/fhir/R4/terminologycapabilities-examples.html

4.12.5 Resource TerminologyCapabilities - Examples

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
General Condition Example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

4.12.6 Resource TerminologyCapabilities - Detailed Descriptions

URL: https://hl7.org/fhir/R4/terminologycapabilities-definitions.html

4.12.6 Resource TerminologyCapabilities - Detailed Descriptions

Vocabulary Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the TerminologyCapabilities resource.

TerminologyCapabilities
Element Id | TerminologyCapabilities
Definition | A TerminologyCapabilities resource documents a set of capabilities (behaviors) of a FHIR Terminology Server that may be used as a statement of actual server functionality or a statement of required or desired server implementation.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element tcp-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}') tcp-2 Rule A Capability Statement SHALL have at least one of description, software, or implementation element. (description.count() + software.count() + implementation.count()) > 0 tcp-3 Rule If kind = instance, implementation must be present and software may be present (kind != 'instance') or implementation.exists() tcp-4 Rule If kind = capability, implementation must be absent, software must be present (kind != 'capability') or (implementation.exists().not() and software.exists()) tcp-5 Rule If kind = requirements, implementation and software must be absent (kind!='requirements') or (implementation.exists().not() and software.exists().not())
Defined on this element
tcp-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
tcp-2 | Rule | A Capability Statement SHALL have at least one of description, software, or implementation element. | (description.count() + software.count() + implementation.count()) > 0
tcp-3 | Rule | If kind = instance, implementation must be present and software may be present | (kind != 'instance') or implementation.exists()
tcp-4 | Rule | If kind = capability, implementation must be absent, software must be present | (kind != 'capability') or (implementation.exists().not() and software.exists())
tcp-5 | Rule | If kind = requirements, implementation and software must be absent | (kind!='requirements') or (implementation.exists().not() and software.exists().not())
TerminologyCapabilities.url
Element Id | TerminologyCapabilities.url
Definition | An absolute URI that is used to identify this terminology capabilities when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this terminology capabilities is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the terminology capabilities is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the terminology capabilities to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
TerminologyCapabilities.version
Element Id | TerminologyCapabilities.version
Definition | The identifier that is used to identify this version of the terminology capabilities when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the terminology capabilities author and is not expected to be

globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different terminology capabilities instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the terminology capabilities with the format [url]|[version].
TerminologyCapabilities.name
Element Id | TerminologyCapabilities.name
Definition | A natural language name identifying the terminology capabilities. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
TerminologyCapabilities.title
Element Id | TerminologyCapabilities.title
Definition | A short, descriptive, user-friendly title for the terminology capabilities.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
TerminologyCapabilities.status
Element Id | TerminologyCapabilities.status
Definition | The status of this terminology capabilities. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of terminology capabilitiess that are appropriate for use versus not.This is not intended for use with actual capability statements, but where capability statements are used to describe possible or desired systems.
TerminologyCapabilities.experimental
Element Id | TerminologyCapabilities.experimental
Definition | A Boolean value to indicate that this terminology capabilities is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level terminology capabilities.
Summary | true
Comments | Allows filtering of terminology capabilitiess that are appropriate for use versus not.
TerminologyCapabilities.date
Element Id | TerminologyCapabilities.date
Definition | The date (and optionally time) when the terminology capabilities was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the terminology capabilities changes.
Cardinality | 1..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the terminology capabilities. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
TerminologyCapabilities.publisher
Element Id | TerminologyCapabilities.publisher
Definition | The name of the organization or individual that published the terminology capabilities.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the terminology capabilities. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the terminology capabilities is the organization or individual primarily responsible for the maintenance and upkeep of the terminology capabilities. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary

point of contact for questions or issues with the terminology capabilities. This item SHOULD be populated unless the information is available from context.
TerminologyCapabilities.contact
Element Id | TerminologyCapabilities.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
TerminologyCapabilities.description
Element Id | TerminologyCapabilities.description
Definition | A free text natural language description of the terminology capabilities from a consumer's perspective. Typically, this is used when the capability statement describes a desired rather than an actual solution, for example as a formal expression of requirements as part of an RFP.
Cardinality | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the terminology capabilities was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the terminology capabilities as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the terminology capabilities is presumed to be the predominant language in the place the terminology capabilities was created).This does not need to be populated if the description is adequately implied by the software or implementation details.
TerminologyCapabilities.useContext
Element Id | TerminologyCapabilities.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate terminology capabilities instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
TerminologyCapabilities.jurisdiction
Element Id | TerminologyCapabilities.jurisdiction
Definition | A legal or geographic region in which the terminology capabilities is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the terminology capabilities to be used in jurisdictions other than those for which it was originally designed or intended.
TerminologyCapabilities.purpose
Element Id | TerminologyCapabilities.purpose
Definition | Explanation of why this terminology capabilities is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the terminology capabilities. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this terminology capabilities.
TerminologyCapabilities.copyright
Element Id | TerminologyCapabilities.copyright
Definition | A copyright statement relating to the terminology capabilities and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the terminology capabilities.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the terminology capabilities and/or its content.
Alternate Names | License; Restrictions
Summary | true
TerminologyCapabilities.kind
Element Id | TerminologyCapabilities.kind
Definition | The way that this statement is intended to be used, to describe an actual running instance of software, a particular product (kind, not instance of software) or a class of implementation (e.g. a desired purchase).
Cardinality | 1..1
Terminology Binding | CapabilityStatementKind (Required)
Type | code
Requirements | Allow searching the 3 modes.
Summary | true
Invariants | Affect this element tcp-3 Rule If kind = instance, implementation must be present and software may be present (kind != 'instance') or implementation.exists() tcp-4 Rule If kind = capability, implementation must be absent, software must be present (kind != 'capability') or (implementation.exists().not() and software.exists()) tcp-5 Rule If kind = requirements, implementation and software must be absent (kind!='requirements') or (implementation.exists().not() and software.exists().not())
Affect this element

tcp-3 | Rule | If kind = instance, implementation must be present and software may be present | (kind != 'instance') or implementation.exists()
tcp-4 | Rule | If kind = capability, implementation must be absent, software must be present | (kind != 'capability') or (implementation.exists().not() and software.exists())
tcp-5 | Rule | If kind = requirements, implementation and software must be absent | (kind!='requirements') or (implementation.exists().not() and software.exists().not())
TerminologyCapabilities.software
Element Id | TerminologyCapabilities.software
Definition | Software that is covered by this terminology capability statement. It is used when the statement describes the capabilities of a particular software version, independent of an installation.
Cardinality | 0..1
Summary | true
Invariants | Affect this element tcp-2 Rule A Capability Statement SHALL have at least one of description, software, or implementation element. (description.count() + software.count() + implementation.count()) > 0
Affect this element
tcp-2 | Rule | A Capability Statement SHALL have at least one of description, software, or implementation element. | (description.count() + software.count() + implementation.count()) > 0
TerminologyCapabilities.software.name
Element Id | TerminologyCapabilities.software.name
Definition | Name the software is known by.
Cardinality | 1..1
Type | string
Summary | true
TerminologyCapabilities.software.version
Element Id | TerminologyCapabilities.software.version
Definition | The version identifier for the software covered by this statement.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | If possible, a version should be specified, as statements are likely to be different for different versions of software.
TerminologyCapabilities.implementation
Element Id | TerminologyCapabilities.implementation
Definition | Identifies a specific implementation instance that is described by the terminology capability statement - i.e. a particular installation, rather than the capabilities of a software program.
Cardinality | 0..1
Summary | true
Invariants | Affect this element tcp-2 Rule A Capability Statement SHALL have at least one of description, software, or implementation element. (description.count() + software.count() + implementation.count()) > 0 tcp-5 Rule If kind = requirements, implementation and software must be absent (kind!='requirements') or (implementation.exists().not() and software.exists().not())
Affect this element
tcp-2 | Rule | A Capability Statement SHALL have at least one of description, software, or implementation element. | (description.count() + software.count() + implementation.count()) > 0
tcp-5 | Rule | If kind = requirements, implementation and software must be absent | (kind!='requirements') or (implementation.exists().not() and software.exists().not())
TerminologyCapabilities.implementation.description
Element Id | TerminologyCapabilities.implementation.description
Definition | Information about the specific installation that this terminology capability statement relates to.
Cardinality | 1..1
Type | string
Summary | true
TerminologyCapabilities.implementation.url
Element Id | TerminologyCapabilities.implementation.url
Definition | An absolute base URL for the implementation.
Cardinality | 0..1
Type | url
Summary | true
TerminologyCapabilities.lockedDate
Element Id | TerminologyCapabilities.lockedDate
Definition | Whether the server supports lockedDate.
Cardinality | 0..1
Type | boolean
Summary | true
TerminologyCapabilities.codeSystem
Element Id | TerminologyCapabilities.codeSystem
Definition | Identifies a code system that is supported by the server. If there is a no code system URL, then this declares the general assumptions a client can make about support for any CodeSystem resource.
Cardinality | 0..*
Alternate Names | 1
Comments | The code system - identified by its system URL - may also be declared explicitly as a Code System Resource at /CodeSystem, but it might not be.
Invariants | Defined on this element tcp-1 Rule If there is more than one version, a version code must be defined version.count() > 1 implies version.all(code.exists())

Defined on this element
tcp-1 | Rule | If there is more than one version, a version code must be defined | version.count() > 1 implies version.all(code.exists())
TerminologyCapabilities.codeSystem.uri
Element Id | TerminologyCapabilities.codeSystem.uri
Definition | URI for the Code System.
Cardinality | 0..1
Type | canonical(CodeSystem)
TerminologyCapabilities.codeSystem.version
Element Id | TerminologyCapabilities.codeSystem.version
Definition | For the code system, a list of versions that are supported by the server.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..*
Comments | Language translations might not be available for all codes.
TerminologyCapabilities.codeSystem.version.code
Element Id | TerminologyCapabilities.codeSystem.version.code
Definition | For version-less code systems, there should be a single version with no identifier.
Cardinality | 0..1
Type | string
Summary | true
Invariants | Affect this element tcp-1 Rule If there is more than one version, a version code must be defined version.count() > 1 implies version.all(code.exists())
Affect this element
tcp-1 | Rule | If there is more than one version, a version code must be defined | version.count() > 1 implies version.all(code.exists())
TerminologyCapabilities.codeSystem.version.isDefault
Element Id | TerminologyCapabilities.codeSystem.version.isDefault
Definition | If this is the default version for this code system.
Cardinality | 0..1
Type | boolean
Summary | true
TerminologyCapabilities.codeSystem.version.compositional
Element Id | TerminologyCapabilities.codeSystem.version.compositional
Definition | If the compositional grammar defined by the code system is supported.
Cardinality | 0..1
Type | boolean
TerminologyCapabilities.codeSystem.version.language
Element Id | TerminologyCapabilities.codeSystem.version.language
Definition | Language Displays supported.
Cardinality | 0..*
Type | code
TerminologyCapabilities.codeSystem.version.filter
Element Id | TerminologyCapabilities.codeSystem.version.filter
Definition | Filter Properties supported.
Cardinality | 0..*
TerminologyCapabilities.codeSystem.version.filter.code
Element Id | TerminologyCapabilities.codeSystem.version.filter.code
Definition | Code of the property supported.
Cardinality | 1..1
Type | code
TerminologyCapabilities.codeSystem.version.filter.op
Element Id | TerminologyCapabilities.codeSystem.version.filter.op
Definition | Operations supported for the property.
Cardinality | 1..*
Type | code
TerminologyCapabilities.codeSystem.version.property
Element Id | TerminologyCapabilities.codeSystem.version.property
Definition | Properties supported for $lookup.
Cardinality | 0..*
Type | code
TerminologyCapabilities.codeSystem.subsumption
Element Id | TerminologyCapabilities.codeSystem.subsumption
Definition | True if subsumption is supported for this version of the code system.
Cardinality | 0..1
Type | boolean
TerminologyCapabilities.expansion
Element Id | TerminologyCapabilities.expansion
Definition | Information about the ValueSet/$expand operation.
Cardinality | 0..1
TerminologyCapabilities.expansion.hierarchical
Element Id | TerminologyCapabilities.expansion.hierarchical
Definition | Whether the server can return nested value sets.

Cardinality | 0..1
Type | boolean
TerminologyCapabilities.expansion.paging
Element Id | TerminologyCapabilities.expansion.paging
Definition | Whether the server supports paging on expansion.
Cardinality | 0..1
Type | boolean
TerminologyCapabilities.expansion.incomplete
Element Id | TerminologyCapabilities.expansion.incomplete
Definition | Allow request for incomplete expansions?
Cardinality | 0..1
Type | boolean
TerminologyCapabilities.expansion.parameter
Element Id | TerminologyCapabilities.expansion.parameter
Definition | Supported expansion parameter.
Cardinality | 0..*
TerminologyCapabilities.expansion.parameter.name
Element Id | TerminologyCapabilities.expansion.parameter.name
Definition | Expansion Parameter name.
Cardinality | 1..1
Type | code
TerminologyCapabilities.expansion.parameter.documentation
Element Id | TerminologyCapabilities.expansion.parameter.documentation
Definition | Description of support for parameter.
Cardinality | 0..1
Type | string
TerminologyCapabilities.expansion.textFilter
Element Id | TerminologyCapabilities.expansion.textFilter
Definition | Documentation about text searching works.
Cardinality | 0..1
Type | markdown
Comments | This documentation should cover things like case sensitivity, use of punctuation if not ignored, what wild cards are supported (if any), whether text is starts with or contains, and whether word order matters.
TerminologyCapabilities.codeSearch
Element Id | TerminologyCapabilities.codeSearch
Definition | The degree to which the server supports the code search parameter on ValueSet, if it is supported.
Cardinality | 0..1
Terminology Binding | CodeSearchSupport (Required)
Type | code
Comments | See notes on the ValueSet resource.
TerminologyCapabilities.validateCode
Element Id | TerminologyCapabilities.validateCode
Definition | Information about the ValueSet/$validate-code operation.
Cardinality | 0..1
TerminologyCapabilities.validateCode.translations
Element Id | TerminologyCapabilities.validateCode.translations
Definition | Whether translations are validated.
Cardinality | 1..1
Type | boolean
TerminologyCapabilities.translation
Element Id | TerminologyCapabilities.translation
Definition | Information about the ConceptMap/$translate operation.
Cardinality | 0..1
TerminologyCapabilities.translation.needsMap
Element Id | TerminologyCapabilities.translation.needsMap
Definition | Whether the client must identify the map.
Cardinality | 1..1
Type | boolean
TerminologyCapabilities.closure
Element Id | TerminologyCapabilities.closure
Definition | Whether the $closure operation is supported.
Cardinality | 0..1
TerminologyCapabilities.closure.translation
Element Id | TerminologyCapabilities.closure.translation
Definition | If cross-system closure is supported.
Cardinality | 0..1
Type | boolean

4.12.7 Resource TerminologyCapabilities - Mappings

URL: https://hl7.org/fhir/R4/terminologycapabilities-mappings.html

4.12.7 Resource TerminologyCapabilities - Mappings

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the terminologycapabilities resource (see Mappings to Other Standards for further information & status).

4.12.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

TerminologyCapabilities
url | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

4.12.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

TerminologyCapabilities
url | Definition.url
version | Definition.version
title | Definition.title
status | Definition.status
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright

4.12.7.3 RIM Mapping (http://hl7.org/v3)

TerminologyCapabilities | n/a

4.12.8 Resource TerminologyCapabilities - Extensions & Profiles

URL: https://hl7.org/fhir/R4/terminologycapabilities-profiles.html

4.12.8 Resource TerminologyCapabilities - Extensions & Profiles

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the TerminologyCapabilities resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

4.12.8.1 Profiles

No Profiles defined for this resource

4.12.8.2 Extensions

replaces | TerminologyCapabilities | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers

replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

4.12.8.3 Search Extensions

No Search Extensions defined for this resource

4.12.9 Resource TerminologyCapabilities - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/terminologycapabilities-version-maps.html

n/a

4.12.9 Resource TerminologyCapabilities - R3/R4 Conversions

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for TerminologyCapabilities.

Functional status for this map: Not Mapped (see documentation)

4.12.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/TerminologyCapabilities3to4" = "R3 to R4 Conversions for TerminologyCapabilities"

uses "http://hl7.org/fhir/3.0/StructureDefinition/TerminologyCapabilities" alias TerminologyCapabilitiesR3 as source
uses "http://hl7.org/fhir/StructureDefinition/TerminologyCapabilities" alias TerminologyCapabilities as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group TerminologyCapabilities(source src : TerminologyCapabilitiesR3, target tgt : TerminologyCapabilities) extends
    DomainResource <<type+>> {
  src.url -> tgt.url;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.copyright -> tgt.copyright;
  src.lockedDate -> tgt.lockedDate;
  src.codeSystem as s -> tgt.codeSystem as t then TerminologyCapabilitiesCodeSystem(s, t);
  src.expansion as s -> tgt.expansion as t then TerminologyCapabilitiesExpansion(s, t);
  src.codeSearch -> tgt.codeSearch;
  src.validateCode as s -> tgt.validateCode as t then TerminologyCapabilitiesValidateCode(s, t);
  src.translation as s -> tgt.translation as t then TerminologyCapabilitiesTranslation(s, t);
  src.closure as s -> tgt.closure as t then TerminologyCapabilitiesClosure(s, t);
}

group TerminologyCapabilitiesCodeSystem(source src, target tgt) extends BackboneElement {
  src.uri -> tgt.uri;
  src.version as s -> tgt.version as t then TerminologyCapabilitiesCodeSystemVersion(s, t);
}

group TerminologyCapabilitiesCodeSystemVersion(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.isDefault -> tgt.isDefault;
  src.compositional -> tgt.compositional;
  src.language -> tgt.language;
  src.filter as s -> tgt.filter as t then TerminologyCapabilitiesCodeSystemVersionFilter(s, t);
  src.property -> tgt.property;
}

group TerminologyCapabilitiesCodeSystemVersionFilter(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.op -> tgt.op;
}

group TerminologyCapabilitiesExpansion(source src, target tgt) extends BackboneElement {
  src.hierarchical -> tgt.hierarchical;
  src.paging -> tgt.paging;
  src.incomplete -> tgt.incomplete;
  src.parameter as s -> tgt.parameter as t then TerminologyCapabilitiesExpansionParameter(s, t);
  src.textFilter -> tgt.textFilter;
}

group TerminologyCapabilitiesExpansionParameter(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.documentation -> tgt.documentation;
}
```

```
group TerminologyCapabilitiesValidateCode(source src, target tgt) extends BackboneElement {
  src.translations -> tgt.translations;
}

group TerminologyCapabilitiesTranslation(source src, target tgt) extends BackboneElement {
  src.needsMap -> tgt.needsMap;
}

group TerminologyCapabilitiesClosure(source src, target tgt) extends BackboneElement {
  src.translation -> tgt.translation;
}
```

4.12.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/TerminologyCapabilities4to3" = "R4 to R3 Conversion for TerminologyCapabilities"

uses "http://hl7.org/fhir/StructureDefinition/TerminologyCapabilities" alias TerminologyCapabilities as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/TerminologyCapabilities" alias TerminologyCapabilitiesR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group TerminologyCapabilities(source src : TerminologyCapabilitiesR3, target tgt : TerminologyCapabilities) extends
    DomainResource <<type+>> {
  src.url -> tgt.url;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.copyright -> tgt.copyright;
  src.lockedDate -> tgt.lockedDate;
  src.codeSystem as s -> tgt.codeSystem as t then TerminologyCapabilitiesCodeSystem(s, t);
  src.expansion as s -> tgt.expansion as t then TerminologyCapabilitiesExpansion(s, t);
  src.codeSearch -> tgt.codeSearch;
  src.validateCode as s -> tgt.validateCode as t then TerminologyCapabilitiesValidateCode(s, t);
  src.translation as s -> tgt.translation as t then TerminologyCapabilitiesTranslation(s, t);
  src.closure as s -> tgt.closure as t then TerminologyCapabilitiesClosure(s, t);
}

group TerminologyCapabilitiesCodeSystem(source src, target tgt) extends BackboneElement {
  src.uri -> tgt.uri;
  src.version as s -> tgt.version as t then TerminologyCapabilitiesCodeSystemVersion(s, t);
}

group TerminologyCapabilitiesCodeSystemVersion(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.isDefault -> tgt.isDefault;
  src.compositional -> tgt.compositional;
  src.language -> tgt.language;
  src.filter as s -> tgt.filter as t then TerminologyCapabilitiesCodeSystemVersionFilter(s, t);
  src.property -> tgt.property;
}

group TerminologyCapabilitiesCodeSystemVersionFilter(source src, target tgt) extends BackboneElement {
  src.code -> tgt.code;
  src.op -> tgt.op;
}

group TerminologyCapabilitiesExpansion(source src, target tgt) extends BackboneElement {
  src.hierarchical -> tgt.hierarchical;
  src.paging -> tgt.paging;
  src.incomplete -> tgt.incomplete;
  src.parameter as s -> tgt.parameter as t then TerminologyCapabilitiesExpansionParameter(s, t);
  src.textFilter -> tgt.textFilter;
}

group TerminologyCapabilitiesExpansionParameter(source src, target tgt) extends BackboneElement {
  src.name -> tgt.name;
  src.documentation -> tgt.documentation;
}

group TerminologyCapabilitiesValidateCode(source src, target tgt) extends BackboneElement {
  src.translations -> tgt.translations;
}

group TerminologyCapabilitiesTranslation(source src, target tgt) extends BackboneElement {
  src.needsMap -> tgt.needsMap;
}

group TerminologyCapabilitiesClosure(source src, target tgt) extends BackboneElement {
  src.translation -> tgt.translation;
}
```

4.12.9.3 R4 Validation Errors

4.12.10 JSON Schema for TerminologyCapabilities

URL: https://hl7.org/fhir/R4/terminologycapabilities.schema.json.html

4.12.10 JSON Schema for TerminologyCapabilities

Vocabulary Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for TerminologyCapabilities, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/TerminologyCapabilities",
  "$ref": "#/definitions/TerminologyCapabilities",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "TerminologyCapabilities": {
      "description": "A TerminologyCapabilities resource documents a set of capabilities (behaviors) of a FHIR
    Terminology Server that may be used as a statement of actual server functionality or a statement of required or
    desired server implementation.",
      "properties": {
        "resourceType": {
          "description": "This is a TerminologyCapabilities resource",
          "const": "TerminologyCapabilities"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
```

```
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this terminology capabilities when it is referenced
in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally
unique and SHOULD be a literal address at which at which an authoritative instance of this terminology capabilities
is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the
terminology capabilities is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the terminology capabilities when it
is referenced in a specification, model, design or instance. This is an arbitrary value managed by the terminology
capabilities author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd)
if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical
 sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the terminology capabilities. This name should be usable
as an identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the terminology capabilities.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this terminology capabilities. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this terminology capabilities is authored for testing
purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the terminology capabilities was published. The date must
 change when the business version changes and it must change if the status code changes. In addition, it should
change when the substantive content of the terminology capabilities changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the terminology capabilities.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the terminology capabilities from a
consumer\u0027s perspective. Typically, this is used when the capability statement describes a desired rather than
an actual solution, for example as a formal expression of requirements as part of an RFP.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate terminology capabilities
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the terminology capabilities is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this terminology capabilities is needed and why it has been designed as it
has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the terminology capabilities and/or its contents. Copyright
statements are generally legal restrictions on the use and publishing of the terminology capabilities.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "kind": {
      "description": "The way that this statement is intended to be used, to describe an actual running instance of
software, a particular product (kind, not instance of software) or a class of implementation (e.g. a desired
purchase).",
      "$ref": "#/definitions/code"
    },
    "_kind": {
      "description": "Extensions for kind",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "software": {
      "description": "Software that is covered by this terminology capability statement.  It is used when the
statement describes the capabilities of a particular software version, independent of an installation.",
      "$ref": "#/definitions/TerminologyCapabilities_Software"
    },
    "implementation": {
      "description": "Identifies a specific implementation instance that is described by the terminology capability
statement - i.e. a particular installation, rather than the capabilities of a software program.",
      "$ref": "#/definitions/TerminologyCapabilities_Implementation"
    },
    "lockedDate": {
      "description": "Whether the server supports lockedDate.",
      "$ref": "#/definitions/boolean"
    },
    "_lockedDate": {
      "description": "Extensions for lockedDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "codeSystem": {
      "description": "Identifies a code system that is supported by the server. If there is a no code system URL,
then this declares the general assumptions a client can make about support for any CodeSystem resource.",
      "items": {
        "$ref": "#/definitions/TerminologyCapabilities_CodeSystem"
      },
      "type": "array"
    },
    "expansion": {
```

```
      "description": "Information about the [ValueSet/$expand](valueset-operation-expand.html) operation.",
      "$ref": "#/definitions/TerminologyCapabilities_Expansion"
    },
    "codeSearch": {
      "description": "The degree to which the server supports the code search parameter on ValueSet, if it is
supported.",
      "enum": [
        "explicit",
        "all"
      ]
    },
    "_codeSearch": {
      "description": "Extensions for codeSearch",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "validateCode": {
      "description": "Information about the [ValueSet/$validate-code](valueset-operation-validate-code.html)
operation.",
      "$ref": "#/definitions/TerminologyCapabilities_ValidateCode"
    },
    "translation": {
      "description": "Information about the [ConceptMap/$translate](conceptmap-operation-translate.html)
operation.",
      "$ref": "#/definitions/TerminologyCapabilities_Translation"
    },
    "closure": {
      "description": "Whether the $closure operation is supported.",
      "$ref": "#/definitions/TerminologyCapabilities_Closure"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"TerminologyCapabilities_Software": {
  "description": "A TerminologyCapabilities resource documents a set of capabilities (behaviors) of a FHIR
Terminology Server that may be used as a statement of actual server functionality or a statement of required or
desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "Name the software is known by.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "version": {
      "description": "The version identifier for the software covered by this statement.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"TerminologyCapabilities_Implementation": {
```

```
  "description": "A TerminologyCapabilities resource documents a set of capabilities (behaviors) of a FHIR Terminology Server that may be used as a statement of actual server functionality or a statement of required or desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "description": {
      "description": "Information about the specific installation that this terminology capability statement relates to.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "url": {
      "description": "An absolute base URL for the implementation.",
      "$ref": "#/definitions/url"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"TerminologyCapabilities_CodeSystem": {
  "description": "A TerminologyCapabilities resource documents a set of capabilities (behaviors) of a FHIR Terminology Server that may be used as a statement of actual server functionality or a statement of required or desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "uri": {
      "description": "URI for the Code System.",
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      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "version": {
      "description": "For the code system, a list of versions that are supported by the server.",
      "items": {
        "$ref": "#/definitions/TerminologyCapabilities_Version"
      },
      "type": "array"
    },
    "subsumption": {
      "description": "True if subsumption is supported for this version of the code system.",
      "$ref": "#/definitions/boolean"
    },
    "_subsumption": {
      "description": "Extensions for subsumption",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"TerminologyCapabilities_Version": {
  "description": "A TerminologyCapabilities resource documents a set of capabilities (behaviors) of a FHIR
Terminology Server that may be used as a statement of actual server functionality or a statement of required or
desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "For version-less code systems, there should be a single version with no identifier.",
      "$ref": "#/definitions/string"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "isDefault": {
      "description": "If this is the default version for this code system.",
      "$ref": "#/definitions/boolean"
    },
    "_isDefault": {
      "description": "Extensions for isDefault",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "compositional": {
      "description": "If the compositional grammar defined by the code system is supported.",
      "$ref": "#/definitions/boolean"
    },
    "_compositional": {
      "description": "Extensions for compositional",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "Language Displays supported.",
      "items": {
        "$ref": "#/definitions/code"
      },
      "type": "array"
    },
    "_language": {
      "description": "Extensions for language",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
```

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      },
      "type": "array"
    },
    "filter": {
      "description": "Filter Properties supported.",
      "items": {
        "$ref": "#/definitions/TerminologyCapabilities_Filter"
      },
      "type": "array"
    },
    "property": {
      "description": "Properties supported for $lookup.",
      "items": {
        "$ref": "#/definitions/code"
      },
      "type": "array"
    },
    "_property": {
      "description": "Extensions for property",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"TerminologyCapabilities_Filter": {
  "description": "A TerminologyCapabilities resource documents a set of capabilities (behaviors) of a FHIR
Terminology Server that may be used as a statement of actual server functionality or a statement of required or
desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "Code of the property supported.",
      "$ref": "#/definitions/code"
    },
    "_code": {
      "description": "Extensions for code",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "op": {
      "description": "Operations supported for the property.",
      "items": {
        "$ref": "#/definitions/code"
      },
      "type": "array"
    },
    "_op": {
      "description": "Extensions for op",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"TerminologyCapabilities_Expansion": {
  "description": "A TerminologyCapabilities resource documents a set of capabilities (behaviors) of a FHIR
Terminology Server that may be used as a statement of actual server functionality or a statement of required or
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desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "hierarchical": {
      "description": "Whether the server can return nested value sets.",
      "$ref": "#/definitions/boolean"
    },
    "_hierarchical": {
      "description": "Extensions for hierarchical",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "paging": {
      "description": "Whether the server supports paging on expansion.",
      "$ref": "#/definitions/boolean"
    },
    "_paging": {
      "description": "Extensions for paging",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "incomplete": {
      "description": "Allow request for incomplete expansions?",
      "$ref": "#/definitions/boolean"
    },
    "_incomplete": {
      "description": "Extensions for incomplete",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "parameter": {
      "description": "Supported expansion parameter.",
      "items": {
        "$ref": "#/definitions/TerminologyCapabilities_Parameter"
      },
      "type": "array"
    },
    "textFilter": {
      "description": "Documentation about text searching works.",
      "$ref": "#/definitions/markdown"
    },
    "_textFilter": {
      "description": "Extensions for textFilter",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"TerminologyCapabilities_Parameter": {
  "description": "A TerminologyCapabilities resource documents a set of capabilities (behaviors) of a FHIR
Terminology Server that may be used as a statement of actual server functionality or a statement of required or
desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
```

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        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "name": {
      "description": "Expansion Parameter name.",
      "$ref": "#/definitions/code"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "documentation": {
      "description": "Description of support for parameter.",
      "$ref": "#/definitions/string"
    },
    "_documentation": {
      "description": "Extensions for documentation",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"TerminologyCapabilities_ValidateCode": {
  "description": "A TerminologyCapabilities resource documents a set of capabilities (behaviors) of a FHIR
Terminology Server that may be used as a statement of actual server functionality or a statement of required or
desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "translations": {
      "description": "Whether translations are validated.",
      "$ref": "#/definitions/boolean"
    },
    "_translations": {
      "description": "Extensions for translations",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"TerminologyCapabilities_Translation": {
  "description": "A TerminologyCapabilities resource documents a set of capabilities (behaviors) of a FHIR
Terminology Server that may be used as a statement of actual server functionality or a statement of required or
desired server implementation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
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    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "needsMap": {
          "description": "Whether the client must identify the map.",
          "$ref": "#/definitions/boolean"
        },
        "_needsMap": {
          "description": "Extensions for needsMap",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    },
    "TerminologyCapabilities_Closure": {
      "description": "A TerminologyCapabilities resource documents a set of capabilities (behaviors) of a FHIR
    Terminology Server that may be used as a statement of actual server functionality or a statement of required or
    desired server implementation.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string
    value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "translation": {
          "description": "If cross-system closure is supported.",
          "$ref": "#/definitions/boolean"
        },
        "_translation": {
          "description": "Extensions for translation",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```

4.12.11 Resource TerminologyCapabilities - Maturity Dependency View

URL: https://hl7.org/fhir/R4/terminologycapabilities-dependencies.html

4.12.11 Resource TerminologyCapabilities - Maturity Dependency View

Vocabulary Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for TerminologyCapabilities FMM level 0

Name | Card. | Type | Dependency Analysis
TerminologyCapabilities | | DomainResource
url | 0..1 | uri
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
date | 1..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
copyright | 0..1 | markdown
kind | 1..1 | code
software | 0..1 | BackboneElement
name | 1..1 | string
version | 0..1 | string
implementation | 0..1 | BackboneElement
description | 1..1 | string
url | 0..1 | url
lockedDate | 0..1 | boolean
codeSystem | 0..* | BackboneElement
uri | 0..1 | canonical(CodeSystem)
version | 0..* | BackboneElement
code | 0..1 | string
isDefault | 0..1 | boolean
compositional | 0..1 | boolean
language | 0..* | code
filter | 0..* | BackboneElement
code | 1..1 | code
op | 1..* | code
property | 0..* | code
subsumption | 0..1 | boolean
expansion | 0..1 | BackboneElement
hierarchical | 0..1 | boolean
paging | 0..1 | boolean
incomplete | 0..1 | boolean
parameter | 0..* | BackboneElement
name | 1..1 | code
documentation | 0..1 | string
textFilter | 0..1 | markdown
codeSearch | 0..1 | code
validateCode | 0..1 | BackboneElement
translations | 1..1 | boolean
translation | 0..1 | BackboneElement
needsMap | 1..1 | boolean
closure | 0..1 | BackboneElement
translation | 0..1 | boolean